

Approved Course Master

S.No.	Branch	Course Title	Courses ID	Content
1	COM	ENGLISH LANGUAGE AND COMMUNICATION (PREP)	ELC001	COURSE CONTENT: Unit-I: Vocabulary-I Word Associations, Formal and Informal Words, New Words(5 Lectures+ 2 Tutorials) Unit-II: Grammar and Usage-I Article, Preposition, Verb Forms, Correct Word Listening(5 Lectures+ 2 Tutorials) Unit-IV: Speaking-I Pronunciation, Voice and Speech, Accent, Volume, Clarity, Variety(7 Lectures+ 2 Tutorials) Unit-V: Reading-I Basic Reading Skills, Comprehension (7 Lectures+ 2 Tutorials) 4.READINGS: 4.1 TEXT BOOKS: 1. Basic English Usage. Michael Swan, OUP, 1984 2. Essential English Grammar. Raymond Murphy, CUP, 2012 3. Successful College Writing. Kathleen T. McWhorter, CUP, 2012 4. The Elements of Style. Strunk and White, 1979
2	CHM	GENERAL CHEMISTRY (PREP)	CHM002	COURSE CONTENT: SECTION: A (19 Lectures) 1.Atomic Structure: Schrodinger wave equation, wave Function, radial and angular wave functions, Aufbau and Pauli exclusion principles, Hund's maximum multiplicity rule, Various types of hybridization and shapes of simple inorganic molecules and ions. Valence Shell Electron Pair Repulsion theory. M.O. theory and its applications to homonuclear and heteronuclear diatomic molecules. Nomenclature of coordination compounds, isomerism in complexes, Sidgwick Effective Atomic number rule SECTION B (20 lectures) 1.Introduction to Analytical Chemistry: Applications of analytical chemistry to various fields. Classification of errors and sampling in analysis. 2.Quantitative Chemical Analysis: Types of quantitative chemical analysis. Classification of titrimetric analyses and the role of pH sensitive indicators in a titration. Interaction of radiation with matter. Fundamental Laws of Analytical absorption Spectroscopy. Validity of Beer's Law. 3.On-Line Analysers: Specification, selection and application of On-line analysers. 4. Structure and reactivity, James Hunecy 3. Analytical Chemistry, Bryan M. Ham, AihuiMaltan
3	BSBE	MOLECULAR VIROLOGY	BSE659	Content updated on 03 Nov, 2023.
4	DES	HUMAN MACHINE INTERACTION	DES638	 1. Introduction to Human Machine Interaction (HMI) § Machines, senses, and perception § Waves of HCI § Human Centred Design § Affordances § Modalities § GUIs, Gaze, and Gesture-based Interaction § Hand Interactions § Pointing and Icons § Haptic and sensing devices § Data Visualization § Interaction 1. HMI Applications § Wearables and Materials § Accessibility and Inclusive Design § Human AI Collaboration § Immersive & Ubiquitous Interaction § Metaverse Interaction (AR / VR / MR / XR) 1. Interaction Technologies Design - Software § Design Thinking § Visual Studio (C#) § Communicate between python and C# using socket programming § Wireless Communication (IoT Modules) § Multithreading for GUI § Actuators § Integrate sensors and actuators signals with the developed GUI § Deploy the entire system on Raspberry Pi or any single board computer (SBC) § Research § Quantitative Research § Qualitative Research § Problem Definition § Experiment Design § User Study § 1. HMI Design Project § Pick up a problem statement § Identify pain points of human users § Design a product which could provide the solution § Additional Exercises based on project requirement Based on students's interest and project requirements, they will be guided in learning and practicing the following in their projects. § Machine Learning § Applied Deep Learning
5	DES	HUMAN MACHINE INTERACTION	DES638A	 1. Introduction to Human Machine Interaction (HMI) § Machines, senses, and perception § Waves of HCI § Human Centred Design § Affordances § Modalities § GUIs, Gaze, and Gesture-based Interaction § Hand Interactions § Pointing and Icons § Haptic and sensing devices § Data Visualization § Interaction 1. HMI Applications § Wearables and Materials § Accessibility and Inclusive Design § Human AI Collaboration § Immersive & Ubiquitous Interaction § Metaverse Interaction (AR / VR / MR / XR) 1. Interaction Technologies Design - Software § Design Thinking § Visual Studio (C#) § Communicate between python and C# using socket programming § Wireless Communication (IoT Modules) § Multithreading for GUI § Actuators § Integrate sensors and actuators signals with the developed GUI § Deploy the entire system on Raspberry Pi or any single board computer (SBC) § Research § Quantitative Research § Qualitative Research § Problem Definition § Experiment Design § User Study § 1. HMI Design Project § Pick up a problem statement § Identify pain points of human users § Design a product which could provide the solution § Additional Exercises based on project requirement Based on students's interest and project requirements, they will be guided in learning and practicing the following in their projects. § Machine Learning § Applied Deep Learning
6	CGS	APPLIED COGNITIVE SCIENCE	CGS402A	Attached
7	CGS	INTRODUCTION TO COGNITIVE SCIENCE	CGS401A	Attached
8	SEE	THERMODYNAMICS OF ENERGY SYSTEMS	SEE604A	<p>Course Objectives This course provides the necessary background of thermodynamics to the students who did not get adequate exposure to the subject at the undergraduate level.</p> <p>Course Content 1. Properties of pure substance, Air standard cycles, Rankine cycle, vapor compression refrigeration cycle, vapor absorption refrigeration, properties of moist air, psychrometric chart, air-conditioning 2. Equilibrium constant, basics of electrochemistry, equilibrium electrochemistry, solar thermal and electrochemical energy conversion</p> <p>Lecture wise breakup S. N. Broad Title Topics No. 1. Properties of pure substance, Air standard cycles, Rankine cycle, vapor compression refrigeration cycle, vapor absorption refrigeration, properties of moist air, psychrometric chart, air-conditioning 2. Equilibrium constant, basics of electrochemistry, equilibrium electrochemistry, solar thermal and electrochemical energy conversion</p> <p>1. 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LEARNING				
15	HSS	POPULATION PROCESSES	SOC733A	<p>BASIC CONCEPTS IN DEMOGRAPHY; SOCIOLOGY OF POPULATION; RECENT DEVELOPMENTS IN POPULATION THEORY; POPULATION POLICIES; POPULATION RESEARCH METHODS; POPULATION
16	BSBE	OPTICAL MICROSCOPY & BIOIMAGING	BSE644A	BSE644A Optical Microscopy and Bioimaging For course content, please see the uploaded PDF file.
17	CSE	INTRODUCTION TO LAMBDA CALCULUS TYPES & MODELS	CS698V	Natural language (NL) refers to the language spoken/written by humans. NL is the primary mode of communication for humans. With the growth of the world wide web, data in the form of text has grown and automation and development of intelligent machines. This course will primarily focus on understanding and developing linguistic techniques, statistical learning algorithms and models for processing large text corpora and develop natural language understanding models from statistical regularities in large corpora of natural language texts while leveraging linguistics theories. Besides regular quizzes, students will work on a research problem in NLP.
18	MTH	TECHNIQUES IN COMBINATORICS	MTH678A	Indian Institute of Technology, Kanpur Proposal for a new course Course No.: MTH 6XXX (PG level) requisites: Instructor’s consent This is an advanced course and the number of students admitted for the course will be limited. Every student interested in the course must meet the prerequisites (MTH201A), Linear Algebra (MTH204A), Probability and Statistics (MSO 201A) as well as in Set theory and Logic (MTH302A) or Set theory and Discrete Mathematics (MTH202A). Some knowledge of combinatorics is recommended. Aims of the course: Combinatorics is an area of mathematics that, at elementary level, deals with problems in counting/enumeration, especially in the realm of finite sets. However, it also ranges from algebra, analysis, number theory, geometry, probability theory, and so on. Leon Mirsky has rightly said: "combinatorics is a range of linked studies which have something in common, namely, the attainment of a certain goal. This course will introduce techniques from three popular branches of combinatorics that are of areas of active research currently, namely extremal combinatorics, extremal solutions of the problems. Credits: 3-0-0-0 [9] Semester: Odd Semester Department/IDP: Mathematics and Computer Science Amit Kuber Other faculty members interested in teaching the course: Arbind Lal Course contents: 1. Extremal combinatorics with set theory, Katona theorem, Erdos Ko-Rado theorems [4 weeks] 2. Extremal graph theory: Mantel’s theorem, Turan’s theorem, Dirac’s theorem, Erdos-Stone-Simonovits theorem, Ramsey's theorem [4 weeks] 3. Additive combinatorics with dyadic groups: Discrete Fourier transform, Plancherel’s theorem, Bogolyubov’s lemma, Arithmetic regularity and removal Sets with small sumsets: Rusza’s covering lemma, Freiman’s theorem [5 weeks] Recommended books: 1. Bollob´s, Béla. Combinatorics. Cambridge University Press, 1986. 2. Bollob´s, Béla. Extremal graph theory. Courier Corporation, 2004. 3. Tao, Terence, and Van H. Vu. Additive combinatorics. Cambridge University Press, 2016.
19	MTH	INTRODUCTION TO HOMOTOPY THEORY	MTH722A	Content can be seen in the attched file
20	CHE	QUANTUM CHEMICAL DESIGN OF ELECTROCHEMICAL CATALYTIC AND SEPARATION MATERIALS	CHE687A	Quantum phenomena and its scope in engineering, Quantum Mechanics of Simple Systems, Models for Properties of Matter from Electronic and Atomic Structure, Ab-Intio Thermodynamics, “Grand Canonical Ensemble, Cells, Sabatier Principle and Industrial Catalysts Design
21	PHY	INTRODUCTORY BIOPHYSICS	PHY309A	<p>INDIAN INSTITUTE OF TECHNOLOGY KANPUR</p> <p>Department of Physics</p> <p>PHY 309/ SE 303: Introductory Biophysics Instructor: Prof. Debashish Chowdhury Pre-requisite: Basic physics from non-living matter. What makes living systems different from non-living ones are some ``processes'' that occur only in living systems. These dynamic processes are governed by the laws of physics. Biological systems are caused by real physical forces (and not by any `vital force'), the laws of thermodynamics impose constraints on which processes are possible, and laws of kinetics govern the complete without insight into the physics of these processes. This course is an elementary introduction to the physical principles underlying the key dynamic processes in living systems. (1) Conservative, dissipative and random forces; molecular binding/unbinding events, probing non-covalent ligandreceptor interactions with molecular force spectroscopy- slip bond versus catch bond in a macromolecule of life; free energy change and generalized chemical force; concepts of thermodynamic equilibrium and non-equilibrium steady-state; coupled processes and engines of Life. (2) Enzymology, fluctuating enzymatic reactions in a living cell- Michaelis-Menten form for the average rate; substrate specificity of an enzyme- induced fit versus conformational selection, kinetic proofreading Diffusion- transport in bacterial cells; transport by motor proteins in eukaryotic cells; cooperativity of motors in vesicular transport; bi-directional movements of organelles by competing motor proteins mitosis. (5) Polymers of life and template-directed polymerization: stretching and entropic spring; bending elasticity and persistence length; transcription and translation as examples of templated polymerization machinery; programmed errors and recoding. (6) Entropy, information, self-organization and emergence of life: Shannon entropy and information; evolution of genetic information and its controversies; plausible pathways for emergence of life- roles of energy and information.</p> <p>Text Books and References: 1. “Physical Biology of the Cell”, by L. D. Landau and E. M. Lifshitz (Elsevier, 1980). 2. “Biophysics: searching for principles”, by W. Bialek (Princeton Univ. Press, 1996). 3. “Mechanics of the Cell”, by D. Boal (Cambridge Univ. Press, 2012). 4. “Biophysics: searching for principles”, by W. Bialek (Princeton Univ. Press, 1996). 5. “Introduction to Biophysics”, by P. D. Bartlett (Cambridge Univ. Press, 2010). 6. “Biophysics: searching for principles”, by W. Bialek (Princeton Univ. 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S.No.	Branch	Course Title	Courses ID	Content
25	CSE	TOPICS IN LEARNING THEORY	CS777A	
26	AE	COMBUSTION DIAGNOSTICS	AE666A	
27	AE	NANO-PARTICLE AEROSOL DYNAMICS	AE667A	
28	HSS	COGNITIVE LINGUISTICS	ENG447A	
29	PHY	SOFT MATTER: CONCEPTS AND METHODS	PHY606A	
30	CE	SURVEY CAMP	CE322A	
31	ME	TRANSPORT IN POROUS MEDIA	ME673A	
32	ME	ENGLISH PROFICIENCY AND SCHOLARLY COMMUNICATION	IDC603A	
33	HSS	TOPICS IN COGNITIVE LINGUISTICS	ENG748A	
34	IME	GOVERNANCE OF GLOBAL VALUE CHAINS	MBA717A.	
35	IME	INTRODUCTION TO CORPORATIONS	MBA716A.	
36	CHE	INTRODUCTION TO PHYSICS OF POLYMERIC SYSTEM	CHE677A	
37	CGS	PROJECT	CGS702A	
38	CSE	INFORMATION RETRIEVAL	CS657A	
39	CGS	COMPUTATIONAL COGNITIVE SCIENCE	CGS786A	
40	MTH	A COURSE IN OPERATOR THEORY	MTH654A	
41	MTH	FINITE ELEMENT ERROR ESTIMATION	MTH711A	
42	ECO	BAYESIAN ECONOMETRICS	ECO545A	
43	ECO	GAME THEORY	ECO754A	Please see uploaded content file attached.
44	CE	MS RESEARCH SEMINAR II	CE898A	
45	MTH	INTRODUCTION TO SHEAVES AND TOPOS THEORY	MTH661A	
46	CE	LABORATORY COURSE IN TRANSPORTATION ENGINEERING	CE780A	
47	CE	PAVEMENT DESIGN	CE684B	
48	CE	SOLID WATER INTERFACIAL PROCESSES	CE767B	
49	CSE	Others	Others	
50	HSS	HUMAN PERCEPTUAL PROCESS	PSY772A	
51	HSS	INTRODUCTION TO COGNITIVE NEUROSCIENCE	PSY760A	
52	PHY	QUANTUM PHYSICS	PHY204A	
53	MTH	INTRODUCTION TO ANALYTIC NUMER THEORY	MTH668A	
54	MTH	AN INTRODUCTION TO BAYESIAN ANALYSIS	MTH535A	
55	MTH	DISCONTINUED	MTH435A	
56	SDS	COMPLEX ANALYSIS	SDS434M	
57	SDS	REAL ANALYSIS	SDS433	

S.No.	Branch	Course Title	Courses ID	Content
58	SDS	INTRODUCTION TO SAMPLING THEORY	SDS432M	
59	MTH	BASIC PROBABILITY & DISTRIBUTION THEORY	MTH431A	
60	IME	CAPSTONE PROJECT IV	MBA704A.	
61	IME	CAPSTONE PROJECT III	MBA703A.	
62	IME	CAPSTONE PROJECT II	MBA702A.	
63	IME	CAPSTONE PROJECT I	MBA701A.	
64	IME	RENEWABLE ENERGY SOURCES:ECONOMICS POLICY AND REGULATION	MBA650A.	
65	EE	ADVANCED RF ANTENNAS	EE638A	
66	ECO	HEALTH ECONOMICS AND POLICY	ECO560A	
67	ECO	MANAGERIAL ECONOMICS & BUSINESS ANALYSIS	ECO507A	
68	CSE	DATA-DRIVEN PROGRAM ANALYSIS	CS686A	
69	CSE	COMPUTER AIDED VERIFICATION	CS652A	
70	CHE	PRACTICAL INTRO. TO QUANTUM MECHANICAL METHODS FOR SCIENTISTS & ENGINEERS	CHE626A	Summary of classical mechanics: conservation laws, Hamiltonian and Lagrangian formulation; The need for quantum mechanics; Basic concepts: physical observables and operators, expectation value, uncertainty principle, Schrodinger equation, Dirac notation, wavefunctions, tunneling, spin, Pauli exclusion principle, many-body systems, Hartree-Fock approximation, approximation, variational principle, the concept of potential energy surface; What makes electronic structure difficult; Hartree-Fock (HF) approach: restricted and unrestricted methods, post-HF methods: configuration interaction, coupled cluster theory (DFT): Hohenberg-Kohn theorems, Kohn-Sham formulation; Exchange and correlations: LDA and GGA approximations; Pure and hybrid DFT functionals; Tricks for self-consistent solution of the Kohn-Sham equations; Basis set convergence, extrapolation, correlation correction and DFT+U approach; Configuration interaction and Møller-Plesset perturbation theory; Applications of quantum-mechanical methods in engineering and sciences; Specialized topics (if time permits)
71	CGS	SEMINAR ON COGNITIVE SCIENCE	CGS700A	
72	CGS	METHODS AND TOOLS IN COGNITIVE SCIENCE	CGS603A	
73	CGS	FOUNDATIONS OF COGNITIVE SCIENCE	CGS601A	
74	CGS	COMPUTATIONAL TOOLS FOR COGNITIVE SCIENCE	CGS600A	
75	HSS	SOCIAL CHANGE IN CONTEMPORARY INDIA: A SOCIOLOGICAL PERSPECTIVE	SOC754A	
76	HSS	SOCIOLOGY OF SOCIAL MOBILITY	SOC753A	
77	HSS	SOCIOLOGY OF MOBILITY	SOC753	
78	HSS	APPLIED STATISTICS FOR SOCIOLOGISTS AND OTHER SOCIAL SCIENTISTS	SOC752	
79	HSS	SOCIOLOGY OF RELIGION	SOC750A	
80	HSS	SOCIOLOGY OF ENVIRONMENT	SOC742	
81	HSS	SOCIOLOGY OF DEVELOPMENT	SOC732A	
82	HSS	URBAN SOCIOLOGY	SOC724A	
83	HSS	SOCIOLOGY OF WORK	SOC490A	
84	HSS	DISASTER MENTAL HEALTH	PSY794A	
85	HSS	LEARNING MEMORY AND COGNITION	PSY789A	
86	HSS	LEARNING MEMORY AND COGNITION	PSY789	
87	HSS	GROUNDRED THEORY METHODOLOGY	PSY771A	

S.No.	Branch	Course Title	Courses ID	Content
88	HSS	BILINGUALISM	PSY770A	
89	HSS	HEALTH AND ILLNESS: PSYCHO-SOCIAL PERSPECTIVES	PSY720A	
90	HSS	PSYCHOLOGY OF LANGUAGE	PSY499A	
91	HSS	PSYCHOLOGY OF LANGUAGE	PSY499	
92	PHY	SPECIAL & GENERAL RELATIVITY	PSO315A	
93	PHY	CLASSICAL MECHANICS	PSO314A	
94	PSE	MS - RESEARCH THESIS	PSE899	
95	PSE	PHD THESIS	PSE799	
96	PSE	M. TECH THESIS	PSE699	
97	PHY	M.SC.RESEARCH PROJECT II	PHY699A	
98	PHY	M.SC.RESEARCH PROJECT II	PHY699	
99	PHY	M.SC.RESEARCH PROJECT I	PHY698A	
100	PHY	M.SC.RESEARCH PROJECT I	PHY698	
101	PHY	INTRODUCTION TO SPINTRONICS	PHY693A	
102	PHY	PLANETARY ASTROPHYSICS	PHY690X	
103	PHY	PHOTONIC DEVICES	PHY690V	
104	PHY	INTRODUCTION TO FUNCTIONAL SPACES & BASIC FORMALISM OF QUANTUM MECHANICS	PHY690D	
105	PHY	CONCEPTS OF PLASMA PHYSICS	PHY682A	
106	PHY	THE STANDARD MODEL OF FUNDAMENTAL PHYSICS	PHY679A	
107	PHY	PHYSICS OF TURBULENCE	PHY672A	
108	PHY	MATHEMATICAL METHODS II	PHY663A	
109	PHY	SPECIAL & GENERAL RELATIVITY	PHY659A	
110	PHY	INTRODUCTION TO X-RAY NEUTRON SCATTERING	PHY653A	
111	PHY	INTRODUCTION TO CONFORMAL FIELD THEORY	PHY652A	
112	PHY	METAMATERIALS & PLASMONICS	PHY645A	
113	PHY	CONDENSED MATTER PHENOMENA IN LOW DIMENSIONAL MATERIALS	PHY642A	
114	PHY	LOW TEMPERATURE PHYSICS	PHY634A	
115	PHY	TRANSPORT IN MESOSCOPIC SYSTEMS	PHY632A	
116	PHY	MAGNETISM IN MATERIALS	PHY624A	
117	PHY	CONDENSED MATTER II	PHY622A	
118	PHY	PHYSICS OF LIFE	PHY618A	
119	PHY	PHYSICS OF LIFE	PHY618	
120	PHY	REVIEW OF MATHEMATICAL METHODS IN PHYSICS	PHY605A	
121	PHY	INTRODUCTION TO	PHY600A	

S.No.	Branch	Course Title	Courses ID	Content
		PROFESSION AND COMMUNICATION SKILLS FOR PHYSICISTS		
122	PHY	M.SC. RESEARCH PROJECT I	PHY598A	
123	PHY	M.SC. RESEARCH PROJECT I	PHY598	
124	PHY	M.SC. PROJECT IV	PHY568A	
125	PHY	M.SC. PROJECT III	PHY566A	
126	PHY	M SC REVIEW PROJECT I	PHY500A	
127	PHY	CLASSICAL MECHANICS II	PHY402A	Short summary for including in the Courses of Study Booklet: Review of essential basic concepts of Classical Mechanics I (PHY401), Integrable and superintegrable systems, systems, Adiabatic invariance, Hannay angle, Poincare map, Homoclinic tangle, Melnikov method, Chaos in horseshoe map and symbolic dynamics; 2 DoF systems: Canonical perturbation theory, KAM theory, Local vs. widespread/global chaos, Chirikov resonance-overlap criterion, Canonical perturbation theory, Problem of small divisors, Area-preserving maps, Poincaré-Birkhoff theorem; 2 DoF systems: Geometry of resonances, Overview (no proofs) of Nekhoroshev theorem and Arnold diffusion, Fermi–Pasta–Ulam–Tsingou problem, Kuramoto model. Reference Books: I. J. V. Jose & E. J. Saletan, Classical Dynamics, Cambridge University Press (1998). II. H. Goldstein, C. Poole, & J. Safko, Classical Mechanics, Wiley (2002). III. L. D. Landau & E. M. Lifshitz, Mechanics, Butterworth-Heinemann (1976). IV. M. Tabor, Chaos and Integrability in Nonlinear Dynamics, Wiley-Interscience (1974). V. S. G. N. da Costa, Chaos and Integrability in Nonlinear Dynamics, Wiley-Interscience (1974). VI. A. Goriely, Integrability and Nonintegrability of Dynamical Systems, World Scientific (2001).
128	PHY	CLASSICAL MECHANICS II	PHY402	
129	HSS	PLATO	PHI785A	
130	HSS	TWENTIETH CENTURY CONTINENTAL PHILOSOPHY	PHI784A	
131	HSS	CONTEMPORARY EPISTEMOLOGY	PHI758A	
132	HSS	CONTEMPORARY EPISTEMOLOGY I	PHI758	
133	HSS	ADVANCED TOPICS IN PHILOSOPHY OF SCIENCE	PHI755A	
134	HSS	MODERN LOGIC	PHI753A	
135	HSS	PROBLEMS IN PHILOSOPHICAL AESTHETICS	PHI752A	
136	HSS	INTRODUCTION TO POLITICAL PHILOSOPHY	PHI143A	
137	NET	NUCLEAR FUEL CYCLE	NT652A	
138	NET	NEUTRON TRANSPORT THEORY	NT631A	
139	NET	NEUTRON TRANSPORT THEORY	NT631	
140	NET	NUCLEAR POWER ENGINEERING III	NT615A	
141	SDS	STATISTICAL RELIABILITY THEORY	SDS784	
142	MTH	STATISTICAL RELIABILITY THEORY	MTH784	
143	MTH	NUMERICAL TECHNIQUES FOR NONLINEAR DYNAMICAL SYSTEMS	MTH770A	
144	MTH	NUMERICAL TECHNIQUES FOR NONLINEAR DYNAMICAL SYSTEMS	MTH770	
145	MTH	PROBABILISTIC TECHNIQUES IN ANALYSIS	MTH765A	
146	MTH	PROBABILISTIC TECHNIQUES IN ANALYSIS	MTH765	
147	MTH	VECTOR BUNDLES AND CHARACTERISTIC CLASSES	MTH761A	
148	MTH	ERGODIC THEORY	MTH729A	
149	MTH	PDE BASED IMAGE PROCESSING	MTH723A	

S.No.	Branch	Course Title	Courses ID	Content
150	MTH	PDE BASED IMAGE PROCESSING	MTH723	
151	MTH	COMPUTATIONAL FINANCIAL MATHEMATICS-A PDE APPROACH	MTH721A	
152	MTH	COMPUTATIONAL FINANCIAL MATHEMATICS-A PDE APPROACH	MTH721	
153	MTH	MS PROJECT III	MTH699	
154	MTH	MS PROJECT I	MTH697A	
155	MTH	SPECTRAL METHODS FOR PARTIAL DIFFERENTIAL EQUATIONS	MTH696A	
156	SDS	LINEAR AND NON-LINEAR MODELS	SDS689	
157	MTH	TOPICS IN ARITHMATIC	MTH688A	
158	MTH	GRAPHS AND MATRICES	MTH687A	
159	SDS	ECONOMETRICS	SDS676	
160	MTH	GEOMETRY OF DIFFERENTIAL FORMS	MTH675A	
161	MTH	VARIATIONAL METHODS IN PDE	MTH674A	
162	SDS	ROBUST STATISTICAL METHODS	SDS673	
163	MTH	HARMONIC ANALYSIS ON POINCARÉ DISC	MTH672A	
164	MTH	INTRODUCTION TO ALGEBRAIC GEOMETRY AND ALGEBRAIC GROUPS	MTH667A	
165	MTH	ANALYTICAL TECHNIQUES FOR PDE	MTH663A	
166	SDS	ADVANCED CALCULUS	MTH652	
167	MTH	SEVERAL COMPLEX VARIABLES	MTH640A	
168	MTH	AN INTRODUCTION TO OPERATOR THEORY	MTH635	
169	MTH	RINGS AND MODULES	MTH613A	
170	MTH	MATHEMATICAL MODELLING	MTH603A	
171	SDS	PROJECT-II	SDS599	
172	SDS	PROJECT-I	SDS598	
173	MTH	STATISTICAL & AI TECHNIQUES IN DATA MINING	MTH552A	
174	MTH	NUMERICAL ANALYSIS & SCIENTIFIC COMPUTING II	MTH430A	
175	MTH	CALCULUS IN $\mathbb{R}^n$	MTH429A	
176	MTH	FOUNDATIONS OF OPTIMIZATION	MTH427A	
177	MTH	DISTRIBUTIONS THEORY IN STATISTICS	MTH425A	
178	MTH	UNDER GRADUATE PROJECT IV	MTH394A	
179	SDS	PROBABILITY THEORY	SDS309	
180	MTH	NUMERICAL ANALYSIS AND SCIENTIFIC COMPUTING-I	MTH308B	

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S.No.	Branch	Course Title	Courses ID	Content
		ROBOTICS		
196	ME	MODELING AND TURBULENT COMBUSTION	ME745A	
197	ME	BOILING AND CONDENSATION	ME742A	
198	ME	MODAL ANALYSIS: THEORY AND PRACTICE	ME730A	
199	ME	MODAL ANALYSIS: THEORY AND PRACTICE	ME730	
200	ME	MODELLING OF MECHANICAL PROPERTIES OF MATERIALS	ME729A	
201	ME	DYNAMIC FRACTURE MECHANICS	ME722A	
202	ME	DYNAMIC FRACTURE MECHANICS	ME722	
203	ME	DESIGN FOR MANUFACTURE & ASSEMBLY	ME701A	
204	ME	MECHANICS OF ENERGY MATERIALS	ME698S	
205	ME	PLASTICITY IN CRYSTALLINE SOLIDS	ME698R	
206	ME	MECHANICS OF ELECTROMAGNETIC MATERIALS	ME698Q	
207	ME	INSTABILITIES IN FLUID DYNAMICS	ME698P	
208	ME	MAGNETOHYDRODYNAMICS AND ITS ENGINEERING APPLICATIONS	ME698O	
209	ME	BIOFLUID MECHANICS	ME684A	
210	ME	BIOFLUID MECHANICS	ME684	
211	ME	NON - LINEAR FINITE ELEMENT METHODS IN SOLID MECHANICS	ME676A	
212	ME	NON - LINEAR FINITE ELEMENT METHODS IN SOLID MECHANICS	ME676	
213	ME	NEW PRODUCT DESIGN AND PROTOTYPING	ME675A	
214	ME	NEW PRODUCT DESIGN AND PROTOTYPING	ME675	
215	ME	MACHINING DYNAMICS	ME668A	
216	ME	RAILROAD VEHICLE DYNAMICS	ME660A	
217	ME	COMPUTER AIDED DESIGN OF THERMAL SYSTEMS	ME648A	
218	ME	SOLAR ENERGY TECHNOLOGY	ME645A	
219	ME	SOLAR ENERGY TECHNOLOGY	ME645	
220	ME	GEOPHYSICAL AND FLUID DYNAMICS	ME632A	
221	ME	CALCULUS OF VARIATIONS	ME624A	
222	ME	ADVANCED THEORY OF TURBOMACHINERY	ME617A	
223	ME	LANGUAGE AND STYLE OF	ME601A	

S.No.	Branch	Course Title	Courses ID	Content
		SCHOLARLY COMMUNICATION		
224	ME	UG PROJECT-IV	ME498A	
225	ME	UNDER GRADUATE PROJECT III	ME498	
226	IME	ADVANCED CORPORATE FINANCE	MBA775A	
227	IME	CREDIT RISK MANAGEMENT & MODELLING	MBA774A.	
228	IME	FINANCIAL INTERMEDIATION & COMMERCIAL BANKING	MBA773A.	
229	IME	DERIVATIVE CONTRACTS	MBA772A.	
230	IME	STRATEGIC ISSUES IN MANUFACTURING II	MBA766A.	
231	IME	STRATEGIC ISSUES IN MANUFACTURING I	MBA765A.	
232	IME	MANUFACTURING PLANNING AND CONTROL II	MBA762A.	
233	IME	MANUFACTURING PLANNING AND CONTROL I	MBA761A.	
234	IME	MARKETING ANALYTICS	MBA751A.	
235	IME	SOCIAL MEDIA ANALYTICS I	MBA749A.	
236	IME	BIG DATA FOR MANAGEMENT	MBA748A.	
237	IME	BUSINESS MANAGEMENT USING CLOUD	MBA747A.	
238	IME	E-SUPPLY CHAIN MANAGEMENT	MBA746A.	
239	IME	BUSINESS PROCESS PATTERNS	MBA745A.	
240	IME	BUSINESS PROCESS MODELLING	MBA744A.	
241	IME	MBA GLOBAL SOURCING OF IT	MBA743A.	
242	IME	ERP: A MANAGERIAL PERSPECTIVE	MBA742A.	
243	IME	INTRODUCTION TO E - COMMERCE	MBA741A.	
244	IME	BRAND MANAGEMENT	MBA733A	
245	IME	PRODUCT STRATEGY AND MANAGEMENT	MBA732A.	
246	IME	INNOVATION AND ENTREPRENEURIAL STARTUPS	MBA722A.	
247	IME	INNOVATION AND MANAGEMENT OF TECHNOLOGY	MBA721A.	
248	IME	EMPLOYMENT RELATIONS UNDER GLOBALISATION	MBA715A.	
249	IME	INTRODUCTION TO BUSINESS MANAGEMENT (IBM)	MBA714A.	
250	IME	CAPSTONE PROJECT	MBA699B.	
251	IME	INTRODUCTION TO UNIVERSAL HUMAN VALUES AND ETHICS	MBA687A.	

S.No.	Branch	Course Title	Courses ID	Content
252	IME	PRODUCT & BRAND MANAGEMENT	MBA685A.	
253	IME	SALES AND DISTRIBUTION MANAGEMENT	MBA684A.	
254	IME	SALES AND DISTRIBUTION MANAGEMENT	MBA684.	
255	IME	POWER SECTOR REFORM & REGULATION: THEORY AND PRACTICE	MBA683.	
256	IME	DISCOUNTINEW	MBA668A.	
257	IME	SOCIAL MEDIA ANALYTICS	MBA653A.	
258	IME	SOCIAL MEDIA ANALYTICS	MBA653.	
259	IME	MBA NEXT GENERATION INFORMATION SYSTEMS	MBA648A.	
260	IME	SOCIOPOLITICAL AND ETHICAL ASPECTS OF BUSINESS	MBA617B	
261	IME	DATA MINING AND KNOWLEDGE DISCOVERY	IME672A	
262	IME	INFORMATION SYSTEMS THEORY	IME640	
263	IME	STOCHASTIC MODELS AND SIMULATION	IME638A	
264	IME	MARKETING MODELS	IME631A	
265	HSS	EXPLORING INDIANNESS	HSS703A	
266	HSS	EXPLORING INDIANNESS	HSS703	
267	HSS	SOFT SKILLS AND PERSONALITY DEVELOPMENT	HSS401	
268	HSS	PHD THESIS	FA799	
269	ES	SPECIAL TOPICS IN SOLID EARTH GEOPHYSICS	ES698A	
270	ES	GEOLOGICAL FIELD TRAINING	ES660A	
271	ES	NATURAL HAZARDS	ES658A	
272	ES	EXPERIMENTAL ROCK MECHANICS & ROCK PHYSICS	ES657A	
273	ES	GEOPHYSICAL METHODS	ES656	
274	ES	SOLID EARTH GEOPHYSICS	ES655	
275	ES	ADVANCED STRUCTURAL GEOLOGY	ES654A	
276	ES	ADVANCED STRUCTURAL GEOLOGY	ES654	
277	ES	APPLIED SEDIMENTOLOGY AND BASIN ANALYSIS	ES653A	
278	ES	SEDIMENTOLOGY AND BASIN ANALYSIS	ES653	
279	ES	MATHEMATICS FOR EARTH SCIENCES	ES651A	
280	ES	COMMUNICATION SKILLS IN EARTH SCIENCES	ES650A	
281	ES	ISOTOPE GEOCHEMISTRY AND APPLICATIONS	ES649A	
282	ES	ISOTOPE GEOCHEMISTRY AND APPLICATIONS	ES649	

S.No.	Branch	Course Title	Courses ID	Content
283	ES	INSTRUMENTATION IN EARTH SCIENCES	ES648A	
284	ES	LABORATORY PRACTICES IN GEOSCIENCES	ES648	
285	ES	GEOLOGY AND GEOCHEMISTRY OF PETROLEUM	ES647A	
286	ES	PETROLEUM EXPLORATION AND PRODUCTION	ES647	
287	ES	RIVER SCIENCE	ES645A	
288	ES	RIVER SCIENCE	ES645	
289	ES	SATELLITE REMOTE SENSING & GIS FOR GEORESOURCE EVALUATION	ES644	
290	ES	AQUEOUS GEOCHEMISTRY	ES643A	
291	ES	RESOURCE EXPLORATION TECHNIQUES	ES643	
292	ES	ENVIRONMENTAL GEOLOGY	ES641A	
293	ES	ENVIRONMENTAL GEOLOGY	ES641	
294	HSS	PERSPECTIVES OF THE POSTHUMAN AND / IN THE ANTHROPOCENE	ENG777A	
295	HSS	PERSPECTIVES OF THE POSTHUMAN AND/IN THE ANTHROPOCENE	ENG777	
296	HSS	HISTORY OF LINGUISTIC THOUGHT	ENG759A	
297	HSS	HISTORICAL LINGUISTICS	ENG754A	
298	HSS	TRANSLATION STUDIES	ENG753A	
299	HSS	RESPONSBILE DISSENT	ENG742	
300	HSS	INDIAN LITERATURE	ENG733A	
301	HSS	AMERICAN RENAISSANCE	ENG719A	
302	HSS	LITERATURE AND SOCIETY	ENG712A	
303	HSS	APPLIED LINGUISTICS	ENG708	
304	HSS	LANGUAGE CURRICULUM DESIGN	ENG707A	
305	HSS	AUTOBIOGRAPHICAL LITERATURE	ENG460A	
306	HSS	COMEDY: FROM SITCOM TO STANDUP!	ENG458A	
307	HSS	COMEDY: FROM SITCOM TO STANDUP!	ENG458	
308	HSS	ACADEMIC WRITING	ENG450	
309	HSS	ENGLISH LANGUAGE AND COMMUNICATION SKILLS	ENG112A	
310	EEM	MS - RESEARCH THESIS	EEM899	
311	EE	MS - RESEARCH THESIS	EE899	
312	EE	INTRODUCTION TO SPIN ELECTRONICS DEVICES	EE698O	
313	EE	SEMICONDUCTOR OPTICAL COMMUNICATION DEVICES	EE685	
314	EE	FIBER-OPTIC COMMUNICATIONS	EE684	
315	EE	QUANTUM AND WAVE PHENOMENA	EE683	

S.No.	Branch	Course Title	Courses ID	Content
316	EE	MIMO WIRELESS COMMUNICATION	EE677A	
317	EE	POWER ELECTRONICS IN SOLAR PHOTOVOLTAIC SYSTEMS	EE663A	
318	EE	POWER ELECTRONICS IN SOLAR PHOTOVOLTAIC SYSTEMS	EE663	
319	EE	PROBABILISTIC MOBILE ROBOTICS	EE652A	
320	EE	NONLINEAR FIBER OPTICS	EE639A	
321	EE	SYNCHROPHASOR TECHNOLOGY AND ITS APPLICATIONS	EE637A	
322	EE	ADVANCED PROTECTIVE RELAYING	EE636	
323	EE	TOPICS IN CRYPTOGRAPHY AND CODING	EE628A	
324	EE	OPTICAL COHERENT IMAGING	EE620A	
325	EE	OPTICAL COHERENT IMAGING	EE620	
326	EE	INDUSTRIAL AUTOMATION AND CONTROL	EE617A	
327	EE	CONVEX OPTIMIZATION IN SIGNAL PROCESSING AND COMMUNICATIONS	EE609A	
328	EE	UNDER GRADUATE PROJECT-IV	EE492A	
329	EE	ELECTRICAL ENGINEERING LAB I	EE480A	
330	EE	MODERN CONVEX OPTIMIZATION FOR ENGINEERING / SCIENCE APPLICATIONS	EE420A	
331	EE	MODERN CONVEX OPTIMIZATION FOR ENGG/ SCIENCE APPLICATIONS	EE420	
332	EE	ELECTRICAL ENGINEERING UNDER GRADUATE PROJECT I	EE391A	
333	ECO	FINANCIAL ECONOMETRICS	EC0764A	
334	ECO	BAYESIAN DATA ANALYSIS	EC0762A	
335	ECO	TOPICS IN APPLIED ECONOMETRICS	EC0761A	
336	ECO	MATHEMATICAL ANALYSIS FOR ECONOMICS	EC0760A	
337	ECO	MATHEMATICAL ANALYSIS FOR ECONOMICS	EC0760	
338	ECO	NONLINEAR DYNAMICS AND CONTROL IN ECONOMICS AND FINANCE	EC0757A	
339	ECO	NONLINEAR DYNAMICS AND CONTROL IN ECONOMICS AND FINANCE	EC0757	
340	ECO	LABOUR ECONOMICS	EC0753A	
341	ECO	QUANTITATIVE ANALYSIS OF INTERNATIONAL TRADE	EC0752A	

S.No.	Branch	Course Title	Courses ID	Content
342	ECO	LAW AND ECONOMICS	ECO751A	
343	ECO	PROGRAM EVALUATION: METHODS AND APPLICATIONS	ECO733A	
344	ECO	M TECH THESIS (FOR DUAL DEGREE ONLY)	ECO699A	
345	ECO	M SC PROJECT	ECO699	
346	ECO	PANEL DATA PROCEDURES AND ANALYSIS	ECO542A	
347	ECO	ECONOMIC GROWTH THEORY AND PRACTICE	ECO523A	
348	ECO	MONETARY ECONOMICS	ECO522	
349	ECO	BEHAVIOURAL AND EXPERIMENTAL ECONOMICS	ECO506A	
350	ECO	BEHAVIOURAL AND EXPERIMENTAL ECONOMICS	ECO506	
351	ECO	UNDER GRADUATE PROJECT-IV	ECO499A	
352	EC	PH D THESIS	EC799	
353	EC	M TECH THESIS	EC699	
354	DES	TOPICS IN DESIGN	DES689A	
355	DES	METHODS FOR DESIGN RESEARCH	DES635A	
356	DES	INTEGRATED PRODUCT DEVELOPMENT-I	DES633A	
357	DES	FUNDAMENTALS OF GRAPHICS	DES630A	
358	DES	MANAGEMENT OF DESIGN AND INNOVATION	DES627A	
359	DES	INTERACTION DESIGN	DES626A	
360	DES	ELEMENTS AND PRINCIPAL OF DESIGN	DES624A	
361	DES	INTERACTION DESIGN FOR PHYSICAL PRODUCTS	DES606A	
362	DES	PRODUCT DESIGN AND PROTOTYPING	DES605A	
363	DES	DESIGN PRACTICE II	DES603A	
364	CSE	MS - RESEARCH THESIS	CS899	
365	CSE	SPECIAL TOPICS IN COMPUTER SCIENCE	CS797A	
366	CSE	OPTIMIZATION TECHNIQUES	CS774A	
367	CSE	ONLINE LEARNING AND OPTIMIZATION	CS773A	
368	CSE	ONLINE LEARNING AND OPTIMIZATION	CS773	
369	CSE	PROBABILISTIC MACHINE LEARNING	CS772A	
370	CSE	ARITHMETIC CIRCUIT COMPLEXITY	CS748A	
371	CSE	RANDOMIZED METHODS IN COMPUTATIONAL COMPLEXITY	CS747A	
372	CSE	RANDOMIZED METHODS IN COMPUTATIONAL COMPLEXITY	CS747	

S.No.	Branch	Course Title	Courses ID	Content
373	CSE	ADVANCED COMPILER OPTIMIZATIONS	CS738A	
374	CSE	TIPICS IN INTERNET TECHNOLOGIES	CS727A	
375	CSE	MACHINE TRANSLATION	CS673A	
376	CSE	ALGORITHMIC GAME THEORY	CS656A	
377	CSE	ALGORITHMIC GAME THEORY	CS656	
378	CSE	TOPICS IN LINEAR PROGRAMMING	CS655A	
379	CSE	TOPICS IN LINEAR PROGRAMMING	CS655	
380	CSE	TOPICS IN DESIGN AND ANALYSIS OF ALGORITHMS	CS645A	
381	CSE	FORMAL METHODS FOR ROBOTICS AND AUTOMATION	CS638A	
382	CSE	EMBEDDED AND CYBER-PHYSICAL SYSTEMS	CS637A	
383	CSE	APPROXIMATION ALGORITHMS	CS635A	
384	CSE	APPROXIMATION ALGORITHMS	CS635	
385	CSE	CYBER SECURITY OF CRITICAL INFRASTRUCTURES	CS631A	
386	CGS	HUMAN CENTERED COMPUTING	CGS616A	
387	CSE	SKYLINE QUERIES IN DATABASES	CS615A	
388	CSE	SKYLINE QUERIES IN DATABASES	CS615	
389	CSE	UNDER GRADUATE PROJECT-IV	CS499A	
390	CHM	CHEMISTRY OF POLYHEDRAL CLUSTERS & METAL-METAL MULTIPL BONDED COMPOUNDS	CHM687A	
391	CHM	MOLECULAR REACTION DYNAMICS	CHM679A	
392	CHM	ORGANIC STRUCTURE DETERMINATION BY SPECTROSCOPIC TECHNIQUES	CHM656A	
393	CHM	ORGANOMETALLIC CHEMISTRY AND CATALYSIS	CHM655A	
394	CHM	ORBITAL INTERACTIONS IN CHEMISTRY	CHM645A	
395	CHM	ENZYME REACTIONS MECHANISM AND ENZYME KINETICS	CHM632A	
396	CHM	CHEMICAL KINETICS	CHM622A	
397	CHM	ORGANIC PHOTOCHEMISTRY	CHM614A	
398	CHM	MS PROJECT	CHM599	
399	CHM	UG PROJECT IV	CHM492A	
400	CHM	ORGANIC QUALITATIVE &	CHM404A	

S.No.	Branch	Course Title	Courses ID	Content
		QUANTITATIVE ANALYSIS		
401	CHE	MS - RESEARCH THESIS	CHE899	
402	CHE	GRADUATE SEMINAR	CHE802A	
403	CHE	GRADUATE SEMINAR	CHE801A	
404	CHE	M TECH SEMINAR	CHE702A	
405	CHE	SPECTROSCOPY OF NANOMATERIALS	CHE675A	
406	CHE	MULTI PHASE TRANSPORT	CHE646A	
407	CHE	INTERMOLECULAR AND SURFACE FORCES	CHE625A	
408	CHE	INTRODUCTION TO GRANULAR MECHANICS	CHE616A	
409	CHE	INTRODUCTION TO HYDRODYNAMIC STABILITY	CHE614A	
410	CHE	RESEARCH METHODS & SKILLS	CHE600A	
411	CHE	UNDER GRADUATE PROJECT IV	CHE498A	
412	CE	MS - RESEARCH THESIS	CE899	
413	CE	MS RESEARCH SEMINAR I	CE897A	
414	CE	PHD SEMINAR II	CE802	
415	CE	PHD SEMINAR II	CE798	
416	CE	PHD SEMINAR I	CE801	
417	CE	PHD SEMINAR I	CE797	
418	CE	SCIENTIFIC DATA ANALYSIS PRESENTATION AND INTERPRETATION	CE794B	
419	CE	SCIENTIFIC WRITING SKILLS	CE792B	
420	CE	AGRICULTURAL SUSTAINABILITY AND CLIMATE CHANGE	CE766A	
421	CE	INDUSTRIAL WASTE MANAGEMENT	CE765B	
422	CE	ENVIRONMENT TOXICOLOGY AND RISK ASSESSMENT	CE764A	
423	CE	SOLID AND HAZARDOUS WASTE MANAGEMENT	CE763A	
424	CE	ATMOSPHERIC MODELING	CE762B	
425	CE	SUBSURFACE CONTAMINANT FATE AND TRANSPORT	CE761B	
426	CE	SURFACE WATER QUALITY MODELING	CE760A	
427	CE	PLASTIC EQUILIBRIUM IN SOILS	CE734A	
428	CE	GEOENVIRONMENTAL DESIGN ASPECTS OF SOLID WASTE MANAGEMENT	CE733A	
429	CE	UNSATURATED SOIL MECHANICS	CE732A	
430	CE	VIBRATION BASED STRUCTURAL HEALTH MONITORING FOR CE APPLICATIONS	CE725A	
431	CE	RELIABILITY ANALYSIS &	CE724A	

S.No.	Branch	Course Title	Courses ID	Content
		RELIABILITY BASED DESIGN OF STRUCTURE		
432	CE	FINITE ELEMENT METHODS IN CIVIL ENGINEERING	CE723A	
433	CE	RANDOM VIBRATIONS	CE721A	
434	CE	UNSATURATED SOIL MECHANICS	CE701A	
435	CE	UNSATURATED SOIL MECHANICS	CE701	
436	CE	M TECH SEMINAR I	CE697A	
437	CE	TRANSPORTATION ECONOMICS	CE686A	
438	CE	URBAN TRANSPORTATION	CE685A	
439	CE	SIGNAL PROCESSING ON THE SPHERE	CE679A	
440	CE	ATMOSPHERIC PHYSICS AND CHEMISTRY	CE669A	
441	CE	ENVIRONMENTAL QUALITY & POLLUTION MONITORING TECHNIQUES	CE668A	
442	CE	PRINCIPLES OF ENVIROMENTAL MANAGEMENT	CE667A	
443	CE	ECOLOGICAL AND BIOLOGICAL PRINCIPLES AND PROCESS	CE665A	
444	CE	PHYSICO-CHEMICAL PRINCIPLES AND PROCESS	CE664A	
445	CE	CHARACTERIZATION AND USE OF AGGREGATES IN PAVEMENT SYSTEMS	CE661A	
446	CE	IGNEOUS AND METAMORPHIC PETROLOGY	CE652	
447	CE	ISOTOPE GEOCHEMISTRY AND APPLICATIONS	CE649	
448	CE	REPAIR AND REHABILITATION OF CONCRETE STRUCTURES	CE648A	
449	CE	CONSTRUCTION METHODS	CE647A	
450	CE	BASIC QUALITY AND SAFETY MANAGEMENT IN CONSTRUCTION	CE645A	
451	CE	INFRASTRUCTURE FINANCING	CE643B	
452	CE	DISCONTINUED	CE643A	
453	CE	INFRASTRUCTURE ASSET MANAGEMENT	CE640A	
454	CE	ANALYTICAL AND NUMERICAL METHODS IN GEOMECHANICS	CE639A	
455	CE	GEOTECHNICAL MEASUREMENTS AND EXPLORATION	CE638A	
456	CE	GEOTECHNICAL EARTHQUAKE ENGINEERING	CE636A	
457	CE	REINFORCED EARTH STRUCTURES	CE633A	
458	CE	BRIDGE ENGINEERING AND	CE626A	

S.No.	Branch	Course Title	Courses ID	Content
		DESIGN		
459	CE	ECOHYDROLOGY	CE619A	
460	CE	VADOSE ZONE HYDROLOGY	CE618A	
461	CE	ADVANCED MODELING OF SUBSURFACE FLOW AND TRANSPORT	CE617A	
462	CE	INTRODUCTION AI TECHNIQUES	CE615A	
463	CE	PROBABILITY AND STATISTICS FOR CIVIL ENGINEERS	CE605A	
464	CE	NUMERICAL METHODS FOR CIVIL ENGINEERS	CE604A	
465	CE	UNDER GRADUATE RESEARCH -IV	CE493A	
466	CE	TRAFFICE SAFETY	CE483A	
467	CE	MOVE TO CE432A	CE472A	
468	CE	CONSTRUCTION MANAGEMENT	CE441B	
469	CE	GEOGRAPHICAL INFORMATION SYSTEM (GIS)	CE432A	
470	CE	PHYSICAL AND ENVIRONMENTAL GEOLOGY	CE422A	
471	BSBE	SEMINAR	BSE802	
472	BSBE	SEMINAR	BSE801	
473	BSBE	SPECIAL TOPICS IN BIOLOGICAL SCIENCES AND BIOENGINEERING	BSE682A	
474	BSBE	RESEARCH METHODS IN BIOLOGICAL SCIENCES	BSE671A	
475	BSBE	FUNCTIONAL GENOMICS	BSE653B	
476	BSBE	BIOLOGY LABORATORY	BSE641A	
477	BSBE	DRUG DESIGN AND METABOLISM	BSE617A	
478	BSBE	TISSUE ENGINEERING	BSE421	
479	HSS	INDIVIDUAL PROJECT	ART710A	
480	HSS	TOPICS IN INDIAN FOLK ART	ART709A	
481	HSS	RESEARCH METHODS IN VISUAL CULTURE	ART708A	
482	HSS	ART CRITICISM: THEORY AND PRACTICE	ART406	
483	HSS	APPRECIATING FAR-EASTERN CINEMA	ART104	
484	AE	NONLINEAR AND ADAPTIVE CONTROL	AE778A	
485	AE	OPTICAL METHODS IN SOLID MECHANICS	AE708A	
486	AE	OPTICAL METHODS IN SOLID MECHANICS	AE708	
487	AE	DEFORMATION AND FRACTURE	AE704A	
488	AE	AIRCRAFT SRUCTURAL INTEGRITY	AE693A	
489	AE	ANALYSIS & COMPOSITE STRUCTURES	AE682A	

S.No.	Branch	Course Title	Courses ID	Content
490	AE	FINITE VOLUME METHODS IN HEAT MASS & MOMENTUM TRANSFER	AE665A	The primary objective of this course is to expose doctoral students to the art and visualculture of the Chinese Cinema. This PG level course will also assist them in comprehending and appreciating the wide range of Chinese cinema from the 1920s to the present. The course will cover the history of Chinese cinema, the major directors and their works, and the current trends in Chinese cinema. The course will also cover the theory and practice of film analysis, and the role of film in Chinese society.
491	AE	FINITE VOLUME METHODS IN HEAT MASS & MOMENTUM TRANSFER	AE665	
492	AE	APPLIED COMPRESSIBLE FLOWS	AE664A	
493	AE	APPLIED COMPRESSIBLE FLOWS	AE664	
494	AE	MISSILES GUIDANCE AND DYNAMICS	AE633A	
495	AE	AUTONOMOUS UNMANNES AERIAL SYSTEMS	AE630A	
496	AE	UNSTEADY GAS DYNAMICS	AE606A	
497	AE	UNDER GRADUATE PROJECT IV	AE473A	
498	AE	EXPERIMENTS IN FLIGHT MECHANICS	AE422A	
499	AE	B TECH PROJECT 1	AE371A	
500	MSE	FRACTURE FATIGUE AND CREEP: MATERIALS AND MODELS	MSE661A	This course is an introduction to many of the theories and methods that have been used by art historians to analyse the modern FarEastern art. The visual foci will include the most popular works of art created in the FarEast (including painting and sculpture) and the common responses of artists to the rapid transformation, industrialization and mechanization of the FarEastern society. Thus the course is based on trends in postmodern FarEastern postmodernist art in the world. This PG level course is designed to introduce students to the questions and debates that propel FarEastern Art history and the methodologies that have shaped it.
501	IME	SECURITY ANALYSIS & PORTFOLIO MANAGEMENT	MBA771A.	
502	CE	AGRICULTURE ENVIRONMENT AND CLIMATECHANGE	EEM615A	
503	DES	DESIGN CULTURE AND SOCIETY	DES628A	
504	CSE	MULTIAGENT SYSTEMS : GAMES ALGORITHMS EVOLUTION	CS785A	
505	CSE	PROGRAMS PROOFS AND TYPES	CS750A	
506	CSE	ADV. OPERATING SYSTEMS FOR EMBEDDED SYSTS PERVASIVE COMP & INT OF THINGS	CS630A	
507	CHE	ENVIROMENTAL POLLUTION CONTROL DESIGN AND MODELLING	CHE673	
508	HSS	INTRODUCTION TO CHINESE CINEMA	ART707A	
509	HSS	MODERN FAR-EASTERN ART	ART705A	
510	HSS	CREATIVE PAINTING: METHODOLOGY & PRACTICES	ART711A	The aim of the course is to expose the postgraduate FineArts/Design students to various postmodern methods and mediums of creative painting. Finearts practices, specially painting is committed to greater understanding of the world through the creation of artworks that reexamines its context, material form and deeper conceptual method. The course aims to broaden the productive, conceptual, and critical faculties of students in their pursuit of careers as practicing artists or as teachers of FineArts at the college level. Studio space (FineArts lab) will be provided to the students for regular lectures as well as their creative pursuit. Here students are expected to develop and understand the creative process in various educational settings.The course assessment will be based on the cumulative performance of the student in written examinations, which would primarily cover the basic painting theory, along with a final evaluation by the student will require students to maintain a rigorous pace of creative research and establish a professional painting practice
511	HSS	STUDIO ART PRACTICES	ART703A	The aim of the course is to expose the students to different mediums and methods of art practice. The course consists of four components based on four different fields: Painting, Cinematography, Graphic Design and Photography. The assessment will be based on a cumulative performance of the student in written examinations, which would primarily cover the theory, along with an exhibition/presentation at the end of each of the two components. The medium with an emphasis on practice.
512	ES	REMOTE SENSING AND GIS FOR GEO-RESOURCE EVALUATION	ES644A	Remote sensing platforms satellite based and airborne sensors; Basic principles of image interpretation; Spectra of earths surface material ; Interpretation of regional geological and geomorphological features using GIS; Remote Sensing Information System(GIS) and its application; Analytical Hierarchy Process (AHP) technique and its integration into GIS. River erosion studies; Identification of groundwater potential zones; criteria for identification of water quality mapping; water quality parameters,indices of water quality monitoring; Vegetation Mapping and forestry applications; Application in glaciology and snow hydrology; Coastal zone mapping and other related topics.

S.No.	Branch	Course Title	Courses ID	Content
				sensing technique for mapping and prediction; Mineral resources evaluation with particular reference to digital remote sensing; Application of thermal infrared data for mapping surface moisture and rock
513	ECO	PRODUCTION ECONOMICS AND EFFICIENCY ANALYSIS	ECO741A	Course Description: This applied course focuses on producer/firm behavior in great detail. This course is designed as a blend of microeconomic theory and quantitative techniques to model production data using econometrics and linear programming. In particular, the course will teach how to estimate production, cost, profit, and input demand functions and compute measures of absolute economic performance (e.g. energy, industry etc.).Course Content (Number of lectures 40):Part 1: Theory of firmProduction function, Elasticity of substitution, Elasticity of scale and size, Cost function, Duality between cost and production measurement, Difference between productivity and efficiency.Part 2: Quantitative analysis of production dataData and measurement issues, Flexible forms of production and cost functions, Empirical estimation of linear models, JustPope type production function to handle risk in production environment, Formulation of linear programming (LP) problems (primal and dual), Graphical and Simplex solution methods, Linear programming Chambers, R.G., Applied Production Analysis: A Dual Approach, Cambridge University Press, 1988. Coelli, T.J., D.S. Prasada Rao, C.J. O'Donnell and G.E. Battese, An Introduction to Efficiency and Productivity Analysis, Kluwer Academic Publishers, 1999. Articles from American Journal of Agricultural Economics, Journal of Econometrics, and The Review of Economics and Statistics.
514	HSS	CURRENT ISSUES IN LINGUISTICS	ENG423A	1.Conceptualising language: Internalist and externalist approaches to language design2.Language Faculty: Modularity, Optimality of design, Constraints on possiblegrammars, Interface systems3.Conceptualising Crosslinguistic variation, Diachronicshifts, Multimodal cognition, Bilingual cognitive and linguistic processing4.Language Acquisition: Perceptual grounding, embodiment, perceptual and linguisticprocesses5.Language boundaries, Language and identity, Language andeducation, Discourse, ideology and control. 1.Chomsky, N. New Horizons in the Study of Language and Mind. Cambridge UniversityPress. 2000.2.Chomsky, N. Architecture of Language. Oxford University Press. 2001.3.Huang, Y. Pragmatics. Oxford University Press. 2002.5.Langacker, R.W. Grammar and Conceptualisation. De Gruyter. 1999.5.Pinker, S. Language Cognition and Human Nature. Oxford University Press. 2013.6.Talmy, L. Cognitive Semantics vol. 1
515	HSS	ACADEMIC WRITING	ENG450A	Course Content: (Number of proposed lectures is given in brackets):1.Getting started (03): Analysing issues, identifying the purpose ofwrite up and the target audience, identifying the relevant materials, generating ideas, paraphrasing, summarising, evaluating, combing available resources3.Paragraphs and paragraphing (06): Why to make paragraphs, how to makeparagraphs, identifying topic sentence and details, understanding active and passive devices, passives, prepositions,referring verbs, tenses, punctuation, relevant style (hedging, complexity, formality, precision)5.Organizing the writeup (07): Parts of a writeup (introduction, mainbody and conclusion)6. Writing Argumentative and descriptive essays, emails,formal letters, invitations, notices, posters7.Common mistakes (04): Mistakes in reasoning, mistakes in grammar andpunctuation8.Revision (03): Getting feedback9.ReferencesReferencesBailey, S. (2004). Academic writing: A practical guide for students.RoutledgeFalmer: New York.Giltrow, J. Gooding, R., Burgoyne, B., & Sawatsky, M. (2009). Academicwriting: An introduction (2nd edn). Routledge (Letts Educational) Ltd: London.Hilton, C. & Hyder, M., (1992b). Getting to Grips with Spelling. BPP (Letts Educational) Ltd: London.University of Essex (2008). How to improve your academic writing.Retrieved from
516	MTH	CATEGORY THEORY	MTH666A	1. Structure vs. property: monoids, groups, preorders, partial orders; structurepreserving maps: homomorphisms, continuous maps; category; homsets andduality; functorcovariant and contravariant; natural transformations2. Russell'sparadox and the category of all categories; skeletons and axiom of choice. (1.lecture)3. Isomorphism; groupoid; monomorphism and epimorphism; full, faithful,essentially surjective functors; commacategories, functor categories, congruence and quotients. (3 lectures)4. Representable functors; Yoneda lemma and Yoneda embedding; separating anddetecting families; injective and projective objects5. Relation tocomma categories; composition of adjoint functors; units, counits andcharacterization of adjoint functors; equivalence gives adjoint functors. (4lectures)6. Categorical properties: initial and terminal objects, regularmono(epi)morphisms; pullbacks and pushouts; direct and inverse limits. (2lectures)7. Constructing all (co)limits from some of them; (co)complete categories;absolute (co)limits; preservation, creation
517	SDS	ASYMPTOTIC STATISTICS	SDS665	Introduction Approximate Statistical Procedures [1]; Asymptotic Optimality Theory [1].Review of Stochastic Convergence Basic Theory, Stochastic o and 0 Symbols, Characteristic Functions[2]; AlmostSure Convergence LindebergFeller Theorem, Convergence in Total Variation [2].Delta Method Basic Result, VarianceStabilizing Transformations [1]; HigherOrder Expansions, UniformDelta Method [1]; Moments [1].M and Z Estimators, [1] Classical Conditions, OneStep Estimators [1]; Rates of Convergence[1]; Argmax Theorem W.Contiguity Likelihood Ratios [1]; Contiguity [1].Local Asymptotic Normality Introduction, Expanded Local AsymptoticNormality [1]; Local AsymptoticNormality [1].Stochastic Convergence in Metric Spaces Metric and Normed Spaces [2]; Basic Properties [1]; BoundedStochastic Processes [2].Empirical Processes Empirical Processes [2]; Maximal Inequalities [1].Functional Delta Method von Mises Calculus [2]; HadamardDifferentiable Functions [1]; Some Examples[1].Bootstrap Introduction, Consistency [1]; HigherOrder Correction
518	CE	CHARACTERIZATION OF PAVEMENT MATERIALS	CE681A	Characterization of pavement materials bituminous mix, aggregates, subgrade, cementedmaterials and cement concrete; Experimental methods; Basic material modelling, Fatigue andPermanent deformation concepts and approaches; Evaluation and useof recycled materials
519	CE	MATHEMATICS FOR CIVIL ENGINEERS	CE603A	Linear Differential Equations: Homogeneous Linear Equations of Second Order; SecondOrder HomogeneousEquations with Constant Coefficients; Case of Complex Roots, Complex Exponential Function; Integrals and Transforms: Fourier Integrals; Fourier Cosine and Sine Transforms; Fourier Transform andProperties; Dirac Delta Function; Convolution Theorem; Parseval's Theorem; Fourier integral to solve Problems of Variables,Use of Fourier Series; Modeling: Membrane, TwoDimensional Wave Equation; Rectangular Membrane, Use ofDouble Fourier SeriesLinear Algebra: Rank of a Matrix, Linear Independence, Vector Spaces, Linear Transformations; Eigenvalues,Eigenvectors; Similarity of Matrices, Basis of Eigenvectors, Diagonalization
520	CE	SCIENTIFIC DATA ANALYSIS PRESENTATION AND INTERPRETATION	CE794A	Basics of data: Primary data, secondary data, data sources and reliability Tools for data analysis: Identifying the right tool based on the project requirement (eg.Matlab, R, MSExcel, Access, ArcGIS, etc) Presentation MSExcel. Interpretation: Interpretation of results and documenting.
521	CE	SCIENTIFIC PRASENTATION SKILLS	CE793A	Basics of scientific presentation Visuals: choice of type and size of fonts, colorcombination, styles, use of sketch and pictures Delivering impressive presentation: Usage of language, clarity, simplicity, speed, summary,acknowledgements Ethics in using contents from other sources
522	CE	SCIENTIFIC WRITING SKILLS	CE792A	
523	CE	SCIENTIFIC WRITING SKILLS	CE792A	Basics of scientific writing Subjects/Actions, Cohesion, Emphasis, Simplicity; Parts of a Scientific papers Abstract, Introduction, Body, Conclusion, Acknowledgements,Reference; Writing styles: Referencing
524	CE	LITERATURE SEARCH AND REVIEW	CE791A	Basics of literature review: Introduction, identifying appropriate search engines Writing style: Styles of citation and referencing Referencing various types of sources: journal articles, conference proceedings, books, from published report,giving proper credit to authors.
525	HSS	ART CRITICISM: THEORY AND PRACTICE	ART406A	
526	ECO	EFFICIENCY AND PRODUCTIVITY ANALYSIS	ECO742A	Course Descriptions: This applied microeconomics course is designed for student who wish to learn about benchmarking analysis. Benchmarking is used to measure performance of decision making units using a metric (e.g. technical efficiency growth) suitable for comparison across similar units. The course offers an introductions to Data Envelopment Analysis (DEA) and Stochastic Frontier Analysis (SFA). In particular, the course will teach how to estimate production frontiers and compute measures of various types of efficiencies and productivity growth. The course is a mixture of advanced theory, quantitative techniques, and computations involving real life production data.
527	CSE	ANALYSIS OF CONCURRENT PROGRAMS	CS636A	*Programing language Basics:Syntax,Semantics,Type*Review of concurrent programming paradigms:sharedmemory messagepassing, partitioned global address space*Synchronizations primitives:locks, mutexes, semaphores, violations,order violations, deadlocks,livelocks*Consistency models: strict consistency, sequential consistency, linearizability (atomic consistency),relaxed memory models,memory fences to enforce orderingConcurrent separations Logic * Statespace reductions techniques:Formal modelling of concurrent systems, Mazurkiewicz traces, Logic
528	MTH	INTRODUCTION TO ARITHMETIC GEOMETRY	MTH671A	padic numbers : nonarchimedean absolute values, valuations, Ostrowskitheorem, Cauchy sequences, padic integers, completions, Hensels lemma, structure of Z and Q. Quadratic forms : definition of quadratic forms, rational points on conics. [8 lectures]Infinite Galois Theory : Profinite groups and profinite topology, Infinite Ga lois extensions and Galois group as profinite groups, absolute Galois groups, the fundamental theorem of Galois theory, absolute Galois group of Q and Qp.Geometry of curves over Q : Affine Varieties and projective varieties, curvesand function fields, divisors on curves, the RiemannRoch theorem (state ment without proof) Q points of elliptic curves, Weak MordellWeil Theorem, MordellWeil Theorem, Faltings Theorem
529	ECO	APPLIED PROBABILITY AND STATISTICS	HSO201A	PROBABILITY THEORY WITH APPLICATIONS: Random Experiments, Events and Probability, Kolmogorov Axioms, Conditional Probability, Independence of Events, Bayes Theorem, with Examples and Applications; The Moment Generating Function; Multivariate Normal Probability Distribution, Independence of Random Variables, Conditional Probability Distributions, Covariance and Correlation, Transformation of Random Variables; Sampling Distributions, Sampling from Normal Distribution, Order Statistics, Convergence of Random Variables, Generating Random Samples, with Examples and Applications; Sufficiency Principle and Sufficiency Statistics; Methods of Evaluating Estimators and Examples of Important Estimators; Fundamentals of Hypothesis Testing, Methods for Evaluating Tests, Error Probabilities and Power Function, NeymanPearson Lemma; Analysis of Variance, with Examples and Applications.

S.No.	Branch	Course Title	Courses ID	Content
530	HSS	VISUAL ETHNOGRAPHY	ART706A	
531	PHY	PHYSICS-II	PHY103A	
532	CHM	ORGANIC CHEMISTRY I	CHM303	
533	CHM	ORGANIC CHEMISTRY I	CHM303	
534	CHM	ORGANIC CHEMISTRY I	CHM303	
535	CSE	ELLIPTIC CURVES AND THEIR APPLICATIONS	CS749A	ELLIPTIC CURVES OVER REALS: WEIERTRASS EQUATION, ENDOMORPHISMSELLIPTIC CURVES OVER COMPLEX NUMBERS:EQUIVALENCE WITH TORI, SYMMETRICS, DIVISORS, AND WEIL PAIRING. EL ALGORATHM. APPLICATIONS: PROOF OF FERMAT'S LAST THEOREM.
536	MTH	AN INTRODUCTION TO OPERATOR THEORY	MTH635A	
537	HSS	LINGUISTIC UNIVERSALS	ENG453A	THEORETICAL AND EMPIRICAL APPROACHES TO LINGUISTIC UNIVERSALS. ARGUMENT FROM LINGUISTIC TYPOLOGY AND EPISTEMOLOGY. ABSOLUTE STATISTICAL AND IMPLICATIONAL UNIVERSALS OF LANGUAGE,PRINCIPAL AND PARAMETERS, THE MINIMALIST PROGRAMME.
538	HSS	LINGUISTIC TYPOLOGY	ENG452A	METHODOLOGY AND UNIVERSAL. PHONOLOGYCAL TYPOLOGY. MORPHOLOGICAL TYPOLOGY. CONSTITUENT ORDER TYPOLOGY. GRAMMATICAL RELATION AND ALIGNMENT. HIERARCHIES AND SEMI RELATIOVE CLAUSES.
539	MTH	NONLINEAR ANALYSIS AND ITS APPLLCATIONS TO PDE	MTH659A	REVIEW OF TOPIC IN FUNCTIONAL ANALYSIS AND SOBOLEV SPACE. MAPPING BETWEEN BANACH SPACES.DEGREE THEORY. BIFURCATION THEORY. VARIATION METHOD: CONSTRAINED CRITICAL P VARIATION PRINCIPAL.
540	PHY	MODERN OPTICS	PHY307A	Review of Max wells and electromagnetic wave equations, wave propagation in anisotropic media, polarized light, diffraction from circular aperture and concept of resolution, Fourier transforms and Fourier optical fibers, optical communication sources (LED, lasers etc.) and detectors, and optical, electro and magneto optic effects, laser matter interaction.
541	HSS	SOFT SKILLS AND PERSONALITY DEVELOPMENT	HSS401A	Personal Skills: SelfAssessment; Identifying Strength & Limitations; Habits, WillPower and Drives; Developing SelfEsteem and Building SelfConfidence, Significance of SelfDiscipline. Understanding Perceptual Excellence; SelfActualisation Need. Goal Setting, Life and Career Planning; Constructive Thinking. Professional Skills: Communicating Clearly: Understanding and Overcoming barriers; Cross gender/Cross Cultural Understanding; Conducting Meetings, Writing Minutes, Sending Memos and Notices. Netiquette: Effective Email Communication; Telephone Etiquette. Body Language in Group Discussion and Interview. Interpersonal Skills: Bonding. Ethics and Etiquettes (Social and Official Settings); Respecting Privacy; Civic Sense and Care for the Environment. Negotiating, DecisionMaking, ConflictResolution, Five Styles. Emotional Literacy: News. Humour, Jokes and Anecdotes in Effective Communication. Management Skills: Managing Time and Beating Procrastination. Managing People: Leading and Working with Team (Coordination and Collaboration) and Maintaining Positive Outlook. Managing Health, Boosting Memory, Enhancing Study Skills. Managing Money and Love; Balancing Personal and Professional Life. Recommended Reading: Dorch, Patricia. What Are Soft Skills? New York: Execu Dress Publisher, 2013.Kamin, Maxine. Soft Skills Revolution: A Guide for Connecting with Compassion for Trainers, Teams, and Leaders. Washington: HarperCollins Ebooks, 2007.Petes S. J., Francis.Soft Skills and Professional Communication. New Delhi: Tata McGrawHill Education, 2011. Stein, Steven J. & Howard E. Book. The EQ Edge: Emotional Intelligence and the Workplace. New York: HarperCollins, 2003.
542	AE	AEROSPACE STRUCTURES-II	AE332A	Buckling of columns. Differential equation approach, energy approach, approximate techniques. Beam columns. Buckling strength of flat sheets in compression, combined stress. Local buckling of composite plates.
543	SDS	PROBABILISTIC THEORY OF PATTERN RECOGNITION	SDS690	Results of convergence in almost sure sense and in probability, DCT, Basic inequalities, Conditional expectation, Methods of resampling. Introduction to discriminant analysis, Basyes' risk, and its probability of its convergence. Parametric methods: Maximum likelihood principle Fisher's linear discriminant function (LDA), quadratic discriminat analysis (QDA). Consistency results. Logistic regression, Linear discriminant analysis (KDA), nearest neighbor classification (kNN), Universal consistencyh results. Idea of curse of dimensionality, and the use of dimension reduction techniques like random projections and PCA. Hybrid classifiesrs, Classification using data depth, Related consistency results. 1. Pattern Calssification by Richard Duda, Peter Hart and David Sstork, Wiley. 2. A Probabilistic Theory of Pattern Recognition by Luc Devroye, Laszlo Gyorfi and Gabor Lugosi. Springer. 3. The Elements of Pattern Recognition by Springer. 05-apr-2016
544	IME	PUBLIC PRIVATE PARTNERSHIP (PPP) IN INFRASTRUCTURE	MBA781A.	Infrastructure, Economic Development and need for PPP. Definition and scope of infrastructure. Economics of infrastructure Natural Monopoly. Modes of PPP. Policy Framework for Infrastructure Investment Roads, WTE, Railways, Metro, Ports, Airports etc. Policy Framework for PPP Across Major Sectors. Global Best Practices for PPP Projects. Dealing with Unsolicited Bids The Swiss Challange. Projects Case Studies. 1. E.R. Yescommbe, Public Private Partnerships Priciples of Policy and Finance, Elsevier Finance (2007). 2. A Guidebook on Public Private Partnership in Infrastructure, UNESCAP, Bangkok, 2011. 3. "Toolkit.pppinindia.com. 6. "Database of Infrastrcture Projects in India", https://infrastructureindia.gov.in/. 7. India Infrastructure Reports 3iNetwork, OUP. 8. Other relevant books, journal articles and studeis and class notes.
545	IME	RENEWABLE ENERGY - ECONOMICS POLICY AND REGULATION	MBA782A	Drivers for Renewable Energy. RES Wind, Solar, Biomass, SHP, Ocean, WTE etc. RE Development National and International Perspective. Economics of Renewable Energy. Policy and Regulatory Instruments in Tariff. Market for Renewable Energy Certificates (RECs). NAPCC and Jawaharlal Nehru National Solar Mission. Competitive Bidding for Renewable Energy. Developing and Implementing a Solar Rooftop Program. Business Models. Projects Caselets on mutually agreed topics. 1. Miguel Mendonca, Feed in Tariff, Earthscan, 2007. 2. Evaluating Policies in Support of the Development of Renewable Power. 3. REN21. Renewables 2015, Global Status Report. 4. Re Policy Database. 5. policy documents, regulations and tariff orders of regulatory commissions, forum of regulators, case studies and class notes. 05-APR-2016
546	IME	PERFORMANCE MANAGEMENT	MBA713A.	Objectives of Performance Mangement, Historical Account, Performance Planning: Synchronization with Organization's vision, mission, strategy and goals, Issues and Problems, Defining and Measuring Performance, Feedback, Rewards and Recognition, Career Management, Role of Performance Management in fostering Employee Engagement, Re designing Jobs for better Performance, Key Implications of Performance Management. 1. Aguinis, H. Performance Management (3rd ed.). McGraw Hill. 2. Cardy, R.L. & Leonard, B. Performance Management: Concepts, Skills and Exercises (2nd ed.). M. E. Sharpe.3. Rao, T.V. Performance Management: Concepts and articles. 29-MAR-2016
547	IME	ORGANIZATIONAL STAFFING	MBA712A.	Introduction to Organizational Staffing: Steps and Strategic Linkages; Manpower Planning; Job Analysis Techniques; Competency Based Staffing; Models and Mapping Techniques; External and Internal Factors Affecting Staffing. Methods and Tools: Utility, Relevance and Applicability; Organizational Socialization and Deployment; Succession Planning and Career Progression; Use of IT in Staffing; Other Emerging Issues. 1. Herbert G. Heneman & T.A. Judge, Staffing Organizations. McGraw Hill. 2. Phillips, J. M. Strategic Staffing. Prentice Hall. 3. Additional readings: relevant cases and articles. 29-MAR-2016
548	IME	CHANGE MANAGEMENT AND ORGANIZATIONAL DEVELOPMENT	MBA711A.	Understanding Organizational Change: Definition and Types, Systems Perspective, Identification and Assessment of Factors Leading to Change, Change Management Process: An Overview, Resistance to Change Management, Business Process Reengineering, Appreciative Inquiry, Change Management through OD Intervention. 1. Grievies, J. (2010). Organizational Change: Themes and Issues. Oxford. 2. Brown, D., & Harvey, D. (2206). An Experiential Approach to Organizational Development. Pearson. 3. Cummings, C.G. & Worley, P. Organizational Development and Change. 29-MAR-2016
549	CE	OPTIMIZATION METHODS FOR CIVIL ENGINEERS	CE606M	Introductionto the course and its importance. Optimization methods: problem formulation, solution techniques for linear and integer problems (both unconstrained and constrained), sensitivity analysis. Bibliography: 1. Deb K, Optimization for Engineering Design: Algorithms and Examples, PHI, Second Edition. 2. Rao, S.S. "Engineering Optimization: Theory and Practice", 4th Edition, Wiley. 3. Hamdy A. Taha. Engineering Optimization: An Introduction to Metaheuristic. 05-MAR-2016

S.No.	Branch	Course Title	Courses ID	Content
				Chong, E. K.P. and Zak, S. H. An Introduction to Optimization: Second Edition, Wiley Student Edition. 01-APR-2016
550	PHY	DYNAMICAL SYSTEMS AND CHAOS	IDC601A	Dynamical systems (map, flow, boolean, automata): Finite vs. infinite dimensional systems, Conservative vs. dissipative systems, Phase space, Fixed points, Linear stability analysis, Poincare section, Numerical methods, Period doubling route to chaos, Intermittency route to chaos, Crisis, Quasi periodic route to chaos. Autonomous flows: Hartman Grobman theorem (proof optional), Poincare Bendixon theorem (proof optional), Bifurcation diagrams, Lyapunov exponents, Routes to chaos. Non Autonomous flows: Floquet theory, Duffing oscillator, Chaos. Introduction to bifurcation theory and normal forms. Fractals, Topological dimension, Similarity dimension, Embedding dimension, Synchronization, Phase space reconstruction, etc. 1. S.H. Strogatz, Nonlinear Dynamics And Chaos: With Applications to Physics, Biology, Chemistry, And Engineering, Westview Press (2001). 2. R.C. Hilborn, Chaos and Nonlinear Dynamics, Oxford University Press (2000). 3. R. Devaney, Introduction to Chaotic Dynamical Systems, Springer (2015). 4. H.G. Schuster & W. Just, Deterministic Chaos: An Introduction, Wiley VCH (2005). 5. D. Jordan & P. Smith, Nonlinear Ordinary Differential Equations: An Introduction for Scientists and Engineers, Dynamical Systems, and Bifurcations of Vector Fields, Springer (1983).
551	PHY	INTRODUCTION TO HIGH PERFORMANCE COMPUTING FOR SCIENTISTS & ENGINEERS	IDC600A	Introduction to HPC and scientific computing. Overview of major applications. Supercomputing architecture; multicores; shared memory; switch etc. Review of basics of C/Fortran programming. Programming in parallel. 1. P.S. Pacheco, An Introduction to Parallel Programming, Elsevier (2011). 2. M. Quinn, Parallel Programming in C and OpenMP, McGraw Hill Education (India) (2003). 3. A. Grama, A. Gupta, G. Karypis, and S. Kulkarni, Introduction to Parallel Computing, Springer (2006).
552	CE	AIR POLLUTION AND ITS CONTROL	CE666A	
553	ES	FIELD GEOLOGY III	ES418A	General overview of geological structures in the field; Concept of orientation and scale in the field; Identification, measurement and presentation of different structural elements (lineation, cleavage, foliation, etc.); Structural mapping of macrostructures; Strain analysis from deformed objects; Technique and ethics of geological samples collection; Large and small scale lithostructural mapping (on toposheet and white paper), crosssectioning and interpretation. 1. Davis, G.H., & Reynolds, S.J., 1996. Structural Geology of rocks and Regions (2nd Ed): NY, John Wiley & sons, 776 p. 2. Lisle, R.J., 1995. Geological Structures & Maps, a Practical Guide. 2nd Ed., Butterworth-Heinemann, London, 129 p. 11-MAR-2016
554	ES	GEOLOGICAL EVOLUTION OF INDIAN PLATE	ES417A	Overview of geologic and tectonic evolution of the Indian plate, major geologic and tectonic features of the Indian subcontinent, Geodynamics and major structural grains in the Indian subcontinent; India's tectonic evolution and their development in the Archean; Proterozoic basins (Chhatisgarh, Cuddapath, Marwar, PranhitaGodavari and Vindhyan), Gondwana basin; Rifting, drifting, Palaeomagnetic interpretation and the evolution of volcanic provinces; Plateau uplift (Deccan, Tibet and Shillong); Phanerozoic stratigraphic records of peninsular India; The Himalaya mountains; northward flight of India and collisional orogenesis; Classification of Indian tectonic provinces. 1. Ramakrishnan, M. and Vaidyanadhan, R., 2008. Geology of India: Vol: I and II, Geological Society of India Publication. 2. Wadia, D.N. 1919. Geology of India for students, McMillan and Co., Ltd. 466 p. 3. Ramakrishnan, M. and Vaidyanadhan, R., 2003. Geology of India: Vol: I and II, Geological Society of India Publication. 942 p. 11-MAR-2016
555	ES	EXPLORATION GEOPHYSICS	ES416A	Introduction to exploration geophysics; Gravity and Magnetic methods: History of gravity magnetic explorations, elementary theories of gravity and magnetic methods, densities and magnetic susceptibility, magnetic interpretation and applications; Electrical and electromagnetic methods: Electrical properties of rocks and minerals, self potential and its origin, concepts of D.C. resistivity, various electrode configurations, resistivity, Maxwells equations, different EM methods, earth's natural electromagnetic field, magnetotellurics, various applications of EM; Seismic methods: Basics of seismic theory, Geometry of seismic wave propagation, seismic processing steps, basic velocity/depth modeling, interpretation of seismic data. Radiometric methods: Principles of radioactivity, radioactivity of rocks and minerals, measuring instruments and applications of radiometric methods, density and sonic logging and well log interpretation. 1. Telford, W.M., Geldart, L.P., and Sheriff, R.E., 1990, Applied geophysics (2nd Edition), Cambridge University Press. 2. Keary, P., Brooks, M., and Hill, I., 2002, An introduction to geophysical exploration (3rd Edition), Cambridge University Press. 3. Serra, O., 1987, Fundamentals of Welllog Interpretation, Elsevier. 5. Sheriff, R.E., and Geldart, L.P., 1995, Exploration Seismology, Cambridge University Press. 11-MAR-2016
556	ES	GEOLOGICAL REMOTE SENSING AND GIS	ES415A	Spectra of earth's surface material; Basic principles of digital image processing point and algebraic operations, filtering and neighbourhood processing, RGB to HIS transformations, image fusion analysis, image classification applications, UAV, and airborne sensors. Basic principles of Geographic Information System (GIS) and its application decision support and uncertainty, multicriteria evaluation. Remote sensing applications in land use/cover Mapping and forestry applications; applications in glaciology and snow hydrology; snow cover mapping and prediction of snowmelt runoff; Coastal zone mapping and other related applications; Natural hazards. 1. Gupta, R.P. (1991), Remote Sensing Geology, Springer Verlag. 2. Lio, J.G. and Mason, P. (2009). Essential image processing and GIS for remote sensing. Wiley Blackwell. 3. Lillesand, T.M. & Kiefer, R.W. (1999), Remote Sensing: The Image-Based Approach, 2nd Edition, John Wiley & Sons. 4. Rhind, D.W. (1999), Geographical Information Systems, Vol. 1 & 2, John Wiley & Sons. 5. Carbonneau, P. and Piegay, H. (2012). Fluvial remote sensing for science and management. Wiley Blackwell. 11-MAR-2016
557	ES	FIELD GEOLOGY II	ES414A	This course is designed to provide practical experience in stratigraphy including field logging of stratigraphic sections, sedimentary facies identification and description, interpretation of sedimentary processes, and also be taught geophysical data acquisition using available geophysical equipments (e.g., gravimeter, magnetometer, VLF electromagnetic equipment, resistivitymeter equipment, GPR, Seismic, and Global Positioning System). 1. Compton, R. (1985) Geology in the field, Wiley. 2. Lahee, F.H. (2002). Field geology, CBS publishers, 3. John Milson and Asger Eriksen, 2011, Field Geophysics (4th Edition): Wiley Blackwell, 4. Reynolds, J.W., 1997, An Introduction to the Interpretation of Geophysical Data, Cambridge University Press. 11-MAR-2016
558	ES	GEOCHEMISTRY	ES413A	Internal structure of atoms, electronic structure, chemical bonding, and chemical properties of elements. Fundamentals of Thermodynamics and its application in Earth Sciences. Aquatic geochemistry, petrogenetic diagrams, Eh pH diagrams, oxidation reduction reactions. The origin and evolution of Earth and the solar system through high temperature chemical processes, trace elements in igneous processes, models of planetary geochemistry. Earth's hydrosphere and its interaction with surficial rocks, sediments, soils, biosphere and the atmosphere. 1. Principles and applications for Geochemistry (1998). Faure, G., Prentice Hall ISBN 10:0023364505, 2. Geochemistry : Pathways and Processes (2004). McSween, H.Y., Richardson, S.M., Uhle, M., Columbia University Press. 11-MAR-2016
559	ES	SEDIMENTARY PROCESSES & STRATIGRAPHIC PRINCIPLES	ES412A	Basinal Sedimentary Systems: Sedimentary basins and production of sediments; Transport of sediment grains; depositional processes and forms; Post depositional changes lithification and diagenesis. Sedimentary facies: facies parameters and distribution, grain shape and form; primary grain fabric; Bedforms & inorganic primary sedimentary structures. Sedimentary Facies analysis: Concept of sedimentary facies; facies relationships in different environments and facies models. Clay Sedimentology: Origin of clay minerals, clay minerals in fluvial, aeolian and lacustrine environments, paleoenvironmental interpretations. Techniques in elastic sedimentology. 1. Allen P.A. and Allen J.R., Basin Analysis: Principles and Application, Blackwell publishing. 2. Leeder M., Sedimentology and Sedimentary Basin: from Turbulence to Tectonics; Wiley Blackwell. 3. Selley R.C., Fundamentals of Stratigraphy, Blackwell. 11-MAR-2016
560	ES	STRUCTURAL GEOLOGY	ES411A	Concepts of deformation and structures in Earth and planetary systems; Concepts of Continuum, Solid, Fluid, Tensor, Force, Stress and Strain; Basics of rheology and deformation mechanisms; Structures and deformation mechanism; Normal, reverse, oblique and strike-slip faults; Fold and thrust belts; Measurement and presentation of 1, 2 and 3D structural elements; Ductile shear zones; Polyphase (superposed) deformation and structures. 1. Ramsay, J.G., & Huber, M.I., 1983. The Techniques of Modern Structural Geology: V.1: Strain analysis. NY, Academic Press, 307 p. 2. Ramsay, J.G., and Huber, M.I., 1987. The Techniques of Modern Structural Geology: V.2: Structural Analysis. Academic Press, 390 p. 3. Ramsay, J.G., and Huber, M.I., 1987. Modern Structural Geology, V.3: Applications of Continuum Mechanics in Structural Geology. Academic Press, 361 p. 11-MAR-2016
561	ES	IGNEOUS & METAMORPHIC PETROLOGY	ES315A	This course will introduce igneous and metamorphic rocks and focus on the processes and principles involved in the generation of these rocks in a wide range of tectonic settings. Emphasis will be on descriptive petrology, the evolution of magma, and metamorphic processes etc. Topics to be discussed in detail are : Classification and Nomenclature of Igneous Rocks; Textures and Petrogenetic Interpretations; Thermodynamic

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				element, and isotopic compositional variations; graphical and mathematical models of magme evolution; Fractionation of trace elements during melting and crystallization; Generation and diversification and chemographic (ADF, AKF, and AFM) giagrams; Metamorphic Facies; Metamorphic Reactions. 1. An Introduction to Igneous and Metamorphic Petrology by John D. Winter (Prentice Hall, 2nd edition). 2. The Interpretation of Igneous Rocks by Cox, Bell and Pankhurst (Chapman and Hall).3. Petrograph 11-MAR-2016
562	ES	FUNDAMENTALS OF GEOPHYSICS	ES314A	Introduction to geophysics, Earth as a planet and member of the solar system, origin and evolution of the Earth, Internal structure of the Earth; Concept of plate tectonics, plate motions and triple junctions paleomagnetism, and geomagnetic reversals; Introduction to seismology, seismic waves P, S and surface waves, seismograph, travel time curves and radial Earth structures, general properties of surface elastic rebound theory, global seismecity and tectonics, focal mechanisms, seismic anistropy; Heat within the Earth, thermal structure of continental and oceanic lithospheres at subduction zones and spr 1. Lowrie, W., &#34;Fundamentals of Geophysics&#34;, Cambridge University Press. 2. Fowler, C.M.R., 2004, &#34;The Solid Earth: An Introduction to Global Geophysics&#34;, Cambridge University Press Geophysicists&#34;, Prentice Hall. 11-MAR-2016
563	ES	GEOMORPHOLOGY AND EARTH SURFACE PROCESSES	ES313A	Fundamental concepts of geomorphic system; Earth&#39;s energy balance, global heat transfer, topography and bathymetry, liberation and flux of sediments, hydrological cycle and water budget. Guiding processes and ages of landscapes. Whole Earth morphology and largescale topography; Exogenic and endogenic processes. The Surface water system: Drainage basins and river systems, river morpholo river dynamics. The Groundwater system: Groundwater in hydrological cycle, groundwater flow and storage; chemistry of groundwater. The Atmospheric System: Atmospheric composition and mixing, atr currents. The relative movement of land and sea; coastal processes and landforms. Cryosphere growth and decay of ice sheets, controlling factors, Himalayan glaciers; Wind activity and geomorphic wor evolution; Landforms and tectonics of plate margins and plate interiors; Tectonic uplifts and denudation rates and controlling factors, Sea level change evidence, mechanism and effects; coupled tectonic 1. Summerfield, S.A. (1991) Global Geomorphology, Longman. 2. Ernst, W.G. (2002) Earth Systems: Processes and Issues, Cambridge University Press. 3. Richards, K.S. (1982) Rivers: forms and processes Dewet, A. and Menking, K. (1998) Environmental Geology; An earth system science approach. W.H. Freeman. 11-MAR-2016
564	ES	FIELD GEOLOGY-I	ES312A	This course will provide an introduction to field work activities. How to use a Brunton Compass, reading of topographic maps, dip and strike measurements, and basics of lithological mapping. 1. Robert R. Compton, Geology in the Field. July, 1985.2. Angela L. Coe, Geological Filed Techniques. October 25, 2010.3. Richard J. Lisle, Peter Brabham, John W. Barnes, Basic Geological Mapping, 5th Ed 11-MAR-2016
565	ES	CRYSTALLOGRAPHY AND MINERALOGY	ES311A	Chemical and physical properties and identification of common rockforming (silicates, carbonates and oxides) minerals; crystallography, Unit cells, Crystal symmetry, classes and rotation. Systems, Plane Stereographic Projection of Crystal Faces, Polymorphis and Pseudomorphs,Twinning; optical mineralogy, uniaxial, biaxial minerals; introductions to xray crystallography; crystal structures, chemistry, and and its Core. 1. Nasse, W. 2011. Introduction to Mineralogy (2 nd Ed.)Oxford University Press. 496 p. 2. Klein, C. & Dutrow, B. 2007. Manual of Mineral Science. Wiley. 716 p. 3. Putnis, A. 1992. An Introduction to Mineral Corporation. 5. Hammond, C. (2001). The basics of crystallography and diffractions, Oxford Science Publications. 6. Lovett, David (1999) Tensor properties of crystals, CRC Press. 11-MAR-2016
566	CE	INTRODUCTION TO ENVIRONMENTAL SCIENCE	CE213A	Introduction: Define environmental science as a subject that draws on learning&#39;s from natural sciences to investigathe human impact on the natural environment. Present concept of hydrosphere, atr particularly the inter relationships between organisms and among organisms and their environments. Water: Hydrologic Cycle and Water Bodies, Water Availability and Use, Water Pollution, Water Manage and the basis on which they are formulated. Air Pollution and Contrlo; : Classify sources and types of airpollution. Pollution monitoring setup including standards for regulated pollutants. Effect of pollution atmosphere and the processes that form climate and climate change. Nazaroff, Jerald L. Environmental Engineering and Science, 2004, John Wiley & Sons, Inc. Environmental Science Towards a Sustainable Future by R.T. Wright and D.F. Boorse, PHI Learning. Environmental Science: A Study of Interrelationships, 12th edition, McGrawHill, 2009. 11-MAR-2016
567	CE	SUSTAINABLE BUILT ENVIRONMENT	CE241A	Module 1: Description of Earth and environment. Earth and Environment Definition of environment; origin of earth, lithosphere, hydrosphere, atmosphere, biosphere. Earth structure, Plate Tectonics theory, of soils, Weathering and Soils; Rocks, rock ecycle, Igneous Rocks, Sedimentary Rocks, Metamorphic Rocks, Rock Properties, Rockwater interaction. Natural disasters: Cyclone, Tornado, Volcanic Eruption, of the world, Tsunami, Land Slides; Sustainability and resilience for natural disasters. Hydrosphere; water cycle, surface and groundwater origin and its quality, occeans, ocean currents, ocean water quality Loftness, V.; Haase, D. (Eds.) Sustainable Built Environments, Springer, 2013. Yao, R. (Ed). Design and Management of Sustainable Built Environments, Springer, 2013. Zetter, R.; Watson, G.B. (Eds.) Designi WileyBlackwell, 2002. 11-MAR-2016
568	HSS	SOCIOLOGY OF GLOBALIZATION	SOC482A	Defining Globalization : a. Flows and Structures, b. Key Debates, c. Related Processes. SocioEconomic Aspects of Globalization : a. Neoliberalism, b. Global Economic Structurlesc. c. Global Economic Flow of people. Environmental Flows. Globalization and Social Stratification : a. UrbanRural, b. Gender and Sexuality, c. Class, d. Race, Ethnicity and Caste. Ritzer, George. 2010. Globalization: A Basic Text. Chichester: Wiley Blackwell.Ritzer, George and Zeynep Atalay, eds. 2010. Readings in Globalization: KeyConcepts and Major Debates. Chichester: Wiley Bla readings will be assigned to students well in advance. 11-MAR-2016
569	HSS	PSYCHOLOGY OF WELLBEING	PSY470A	Conceptualizing Wellbeing : (a) Nature, definitione and lived experience of wellbeing, (b) Methods in the studey of wellbeing: Quantitative and qualitative approaches.2) Types of Wellbeing: a) Objective well Wellbeing: Experiences and Constructions within the Socio]Cultural Contexts 4) Wellbeing among Specific Groups: Youth, Elderly, Women, Disadvantaged Groups and Disaster Survivors 5) Indian Psycholo 1. Denzin, N. (1985). Emotions as lived experiences. Symbolic Interaction, 8, 223]240.2. Diener, E., & Suh, E.M. (2000). Culture and subjective well]being (pp. 185]218). Cambridge, MA: MIT Press. 3. Diener, life. Annual Review of Psychology, 2003, 54, 403]425. 4. Kahneman, D., Diener, E., & Schwarz, N. (Eds.). (1999). Well]being: The foundations of hedonic psychology. New York: Russell Sage Foundation. 5. K US: American Psychological Association.6. Kitayama, S., Markus, H.R. & Kurokawa, M. (2000). Culture, emotion, and wellbeing: Good feelings in Japan and the United States, Cognition and Emotion, 14, 93 11-MAR-2016
570	IME	INVESTMENT VALUATION AND REAL OPTIONS	MBA610A.	Introduction to Valuation methods, Investment Valuation, Estimating Cost of Equity and Cost of Capital, Derivatives, Option Pricing Theory, Option Pricing Applications ain Valuation, Real Options in Manag Abandon, Valuing Natural Resources Using Real Options, Mini Project Identifying Real Options in Practice, Appraising Projects with Real Options. Damodaran, Ashwath (2002), Investment Valuation, (Second Edition), Wiley.Broyles, Jack (2003), Financial Managemnt and Real Options, Wiley.Mun, Johnathan (2005), Real Options Analysis: Tools and Te (eds.) (2001), Real Options and Investment Under Uncertainty, MIT.Copeland, Tom & Antikarov, Valdimir (2001), Real Options: A Practitioner&#39;s Guide, Textre. Selected Case Studies on Valuation and Re
571	HSS	INTRODUCTION TO QUANTITATIVE STATISTICS USING R	PSY792A	
572	HSS	INTRODUCTION TO QUANTITATIVE STATISTICS USING R	PSY792	
573	ME	PROJECT II	ME452A	Fabrication of a prototype and appropriate modification in the design (if necessary) to meetthe qualitative and quantitative performance parameters as envisaged in the Project I

S.No.	Branch	Course Title	Courses ID	Content
574	ME	MANUFACTURING SYSTEMS	ME461A	Introduction to manufacturing, Manufacturing system concept.Manufacturing automation, FMS, CIMS, Flow lines and assembly systems, Automated storage /retrieval systems, AGV. Introduction to CAD/ application inmanufacturing. Process planning and Computer Aide Process planning. Group Technology,Opitz System and GT benefits. Material Management and Inventory control, MRP and MRP II.Just i charts, sampling. Total Quality Management. Manufacturingsystem simulation. 1. Computer Integrated Design and Manufacturing by Nanua Singh, john Wiley. 2. Computer Aided Manufacturing by Chang, Wysk, Wang, Prentice Hall. 3. Computer Aided Manufacturing by Rao, Tewari, K
575	ME	VIBRATION & CONTROL	ME354	Introduction to modeling of dynamical systems. Single Degree of FreedomSystems Free undamped vibration, Free damped vibration, Forced vibration, Transmissibility, Convolution method, Mechanisms o Multi Degree of Freedom Systems(undamped and proportional damping) Matrix methods, Modal analysis. Approximate Methods.Vibration of continuous systems (free vibration only).Introduction to contr Bode plots. Controller performance and types.Steady state errors and constants. Types of feedback control systems Derivative errorcompensation, Integral error compensation, Proportional error compen 1. Theory of Vibrations. W. T. Thomson, Prentice Hall.2. Control Systems Engineering. N. S. Nise, john Wiley & Sons.3. Vibration Problems in Engineering. W. Weaver, S. P. Timoshenko and D. H. Young, john Franklin,). D. Powell, and A. EmamiNaeini,Prentice Hall.6. Modern Control Engineering. K. Ogata, Prentice Hall.
576	CE	CHARACTERIZATION AND USE OF AGGREGATES IN PAVEMENT SYSTEMS	CE661	
577	ES	GEOPHYSICAL METHODS	ES656A	
578	ES	SOLID EARTH GEOPHYSICS	ES655A	
579	ES	FUNDAMENTALS OF EARTH SCIENCES	ESO213A	universe and its characteristics Solar System and Earth The primitive Earth Geological Time scale Origin of the life and major geological events Numerical Dating. Rocks, minerals and soils; Plate Tectonic Atmosphere, Biosphere, Cryosphere and Climate; Energy budget; Carbon Cycle; Hydrological Cycle; Weathering and erosion.Coupled processoes in Earth System; climate change,Geological resourses (min 1. D.R.Prothero and R.H. Dott, Jr. Evolution of the Earth. 2010 (8th Ed.), McGraw Hill, 576p. 2. E.J. Tarbuck, F.K. Lutgens and D.G. Tasa. Earth: An introduction to Physical Geology, 2013 (11th Ed.). Prentice
580	IME	ENERGY AND CARBON MARKETS: ECONOMICS POLICY AND REGULATION	MBA681.	Energy and Economic Development: National and International Perspective, Structure of Energy Demand and Supply,, Energy Value Chain and Energy Accounting, India Energy Scenario 2047, Economics o Applications in General Algaebric Modelling System (GAMS), International Markets for Energy: Oil, Coal, Natural Gas and Uranium, Indian Energy Markets: Oil & Gas, Coal and Electricity, Private Investment Sectors, Pricing in Energy Markets: Electricity, Coal, Oil and Natural Gas, Functioning of Power Exchange and Commodity Exchanges (Energy), Cross Border Energy Cooperation, Energy and Environment, Carbon Markets and Carbon Finance, National Action Plan on Climate Change, JNN National Solar Mission, Renewable Energy: Technology, Economics and Policy, Market for Renewable Energy Certificate on relevant topic. Recommended Test: 1. Environmental and Natural Resource Economics, Tom Tietenberg, AddisonWesley, 2000. 2. International Energy Markets: Understanding Pricing, Policies, and Profits, Carol Dahl, Pe 9781593701789, 2009.4. Economics of Regulation and Antitrust Viscusi, Vernon & Harrington, MIT Press, 2000.5. Integrated Energy Policy, Report of the Expert Committee, Govt. of India, Planning Comm Bank and Asian Developoment Bank.7. Relevant reports of respective Ministries.8. Other relevant journal articles, reports, policy documents, regulations of regulatory commissions, case studies and class r
581	MSP	ELECTRONIC MATERIALS	MS606A	Classification, Crystal growth techniques, Wafer processing, Doping methods, Formation of oxide layer, CVD, MOCVD and MBE, Metallic contacts and interconnects, Lithography, Processing integration. Pl and high temperature devices, Application of linear and nonlinear dielectric materials, Electrooptic ceramics, Materials for signal processing, transducers and digital data storage, Superconducting materia
582	PHY	NUCLEAR AND PARTICLE PHYSICS	PHY526A	Nuclear Physics: General properties of nuclear two body problem, Nuclear force and nuclear models, Nuclear decay, Nuclear reaction kinematics, Scattering and reaction cross section, Optical Model, Clas fusion. Particle Physics: Natural Units, Evidence for four fundamental interactions, Leptons and hadrons, Historical introduction to the particle zoo, introduction to cross sections and decay rates, Particle a CP, isospin, Strangeness. 1. I. Kaplan, Nuclear Physics (Narosa). 2. S. Krane, Introduction to Nuclear Physics (Wiley). 3. D.H. Perkins, Introduction to High Energy Physics (Cambridge University Press).
583	IME	MANAGEMENT INFORMATION SYSTEMS	MBA645A.	Foundation Concepts: Basic information systems concepts about the components and the operations, managerial, and strategic roles of information systems; Technology: Major concepts, developments, technologies; Applications: How the Internet, intranets, extranets and other information technologies are used in modern information systems to support electronic commerce, enterprise collaboration, bus solutions to business problems using a systems approach to problem solving and variety of business application development methodologies; Management: The challenges of managing information syst and security and ethical challenges. This course also includes case presentations/discussions along with a final term paper.
584	IME	MANAGEMENT OF TECHNOLOGY	MBA626A.	Policy Technology Choice: Linkage; National Technology Policies; Technology,Competition and Industrial Structure; formulating the technology strategy,Technology Development and Acquisition process; Licensing and joint Ventures, Managing Technology Spillovers;Justification of new technology; management accounting and technology;Integration of New with Old technology, Assimilation of Technology
585	ME	COMBUSTION ENGINES & AIR POLLUTION	ME741A	Thermochemistry, Chemical Equilibrium, Kinetics; Laminar and Tarbulent Flamepropagation in SI Engines, Unburned and Burned Mixture States, Flame Quenching; Fuel Injection, Spray Atomization, Penetr Mechanism, Soot Formation; Vehicle Emissions and standards; Emission control Technologies, Catalytic Control, Engine Design and Fuel effects,New Advances; Emission Measurement.
586	ME	COMBUSTION	ME643	Flame phenomena in premixed combustible gases. Diffusion flamesanalysis of single fuel droplet, chemical reactions. Burning in convective atmosphere, spray combustion, fire modelling, radiation in flar
587	ME	COMBUSTION	ME643A	Flame phenomena in premixed combustible gases. Diffusion flamesanalysis of single fuel droplet, chemical reactions. Burning in convective atmosphere, spray combustion, fire modelling, radiation in flar
588	EE	ADVANCED PROTECTIVE RELAYING	EE636A	Advanced protective relaying, basic protection schemes, relay terminology, relays as comparators, static relays, application of solid state devices, differential relaying systems, distance relaying schemes
589	EE	OPTO-ELECTRONICS	EE416A	LEDs, semiconductor lasers, modulation of laser sources. Avalanche and PIN photodetectors and their characteristics. Solar cells. Optical fibers and their characteristics. Integrated optics. Fiber optic c
590	CSE	PSEUDO-RANDOM GENERATORS	CS744A	Pseudorandom generators are efficiently computable functions that stretch and input random string to a much bigger sized string such that the output string appears random to resourcebounded comput utility. They are used to derandomize randomized algorithms, formalize notions of cryptographic security, obtain lower bounds on the complexity of problems etc. (unfortunately, as of now very few constr pseudorandom generators and their connections in depth. The topics covered in the course are as follows :Pseudorandom generators: definitions and Existence. Pseudorandom generators of small functions conjectured to be oneway functions, Pseudorandom function generators. Pseudorandom generators of large stretch: Equivalence of lower bounds and pseudorandom generators, Known pseud generators.Pseudorandom generators against arithmetic circuits: Equivalence of lower bounds and pseudorandom generators, A function conjectured to be pseudo random.
591	CHE	POLYMER PROCESSING	CHE672A	Review of equations of motion, constitutive equations; Calendaring, extrusion;molding; mixing; fibre spinning.
592	CHE	THERMODYNAMICS OF FLUIDS AND FLUID MIXTURES	CHE623A	Classical thermodynamics of phase equilibria: Thermodynamic properties from volumetric data: Nature of intermolecular forces, Theory of corresponding states; Fugacities in gas mixtures and liquid solu
593	CE	GROUND IMPROVEMENT TECHNIQUES	CE634A	Engineering properties of soft, weak and compressible deposits; Principles of treatmentloading (static and Dynamic); Accelerated flow; Reinforcement; Drainage and filters, Injections, thermal, electrical a methods and case studies.
594	CE	FLUID MECHANICS LABORATORY	CE612A	Verification of momentum equation; Friction loss in pipes; Rainfallrunoff relationship; Flow over sharp crested weir; Flow in pipe networks; Bernoulli theorem; Fall velocity of objects; Point velocity measur abstraction; Hydrogen bubble flow visualization; Hydraulic jump; Flow past a cylinder.
595	HSS	INDIAN WRITING IN ENGLISH	ENG711A	Selected Indian writings in English or translated from Indian Languages will be used for an extensive examination of some significant themes, e.g., rejection of old taboos, industrialization and its attendan

S.No.	Branch	Course Title	Courses ID	Content
596	HSS	RESEARCH METHODS	SOC720A	Nature and types of scientific explanation, Values and objectivity in social science research, Various research designs: (i) Descriptive, (ii) Exploratory, (ii) Experimental (two and multigroup designs). Tools and scaling techniques (Differential scales, Summated scales and Cumulative scales, etc), problems of measurement, Various types of reliability and validity of measures, Qualitative research.
597	HSS	HUMAN COGNITIVE PROCESSES	PSY790A	The course deals with scientific study of human mind and explores cognitive processes involved in perception, memory, pattern recognition, psycholinguistics, and bilingualism. Recent research techniques
598	HSS	TWENTIETH CENTURY PHILOSOPHY II	PHI765A	An Analytical Study of the Logic of Ordinary Language; Problems of Meaning, Reference and Truth; the AnalyticSynthetic Distinction; the SchemeContent Distinction; Antifoundationalism and Neopragmatism; Ludwig Wittgenstein; Martin Heidegger; Hans Nagel.
599	HSS	POSTWAR AMERICAN FICTION	ENG750A	Postwar American fiction is characterized by a complex sensibility that is oftenpervasive in the novels of the period from 1945 through the Cold War of the seventies and eighties to the present. With American writers to give way to introspection and selfdebate concerning the problematic of defining American identity and nationhood against a distinct multicultural presence, and the unviability of "American Dream" and the investments. Interestingly, fiction of this period frames war both as a trope and a realistic concern. The protracted cultural wars which began with the Civil Rights era of the 1960s infused new life into the novel, and the immigrants, began to be heard. And the variegated literary movements (postmodernism, humanism, and feminism to name a few) and fictional strategies (such as protest, fantasy, black humor) employed to negotiate reality into their imaginative artistic vision. The fiction of this period is marked by thematic concerns such as politics, paranoia, race, money, technology, sex, suburbia, urban decay, immigration, and social interventions.
600	HSS	FEMINIST THEORY AND LITERATURE	ENG743A	Definitins: the Sex / Gender debaate, Masculinity and femininity, patriarchy, the feminism/ gender studies debate.Gender and Identity: Processes of Identity formation, gender roles ain different social contexts, internal vs external selfrepresentations, the media and representation.The Politics of Feminism: The reasons and implications of focusing on gender as a subject of inquirey, the relationship between theory and literature.
601	HSS	INDIAN LITERATURE	ENG437A	The course attempts to g1ve a crosssection of Indian literature which isdiverse, multilingual, and spans a vast period of time. The broad areas that will be covered arethe origin and development of genres, the literature written in the variousIndian languages. These will be exemplified with the help of relevant literary texts from differentages and different parts of the country. It will also critically examine the attention to Indianlanguage texts, including fiction, poems and plays in English translation. No specific textbook. Selections can include Sangampoetry, Bhakti poetry, and works of writers like Premchand, Rabindranath Tagore, IsmatChughtai, Dilip Chitre, M. T. Vasudevan Nair, and Mahasweta Devi.
602	HSS	ART AS A MEDIUM OF COMMUNICATION	ART704A	Theory: Prehistoric Art; Traditional Art as Medum of Communication Religious Art, Buddhist Art, Christian Art, and Hindu Art; Critical analysis of the following theories/ articles Plato's "Art as Immitation"; Communication of Feeling"; David Hume's "Of the Standard of Taste"; susanne Langer's "Art as Symbolic Expression : From Feeling and Form"; Arthur Danto's, "The End of Art";
603	ECO	ENVIRONMENTAL ECONOMICS LEGISLATION AND SOCIAL IMPACT	ECO747A	With a view to establish sustainable development and to overcome the dilemma of development, the course proposes to focus on the aspects of environtal economics, environmental legislations and the effect of the leadership of U.N. and World Bank will be discussed along with a number of case studies. The legal aspects of various environmental excercise through project assignment.
604	ECO	DEVELOPMENT ECONOMICS	ECO735A	The international trends in economic development, various development theories and operational strategies towards capital formation, international capital flows, foreign trade, agriculture, industry, HRD, technology, economic policies and emerging new international structures. Besides, the shidents will undertake some empirical project as group assign many.
605	HSS	SECOND LANGUAGE ACQUISITION	ENG408A	1. The foundations of SLA First language and Second language, acquiring L1 and L2, diversity in learning and learners, logical problem of language acquisition, differentframeworks of SLA. 2. Interlanguage, the hypothesis, functional approaches, functiontoform mapping. 3. Psycholinguistic aspectsLanguage and brain, learning processes, competition models, connectionistapproaches, differences in learners, learning, social factors, role ofinput and interaction, role of output. 5. Acquiring knowledge for L2 use Linguistic competence, linguistic performance, academic vs interpersonal competence, receptive activities, production, instruction, teachability/ learnability, input manipulation, input enhancement.
606	MTH	A FIRST COURSE IN ALGEBRAIC NUMBER THEORY	MTH712A	A brief review of commutative algebra Localization, Noetherian rings and modules, integral extensions, Dedekind domains and discrete valuation ring, Spec of ring. Number field, ring of integer, primes and prime ideals, valuations. Cyclotomic fields. Zeta functions and Lfunctions, class number formula. Adeles and ideles. 1. Number Fields, D. Marcus, Universitext, SpringerVerlag.2. Problems in Algebraic Number Theory, E. Jody and M.Ram Murty, GTM 190, SpringerVerlog. 3. Algebraic Number Theory, J.S. Milne, Course notes, Mathematische Wissenschaften 322, SpringerVerlog. 5. Algebraic Number Theory, S. Lang, GTM 110, SpringerVerlog. 6. A course in Arithmetic, J P Serre, GTM 7, SpringerVerlog. 7. Algebraic Number Theory, R. Narasimha Murthy, algebra, M.F. Atiyah, and I.G. Macdonald, AddisonWesley Publishing Co.
607	MTH	REPRESENTATION THEORY OF FINITE GROUPS	MTH732A	Basic representation theory, Irreducible representation, Equivalence and unitary equivalence, Construction of new representation, Character of a representation, Schur's lemma and its applications, Simple groups, group algebra of S and A for small values of n.
608	MTH	BRANCH ALGEBRAS C* ALGEBRAS AND SPECTRAL THEORY	MTH734A	Elementary properties of Banach Algebras and examples; Ideals and quotients, the Spectrum, the Riesz Functional Calculas. Abelian Banach Algebras, C* Algebras; Representations of C* Algebras and the GNS construction; abelian C* algebras; The Spectral theorem; Some applications.
609	MTH	REPRESENTATION THEORY OF LINEAR LIE GROUPS	MTH733A	Representation theory of Compact groups; Peter Weyl Theorem, Linear Lie Group/ The Exponential map, Lie Algebra, Invariant Differential Operaters. Representations of the group and its Lie Algebra, Four Series, Representation of the Euclidean Motion Group.
610	SDS	ORDER STATISTICS	SDS682	Basic distribution theory, Moments of order statistics including recurrence relations, Bounds and approximations, Estimation of parameters, Life testing, Short cut procedures, Treatment of outliers, Asymptotic theory
611	MTH	TOPICS IN OPERATOR THEORY AND HARMONIC ANALYSIS	MTH637A	Operators on Hilbert spaces: Compact operators, Schatten class and Hilbert Schmidt operators, Spectral theore. Fourier series, Smooth functions and distributions. Hardy spaces, Carleson measures, H1BMO, and SO (3).
612	ES	ACTIVE TECTONICS AND PALEOSEISMOLOGY	ES659A	Crustal deformation and earthquakes (02); significance of seismicity (01); Identification of Prehistoric Earthquakes based on Primary and Secondary signatures preserved in landforms and sediment succession onfault and offfault landforms, indentification and mapping of active faults and associated landforms in field, structural analysis of active faults & its implication to regional scale tectonics (04); Field Techniques to identify earthquake by trenching, mapping of deformed sedimentary succession by faulting, estimation of net displacement during single event, slip rate, magnitude of historic earthquake, recurrence interval, and peak ground shaking identification of paleoliquefaction features (02); Dating techniques (01); Correlation of paleoseismic data with existing geodetic and geophysical data (01); Delineation of seismogenic faults (01).Identification of region along subduction zones (05); Understanding landlevel change caused by major earthquakes (02); decoupling the role of climate and tectonics (01); Understanding the effect of nearfield and farfield earthquakes. 1. McCa1pin, J.P., (1996), Paleoseismology, Academic Press, New York, p. 588. 2. Yeats, R.S., Sieh, K., Allen, C.R., (19997), Geology of Earthquakes, Oxford Univ. Press, 568. 3. Shiki, T., Tsuji, Y., Yamasaki, T., and others, Sciences, 287. 5. Bull, W. B. Tectonic Geomorphology of Mountains; A new Approach of Paleoseismology, Blackwell Sciences, 326.
613	EE	SEMICONDUCTOR OPTICAL COMMUNICATION DEVICES	EE685A	1. Introduction. 2. Review of Semiconductors. 3. Epitaxial Growth of Semiconductors. 4. Semiconductor Optical Waveguides. 5. LED. 6. Diode Lasers. 7. Fabrication and Packaging. 8. Single mode Laser diodes. 9. Fiber Optic Communication Systems. 1. Modular Series on Solid State Devices, Vol. VI, Ed: G.W. Neudek & R.F. Pierret: R.F. PIERRET, AddisonWesley. 2. Optical Fiber Communication Systems: W.B. JONES, HRW (1988). 3. Semiconductor Optoelectronics: Irwin. 5. Handbook of Semiconductor Lasers and Photonic Integrated Circuits: Y. SUEMATSU & A.R. ADAMS, Chapman & Hall. 6. Optical Electronics in Modern Communication: A. YARIV, Oxford University Press, 07-OCT-2015
614	EE	FIBER-OPTIC COMMUNICATIONS	EE684A	1. Introduction. 2. Light propagation in optical fibers. Single and multimode fibers, light guiding by fibers, material and waveguide dispersion, Polarization mode dispersion, Nonlinear effects: self and cross phase modulation. External modulators: phase and intensity, bias control, Pulse shaping, pulse carving, Modulation formats Intensity modulation, RZ and NRZ amplitude modulation, Many modulation using MachZehnder modulator. 3. Signal recovery, signal to noise ratio, performance calculations for binary digital optical systems. Coherent detection: heterodyne, homodyne, DSP assisted coherent optical receiver, performance analysis. 5. Optical link design. 6. Optical calculations. 6. Optical link design. Power budget under linear and nonlinear effects, power penalty, dispersion tolerance in DWDM systems. 1. M. Cvijetic and I.B. Djordjevic, Advanced optical communication systems and networks, Artech House, 2013. 2. S. Kumar and M.J. Deen, Fiberoptic communications: Fundamentals and applications, Wiley, 2013.

S.No.	Branch	Course Title	Courses ID	Content
				07-OCT-2015
615	EE	QUANTUM AND WAVE PHENOMENA	EE683A	1. Introduction to quantum mechanics and its tools: Motivation for quantum mechanics: early experiments; general principles of quantum mechanics: operator algebra, eigenstates, superposition, observables, barriers, harmonic oscillator, Hydrogenic atom; time independent and dependent perturbation theory. 2. Device applications of quantum and wave phenomena : Density of states; practical examples of low dimensional engineered electronic and optical properties of these lowdimensional materials; application in electronic, optoelectronic and photonic devices; current research efforts towards using quantum mechanical materials. 1. R.L. Liboff, Introductory Quantum Mechanics (Addison Wesley). 2. A. Yariv, An Introduction to Theory and Applications of Quantum Mechanics (John Wiley & Sons). 3. S. Gasiorowicz, Quantum Physics of Lowdimensional Semiconductors (Cambridge University Press). 07-OCT-2015
616	CHE	MATHEMATICAL MODEL IN BIOLOGY	CHE603A	Continuous Population Models for single species : Continuous growth models, Insect outbreak model, Delay models, Linear analysis of Delay population models, Delay models in Physiology, Harvesting and control models: LotkaVolterra systems, Complexity and stability, Realistic predatorprey models, Analysis of a predatorPrey model with limit cycle periodic behaviour: Parameter domains of stability, Competition models. Phenomena. Reaction Kinetics : Enzyme kinetics: Basic enzyme reaction, Transient time estimates and nondimensionalisation, MichaelisMenten quasisteady state analysis, Suicide substrate kinetics, Cooperative binding. Oscillators and Switches : Motivation, brief history and background, Feedback control mechanisms, Oscillators and switches with two or more species: General qualitative results, Simple two species oscillation model. 1. Mathematical Biology I : An Introduction, Author: J.D. Murray, Publisher Springer. 2. Mathematical Models in Biology: AuthorLeah Edelstein, Keshet, Publisher : SIAM 07-OCT-2015
617	SDS	ADVANCED CALCULUS	SDS652	Least upper bound principle; limits; monotone sequence; subsequences, BolzanoWeierstrass, Cauchy sequence, completeness; countable and uncountable sets; convergence of series, conditional convergence, Rearrangement Theorem; convergence in R, open sets and closed sets on R; Cantor intersection Theorem, Cantor set; limits and continuity; discontinuous functions; properties of continuous functions; uniform continuity; Theorem of Calculus; sequence and series of functions; point wise convergence; uniform convergence, Weierstrass Mtest, Dedekind test; uniform convergence and continuity; term by term integration and differentiation of f:R→R; partial derivatives; chain rule; higher derivatives, local extrema; Taylor expansion; multiple integrals, determinant and volume, Jacobians. [1]. K R Davidson and A P Donsig: Real Analysis and Applications, Springer, 2010. [2]. R S Strichartz: The Way of Analysis, Jones and Bartlett Mathematics, 2010. 27-MAR-2014
618	NET	RADIATION INTERACTION DETECTION AND SHIELDING-THEORY ON NUCLEAR MEASUREME	NT620A	Applications of Radiation. Half Life Determination. Radiation Dose Levels and Calculation :Inverse Square Law Verification, ALARA Principle, Linearthreshold model. Basic Principles of Radiation Detection and Linear Attention Coefficient Measurement. Evaluation of GM Detector Characteristics: Plateau determination, Dead time calculations, paralyzable and nonparalyzable detectors. Measurements and Counting rate correction for detection for security applications. Mechanisms of neutron interaction: capture, scintillation etc. Scintillation detectors, Capture detectors, Fission chambers. Semiconductor Detectors. Pulse shaping and gamma dose and shielding calculations. Measuring neutron spectrum and dose and shielding calculations. Measuring other forms of radiation, dose and shielding calculations. Principles of particle acceleration in nuclear power plants* 1. G.F. Knoll, Radiation Detection and Measurements, John Wiley & Sons, Hoboken, New Jersey (2010).2. S.S. Kapoor and V.S. Ramamurthy, Nuclear Radiation Detectors, Wiley Eastern Limited (1986).3. K. S. G. 27-MAR-2014
619	NET	RADIATION INTERACTION DETECTION AND SHIELDING-THEORY ON NUCLEAR MEASUREME	NE620A	Applications of Radiation. Half Life Determination. Radiation Dose Levels and Calculation :Inverse Square Law Verification, ALARA Principle, Linearthreshold model. Basic Principles of Radiation Detection and Linear Attention Coefficient Measurement. Evaluation of GM Detector Characteristics: Plateau determination, Dead time calculations, paralyzable and nonparalyzable detectors. Measurements and Counting rate correction for detection for security applications. Mechanisms of neutron interaction: capture, scintillation etc. Scintillation detectors, Capture detectors, Fission chambers. Semiconductor Detectors. Pulse shaping and gamma dose and shielding calculations. Measuring neutron spectrum and dose and shielding calculations. Measuring other forms of radiation, dose and shielding calculations. Principles of particle acceleration in nuclear power plants* 1. G.F. Knoll, Radiation Detection and Measurements, John Wiley & Sons, Hoboken, New Jersey (2010)2. S. S. Kapoor and V.S. Ramamurthy, Nuclear Radiation Detectors, Wiley Eastern Limited (1986)3. K. S. G. 27-MAR-2014
620	ME	RADIATION INTERACTION DETECTION AND SHIELDING- THEORY ON NUCLEAR MEASUREM	ME620A	Applications of Radiation. Half Life Determination. Radiation Dose Levels and Calculation :Inverse Square Law Verification, ALARA Principle, Linearthreshold model. Basic Principles of Radiation Detection and Linear Attention Coefficient Measurement. Evaluation of GM Detector Characteristics: Plateau determination, Dead time calculations, paralyzable and nonparalyzable detectors. Measurements and Counting rate correction for detection for security applications. Mechanisms of neutron interaction: capture, scintillation etc. Scintillation detectors, Capture detectors, Fission chambers. Semiconductor Detectors. Pulse shaping and gamma dose and shielding calculations. Measuring neutron spectrum and dose and shielding calculations. Measuring other forms of radiation, dose and shielding calculations. Principles of particle acceleration in nuclear power plants* 1. G.F. Knoll, Radiation Detection and Measurements, John Wiley & Sons, Hoboken, New Jersey (2010)2. S. S. Kapoor and V.S. Ramamurthy, Nuclear Radiation Detectors, Wiley Eastern Limited (1986)3. K. S. G. 27-MAR-2014
621	ECO	ECONOMIC GROWTH THEORY AND PRACTICE	ECO523	
622	MSE	MICROSTRUCTURE MODELLING AND SIMULATION	MSE608A	
623	MSE	MICROSTRUCTURE MODELLING AND SIMULATION	MSE608	
624	CHE	SPECTROSCOPY OF NANOMATERIALS	CHE675	
625	HSS	FILM AND THEORY	ENG457A	•The Uncanny•: omnipotence of thought; the doppelganger, the evil eye, Freud's theory of the Oedipus complex, Mulvey's concept of scopophilia. Analysis of •Vertigo• and •Psycho• events; function of fate, determinism. The use of events happening •too late•; Analysis of Douglas Sirk's •All That Heaven Allows• and •Written on the Wind•; Discussion of the consequence in the creation of •linkage•; Movement toward closed and legible wholes under •the mandate of desire•; Analysis of •Picnic• and •Splendor in the Grass•; Discussion of the internalization, the ego ideal. Analysis of •Strangers on a Train•; •The Day After Tomorrow•. Brooks, Peter. •Freud's masterplot•; Yale French Studies (1977): 280-300. Cook, Pam. •Duplicity in Mildred Pierce•; Women in film noir (1978): 68-82. Creed, Barbara. The monstrous-feminine in psychoanalysis. •The Oxford Guide to Film Studies (1998): 77-90. De Lauretis, Teresa. Technologies of gender: Essays on theory, film, and fiction. Vol. 44. Indiana University Press, 1987. Heung, Maria. •Life•; Cinema journal (1987): 21-43. Krips, Henry. •The politics of the gaze: Foucault, Lacan and Žižek•; Culture Unbound 2 (2010): 91-102. Loren, Scott. •Self-fashioning, Freedom, and the
626	CSE	CYBER SECURITY OF CRITICAL INFRASTRUCTURES	CS631	
627	HSS	CLIMATE FICTION AND FILMS	ENG456A	I.Introduction to Climate Change. II. Geological Causes of Climate Change. III. Anthropogenic Causes of Climate Change. IV. Climate Skepticism; Threat Perception and Underlying Politics. V. Effects of Climate Change: Ecofiction/Ecofabulism. IX. Film Representations of Climate Change: Enviropocalypse, Postapocalypse. X. Concepts, Themes and Issues in Climate Fiction. XI. What is Wrong with Climate Change? XII. Possibilities: Gaia and Geoengineering. XIV. Sustainability and Climate Responsibility. XV. Reality Check: Climate Politics that Avert Possible Solutions. Atwood, Margaret. Oryx and Crake. Toronto: McClelland & Stewart Limited, 2003._____. Maddaddam. Toronto: McClelland & Stewart Limited, 2013. Budyko, M.I. and Yu. A. Izrael, Eds. Anthropogenic Climate Change. New York: SpringerVerlag, 1986.Bacigalupi, Paolo. The Windup Girl. San Francisco, CA: Night Shade Books, 2009. Clark, Timothy, The Cambridge Introduction to Literature and the Environment. Cambridge University Press, 2010. 27-MAR-2014

S.No.	Branch	Course Title	Courses ID	Content
628	ME	MATHEMATICS FOR ENGINEERS	ME681	Fixing the Sky. New York: Columbia Univ. Press 2010.Fraser, Evan D.G. #foodcrisis: A Graphic Novel about Global Food Security. Guelph, Ontrario: Evan Fraser, 2014. Glass, Matthew. Ultimatum. New York: (I)Introduction. 1. Introduction to the course.(II) Linear Algebra. 1.Vector spaces: definition, linear independence of vectors, basis, inner product and inner product space, orthogonality, GramSchmidt procedure, equations: existence and uniqueness of solution, elementary row/column operations, Gauss elimination and Gauss Jordan methods, Echelon form, pivoting, LU decomposition and Cholesky method, Gauss-Jordan method and eigenvectors of matrices: properties like multiplicity, eigenspace, spectrum and linear independence of eigenvectors, similarity transformation and Jordan canonical form, eigenvalues/eigenvectors of matrices: forward iteration and Mises power method, inverse iteration. (III) Tensor Algebra. 1. Index Notation and Summation Convention. 2.Tensor algebra: tensor as a linear vector transformation, dyadic product, symmetric and antisymmetric parts, invariants, decomposition into isotropic and deviatoric parts, inner product and norm, inverse, orthogonal tensors, eigenvalues and eigenvectors, squareroot, positive definite symmetric tensors. 1.Advanced Engineering Mathematics by E. Kreyszig, John Wiley and Sons, International 8th Revised Edition, 1999,2.Applied Mathematical Methods by B. Dasgupta, Pearson Education, 2006.
629	IME	INFORMATION SYSTEMS THEORY	IME640A	Introduction: Information and its characteristics including entropy, System and its characteristics. IS Cycle Theories: Delone and McLean’s Success Model, Technology Acceptance Model, Unified Theory of Acceptance and Use of Technology, Virtualization Theory, Theory of Deferred Action. Strategic and Economic Theories: Resource based view, Theory of Slack Resources, Portfolio Theory, Theory of Lemon Markets, Technology Organization Theory, Institutional Theory, A Multilevel Social Network Perspective, Agency Theory. Socio Psychological Theories: Actor network theory, Theory of Planned Behaviour, Structuration Theory. 1. Dwivedi, Y.K.,Wade, M.R., & Schneberger, S.L. (2012) Information Systems Theory: Explaining and Predicting Our Digital Society, Vol 1. Springer 14-JUL-2015
630	AE	FUNDAMENTALS OF COMBUSTION	AE663A	Introduction, Combustion and Thermochemistry: Motivation and objective, Property relations, Thermodynamic laws, Reactant & Product mixtures, Adiabatic flame temperature, Chemical equilibrium and equilibrium constants, multistep mechanismsConservation equations: Simplified governing equations, Momentum, energy, mass conservation, Multicomponent diffusion, Concept of conserved scalar Laminar Premixed and Nonpremixed flames, flammability limits, Flame stabilization, Nonremixed flames, laminar jets, Droplet Combustion, Solid fuel Combustion
631	ME	THEORY OF PLASTICITY	ME721A	Yield surfaces. Deformation and flow theories; Theory of plastic constitutive equations; Axisymmetric and spherically symmetric problems; Slipline theory and application to problems of extrusion, drawing and rolling
632	HSS	FILM AND THEORY	ENG457	•The Uncanny•: omnipotence of thought; the doppelganger, the evil eye, Freud’s theory of the Oedipus complex, Mulvey’s concept of scopophilia. Analysis of •Vertigo• and •Psycho• events; function of fate, determinism. The use of events happening •too late•; Analysis of Douglas Sirk’s •All That Heaven Allows• and •Written on the Wind•; Discontinuity and consequence in the creation of •linkage•; Movement toward closed and legible wholes under •the mandate of desire•; Analysis of •Picnic• and •Splendor in the Grass•; Psychoanalysis and internalization, the ego ideal. Analysis of •Strangers on a Train•; Brooks, Peter. •Freud’s masterplot•; Yale French Studies (1977): 280300. Cook, Pam. •Duplicity in Mildred Pierce•; Women in film noir (1978): 6882. Creed, Barbara. The monstrous female in psychoanalysis. •The Oxford Guide to Film Studies (1998): 7790. De Lauretis, Teresa. Technologies of gender: Essays on theory, film, and fiction. Vol. 441. Indiana University Press, 1987. Heung, Maria. •Life•; Cinema journal (1987): 2143. Krips, Henry. •The politics of the gaze: Foucault, Lacan and Zizek•; Culture Unbound 2 (2010): 91102. Loren, Scott. •Selffashioning, Freedom, and the
633	HSS	CLIMATE FICTION AND FILMS	ENG456	I.Introduction to Climate Change. II. Geological Causes of Climate Change. III. Anthropogenic Causes of Climate Change. IV. Climate Skepticism; Threat Perception and Underlying Politics. V. Effects of Climate Change on Society. VI. Climate Change: Ecofiction/Ecofabulism. IX. Film Representations of Climate Change: Enviropocalypse, Postapocalypse. X. Concepts, Themes and Issues in Climate Fiction. XI. What is Wrong with Climate Change? XII. Possible Solutions: Gaia and Geoengineering. XIV. Sustainability and Climate Responsibility. XV. Reality Check: Climate Politics that Avert Possible Solutions. Atwood, Margaret. Oryx and Crake. Toronto: McClelland & Stewart Limited, 2003._____. Maddaddam. Toronto: McClelland & Stewart Limited, 2013. Budyko, M.I. and Yu. A. Izrael, Eds. Anthropogenic Climate Change. New York: SpringerVerlag, 1986.Bacigalupi, Paolo. The Windup Girl. San Francisco, CA: Night Shade Books, 2009. Clark, Timothy, The Cambridge Introduction to Literature and the Environment. Cambridge University Press, 2010. Fixing the Sky. New York: Columbia Univ. Press 2010.Fraser, Evan D.G. #foodcrisis: A Graphic Novel about Global Food Security. Guelph, Ontrario: Evan Fraser, 2014. <https://feedingninebillion.com/foodcrisis> MacArthur & Co., 2009.
634	ME	INTRODUCTION TO SOLID MECHANICS	ME621	I. Introduction. 1. Review of strength of Materials and its limitations. II. Mathematical Preliminaries. 1.Vector and tensor calculus. 2. Indicial notationIII. Strains. 1. Definition of small strain, StrainDisplacement relations. 2. Strain components in 3D and their physical interpretations. 2. Principal Stresses. 3. Cauchy’s principle and derivations of stress equilibrium equations in stress components. V. Constitutive law, Navier’s equations for isotropic solid. 2. Navier’s equations. 3. Stress and displacement approaches. 4.Compatibility equations. VI. Formulation of boundary value problems and solution methods. 1. Formulation of boundary value problems. 2. Examples of plane problems: Stress function approach, Series solutions. 4.Fourier transform methods with examples. 5. Superposition principle: Flamant’s solutions; Kelvin’s solution; Boussinesq’s solutions for plane problems; Wedge problems; Dislocations and inclusions; Cracks; Thin film problems. 2. Further methods: Advanced transform methods; Complex variable techniques; Potential methods. 3.Further ideas: Elasticity. 1.Elasticity, J. R. Barber2.The Linearized Theory of Elasticity, W. L. Slaughter3.Continuum Mechanics for Engineers, G. T. Mase and G. E. Mase4.Theory of Elasticity, S. Timoshenko and J. N. Goodier5.Elasticity of Solids, L. D. Landau and E. M. Lifshitz
635	ME	CONVECTIVE HEAT AND MASS TRANSFER	ME642	Introduction: Convective heat transfer and its applications; Forced, free and mixed convection; internal and external flow; heat transfer coefficient and its physical significance; dimensional analysis in convection. Laminar Forced Convection: Boundary layer equations; energy equation for flow over flat plate; similarity solution for flow over a flat plate having various boundary conditions and Prandtl numbers; Scale analysis. Developing and developed flow and heat transfer in a duct and circular pipe having various boundary conditions. Natural/Free and Mixed Convection: Boussinesq approximations; Similarity solution for flow over a vertical plate. Course Reference : Suggested text books:Introduction to Convective Heat Transfer Analysis by Patrick H. Oosthuizen and David Laylor (McGrawHill)Convective Heat and Mass Transfer by Kays, Crawford and Weigand (4th Edition, Wiley)reference books:Convective Heat Transfer by M FavreMarinet and S Tardu (John Wiley and Sons)Principles of Convective Heat Transfer by Massoud Kaviany (2nd Edition, Springer)Convective Heat Transfer by J. P. Holman (McGrawHill)Convection by Latif M Jiji (Springer)Viscous Fluid Flow by Frank M White (McGrawHill)Boundary Layer Theory by H Schlichting (McGrawHill)
636	ME	CONVECTIVE HEAT AND MASS TRANSFER	ME642A	Introduction: Convective heat transfer and its applications; Forced, free and mixed convection; internal and external flow; heat transfer coefficient and its physical significance; dimensional analysis in convection. Laminar Forced Convection: Boundary layer equations; energy equation for flow over flat plate; similarity solution for flow over a flat plate having various boundary conditions and Prandtl numbers; Scale analysis. Developing and developed flow and heat transfer in a duct and circular pipe having various boundary conditions. Natural/Free and Mixed Convection: Boussinesq approximations; Similarity solution for flow over a vertical plate. Suggested text books:Introduction to Convective Heat Transfer Analysis by Patrick H. Oosthuizen and David Laylor (McGrawHill)Convective Heat and Mass Transfer by Kays, Crawford and Weigand (4th Edition, Wiley)reference books:Convective Heat Transfer by M FavreMarinet and S Tardu (John Wiley and Sons)Principles of Convective Heat Transfer by Massoud Kaviany (2nd Edition, Springer)Convective Heat Transfer by I. P. Holman (McGrawHill)Convection by Latif M Jiji (Springer)Viscous Fluid Flow by Frank M White (McGrawHill)Boundary Layer Theory by H Schlichting (McGrawHill)
637	ME	MATHEMATICS FOR ENGINEERS	ME681A	(I)Introduction. 1. Introduction to the course.(II) Linear Algebra. 1.Vector spaces: definition, linear independence of vectors, basis, inner product and inner product space, orthogonality, GramSchmidt procedure, equations: existence and uniqueness of solution, elementary row/column operations, Gauss elimination and Gauss Jordan methods, Echelon form, pivoting, LU decomposition and Cholesky method, Gauss-Jordan method and eigenvectors of matrices: properties like multiplicity, eigenspace, spectrum and linear independence of eigenvectors, similarity transformation and Jordan canonical form, eigenvalues/eigenvectors of matrices: forward iteration and Mises power method, inverse iteration. (III) Tensor Algebra. 1. Index Notation and Summation Convention. 2.Tensor algebra: tensor as a linear vector transformation, dyadic product, symmetric and antisymmetric parts, invariants, decomposition into isotropic and deviatoric parts, inner product and norm, inverse, orthogonal tensors, eigenvalues and eigenvectors, squareroot, positive definite symmetric tensors. 1.Advanced Engineering Mathematics by E. Kreyszig, John Wiley and Sons, International 8th Revised Edition, 1999,2.Applied Mathematical Methods by B. Dasgupta, Pearson Education, 2006.
638	ME	APPLIED DYNAMICS AND VIBRATIONS	ME625A	I: NewtonEuler mechanics. 1. Mathematical preliminaries: Coordinate systems, Vectors, Tensors, Outer product; Coordinate transformation. 2. Rotating frames; Rotation tensor; Euler angles; Angular velocity and angular momentum; Inertia tensor; Kinetic energy. 5. Rigidbody kinetics: Balance laws; Governing equations; Euler’s equations. 6. Examples: Rigid body in free space; Gyroscopes. Part II: Analytical mechanics. 1. Virtual work; Constraint forces; Workless constraints; Principle of virtual work; Lagrange multipliers; Equilibria and stability of conservative systems; Examples. 3. Dynamics: d’Alembert’s principle; Lagrangian mechanics. Conservative systems. Legendre transformation; Hamiltonian mechanics; Energy theorem; Examples. Part III: Vibrations. 1. Single degree of freedom system: free, damped, forced. 2. Convolution integral. Laboratory demos/sessions: Summary: Mathematical preliminaries: Vectors; Tensors; Coordinate transformations. NewtonEuler Mechanics: Rotation; Threedimensional Rigidbody kinematics and dynamics. Equations of motion; Holonomic and nonholonomic systems; Hamiltonian mechanics. Vibrations: Free, damped and forced single degree of freedom system; Two degree of freedom system; Normal modes; Multidegree of freedom systems. 1. Greenwood, D. T. 1987. Principles of Dynamics 2nd edition. Pearson Education.2. Beatty, M. F. 1986. Principles of Engineering Mechanics: Part I, II. Springer.3. Meirovitch, L. 1986. Elements of Vibration Analysis. Wiley publications.5. Thomson, W. T. 2002. Theory of Vibrations with Applications 3rd edition. CBS publishers.6. Hartog, D. 1985. Mechanical Vibrations. Dover publishers.7. Lanczos, C. 1986. The Variational Principles of Dynamics. (under preparation)

S.No.	Branch	Course Title	Courses ID	Content
639	ME	INTRODUCTION TO SOLID MECHANICS	ME621A	I. Introduction. 1. Review of strength of Materials and its limitations. II. Mathematical Preliminaries. 1.Vector and tensor calculus. 2. Indicial notationIII. Strains. 1. Definition of small strain, StrainDisplacement components in 3D and their physical interpretations. 2. Principal Stresses. 3. Cauchys principle and derivations of stress equilibrium equations in stress components. V. Constitutive law, Naviers equations for isotropic solid. 2. Naviers equations. 3. Stress and displacement approaches. 4.Compatibility equations. VI. Formulation of boundary value problems and solution methods. 1. Formulation of boundary value problems. Examples of plane problems: Stress function approach, Series solutions. 4.Fourier transform methods with examples. 5. Superposition principle: Flamants solutions; Kelvins solution; Boussinesqs solutions for plane problems; Wedge problems; Dislocations and inclusions; Cracks; Thinfilm problems. 2. Further methods: Advanced transform methods; Complex variable techniques; Potential methods. 3.Further ideas: Elasticity. 1.Elasticity, J. R. Barber2.The Linearized Theory of Elasticity, W. L. Slaughter3.Continuum Mechanics for Engineers, G. T. Mase and G. E. Mase4.Theory of Elasticity, S. Timoshenko and J. N. Goodier5.Elasticity, J. N. Goodier
640	ME	METAL FORMING	ME663A	(I) Introduction. 1. Introduction to the course: different metal forming processes, importance of plasticity in the course. (II)Fundamentals of Plasticity. 1. ReviewAnalysis of stress: transformation relations, principal strains, invariants, hydrostatic and deviatoric parts; (Infinitesimal) rotation, Stressesstrain relations for isotropic, linearly elastic material. 2. Plasticity: yield effect, hysteresis, incompressibility of plastic deformation, anisotropy, plastic instability . 3. Yield criterion for isotropic materials: von Mises and Tresca yield criterion, their geometric interpretation, conversion of stress tensor to linear incremental strain tensor, strain rate (i.e. the rate of deformation) tensor and their relation, incremental rotation tensor and spin tensor. 5. Change in yield criteria due to isotropic hardening: strain hardening for isotropic materials: plastic potential and associated flow rule, incremental and rate forms of elastoplastic stressesstrain relations, simplifications for nonhardening and rigidplastic materials (Prandtl Reuss model) of plasticity: updated Lagrangian and Eulerian formulations, boundary and initial conditions, examples. References: 19502. Engineering Plasticity by W. Johnson and P.B. Mellor, von Nostrand Co. Ltd,19723. Theory of Plasticity by J. Chakrabarty, McGrawHill Book Co., InternationalEdition, 19874. Metal Forming: Processes and Machines, John Wiley andSons Inc., 1995.
641	ME	MACHINING SCIENCE II	ME662A	I. Introduction to Advanced Manufacturing Processes. Introduction to manufacturing processes. Overview of non conventional machining processes with (AJM, USM, ECM, EDM, EBM, LBM, AFM, MRF, MA) Removal Processes (AJM, USM and WAJM) Abrasive Jet Machining (AJM): Introduction to abrasive jet machining (AJM), Mechanics of AJM, AJM process parameters Components of AJM (Abrasive, Gas, Spindle Speed, Rate (MRR). Ultrasonic Machining (USM): Basics of USM processes, Mechanics of USM, Process parameters of USM, Shaws model of USM mechanics, Compressed grain modelling and direct throw mode Ultrasonic machining setup, Design of acoustic ultrasonic head and feed mechanism in USM. Water Abrasive jet machining (WAJM): Introduction to WAJM (Basic principle and MRR estimation), WAJM process parameters Finishing (AFF), Introduction to AFF and self deformable feature, AFF machine elements, Magnetic Abrasive Finishing (MAF), Introduction to MAF, Elements of MAF, Setup and process parameters for AFF. References: Advanced manufacturing processes, Hassan Abdel, Gabad El Hoffy, McGraw Hill. V.K.Jain, Advance Machining Processes, Allied Publisher Bombay. Ghosh and Mallik, Manufacturing Science, 2000. Traditional machining processes, Society of Manufacturing Engineers, Publications. The Science and Engineering of Microfabrication, Stephen P. Campbell, Oxford university press.
642	ME	MACHINING SCIENCE I	ME661A	1. Introduction : Machining; Plastic Deformation, Tensile Test, Stress and Strain; Mechanism of Plastic Deformation: Slips, defects, plastic deformation on atomic scale.2. Machining Process : Types of machining, Tool Geometry : Reference planes; Tool specification: American System (ASA), continental or Orthogonal System (ORS), International or Normal Rake system (NRS); Tool angle relationships in ORS, ASA and NRS Merchant's Circle Diagram; Coefficient of Friction: Determination of stress, strain and strain rate; Measurement of shear angle; Thin Zone model: Lee and Shaffer's Relationship; Thick Zone model: Sticking and Sliding Zones, Analysis of Stress Distribution on the tool face: Zorevs model; Determination of mean angle of friction. 6. Mechanism of Oblique cutting : Rake angles in oblique cutting: Analytical model of oblique cutting; velocity relationship; Force relationships in oblique cutting. References:1. E.J.A. Armarego and R.H.BrownThe machining of Metals2. G BoothroydFundamentals of Metal Machining and Machine tools3. A.Ghosh and Asok Mallik Machining Science4. G.K.Lal and S. Chakrabarti Journal of Manufacturing Science and Engineering.2. International Journal of Machine Tools and Manufacturing.3. Annals of CIRP.4. Journal of Materials processing technology.5. Precision Engineering.6. International Conference.
643	ME	CONDUCTION AND RADIATION	ME641A	Lec 1: Conduction: Derivation of Heat Conduction Equation for Heterogeneous, Isotropic Materials in Cartesian Coordinates. Heat conduction equation for homogeneous, isotropic materials in Cartesian, Cylindrical and Spherical coordinates. Thermal resistances. Lec 2: Heat transfer from a fin of uniform crosssection. Fin efficiency and fin effectiveness. Fin with variable crosssection. Lec 3: Twodimensional Steady State Heat Conduction: Illustration 1 of Separation of Variables. Isotherms and Heat Flux Lines. Lec 4: Illustration 2: 2D Steady State Heat Conduction with Constant Heat Generation in a Long Rod of Rectangular Crosssection with Boundary Conditions. Radiation in participating media: Gas radiation. Combined Conduction and Radiation: Example of a spacecraft radiator. Solar radiation. Greenhouse effect.
644	NET	INTRODUCTION TO COMPUTERIZED TOMOGRAPHY	NT641A	Overview, medical imaging, nondestructive testing, radiographic techniques,various applications, data collection, design of CT scanners for materials testing,flow measurement, related instrumentation, Radiographic transformmmethods, seriesexpansion methods, convolution algorithms, error estimates,direct theorems, inverse theorems.
645	HSS	SOCIOLOGICAL THEORY	SOC721A	Relation between theory and research. The nature and use of theory, basic concepts, fundamental perspective of society, conceptualization of social system, social structure and culture, Various perspectives on society. Phenomenology, Ethnomethodology & Structuration theory.
646	PSE	RESEARCH IN PHOTONICS AND LASERS	PSE606A	Biophotonics, nanobiophotonics,Optical communication, Quantum cryptographyNonlinear opticsTomographyQuantum dots, photonic crystalsLaser manufacturing and materials processingLaser instrumentation and materials processing
647	PSE	PHOTONICS LAB TECHNIQUES	PSE605A	*Electrooptic effect using LiNbO3 crystal* Acoustooptic modulator* Study of effects of loss, dispersion, amplifier noise on 10Gbps links* 40Gbps QAM modulation and coherent demodulation* Nonlinearities in optical crystal* Fresnel and Fraunhofer Diffraction* HeNe laser beam parameters* Laser diode characteristics: LI characteristic, beam profile measurement, modes and spectrum using FP cavity* Michelson interferometer* Holography
648	PSE	PHOTONICS SYSTEMS AND APPLICATIONS	PSE604A	Principles and Applications of Solid State Laser SystemsLaser diode Structures, Mechanism of photon emission in semiconductor laser, Tunablesemiconductor diode laser, Rare earth doped lasers, Nd:Glass Lasers,Quantum cascade Laser.Principles and Applications of Liquid and Gas Laser SystemsDye laser, Tunable Lasers, Frequency stabilization, Tuning Techniques, Ar+ lasers, HeNe laser,CO2 lasers. Course Texts: Principles of lasers, 4th Ed., 1998* R W Boyd, Nonlinear optics, Academic Press, 2nd Ed, 2003* M. H. Niemz, Laser Tissue Interaction Fundamentals and Applications, Springer, 2007
649	PHY	PARTICLE PHYSICS	PHY680A	Natural units; evidence for 4 fundamental interactions, leptons and hadrons,historical introduction to particle zoo, relativistic kinematics, Lorentzinvariantphase space, calculation of 2 and 3body phase space integrals, representations and Young tableaux, the GellMann eightfold way,prediction of Λ , quark model, construction of hadronic wave functions, magneticmoment of the neutron, statistics of baryons & concept of confinement, theory, local U(1) gauge theory andMaxwell equations, YangMills theories, SU(2) and SU(3) gauge theories,construction of SU(2)xU(1) gauge theory, gauge boson selfinteractions,spontaneous breaking of symmetry, construction ofthe GlashowSalamWeinberg model (outline only).
650	PHY	GENERAL RELATIVITY & COSMOLOGY	PHY660A	Machs principle. Riemanniangometry. energymomentum tensor and Einsteinsequations. Schwarzschild metric and singularities of space time. postNewtonianapproximations. spherically symmetric solutions. Cosmology. Friedmann equations. Big Bang model. Dark matter and Dark energy.
651	PHY	QUANTUM ELECTRONICS	PHY644A	Semiclassical theory of lasers, single and multimode operation, gas laser theory,ring and Zeeman lasers, coherence in lasers. nonlinear optical phenomena,Feynman diagrams in multiphoton problems.
652	PHY	ELECTRONIC STRUCTURE OF MATERIALS	PHY621A	One electron model, BornOppenheimer approximation, Hartree & HartreeFockapproximation, density functional theory, local density approximation, beyondLDA. electrons in periodic solids, Blochs theorem, pseudopotential method, KKR method, LMTO method, the fullpotentialmethods. applications to different types of solids; electron in disorderedsolids, meanfield theories, coherent potential approximation, applications to clustersand amorphous semiconductors, applications of electronic structure methodsto materials design.
653	NET	FAST REACTOR TECHNOLOGY	NT651A	Introduction, Core design, Fuelelement design, Fuel management, Heat transportsystems, Steamgenerators, IHX design, Sodium pumps & piping, Instrumentation& controls, safety, extractive and physical chemistry of environment, advanced NDT techniques, corrosion.
654	NET	NUCLEAR POWER ENGINEERING II	NT614A	Introduction to control theory. Point reactor kinetics with introduction to feedbackeffects. Nonlinear effects. Shielding. Introduction to reactor reliability and safety analysis. Radioactive waste disposal. Economics of nuclear power.
655	MSE	MULTI FUNCTIONAL OXIDES:	MSE689A	Fundamentals of oxides : crystal structure, defect chemistry, and properties;focus on various material systems methods of fabrication e.g. solid statechemistry. Oxide thin films. polycrystalline versus epitaxial films.

S.No.	Branch	Course Title	Courses ID	Content
		THIN FILMS & DEVICE		(fundamentals of glow discharge processes and film deposition RFand DC magnetron sputtering new approaches), laser ablation (basic science,applications, various approaches), science and technology of chemical vapor deposition atomic layer deposition; PVD visavischemical processes; issues related to epitaxy and case studies. Characterizationmethods : Structural techniques uses of X ray diffraction, and SEM Measurements. Devices types of devices, fabrication: fundamentalsand issues; Lithographic methods: conventional and next generation, FIB (fieldion) techniques, Nanofabrication: principles, processes and applications
656	MSE	MATERIALS FAILURE: ANALYSIS AND PREVENTION	MSE676A	1. Introduction to Failure analysis and prevention: Concepts, root causes analysis, primary rootcauses, design deficiencies, material defects, manufacturing/installation defects, categoriesof failure, failure modes (3 hours)2. Fracture mechanics: Fundamentals of fracture, stress intensity factor, crack growth rates, fatigue crack growth, fracture toughness (3 hours)3. Fracture modes, Ductile fracture of metallic . materials and their interpretations, factorsaffecting ductilebrittle relationships. (3 hours)4. Failure characteristics of Ceramics and Plastics (2 hours)5. Failure characteristics of Composites (2 hours)6. Failure characteristics of Polymers (2 hours)7. Macroscopic and microscopic characteristics, statistical aspects of fatigue,Fatigue failure prediction and life assessment. (3 hours)7. Wear Failures and Prevention (2 hours)8. Corrosion related failure modes: galvanic corrosion, stress corrosion cracking, hydrogen embrittlement, corrosion fatigue, high temperature failures, creep and stress rupture, metallurgical instabilities(2 hours)9. Distortion failures and deformations (2 hours)10. Structural life assessment methods, Nondestructive techniques. (2 hours)11. Fractography,, (4 hours)12. Illustrative Case studies of engineering failure ; due to: improper processing practice,improper treatment procedure, improper design, unanticipated service conditions13. Source book in failure analysis, American Society of Metals, Metals Park, Ohio, 1974.2. Understanding how components fail , D.J Wulpi, ASM International, The MaterialsInformation Society, 1 999.3. A practical approach to failure analysis, H.M. Tawancy, A. UlHamid and N.M. Abbas,Marcel Dekker, New York, 2004.5. Failure analysis and prevention, Volume II , ASM Handbook, The MaterialsInformation Society, 2002.6. Failure analysis and prevention, Volume I, ASM Handbook, ASM International,2005.
657	CE	ANALYSIS OF PAVEMENT STRUCTURES	CE682A	Introduction to supply and demand sides of transportation engineering, analysisof transportation demand (including topics such as category analysis, gravitymodel, entropy models, choice models, user equilibrium, etc.). (includingtopics such as route development, schedule development, pricing strategies,etc.). Concept of structural, functional and drainage design of pavementstructure. Design of flexible and rigid pavement structures. Pavement evaluation, maintenance measures, and network level maintenance strategy.
658	CE	MACHINE PROCESSING OF REMOTELY SENSED DATA	CE672A	Image processing system; Preprocessing of remotely sensed data; Radiometricand Geometric distortions and corrections; Image enhancement; Imagetransformations; Pattern recognition.
659	AE	UNDERGRADUATE PROJECT III	AE472A	
660	ME	MICROSCALE THERMAL ENGINEERING	ME689A	MicroMechanical systems (MEMS), Micro Channels, Heat pipes, jets, valves,Heat Sinks, Solar cells, Bearings, Pumps, Heat pipes, Jets, valves, Heat sinks,Solar Cells, Bearings, Pumps, Flow Sensors and actuators. CharacteristicNondimensional parameters, Microscale Heat conduction, Heat transport in thinfilms and at solidsolid interfaces, Convective diffusion phenomena, Enzymesubstratereactions, channel flow in microchannels, Electroosmosis,Electroosmotic pumps, Surface tension driven flows, Coating flows,Thermocapillary flows, Molecular dynamics simulations.
661	ME	THERMAL ENVIROMENTAL CONTROL	ME613A	Moist air and psychrometric processes. Physiological principles of thermal comfort, calculation of cooling and heating loads; ADP determination, solar radiation and shading devices, duct design; Heat and mass transfer in buildings.
662	ME	THERMAL ENVIROMENTAL CONTROL	ME613	Moist air and psychrometric processes. Physiological principles of thermal comfort, calculation of cooling and heating loads; ADP determination, solar radiation and shading devices, duct design; Heat and mass transfer in buildings.
663	ME	AN INTRODUCTION TO PROFESSION AND COMMUNICATION	ME600A	COMMUNICATION SKILLS FOR ENGINEERS
664	IME	RESEARCH METHODOLOGY	IME700A	Introduction to Social Science Research Perspective, Different Approaches to Social Research, Approaches to Theory Building, Sampling , Measurement Issues & Scale construction, Research Design , Quantitative and Qualitative Research and Presentation, Research Evaluation and Critique, Issues in Current Research Practice.
665	DES	DESIGN PROJECT II	DES682A	Manufacturability Studies, Embedded products, Product styling, Package Designing,Information design, HCI, GUI, Animation, Film appreciation, Print Making.
666	CHM	M. Sc. PROJECT	CHM699A	Fundamentals of Lasers, laserinduced fluorescence and multiphoton ionization processes of molecules, probing IVR and dynamics of chemical reactions in liquid and molecular beam, spectroscopy of single molecules, organelles, confocal microscopy and fluorescence correlation spectroscopy, applications to diagnostics and biotechnology.
667	CE	FOUNDATION DYNAMICS	CE635A	Dynamics of elastic systems; Single and multidegrees of freedom systems;Empirical and semiempirical approaches to the theory of soil dynamics; Elastictheories of soil dynamics; Wave propagation; Dynamic analysis of structures.
668	CE	ROCK MECHANICS	CE630A	Physical properties and classification of intact rock and rock masses, rockexploration, engineering properties of rock, stresses in rock near undergroundopenings; Rock tunneling, rock slope stability, bolting and rock anchors.
669	CE	DURABILITY OF CONCRETE STRUCTURES	CE628A	Concrete and the environment: interaction; Overview of concrete deterioration:alkaliaggregate reaction, corrosion, carbonation; Permeability of concrete andits measurement: penetration of carbon dioxide, chloride ions, water, etc.; corrosion cells and currents, role of concrete, prevention ofcorrosion; Corrosion induced longitudinal cracks: nature and properties ofcorrosion products; Alkali aggregate reaction: reactive minerals, mechanisms, prevention and rehabilitation of structures.
670	CE	MASONARY STRUCTURES	CE625A	Properties of constituents: units burnt clay, concrete blocks, mortar, grout,reinforcement; Masonry bonds and properties: patterns, shrinkage, differentialmovement, masonry properties compression strength, tensile strength, shear strength, inplane, outofplane; Behaviour of masonry walls and piers:axial and flexure, axial shear and flexure, Behaviour of Masonry Buildings:unreinforced masonry buildings importance of bands and corner & vertical joints in RC frames: strut action; Structuraldesign of masonry in buildings: methods of design WSD, USD, seismic design seismic loads, code provisions, infills, connectors, ties; Seismic evaluationand strengthening of masonry structures.
671	MSP	CHARACTERIZATION OF MATERIALS	MS604A	Crystallography, Reciprocal lattice, Diffraction methods, Electron microscopy, Metallography, Thermal analysis, Chemical analysis, Spectroscopic techniques,Laboratory sessions.
672	MSP	ELECTRICAL AND DIELECTRIC MATERIALS	MS602A	Metallic conduction, Energy bands, Brillouin zones, Temperature dependence of metallic conductivity, Impurity contributions, Semiconductor materials, Doping effects, Law of mass action, Electrical resistivity, Superconductivity, technology, Point defects, Diffusion phenomenon,Ionic conduction, Temperature and (aliovalent) impurity effects, Superionic conductors and devices, Di, piezo and ferroelectric materials, Mechanisms of piezoelectricity.
673	PHY	M.SC.RESEARCH PROJECT II	PHY599A	M. Sc. Project II
674	PHY	M.SC. PROJECT II	PHY565A	Experimental project in a research laboratory: 2. Development and testingof experimental setup.
675	PHY	M.SC. PROJECT I	PHY563A	Experimental project in a research laboratory: 1. Literature survey andpreparation for the project.
676	PHY	M.SC. REVIEW PROJECT III	PHY502A	M.Sc. Review Project III
677	PHY	M SC REVIEW PROJECT II	PHY501A	M.Sc. Review Project II
678	CSE	LEARNING WITH KERNELS	CS678	Kernel based methods in machine learning have become a major paradigm in machine learning in the last decade. The methods have also found widespread application in pattern classification problems. areas of current research like : techniques for finding optimal kernels, error bound analysis novelty detection etc.
679	CSE	LEARNING WITH KERNELS	CS678A	Kernel based methods in machine learning have become a major paradigm in machine learning in the last decade. The methods have also found widespread application in pattern classification problems. areas of current research like : techniques for finding optimal kernels, error bound analysis novelty detection etc.
680	CSE	PROGRAM ANALYSIS VERIFICATION AND TESTING	CS639	Data flow Analysis; Interprocedural Analysis: functional, call string and graph reachability based approaches Abstract Interpretation; Weakest Precondition, Floyd Hoare Logic, Separation Logic; Software model checking, abstraction refinement, model checking of concurrent programs, Program Testing: program testing basics, automatic test case generation, directed testing. 1. Edsger Wybe Dijkstra. A Discipline of Programming. Prentice Hall PTR, Upper Saddle River, NJ, USA, 1997. 2. David Gries. The Science of Programming. SpringerVerlag New York, Inc., Secaucus, NJ, USA, 1988. 3. Robert S. Boyer. The Art of Computer Programming. Englewood Cliffs, NJ, 1981. 4. Flemming Nielson, Hanne R. Nielson, and Chris Hankin. Principles of Program Analysis. SpringerVerlag New York, Inc., Secaucus, NJ, USA, 1999. 5. Alfred V. Aho, Monica S. Lam, Ravi Sethi, and Jeffrey D. Ullman. Compilers: Principles, Techniques, and Tools. Addison-Wesley, Reading, MA, USA, 1986.

S.No.	Branch	Course Title	Courses ID	Content
				Longman Publishing Co., Inc., Boston, MA, USA, 2006. 6. Michael Huth and Mark Ryan. Logic in Computer Science: Modelling and Reasoning about Systems. Cambridge University Press, New York, NY, USA, 1999.
681	CSE	COMPUTING LABORATORY II	CS252	1. System administration: Students will learn to setup and manage a network server including a web server andan email server. They should also be familiar with various network protocols. Further, students will learn simpleloadbalancing tools.2. Security: While setting up different network and database servers, students will learn to manage the securityissues including different attacks. Public key infrastructure and certificates.3. Troubleshooting: While of maintaining sys terns, students willneed to use system vulnerability and intrusion testing tools as well.3. Compiler tools: Students will need to learn the various lowlevel tools used routinely in compilers and linkers.4. Debugging: Students will learn to use various integrateddevelopment environments (IDEs) such as eclipse and debuggers such as gdb. They should also use tools foreffective tagging and browsing of source code. Finally, they will learn to use various tools for performance analysis. Course Reference : 1. Manuals.2. Online help that is available for different tools.
682	CHE	INTRODUCTION TO POLIMER SCIENCE & TECHNOLOGY	CHE670A	Polymer Fundamentals: Chemistry of Polymer synthesis, Polymer Reaction Kinetics:Stepgrowth polymerization, freeradical chain growth polymerization, EmulsionPolymerization, Ionic and cationic polymerization.
683	HSS	SOCIOLOGY OF ENVIRONMENT	SOC742A	The new environmental paradigm in sociology. The relationship between environment and ‘social complex’. Environmentalism and Development (issue of socioeconomic equities). The concept of sustainable development. Environmental policy and planning (structure, normative and attitudinal patterns in communities). Social Institutions in the context of environmentalis, e.g. Roligion, caste, tribe, local communities and voluntary associations. Environmental movements and their impact on policy and planning.
684	HSS	EXISTENTIALISM	PHI782A	The course seeks to go into the sources of modern existentialist movement in the thoughts of Hegel, Husserl and Kierkegaard. A detailed study of Sartre’s philosophy will be undertaken in the light of his major works.
685	HSS	SOCIAL AND POLITICAL PHILOSPHY	PHI774A	Nature and Method of Social Philosophy; Nature of Society; Theories of Origins of Society; Place and Role of Social Institutions; Social Values and Ends of Political Authority; Sources of Justification and Criticism of Political Authority.
686	HSS	CONTEMPORARY CRITICAL THEORIES & LITERARY PRAXES	ENG751A	Contemporary critical theories particularly those of the post1960s, have problematised the writing, rading and receiving of literature. This course examines complex aspects of recent critical theories associated with poststructuralism to expose students to contemporary theories that would enhance their research into chosen areas of literature. Delving on relevant issues and debates, the course acquaints students with various strategies of reading and writing. Important topics of discussion include semiotics, deconstruction, intertextuality, reader response theory, autobiographical theory, postLacanian psychoanalytic criticism, postcolonialism, postmodernism, etc.
687	HSS	RESPONSBILE DISSENT	ENG742A	Concept and practice of responsible dissent in various sociopolitical contexts: Notions of dissent and responsible dissent, ethical roots of dissent, contexts and forms of dissent, profiles of dissenties (Chomsky, Gandhi, etc.), the supression of facts and dissemination of misinformation, strategies of manufacture of consent in totalitarian and democratic structures.
688	HSS	LINGUISTIC ANALYSIS	ENG709A	This course attempts to apply the principles of lingustic analysis to real language data. It covers phonologica, lexical, syntactic and semantic analysis and involves some amount of fieldwork.
689	HSS	APPLIED LINGUISTICS	ENG708A	Modern developments in applied linguistics particularly in the fields of language learning and teaching; psychological, sociological, lingustic, and pedagogical aspects of language learning, second language acquisition, contrastive analysis, erroranalysis, programmed instruction, audiovisual aids, language testing etc.
690	ECO	ADVANCED MACROECONOMICS	ECO750A	Discussions on macroeconomic theory within the ISLM frame work, and the supply considerations, and then review some of the new classical and new keynesian contributions; budget financing, interaction between macro and microeconomics.
691	ECO	ADVANCES IN MICROECONOMIC THEORY	ECO737A	Developments in monopolistic competition (nonprice decisions in particular), the microeconomic theory of modern organizations (discretionary managerial behaviour, the organizational dimension) as environments for strategic behaviour.
692	ECO	PUBLIC ECONOMICS	ECO535A	Scope of public economics; Equity, social welfare and taxation; Taxation, income support and social insurance; Taxation and individuals; Market failure and government intervention; Optimal provision of public goods.
693	CHM	ORGANIC PREPARATION LAB	CHM503A	Preparations of various organic compounds employing different reactions willbe carried out, with a view to give the student sufficient training in syntheticorganic chemistry.
694	PHY	ELECTRONICS	PHY647A	Survey of network theorems and network analysis, basic differential amplifier circuit, op amp characteristics and applications, simple analog computer, analog integrated circuits, PLL, etc. digital electronics, microprocessors, digital signal analyzer etc., introduction to microprocessors.
695	PHY	NON-EQUILIBRIUM STATISTICAL MECHANICS	PHY615A	Linear response theory, FokkerPlanck and Langevin equations, master equation; nucleartion and spinodal decomposition, critical dynamics, Boltzmann equation.
696	PHY	ADVANCED STATISTICAL MECHANICS	PHY613A	Equilibrium statistical mechanics, phase transitions, critical phenomena, superfluidity, superconductivity, nonequilibrium statistical mechanics, Langevin equations, FokkerPlanck equations, ergodic hypothesis.
697	SDS	STATISTICAL PATTERN RECONGNITION	SDS781	Introduction to pattern recongnition supervised and unsupervised classification. Dimension reduction techniques: principal component analysis, multidimensional scaling features for maximum linear separation. Linear and quadratic discriminant analysis regularized discriminant analysisi. Linear and nonlinear support vector machines. Cluster analysis: hierarchical and nonhierarchical techniques classification using supervised and unsupervised classification.
698	MTH	STATISTICAL PATTERN RECONGNITION	MTH781	Introduction to pattern recongnition supervised and unsupervised classification. Dimension reduction techniques: principal component analysis, multidimensional scaling features for maximum linear separation. Linear and quadratic discriminant analysis regularized discriminant analysisi. Linear and nonlinear support vector machines. Cluster analysis: hierarchical and nonhierarchical techniques classification using supervised and unsupervised classification.
699	MTH	APPLIED MATRIX THEORY	MTH610A	Review of basic lin.alg. canonical factorization. QForms. Courant Fischerminmax & related theorems. PerronFrobenius theory. Matrixstability. Inequalities,ginverse (A, Am, A+). Direct, iterative, projection and iterative methods.
700	ME	ADVANCED TOPICS IN ROBOTICS	ME769A	Advanced techniques of kinematics and dynamics of mechanical systems. Parallelactuated and closedloop manipulators. Redundant manipulators. Mobile robotes and path planning. Complaint motion and control.
701	MSE	PROCESS PLANT DESIGN FOR METT. ENGG. OPERATIONS	MSE425A	Identification of process flow sheet: Preliminary estimate of resources and facilities: Materials and energy balance, detailed plant flow sheet: Equipment selection and specification, economic selection and optimization.
702	MSE	PHYSICAL & MATHEMATICAL MODELLING OF STEELMAKING PROCESSES	MSE629A	Brief review of scientific fundamentals such as thermodynamics, kinetics and trasport phenomena of relevance to steel making: Mathematical modeling techniques: Principles of physical modeling: Successive approximation and continuous casting.
703	PHY	PHYSICS OF UNIVERSE	PHY305A	Astronomical observations and instruments, photometry, stellar spectraand structure; stellar evolution, nucleosynthesis and formation of elements,variable stars, compact stars, star clusters and binary systems.
704	IME	POWER SECTOR REFORM & REGULATION: THEORY AND PRACTICE	MBA683A.	Electricity value chain and economic development. Structure of Electricity Demand and Supply: Power Sector Scenario in India. Theories of Regulation and Economics of Regulation. Power Sector Reform and Regulation: Electricity Policy, National Tariff Policy, Rural Electrification Policy.Regulatory Process: Functions of Electricity Regulatory Commissions and APTEL. Rate of Return Regulation and Performance Based Regulation. Multiyear Tariff. Principles of Retail Tariff Design Single Part, Multipart, TOD Tariffs etc. Unbundling, Privatization and Franchisee Development. Power Purchase Agreement. Competitive Bidding Guidelines. Power Exchanges and Market Monitoring. Distribution Reforms and Performance incl. RAPDRP and RGGVY. Regulatory Approach to Promote Renewable Energy: Renewable Portfolio Obligation, FeedinTariff. 1.Electricity Markets, Harris,Chris, John Wiley, 2006. 2.Electricity Economics: Electricity Economics: Regulation and Deregulation, Geoffrey Rothwell, Toms Gmez, Wiley IEEE Press, 2003. 3. Economics of Power Systems, Wiley Press, 1998.5. India Infrastructure Reports 3iNetwork, OUP. 6. Power System Economics: Designing Markets for Electricity, Steven Stoft, Wiley IEEE Press, 2001. 7. Privatisation, Restructuring and Regulation of the Electricity Sector, Wiley Press, 2001.

S.No.	Branch	Course Title	Courses ID	Content
				documents, regulations and tariff orders of regulatory commissions, forum of regulators, case studies and class notes.
705	ME	ADDITIVE MANUFACTURING	ME670	
706	CHE	ENVIROMENTAL POLLUTION CONTROL DESIGN AND MODELLING	CHE673A	1. Air Pollution: a. Introduction: Atmospheric pollutants: definition, sources, concentration levels, units. b.Gaseous phase chemistry: photochemical smog in troposphere; O3 depletion in stratosphere; NOx equilibria.d. Aerosols: size, distribution, deposition, visibility degradation, nucleation. f. Mass transfer aspects: diffusion, mass transfer coefficient, characteristics times. g. Troposphere energy balance: production and destruction of ozone. h. Air quality criteria: health effects, standards. i. Air pollution control: dispersion: puff and plume dispersion, Gaussian models. j. Control of Pollutants: (1) Absorption: design of an absorber (SO2, CO2, NO2)(2)Adsorption: design of an adsorber (SO2, VOC), breakthrough analysis. k. Biological and chemical treatment of organic/inorganic/biological pollutants, water quality and parameters. b. Waste water treatment: primary and secondary treatments, sludge disposal (aerobic and anaerobic digesters. c. Biological organic waste treatment equipment design: (1) Activated sludge process reactor(2)Trickling filter (3)Biotower reactor. e. Advanced waste water reactors: (1) Continuous counter current multi stage fluidized bed. (2) Moving bed reactor. f. Environmental impact assessment. 1. Atmospheric Chemistry and Physics by Seinfeld and Pandis (Wiley.) 2. Environmental Engineering by Peavy, Rowe, and Tchobanoglous (McGraw Hill.)
707	CE	INSTRUMENTATION, LABORATORY AND FIELD PRACTICES IN GEOINFORMATICS	CE673A	Use of automatic and digital levels, electronic theodolites, total stations, planetabing; Control surveys using GPS, Total station and triangulation methods(adjustment and computations of coordinates); Control
708	CSE	PROGRAM ANALYSIS VERIFICATION AND TESTING	CS639A	Data flow Analysis; Interprocedural Analysis: functional, call string and graph reachability based approaches Abstract Interpretation; Weakest Precondition, Floyd Hoare Logic, Separation Logic; Software abstraction refinement, model checking of concurrent programs, Program Testing: program testing basics, automatic test case generation, directed testing. 1. Edsger Wybe Dijkstra. A Discipline of Programming. Prentice Hall PTR, Upper Saddle River, NJ, USA, 1997. 2. David Gries. The Science of Programming. SpringerVerlag New York, Inc., Secaucus, NJ, USA, 1997. 3. Robert S. Milner. Principles of Computer System Design. Prentice Hall PTR, Englewood Cliffs, NJ, 1981. 4. Flemming Nielson, Hanne R. Nielson, and Chris Hankin. Principles of Program Analysis. SpringerVerlag New York, Inc., Secaucus, NJ, USA, 1999. 5. Alfred V. Aho, Monica S. Lam, Ravi Sethi, and John D. Ullman. Compilers: Principles, Techniques, and Tools. Longman Publishing Co., Inc., Boston, MA, USA, 2006. 6. Michael Huth and Mark Ryan. Logic in Computer Science: Modelling and Reasoning about Systems. Cambridge University Press, New York, NY, USA, 1999.
709	CE	STOCHASTIC HYDROLOGY	CE614A	Statistical methods in hydrology, probability distribution of hydrologic variables, hypothesis testing and goodness of fit, flood frequency analysis, single and multiple regression analysis, classification of time series, time series modelling, time series modelling of annual and periodic hydrologic time series (including AR, ARMA, ARIMA, and DARMA models), multivariate modelling of hydrologic time series, practical considerations
710	CE	AIRPORT SYSTEMS PLANNING AND DESIGN	CE688A	Air Transport structure and organization, the challenges and the issues, Forecasting air travel demand trend forecasts and analytical methods; Air freight demand, Characteristics of the aircraft as they affect airport design; Airport capacity and configuration; Geometric design of runway, taxiway and aprons; passenger terminal functions, passenger and baggage flow, design concepts, analysis of flow through terminal buildings; Airport access problem; Environmental impact of airports.
711	IME	ANALYTICS IN TRANSPORT AND TELECOM	IME639A	Introduction. Commonality in modeling problems across Transport and TelecomIntroduction to graph TheoryReview of linear and integer linear programming Minimum Spanning Tree Problem Steiner Tree Problem ApplicationsIntroduction to complexity theory and NPcompletenessNetwork Flow Models, Max Flow Min Cut Problem, Minimum Cost FlowsKnapsack Problem and applicationsBin Packing and applicationsPartitioning/Packing Models and ApplicationsTraveling Salesman Problem with applicationsFixed Charge Transportation ProblemTelecom Network Design, Access Networks, Backbone NetworksDesign of NetworksRecommended Books:oR.K. Ahuja, T.L. Magnanti, J.B. Orlin, Network Flows, Prentice HalloE. Lawler, Combinatorial Optimization: Networks and Matroids, DoveroH.M. Salkin, K. Mathur Foundations of Integer Programming
712	IME	ANALYTICS IN TRANSPORT AND TELECOM	IME639	Introduction. Commonality in modeling problems across Transport and TelecomIntroduction to graph TheoryReview of linear and integer linear programming Minimum Spanning Tree Problem Steiner Tree Problem ApplicationsIntroduction to complexity theory and NPcompletenessNetwork Flow Models, Max Flow Min Cut Problem, Minimum Cost FlowsKnapsack Problem and applicationsBin Packing and applicationsPartitioning/Packing Models and ApplicationsTraveling Salesman Problem with applicationsFixed Charge Transportation ProblemTelecom Network Design, Access Networks, Backbone NetworksDesign of NetworksRecommended Books:oR.K. Ahuja, T.L. Magnanti, J.B. Orlin, Network Flows, Prentice HalloE. Lawler, Combinatorial Optimization: Networks and Matroids, DoveroH.M. Salkin, K. Mathur Foundations of Integer Programming
713	CSE	VLSI TESTING AND FAULT TOLERANCE	CS720A	The course is primarily intended to familiarize students with the problemof testing large and complex electronic circuits. Various techniques to solvethis problem and concepts of design for easy testability will be covered in detail. The problem of reducedyield and reliability of circuits in presence of faults will be discussed andtechniques to improve the yield and reliability of these circuits by introducingfault tolerance will be discussed in detail.
714	ECO	PRODUCTIVITY AND EFFICIENCY ANALYSIS	ECO572	This course is designed for students who wish to learn about production economics and benchmarking analysis. Benchmarking is used to measure performance of a producer using a specific indicator (productivity) to estimate production and cost functions and compute measures of absolute and relative economic performance using dataenvelopment analysis (DEA), deterministic frontier analysis (DFA), and stochastic frontier analysis (SFA). DEAP, Excel solver, FRONTIER, and Stata. 01-APR-2015
715	ECO	PRODUCTIVITY AND EFFICIENCY ANALYSIS	ECO572A	This course is designed for students who wish to learn about production economics and benchmarking analysis. Benchmarking is used to measure performance of a producer using a specific indicator (productivity) to estimate production and cost functions and compute measures of absolute and relative economic performance using dataenvelopment analysis (DEA), deterministic frontier analysis (DFA), and stochastic frontier analysis (SFA). DEAP, Excel solver, FRONTIER, and Stata. 01-APR-2015
716	DES	DESIGN THINKING & RESEARCH METHODS	DES604A	
717	AE	COMBUSTION PROBLEMS IN ROCKET PROPULSION	AE654A	
718	AE	COMBUSTION PROBLEMS IN ROCKET PROPULSION	AE654	
719	AE	COMBUSTION PROBLEMS IN ROCKET PROPULSION	AE654	
720	AE	HEAT TRANSFER IN AEROSPACE APPLICATIONS	AE608A	Importance of Heat transfer in Aerospace Applications, Different Modes of Heat Transfer, Governing Equations of fluid Dynamics and Heat Transfer, Material and Transport Properties in Heat Transfer. (TPS) design and analysis. Introduction to the thermo-structural Aspects of high speed vehicles, Introduction to the Experimental investigation of High speed flows.
721	AE	HEAT TRANSFER IN AEROSPACE APPLICATIONS	AE608	
722	ME	MODELING THERMAL TRANSPORT IN MANUFACTURING PROCESSES	ME669A	1. Review of fundamentals of thermal transport in manufacturing : 1. Introduction to the course Importance of heat transfer in manufacturing and applications. 2. Steady and transient heat conduction, Conduction based modeling of heat transfer in manufacturing and numerical implementation: 1. Basic introduction to FDM and FVM techniques, Mathematical formulation of thermal transport, Governing equations and numerical solution of Convectiondiffusion problems, 3. Mesh terminology, Accuracy, Consistency, Stability and Convergence, 4. Phase change Enthalpy based algorithm for Melting/solidification, Twophase mushy zone flows, Heat transport in manufacturing processes: 1. Solidification processing Modeling of moving melting/solidification phase change interface, Alloy solidification, Segregation, Twophase mushy zone flow, Modeling of heat transport in manufacturing processes: 2. Modeling using enthalpy method for solid liquid and/ or liquidvapour phase change interface. Melt pool formation and flow behaviour, Beam heat flux models, Eample problems of thermal modeling in Laser assisted manufacturing process, 3. Thermal deposition process Modeling of free surface evolution, Modeling of droplet impact and deposition on substrates. C.H. Doumanidis, R. Pitchumani, Chapter 17, Heat Transfer in Manufacturing and Materials Processing, John Wiley & Sons Inc., 2006. 2. T.L. Bergman, A.S. Lavine, F.P. Incropera, D.P. DeWitt, Fundamentals of Heat and Mass Transfer, 7th Edition, Wiley, 2011.

S.No.	Branch	Course Title	Courses ID	Content
723	AE	ACOUSTICS IN FLUIDS	AE694A	Flow, McGraw Hill, New York, 1980. 4. J. Dowden (Ed.), The Theory of Laser Materials Processing in Heat and Mass Transfer in Modern Technology, Springer, 2008. 5. D.M. Stefanescu Science and Engineering Series, Wiley & Sons Inc., 2010. 7. Larry F. Jeffus, Welding: Principles and Applications, 4th ed., Thomson Learning, 1999. 8. L. Pawlowski, The Science and Engineering of Thermal Spray Coatings, 2nd ed., John Wiley & Sons, 2008. Itroduction to the course. Fundamentals of acoustics: Derivation of wave equation, speed of sound, harmonic waves, acoustic energy/intensity, decibel scale, acoustic impedance, reflection and transmission coefficient, wave propagation. Acoustics of resonantors: Travelling and standing waves, boundary conditions, eigenfrequency and eigenmodes, effects of area variation, reflection and transmission of waves in pipes. Acoustic reciprocity, aeroacoustic analogies. Attenuation of sound: Viscous and thermal conduction losses, absorption coefficient, sound absorption in pipes. Application of principles of acoustics: Aeroacoustic jet noise. 1. Lawrence E. Kinsler, Austin R. Frey, and Alan B. Coppens, 2000. Fundamentaals of acoustics, 4th edn. JohnWiley & Sons, Inc. 2. Philip M. Morse and K. Uno Ingard, 1986. Fundamentals of acoustics. 1st edition. www.win.tue.nl/sjoerder/papers/boek.pdf 4. S.W. Rienstra & A. Hirschberg, 2004. An introduction to aeroacoustics. http://www.win.tue.nl/sjoerder/papers/lesswrmh.pdf 24-MAR-15
724	PHY	INTRODUCTION TO PROFESSION AND COMMUNICATION SKILLS FOR PHYSICISTS	PHY888	1. Introduction to HPC and scientific computing. Overview of major applications. 2. Supercomputing architecture; multicores; shared memory; switch etc. 3. Review of basics of C/Fortran programming. 4. Parallel programming. 1. P.S. Pacheco, An Introduction to Parallel Programming, Elsevier (2011). 2. M. Quinn, Parallel Programming in C and OpenMP, McGraw Hill Education (India) (2003). 3. A. Grama, A. Gupta, G. Karypis, and D. A. Padua, Introduction to Parallel Computing, Morgan Kaufmann Publishers, 2003. 4. S. Rajkumar, Parallel Programming, Pearson Education, 2004. 5. S. Rajkumar, Parallel Programming, Pearson Education, 2004. 6. S. Rajkumar, Parallel Programming, Pearson Education, 2004. 7. S. Rajkumar, Parallel Programming, Pearson Education, 2004. 8. S. Rajkumar, Parallel Programming, Pearson Education, 2004. 9. S. Rajkumar, Parallel Programming, Pearson Education, 2004. 10. S. 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S.No.	Branch	Course Title	Courses ID	Content
738	AE	VISCOUS FLOWS	AE614A	Basic concepts of BL theory; similar flows: generalized techniques of solvingBL eqns. for incompressible fluids : thermal BL. BL control. Intro. to turbulentshear flows.
739	CHM	MATHEMATICS FOR CHEMISTRY	CHM600A	Error Analysis, Scalars, vectors, curl, divergence and gradient, ordinary3000[4] differential equations, symmetry and group theory, matrices, etc.
740	BSBE	TISSUE ENGINEERING	BSE614A	What is tissue engineering? Scope and objective of tissue engineering; Principles of tissue engineering; Essential components of tissue engineering; Materials Science/Engineering aspects (degradable m aspects (choice of cell type, progenitor cells and cell differentiations); Molecular biology aspects (cell signaling moleculesgrowth factors, cell attachmentintegrins); Drug delivery in tissue engineering; Cor
741	MSE	SEMINAR PARTICIPATION	MSE690A	Seminar Participation
742	BSBE	BIOLOGY LABORATORY	BSE681A	
743	IME	INDEPENDENT STUDY	IME797A	Market Efficiency, Capital Structure, Asset Pricing, Corporate Governance, PortfolioTheory, Financial Intermediation, Credit Risk Modeling & Market Microstructure 1. Ananth Madhavan, Market microstructure: A survey, Journal of Financial Markets,Volume 3, Issue 3, August 2000, Pages 2052582. Bhattacharya Sudipto & Thakor Anjan V., 1993."Contemporary Ba (1973)., The Pricing of Options, and Corporate Liabilities, Journal of Political Economy, MayJune, 637659.4. Burton G. Malkiel., The Efficient Market Hypothesis and Its Critics, Princeton University CEPS W Fama, "Random Walks in Stock Market Prices," Financial Analysts Journal,September/October 1965 (reprinted JanuaryFebruary 1995)7. Fama, E. F., 1970. "Efficient Capital Markets: A Revi 24-DEC-14
744	EE	INFORMATION THEORY	EE667	Introduction: Entropy, Relative Entropy, Mutual Information Inequalities, Entropy rate. Ansymptotic Equipartition Property(AEP): Consequences of the AEP, Typical Sequences, ShannonMcMillanBreiman TH coding Tunstal code, variable to variable length codes/ arithmetic code. Channel capacity: Discrete Memoryless Channel, Joint Typicality, Channel Coding Theorem and its converse, Feedback capacity, Sc Gaussian Channels, Channels with Colored Gaussian Noise, Gaussian Channels with Feedback. Rate Distortion Theory: Rate Distortion Function, Rate Distortion theorem and its converse, BlahutArimoto A algorithm (LZ77). Network Information Theory: Gaussian MultiUser Channels, Multiple Access Channel, Broadeast Channel, Encoding of Correlated Sources. 1. Thomas M. Cover, Joy A. Thomas, "Elements of Information Theory", 2nd Edition , John Wiley & Sons, 2006. 2. James L. Massey, Lecture notes on "Applied Digital Information Theory 1&# David J.C. MacKay, "Information Theory, Inference, and Learning Algorithms", Cambridge University Press. 5. Robert Ash, "Information Theory", Dover Publications, 1965. 6. Raymond V "Network Information Theory", Cambridge University Press, 2012. 8. I. Csiszar and J. Korner, "Information Theory: Coding Theorems for Discrete Memoryless Systems", 2nd edition, Car
745	HSS	TWENTIETH CENTURY PHILOSOPHY-I	PHI751	An analytical study of Russells The Philosophy of Logical Atomism andWittgensteins Tractatus LogicoPhilosophicus: Facts and Propositions; Namesand Objects; Definite Description; Picture Theory of M 21-jul-2014
746	AE	AUTONOMOUS NAVIGATION	AE640A	Course introduction, basic definition, notion, guidance, navigation, and control loops.Review to linear algebra. Coordinated frames, kinematics and dynamics, trim conditions.Linear control and autopilot de inertial measurement units (IMUs), global positioning systems (GPS). State estimation: Kalman filter (KF), Extended Kalman filter(EKF), Unscented Kalman filter (UKF), Cubature Kalman filter (CKF), Inform guided navigation. Cooperative control 1. D. P. Bertsekas and J. N. Tsitsiklis, Introduction to Probability, Athena Scientific, 2008.2. S. Thrun, W. Burgard, and D. Fox, Probabilistic Robotics, MIT Press, 2005.3. S. M. LaValle. Planning Algorithms. C
747	CE	TRANSPORTATION FACILITIES DESIGN	CE481A	Any two of the three modules listed below will be taught in any semester :Module 1Traffic Design : Introduction, freeway and toll booths, intersections/interchanges, signs and lighting, arterials/ weaving s concrete pavement, composite pavement.Module 3Geometric Design: Introduction, design controls and criteria, freeway design, arterial/collector design, atgrade intersections, terminals. Chakroborty, P. and Das, A., "Principles of Transportation Engineering", Prentice Hall of India, 2003Garder, N.J. and Hoel, L.A., "Traffic & Highway Engineering", 3rd Ed., Brooks/Cole, Pac 1998Americal Association of State Highway and Transportation Officals (AASHTO), "A Policy on Geometric Design of Highways and Streets", 5th Ed., AASHTO, 2004
748	BSBE	DRUG DESIGN AND METABOLISM	BSE617	Physicochemical Principles of Drug Action; Partition Coefficients; ReceptorEffector Theories;Role of Second Messengers in Drug Action; Methods of Receptor Isolation,Characterization and Modeling (5)F Drugs and Regulatory Implications; TheoreticalApproaches: QSAR, Topliss Tree, MSA, CoMFA (2)Neuroactive Drugs: Neurons and Neurotransmitters; Brainrelated Disorders andChemotherapy; Drugs Inter Antimalarial, Antiviral, and Cardiovascular Drugs; Emerging Trends in DrugDesign: Inhibitors of DNA Topoisomerase and Protein Farnesylation & Prenylation; GeneBased Medicines (8)Biopharmaceuticals Conventional and GeneticDrugs; TissueSpecific Delivery of Antitumor Agents (8)Drug Administration, Distribution, Metabolism and Elimination (ADME); Pathways of DrugMetabolism: Enzymology and Mo Metabolism; Toxicological Aspects of Metabolism: MetabolicActivation of Environmental Carcinogens and DNA Damage; Drug Pharmacokinetics andFinal Body Clearance (2) 1. Medicinal Chemistry: A Biochemical Approach, Thomas Nogrady2. Principles of Medicinal Chemistry, William O. Foye3. The Pharmacological Basis of Therapeutics: Goodman and Gilman4. Introduction 15-SEP-14
749	PHY	SOFT MATTER PHYSICS	PHY616A	Introduction to Soft Matter, Forces, energies and timescales in soft matter, Thermodynamic aspects of intermolecular forces, van der Waals force, Hydrophobic and hydrophilic interaction, Interfacial phen hydrodynamic instabilities, Foams and emulsions, Polymers, colloids and Surfactants, Liquid crystals, Selfassembly in soft matter, Experimental tools for soft matter. Books recommended:1. Intermolecular and Surface Forces by Jacob N. Israelachvili (Academic Press, 1998)2. Soft Condensed Matter by R. A. L. Jones (Oxford University Press, 2002)3. Principles of Con hydromagnetic stability by S. Chandrashekhar (Oxford University Press, 1981)5. Structured Fluids by Thomas A. Witten (Oxford University Press, 2004)6. Structure and Rheology of Complex Fluids by Ron
750	MTH	INTRODUCTION TO LIE ALGEBRAS AND REPRESENTATION THEORY	MTH641A	Definitions and first examples. Classical Lie algebras. Ideals and homomorphisms. Niporent Lie algegras. Engel's theorem. Solvable Lie algebras. Lie's theorem. JordanChevalley Decomposition Complete reducibility and Weyls theorem. Representation theory of the Lie algebra sl(2). Total subalgebras and root systems. Integrality properties. Simple Lie algebras and irreducible root systems. Course Reference : 1. Humphreys, James E. (1972), Introduction to Lie Algebras and Representation Theory, Berlin, New York.2. J.P. Serre, Complex semisimple Lie algebras (trnslated from French: Algebr 24.09.14
751	ES	IGNEOUS AND METAMORPHIC PETROLOGY	ES652A	This course will focus on the processes and principles involved in the generation of Earth materials, in particular, igneous and metamorphic rocks. Students will gain a broad overview of the Earth as dyna emphasis on developing skills necessary to understand and evaluate melt generation and crystallization, differentiation and chemical evolution of magma, and metamorphic processes etc. Topics to be di Thermodynamic evaluation of phase diagrams; Phase Rule and One Component System; Phase Diagrams for Binary Systems (Solid solution, Eutectic, and Peritectic systems) such as ForsteriteFayalite, A (AnorthiteDiopsideForsterite, and AnorthiteForsteritesilica); Effect of P, T & fluids on Melting; Chemical Petrology: Major and minor element, and isotopic compositional variations; graphical and mathemat (REE) patterns and modeling source magma composition; Generation and diversification of magmas; MidOcean Ridge Volcanism; Mantle Plumes and Ocean Island Basalts; Subduction related Arc magm assemblagesand chemographic (ACF, AKF, and AFM) diagrams; Metamorphic Facies; Metamorphic Reactions. Introduction to Igneous and Metamorphic Petrology by John D. Winter (Prentice Hall, 2nd edition)Recommended readings:1. The Interpretation of Igneous Rocks by Cox, Bell and Pankhurst (Chapman and Press; 2nd edition).3. Petrology: Igneous, Sedimentary and Metamorphic by Blatt, Tracy, and Owens (Freeman and Company)
752	ES	GLOBAL CLIMATE CHANGE	ES646A	Introduction to global climate: earths climate system, time scales of climate change, climate forcings and response systems, climate feedbacks and interactions.Earths radiation budget and circulation sy circulation, oceanic conveyor belt.Climate archives and proxies: Methods of reconstructing climate, Ice sheet, sedimentary archives, biotic proxies, geological and geochemical proxies.Global climatic mod geochemical modelsQuaternary climates: Sea level changes, glacial/interglacial cycles, tectonicsclimate coupling, sea floor spreading, BLAG hypothesis, Upliftweathering hypothesis, carbon reservoir, veg Sedimentology, stable isotopes, geochemistry, geochronology relative and numerical methods, PreQuaternary climates, evolution of climate through geological time.Impacts of climate change: impacts on 1.Ruddiman, W.F. 2001. Earths Climate: past and Future. W.H. Freeman & Co.2.Bradley, R. S., 1999, Quaternary paleoclimatology. 3.Williams, M., Dunkereley, D., Decker, P.D., Kershaw, P. & Chappel, J. (1998) and Menking, K., (1998) Environmental geology; an earth system science approach, Freeman.
753	ES	EARTH SYSTEM PROCESSES	ES640A	Principles: Earth System processes, geomorphic systems, Threshold and equilibrium; scale in geomorphology, key concepts in geomorphology, hydrologic cycle.Processes and Products: Exogenetic and e Drainage basins and river systems, river morphology and hydrology, fluvial erosion, transport and sedimentation, fluvial depositional landforms, geomorphometric concepts and drainage basin morphome permeability, aquifers, chemistry of groundwater.The Atmospheric System: Atmospheric composition and mixing, atmospheric circulation, greenhouse effect.The Ocean and Coastal system: Coastal enviro resulting topographic features, coastal deposition and landforms.Global geomorphology and tectonics: Earths physiography and landscape evolution; Landforms and tectonics of plate margins and plate

S.No.	Branch	Course Title	Courses ID	Content
				effects; coupled tectonicssurface process models. 1.Summerfield, S.A. (1991) Global Geomorphology, Longman.2.Ernst, W.G. (2000) Earth Systems: Processes and Issues, Cambridge University Press. 3.Richards, K.S. (1982) Rivers forms and processes in W.D. (1969) Principles of geomorphology, John Wiley & Sons.6.Williams, M.J. (1998) Quaternary Environments, Arnold Publishers7.Reading, H.G. (1996) Sedimentary Environments. Blackwell.8.Merriots, D,
754	MSP	MATERIALS ENGINEERING	MS605A	Solidification, Powder processing, Crystal growth, Heat treatment andmicrostructures, Non destructive evaluation, Processing of glasses and polymers,Novel processing methods, Thin films, Surface phen
755	BSBE	NEUROBIOLOGY	BSE656A	Cell and molecular biology of neurons. Membrane potential, local signaling andgeneration of action potential. Mechanism of synaptic transmission. Sensorypereception : Vision, hearing, taste, touch and s system and the Cerebral cortex learning and memory, sleep and dream.
756	BSBE	HUMAN MOLECULAR GENETICS	BSE654	Simple Mendelian traits; Lossoffunction mutations; Gainoffunction mutations;Gene interactions; Dynamic mutations; Genetics of neoplasia; Genomic imprintingand human disease; Xinactivation and DNA disorders;Pharmacogenetics and biochemical genetics; Animal models in human genetics:Methods used for diagnosis and detection of gene mutations; Gene Therapy.
757	AE	AUTONOMOUS NAVIGATION	AE640	Course introduction, basic definition, notion, guidance, navigation, and control loops.Review to linear algebra. Coordinated frames, kinematics and dynamics, trim conditions.Linear control and autopilot de inertial measurement units (IMUs), global positioning systems (GPS). State estimation: Kalman filter (KF), Extended Kalman filter(EKF), Unscented Kalman filter (UKF), Cubature Kalman filter (CKF), Inform guided navigation. Cooperative control 1. D. P. Bertsekas and J. N. Tsitsiklis, Introduction to Probability, Athena Scientific, 2008.2. S. Thrun, W. Burgard, and D. Fox, Probabilistic Robotics, MIT Press, 2005.3. S. M. LaValle. Planning Algorithms. C
758	AE	SPACECRAFT GUIDANCE NAVIGATION AND CONTROL	AE645A	Course Details: Attitude dynamics and stability ofthreeaxis stabilized, singlespin, dualspin, and multibodyspacecraft with a1ticulated antennas, sensors, and solar arrays.Design of control ofthreeaxis stabl gyrosLargeangle threeaxis attitude maneuver controllers using reaction wheels and thrustersControl of spinning spacecraft in transfer orbit during delta_ v firing and in operational orbitsaround the Earth, thrustersDynamics and control of dualspin spacecraftPrecision pointing and tracking controllers for tracking landmarks, moving objects, and othersatellites for crosslink communicationSolar array contro solar arrays, its interaction with spacecraft dynamics andcontrol systemsAttitude determination with gyros, star trackers, sun sensors, and horizon sensors usingalgorithms such as TRIAD and QUEST (qu rendezvousThe above control techniques will be related with the control of Indian communication, remotesensing, and other special purpose satellites (Ca1tosat, Edusat, telemedicine). Course Reference : * Hughes, P.C., Spacecraft Attitude Dynamics, John Wiley, 1986.* Sidi, M.J., Spacecraft Dynamics and Control, Cambridge University Press, 1997* Noton, M., Spacecraft Navigation and of Geosynchronous Spacecraft, Prentice Hall, 1986* Bryson, A.E., Control of Spacecraft and Aircraft, Princeton University Press, 1994* Pocha, J.J., An Introduction to Mission Design for Geostationary Sate Wie, B., Space Vehicle Dynamics and Control, AIAA Education Series, 1998
759	MTH	COMPUTER PROGRAMMING AND DATA STRUCTURES	MTH409A	Fortran 77: Integer and real operations, logic and complex operations, Controlstatements, Do statement, arrays subroutines and functions. Introduction tod data structures in C Programming Language; Arra circular linked lists, Application of Linked Lists; Polynomial addition,sparse matrices, Trees: binary trees, redblack trees, Hash tables. somediscussion about data structures in F90F95 with examples. 21-JUL-2014
760	CHM	PHYSICAL CHEMISTRY I	CHM421A	Atomic Structure, chemical binding and molecular structure. Elements ofmolecular spectroscopy. 21-JUL-2014
761	AE	DETERMINISTIC & RANDOM VIBRATION	AE685A	Free and forced vibration of discrete multidegree of freedom systems with andwithout viscous damping; impulse and frequency response methods. Continuoussystem systems; natural modes, free & forced vibra processes,response of system to random excitations
762	HSS	METHODS OF SOCIOLOGICAL RESEARCH	SOC751A	Relation between theory and method and the concept of paradigmQuantitative Research Methods: Experimental design and surveys; different types of experimental designs; longitudinal studies; sampling sources and their use in the historical and documentary study of Indian society and culture: Census, Archival material, National Sample Survey, Gazetteers and District Handbooks and other large data set Brewer, Jonhn and Albert Hunter. Multimethod Research, Sage, 1989.Clifford, James and G. Marcus, Writing Cultures: Poetics and Politics of Ethnography, University of California Press, 1986. Cochran, Wil pp. 145294)Epstein, A. L. (ed) The Craft of Social Anthropology, Tavistock, 1967. Flick, Uwe. An Introduction to Qualitative Research, Sage, 2002.Geertz, C. The Interpretation of Cultures. Basic Books, 1963 Tests, Sage,1993. LeviStrauss, C. Structural Anthropology. Basic Books 1963. (Part 1 pp. 3791)Luckacs, George. History and Class Consciousness. Merlin Press, 1971. (Chapter 1)Margolis, Eric (ed), The S
763	IME	PROBABILITY AND STATISTICS	IME602A	Probability: Sample space, events and probability; Conditional probability and independence; Random variable (RV); Expectation, variance and higher moments; Some standard probability distributions; Fu Population and sample; Sampling distributions; Properties of point estimators; Maximum likelihood method; Interval estimation; Hypothesis testing; Some standard hypothesis tests; Analysis of variance; Probability Theory and Statistical Inference, John Wiley & Sons.2)R.V. Hogg, A.T. Craig, and J.W. McKean, Introduction to Mathematical Statistics, Pearson Education. Course Reference : 1)V.K. Rohatgi and A.K.Md.E. Saleh, An Introduction to Probability and Statistics, Wiley India Private Limited.2)R.J. Larsen and M.L. Marx, An Introduction to Mathematical Statistics and Mifflin.4)P.G. Hoel, S. Port, and S. Stone, Introduction to Statistical Theory, Houghton Mifflin.5)W. Feller, An Introduction to Probability Theory and its Applications (Vol. 1 & 2), Wiley India Private Limited.6)A Goon, M.K. Gupta, and B. Dasgupta, Fundamentals of Statistics (Vol. 1 & 2), World Press Private Limited.8)J.P. Romano and E.L. Lehmann, Testing Statistical Hypotheses, Springer (SIE).9)N.R. Draper and I
764	HSS	FUNDAMENTAL OF NEUROPSYCHOLOGY	PSY775A	Development and evolution of brain : Brain organization and function; Cerebralsymmetry; Neuropsychology of higher order functions memory language,emotional processes, spatial behaviour ; Applied h psychology,Palgrave Publishers Ltd.2. Kolb, B. & Whishaw, I.Q. (1990) Fundamentals of human neuropsychology,W.H. Freeman & Company.3. Mandal, M.K., BulmanFleming , M.B., & Tiwari, G.(2000). Side
765	HSS	QUALITATIVE RESEARCH : THEORY AND PRACTICE	PSY774A	In the last few years, research in psychology has witnessed a remarkable shiftto a post positivist paradigm. A large number of qualitative methods haveemerged in consonance with this shift. For a resear understanding of the qualitative methods. This course isaimed at providing an extensive overview of qualitative research methods,methodological roots, major theoretical principles and issues in qualitativ developmental psychology discursive psychology,counseling and psychotherapy.
766	HSS	COMPARATIVE AESTHETICS	ENG752A	The course explores the convergence and divergence of eastern and westernaesthetic practices, in an attempt to two. Its scope includes classical Greekand Sanskrit texts like Bharata s Natyasastra and A concept of dhvani and postulates of New Criticism.
767	ES	GEOCHEMISTRY	ES642	Course outline, grading policy, introduction to geochemical cycles, periodic table, Origin of the universe and Earth, Earths history from its creation to Anthropocene, The Basics: Nuclear (in) stability, isotop Thermodynamics, Radioactive decay, principles of geochronology and isochrons, radiogenic isotopes as geochronometer and process tracers, mixing and modeling calculations, RbSr, SmNd, UThPb and l assignments2 & 3, as well as reading material, Fractionation of stable isotopes, H, O, C, Discussion on assignment4 and reading material, Mass spectrometry: Measurement principles and methodology, te Principles and Applications of Geochemistry by Gunter Faure (2nd edition, Prentice Hall)Geochemistry: pathways and Processes by McSween, Richardson, Uhle (Columbia University Press)Geochemistry (McGrawHill)
768	CE	INTRODUCTION AI TECHNIQUES	CE615	Expert Systems (ES): history of ES, basic concepts of ES, definition and components of ES, inference engines and reasoning mechanisms e.g. forward reasoning, backward reasoning, and mixed reasonin selected case studies of ES applications to engineering and sciences; Artificial Neural Networks (ANNs): background and history of ANNs, definitions and basic concepts of ANNs, biological and artificial backpropagation (BP), conjugate gradients BP, self organizing networks, etc., development of ANN models for specific problems and selected case studies; Genetic Algorithms (GAs): fundamentals and p mutation, binary and realcoded GAs, constraint handling in GAs, and selected case studies involving GA applications to engineering.
769	CE	SOIL MECHANICS	CE351A	Stresses within a soil, effective stress principle, stress point and stress path, Soil water systems capillarity, flow(4)Darcys law, permeability, and tests for its determination, different heads, piping, quicksar sands, NC and OC soils, dilation, pore pressures, Skemptions coefficients, etc.(10) Compaction characteristics, water content dry unit weight relationships, OMC, max. dry unit weight, field compaction con Craig RF., Craigs Soil Mechanics., 2004, Taylor and Francis, New York, USA.Das BM., Advanced Soil Mechanics., 2008, Taylor and Francis, New York, USA.LambeTW., and Whitman RV., Soil Mechanics., 20
770	CE	STRUCTURAL ANALYSIS	CE272A	Stability and Determinacy of Structures. Review of shear force and bending moment diagrams in beams and frames. Plane trusses: method of joints and method of sections. Deflection of trusses: method Influence line diagrams and moving loads. Force and stiffness methods of analysis. Plane trusses by using method of consistent deformations. Beams and frames: method of consistent deformations, sle TEXT : Norris, C.H., Wilbur, J.B., and Utku, S., Elementary Structural Analysis, McGraw Hill, Hibbeler, R. C. (2002). Structural Analysis, 6/e, Pearson EducationREFERENCE : Hsieh, Y., Elementary Theory of St Structural Analysis, McGrawHill.Wang, C.K., Statically Indeterminate Structures, McGrawHill.West, H.H., Analysis of Structures, John Wiley.

S.No.	Branch	Course Title	Courses ID	Content
771	ECO	ADVANCED MACRO ECONOMICS	ECO408A	Introduction; The Solow growth model; Infinite horizon and overlapping generation models; New growth theory; Real business cycle theory; Traditional Keynesian theories of fluctuations; Incomplete nominal
772	HSS	APPRECIATING FAR-EASTERN CINEMA	ART401A	This is a (theoretical) cinema study course, which presents a survey of the Far Eastern films both as an art form for creative expression and as a medium of mass communication. The course will broadly orient students to the countries of the Far East Japan, Korea, Mainland China, Hong Kong and Taiwan. The films included in this course, span the decades between the post Second World War era to the 21st century and they are not only aesthetically rich but also culturally significant. The course content comprises of vast level of theoretical and textual study. References: Berggreen, Shu Ling Chen, and Rob Peaslee. "Trans Chinese imagination: film and cross Strait perception as a historical case study for contextual journalism education." Asia Pacific Media Education Review 9.1 (2008): 239-250. Fang, Karen. "Arresting Cinema: Surveillance and the City State in the Representation of Hong Kong." New Formations 18-SEP-14
773	MME	SEMINAR PARTICIPATION	MME698	SEMINAR PARTICIPATION
774	MME	M TECH THESIS	MME699	M TECH THESIS
775	MME	SEMINAR PRESENTATION	MME691	SEMINAR PRESENTATION
776	MME	SEMINAR PARTICIPATION	MME690	SEMINAR PARTICIPATION
777	DES	M DES THESIS	DES699B	M DES THESIS
778	CSE	MACHINE LEARNING FOR COMPUTER VISION	CS679	1. Probability basics and common probability distributions. 2. Fitting Probability models (ML, MAP, Bayesian). 3. Normal distribution. 4. Regression 5. Classification. 6. Graphical models. 7. Temporal models. Text Book : Computer Vision: Models, Learning and Inference by Simon J.D. Prince. References : Pattern Recognition and Machine Learning by Christopher M. Bishop. Probabilistic Graphical Models: Principles and Techniques by Daphne Koller and Andrew Y. Ng.
779	ECO	APPLIED GAME THEORY	ECO502	Introduction: Optimal contracts under uncertainty, hidden information (adverse selection), or hidden action (moral hazard); Bilateral contracting: Hidden information screening and signaling, hidden action and private information; moral hazard in teams, tournaments and organizations; Incomplete contracts: Institution design, implementation theory, bilateral and multilateral contracts
780	AE	NUMERICAL MODELING OF CHEMICALLY REACTING FLOWS	AE658A	1. Patrick Bolton and Mathias Dewatripont, Contract Theory (2005), The MIT Press. 2. T.V. S. Rammohan Rao, Contract Economics (2004), New Age International. Introduction, Governing equations, Modeling of laminar premixed and nonpremixed flames, Modeling of turbulent premixed and nonpremixed flames, Advanced modeling aspect. Introduction (3 hrs) Motivation and objectives Conservation equations and numerical solutions Steady 1 D flames Theoretical solution methods Calculation of flame speed, thickness and stretch Modeling of Laminar nonpremixed flames (6 hrs) Nonpremixed flames of other flame structures Modeling of turbulent premixed flames (10 hrs) Phenomenological description Premixed turbulent combustion regime RANS modeling LES modeling DNS modeling Modeling of turbulent combustion LES modeling DNS modeling Advanced modeling aspect (4 hrs) Combustion in two phase flows Boundary conditions Flame/wall interactions Flame/acoustics interaction Course Reference : * Peters, N., Turbulent Combustion, Cambridge University Press, 2000. * Warnatz, J., Mass, U., Dibble, R.W., Combustion: Physical and Chemical Fundamentals, Modeling and Simulation, Springer-Verlag, 2nd Edition, 2005. * Chung, T. J., Computational Fluid Dynamics, Cambridge University Press, 2002. * Law, C. K., Combustion Physics, Cambridge University Press, 2006.
781	CHM	PRINCIPLES OF INORGANIC CHEMISTRY	CHM649A	Principles of modern inorganic chemistry discussion of the chemistry of nontransition elements, Coordination Chemistry, organometallic chemistry, Inorganic chemistry of biological systems.
782	MSE	ELECTRICAL AND MAGNETIC PROPERTIES OF CERAMIC MATERIALS	MSE663	Structure of oxides: Ionic diffusion in oxides: Defect structure of nonstoichiometric compounds: Conductivity dependence on partial pressure of oxygen: Macroscopic characterization of dielectric materials: Ferroelectrics: Diamagnetism, paramagnetism and ferromagnetism, exchange ferromagnetic domain: Structure and properties of ferrites.
783	MSE	MODERN TRENDS IN METAL FORMING PROCESSES	MSE655	Limitation of conventional metal forming methods: Powder rolling and its various variants, spray rolling, direct strip process: Powder, spray, rotary and isothermal forging: Hydrostatic and powder extrusion.
784	MSE	FUNDAMENTALS OF STEREOLOGY & APPLICATIONS TO MICROSTRUCTURAL ANALYSIS	MSE650A	Concepts and language of stereology; geometrical probability; fundamental operations in stereology; averaging with respect to orientation; basic stereological parameters on true 2D sections and thick sections and transmission electron micrographs; numerical density and size distribution of particles and grains of various shapes and sizes; stereological analysis of anisotropic microstructures; fractal description of polycrystalline and other structures and fracture surfaces.
785	MSE	FUNDAMENTALS OF STEREOLOGY & APPLICATIONS TO MICROSTRUCTURAL ANALYSIS	MSE650	Concepts and language of stereology; geometrical probability; fundamental operations in stereology; averaging with respect to orientation; basic stereological parameters on true 2D sections and thick sections and transmission electron micrographs; numerical density and size distribution of particles and grains of various shapes and sizes; stereological analysis of anisotropic microstructures; fractal description of polycrystalline and other structures and fracture surfaces.
786	MSE	X-RAY CRYSTALLOGRAPHY - I	MSE646	Elemental compound and alloy crystals, modes of bonding, crystal types, density of packing, atomic stacking, inter atomic voids, coordination polyhedra, Pauling's rules, symmetry elements, space and point groups.
787	MSE	ADVANCES IN IRON AND STEEL MAKING	MSE630	Recent trends in iron and steel making: Gas solid and slag metal reaction: Sponge iron making: Continuous steel making: Continuous casting: Vacuum degassing and electroslag remelting: Advances in alloying and impurity control: Emphasis on application of physical chemistry and transport phenomena.
788	MSE	TRANSPORT PHENOMENA	MSE626	Review of the basic concepts in heat, mass and momentum transfer: Advanced topics in convective heat and mass transfer: Radiative heat transmission: Simultaneous heat and mass transfer: Selected topics in microscale transport.
789	MSE	COMPUTING APPLICATIONS IN METALLURGY	MSE607	Fortran fundamentals: Applications of regression analysis and curve fitting techniques, computer calculations of phase diagrams: Numerical of partial differential equations pertinent to heat, mass and momentum transfer.
790	MSE	NON EQUILIBRIUM PROCESSING OF MATERIALS	MSE603	Introduction to nonequilibrium processing Thermodynamics and kinetics of metastable phase formation ; Rapid solidification : Undercooling. Phase diagram metastable states, Methods of rapid solidification mechanical alloying, Mechanism of alloying Energy criteria for mechanical alloying, Synthesis of nonequilibrium phases, Application of mechanical alloying, Metallic glass : Understanding of glass formation and properties, glass, application, Special nonequilibrium processing and phase transformations
791	AE	ADVANCED COMPUTATIONAL METHODS IN CFD	AE615A	Main issues of space time resolution: Computing time averaged and unsteady problems. Discretization with operators. Problems in physical and transformed plane: Jacobians and flux vector splitting. Generalized aerodynamic problems. Spectral tools of analysis for discrete schemes: FDM, FVM & FEM. High order and high accuracy schemes of FDMs and FVMs. Design of Dispersion Relation Preservation schemes for high order upwinding. Computing equations with discontinuous solutions and Gibbs phenomenon. Applications to incompressible viscous and compressible flows. DNS of turbulence and acoustic problems.
792	EE	SIMULATION OF COMMUNICATION SYSTEMS	EE669	Introduction to simulation. Discrete time signals and systems. Modeling linear time invariant system. Modeling linear time varying system. Modeling memoryless nonlinear systems. Modeling nonlinear systems. Simulation and random number generation. Testing of random number generators. Modeling of functional blocks in communication systems. Wireless fading channel models. Discrete Markov fading channels. Other performance estimation methods. Variance reduction techniques. Simulation optimization. Case Study. Course Reference : 1. P. Balaban, K.S. Shanmugan, and B.W. Stuck, eds., Computer Aided Modeling, Analysis, and Design of Communication Systems, vol. 6, IEEE Journal on Selected Areas in Communications, Jan. 1988. 2. P. Balaban, K.S. Shanmugan, and B.W. Stuck, eds., Computer Aided Modeling, Analysis, and Design of Communication Systems II, vol. 6, IEEE Journal on Selected Areas in Communications, Jan. 1988. 3. J.K. Townsend, A.F. Elrefaie, H. Meyr, and M. Pent, eds., Computer Aided Modeling, Analysis, and Design of Communication Systems III, vol. 6, IEEE Journal on Selected Areas in Communications, Jan. 1988.

S.No.	Branch	Course Title	Courses ID	Content
				Communications Apr. 1993. 4. K.S. Shanmugan, eds., Simulation and Implementation of Communication and Signal Processing Systems, vol. 32, IEEE Communication Magazine, July 1994.
793	AE	ADVANCED COMPUTATIONAL FLUID MECHANICS	AE605	Issues of spacetime Resolution, Computing timeaveraged unsteady problem, Defference type of high Modeling RANS, URANS, LES, DES, DNS, Generalized transformation: Orthogonal/Nonorthogonal grid, 1. Computational Fluid Dynamics : Charles Hirsch, Wiley, Chichester, U.K. (1990)2. Computational Fluid Dynamics and heat transfer: J.C. Tannehill, D.A. Anderson and R.H. Fletcher, Taylor and Francis (199
794	AE	ADVANCED COMPUTATIONAL FLUID MECHANICS	AE605A	Issues of spacetime Resolution, Computing timeaveraged unsteady problem, Defference type of high Modeling RANS, URANS, LES, DES, DNS, Generalized transformation: Orthogonal/Nonorthogonal grid, 1. Computational Fluid Dynamics : Charles Hirsch, Wiley, Chichester, U.K. (1990)2. Computational Fluid Dynamics and heat transfer: J.C. Tannehill, D.A. Anderson and R.H. Fletcher, Taylor and Francis (199
795	AE	INTRODUCTION TO SCIENTIFIC COMPUTING	AE603A	Basics of Computing & Discretization, Errors: Different types of error, Interpolation and extrapolation, Root finding: Polynomials; NewtonRaphson Method, Secant Method,ODE and their computations, Stiff eigenvector 1. Computational Fluid Dynamics : Charles Hirsch, Wiley, Chichester, U.K. (1990)2. Computational Fluid Dynamics and heat transfer: J.C. Tannehill, D.A. Anderson and R.H. Fletcher, Taylor and Francis (199
796	AE	INTRODUCTION TO SCIENTIFIC COMPUTING	AE603	Basics of Computing & Discretization, Errors: Different types of error, Interpolation and extrapolation, Root finding: Polynomials; NewtonRaphson Method, Secant Method,ODE and their computations, Stiff eigenvalue/eigenvector 1. Computational Fluid Dynamics : Charles Hirsch, Wiley, Chichester, U.K. (1990)2. Computational Fluid Dynamics and heat transfer: J.C. Tannehill, D.A. Anderson and R.H. Fletcher, Taylor and Francis (199
797	MSE	DISLOCATIONS AND PLASTICITY	MSE658A	Overview of defects in Materials: (point, line, planar and volume defects) and their classification. Overview of plastic deformation mechanisms. Point defects: interaction and distributions, statistical therm the Volterra cut, Burgers vector and the Burgers circuit, the line vector, edge, screw and mixed locations, Role of dislocations in weakening the crystal and in plasticity. Elasticity theory of dislocations: Stre between dislocations, Core of a dislocation. Motion of dislocations: The Peierls stress, role of the core structure, interaction of dislocations with other defects (including yield point phenomenon); kinks; jo stressstrain curve. Dislocations in FCC Metals: Partial dislocations (Shockley and Frank partials) stacking faults, Thompson's tetrahedron, LomerCottrell sessile dislocation. Overview of dislocations i multiplication of dislocations: dislocations in freshly grown crystals, nucleation of dislocations, multiplication of dislocations (by FrankRead sources, cross slip and climb), Grain boundary sources, Recove indentation, interfacial dislocations, Twinning including incoherent twins. Specific examples of role of dislocations and case studies:Dislocations in nanocrystals, The HallPetch relation and the Inverse Ha and Fracture. 1. Introduction to Dislocations, D. Hull and D.J. Bacon, Pergamon Press, Oxford, 1984.2. Theory of Dislocations, J. P. Hirth and J. Lothe, McGrawHill, New York, 1968.3. Cry Weertman and J. Weertman, The MacMillian Company, New York, 1964.5. http://www.tf.unikel.de/matwis/amat/def_en/
798	MSE	DISLOCATIONS AND PLASTICITY	MSE658	Overview of defects in Materials: (point, line, planar and volume defects) and their classification. Overview of plastic deformation mechanisms. Point defects: interaction and distributions, statistical therm the Volterra cut, Burgers vector and the Burgers circuit, the line vector, edge, screw and mixed locations, Role of dislocations in weakening the crystal and in plasticity. Elasticity theory of dislocations: Stre between dislocations, Core of a dislocation. Motion of dislocations: The Peierls stress, role of the core structure, interaction of dislocations with other defects (including yield point phenomenon); kinks; jo stressstrain curve. Dislocations in FCC Metals: Partial dislocations (Shockley and Frank partials) stacking faults, Thompson's tetrahedron, LomerCottrell sessile dislocation. Overview of dislocations i multiplication of dislocations: dislocations in freshly grown crystals, nucleation of dislocations, multiplication of dislocations (by FrankRead sources, cross slip and climb), Grain boundary sources, Recove indentation, interfacial dislocations, Twinning including incoherent twins. Specific examples of role of dislocations and case studies:Dislocations in nanocrystals, The HallPetch relation and the Inverse Ha and Fracture. 1. Introduction to Dislocations, D. Hull and D.J. Bacon, Pergamon Press, Oxford, 1984.2. Theory of Dislocations, J. P. Hirth and J. Lothe, McGrawHill, New York, 1968.3. Cry Weertman and J. Weertman, The MacMillian Company, New York, 1964.5. http://www.tf.unikel.de/matwis/amat/def_en/
799	PHY	INTRODUCTION TO THE DEPARTMENT	PHY400A	The course will expose the students to research areas being pursued in theDepartment, and issues relevant to research as a profession. Faculty membersfrom different subdisciplines would deliver lecture to be given. 21-JUL-2014
800	HSS	UNDERSTANDING ORGANIZATIONAL BEHAVIOUR	PSY786	A coherent introduction to organizational psychology, historically takingoff from industrial psychology and human relations movement. The coursetakes a look at manin organizations in a social environm 21-JUL-2014
801	CE	GLOBAL NAVIGATION SATELLITE SYSTEMS(GNSS)	CE674A	1.Overview of GNSS and Introduction to GPS, GLONASS, GALILEO, COMPASS, IRNSS systems. 2. GPS: Basic concepts, signal structure and code modulation Pseudorange measurements and navigation p degradation of GPS signals: Selective availability (SA) and Antispoofing (AS), Differential GPS: Space based augmentation systems (e.g., SBAS, GAGAN) and Ground based augmentation systems (e.g., WA GPS measurements. Ambiguity resolution, multi path and other observational errors, Doppler effect on GPS signals, Code and Phase combinations for Ionosphere free, Geometry free, Multipath reduction, Kalman filtering and adjustment computation for GPS. 6. Surveying with GNSS: Point positioning, Relative positioning, Static and Kinematic positioning. Planning and field observations, Networking, Data l 1. Diggelen, Frank van, 2009, AGPS: Assisted GPS, GNSS, and SBAS, Artech House: Boston, pp: 400. 2. Guochang, X., 2007. GPS: Theory, Algorithms and Applications, Springer: Berlin, pp. 355. 3. Hafmann, 382. 4. Hafmann, B., Lichtenegger H. and Wasle, E., 2008, GNSS Global Navigation Satelite Systems: GPS, GLONASS, Galileo, and more, Springer: Berlin, pp.546. 5. International Committee on Global Navig Augmentations Systems, United Nations: NY, pp.70.
802	AE	PRINCIPLES OF ACOUSTICSV	AE752A	Wave theory of sound: Plane waves, Harmonic waves and complex Algebra, Speed ofsound, Energy, Intensity and Power, Spherical wavesb. Quantitative measurement of sound: Frequency content and ba Propagation of Plane waves: Reflection from a rigid surface, Propagation in a tube,Radiation due to waves on the wall, Oblique reflection and transmission at a planarinterfaced. Radiation from Vibrating B
803	MSE	SEMINAR PRESENTATION	MSE691A	Seminar Participation
804	CHM	MATHEMATICS FOR CHEMISTRY	CHM521A	Preparations of various organic compounds employing different reactions willbe carried out, with a view to give the student sufficient training in syntheticorganic chemistry. 21-JUL-2014
805	CHM	INORGANIC CHEMISTRY I	CHM441A	Principles of modern inorganic chemistry, discussion of the chemistry of nontransitionelements.
806	HSS	POSTCOLONIAL LITERATURE	ENG438	This course will unpack the many meanings of the term 'postcolonial' through a study of literature that has emerged from the colonial encounter in various parts of the world.Focused mainly on from Africa, the Caribbean, and India, in addition to British literature itself, to understand how such literature deals with and/or resists the experience and legacy of colonialism. In addition, the course will e within current global relations between the North and the South. l) Loomba, Ania. Colonialism/Postcolonialism. New York: Routledge, 1998. 2) The Empire Writes Back: Theory and Practice in PostColonial Literatures. Edited byBill Ashcroft, Gareth Griffiths, Helen Tiffin. l
807	EE	SPECIAL TOPICS IN MICROELECTRONICS	EE698D	Solar radiations, solar PV modules, maximum power point tracking, partial shading,leakage currents, grounding techniques, panel optimizers, grid integration issues,module integrated inverters, standalon
808	CE	M.TECH SEMINAR 2	CE698A	Components of a pavement structure. Experimental characterization of pavementmaterials bituminous mix, aggregates, subgrade, cemented material. Materialmodeling viscoelastic, viscoplastic behavior dynamic analysis. Fatigue and rutting modeling and calibration. Temperaturestresses in pavements. Estimation of cumulative damage. 21-JUL-2014
809	CE	PHYSICAL GEODESY	CE678	Geometric geodesy: DatumsHorizontal & Vertical, GRS80, WGS84, ITRF. Transformation of Coordinates from one datum to another. Mean Sea level, Geoid and MSL in India. Coordinate Systems in Geodesy Astrogeodetic deflection.Physical geodesy: Fundamentals of potential theory, Laplace Equation in spherical coordinates, Solution of Laplace Equation, Brun's formula, Fundamental equation of Phys

S.No.	Branch	Course Title	Courses ID	Content
				Anomaly, Gravity reductions: Free Air, Bouger and Isostatic reduction, Earth Gravity Models, Potential number and different height systems.Map Projection: Introduction to Map projection, Purpose and me Indian Grid System and UTM.Astronomy: Celestial Sphere, Definition of terms in Astronomy, Celestial coordinate systems, Variations in Celestial coordinates. Precession and Nutation. Time systems Sider parameters and Leap second.Satellite geodesy: Introduction to Satellite Geodesy, Keplerian Motion, Geometry of ellipse, Kepler ellipse in space, Perturbed satellite motion, Lagrange’s and Gaussian f
810	EE	SIMULATION OF COMMUNICATION SYSTEMS	EE669A	Introduction to simulation. Discrete time signals and systems. Modeling linear time invariant system. Modeling linear time varying system. Modeling memoryless nonlinear systems. Modeling nonlinear sy simulation and randon number generation. Testing of random number generators. Modeling of functional blocks in communication systems. Wireless fading channel models. Discrete Markov fading char sampling. Other performance estimation methods. Variance reduction techniques. Simulation optimization. Case Study. 1. P. Balaban, K.S. Shanmugan, and B.W. Stuck, eds., ComputerAided Modeling, Analysis, and Design of Communication Systems, vol. 6, IEEE Journal on Selected Areas in Communications, Jan. 1984. 2. F Analysis, and Design of Communication Systems II, vol. 6, IEEE Journal on Selected Areas in Communications, Jan. 1988. 3. J.K. Townsend, A.F. Elrefaic, H. Meyr, and M. Pent, eds., ComputerAided Modeli 4. K.S. Shanmugan, eds., Simulation and Implementation of Communication and Signal Processing Systems, vol. 32, IEEE Communication Magazine, July 1994.
811	PHY	CORRELATED ELECTRONS AND QUANTUM MAGNETISM	PHY690N	
812	PSE	INTRODUCTION TO PHOTONICS	PSE601A	Photonics deals with light generation, amplification, guiding, manipulation, and detection forharvesting information. This course introduces some of the fundamental aspects of photonicsexcluding genera Gaussian Beams, Absorption and DispersionSpatial and Temporal CoherenceBoundary conditions, Fresnels equations and coefficients, Brewster and critical angles, Course Reference : * E. Hecht, Optics, 4th ed., Pearson Education (2001).* G. R. Fowles, Introduction to Modern Optics, Dover (1989).* Smith, F.G. & King, T.A. Optics and Photonics: An introduction, John W
813	CHM	ORGANIC CHEMISTRY I	CHM401A	Stereochemistry, Dynamic stereochemistry, Mechanistic and Stereochemicalaspects, Reactive Intermediates: Carbenes, Nitrenes, Radicals, Carbocations. 21-JUL-2014
814	EE	CODING THEORY	EE668A	1.Introduction: Types of codes, channel models, maximum likelihood decoding, Shannon’s noisy channel coding theorem, FEC, ARQ, HARQ. 2. Linear Blaock Codes : Generator matrix, parity check ma Williams theorem. Examples of simple linear block codes. 3. Some linear block codes : Construction, properties and decoding of some popular block codes; Hamming codes, ReedMuller codes. 4. Bonds c bounds. 5. New codes from old codes : Extending a code, puncturing a code, expunging a code, augmenting a code, shortening a code, direct sum construction. Course Reference : 1. Shu Lin and Daniel J. Costello, Jr., "Error Control Coding", second ed., Prentice Hall, 2004. 2. F.J. Mac Williams, N.J.A. Sloance, "The Theory of ErrorCorrecting Codes&# Cambridge Univ. Press 2003. 4. Todd K. Moon, "Error Correction Coding", 1st Ed., WileyInterscience, 2006. 5. Cary W. Huffman, Vera Pless, "Fundamentals of ErrorCorrecting Codes", 1s Richardson and Rudiger Urbanke, "Modern Coding Theory", Cambridge Univ. Press, 2008.
815	ES	GEOCHEMISTRY	ES642A	Course outline, grading policy, introduction to geochemical cycles, periodic table, Origin of the universe and Earth, Earths history from its creation to Anthropocene, The Basics: Nuclear (in) stability, isotop Thermodynamics, Radioactive decay, principles of geochronology and isochrons, radiogenic isotopes as geochronometer and process tracers, mixing and modeling calculations, RbSr, SmNd, UThPb and d assignments2 & 3, as well as reading material, Fractionation of stable isotopes, H, O, C, Discussion on assignment4 and reading material, Mass spectrometry: Measurement principles and methodology, te Principles and Applications of Geochemistry by Gunter Faure (2nd edition, Prentice Hall)Geochemistry: pathways and Processes by McSween, Richardson, Uhle (Columbia University Press)Geochemistry (McGrawHill)
816	ES	IGNEOUS AND METAMORPHIC PETROLOGY	ES652	This course will focus on the processes and principles involved in the generation of Earth materials, in particular, igneous and metamorphic rocks. Students will gain a broad overview of the Earth as dyna emphasis on developing skills necessary to understand and evaluate melt generation and crystallization, differentiation and chemical evolution of magma, and metamorphic processes etc. Topics to be d Thermodynamic evaluation of phase diagrams; Phase Rule and One Component System; Phase Diagrams for Binary Systems (Solid solution, Eutectic, and Peritectic systems) such as ForsteriteFayalite, A (AnorthiteDiopsideForsterite, and AnorthiteForsteritesilica); Effect of P, T & fluids on Melting; Chemical Petrology: Major and minor element, and isotopic compositional variations; graphical and mathemat (REE) patterns and modeling source magma composition; Generation and diversification of magmas; MidOcean Ridge Volcanism; Mantle Plumes and Ocean Island Basalts; Subduction related Arc magm assemblagesand chemographic (ACF, AKF, and AFM) diagrams; Metamorphic Facies; Metamorphic Reactions. Introduction to Igneous and Metamorphic Petrology by John D. Winter (Prentice Hall, 2nd edition)Recommended readings:1. The Interpretation of Igneous Rocks by Cox, Bell and Pankhurst (Chapman and Press; 2nd edition).3. Petrology: Igneous, Sedimentary and Metamorphic by Blatt, Tracy, and Owens (Freeman and Company)
817	HSS	ENGLISH LANGUAGE AND COMPOSITION	ENG112N	This course imparts training in the use of English language for communicativepurposes, and aims to develop reading comprehension, writing, listening, andspoken language skills of the student. The Lang tutorial groups for effective individual attention.
818	ECO	EMPIRICAL METHODS IN APPLIED MICROECONOMICS	ECO541	Correlation vs Causality, Omitted Variables, Measurement Error, Randomized Experiments, Introduction to the course; Random Variable, Probability Review2. Observational Studies and Regression, Overvie Designs , Applications5. QuasiExperimental Designs, Applications4. Selection on Observables: Matching and Propensity Scores, Applications4. Selection on Observables: Fixed Effects, PrePost Designs a Applications4. Recent Developments in the field: Discussion of papers on new methods and applications5. Joshua D. Angrist, JornSteffen Pischke, Mostly Harmless Econometrics: An Empiricist’s Companion, Princeton University Press, 2008A. Colin Cameron, Pravin K. Trivedi , Microeconometrics: Method Quarterly Journal of Economics, Journal of Labor Economics, Journal of development Economics etc. 18-SEP-14
819	HSS	METHODS OF SOCIOLOGICAL RESEARCH	SOC751	Relation between theory and method and the concept of paradigmQuantitative Research Methods: Experimental design and surveys; different types of experimental designs; longitudinal studies; sampling sources and their use in the historical and documentary study of Indian society and culture: Census, Archival material, National Sample Survey, Gazetteers and District Handbooks and other large data set Brewer, Jonhn and Albert Hunter. Multimethod Research, Sage, 1989.Clifford, James and G. Marcus, Writing Cultures: Poetics and Politics of Ethnography, University of California Press, 1986. Cochran, Wil pp. 145294)Epstein, A. L. (ed) The Craft of Social Anthropology, Tavistock, 1967. Flick, Uwe. An Introduction to Qualitative Research, Sage, 2002.Geertz, C. The Interpretation of Cultures. Basic Books, 1963 Tests, Sage,1993. LeviStrauss, C. Structural Anthropology. Basic Books 1963. (Part 1 pp. 3791)Luckacs, George. History and Class Consciousness. Merlin Press, 1971. (Chapter 1)Margolis, Eric (ed), The S SEP-14
820	CHE	UG PROJECT (UGP-I)	CHE349A	UG PROJECT (UGPI)
821	EE	INDUSTRIAL AUTOMATION AND CONTROL	EE617	
822	HSS	HEALTH AND ILLNES: PSYCHO-SOCIAL PERSPECTIVES	HSS702A	1.Health and illness: biomedical and sociopsychological models2.Social cognitive models of health3.Culture and health; cross cultural and comparative research in mental health4.Illness, distress and sel health7.Stress and coping8.Personal and social meanings of illness; Illness narratives and cultural construction of illness9.Postmodernity, health policy and its impact on peoples health10.Major national approaches12.National health policy in India and issues for research; Caregiving in the Indian context The main readings are various journal articles and survey reports and policy documents that are available online. Reference books include (i) Margaret Konz Snooks, Health Psychology: Biological, Psych Illness, and Healing: A Blueprint for the 21st Century. (iii) K. Charmaz and D. A. Paterniti, Health. Illness and Healing Social Context and Self. (iv) David Wainwright (ed.), A Sociology of Health. (v) Poul Roh Sociology of Health and Illness. (viii) L. D. Cameron and H. Leventhal, The Selfregulation of Health and Illness Behaviour. (ix) A. Kleinman, The illness narratives: Suffering, healing and the human condition
823	HSS	HEALTH AND ILLNES: PSYCHO-SOCIAL PERSPECTIVES	HSS702	1.Health and illness: biomedical and sociopsychological models2.Social cognitive models of health3.Culture and health; cross cultural and comparative research in mental health4.Illness, distress and sel health7.Stress and coping8.Personal and social meanings of illness; Illness narratives and cultural construction of illness9.Postmodernity, health policy and its impact on peoples health10.Major national approaches12.National health policy in India and issues for research; Caregiving in the Indian context

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				The main readings are various journal articles and survey reports and policy documents that are available online. Reference books include (i) Margaret Konz Snooks, Health Psychology: Biological, Psychological, and Sociological Aspects of Health and Illness, and Healing: A Blueprint for the 21st Century. (iii) K. Charmaz and D. A. Paterniti, Health, Illness and Healing Social Context and Self. (iv) David Wainwright (ed.), A Sociology of Health. (v) Poul Rohlf, The Sociology of Health and Illness. (viii) L. D. Cameron and H. Leventhal, The Selfregulation of Health and Illness Behaviour. (ix) A. Kleinman, The illness narratives: Suffering, healing and the human condition of illness.
824	DES	DESIGN PRACTICE I	DES602A	Stages of a Product and Concurrent Engineering; Problem Formulation, Specificationsand Constraints; Creating Forms; Configuration Optimisation; Coupled, Decoupledand Uncoupled designs; Product of Design; Design for Manufacturing,Addressing Failures and Courage to Create; Interpersonal Skills; Robust Design,Incubation; Economic Considerations; Micro and Macro Designs; Introduction toElectronics; Laboratory on Problem Solving and Design; Journals Review.
825	CE	GEOSPATIAL DATA PROCESSING	CE677A	Geodetic reference systems: ICRF and ITRF, Geodetic datums, Earth ellipsoid;basic geometric geodesy; Coordinate systems and transformation; Map projections,geoid and geoidal heights and undulation; Gravity anomalies, cofactors,correlation and covariance, propagation of errors and variancecovariance; Leastsquares adjustment computations; Sequential processing and Kalman Filtering;Variancecovariance of adjusted coordinates; Error analysis of measurements;Models for single point positioning and relative positioning using code and phasedata; Methods of interpolation; Geostatistical tools: variogram and krigging. 21-JUL-2014
826	CE	ADVANCED GEOTECHNICAL ENGINEERING	CE631A	Soil composition and soil structure, Steady State flow, 2D and 3D seepage, Transientflow; Compressibility and Rate of consolidation, One, two, and three dimensionalconsolidation theories; Shear strength and failure envelopes; Soil strength and failure envelopes; Soil strength and failure envelopes; Soil strength and failure envelopes.
827	CE	INTRODUCTION TO REMOTE SENSING	CE671A	Remote sensing system; Physics of remote sensing, EMR characteristics andinteraction in atmosphere and with ground objects; Sensor types characteristics:types of resolution, FOV, IFOV, PSF; RS satellite systems; Data processing: Georeferencing, resampling methods; Atmospheric errors andremoval; Satellite orbits and characteristics; Applications of optical and microwaveremote sensing techniques in Civil Engineering.
828	BSBE	BIOPHARMACEUTICALS	BSE616A	
829	BSBE	PROFESSIONAL COMMUNICATION	BSE601A	1. CRITICAL READING : Context, How is the text argued, Central claims, Kinds of reasoning, Examine Evidence, Examine Interpretations, Evaluations,Limitation of study2. CRITICAL WRITING : Learning to write a research paper, Books, Proposals,Patents.3.ORAL PRESENTATION : Conference, Scientific, Teaching, Popular, 4. ETHICS : Experimentation, Writing No prescribed textbook.
830	MTH	TECHNICAL COMMUNICATION SKILLS	MTH399A	Technical Communication
831	MSE	SEMINAR PARTICIPATION	MSE698	SEMINAR PARTICIPATION
832	MSE	MANUFACTURING PROCESSES LABORATORY	MSE315A	1Deformation Behavior of Metals during Rolling and study of the associated microstructural changes2Fluidity Measurement during Casting3Permanent mold casting and casting defect evaluation4Effect of casting conditions on the properties of castings5Study Various Characteristics of Metal Powders and Evaluate the effect of particle size and shape on the green density, apparent density and green strength of coldcompacted powders7Conventional and die casting8Microstructural Analysis of Castings9Structural NanoMaterials through ECAP10Spray Forming of Alloys and MMCs11MicroExtrusion of Alloys12Injection molding of thermoplastic polymers (e.g. PE, PP)133Dimensional Printing (3DP)
833	BSBE	UG PROJECT (UGP-III)	BSE498A	UG PROJECT (UGPIII)
834	AE	ADVANCES IN WIND ENERGY CONVERSION	AE629A	Earths atmosphere & rotation, Coriolis force, geostrophic winds. Terrain, mettheories, measuring technique. Wind power site. Atmosph BL. Aerodyn in windpower. Sails, airscrews. Vertical & horizontal axis windmills.
835	CSE	MACHINE LEARNING FOR COMPUTER VISION	CS679A	1. Probability basics and common probability distributions. 2. Fitting Probability models (ML, MAP, Bayesian). 3. Normal distribution. 4. Regression 5. Classification. 6. Graphical models. 7. Temporal models. Text Book : Computer Vision: Models, Learning and Inference by Simon J.D. Prince.References : Pattern Recognition and Machine Learning by Christopher M. Bishop. Probabilistic Graphical Models: Principles and Techniques by Daphne Koller.
836	IME	INDEPENDENT STUDY	IME797A	
837	ES	EARTH SYSTEM PROCESSES	ES640	Principles: Earth System processes, geomorphic systems, Threshold and equilibrium; scale in geomorphology, key concepts in geomorphology, hydrologic cycle.Processes and Products: Exogenetic and endogenetic processes, Drainage basins and river systems, river morphology and hydrology, fluvial erosion, transport and sedimentation, fluvial depositional landforms, geomorphometric concepts and drainage basin morphometric analysis, permeability, aquifers, chemistry of groundwater.The Atmospheric System: Atmospheric composition and mixing, atmospheric circulation, greenhouse effect.The Ocean and Coastal system: Coastal environment, resulting topographic features, coastal deposition and landforms.Global geomorphology and tectonics: Earths physiography and landscape evolution; Landforms and tectonics of plate margins and plate tectonics; coupled tectonicsurface process models. 1.Summerfield, S.A. (1991) Global Geomorphology, Longman.2.Ernst, W.G. (2000) Earth Systems: Processes and Issues, Cambridge University Press. 3.Richards, K.S. (1982) Rivers forms and processes in landscape evolution, Longman.5.Thronbury, W.D. (1969) Principles of geomorphology, John Wiley & Sons.6.Williams, M.J. (1998) Quaternary Environments, Arnold Publishers7.Reading, H.G. (1996) Sedimentary Environments, W.H. Freeman.
838	MSE	CRYSTALLOGRAPHIC TEXTURE & MECROSTRUCTURAL ENGINEERING	MSE683A	Introduction to crystallographic texture.Refreshers on Xray diffraction: basic diffraction concepts, reciprocal space,instrumentation and geometry.Basics of neutron and synchrotron diffraction and comparison of diffraction figures: experimental details, data processing, indexingDetermination of Orientation Distribution Function and Misorientation.distribution Function: calculation techniques, different notations, 3D and 2DReconstruction of texture from data acquisition and analysis.Mechanisms of evolution of texture during processing: solidification, phasetransformation, deformation, annealing.Modelling texture evolution.Texture evolution and measurement of texture.Functional properties and texture. 1. An Introduction to Texture in Metals; M. Heatherly, W. B. Hutchinson, Monograph no. 5, The Institution of Metallurgist London 1979.2. Introduction to Texture Analysis: Macrotexture, Microtexture, and Orientation Texture and their Effect on Materials Properties; U. F. Kocks, C. N. Tome, H.R. Wenk, Cambridge University Press, 2000.4. Research publications on: Texture manipulation and control; Microstructural engineering ;
839	HSS	COMMONWEALTH LITERATURE	ENG735A	The course covers literature in English in the Commonwealth countries of Africa,Australia, Canada, India, New Zealand, and the West Indies. An attempt willbe made to show that commonwealth writers, who are often called what William Walsh calls other silent but active aims.
840	HSS	INDIVIDUAL PROJECTS	ENG716A	This course is intended to provide the students an opportunity to take up ontheir own a shortterm fieldwork or library project, with constant guidancefrom the Instructor. Evaluation will depend on the final report.
841	CSE	COMPUTATIONAL NUMBER THEORY AND ALGEBRA	CS681A	Elementary operations: the complexity of basic operations like additions,multiplications for integers and polynomials. Polynomials: The complexity offactorization, irreducibility testing, ideal membership and short vector in an integer lattice. Motivating example;polynomial factorization. Integers: The complexity of factorization, primalitytesting, discrete log computation etc for integers. Motivating examples: RSA and Elliptic curve cryptosystemsand integer factoring.
842	MSE	SYMMETRY AND PROPERTIES OF CRYSTALS	MSE638A	Introduction 1 Symmetries in 1 D, 2D and 3D, Examples of patterns showing various, symmetries, Symmetries and Lattices in 2D space : Operations ofTranslation, Rotation and Reflection, standard symbols for symmetry operations, rectangular, square, hexagonalPoint Groups in 2D 4 : Set of symmetry operations, Group Theory Essentials, Evolution of 2D crystallographic point groups, 2D Space Groups (Plane Groups) Glide Planes: common glide planes, symmetry operations, entries in the International Tables, of Crystallography3D Point Groups 4 : Combination of rotation axes in 3D, Development of the 32 point groups, Laue Groups3D Bravais Lattices Addition of a third translation and rotation, Development of the 230 space groups, Understanding the Space Groups entries in the International Tables of Crystallography I. M.J. Buerger, Elementary Crystallography, 1962. 2. International Tables of Crystallography A, International Union of Crystallography3. J. F. Nye, Physical Properties of Crystals (1995), Oxford Science Publications4. Properties of Materials: Anisotropy, Symmetry, Structure", Oxford Pr.
843	MSE	TRANSPORT PHENOMENA	MSE626A	I. Fluid dynamics : Introduction to Transport phenomena in materials processing, Newton's law of viscosity, equation of continuity, Navier Stokes equations, Macroscopic mass and energy balance ; Conduction Heat transfer : Fundamentals of conduction heat transfer; Laws and equations; Steady and unsteady heat, conduction Numerical problems on conductive heat transfer, Fundamentals of convective heat transfer Problems on Convective heat transfer, Fundamentals of Radiation heat transfer and rate laws; view factors, Problems on Radiation heat transfer, Application of heat transfer in: Heat treatment, solidification, Diffusion and Kirkendal's effect, steady and unsteady diffusion, Numericalproblems on diffusion mass transfer, Fundamentals of convective mass transfer; free and forced convective mass transfer t

S.No.	Branch	Course Title	Courses ID	Content
				Application of mass transfer in: case hardening, doping of semi conductors, homogenization, oxidation, absorption/desorption of gases in liquid metals. 1. Transport phenomena: D. R. Geiger and G. H. Poirier2. Transport phenomena: D. R. Gaskel3. Engineering in process metallurgy: R.Guthrie4. Mass transport in solids and fluids: D. S. WilkinsonRecommended Reading: J. Philibert3. Diffusion in solids: field theory, solidstate principles, and applications: M. E. Glicksman
844	MSE	TRANSPORT PHENOMENA	MSE626N	I. Fluid dynamics : Introduction to Transport phenomena in materials processing, Newton's law of viscosity, equation of continuity, Navier Stokes equations, Macroscopic mass and energy balance ; Conduction Heat transfer : Fundamentals of conduction heat transfer; Laws and equations; Steady and unsteady heat, conduction Numerical problems on conductive heat transfer, Fundamentals of convective heat transfer, Problems on Convective heat transfer, Fundamentals of Radiation heat transfer and rate laws; view factors, Problems on Radiation heat transfer, Application of heat transfer in: Heat treatment, solidification, mass diffusion and Kirkendall's effect, steady and unsteady diffusion, Numericalproblems on diffusion mass transfer, Fundamentals of convective mass transfer; free and forced convective mass transfer problems. Application of mass transfer in: case hardening, doping of semi conductors, homogenization, oxidation, absorption/desorption of gases in liquid metals. 1. Transport phenomena: D. R. Geiger and G. H. Poirier2. Transport phenomena: D. R. Gaskel3. Engineering in process metallurgy: R.Guthrie4. Mass transport in solids and fluids: D. S. WilkinsonRecommended Reading: J. Philibert3. Diffusion in solids: field theory, solidstate principles, and applications: M. E. Glicksman
845	MSE	PROCESS ENGINEERING LABORATORY	MSE314A	1Measurement techniques 2Laminar fluid flow 3Macroscopic energy balance (Bernoulli's equation) 4Steady and unsteady conduction heat transfer 5Convective heat transfer 6Radiation heat transfer7Thermal analysis using DSC8Hydrometallurgy II 13Process metallurgy I 14Process metallurgy II 15Electrochemistry
846	MSE	PHYSICAL METALLURGY LABORATORY	MSE311A	1 Metallographic specimen preparation 2 Optical microscopy of illustrative Ferrous samples 3 Optical microscopy of NonFerrous samples 4 Optical microscopy of Ceramic samples: Colour metallography 5 X-ray powder diffraction in materials analysis 7 Study of nucleation and growth in eutectoid steel 8 Carburization of Steel and Hardenability of steel 9 Recovery and Recrystallisation 10 Thermal analysis using DSC
847	MSE	FUNCTIONAL MATERIALS LABORATORY	MSE312A	1Ionic conductivity Conductivity measurement as a function of temperature for different samples2Dielectric and ferroelectric/piezoelectric materials Measurement of dielectric constant Hysteresis loop3Capacitors and voltage4Fabrication of organic light emitting diodes Partial fabrication of organic polymer light emitting diodes Characterization of an OLED5Magnetic materials Magnetoresistance MH curves6Semiconductors and characterization Fabrication of organic photovoltaic cells (PV) cells Characterization of solar cells9Processing of biomaterials Fabricate biomaterials using(i) compression molding (for entire component) (ii) polymer without modifier for comparison purpose. Comparison with metallic substrates (demo).11Biomimetics/ Surface modification22Effect of surface energy on wetting of surfaces Sample calculation of wettability of surface.12Effect of surface modification on Bacteria/ Cell growth Effect of surface modification/functionalization characterized by comparing cell growth/proliferation on treated versus pristine surfaces.13 chemistry (hydrophobic/ hydrophilic and hydrophobic/ hydrophilic + modifier)(ii) Role of surface roughness
848	CHM	ATOMS MOLECULES AND PHOTONS	CSO202A	1. Landmark experiments in physical chemistry2. Interplay of theory and experiments in modern physical chemistry3. Structure of atoms and molecules4. Dynamics of atoms and molecules5. Structure and properties of molecules 1. R. S. Berry, S. A. Rice and J. Ross, Physical Chemistry2. D. A. McQuarrie, J. D. Simon, Physical Chemistry: A molecular approach3. K. J. Laidler, The World of Physical Chemistry4. C. E. Dykstra, Physical Chemistry
849	CHM	ATOMS MOLECULES AND PHOTONS	CSO202	1. Landmark experiments in physical chemistry2. Interplay of theory and experiments in modern physical chemistry3. Structure of atoms and molecules4. Dynamics of atoms and molecules5. Structure and properties of molecules 1. R. S. Berry, S. A. Rice and J. Ross, Physical Chemistry2. D. A. McQuarrie, J. D. Simon, Physical Chemistry: A molecular approach3. K. J. Laidler, The World of Physical Chemistry4. C. E. Dykstra, Physical Chemistry
850	CHM	ORGANIC CHEMISTRY: FUNDAMENTALS AND APPLICATIONS	CSO201A	Basic Organic Chemistry Concepts: introduction to organic molecules and functional groups,understanding organic reactions, stereochemistry and carboncarbon bond forming reactions in organic synthesis, dyes and flavors (3 Lectures)Green Chemistry introduction, principles, sustainability, atom economy, some green initiatives, management of resources and its effect on health and environment. Catalysis and drug targets and their inhibitors as model inhibitors. (4 Lectures)Solid phase synthesis and strategies for futuristic designs in organic chemistry.Photochemistry: simple concepts and applications (semiconductors, photochemistry,(4 Lectures)Organic Materials: polymers (biodegradable polymers, conducting polymers, etc.), smart materials,OLEDs, intelligent gels, dyes, etc 1. Organic Chemistry: Structure, Mechanism,and Synthesis by R. J. Ouellette and J. D. Rawn2.Green Organic Chemistry: Strategies, Tools, and Laboratory Experiments by K. Doxsee and J.Hutchinson3. Biocatalysis: strategies and applications by P. H. Toy and Y. Lam5. General Organic Chemistry, Janice Smith, McGrawHill, New York, USA, 20116. Organic Chemistry, W. H. Brown, C. S. Foote, B. L. Iverson and E. V. Anslyn, Wiley, Chichester, UK., 2009.8. Chemistry of New Materials, FactsOn File, Inc. New York, USA, 2007.9. Love Drugs, Otto Snow, Thoth Press, USA, 2005.10. Molecules of Death, R. H. Waring, G. B. Steventon, S. C. M. Lea, Wiley, Chichester, UK., 2009.
851	MSE	PHASE TRANSFORMATIONS	MSE301A	1. Introduction 2. Gibbs free energy change calculations 3. Interfaces 4. Nucleation 5. Growth 6. Transformation kinetics 7. Precipitation 8. Recrystallisation and grain growth 9. Martensitic Transformation 1.Phase Transformations in Materials by R. C. Sharma, CBS Publishers, New Delhi 2. Solid State Transformations by V. Raghavan, PrenticeHall of India, New Delhi
852	MSE	NANOMATERIALS: PROCESSING AND PROPERTIES	MSE688	Definition and Classification of Nanomaterials, Fundamental Properties of variousprimary material classes (Metals, ceramics and Polymers), Size dependentproperties and various characterization techniques, Synthesis of nanomaterials to produce nanosized precursor powders, Various routes to produce Nanometallicalloys (Rapid solidification), Challenges in processing bulk ceramic nanomaterials,Various densification routes for nanoceramics, MechanicalProperties, Thermal properties, Tribological Properties, Biological Properties(Biomedical applications), Applications of bulk nanomaterials, Critical issuesrelated to understanding properties of nanomaterials
853	EE	NEURAL NETWORKS	EE671A	Theory of representation; Two computational paradigms; Multilayer networks;Autoassociative and heteroassociative nets; Learning in neural nets: Supervisedand unsupervised learning; Application of neural networks
854	EE	MICROWAVE CIRCUITS	EE648A	Transmission lines for microwave circuits; waveguides, stripline, microstrip, slotline; microwave circuit design principles; passive circuits; impedance transformers,filters, hybrids, isolators etc., active circuits
855	CE	STABILITY OF STRUCTURES	CE622A	Criteria for design of structures: stability, strength, and stiffness; Classicalconcept of stability; Stability of discrete systems: linear and nonlinear behaviour;Stability of continuous systems: stability of columns, plates, shells, and beams; member buckling versus global buckling, slenderness ratio of framemembers; Stability of beams: lateraltorsion buckling; Stability of plates: axialflexural buckling, shear flexural buckling, buckling under combined loading
856	PHY	SOFT MATTER PHYSICS	PHY616	Introduction to Soft Matter, Forces, energies and timescales in soft matter, Thermodynamic aspects of intermolecular forces, van der Waals force, Hydrophobic and hydrophilic interaction, Interfacial phenomena, hydrodynamic instabilities, Foams and emulsions, Polymers, colloids and Surfactants, Liquid crystals, Selfassembly in soft matter, Experimental tools for soft matter. Books recommended:1. Intermolecular and Surface Forces by Jacob N. Israelachvili (Academic Press, 1998)2. Soft Condensed Matter by R. A. L. Jones (Oxford University Press, 2002)3. Principles of Condensed Matter Physics by N. M. Johnson (Oxford University Press, 2002)4. Hydrodynamic stability by S. Chandrashekhar (Oxford University Press, 1981)5. Structured Fluids by Thomas A. Witten (Oxford University Press, 2004)6. Structure and Rheology of Complex Fluids by Ron Larson (Oxford University Press, 2003)
857	AE	THERMAL TURBOMACHINERY	AE653A	Axial compressors, stage dynamics, degree of reaction, pressure rise limitations,secondary flows, performance: design and offdesign, starting problems, centrifugalcompressors; inlet flow, slip, sweep, diffuser
858	CE	STATISTICAL ANALYSIS FOR CIVIL ENGINEERS	CE601A	1. Basic Concept of Probability and Distributions2. Experimental Error and their Characteristics(a) Random error(b) Bias(c) Propagation of random error(d) Error due to roundingoff(e) Calibration3. Adjustment of Observations(a) Sample(a) Sample designs(c) Sample statistics(d) Sampling distribution of mean and variance(e) Question of sample size5. Estimation and Hypothesis Testing(a) Properties of good estimates(b) Interval Estimation(c) Type I and Type II errors(g) One and two tailed tests(h) Tests on mean and variance from samples under different assumptions and knowledge of the underlying distribution6. Regression Analysis and Correlation(a) Simple and multiple regression(b) Detection, effect, and remedy of multicollinearity(d) Detection, effect, and remedy of heteroskedasticity(e) Detection, effect, and remedy of autocorrelation(f) Misspecification errors and regression model building(g) Hypothesis testing on regression coefficients and effect of outliers7. Miscellaneous Topics(a) Fitting theoretical distributions to observed frequency distributions (b) Tests of goodnessoffit (chisquare test, KolmogorovSmirnov test) (c) Identification of outliers 1 . Experimentation and Uncertainty Analysis: Hugh M. Coleman and W.G. Steele2. Handbook of statistical Methods for Scientists and Engineers: H.M. Wadsworth3. Statistical Treatment of Experimental Data: R. A. Fisher
859	CSE	ADVANCED TOPICS IN ALGORITHMS & DATA STRCUTURES	CS647A	The course intends to deal with advanced aspects of algorithm: design andanalysis including data structures, analysis and lower bound proofs, amortizedcomplexity of algorithms. Fibonacci heaps and shortest path problem. Network flows preflowpush algorithms,max flow algorithm, and scaling algorithms. Matching, blossoms, MicaliVaziranialgorithm. Lower bound theory for parallel computations.
860	CSE	INDEXING AND SEARCHING TECHNIQUES IN DATABASE	CS618A	Index structures Btree, Mtree, Vfile, etc., Spacepartitioning versus datapartitioningmethods; Similarity queries Range search, kNN search, Selfjoin;Retrieval techniques Fagin's Algorithm, Threshold algorithms, Wavelets, Fourier transform, etc.; Distance measures Lp norm, Mahalanobis distance, KullbackLeibler divergence measure, Earth Mover'sDistance, etc.; Data compression Wavelets, Fourier, Voptimal
861	EE	COMPACT MODELING	EE681A	Introduction to Simulation. Introduction to SPICE and Compact Modeling. Integrated Resistor Modeling. Integrated MOS Varactor Modeling. MOSFET Modeling approaches : Threshold Voltage based model, Charge based model, and Hybrid model. MOSFET Compact Models and Benchmark Tests. Layout Effects and Parameter Extraction. High Frequency Effects and RF Modeling. SOI MOSFET Modeling. Noise Modeling. Multigate MOSFETs. C. Hu, "Modern Semiconductor Devices for Integrated Circuits,"; Pearson/Prentice Hall, New Jersey.Compac Modeling: Principles, Techniques and Applications by Gennady Gildenblat, Springer. MCMOS Modeling: The EKV Model for LowPower and RF IC Design by Christian C. Enz and Eric A. Vittoz, John Wiley & Sons.BSIM4 and MOSFET Modeling for IC Simulation by W. Liu and C. Hu, World Scientific

S.No.	Branch	Course Title	Courses ID	Content
				24.09.14
862	CE	UNDER GRADUATE RESEARCH -II	CE491A	UNDER GRADUATE RESEARCH I
863	CE	SURVEY AND GEOLOGY CAMP	CE373	Survey Camp: Reconnaissance and establishing the stations; Base line measurements, Triangulation readings on various stations; computation and preparation of triangulation map; contouring; preparation of geological map; formations and structural details; Geomorphic processes Preparation of report.
864	SPEC	EVENING EXERCISE	PE102AA	There are two components in this course, namely: 1. Physical exercise 2. Personality Development Activities. Activities under physical exercise: Warming up exercises: jogging, running, stretching, simple exercises like: 1. Strength 2. Speed 3. Endurance 4. Flexibility 5. Coordinate Activities. Intramural activities: Group sports. Personality development activities: There are five streams in this component: 1. Games and sports, the students choose one of the following 12 sports: 1. Athletics 2. Badminton 3. Cricket 4. Hockey 5. Table Tennis 6. Tennis 7. Football. 8. Swimming 9. Basketball 10. Volleyball 11. Weightlifting 12. Table Tennis
865	STA	RESEARCH	STA800	RESEARCH
866	MSE	INTRODUCTION OF BIOMATERIALS	MSE204A	1. Introduction to Biomaterials 2. Classification of biomaterials 3. Structure/Property correlation 4. Processing and properties of biocompatible materials 5. Surface engineering & case studies * Biomaterials Science: An introduction to Materials in Medicine, Edited by Ratner, Hoffman, Schoen and Lemons, Second Edition: Elsevier Academic Press, 2004. * Biological Performance of Materials: Fundamentals and Applications, Edited by Ratner, Hoffman, Schoen and Lemons, Second Edition: Elsevier Academic Press, 2004.
867	MSE	SOLAR ENERGY TECHNOLOGIES AND MATERIALS	MSE681	1. Introduction to the course. 2. Solar Spectrum. 3. Available solar energy technologies. 4. Solar Thermal Energy Conversion. a. Fundamentals. b. Materials and Technologies. c. Applications. d. Present status and Future outlook Technologies _b. Materials, devices and issues, 1. First generation technologies Si based. u. Second generation technologies (low cost) thin films (aSi, CdTe, CIGS); solar concentrators. m. Third generation (Heterojunction, Quantum Dots, Nanowires, Organic PV) Performance 7. PV Processing with emphasis on migration from solar cells to modules to systems 8. Present status and Future outlook 1. Thin Films Solar Cells, K.L. Chopra, McGraw Hill 2. Physics and Technology of Solar Energy, H. P. Garg, M. Dayal, G. Furlan (1987) a. Volume 1: Solar Thermal Applications b. Volume II: Photovoltaic And Solar Thermal Hybrid Systems c. Volume III: Solar Energy Conversion Nelson 4. Third Generation Photovoltaics: Advanced Solar Energy Conversion (Springer Series in Photonics) by M.A. Green 5. Flexible Solar Cells by Mario Pagliaro, Giovanni Palmisano, and Rosaria Cirrinio 6. Thin Film Solar Cells by Peter Würfel (Paperback April 13, 2009)
868	CE	SPECIAL CONCRETES	CE651A	Production & properties of normal concrete : Introduction to portland cement concrete concrete production operations Indian Standard and ACI Mix design of concrete Fresh and hardened properties of concrete Properties of concrete admixtures and additives in concrete Experimental test parameters and measurements during concrete testing, Special cements : Need Classifications Blended cements, modified hydraulic cements, calcium sulfoaluminate cements, shrinkage compensating (or) expansive cements Other special cements: macrodefect free cements, phosphate cements, expansive cements, fast setting cements, oil well cements Performance and prescriptive specifications classifications. Special concretes: Mass concrete, self compacting or self consolidating concrete, fiber reinforced concrete, high strength concretes, high volume fly ash concretes, geopolymer concrete, roller compacted concrete, ultra high performance concretes and others. Mixture proportioning and parameters in the development of special concretes. Course Reference : [1] Neville, A.M., Properties of concrete, 5th Edition, Pitman Publishers, 1996. [2] Mehta, P.K., and Monteiro P.J.M., Concrete Structure, Properties and Materials, 2nd Edition, Prentice Hall, New Jersey, 1993 [3] Neville, A.M., Concrete Technology, 3rd Edition, Pitman Publishers, 1996 [4] Bapat, J.D., Mineral admixtures in cement and concrete, 2013 [5] High Performance Concrete (Modern Concrete Technology) by Pierre Claude Atcin (Aug 25, 1998) [6] High Performance Concrete: From Materials to Structures, Edited by Pierre Claude Atcin, Marcel Dekker, New York, 1998
869	PSE	NUMERICAL METHODS IN OPTICS	PSE603A	Linear algebra: matrices, matrix inversion; QR, Singular value decomposition, systems of equations, eigenvalues, eigenvectors, orthonormalization, condition number Laplace and Fourier transforms Vector calculus: gradient, divergence, curl, Gauss theorem, Stokes theorem, Helmholtz theorem direct methods for linear algebraic equations; generalized inverses, least squares Numerical differentiation and integration; Numerical solution of 1st and second order ODEs, Runge Kutta method, stability, stiffness Course Reference : * E. Kreyszig, Advanced Engineering mathematics, 8th edition, McGraw Hill, New York, 2000. * M.D. Greenberg, Advanced Engineering Mathematics, Prentice Hall, New Jersey, International Edition, 1998
870	CHE	M. TECH. SEMINAR	CHE701A	M. Tech. Seminar
871	ECO	PUBLIC ECONOMICS AND PUBLIC POLICY	ECO371A	Unbounded Operators, Matrix representation, Self adjointness Criterion, Quadratic Forms, Differential Operators, Self adjoint Extensions, Functional Calculus, Spectra of Self adjoint Operators, Semianalyticity, Perturbation Theory, Uncertainty Principle of Bargmann, Canonical Commutation Relations, Schrodinger representations, Schrodinger Operators, Self adjointness, A Theorem of Kato, Spectral Theory for Schrodinger Operators * N. Akhiezer, I. Glazman, Theory of Linear Operators in Hilbert Space II, Dover, 1961. * J. Blank, P. Exner, M. Havlicek, Hilbert Space Operators in Quantum Physics, Springer, 2008. * T. Kato, Perturbation Theory for Linear Operators, Springer, 1978. * M. Reed and B. Simon, Methods of Modern Mathematical Physics II, Academic Press, 1975.
872	BSBE	MODERN INSTRUMENTAL METHODS IN BIOLOGICAL SCIENCES	BSE611A	1. Applications of spectroscopic and other techniques to the study of biomolecules: UV-Vis spectroscopy, Circular dichroism, Fluorescence, NMR, Mass, IR and Raman spectroscopy, X-ray diffraction. 2. Analytical techniques (e.g. electrophoretic analysis), detection (staining, blotting and immunodetection) specialized applications (in vitro synthesis of protein, labeling and Mass spectrometry), ultracentrifugation. 3. Techniques with applications in molecular biology: Gel electrophoresis, DNA sequencing, PCR, Microarrays, etc. 4. Techniques with applications in cell biology: Autoradiography, Biological Uses of Radioisotopes, Tracer Dilution Technique, Radioimmunoassay 5. Cellular Imaging Techniques: Microscopy: Phase contrast, Fluorescence, Atomic Force and Scanning Tunneling Microscopy
873	MSE	MECHANICAL BEHAVIOUR LABORATORY	MSE313A	1. Determination of tensile properties of different classes of materials 2. Principles of Hardness Testing: comparison of different hardness measurement techniques 3. Impact Testing of Materials: Charpy Impact Test 4. Dislocations by using the etch pitting technique 5. Effect of Work Hardening on Tensile Properties of Metals. 6. Plastic Anisotropy 7. Project
874	MSE	PROFESSIONAL & TECHNICAL COMMUNICATION	MSE300A	* Importance of professional/technical communication: * Domains: Thesis/Report writing, Paper writing, Business letters, job letters, resume, etc. * Plagiarism: Importance, what is it and how to keep a check on it * Selecting contents, highlighting novelty, using visual aids and providing illustrations)
875	MSE	STRUCTURE & CHARACTERIZATION OF MATERIALS	MSE203A	1. Introduction 2. Crystalline State 3. Noncrystalline state 4. Microstructures 5. X-ray Diffraction 6. Electron diffraction 7. Optical microscopy 8. Scanning electron microscope 9. Transmission electron microscope 1. Elements of X-ray diffraction, B.D. Cullity and S.R. Stocks, Addison-Wiley Publishing Co. 2. Introduction to solids, L.V. Azaroff, McGraw-Hill Book Company 3. Elementary Crystallography by M.J. Buerger 4. Introduction to Crystallography, R.J.D. Tilley, John Wiley and Sons, 2006 6. Fundamentals of Materials Sciencethe microstructure-property relationship using metals as model systems, E.J. Mittemeijer, Springer, 2010 7. Microstructural Characterization of Materials, R. W. Cahn, P. Haasen, E. J. Kramer, Cambridge University Press, 1996
876	MSE	RATE PROCESSES	MSE202A	1 Introduction to Fluid flow 2. Introduction to heat transfer 3. Introduction to mass transfer 4. Mass transfer with Chemical reaction 5. Introduction to Heterogeneous Reaction Kinetics 6. Simultaneous Heat and Mass Transfer 1. Engineering in Process Metallurgy: R. Guthrie, Oxford Scientific Publications 2. Transport Phenomena in Metallurgy: G.H. Geiger and D.R. Poirier, TMS publication 3. Kinetic and metallurgical processes: Fundamentals of Metallurgy, R. W. Cahn, P. Haasen, E. J. Kramer, Cambridge University Press, 1996
877	PHY	BACHELOR OF SCIENCE PROJECT -II	PHY556A	BACHELOR OF SCIENCE PROJECT II
878	CHE	GRADUATE SEMINAR	CHE801	Ph. D. Seminar
879	CHM	INORGANIC CHEMISTRY I	CHM345A	1. Representative Chemistry of Main Group Elements (15) (a) Organometallic Chemistry of Lithium and Magnesium: synthesis, structure and reactivity (b) Chemistry of Boron: boranes, bonding in boranes, transition metal complexes supported Lewis acids (c) Chemistry of Aluminum: Aluminum alkyls, use of aluminum alkyls in polymerization of olefins (d) C60 and carbon nanotubes: discovery, preparation and selected reactions 2. Transition Metal Chemistry (10) (a) Multiple bonding in heavier main group elements, unusual compounds of main group elements: (i) Si-Si double bond, Si-Si triple bond, Pb-Pb double bond, Bi-Bi double bond, synthesis, structure and properties (ii) low valent compounds (c) Chemistry of stable N-heterocyclic carbenes, use of carbenes in catalysis (d) Inorganic rings and polymers: cyclo and heterocyclophosphazenes, polysilanes, borazine and boron nitride (a) bonded systems: metal alkyls, aryls and hydrides, stability, preparation and reactivity, metal carbonyls, metal phosphines, metal nitrosyls, metal isocyanides: structures, reactivity and bonding Metal carbonyls (b) bonded systems: metal olefins, alkyls, alkynes, dienes, Cp and Cp*, structure, bonding and reactivity (c) Applications of organometallics in organic synthesis: C-C bond coupling reactions (Heck, Sonogashira, Suzuki, Stille, Negishi, etc.) 1) Elschenbroich, C.; and Salzer, A., Organometallics: A Concise Introduction, 3rd Edn. 1999. 2) Greenwood, N. N.; Earnshaw, A., Chemistry of the Elements, Pergamon Press, 2nd Edn., 2002. 3) Douglas, B.; McVey, R. H. The Organometallic Chemistry of the Transition Metals, 5th Edn., John Wiley and Sons, 2009
880	CHM	ORGANIC QUALITATIVE & QUANTITATIVE ANALYSIS	CHM305A	1. Experimental Techniques (A) Purification of Organic Compounds a. Recrystallisation b. Sublimation at atmospheric pressure and under reduced pressure c. Separation of organic compounds by Steam distillation d. Column Chromatography e. Thin layer chromatography (TLC) and calculation of Rf values f. Column Chromatography: separation of organic mixture g. Preparative TLC: preparation of plates and separation of organic compounds h. Gas chromatography i. Distillation j. Extraction k. Molecular rotation and molecular rotation (D) Spectroscopic Methods a. Preparation of an ester and its confirmation by IR and NMR b. Structure elucidation of unknown compounds based on the given spectral data 2. Organic Chemistry (10) (a) Multiple bonding in heavier main group elements, unusual compounds of main group elements: (i) Si-Si double bond, Si-Si triple bond, Pb-Pb double bond, Bi-Bi double bond, synthesis, structure and properties (ii) low valent compounds (c) Chemistry of stable N-heterocyclic carbenes, use of carbenes in catalysis (d) Inorganic rings and polymers: cyclo and heterocyclophosphazenes, polysilanes, borazine and boron nitride (a) bonded systems: metal alkyls, aryls and hydrides, stability, preparation and reactivity, metal carbonyls, metal phosphines, metal nitrosyls, metal isocyanides: structures, reactivity and bonding Metal carbonyls (b) bonded systems: metal olefins, alkyls, alkynes, dienes, Cp and Cp*, structure, bonding and reactivity (c) Applications of organometallics in organic synthesis: C-C bond coupling reactions (Heck, Sonogashira, Suzuki, Stille, Negishi, etc.) 1) Elschenbroich, C.; and Salzer, A., Organometallics: A Concise Introduction, 3rd Edn. 1999. 2) Greenwood, N. N.; Earnshaw, A., Chemistry of the Elements, Pergamon Press, 2nd Edn., 2002. 3) Douglas, B.; McVey, R. H. The Organometallic Chemistry of the Transition Metals, 5th Edn., John Wiley and Sons, 2009

S.No.	Branch	Course Title	Courses ID	Content
				compound.b. Identification of functional groups in a given organic compound.c. Identification of unknown organic compounds.d. Separation of organic mixture by chemical methods, preparation of deriva
881	CHM	BASIC ORGANIC CHEMISTRY-I	CHM203A	Nomenclature o f Organic molecules : (1)Brief revision, Nomenclature of polycyclic compounds including bridged, spiro and other special structures.Structure and Bonding : (2)Nature of bonding in aliphatic and nonalternant hydrocarbons; Dipole moment; Resonance; Inductive and Field effects, hyperconjugation, Steric inhibition of resonance, structural effects on acidity and basicity.Stereochemistry: (10) Conformational analysis of decalins and other polycyclic compounds related to steroids. A brief introduction to compounds with more than one chiral centre and (iii) in biphenyls, allenes and spiro compounds. Optical isomerism in compounds without an asymmetric atom, Racemic modifications. Conformation of a transformations of heterotopic ligands and faces. Conformations of cyclic, fused and bridged ring compounds. Allylic strain (A1,2 and A1,3) and other strains. 1. March, J., Advanced Organic Chemistry, 4th ed, 1999.2. Nasipuri, D., Stereochemistry of Organic Compounds, 2nd ed., 1995.3. Solomons, T. W. G., Organic Chemistry 6th ed, 1996.4. Sykes, Peter, A guide to Organic Reaction Mechanisms, 20107. M. B. Smith, Organic Synthesis, 3rd Ed. 2010
882	EE	CODING THEORY	EE668	1.Introduction: Types of codes, channel models, maximum likelihood decoding, Shannon's noisy channel coding theorem, FEC, ARQ, HARQ. 2. Linear Blaock Codes : Generator matrix, parity check matrix, Syndrome decoding, Shannon's theorem, Williams theorem. Examples of simple linear block codes. 3. Some linear block codes : Construction, properties and decoding of some popular block codes; Hamming codes, ReedMuller codes. 4. Bonds of codes and bounds. 5. New codes from old codes : Extending a code, puncturing a code, expunging a code, augmenting a code, shortening a code, direct sum construction. 1. Shu Lin and Daniel J. Costello, Jr., "Error Control Coding", second ed., Prentice Hall, 2004. 2. F.J. Mac Williams, N.J.A. Sloance, "The Theory of ErrorCorrecting Codes", NorthHolland, 1977. 3. J. H. van Tilborg, 2003. 4. Todd K. Moon, "Error Correction Coding", 1st Ed., WileyInterscience, 2006. 5. Cary W. Huffman, Vera Pless, "Fundamentals of ErrorCorrecting Codes", 1st Edi. Cambridge Univ. Press, 2003. 6. J. Urbanke, "Modern Coding Theory", Cambridge Univ. Press, 2008.
883	AE	MECHANICS OF HIGHLY DEFORMABLE STRUCTURES	AE692A	Definition of large deformation of structures as large deflection, large rotation with small strains.Linear and nonlinear structural responses. Theory of elastica with exact and numerical solutions.Elements of stress, work conjugate stress and strain measures. Governingequations in strong and weak form. Hyperelastic material constitutive law. Incrementalnumerical solution oflarge deformation boundary value problems. Total Lagrangianmethod. Corotational scheme. Course Reference : * JamesF. Doyle. Nonlinear analysis of thin walled structures. Springer Verlag. 2001 .* M. A. Crisfield. Nonlinear finite element analysis of solids and structures. Vols. I and II. JohnWiley & Sons, 1999.
884	CE	ENGINEERING MECHANICS	CE621A	Stress analysis: forces and moments, theory of stress, principal stresses andstress invariants, compatibility equations, equilibrium equations; Strain:deformation and velocity gradients, Lagrangian and Eulerian descriptions; conditions; Fundamental physical principles: conservation of mass,linear momentum, angular momentum, and energy, second law of thermodynamics;Constitutive theory: St. Venants principal, linear elastic, viscoelasticity, yield criteria; Applications: Airy stress function,twodimensional elastostatics problems, torsion.
885	MSP	GRADUATE SEMINAR	MS698	Graduate Seminar
886	CHM	ORGANIC CHEMISTRY: FUNDAMENTALS AND APPLICATIONS	CSO201A	
887	CHM	INORGANIC CHEMISTRY LAB	CHM443A	
888	MME	NANOSTRUCTURES AND NANOMATERIALS: CHARACTERIZATION AND PROPERTIES	MME694	Overview of Nanostructures and Nanomaterials: classification (Dimensionality, Morphology/ shape/structure of nanoentities, New Effect/ Phenomena). Crystallinenanomaterials and defects therein. Hybrid nanostructures built out of nanosized building blocks (nano to macro). Eucledian, Hyperbolic and Spherical space structures. Nanostructures: Carbon Nanotubes, Fullerenes, Nanowires, Graphene, Quantum Dots, Nanoparticles, Nanoporous materials, Nanocomposites, Nanosensors, Nanomedicine, Nanoelectronics, Nanophotonics, Nanobiotechnology, Nanofabrication, Nanoreconstruction. Surface reconstruction and reconfiguration. Adsorption and Absorption. 1. Nanomaterials, Nanotechnologies and Design: an Introduction to Engineers and Architects, D.Michael Ashby, Paulo Ferreira, Daniel L. Schodek, ButterworthHeinemann, 2009.2. Handbook of Nanophase Materials, Vol. 1, Ed. by H. Gleiter, 2003.3. Encyclopedia of Nanoscience and Nanotechnology, Ed.: Hari Singh Nalwa, American ScientificPublishers, 2004.4. Handbook of Nanoceramics and their Based Nanodevices (Vol. 2) Edited by Tseung-Yee Duttu, H. F. Tibbals, A.K. Rao, CRC Press (2008).
889	MME	DISLOCATIONS AND PLASTICITY	MME658	Overview of defects in Materials: (point, line, planar and volume defects) and their classification. Overview of plastic deformation mechanisms. Point defects: interaction and distributions, statistical thermodynamics, the Volterra cut, Burgers vector and the Burgers circuit, the line vector, edge, screw and mixed locations, Role of dislocations in weakening the crystal and in plasticity. Elasticity theory of dislocations: Stress fields around dislocations, interaction between dislocations, Core of a dislocation. Motion of dislocations: The Peierls stress, role of the core structure, interaction of dislocations with other defects (including yield point phenomenon); kinks; jog formation; stressstrain curve. Dislocations in FCC Metals: Partial dislocations (Shockley and Frank partials) stacking faults, Thompson's tetrahedron, LomerCotrell sessile dislocation. Overview of dislocations in BCC metals: multiplication of dislocations: dislocations in freshly grown crystals, nucleation of dislocations, multiplication of dislocations (by FrankRead sources, cross slip and climb), Grain boundary sources, Recovery, Strain rate sensitivity, indentation, interfacial dislocations, Twinning including incoherent twins. Specific examples of role of dislocations and case studies:Dislocations in nanocrystals, The HallPetch relation and the Inverse HallPetch effect, Dislocation and Fracture. 1. Introduction to Dislocations, D. Hull and D.J. Bacon, Pergamon Press, Oxford, 1984.2. Theory of Dislocations, J.P. Hirth and J. Lothe, McGrawHill, New York, 1968.3. Crystal Dislocations, J. Weertmen and J. Weertman, The MacMillian Company, New York, 1964.5. http://www.if.unikiel.de/matwis/amat/defenl
890	MSE	SURFACE PHENOMENA AND CHARACTERIZATION	MSE605A	Surface Chemistry/Interaction (surface charge, dipoles, energies, interfacial chemical reactions), Surface Phonons/Plasmons (Quantisation of plasma, light and, sound, Brillioun Zones), Elastic/Inelastic scattering, Surface Lattice, Brillouim Zone, Surface Diffusion (Fick's law, intergraular/amorphous layer formation), Surface Energy/Young's Equation (Capillary/surface tension in fluids), Surface Sensitive Properties (Contact angle in wetting surfaces), Density functional theory of atomic equilibrium, Case Studies: FeB49system, and Al2O3carbon nanotube, Surface Modification/UncionalizationNanostructures: Self Assembly (Molecular sieves, filters : purification of gases/chemicals etc). I. Materials Science and Tedmology. Ed. R.W. Kahn, P. Hassen, E.J. Kramer, Vol. 2n/2b, Wiley YCII (2005).2. Physics of Atoms and Molecules. B.H. Bransden, and C.J. Joachain, Longman (1996).3. Surface Science: Fundamentals and Physics and Chemistry of Materials, J. I. (Gersten, F. W. Smith. Wiley (2001).5. Elementary Solid State Physics. M. A, Omar, AddisonWesley {2001}.6. Course materials wlll be supplmemmed with hundouts,
891	MSE	SURFACE PHENOMENA AND CHARACTERIZATION	MSE605	Surface Chemistry/Interaction (surface charge, dipoles, energies, interfacial chemical reactions), Surface Phonons/Plasmons (Quantisation of plasma, light and, sound, Brillioun Zones), Elastic/Inelastic scattering, Surface Lattice, Brillouim Zone, Surface Diffusion (Fick's law, intergraular/amorphous layer formation), Surface Energy/Young's Equation (Capillary/surface tension in fluids), Surface Sensitive Properties (Contact angle in wetting surfaces), Density functional theory of atomic equilibrium, Case Studies: FeB49system, and Al2O3carbon nanotube, Surface Modification/UncionalizationNanostructures: Self Assembly (Molecular sieves, filters : purification of gases/chemicals etc). I. Materials Science and Tedmology. Ed. R.W. Kahn, P. Hassen, E.J. Kramer, Vol. 2n/2b, Wiley YCII (2005).2. Physics of Atoms and Molecules. B.H. Bransden, and C.J. Joachain, Longman (1996).3. Surface Science: Fundamentals and Physics and Chemistry of Materials, J. I. (Gersten, F. W. Smith. Wiley (2001).5. Elementary Solid State Physics. M. A, Omar, AddisonWesley {2001}.6. Course materials wlll be supplmemmed with hundouts,
892	MSE	ELECTROCERAMIC MATERIALS AND APPLICATIONS	MSE631A	Introduction, FCC packed structures (MgO, CeO2 etc),HCP packed structures (LiNbO3 etc), Other structures such as Perovskite (BaTiO3 etc) and Rutile structures, Defects in Elemental Solids and Ionic Compounds, Charge and electrical neutrality, Thermodynamics of Intrinsic Defects and Defect Reactions. Complexes Containing an Impurity Center and an Ionic. Defect, Intrinsic Ionic Defect Associates and Effect of Impurities on Properties with Schottky Defects, Frenkel Defect Pairs and Intrinsic Ionization of Electrons, 1 L.L. Hench and West, Electroceramics, Wiley2 D. M. Smyth, "The Defect Chemistry of Metal Oxides", Publisher: Oxford University Press, ISBN10:01951101453 Wei Gao and Nigel M. Sammes, & "Electroceramics: Materials, Properties, Applications", Publisher: Wiley5 M. W. Barsoum, "Fundamentals of Ceramics", Publisher: Institute of Physics6 "Impedance Spectroscopy: Theory and Applications", Sons.7 Robert Huggins, "Use of defect equilibrium diagrams to understand minority species transport in solidelectrolytes", Solid State Ionics, 143 (2001) 316.
893	MSE	ELECTROCERAMIC MATERIALS AND APPLICATIONS	MSE631	Introduction, FCC packed structures (MgO, CeO2 etc),HCP packed structures (LiNbO3 etc), Other structures such as Perovskite (BaTiO3 etc) and Rutile structures, Defects in Elemental Solids and Ionic Compounds, Charge and electrical neutrality, Thermodynamics of Intrinsic Defects and Defect Reactions. Complexes Containing an Impurity Center and an Ionic. Defect, Intrinsic Ionic Defect Associates and Effect of Impurities on Properties with Schottky Defects, Frenkel Defect Pairs and Intrinsic Ionization of Electrons,

S.No.	Branch	Course Title	Courses ID	Content
				1 L.L. Hench and West, Electroceramics, Wiley2 D. M. Smyth, &#34;The Defect Chemistry of Metal Oxides&#34;, Publisher: Oxford University Press, ISBN10:01951101453 Wei Gao and Nigel M. Sammes, &#34;Electroceramics: Materials, Properties, Applications&#34;, Publisher: Wiley5 M. W. Barsoum, &#34;Fundamentals of Ceramics&#34;, Publisher: Institute of Physics6 &#34;Impedance Spectroscopy: Sons.7 Robert Huggins, &#34;Use of defect equilibrium diagrams to understand minority species transport in solidelectrolytes&#34;, Solid State Ionics, 143 (2001) 316.
894	BSBE	UG PROJECT (UGP-II)	BSE399A	UG PROJECT (UGPII)
895	MTH	UG PROJECT (UGP-I)	MTH391A	UG PROJECT (UGPI)
896	ECO	UG PROJECT (UGP-I)	ECO398A	UG PROJECT (UGPI)
897	CSE	UG PROJECT (UGP-I)	CS395A	UG PROJECT (UGPI)
898	MSE	STRUCTURE AND CHARACTERIZATION OF MATERIALS	MSE615A	Basic crystallo2raphy and crystal structures (8 Lectures hour)Periodic patterns, Lattices, Motif, Unit cells, Crystal structure, Primitive and Non primitive cells 1, Symmetry elements and point group notation Streographic projections 1, Bonding in materials and atomic packing in metals, coordination number concepts 1, Covalent bonding, glasses and polymers 2,Crystal defects and their significance (12 Lectu Pauling&#39;s rules and some examples 2, Point defects: thermodynamics, schottkey and Frenkel defect, KrogerVink notation, defect interactions 2, Dislocations, burgers vector, types of dislocations 1, Di grain boundaries (low angle and high angle), antiphase 2 1. Crystals and Crystal structures, R.J.D. Tilley, John Wiley and Sons, 20062. Materials Science and Engineering W.D. Callister, Jr. Wiley India(P) Ltd., 20073. Materials Science and Engineering, G.S. Upadhy relationship using metals as model systems, E.J. Mittemeijer, Springer, 20105. Microstructural Characterization of Materials D. Brandon and W.D. Kaplan, John Wiley and Sons, 20086. Science of Microsco et.al, Springer, 20038. Transmission Electron Microscopy B.D.Williams & C.B.Carter, Springer, 20099. Surface Analysis methods in materials science, Editors: D.J.O&#39;Connor, B.A. Sextton, R.St. C. Smar
899	CSE	FUNDAMENTAL OF COMPUTING	ESC101N	Stored program concept (with simple computer simulator), machine language and instruction formats,assembly language for the simple computer. Integer representation, finite representation of real num unary, binary, ternary operations. Conditionals, ifthen, ifthenelse, nested conditionals, switchcase.Loops, for, while, repeat, loopinvariants, precondition, postcondition. Functions and return values,argumen lists, dynamic data structures, stack, queue, graphs, trees related algorithms,memory and its management. Elementary complexity motivation, concrete complexity, big O notation.Linux tools, introduction solutions of systems of linear equations, integration, solution ofODEs. 1. Brian W Kernighan and Dennis M Ritchie, The C Programming Language,2nd Ed. ANSI C version,Pearson, 2006.
900	ECO	BAYESIAN DATA ANALYSIS	ECO543	Course Description: This is an introductory level course in Bayesian methods aimed to introduce students to the Bayesian framework (originating from Bayes rule), that formally allows to incorporate nons techniques that will be useful in academia as well as industry. The course is targeted for advanced undergraduate students or any individual seeking to learn Bayesian techniques.The course will be heavil linear regression. Consequently, students who have taken the course Econometric Methods (ECO341A) will have an advantage. However, students who are willing to take this course but have not taken eco linear regression.Softwares: Bayesian methods are highly computational, hence proficiency in coding is required or one should be willing to put in extra hours to code programs. The official software for th
901	MSE	THIN FILM PHYSICS & APPLICATIONS	MSE685	Surface science; experimental techniques to study surfaces; kinetics of surfaceprocesses impingement of atoms, scattering, adsorption, sticking coefficient;Film nucleation and growth mechanisms, critic chemical vapour deposition, atomic layer epitaxy, liquid phaseepitaxy, solgel technique etc, Electrical, optical, magnetic and mechanicalproperties of thin films and their applications.
902	MSE	GRAIN BOUNDARY ENGINEERING	MSE680A	Grain boundary structure : Geometrical aspects, Degress of freedom, Principlesgoverning grain shape and size their orientation. Theoretical formulations :Structurals units model, Plane matching model, C mobility and boundary solute interactions. GB structure and Properties:mechanical strength wear, creep magnetic, electrical etc. Simulation andmodeling . Grain boundary engineering strategy : Deformati distribution, Character distribution. Boundary CharacterizationTools : Xray, EBSDOIM, CTEM, AEM, HRTEM, etc. Macrotexture analysis : Polefigure measurement, Xray diffraction, neutron diffraction metho representation of Data Prospective applications:Superplasticity. Creep resistance, Corrosion Resistance, Superconductivity,Electronic ceramics etc.
903	MSE	GRAIN BOUNDARY ENGINEERING	MSE680	Grain boundary structure : Geometrical aspects, Degress of freedom, Principlesgoverning grain shape and size their orientation. Theoretical formulations :Structurals units model, Plane matching model, C mobility and boundary solute interactions. GB structure and Properties:mechanical strength wear, creep magnetic, electrical etc. Simulation andmodeling . Grain boundary engineering strategy : Deformati distribution, Character distribution. Boundary CharacterizationTools : Xray, EBSDOIM, CTEM, AEM, HRTEM, etc. Macrotexture analysis : Polefigure measurement, Xray diffraction, neutron diffraction metho representation of Data Prospective applications:Superplasticity. Creep resistance, Corrosion Resistance, Superconductivity,Electronic ceramics etc.
904	MSE	DESIGN OF SINTERED PRODUCTS	MSE674	Factors affecting designmaterials and geometry: Specific design of productslike permeable materials, structural parts, bearings and cutting tool materials:conditioning of metal powders to influence proce
905	MSE	MATERIALS FOR BIOMEDICAL APPLICATIONS	MSE668	Introduction to basic concepts of Materials Science; Salient properties ofimportant material classes; Property requirement of biomaterials; Concept ofbiocompatibility; cellmaterial interactions and foreign CoCrMo alloys; Bioinert, Bioactive and bioresorbable ceramics;Processing and properties of different bioceramic materials with emphasize onhydroxyapatite; synthesis of biocompatible coatings on struc developing new materials for bioimplant applications.
906	MSE	SELECTION AND DESIGNING WITH ENGINEERING MATERIALS	MSE667	Overview of the design process: concepts and stages of engineering design anddesign alternatives to develop materials with tailored properties; Performanceindices of materials; function, objective and c for thermal, mechanical, thermomechanical applications,damage tolerant designs for structural applications; Basic concepts of materialsscience: processingstructurepropertyperformance correlation; ove defect structure of different material classes, Briefintroduction to the manufacturing processes for metals, polymers, ceramics,glasses and composite materials; design for manufacturability, Ashbys mate materials selection and processing; Case studies:designing of Metals and alloys, ceramics and glasses, composite materials (MMC,CMC and PMC/ FRC) for specific applications.
907	MSE	SCIENCE AND TECHNOLOGY OF MAGNETIC MATERIALS	MSE666	Magnetic units: Magnetic moments: Dia, para and paulipara magnetism:Molecular field: Ferro, antiferro and ferrimagnetism: Alloying effect on transitionmetals and intermetallics: Stability of domain struc orientation on magnetization: Susceptibility and coercivitycalculations: Magnetic thin filmsamorphous and crystalline, soft and permanentmagnets: Technological aspects of magnetic materials.
908	MSE	MATERIALS DEGRADATION & ITS PREVENTION	MSE480	Types of processes leading to degradation of materials, viz Oxidation Corrosion,Wear, Creep and fatigue review of basics of thermodynamics and kinetics relatedto oxidation and corrosion studies, Pourba atmospheric galvanic, intergranular, dealloying, crevice and pittingcorrosion, microbiological, stress corrosion cracking, hydrogen damage, radiationdamagte; Oxidation and hot corrosion of materials at hig environment conditioning design modification cathodic and anodicprotection, metallic coating inorganic coating organic coating, inhibitors andpassivators wear resistant materials structural modification
909	ECO	INDUSTRIAL ORGANISATION AND POLICY	ECO734A	Scope of Industrial organization; Industrial Efficiency; Basic Framework forthe Study of Industrial Organization; Recent Approaches to Industrial Organization;Market Structure and Its Elements; Market Co Heavy Industries, Consumer Goods Industries and Public Utilities.
910	BSBE	HUMAN MOLECULAR GENETICS	BSE654A	Simple Mendelian traits; Lossoffunction mutations; Gainoffunction mutations;Gene interactions; Dynamic mutations; Genetics of neoplasia; Genomic imprintingand human disease; Xinactivation and DNA disorders;Pharmacogenetics and biochemical genetics; Animal models in human genetics:Methods used for diagnosis and detection of gene mutations; Gene Therapy.
911	HSS	INTRODUCTION TO THE ART OF VIDEO MAKING	ART105	TheoryCinematography, Production Process, Writing for Screen, Basics Picture Making, CameraAngles, Direction Time and Space continuity, Cuts, Art of Editing The 5C&#39;s of Cinematography, MascelliDirecting the Documentary, Michail RabigerA History ofthe Cinema, Eric RhodeWriting for Screen and Television, TobiasSingle camera video production, Robert L Short Film & Video, Irving & ReaThe Art of Cinema, B.D.GargaA compauion to Film Theory, Miller & Starn
912	HSS	INTRODUCTION TO ART CRITICISM AND APPRECIATION	ART102	What is Art, What is art appreciation, History of Art Criticism, Methodology of Critical ArtAppreciation, , Elements and Principles of Visual Art, Characteristics of art of Prehistoric,Ancient, middle age, Dark A methods and materials.Practice: Still life, Nature Study, Composition, Symmetry, Narrative Continuity, 2D and 3D Exercises Jaoson&#39;s Basic History of Western Art by Penelope CaviesArt of Twentieth Century by Loredana ParmesaoiIndiaao Art A Concise History by Roy C. CravenEarly India From the Origins to AD 1300 by Ro Eastern Art (Chinese aod Japaoese Art) Skira, Sharman Lee PublishersMoving Focus: Essays on Indian Art by K.G. SubramaoyaoThe Making of a New &#39;Indiaao&#39; Art by Tapati Guha ThakurtaIndiaao
913	MME	PROJECT II	MME499	PROJECT II
914	CE	M.TECH THESIS (FOR DUAL	CE699A	M.TECH THESIS (FOR DUAL DEGREE ONLY)

S.No.	Branch	Course Title	Courses ID	Content
		DEGREE ONLY)		
915	CE	CIVIL ENGINEERING COMMUNICATION SKILLS	CE341A	CIVIL ENGINEERING COMMUNICATION SKILLS
916	CSE	UNDER GRADUATE PROJECT-III	CS498A	UG PROJECT (UGPIII)
917	MSE	SYMMETRY AND PROPERTIES OF CRYSTALS	MSE638	Introduction 1 Symmetries in 1 D, 2D and 3D, Examples of patterns showing various, symmetries, Symmetries and Lattices in 2D space : Operations ofTranslation, Rotation and Reflection, standard symbols for rectangular, square, hexagonalPoint Groups in 2D 4 : Set of symmetry operations, Group Theory Essentials, Evolution of 2D crystallographic point groups, 2D Space Groups (Plane Groups) Glide Planes: common and unique entries in the International Tables, of Crystallography3D Point Groups 4 : Combination of rotation axes in 3D, Development of the 32 point groups, Laue Groups3D Bravais Lattices Addition of a third translation and rotation, Development of the 230 space groups, Understanding the Space Groups entries in the International Tables of Crystallography I. M.J. Buerger, Elementary Crystallography, 1962. 2. International Tables of Crystallography A, International Union of Crystallography3. J. F. Nye, Physical Properties of Crystals (1995), Oxford Science Publications4. Properties of Materials: Anisotropy, Symmetry, Structure", Oxford Pr.
918	MSE	CRYSTALLOGRAPHIC TEXTURE & MECROSTRUCTURAL ENGINEERING	MSE683	Introduction to crystallographic texture.Refreshers on Xray diffraction: basic diffraction concepts, reciprocal space,instrumentation and geometry.Basics of neutron and synchrotron diffraction and comparison of diffraction figures: experimental details, data processing, indexingDetermination of Orientation Distribution Function and Misorientation.distribution Function: calculation techniques, different notations, 3D and 2DRepresentation of data acquisition and analysis.Mechanisms of evolution of texture during processing: solidification, phasetransformation, deformation, annealing.Modelling texture evolution.Texture evolution and measurement of texture.Functional properties and texture. 1. An Introduction to Texture in Metals; M. Heatherly, W. B. Hutchinson, Monograph no. 5, The Institution of Metallurgist London 1979.2. Introduction to Texture Analysis: Macrotexture, Microtexture, and Orientation Texture and their Effect on Materials Properties; U. F. Kocks, C. N. Tome, H.R. Wenk, Cambridge University Press, 2000.4. Research publications on: Texture manipulation and control; Microstructural engineering ;
919	MSE	MATHEMATICS AND COMPUTATIONAL METHODS	MSE617A	1 Introduction of functions, vectors, matrices2 Partial Differentiation (Total differentiation, Maximum and minimum: method of Lagrange multipliers, Change of variables: Legendre transformation, Differentiation of Vectors (Geometry: Lines and planes, Directional derivative, gradients (fields, equipotential, grad, normal to surface, curl, div), Line integration (conservative fields, potential, exact differentiation), Green, Stokes theorem, diagonalization of matrix)6 Ordinary differential equations (Linear first order, Second order: constant coefficient and zero right hand side, Second order: constant coefficient and non zero right hand side)7
920	MSE	THERMODYNAMICS OF MATERIALS	MSE616A	Thermodynamic systems and variables. 1 First, second and third laws of thermodynamics. 7Statistical interpretation of entropy. 2 Free energy functions and criteria for equilibrium. 2 Thermodynamics of solutions. 2 Polynomial expressions for excess Gibbs energy of mixing for binary and higher order solutions. Multicomponent dilute solutions and interaction parameters. 2 Chemical reaction equilibrium, equilibrium constants, Formation and concentrations cells. 3 Phase rule and binary phase diagrams 2Free energy composition diagrams 3Phase equilibrium calculations 5Introduction to ternary phase diagrams.1
921	MSE	THERMODYNAMICS OF MATERIALS	MSE616	Thermodynamic systems and variables. 1First, second and third laws of thermodynamics. 7Statistical interpretation of entropy. 2Free energy functions and criteria for equilibrium. 2Thermodynamics of solutions. 2Polynomial expressions for excess Gibbs energy of mixing for binary and higher order solutions. Multicomponent dilute solutions and interaction parameters. 2Chemical reaction equilibrium, equilibrium constants, Formation and concentrations cells. 3Phase rule and binary phase diagrams 2Free energy composition diagrams 3Phase equilibrium calculations 5Introduction to ternary phase diagrams.1
922	MSE	MATHEMATICS AND COMPUTATIONAL METHODS	MSE617	1 Introduction of functions, vectors, matrices2 Partial Differentiation (Total differentiation, Maximum and minimum: method of Lagrange multipliers, Change of variables: Legendre transformation, Differentiation of Vectors (Geometry: Lines and planes, Directional derivative, gradients (fields, equipotential, grad, normal to surface, curl, div), Line integration (conservative fields, potential, exact differentiation), Green, Stokes theorem, diagonalization of matrix)6 Ordinary differential equations (Linear first order, Second order: constant coefficient and zero right hand side, Second order: constant coefficient and non zero right hand side)7
923	ME	DYNAMICS	ESO209A	Introduction to coordinate system and phase space, review of vectors and tensors, dynamics ofsystem of particles, steady mass flow and variable mass problems, coordinate transformationinvolving 3d motions, general 3D motions, advanced examples.
924	MSE	STRUCTURE AND CHARACTERIZATION OF MATERIALS	MSE615	Basic crystallo2raphy and crystal structures (8 Lectures hour)Periodic patterns, Lattices, Motif, Unit cells, Crystal structure, Primitive and Non primitive cells 1, Symmetry elements and point group notation 1, Stereographic projections 1, Bonding in materials and atomic packing in metals, coordination number concepts 1, Covalent bonding, glasses and polymers 2,Crystal defects and their significance (12 Lectures)1, Pauling's rules and some examples 2, Point defects: thermodynamics, schottkey and Frenkel defect, KrogerVink notation, defect interactions 2, Dislocations, burgers vector, types of dislocations 1, Dislocations and grain boundaries (low angle and high angle), antiphase 2 1. Crystals and Crystal structures, R.J.D. Tilley, John Wiley and Sons, 20062. Materials Science and Engineering W.D. Callister, Jr. Wiley India(P) Ltd., 20073. Materials Science and Engineering, G.S. Upadhyay, Wiley India, 20084. Relationship between microstructure and properties using metals as model systems, E.J. Mittemeijer, Springer, 20105. Microstructural Characterization of Materials D. Brandon and W.D. Kaplan, John Wiley and Sons, 20086. Science of Microscopy, J. E. Enderby et.al, Springer, 20038. Transmission Electron Microscopy B.D.Williams & C.B.Carter, Springer, 20099. Surface Analysis methods in materials science, Editors: D.J.O'Connor, B.A. Sextton, R.St. C. Smart
925	EE	MICROWAVE CIRCUITS	EE648	Transmission lines for microwave circuits; waveguides, stripline, microstrip, slotline; microwave circuit design principles; passive circuits; impedance transformers, filters, hybrids, isolators etc., active circuits
926	CHM	CHEMICAL APPROACHES TO THE SYNTHESIS OF ADVANCED MATERIALS	CHM693	Chemical methods of synthesis play a crucial role in designing materials,discovering novel materials, metastable phases, nanomaterials and provide lesscumbersome routes for the known materials. Chemical synthesis: multidisciplinary nature of the subject, a rational approach to thesynthesis of materials is evolved. Indeed, soft chemistry routes/techniques arepursued with greater vigour. These include precursor techniques, chemical vapour deposition, electron beam evaporation, freeze drying, spraydrying, topochemical reactions, intercalation, electrochemical methods, CVDlaser ablation, arcmethod, molten salt method, intergrown structures. Solidstate synthesis
927	MSE	SOLID STATE IONICS	MSE664A	Perfect Structure, Defects in Elemental Solid and Ionic Compound, Defect Classes,Point Defects, KrogerVink Notation for Point Defects, Point Defect Formation & Equilibrium, Law of Massaction, Thermodynamic Properties of Intrinsic Ionic Defect Associates and Effect of Impurities on the Concentration of Defect Complexes and Associate. Basic Concepts of Diffusion, Tracer Diffusion, Self Diffusion, Chemical Diffusion, Ambipolar Diffusion, NernstEinstein Relationship, and ConductivityDiffusion Relationship. Defect Equilibria in Pure and Stoichiometric Compounds with Schottky Defects,Frenkel Defect Pairs and Intrinsic Ionization of Electroneutral Deficient Oxide, Metal Oxide with Excess Oxygen. Brouwer Diagrams for YSZ, Undoped and Doped CeO2, TiO2 and BaTiO3 Electrical Characterization Techniques such as AC Electrochemical Impedance Spectroscopy, Electrode Method. OpenCircuit Potential, Efficiency, Nernst Equation Analysis, Activation Losses (Tafel Equation), Ohmic Losses,Concentration Losses. Description of Operation, Configurations, Cell Components, Materials, Solid Oxide Fuel Cells, Gas Sensors and Batteries. Publisher: Oxford University Press,ISBNIO: 01951101452 A.J. Moulson & J. M. Herbert, "Electroceraamics: Materials, Properties, Applications", Publisher:Springer3 M. W. Barsoum, "Fundamentals of Electroceramics", Edited by J. Ross Macdonald & Evgenij Barsoukov, Publisher: John Wiley and Sons.5 Robert Huggins, "Use of defect equilibrium diagrams to understand minority species transport", Publisher: John Wiley and Sons.6 Electrochemistry", Edited by P. J. Gellings & H. J. M.Bouwmeester, Publisher: CRC Press7 "High Temperature Solid Oxide Fuel Cells, Fundamental, Design and Applications ", Edited by Subhan
928	CSE	DATA STRUCTURE & ALGORITHM	ESO207A	Order Analysis: Objectives of time analysis of algorithms; Bigoh and Theta notations, Data Structures: Arrays, Linked lists, Stacks (example: expression evaluation), Binary search trees, RedBlack trees, Heapsort, QuickSort with average case analysis, Heaps and heapsort, Lower bound on comparisonbased sorting and Counting sort, Radix sort, Btrees, Dynamic Programming: methodology and examples (Fibonacci numbers, Matrix Multiplication, Method: Methodology, examples (lecture scheduling,process scheduling) and comparison with DP (more examples to come later in graph algorithms), Graph Algorithms: Basics of graphs and their representations, Shortest of disjoint set and Fibonacci heap data structures), Shortest Paths (Dijkstra, BellmanFord, FloydWarshall).
929	MSE	NATURE AND PROPERTIES OF MATERIALS	ESO205A	Examples of materials highlighting StructureProcessingPropertyperformancerelations. 14 space lattices, unit cells, cubic and HCP structures, Miller indices,Packing, interstitials, different ceramic structures, polymerstructures.LTPD[C]3131[5]LTPD[C]3101[4]26Structure determination using Xray diffraction (Braggs diffraction and structurefactor for cubic lattices); Point defects, edge and screw dislocationstheir interaction with EVOLUTION: Definition of diffusivity, concept of activation energy,examples of diffusion process; Definition of a phase, phase rule, unary andbinary (eutectic, eutectic with terminal solid solutions) systems and phase transformations (heterogeneous), Introduction to TTT curves, examples ofvarious transformations;
930	PHY	M SC EXPERIMENTAL PROJECT III	PHY502N	Student must carry out review of an advanced topic of current interest andmake a presentation to an Evaluation Committee. Letter Grades will be awarded.PHY 500 : M.Sc. Review Project IPHY 501 : M.Sc. Review Project material, Detailedstudy of the material, verification of results and writing of the review. ReviewProjects will include exposure to and conduct of Experiments as required.

S.No.	Branch	Course Title	Courses ID	Content
				number of case studies there will be an extended term project for Case Building.
948	AE	PRELIMINARY DESIGN OF HELICOPTER	AE660A	1. Introduction to: design process, design goals, types of rotorcraft (2 Lec.)2. Understanding mission requirements, use of Analytical Hierarchy Process in configuration selection (2 Lec.) (1 Lab.)3. Conceptual design (1 Lab.)4. Generating design alternatives: preliminary sizing using Tischenko's Method, preliminaryweight estimation, rotor propulsive efficiency, Lift/Drag ratio, engine performance, main rotor speed, maximum level speed, speed forbest endurance, best range, autorotative performance (3 Lec.) (2 Lab.)6. Main rotor configuration design: rotor structural and aerodynamic design (number ofblades, pitch, offset, air foils, frequency placement, material selection) (5 Lec.) (3 Lab.)7. Rotor component design: hub design, control power, helicopter stability considerations (3 Lec.) (2 Lab.)8. Tail rotor/ antitorque system design (4 Lec.) (2 Lab.) 10. Vibration sources, vibration reduction (2 Lec.) (1 Lab.) 11. Life cycle cost estimation: environmental cost, purchase cost. operating cost(2 Lec.) (1 Lab.) Course Reference : * Leishman, J. G., Principles of Helicopter Aerodynamics, Cambridge Aerospace Series,2000.* Prouty, R. W., Helicopter Performance, Stability, and Control, Krieger Publishing Company, 2000.* Department of AerospaceEngineering, IIT Kanpur, 2000.* Filippone, A., Flight Performance of Fixed and Rotary Wing Aircraft, AIAA EducationSeries, 2000.
949	AE	COMPUTATIONAL FLUID MECHANICS	AE604	Basics of Governing Equations, Spacetime discretization for PDE, Classification of PDE, Grid Generation, Waves and disturbances in fluid flow, Spacetime scales in fluid flow, Classical methods for solving PDE, Solution of NavierStokes equations. 1.Computational Fluid Dynamics: C. Hirsch, Wiley (1998) 2.Computational Fluid Flow and Heat Transfer, Tannehill, Anderson, Pletcher3.High Accuracy Computing Method: Fluid flow and wave phenomena in fluid flow, Hyderabad, India (2004)
950	AE	COMPUTATIONAL FLUID MECHANICS	AE604A	Basics of Governing Equations, Spacetime discretization for PDE, Classification of PDE, Grid Generation, Waves and disturbances in fluid flow, Spacetime scales in fluid flow, Classical methods for solving PDE, Solution of NavierStokes equations, 1.Computational Fluid Dynamics: C. Hirsch, Wiley (1998) 2.Computational Fluid Flow and Heat Transfer, Tannehill, Anderson, Pletcher3.High Accuracy Computing Method: Fluid flow and wave phenomena in fluid flow, Hyderabad, India (2004)
951	PHY	GREEN NANOTECHNOLOGIES	PHY690A	The course will deal with specialized topics of current interest in solid state,theoretical physics, molecular physics, or structure of matter. Detailed contentswill be given by the instructor when the course is offered. DPGC.Every new course, other than Reading Courses, offered is numbered PHY 690A,PHY 690B, and so on, until PHY 690Z is reached. After that the cycle repeatsfrom PHY 690A onwards.
952	IME	INDEPENDENT STUDY	IME797	Market Efficiency, Capital Structure, Asset Pricing, Corporate Governance, PortfolioTheory, Financial Intermediation, Credit Risk Modeling & Market Microstructure 1. Ananth Madhavan, Market microstructure: A survey, Journal of Financial Markets,Volume 3, Issue 3, August 2000, Pages 2052582. Bhattacharya Sudipto & Thakor Anjan V., 1993."Contemporary Banking and Finance (1973)., The Pricing of Options, and Corporate Liabilities, Journal of Political Economy, MayJune, 637659.4. Burton G. Malkiel., The Efficient Market Hypothesis and Its Critics, Princeton University CEPS Working Paper, 1985.5. Fama, E. F., "Random Walks in Stock Market Prices," Financial Analysts Journal,September/October 1965 (reprinted JanuaryFebruary 1995)7. Fama, E. F., 1970. "Efficient Capital Markets: A Review of Theory and Evidence, 24-DEC-14
953	CSE	ADVANCED COMPUTER NETWORKS	CS625A	Introduction: Overview of computer networks, sevenlayer architecture, TCP/IP suite of protocols, etc. MAC protocols for highspeed LANs, MANs, andwireless LANs. (for example, FDDI, DQDB, HIPPI, Gigabit Ethernet, etc.) ATM layer. ATM Adaptation Layers. Congestion control.Signalling, Routing, QoS support, Neighbourdiscovery, Autoconfiguration. Changes to other protocols. Application Programming Interface for IPv6. Multicast, session discovery, etc. TCP extensions forhighspeed networks, transactionoriented applications. Other new optionsin TCP. Network security at various layers. SecureHTTP, SSL, ESP, Authenticationheader, etc.
954	MTH	ABSTRACT HARMONIC ANALYSIS	MTH638	Banach Algebras and Spectral theory, Locally compact groups, Basic representation theory, Analysis on Locally compact abelian group, Analysis on compact groups, Group C* algebra and structure of dual groups. Recommended books: 1. G.B. Folland: Course in Abstract harmonic analysis, Studies inAdvanced Maths., CRC Press, Boca Raton, etc., 1995.
955	MSE	XPS AND AES IN STUDY OF SURFACE AND THIN FILMS	MSE621	Solid surfaces, their structure and composition: Importance of the surfaces/ Surfaces in different materials/ Structural imaging/ Composition of surface selvages; Practical detection and spatial limits; Characterization of surfaces; Principles; Instrumentation; Vacuum systems, Xray sources, synchrotron radiation, electron energy analyzers; Spectral information, chemical shifts and interpretations; Quantification, depth profiling, imaging; Surface analysis; Thin films; Semiconductors; metallurgy/ tribology; thin films; Biomaterials, Case studies and lab visit/demo. Auger Electron SpectroscopyBasics: Principles; Instrumentation; Vacuum requirements, Electron sources, etc. 1. Surface and Thin Film Analysis, Editors; G. Friedbacher and H. Bubert, WileyVCH, 2011. 2. Encyclopedia of Materials Characterisation, C.R. Brundle, C.A. Evans and S. Wilson, ButterworthHeinemann, 1997. 3. Surface Analysis Methods in Materials Science, D.J. O'Connor, B.A. Sexton, and R. St. C. Smart, Springer (2003). 5. Photoelectron and Auger spectroscopy. Thomas A Carlson, Plenum, New York. 4. Surface Analysis Methods in Materials Science, D.J. O'Connor, B.A. Sexton, and R. St. C. Smart, Springer (2003). 5. Photoelectron and Auger spectroscopy. Thomas A Carlson, Plenum, Berlin, 1977. 7. Scanning Auger electron microscopy, Editors: M. Prutton and Mohamed M. El Gomati, John Wiley & Sons, 2006. 8. Practical guide to surface science and spectroscopy. YipWah Chung, Academic Press, 1997.
956	AE	COMPUTATIONAL FLUID DYNAMICS	AE622A	Eqns of fluid dynamics & its classifications. Boundary conditions. Stabilityanalysis & concept of feedback. Various explicit & implicit schemes. Gridgeneration. Solving parabolic, elliptic PDEs by explicit, implicit, and hybrid methods.
957	EE	COMPACT MODELING	EE681	Introduction to Simulation. Introduction to SPICE and Compact Modeling. Integrated Resistor Modeling. Integrated MOS Varactor Modeling. MOSFET Modeling approaches : Threshold Voltage based modeling, Charge based modeling, and Hybrid modeling. MOSFET Compact Models and Benchmark Tests. Layout Effects and Parameter Extraction. High Frequency Effects and RF Modeling. SOI MOSFET Modeling. Noise Modeling. Multigate MOSFETs. Modeling of Emerging Devices. C. Hu, "Modern Semiconductor Devices for Integrated Circuits," Pearson/Prentice Hall, New Jersey.Compact Modeling: Principles, Techniques and Applications by Gennady Gildenblat, Springer. Modeling of Emerging Devices: Transistor Modeling: The EKV Model for LowPower and RF IC Design by Christian C. Enz and Eric A. Vittoz, John Wiley & Sons.BSIM4 and MOSFET Modeling for IC Simulation by W. Liu and C. Hu, World Scientific.
958	AE	FOUNDATION OF SCIENTIFIC COMPUTING	SE422	Basics of Computing: discretization and numerical errors. ODEs and their computations. Stiff ODEs & parasitic error, solving stiff equation via orthogonalization and compound matrix method. Governing equations for fluid flow, relation and spacetime resolution. Finite difference methods for wave equation. Discretization of spatial derivatives by polynomial expansion and operators explicit methods. Spectral theory of discrete continuous systems. Finite difference methods (FDM) for wave equation. Designing high accuracy, high fidelity methods via error control as anoptimization problem. Spectral method visvis discrete computing methods:Aliasing error, stability, etc. Numerical Methods for Partial Differential Equations WF AmesFundamentals of Computational Fluid Dynamics: T. K. Sengupta.Waves in Fluids M. J. LighthillComputational Aeroacoustics: M. J. Lighthill. 17-SEP-14
959	CSE	CATEGORY THEORY AND APPLICATIONS IN COMPUTING	CS680A	Introduction: Basic definition and diagram; sources and sinks; monicity andepicity; isomorphisms of objects and morphisms; duality; Universal Structures:Initial terminal and zero; Category of sources and sinks; Monoidal categories; hierarchy; extension of categories; factorization; chains and exactness;Morphism algebra: Biproduct; semiadditive category; Additive category; Functors:Natural transformation; categories on natural transformations; and colimit; Hreflection morphism and adjoint functor; Representablefunctors Category in context of another category; Application to Logic (Topoi);application to Programming Languages.
960	CHE	THERMODYNAMICS	CHE621A	Postulational Thermodynamics: Postulates; Equilibrium criteria; Gibbs Duhemrelation; Energy minimum principle; Thermodynamic potentials; Stability andphase Transition. Statistical Thermodynamics: Statistical mechanics; equilibriumconstant in reacting systems.
961	AE	FUNDAMENTALS OF LIQUID ATOMIZATION	AE751A	Course Details: Introduction to atomization, Physical processes in atomization, Types of atomizers,Classical Theories of atomization, Numerical modeling of atomization process,Theory of multiphase flow, Characterization, Applicationsof Atomizers Metal forming Chemical Industry Combustion.
962	IME	ADVANCED DECISION MODELS	IME637A	Review of linear and integer linear programming. Multistage decision models: Dynamic programming. Network flow problems: Shortest path, maximum flow and minimum cost flow problems; Network optimization: Unconstrained optimization; Lagrangian relaxation and KKT conditions; Convex optimization; Search, gradient and penalty based methods; Quadratic programming. Metaheuristics and their applications to combinatorial optimization. Markov chains; Queues and queuing networks. Katta G. Murty, Linear Programming, Wiley. Laurence A. Wolsey, Integer Programming, Wiley. Richard Bellman, Dynamic Programming, Dover. R.K. Ahuja, T.L. Magnanti, J.B. Orlin, Network Flows, Prentice Hall. Optimization Theory, Cambridge. S. Boyd and L Vandenberghe, Convex Optimization, Cambridge. J. Nocedal, S.J. Wright, Numerical Optimization, Springer. ElGhazali Talbi, Metaheuristics: From design to implementation, Wiley. Harris, Fundamentals of Queueing Theory, Wiley. Wayne L. Winston, Operations Research: Applications and Algorithms.
963	CE	REINFORCED CEMENT CONCRETE DESIGN	CE372A	Reinforced concrete (RC) structures, Loadings, analytical models for analysis and design of RC structures, Design Methodologies: Working Stress Method and Limit State Method; Behaviour of RC members (one way and two way) for flexure; Singly and doubly reinforced sections; rectangular and flanged sections; Shear and torsion; Bond and anchorage; Short columns under axial compression, Short columns under eccentric compression. Types of footings; design of isolated / combined footing.

S.No.	Branch	Course Title	Courses ID	Content
				Text : Sinha, S.N., Reinforced Concrete Design, Tata McGraw HillPillai, S.U., and Menon, D., Reinforced Concrete Design, Tata McGraw Hill, 2009References : Nilson, A.H., and Winter, G., Design of Concrete Bren, J.E. and Jirsa, J.O., Reinforced Concrete Fundamentals, John Wiley and Sons, New YorkMcGregor, J.M., Reinforced Concrete Mechanics and Design, Prentice Hall, New York
964	CE	ADVANCED HYDROLOGY	CE610A	Hydrologic cycle, systems concept, hydrologic model classification; Reynold'sTransport Theorem, continuity, momentum, and energy equations; Atmospheric hydrology:atmospheric circulation, water model, IDF relationships; factorsaffecting evaporation, estimation and measurement of evaporation, energy balance method,aerodynamic method, PriestleyTaylor method, and pan evaporation; Surface Water index, ERH & DRH, algorithm for abstraction using GreenAmptequation, SCS method, overland and channel flow modeling, time area concepts, and streamnetworks; Unit Hydrograph: General hydrologic sy definition andlimitations of a UH; UH derivation from single and complex storms; UH optimization usingregression, matrix, and LP methods; Synthetic unit hydrograph, SCurve, IUH; SubsurfaceWater: Soil Green Ampt methods, parameter estimation, ponding timeconcepts; Groundwater Hydrology: Occurrence of groundwater, aquifers & their properties,Darcy's law, permeability, transmissibility, stratific flow intoconfined and unconfined wells; Unsteady flow in a confined aquifer. 21-JUL-2014
965	HSS	ADVANCED COURSE IN SOCIAL COGNITION	PSY798A	The course aims at providing an understanding of the concepts and issues insocial cognition research. The course contents include historical roots of socialcognition; social cognitive principles a cost socialrepresentations; rituals and rhetoric; knowledge and social process; socialcognition and the study of stereotyping, prejudice and discrimination, socialcognition and discourse; social sensibility and
966	DES	SPECIAL STUDIES/PROJECT COURSES IN DESIGN	DES698A	Cognitive Design; Design Management; Human factors in Ergonomics Design;Usability & Usercentric Design; Axiomatic Design; Human Computer InterfaceDesign (HCI), etc.
967	CE	PHYSICAL GEODESY	CE678A	Geometric geodesy: DatumsHorizontal & Vertical, GRS80, WGS84, ITRF. Transformation of Coordinates from one datum to another. Mean Sea level, Geoid and MSL in India. Coordinate Systems in Geodesy; Astrogeodetic deflection.Physical geodesy: Fundamentals of potential theory, Laplace Equation in spherical coordinates, Solution of Laplace Equation, Brun's formula, Fundamental equation of Phys Anomaly, Gravity reductions: Free Air, Bouger and Isostatic reduction, Earth Gravity Models, Potential number and different height systems.Map Projection: Introduction to Map projection, Purpose and me Indian Grid System and UTM.Astronomy: Celestial Sphere, Definition of terms in Astronomy, Celestial coordinate systems, Variations in Celestial coordinates. Precession and Nutation. Time systems Sider parameters and Leap second.Satellite geodesy: Introduction to Satellite Geodesy, Keplerian Motion, Geometry of ellipse, Kepler ellipse in space, Perturbed satellite motion, Lagrange's and Gaussian f
968	MTH	PROBABILITY AND STATISTICS	MSO201A	Probability: Axiomatic definition, properties, conditional probability, Bayes' rule andindependence of events. Random variables, distribution function, probability mass and densityfunctions, expectation geometric, negative binomial, hypergeometric, Poisson,exponential, gamma, Weibull, beta, Cauchy, double exponential, normal. Reliability and hazardrate, reliability of series and parallel systems. Joint dis correlation. Functionsof random variables. Weak Law of large numbers and Central limit theorems.Statistics: Descriptive statistics, graphical representation of the data, measures of location andvariability consistency. Confidence intervals for mean, difference ofmeans, proportions. Testing of hypothesis; Null and alternate hypothesis, Neyman Pearsonfundamental lemma, Tests for one sample and two sam 1. Introduction to Mathematical Statistics, by R V Hogg, A Craig and J W McKean.2. An Introduction to Probability and Statistics by V.K. Rohatgi & A.K. Md. E. Saleh.3. "Introduction to Probability and Statistics Introduction to Probability and Statistics for Engineers and Scientists by S.M. Ross
969	ECO	PROGRAMME EVALUATION: METHODS AND APPLICATIONS	ECO756	Experimental Design, Measurement Error, Omitted Vairables Bias, Quasi_Experimental Designs, Selection on Observables: Matching and Propensity Scores, Difference in Differences, Instrumental Variables 1. Wooldridge, J.M. (2010), Econometric Analysis of Cross Section and Panel Data, 2nd edition. 2. Cameron, A.C. and Trivedi, P.K. (2005), Microeconometric: Methods and Applications. 3. Cameron A.C. and "Mostly Harmless Exonometrics: An Empiricist Companion". 5. Journal Papers to becrecommended by the Instrucor.
970	MTH	ABSTRACT HARMONIC ANALYSIS	MTH638A	Banach Algebras and Spectral theory, Locally compact groups, Basic representation theory, Analysis on Locally compact abelian group, Analysis on compact groups, Group C* algebra and structure of dual Recommended books: 1. G.B. Folland: Course in Abstract harmonic analysis, Studies inAdvanced Maths., CRC Press, Boca Raton, etc., 1995.
971	BSBE	SPECIAL TOPICS IN BIOLOGICAL SCIENCES AND BIOENGINEERING	BSE682	
972	CHM	PRINCIPLES OF PHYSICAL CHEMISTRY	CHM629A	Atomic and Molecular structure, Molecular Spectroscopy, Concepts of StatisticalThermodynamics, Electrochemistry, Chemical Kinetics, Photochemistry
973	CHM	PRINCIPLES OF ORGANIC CHEMISTRY	CHM609A	Stereochemistry, mechanisms of selected reactions, Reactive intermediates,oxidation, Reduction, CC bond formations, synthesis of some useful naturalproducts
974	BSBE	LABORATORY METHODS	BSE602	Demonstration based course. Usage of different equipments will be demonstrated. Demonstration based course. No text book.
975	BSBE	PROFESSIONAL COMMUNICATION	BSE601	1. CRITICAL READING : Context, How is the text argued, Central claims, Kinds of reasoning, Examine Evidence, Examine Interpretations, Evaluations,Limitation of study2. CRITICAL WRITING : Learning to write Books, Proposals,Patents.3.ORAL PRESENTATION : Conference, Scientific, Teaching, Popular, 4. ETHICS : Experimentation, Writing No prescribed textbook.
976	BSBE	BIOCHEMICAL ENGINEERING	BSE612A	A. Introduction to biochemical engineering, bioprocesses, bioproducts and biochemical technology with specific examples Three lectures.B. Upstream process: Microbial, mammalian and plant systems for Five lectures.C. Enzyme catalyzed reactions and processes. Cell and enzyme immobilization. Bioreactor design and applications. Instrumentation and control Six lectures. D. Downstream process: Biosep centrifugation, precipitation and extraction Four lectures. E. Liquid chromatography principles, plate and rate theory, ion exchange, gel filtration, affinity chromatography, hydrophobic interaction and reverse recovery, expanded bed separations, affinity precipitations, aqueous twophase processes, monolithic chromatographic separations Six lectures. G. Polishing, crystallization, drying, scaleup consideration, population, biological wastewater treatment, anaerobic digesters, bioremediation Three lectures.I. Case studies and new developments of bioprocesses paper readings and presentations. H. W. and Clark, D. S. Biochemical Engineering. Marcel Dekker, Inc., 2.Bailey, J. E. and Ollis, D. F. Biochemical Engineeing Fundamentals. McGrawHill, Inc., 3.Belter, P. A., Cussler, E. L. and Hu, W. S. Bioseparation Proteins: Methods and Protocols. Humana Press5.Shuler, M. L. and Kargi, F. Bioprocess Engineering Basic Concepts
977	MTH	LOCALLY CONVEX SPACES	MTH639A	Topological linear spaces, Equicontinuity, Function spaces, Convexity & convextopological spaces, HahnBanach theorem, Barrelled spaces, Principle of uniformboundedness, Bornological spaces, Duality
978	IME	ADVANCED DECISION MODELS	IME637	Review of linear and integer linear programming. Multistage decision models: Dynamic programming. Network flow problems: Shortest path, maximum flow and minimum cost flow problems; Network optimization Uncostrained optimization; Lagrangian relaxation and KKT conditions; Convex optimization; Search, gradient and penalty based methods; Quadratic programming. Metaheuristics and their applications to Markov chains; Queues and queuing networks. Course Reference : Katta G. Murty, Linear Programming, Wiley. Laurence A. Wolsey, Integer Programming, Wiley. Richard Bellman, Dynamic Programming, Dover. R.K. Ahuja, T.L. Magnanti, J.B. Orlin, Network First Course in Optimization Theory, Cambridge. S. Boyd and L Vandenberghe, Convex Optimization, Cambridge. J. Nocedal, S.J. Wright, Numerical Optimization, Springer. ElGhazali Talbi, Metaheuristics: Wiley. D. Gross, C.M. Harris, Fundamentals of Queueing Theory, Wiley. Wayne L. Winston, Operations Research: Applications and Algorithms.
979	MTH	GRAPH THEORY	MTH657A	Basic definitions. Blocks. Ramsey Numbers. Degree sequences. Connectivity.Eulerian and Hamiltonian Graphs. Planar graphs and 5colour theorem. Chromaticnumbers. Enumeration. MaxFlow MinCut The
980	HSS	ELEMENTS OF VISUAL REPRESENTATION	ART106AA	1. Space Two dimensional Spaces Illusion of Space One, two and multiple point perspective Three dimensional Spaces Space distribution Negative and Positive Space Object and ground relationship: Form scale Scale Confusion in surreal space3. Illusion of motion Illusion of motion Anticipated and optical movement4. Line Line quality Line and Shape delineation5. Style of Visual Representation Form and Colour Balance Rhythm7. Visual Perception Compositional Arrangement Gestalt Principle of universal whole8. Shape and Volume Naturalism and distortion Naturalism and Idealism Abstract Expression Form and and Value Colour theory Colour Combination Hue and Saturation Colour Characteristics Cognitive perception Optical Colour Mixing11. Visual Culture Cultural Identity through Visual Interface Refinement of

S.No.	Branch	Course Title	Courses ID	Content
				Applied and Commercial Art 12. Medium of Representation (Method and Material) Process of Visual Communication (Thinking, looking and doing) 13. Visual Analysis (Conclusion) Evaluation Assignments, Projects Varieties of Visual Experience Art as image and idea Vol.6 Edmund Burke Feldman Janson, H. W. History of Art New York: Prentice Hall and Harry N.Abrams, Inc 1977 De Lucio Meyer, J. Visual Aesthetics, New York: Dover Publications, 1980 Visual Perception, A Psychology of the Creative Eye, California E. H.Gombrich, Art and Illusion A study in the psychology of pictorial representation The A. W. Mellon lectures in the fine arts, 1956 Bollingen series Movements, Theories, Schools and Tendencies 1900-2000, SKIRA
981	AE	OPTIMAL SPACE FLIGHT CONTROL	AE777A	1. Control Systems (Basic definitions, notation, tracking systems.) 2. Guidance and navigation(Basic concepts, linear regulation and tracking, proportional navigation, crossproduct steering). 3. Optimal control and the minimum principle, HamiltonJacobiBellman formulation, endpoint constraints, EulerLagrange formulation, twopoint boundaryvalue solution techniques, optimal terminal control with interior constraints and Hinfinity robust optimal control.) 4. Optimal guidance and control of rocket flight (Terminal guidance of interceptors, nonplanar tracking systems (3DPN),Goddards problem, 2PBVP solutions for gravitation control (Introduction to orbital mechanics, HillClohesseyWiltshire model, autonomous rendezvous and docking, minimum energy transfer, Lamberts problem, optimal guidance of reentry vehicles, nonplanar trajectories) Textbook:1. Tewari, A., Advanced Control of Aircraft, Spacecraft, and Rockets, John Wiley & Sons, Chichester, 2011. References:1. Tewari, A., Atmospheric and Space Flight Dynamics, Birkhauser, Boston, 2006 Control. Dover, 2007.
982	MME	SURFACE PHENOMENA AND CHARACTERIZATION	MME605	Surface Chemistry/Interaction (surface charge, dipoles, energies, interfacial chemical reactions), Surface Phonons/Plasmons (Quantisation of plasma, light and, sound, Brillouin Zones), Elastic/Inelastic scattering, Lattice, Brillouin Zone, Surface Diffusion (Fick's law, intergranular/amorphous layer formation), Surface Energy/Young's Equation (Capillary/surface tension in fluids), Surface Sensitive Properties (Contact angle in wetting surfaces), Density functional theory of atomic equilibrium, Case Studies: Fe-B system, and Al ₂ O ₃ /carbon nanotube, Surface Modification/Unfunctionalization Nanostructures: Self Assembly (Block copolymers, dendrimers, micelles, etc.), Nanofabrication filters : purification of gases/chemicals etc). I. Materials Science and Technology. Ed. R.W. Kahn, P. Hassen, E.J. Kramer, Vol. 2n/2b, Wiley YCII (2005). 2. Physics of Atoms and Molecules. B.H. Bransden, and C.J. Joachain, Longman (1996). 3. Surface Phenomena. I. Materials Science and Technology. Ed. R.W. Kahn, P. Hassen, E.J. Kramer, Vol. 2n/2b, Wiley YCII (2005). Physics and Chemistry of Materials, J. I. (Gersten, F. W. Smith. Wiley (2001). 5. Elementary Solid State Physics. M. A. Omar, Addison Wesley (2001). 6. Course materials will be supplemented with handouts, references and projects.
983	PHY	REVIEW OF STATISTICAL MECHANICS	PHY604A	Problem oriented review of Statistical Mechanics. Review of thermodynamics: Laws of thermodynamics; thermodynamics of phase transitions and phase diagram. Review of Ensembles and rules of calculation for quantum gases. Models of classical interacting systems : Ising model in 1 dimension: exact solution by transfer matrix ; PeierlsGriffiths argument for Ising model in 2 dimensions; Meanfield approximation for interacting systems : Density matrix, Transverse Ising model, exact solution by JordanWigner transformation, Heisenberg model magnons; MerminWagner theorem; general theory of quantum phase transition equation; Systems near equilibrium Linear Response Theory, FluctuationDissipation Theorem; Escape over a barrier relaxation phenomena; critical dynamics. Supplementary reading materials for term paper: Various types of series expansions, Boltzmann equation, Molecular hydrodynamics, BBGKY hierarchy; Random and glassy systems, Linear and branched Polymers, Percolation; XY model and vortices superconductivity 1. M. Kardar, "Statistical Physics of Particles"; (CUP, 2007). 2. R.K. Pathria, "Statistical Mechanics"; (Academic Press, 2007). 3. D. Chowdhury and D. Stauffer, "Principles of Equilibrium Statistical Mechanics"; (Springer, 1996). 5. S.K. Ma, "Statistical Mechanics"; (World Scientific, 1985). 6. L.D. Landau and E.M. Lifshitz, "Statistical Mechanics"; (Academic Press, 1975) 7. K. Huang, "Statistical Mechanics"; (Wiley, 1963).
984	PHY	REVIEW OF QUANTUM MECHANICS	PHY602A	Problem oriented review of Quantum Mechanics. Historical development of quantum mechanics, wave packets, Schrodinger's equation, two level systems. Solution (analytical and numerical) of time independent problems. Numerical solution of the radial Schrodinger equation for arbitrary spherically symmetric potential. Equivalence of Heisenberg approach and Schrodinger approach; matrix mechanics. Quantization of angular momentum, uncertainty principle for time independent problems, WKB approximation. Time dependent Schrodinger equation. Time dependent perturbation theory and matter radiation interaction. Selection rules for dipole radiation. Transition from path integral formulation, (iv) identical particles. Problems of current interest, many body physics. 1. J. J. Sakurai, Modern Quantum Mechanics. 2. L.I. Schiff, Quantum Mechanics 3. E. Merzbacher, Quantum Mechanics 4. R. Shankar, Principles of Quantum Mechanics 5. Loudon, Quantum theory of light 6. Messiah, Quantum Mechanics
985	HSS	TRANSLATION STUDIES	ENG753	The course covers the theoretical aspect of the practice of translation in the west and in India. It includes the history of translation and various forms of translation like retellings and intersemiotic adaptation. It also deals with the translation between unequal languages in a globalized world.
986	CE	APPLICATION OF GEOTECHNICAL ENGINEERING	CE451A	Earth and Earth retaining structures: Slope stability analysis, flexible and rigid retaining wall, gravity, cantilever, counter fort, reinforced earth etc., design and check for stability. Introduction to ground improvement techniques: collapsible soils etc., preloading, vertical drains, stone columns, heavy tamping, grouting etc., Machine foundation and design. Das BM., Principles of Geotechnical Engineering, 2007, Thomson, India Das BM., Theoretical Foundation Engineering, 2008, Cengage Learning India Private limited, India. Bowles JE., Foundation Analysis and Design, 2010, I.K. International publishing house private limited, India. Mittal S., An introduction to Ground Improvement Engineering, 2013, Scientific International private limited, India. Research papers from Journals and Conference Proceedings
987	PSE	PRINCIPLES OF LASERS AND DETECTORS	PSE602A	Introduction to light sources, Lasers, principle of lasing Optical cavities, longitudinal, transverse modes, Stability Interaction of radiation with matter, Spontaneous emission Absorption and stimulated emission Two level system (rate equation based Lasing model) Three and four level lasers CW and pulsed lasers, Q-switching and mode locking Course Reference : *Principles of Lasers, O. Svelto and D. C. Hanna, 5th edition, 2010 *Laser Physics, Peter W. Milonni and Joseph H. Eberly, Wiley, 2nd edition, 2010 *Lasers, Anthony E. Siegman, University of California Press, 1986 spectroscopy, W. Demtroder, 3rd edition, 2004
988	DES	TOPICS IN MOTION PICTURES	DES623A	Topics in motion pictures require a combined practical and theoretical approach for realizing the creative media in its totality. The proposed course would offer students the opportunity to acquire a range of knowledge about film and television media and a general approach to the criticism of film and television media; Approach and methods in these forms of media productions; Overview of the digital media arts field with an emphasis on technological developments; historical directions, key concepts and methodologies through seminar lectures, research presentations, practical exercises and a final project. Classes would be supplemented with viewing a range of products and discussing them in relation to modern methods and explore both technical and creative approaches to the medium. The course will also include short workshops supported by specialized professionals in the related fields.
989	HSS	SEMINAR COURSE IN PSYCHOLOGY	PSY776A	Seminar Course in Psychology
990	CHE	ADVANCED PROCESS DYNAMICS & CONTROL	CHE681A	Process Identification Techniques for SISO & MIMO systems off line & on line; Generalized Predictive Control (GPC); Model Predictive Control (MPC): Dynamic Matrix Control (DMC), Internal Model Control (IMC); MATLAB & SIMULINK software.
991	CHE	NUMERICAL METHODS IN CHEMICAL ENGINEERING	CHE642A	Systems of Linear and NonLinear Algebraic Equations: Successive Substitution, Newton Raphson, Eigenvalues and Eigenvectors of Matrices, Interpolation, Solutions of ODEs (IVP): Runge Kutta, Multistep Methods, Shooting Methods.
992	CHE	ELECTRONIC POLYMERIC & CERAMIC MATERIALS & PROCESSING	CHE463A	Atomic structure, semiconductor materials, solar cells, Transistors and Process Sequence of Fabrication, Control of Microcontamination, Microlithography, Doping, Etching, Oxidation, Chemical Vapour Deposition Molecular Chemistry, Flow Behaviour and Polymer Processing, Polymer Blends and Composites, Applications of Polymers (Exchange Resin etc.), Ceramic Raw Materials and Characterization, Ceramic Processing, Drying, Surface Processing and Shaping.
993	PHY	GAUGE THEORY AND RENORMALIZATION	PHY6900	
994	PHY	REVIEW OF CLASSICAL ELECTRODYNAMICS	PHY603A	Problem oriented review of Classical Electrodynamics. Electrostatics and Magnetostatics: Methods of solving electrostatic problems in cartesian, spherical and cylindrical coordinates, Green's functions, Dielectrics and Magnetic media. Electrodynamics: Faraday's law, Displacement current, Poynting Vector, Conservation laws. Electromagnetic waves in free space and different media, waveguides. Radiation of electromagnetic fields. Scattering and diffraction, Resonant cavities, Optical fibers, Dispersion. 1. J. D. Jackson, Classical Electrodynamics. 2. Landau and Lifshitz, Electrodynamics of continuous media. 3. Griffiths, Electrodynamics. 4. Zangwill, Electrodynamics. 5. Reitz, Christy and Millford, Electrodynamics 6. Purcell and Morfitt, Electricity and Magnetism
995	PHY	REVIEW OF CLASSICAL MECHANICS	PHY601A	Problem oriented review of Classical Mechanics, Newton's laws of motion, Galilean transformations, Particle mechanics, System of particles, Noninertial frames, Pseudoforces. Small oscillations and normal modes, Rigid body motion, Hamiltonian formulation. Phase space, Liouville's theorem, Canonical transformations, Poisson brackets, HamiltonJacobi theory, Actionangle variables. Integrability, Perturbation theory and chaos, relativity. 1. J. V. Jose & E. J. Saletan, Classical Dynamics, Cambridge University Press (1998). 2. I. C. Percival & D. Richards, Introduction to Dynamics, Cambridge University Press (1982). 3. L. D. Landau & E. M. Lifshitz, Mechanics, Butterworths, London (1976). 4. Strogatz, Nonlinear Dynamics and Chaos, Westview Press (2001). 6. M. Tabor, Chaos and Integrability in Nonlinear Dynamics, Wiley Interscience (1974).

S.No.	Branch	Course Title	Courses ID	Content
996	AE	FOUNDATION OF SCIENTIFIC COMPUTING	SE371	Basics of Computing: discretization and numerical errors. ODEs and their computations. Stiff ODEs & parasitic error, solving stiff equation via orthogonalization and compound matrix method. Governing PDEs, wave equation, relation and spacetime resolution. Finite difference methods for wave equation. Discretization of spatial derivatives by polynomial expansion and operators explicit methods. Spectral theory of discrete coarsening. Finite difference methods (FDM) for wave equation. Designing high accuracy, high fidelity methods via error control as an optimization problem. Spectral method visvis discrete computing methods:Aliasing error, aliasing filter. Numerical Methods for Partial Differential Equations WF AmesFundamentals of Computational Fluid Dynamics: T. K. Sengupta.Waves in Fluids M. J. LighthillComputational Aeroacoustics: M. J. Lighthill.L17-SEP-14
997	MSE	DIFFRACTION AND SPECTROSCOPY TECHNIQUES FOR SURFACE CHARACTERISATION	MSE695	DiiTraction and Spectroscopy Techniques for Surface Course Details: Importance of Smfacc Characterization. Present status sensitivity and resolution achievable.Diffrraction Techniques: Revicv.,: of basic electron cliftl'action,LEED and RHEED. Properties of neutron radiation: neutron sources: Small angle neutron scattering; Illustrative analysis using diffraction techniques Spectroscopy: Basic principles of XPS imaging; Auger electron generation: Principle of AES. Chemical e1Tect, Quantitative analysis, Depth profiling,Applications; Static and Dynamic SIMS, Common modes of analysis, quantitative analysis. I. Encyclopedia ofMaterial's Characterisation, C. R. Brunelle. C. A. Evans and S. Vilson,ButterworthHennmann, 1992. Boston.2. Characterisation ofMaterials Volume 2. Editor:E.N. Kaufmann, WileyInterscience, 1992.3. Springer (2003).4. Materials Charocterization. published in 1986 as Volume I 0 of the 9th Edition MetalsHandbook, ASM International. 1986 (Fifth printing 1998).
998	MTH	INTRODUCTION TO LIE ALGEBRAS AND REPRESENTATION THEORY	MTH641	Definitions and first examples. Classical Lie algebras. Ideals and homomorphisms. Niporent Lie algegras. Engel's theorem. Solvable Lie algebras. Lie's theorem. JordanChevalley Decomposition. Complete reducibility and Weyls theorem. Representation theory of the Lie algebra sl(2). Total subalgebras and root systems. Integrality properties. Simple Lie algebras and irreducible root systems. 1. Humphreys, James E. (1972), Introduction to Lie Algebras and Representation Theory, Berlin, New York.2. J.P. Serre, Complex semisimple Lie algebras (trnslated from French: Algebras de Lie complex s1965) 24.09.14
999	HSS	TWENTIETH CENTURY PHILOSOPHY-I	PHI751A	An analytical study of Russells The Philosophy of Logical Atomism andWittgensteins Tractatus LogicoPhilosophicus: Facts and Propositions; Namesand Objects; Definite Description; Picture Theory of M
1000	HSS	METHODOLOGY OF TEACHING AND RESEARCH IN LITERATURE	ENG715A	This course will study selected pedagogical material on the teaching of literaturewith a view to acquainting the student with the major theories and techniquesof literary study as an academic discipline. It
1001	BSBE	BIOCHEMISTRY STRUCTURAL BIOLOGY	BSE631	A. Protein structure and folding : Building blocks of biopolymers, conformational studies of biomolecules, Ramachandran map, protein secondary and supersecondary structures, different classes of tertiary structures, protein foldingB. Enzymes and the structurefunction relationship: Enzyme kinetics, structural basis of catalytic activity, MichaelisMenten kinetics, Enzyme inhibitors and complex reaction schemesC. Structures of RNA and DNA, functional versatility of RNA, proteinDNA interactionsD. Membrane proteins: Lipids, bilayer assembly, biological membranes as solvent for membrane proteins, structural principlesE. The Molecules of Life: Physical and Chemical Principles. John Kurian, Boyana Konforti and David Wemmer, Garland Science, New York (2013)Biochemistry. L. Stryer W. H. Freeman & Company, New York (1995)
1002	BSBE	BIOCHEMICAL ENGINEERING	BSE612	A. Introduction to biochemical engineering, bioprocesses, bioproducts and biochemical technology with specific examples Three lectures.B. Upstream process: Microbial, mammalian and plant systems for production of biochemicals Five lectures.C. Enzyme catalyzed reactions and processes. Cell and enzyme immobilization. Bioreactor design and applications. Instrumentation and control Six lectures. D. Downstream process: Bioseparation techniques: centrifugation, precipitation and extraction Four lectures. E. Liquid chromatography principles, plate and rate theory, ion exchange, gel filtration, affinity chromatography, hydrophobic interaction and reversed phase recovery, expanded bed separations, affinity precipitations, aqueous twophase processes, monolithic chromatographic separations Six lectures. G. Polishing, crystallization, drying, scaleup considerationsH. Microbial population, biological wastewater treatment, anaerobic digesters, bioremediation Three lectures.I. Case studies and new developments of bioprocesses paper readings and presentations Four lectures. 1.Blanch, H. W. and Clark, D. S. Biochemical Engineering. Marcel Dekker, Inc., 2.Bailey, J. E. and Ollis, D. F. Biochemical Engineering Fundamentals. McGrawHill, Inc., 3.Belter, P. A., Cussler, E. L. and Hu, W. S. Bioprocess Engineering Processing of Proteins: Methods and Protocols. Humana Press5.Shuler, M. L. and Kargi, F. Bioprocess Engineering Basic Concepts
1003	ECO	EMPIRICAL METHODS IN APPLIED MICROECONOMICS	ECO541A	Correlation vs Causality, Omitted Variables, Measurement Error, Randomized Experiments, Introduction to the course; Random Variable, Probability Review2. Observational Studies and Regression, Overview of Regression Designs , Applications5. QuasiExperimental Designs, Applications4. Selection on Observables: Matching and Propensity Scores, Applications4. Selection on Observables: Fixed Effects, PrePost Designs and Applications4. Recent Developments in the field: Discussion of papers on new methods and applications5. Joshua D. Angrist, JornSteffen Pischke, Mostly Harmless Econometrics: An Empiricist's Companion, Princeton University Press, 2008A. Colin Cameron, Pravin K. Trivedi , Microeconometrics: MethodologyB. Quarterly Journal of Economics, Journal of Labor Economics, Journal of development Economics etc.
1004	HSS	CONTEMPORARY SOCIAL THEORY	SOC749	Theories of risk society, structuration, disorganized capitalism, social and cultural capital, standpoint theory, postmodernism, and complexity theory. Major contributions of Ulrich Beck, Anthony Giddens, S18-SEP-14
1005	EE	RF MICROELECTRONICS	EE698F	Solar radiations, solar PV modules, maximum power point tracking, partial shading,leakage currents, grounding techniques, panel optimizers, grid integration issues,module integrated inverters, standalone
1006	MSE	SEMI CONDUCTOR DEVICES AND PROCESSING	MSE686	contents: Review of semiconductor physics, carrier statistics, generationrecombination and carrier transport. Devices: PN junctions, Schottky barrier diodes, MOS capacitors, field effect transistors.Planar technology (surface transient enhanced diffusion), ionimplantation, deposition (chemical and physical vapour techniques), etching; Lithography; Device and process integration, withMOSFET as an example. I . Nanomaterials, Nanotechnologies and Design: an Introduction to Engineers and Architects, D.Michael Ashby, Paulo Ferreira. Daniel L. Schodek , ButterworthHeinemann, 2009.2. Handbook ofNanophase Materials, 2003.3. Encyclopedia ofNanoscience and Nanotechnology, Ed.: Hari Singh Nalwa, AmericanScientific Publishers, 2004.4. Handbook ojNanoceramics and their Based Nanodevices (Vol. 2) Edited by T eung
1007	ME	M TECH THESIS (FOR DUAL DEGREE ONLY)	ME699A	M TECH THESIS (FOR DUAL DEGREE ONLY)
1008	CSE	M TECH THESIS (FOR DUAL DEGREE ONLY)	CS699A	M TECH THESIS (FOR DUAL DEGREE ONLY)
1009	CSE	UNDER GRADUATE PROJECT-III	CS498	B TECH PROJECT
1010	CHM	CHEMISTRY COMMUNICATION SKILLS	CHM399A	CHEMISTRY COMMUNICATION SKILLS
1011	CHE	M TECH THESIS (FOR DUAL DEGREE ONLY)	CHE699A	M TECH THESIS (FOR DUAL DEGREE ONLY)
1012	CHE	UNDER GRADUATE PROJECT I	CHE496	PROJECT II
1013	CHE	CHEMICAL ENGINEERING COMMUNICATION SKILLS	CHE399A	CHEMICAL ENGINEERING COMMUNICATION SKILLS
1014	CHE	UG PROJECT II	CHE398A	UG PROJECT (UGP II)
1015	EE	UNDER GRADUATE PROJECT -III	EE491A	UG PROJECT (UGPIII)
1016	EE	ELECTRICAL ENGINEERING	EE395A	ELECTRICAL ENGINEERING UNDER GRADUATE PROJECT III

S.No.	Branch	Course Title	Courses ID	Content
		UNDER GRADUATE PROJECT III		
1017	ECO	PROJECT-II	ECO599	PROJECTII
1018	CHM	UNDER GRADUATE PROJECT-III	CHM491A	UG PROJECT (UGPIII)
1019	BSBE	UNDER GRADUATE PROJECT III	BSE498	PROJECT I
1020	AE	CONTINUUM HYPERSONIC AERODYNAMICS	AE628A	Continuum hypersonics in entry flight. Gen features, Mach no. Small disturbtheory, similitude. Large diflecn similitude; wedge, cone, wing. Unified similitude.Lighthill & other piston analogies, NewtonBuse
1021	ECO	TECHNICAL COMMUNICATION	ECO397A	TECHNICAL COMMUNICATION
1022	CSE	TECHNICAL COMMUNICATION	CS300A	TECHNICAL COMMUNICATION
1023	CHM	GRADUATE SEMINAR	CHM801A	GENERAL SEMINAR
1024	CHM	GENERAL SEMINAR	CHM800B	GENERAL SEMINAR
1025	CHM	GENERAL SEMINAR	CHM800A	GENERAL SEMINAR
1026	CHE	CHEMICAL ENGINEERING COMMUNICATION SKILLS	CHE300A	CHEMICAL ENGINEERING COMMUNICATION SKILLS
1027	PHY	BACHELOR OF SCIENCE PROJECT -I	PHY555A	BATCHLOR OF SCIENCE PROJECT I
1028	PHY	M.SC.RESEARCH PROJECT II	PHY599	M. Sc. Project I
1029	MSE	UNDER GRADUATE PROJECT IV	MSE499	PROJECT II
1030	MSE	UNDER GRADUATE PROJECT III	MSE498	PROJECT I
1031	ECO	UNDER GRADUATE PROJECT-III	ECO498A	UG PROJECT (UGPIII)
1032	CHE	UNDER GRADUATE PROJECT-III	CHE497A	UG PROJECT (UGPII)
1033	CHE	ENGINEERING APPLICATIONS OF RHEOLOGY	CHE676A	Classification of fluid behaviour: Constitutive relations; Rheometry: Flow ofnonNewtonian fluids in closed conduits: Flow in complex geometries: Fixed andfluidized beds; two phase flows, Mixing and agita
1034	CHE	OPTIMIZATION	CHE652A	Mathematical formulation of optimization problems; single variable problems:search techniques; Multivariable problems without constraints: direct methods,first and second order methods; Multivariable
1035	MTH	SPECTRAL THEORY FOR SELF-ADJOINT OPERATORS	MTH632A	Unbounded Operators, Matrix representation, Selfadjointness Criterion, Quadratic Forms, Differential Operators, Selfadjoint Extensions, Functional Calculus, Spectra of Selfadjoint Operators, Semianalytic Uncertainty Principle of Bargmann, Canonical Commutation Relations, Schrodinger representations, Schrodinger Operators, Selfadjointness, A Theorem of Kato, Spectral Theory for Schrodinger Operators N. Akhiezer, I. Glazman, Theory of Linear Operators in Hilbert Space II, Dover, 1961.* J. Blank, P. Exner, M. Havlivcek, Hilbert Space Operators in Quantum Physics, Springer, 2008.* T. Kato, Perturbation The M. Reed and B. Simon, Methods of Modern Mathematical Physics II, Academic Press, 1975.
1036	MTH	NUMBER THEORY	MTH215A	Divisibility, Primes, Congruences, Residue systems, Primitive roots; Quadraticreciprocity, Some arithmetic functions, Farey fractions, Continued fractions,Some Diophantine equations, Bertrands postulate
1037	MTH	SOBOLEV SPACES AND APPLICATIONS	MTH656A	Elements of operator theory and Hilbert spaces; Introduction to the theoryof distributions. Sobolev Spaces : Imbedding and compactness theorems,Fractional spaces and elements of trace theory. Applica
1038	PHY	CLASSICAL ELECTRODYNAMICS II	PHY553A	Special relativity, Minkowski space and four vectors, concept of fourvelocity,four acceleration and higher rank tensors, relativistic formulation ofelectrodynamics, Maxwell equations in covariant form, gau potentials, radiation from an accelerated charge,Larmor formula, bremsstrahlung and synchrotron radiation, multipole radiation,dispersion theory, radiative reaction, radiative damping, scattering byfree ch
1039	MSE	NANOSTRUCTURES AND NANOMATERIALS: CHARACTERIZATION AND PROPERTIES	MSE694	Overview of Nanostructures and Nanomaterials: classification (Dimensionality, Morphology/ shape/structure of nanoentities, New Effect/ Phenomena). Crystallinenanomaterials and defects therein. Hybr structures built out of nanosized building blocks (nano to macro). Eucledian, Hyperbolic and Spherical space structures. Nanostructures: Carbon Nanotubes, Fullerenes, Nanowires, Graphene, Quantum D reconstruction. Surface reconstruction and reconfiguration. Adsorption and Absorption. 1. Nanomaterials, Nanotechnologies and Design: an Introduction to Engineers and Architects, D.Michael Ashby, Paulo Ferreira, Daniel L. Schodek, ButterworthHeinemann, 2009.2. Handbook of Nanophase 2003.3. Encyclopedia of Nanoscience and Nanotechnology, Ed.: Hari Singh Nalwa, American ScientificPublishers, 2004.4. Handbook of Nanoceramics and their Based Nanodevices (Vol. 2) Edited by Tseu Dutta, H. F. Tibbals, A.K. Rao, CRC Press (2008).
1040	MTH	TOPICS IN TOPOLOGY	MTH628A	Classification of 2dimensional surfaces; Fundamental group; Knots and coveringspaces; Braids and links; Simplicial homology groups and applications; Degreeand Lefschetz Number; BorsukUlam Theore
1041	MTH	ALGORITHMS	MTH524A	Preliminaries: Introduction to algorithms; Analyzing algorithms: space and timecomplexity; growth of functions; summations; recurrences; sets, etc. GreedyAlgorithms: General characteristics; Graphs: mi qucksort. Dynamic Programming: Elements of dynamicprogramming; The principle of optimality; The knapsack problem; Shortestpaths; Chained matrix multiplication. Graph Algorithms: Depthfirst search FFT; Efficient FFT implementation.Number Theoretic Algorithms: Greatest common divisor; Modular arithmetic;Solving modular linear equations. Introduction to cryptography. ComputationalGeometry: Li Heuristic andApproximate Algorithms: Heuristic algorithms; Approximate algorithms; NPhardapproximation problems.
1042	HSS	HISTORICAL LINGUISTICS	ENG754	This subject is an introduction to historical linguistics, the study oflanguage change and linguistic relationships. We will explore the kinds ofchanges that languages undergo throughout their history, draw and demonstrated, togetherwith methods of language classification and subgrouping. We will alsoinvestigate language change in progress and relations between language andcultural history.Topics to be changeMorphological changeSyntactic changeLanguage classification, subgrouping and regroupingLinguistic reconstruction: comparative and other methodsModels of language changeLanguage chang 1998. Historical Linguistics: An Introduction. EdinburghUniversity PressMillar, Robert McColl (ed.) 2007. Trasks Historical Linguistics. HodderArnold Bhat, D.N.S. 2001. Sound Change. MLBD.Croft, William 2000. Explaining Language Change: An Evolutionary Approach.Longman.Crowley, Terry 2010. An Introduction to Historical Linguistics. OxfordUniver

S.No.	Branch	Course Title	Courses ID	Content
				Method Reviewed:Regularity and Irregularity in Language Change. Oxford University Press[D&R below]Fortson, Benjamin W. 2004. IndoEuropean Language and Culture. Blackwell.Hock, Hans Henrich 1999. HistoricalLinguistics. BlackwellLightfoot, D.W. 1979. Principles of Diachronic Syntax. CUPMcMahon, April S. 1994. Understanding Language Change. CUPSchendl, Herbert 2001. Historical Linguistics. Oxf
1043	MSP	GRADUATE SEMINAR	MS801	Graduate Seminar
1044	MSP	STRUCTURAL & MAGNETIC PROPERTIES OF MATERIALS	MS601A	Crystal structure, Bonding of atoms, Crystal chemistry, Equilibrium thermodynamics,Phase equilibria, Phase transformations, Dia, Para, Ferro, Ferri and Antiferromagnetism,Magnetic domains, Anisotropy
1045	HSS	APPLIED STATISTICS FOR SOCIOLOGISTS AND OTHER SOCIAL SCIENTISTS	SOC752A	1. Styles of causal thought and models of controlled experiment: meaning of statistical inference concepts of relative error and confidence interval; simple bivariate approaches and unmatched difference flowcharts. 3. Models and explanatory fit; modelling social phenomena; measures of life expectancy, disease burden and health inequality. 4. Analysis of frequencies - chisquare, odds ratios and relative risk coefficients; uses of dummy variables; role of proxies; structural equation models; analysing longitudianl and cohort data; tempo effects. 6. Standardized rates and ratios as dependent vairable; ratios of rates. Articles from journals : Americal Sociological Review, American Journal of Sociology, British Journal of Sociology, and Population and Development Review. (i) Emile Durkheim, Suicide: Studey in Sociology of Social Sciences, Wadsworth, 2011. (iii) Thomas J. Linneman, Social Statistics Routledge, 2011. (iv) Jay Alan Weinstein, Applying Social Statistics: An Introduction to Quantitative Reasoning in Sociology, Rowman
1046	AE	BOUNDARY LAYER INSTABILITY AND TRANSITION	AE617A	NavierStokes eqn. and its various forms, thin shear layer approxn. Varioustypes of flows. Instabilities in laminar flows. Relationship of instability theorywith transition to turbulence. Transition prediction. F
1047	SDS	STATISTICAL INFERENCE	SDS755	Population and samples; Parametric and nonparametricmodels; Exponential andlocationscale families; Sufficiency and minimal sufficiency; Complete statistics;Unbiased and UMVU estimation; Asymptotic theory of maximum likelihood; Asymptotically efficient estimation; Varianceestimation; The jackknife; The bootstrap; The NP lemma; MLR; UMP tests forone and two sided hypotheses; Unbiased and similarity; Uniformly most powerful tests; Bayes tests; Pivotal quantities; Inverting acceptanceregions of tests; The Bayesian confidence interval; Prediction sets; Length ofconfidence intervals; UMA and UMAU confidence sets; Invariant con
1048	HSS	MORAL JUDGEMENT	PHI757A	The details of the studies in this course will be designed for advanced studentsdepending on their fields of research.
1049	HSS	SOCIAL MOVEMENTS : OLD AND NEW	SOC746A	The aim of the course is to define the various types of social movements,and understand their role in striving for social change or transformation. Itwill examine theories of social movements in the context of various actors, will be studied with special reference toIndia.
1050	HSS	UNDERSTANDING ORGANIZATIONAL BEHAVIOUR	PSY786A	A coherent introduction to organizational psychology, historically takingoff from industrial psychology and human relations movement. The coursetakes a look at manin organizations in a social environm
1051	MSE	MATERIALS PROCESSING	MSE305A	1. Solidification Processing 2. Mechanical Working 3. Metal and Ceramic Powder Processing 4. ThinFilm and Coating Techniques 1. R.W. Heine, C.R. Loper, and P.C. Rosenthal, Principles of Metal Casting, 2nd ed., 1967.2. A. Upadhyaya, G.S. Upadhyaya: Powder Metallurgy Science, Technology and Materials (2011)3. Donald Leonard S Theory & Practice, Pergamon Press, Exeter, UK, 1983.
1052	MSE	PRINCIPLES OF METAL EXTRACTION & REFINING	MSE304A	1. Thermodynamics and Kinetics in metals extraction and refining 2. Principles of Mineral Beneficiation 3. Principles of Pyrometallurgy 4. Principles of Electrometallurgy 5. Principles of Hydrometallurgy 1. DR Gaskel: An Introduction to Metallurgical Thermodynamics, McMillan Publishing Co., 1992 2. Barry Willis: Mineral Processing Technology, Elsevier 3. T.A.Engh: Principles of metal refining, Oxford Scie Metallurgy, 2nd Edition, Tapiar Academic Press
1053	MSE	ELECTRONIC & MAGNETIC PROPERTIES OF MATERIALS	MSE303A	1.Introduction to electronic structure 2. Electronic structure in crystalline materials 3. Electron Dynamics 4.Semiconductors 5. Ionic conductors 6. Dielectric materials 7. Magnetic materials 8. Optical materials 1. Electronic Properties of Materials: An Introduction for Engineers, Rolf E. Hummel, Springer Verlag, 19852. Physical Properties of Semiconductors, Charles M. Wolfe, Nick Holonyak and Gregory E. Stillman, Wiley, 19764. Advanced Theory of Semiconductor Devices, Karl Hess, Prentice Hall, 19885. Advanced Semiconductor Fundamentals, Robert F. Pierret as part of Modular Series on Solid State Devices Vol. 6, Addison Wesley, 19906. Semiconductor Materials, L. Solymar and D. Walsh, Oxford University press, 19988. Physics of Solids, C. A. Wert and R.M. Thomson, McGrawHill Book Company, 1970 or later9. Physics of Semiconductor Devices by JP Cardona, Wiley, 2011
1054	MSE	MECHANICAL BEHAVIOUR OF MATERIALS	MSE302A	1. Elasticity 2. Plasticity 3. Dislocations 4. Applications Suggested text books:1. Mechanical Behaviour of Materials, M.A. Meyers and K.K. Chawla2. Introduction to Dislocations, Hull and BaconReference Material:3. Mechanical Metallurgy, G.E. Dieter4. Mechanical Properties of Metals, C. Crandall and N.C. Dahl7. Deformation and Fracture Mechanics, R.W. Hertzberg8. Mechanical Testing, Metals Handbook9. Recrystallization and Related Annealing Phenomena, F.J. Humphreys
1055	PHY	REVIEW OF MATHEMATICAL METHODS IN PHYSICS	PHY605	Problem oriented review of Mathematical Methods in Physics.Vector spaces Discrete and continuous: orthogonality, operatoralgebra. Hermitian and unitary operators, projection operators,matrices and adjoint operators. Orthogonal and orthogonal polynomials, generating functions. Integraltransforms (e.g Fourier, Laplace, etc.) and applications to physics. Differential equations: General introduction to ordinary differentialequations, linear and nonlinear equations, exact solution, inhomogeneous equationsGreen's function, SturmLiouville theory, partial differential equations, characteristics,Boundary conditions. Special functions and applications in Physics. Complex analysis andLaurent Series, The residue theorem. Applications of complexanalysis to physics problems. 1. Sadri Hassani, Mathematical Physics: a modern introduction to its foundations (Springer) 2. Arfken, Weber Mathematical Methods for Physicists (Academic Press) 3. Tulsı Dass and S. K. Sharma, Mathematical Physics (Springer) 4. T. Arfken, Mathematical Physics (Scientific) 5. Mathews, Walker Mathematical Methods of Physics (AddisonWesley) 6. Schaum Series Vector Analysis 7. A. W. Joshi, Matrices and Tensors in Physics (New age international)
1056	MSP	INTRODUCTION TO NANOMATERIALS AND NANOTECHNOLOGY	MS617A	Effects of confinement and finite sizezero, one and two dimensional nanostructures(concepts of surface and interfacial energies), Intermolecular andinterfacial forces in organic, polymeric, biological and inorganic materials, mesoscale thermodynamics, Gibbs treatment of interfaces,mesoscale fluid dynamics, thin soft films, mesoscale phenomena in soft matterand applications: adhesion, wetting, nucleation, Nanofabrication of nanomaterials: cluster fabrication, LangmuirBlodget growth, Nanolithography,Scanning probe lithography, Micro contact printing, Synthesis of nanoparticlesand films: solgel, hydrothermal, freeze drying, intercalation, attachment of nanocrystlline materials, Carbonnanotubes, short and long term applications and perspectives, Demonstrationof some techniques in preparation and characterization of nanomaterials.
1057	BSBE	DEVELOPMENTAL BIOLOGY	BSE652A	1. Concepts in Developmental Biology :A. Axis specification and patterning SIX lecturesB. Cell fate Determination, Specification, Commitment, Differentiation, Proliferation Vs Differentiation, Induction, Control of gene expression Classical techniques e.g. Cell labeling, grafting ONE lectureB. Candidate gene approach i. Saturated mutagenesis screens FOUR lectures ii. Lessons from other species/contexts FOUR lectures3. Developmental Biology Recommended Text Book Developmental Biology, Scott F. Gilbert Ninth Edition
1058	HSS	MANAGEMENT OF ORGANIZATIONAL BEHAVIOUR	PSY787A	The course focuses on a practical training to understand the human behaviour at workin order to predict the effectiveness and wellbeing in various types of organizations,and to enhance the understanding of
1059	HSS	SOCIAL THEORY IN LATE TWENTIETH CENTURY	SOC745A	Canons in sociological theory, breaking with modernity, post positivist and poststructural social thought, post modernism in social theory: present images andfuture possibilities, Writings of J Baudrillard, Michel Foucault, Zygmunt Bauman, Pierre Bourdieu, Anthony Giddens, Jürgen Habermas, Hans Jürgen
1060	HSS	THE WRITERS VISION OF THE FUTURE	ENG436A	This course identifies and defines visionary literature through a broadrange of disciplines and/or literary genres. Open to engaging with texts from differentdomains, from philosophy to economics to science to religion to art to literature to asprophecy. In the process, the course examines the ways in which the visions articulatedby such texts intersect with their and our realities, their purpose and effectiveness, as well as their relevance for society. No specific textbook or reference material. Thespecific texts chosen for the course in a particular semester will determine the appropriatereference materials.
1061	MTH	PARTIAL DIFFERENTIAL EQUATIONS	MSO203M	Sturm Liouville BVP: introduction, examples. Sturm Liouville BVP: orthogonal functions, Sturm Liouville expansions. Fourier series, convergence of Fourier series, Fourier series with arbitrary period. Fourier series with arbitrary period. Introduction to PDE, linear, nonlinear (semi linear, quasi linear) examples, order of PDEs. First order (linear, semi linear) PDEs, interpretation, method of characteristics. First order (linear, semi linear) PDEs, method of characteristics. Classification of 2nd order PDEs, Canonical form: hyperbolic equations. Canonical form: parabolic equations, elliptic equations. Wave equations: D'Alembert's formula, Duhamel's principle. Wave equation

S.No.	Branch	Course Title	Courses ID	Content
				equation: solutions for initial boundary value problems. Laplace and Poisson equations: Uniqueness and maximum principle for Dirichlet problem. Laplace and Poisson equations: BVP in 2D (rectangular, 1. E. Kreyszig. Advanced Engineering Mathematics. (8th Edition) 2. T. Amarnath. An Elementary Course in Partial Differential Equations.
1062	ME	FLOW HEAT & MASS TRANSFER THROUGH POROUS MEDIA	ME673	1. Fundamentals: REV, Mass, momentumand energy transport, Darcy and NonDarcy equations,equilibrium and nonequilibrium conditions, species transport, radioactive decay. 2. Effective medium approximation for porous media. 3. Flow in porous media: cylindrical, boundarylayers, reservoir problems. 4. Special topics: Field scale and stochastic modeling, Turbulent flow, compressible flow, multiphase flow, numerical techniques, hierarchical porous media, multiscale storage systems. 6. Experimental techniques: Flow visualization, quantitative methods, inverse parameter estimation.
1063	ME	MICROMACHINING	ME665	1. Principles of Heat Transfer in Porous Media, by M. Kaviany, Springer New York(1995).2. Transport Phenomena in Porous Media, Volumes 1111, edited by D. R. Ingham and LPop, Elsevier, New York (1999).3. Transport Phenomena in Porous Media, J. Bear and Y.Bachmat, Kluwer Academic Publishers, London (1990).5. Enhanced Oil Recovery, L.W. Lake, GulfPublishing Co. Texas (1989).6. The Mathematics of Reservoir Simulation, R. S. Verrill, Academic Press, New York (1990).
1064	ME	BIOMEMS AND MICROSYSTEMS TECHNOLOGY	ME774A	1. Introduction to the Course & Classification of MMPs, Part1: Mechanical Type Advanced Micromachining Processes, 2. Abrasive Jet Micro Machining (AJMM), 3. Ultrasonic Micro Machining (USMM), 4. Laser Micro Machining (LMM), 5. Chemical Mechanical Polishing (CMP), 6. Chemo mechanical Polishing (CMP), 7. Magnetic Abrasive Finishing (MAF), 8. Magnetorheological Finishing (MRF), 9. Magnetorheological Abrasive Flow Finishing (MRAFF), 10. Magnetic Abrasive Finishing (MAF), 11. Electric Discharge Micro Machining (EDMM), 13. Wire EDM, EDDG, ELID, 14. Laser Beam Micro Machining (LBMM), 15. Electron Beam Micro Machining (EBMM),16. Ion Beam Machining
1065	AE	BOUNDARY LAYER THEORY	AE481A	1. Introduction to Micromachining, V. K. Jain (Ed.), Narosa publisher, 2010.2. Micromachining, J. A. McGeough,3. Micromanufacturing, V. K. Jain (Ed.), CRC press, 2012.4. Micromanufacturing & Nanotechnology, V. K. Jain (Ed.), Narosa publisher, 2010.
1066	CE	GLOBAL NAVIGATION SATELLITE SYSTEMS(GNSS)	CE674	Introduction to BioMEMS and Microsystems technology, Biochips/ biosensors and introduction to device fabrication, Introduction to Cell biology, DNA & Protein chemistry, Microfluidics, Biochip Sensors & Biochips andLabonchip devices, Introduction to MEMS Design. Microfluidics: Continuum mechanics at small scales, Basics of microfluidics, Gas flows, liquid flows, boundary conditions, low Reynold's no. flow, dielectrophoresis, microfluid'cs for internal flow eontrol (micropumps and microvalves, cfevice building and characterization),
				1. Fundamentals ofMicrofabrication (Second Edition), Marc J. Madou, CRCpress Taylor and Francis Group, 6000 Broken Sound Parkway NW, Suite 300,Boca Raton, FL334872724, 2002.2. BioMEMS Technology, Marc J. Madou, CRCpress Taylor and Francis Group, 6000 Broken Sound ParkwayNW, Suite 300, Boca Raton, FL334872724, 2006.3. Biomolecular sensing, processing and analysis, Rashid Bashir, Steve T.Werely, Mauro Ferrari, Springer Science and Business Media LLC, 2010.
1067	CE	ENVIRONMENTAL QUALITY AND POLLUTION	CE311A	Conclusions from small BL thickness, BL eqns, exact & similar solns: Blasius,Howarth & Merk. Methods: continuation & integral conditions, Polhausen, Walz,Weighardt. Axisymm BL. Mangler transformation.
1068	CE	SEDIMENT TRANSPORTATION	CE616A	Properties of sediment, incipient motion, bed load, suspended load, totalload, sediment measurements, regime concept, bed form mechanics, plan formand stream bed variations of rivers, reservoir sedimentation
1069	CE	SEDIMENT TRANSPORTATION	CE616	Properties of sediment, incipient motion, bed load, suspended load, totalload, sediment measurements, regime concept, bed form mechanics, plan formand stream bed variations of rivers, reservoir sedimentation
1070	DES	DESIGN PROJECT I	DES681A	Introduction to Graphics, Introduction to Graphics Software, Introduction toStyle, Introduction to Multimedia Application, Introduction to CAD Application,User Consumer Interaction Study, Media Communication
1071	DES	FORM AND STYLE	DES625A	Theory : Form envelops and assists function and in the process creates a networkof values that is termed as Style. Studies in form and style concerns theconceptualization, exploration, and development of form. Visualization like Fantasy. Metaphors, Cultural connotations andBionics in the context of form making.From analysis Analysis of principles of form in relation with society study ofevolution of forms in product design andform, Light space analysisFor Visual communication form will ne explored in the context of conceptcommunicationand the application of information theory to visual structures.This will include case studies on market behaviorand towards forms.Studio: Form development assignments in both Product and visual communicationswill deal with the evolution of form in relation with exploration and function.Each assignment will be related to market trendsas related to the study of Form. The study of the design process will help inunderstanding the shifts between analytical and creative phases in differentstages of design that is conducted throughthe assignments.The course is envisaged in collaboration with assignments from the industrywherever possible in which the primary work will take place in the Lab andthe critical evaluation in the studio.
1072	DES	DESIGN THEORY	DES601A	Design Philosophy, Art (aesthetics) in ID, History of Design, Human Experiencein Design, Design Elements, Design Principles, Theory of Colour, Colour Aesthetics,Subject of Colour, Design Paradigm, Art Design, Texturein 2D and 3D surface; Colour Aesthetics; Product Analysis and Ergonomics,Product Design and Developments.
1073	HSS	ELEMENTS OF VISUAL REPRESENTATION	ART106A	1. Space Two dimensional Spaces Illusion of Space One, two and multiple point perspective Three dimensional Spaces Space distribution Negative and Positive Space Object and ground relationship: Form and space Scale Scale Confusion in surreal space3. Illusion of motion Illusion of motion Anticipated and optical movement4. Line Line quality Line and Shape delineation5. Style of Visual Representation Form and Colour Balance Rhythm7. Visual Perception Compositional Arrangement Gestalt Principle of universal whole8. Shape and Volume Naturalism and distortion Naturalism and Idealism Abstract Expression Form and Colour and Value Colour theory Colour Combination Hue and Saturation Colour Characteristics Cognitive perception Optical Colour Mixing11. Visual Culture Cultural Identity through Visual Interface Refinement of Visual Communication Applied and Commercial Art12. Medium of Representation (Method and Material)Process of Visual Communication (Thinking, looking and doing)13. Visual Analysis (Conclusion)EvaluationAssignments, Final ProjectVarieties of Visual ExperienceArt as image and idea Vol.6Edmund Burke FeldmanJanson, H. W. History of ArtNew York: PrenticeHall and HarryN.Abrams, Inc 1977De LucioMeyer, J. Visual Aesthetics,New York: Dover Publications, 1969Visual Perception, A Psychology of the Creative Eye, CaliforniaE. H.Gombrich, Art and IllusionA study in the psychology of pictorial representationThe A. W. Mellon lectures in the fine arts, 1956Bollingen series in art and architecture, Princeton University Press, 1956Movements, Theories, Schools and Tendencies 19002000, SKIRA
1074	BSBE	TISSUE ENGINEERING	BSE614	What is tissue engineering? Scope and objective of tissue engineering; Principles of tissue engineering; Essential components of tissue engineering; Materials Science/Engineering aspects (degradable materials, biomaterials aspects (choice of cell type, progenitor cells and cell differentiations); Molecular biology aspects (cell signaling moleculesgrowth factors, cell attachmentintegrins); Drug delivery in tissue engineering; Correlation of tissue engineering with other disciplines
1075	BSBE	DEVELOPMENTAL BIOLOGY	BSE652	1. Concepts in Developmental Biology :A. Axis specification and patterning SIX lecturesB. Cell fate Determination, Specification, Commitment, Differentiation, Proliferation Vs Differentiation, Induction, Cooperation, Classical techniques e.g. Cell labeling, grafting ONE lectureB. Candidate gene approach i. Saturated mutagenesis screens FOUR lectures ii. Lessons from other species/contexts FOUR lectures3. Developmental Biology Recommended Text Book Developmental Biology, Scott F. Gilbert Ninth Edition
1076	MTH	PARTIAL DIFFERENTIAL EQUATIONS	MSO203B	Sturm Liouville BVP: introduction, examples. Sturm Liouville BVP: orthogonal functions, Sturm Liouville expansions. Fourier series, convergence of Fourier series, Fourier series with arbitrary period. Fourier series with arbitrary period. Introduction to PDE, linear, nonlinear (semi linear, quasi linear) examples, order of PDEs. First order (linear, semi linear) PDEs, interpretation, method of characteristics. First order (linear, semi linear) PDEs, Classification of 2nd order PDEs, Canonical form: hyperbolic equations. Canonical form: parabolic equations, elliptic equations. Wave equations: D'Alembert's formula, Duhamel's principle. Wave equation: solutions for initial boundary value problems. Laplace and Poisson equations: Uniqueness and maximum principle for Dirichlet problem. Laplace and Poisson equations: BVP in 2D (rectangular, 1. E. Kreyszig. Advanced Engineering Mathematics. (8th Edition) 2. T. Amarnath. An Elementary Course in Partial Differential Equations.
1077	CHM	BASIC INORGANIC CHEMISTRY	CHM242A	Basic Concepts (10)1. Molecular symmetry, point groups and character tables2. Bonding models3. Chemical forces4. Acids and BasesMainGroup Chemistry (12)5. Chemistry of selective maingroup elements6. Coordination metal complexes, Bonding in transition metal complexes Valence Bond, Crystal Field and Molecular Orbital theories, effects of d-orbital splitting7. Organometallic compounds: EAN rule, metal carbonyls, metal complexes (5)9. Chemistry of f-block elements: special features, lanthanide contraction, coordination number, structure and reactions

S.No.	Branch	Course Title	Courses ID	Content
				1. Shriver, D. F.; Atkins, P. W. and Langford, C. G. Inorganic Chemistry, 3rd Edn., Oxford University, Oxford, 1999.2. Jolly, W. L.: Modern Inorganic Chemistry, 2nd Edn., 1991.3. Cotton, F. A. Chemical Applications of Group Theory, 3rd Edn., John Wiley and Sons1998.5. Cotton, F.A.; Carlos, A. M.; and Bochmann, M. Advanced Inorganic Chemistry, 6th Edn .Wiley Interscience Publication, 2001.
1078	ECO	MACROECONOMICS II	ECO402	Microeconomic foundations of Macroeconomics, Representativeagent model, Application of representative agent model in asset pricing, consumption theoriesetc., Longrun economic growth models, Monetarism and rational expectations cycle models.
1079	BSBE	CELLULAR MOLECULAR BIOLOGY	BSE651A	1. Aghion, A. and Howitt P. The Economics of Growth, MIT Press, 2009.2. Barro, R. Macroeconomics A Modern Approach, 5th Edition, MIT Press.3. Mankiw, G. Macroeconomics, 7th Edition, Worth Publishers.4. Dornbusch, R. and F. P. Martin. Macroeconomics, 2nd Edition, Oxford University Press.5. Blanchard, O. J. and P. A. Johnson. Macroeconomics, 2nd Edition, Pearson Education, Inc.6. A. Molecular Genetic Techniques and Genomics: Genetic analyses of mutations to indentify and study genes; DNA cloning and characterization; Genome wide analyses of gene structure and gene expression.7. Molecular Structure of Genes and Chromosomes: Chromosomal organization genes and noncoding DNA; Mobile DNA; Structural organization of eukaryotic chromosomes; organelle DNAsC. Transcriptional Control: Regulation of gene expression; activators and repressors of transcription; mechanism of transcription activation and repression.D. Posttranscriptional Gene Control: Processing of eukaryotic premRNA; transport across nuclear envelope. E. Signalling molecules and cell surface receptors; intracellular signal transduction; G protein coupled receptorsF. Membrane trafficking: Translocation of secretory proteins across the ER membrane; protein modification. G. Cell cycle and genetics studies on cell cycle; mechanisms regulating mitotic events; meiosis a special type of cell division.8. Molecular Cell Biology, by Lodish et al (5th edition or recent), W.H. Freeman and Company, New YorkMolecular Biology of the Cell, by Alberts et al (4th edition or later), Garland Sciences, New York
1080	BSBE	CELLULAR MOLECULAR BIOLOGY	BSE651	A. Molecular Genetic Techniques and Genomics: Genetic analyses of mutations to indentify and study genes; DNA cloning and characterization; Genome wide analyses of gene structure and gene expression.7. Molecular Structure of Genes and Chromosomes: Chromosomal organization genes and noncoding DNA; Mobile DNA; Structural organization of eukaryotic chromosomes; organelle DNAsC. Transcriptional Control: Regulation of gene expression; genes; activators and repressors of transcription; mechanism of transcription activation and repression.D. Posttranscriptional Gene Control: Processing of eukaryotic premRNA; transport across nuclear envelope. E. Signalling molecules and cell surface receptors; intracellular signal transduction; G protein coupled receptorsF. Membrane trafficking: Translocation of secretory proteins across the ER membrane; protein modification. G. Cell cycle and genetics studies on cell cycle; mechanisms regulating mitotic events; meiosis a special type of cell division.8. Molecular Cell Biology, by Lodish et al (5th edition or recent), W.H. Freeman and Company, New YorkMolecular Biology of the Cell, by Alberts et al (4th edition or later), Garland Sciences, New York
1081	BSBE	NEUROBIOLOGY	BSE656	Cell and molecular biology of neurons. Membrane potential, local signaling andgeneration of action potential. Mechanism of synaptic transmission. Sensoryperception : Vision, hearing, taste, touch and smell. Motor system and the Cerebral cortex learning and memory, sleep and dream.
1082	MSE	SOLAR ENERGY TECHNOLOGIES AND MATERIALS	MSE681A	1. Introduction to the course. 2. Solar Spectrum. 3. Available solar energy technologies.4. Solar Thermal Energy Conversion. a. Fundamentals. b. Materials and Technologies.c. Applications. d. Present status and Future outlook.5. Solar Technologies _b. Materials, devices and issues, 1. First generation technologies Si based.u. Second generation technologies (low cost) thinfilms (aSi, CdTe, CIGS); solarconcentratorsm. Third generation (Heterojunctions) Performance7. PV Processing with emphasis on migration from solar cells to modules to systems8. Present status and Future outlook
1083	MSE	NANOSTRUCTURES AND NANOMATERIALS: CHARACTERIZATION AND PROPERTIES	MSE694A	1. Thin Films Solar Cells, K.L. Chopra, McGraw Hill2. Physics and Technology of Solar Energy, H. P. Garg, M. Dayal, G. Furlan (1987)a. Volume 1: Solar Thermal Applicationsb. Volume II: Photovoltaic And Solar Cells3. Nelson4. Third Generation Photovoltaics: Advanced Solar Energy Conversion (Springer Series inPhotonics) by M.A. Green5. Flexible Solar Cells by Mario Pagliaro, Giovanni Palmisano, and Rosaria Cirirninelli6. Solar Cells by Peter Würfel (Paperback April 13, 2009)
1084	MME	SOLAR ENERGY TECHNOLOGIES AND MATERIALS	MME681	Overview of Nanostructures and Nanomaterials: classification (Dimensionality, Morphology/ shape/structure of nanoentities, New Effect/ Phenomena). Crystallinenanomaterials and defects therein. Hybrid nanostructures built out of nanosized building blocks (nano to macro). Eucledian, Hyperbolic and Spherical space structures. Nanostructures: Carbon Nanotubes, Fullerenes, Nanowires, Graphene, Quantum Dots, Nanoparticles, Nanocomposites, Nanosensors, Nanomedicine, Nanoelectronics, Nanophotonics, Nanofabrication. Surface reconstruction and reconfiguration. Adsorption and Absorption.
1085	EE	INFORMATION THEORY	EE667A	1. Nanomaterials, Nanotechnologies and Design: an Introduction to Engineers and Architects, D.Michael Ashby, Paulo Ferreira, Daniel L. Schodek, ButterworthHeinemann, 2009.2. Handbook of Nanophase Materials, Vol. 1, Ed. by H. Gleiter, 2003.3. Encyclopedia of Nanoscience and Nanotechnology, Ed.: Hari Singh Nalwa, American ScientificPublishers, 2004.4. Handbook of Nanoceramics and their Based Nanodevices (Vol. 2) Edited by Tseung-Chieh Wei, H. F. Tibbals, A.K. Rao, CRC Press (2008).
1086	IME	SUMMER PROJECT	MBA697A.	1. Introduction to the course. 2. Solar Spectrum. 3. Available solar energy technologies.4. Solar Thermal Energy Conversion. a. Fundamentals. b. Materials and Technologies.c. Applications. d. Present status and Future outlook.5. Solar Technologies _b. Materials, devices and issues, 1. First generation technologies Si based.u. Second generation technologies (low cost) thinfilms (aSi, CdTe, CIGS); solarconcentratorsm. Third generation (Heterojunctions) Performance7. PV Processing with emphasis on migration from solar cells to modules to systems8. Present status and Future outlook
1087	ECO	PUBLIC ECONOMICS AND PUBLIC POLICY	ECO731A	1. Thin Films Solar Cells, K.L. Chopra, McGraw Hill2. Physics and Technology of Solar Energy, H. P. Garg, M. Dayal, G. Furlan (1987)a. Volume 1: Solar Thermal Applicationsb. Volume II: Photovoltaic And Solar Cells3. Nelson4. Third Generation Photovoltaics: Advanced Solar Energy Conversion (Springer Series inPhotonics) by M.A. Green5. Flexible Solar Cells by Mario Pagliaro, Giovanni Palmisano, and Rosaria Cirirninelli6. Solar Cells by Peter Würfel (Paperback April 13, 2009)
1088	MSE	XPS AND AES IN STUDY OF SURFACE AND THIN FILMS	MSE621A	Introduction: Entropy, Relative Entropy, Mutual Information Inequalities, Entropy rate. Ansymptotic Equipartition Property(AEP): Consequences of the AEP, Typical Sequences, ShannonMcMillanBreiman Theorem. Channel coding Tunstal code, variable to variable length codes/ arithmetic code. Channel capacity: Discrete Memoryless Channel, Joint Typicality, Channel Coding Theorem and its converse, Feedback capacity, Source coding, Gaussian Channels, Channels with Colored Gaussian Noise, Gaussian Channels with Feedback. Rate Distortion Theory: Rate Distortion Function, Rate Distortion theorem and its converse, BlahutArimoto Algorithm, Lempel-Ziv algorithm (LZ77). Network Information Theory: Gaussian MultiUser Channels, Multiple Access Channel, Broadeast Channel, Encoding of Correlated Sources.
1089	CSE	SPECIAL TOPICS IN COMPUTER SCIENCE	CS697A	Course Reference : 1. Thomas M. Cover, Joy A. Thomas, "Elements of Information Theory"; 2nd Edition , John Wiley & Sons, 2006. 2. James L. Massey, Lecture notes on "Applied Digital Information Theory", John Wiley & Sons, 1968. 4. David J.C. MacKay, "Information Theory, Inference, and Learning Algorithms"; Cambridge University Press. 5. Robert Ash, "Information Theory"; Dover Publications, 1991. 6. YoungHan Kim, "Network Information Theory"; Cambridge University Press, 2012. 8. I. Csiszar and J. Korner, "Information Theory: Coding Theorems for Discrete Memoryless Systems"; Cambridge University Press, 1982.
1090	CSE	DESIGN AND ANALYSIS OF ALGORITHMS	CS602A	During the summer after first two semesters, each student will take up a summerproject in an industrial or service organization for 810 weeks. During this period,the student will work under the guidance of a faculty member and present the project during the third semester.
1091	AE	INTRODUCTION TO THE THEORY OF SMART	AE689A	Unbounded Operators, Matrix representation, Selfadjointness Criterion, Quadratic Forms, Differential Operators, Selfadjoint Extensions, Functional Calculus, Spectra of Selfadjoint Operators, Semianalytic Operators, Uncertainty Principle of Bargmann, Canonical Commutation Relations, Schrodinger representations, Schrodinger Operators, Selfadjointness, A Theorem of Kato, Spectral Theory for Schrodinger Operators. Course Reference : * N. Akhiezer, I. Glazman, Theory of Linear Operators in Hilbert Space II, Dover, 1961.* J. Blank, P. Exner, M. Havlivcek, Hilbert Space Operators in Quantum Physics, Springer, 2008.* T. Kato, Perturbation Theory for Linear Operators, World Scientific, 1998.* M. Reed and B. Simon, Methods of Modern Mathematical Physics II, Academic Press, 1975.
				Solid surfaces, their structure and composition: Importance of the surfaces/ Surfaces in different materials/ Structural imaging/ Composition of surface selvedges; Practical detection and spatial limits; Characterization of surfaces; Principles; Instrumentation; Vacuum systems, Xray sources, synchrotron radiation, electron energy analyzers; Spectral information, chemical shifts and interpretations; Quantification, depth profiling, imaging of surfaces; semiconductors; metallurgy/ tribology; thin films; Biomaterials, Case studies and lab visit/demo. Auger Electron SpectroscopyBasics: Principles; Instrumentation; Vacuum requirements, Electron sources, Auger spectroscopy. 1. Surface and Thin Film Analysis, Editors; G. Friedbachaer and H. Bubert, WileyVCH, 2011. 2. Encyclopedia of Materials Characterisation, C.R. Brundle, C.A. Evans and S. Wilson, ButterworthHennmann, 1996. 3. New York. 4. Surface Analysis Methods in Materials Science, D.J. O'Connors, B.A. Sexton, and R. St. C. Smart, Springer (2003). 5. Photoelectron and Auger spectroscopy. Thomas A Carlson, Plenum, 1977. 6. Berlin, 1977. 7. Scanning Auger electron microscopy, Editors: M. PLrutton and Mohamed M. El Gomati, John Wiley & Sons, 2006. 8. Practial guide to surface science and spectroscopy. YipWah Chung, Academic Press, 1999.
				Special and advanced topics in different areas of Computer Science andEngineering will be covered under this course.
				Basic concepts of operating systems, compilers, and data base managementsystems.
				Introduction to smart materials, piezo, pyro and ferro electric effects; hysteresiseffects:electric field in solids: fundamentals of continuum mechanics; basicconservation laws; thermodynamic principles; characterization of smart structures; applications to aerospace vehicles.

S.No.	Branch	Course Title	Courses ID	Content
		STRUCTURES		
1092	BSBE	LABORATORY METHODS	BSE602A	Demonstration based course. Usage of different equipments will be demonstrated. Demonstration based course. No text book.
1093	ME	FLUID MECHANICS AND RATE PROCESSES	ESO204A	FLUID MECHANICS: Introduction to fluids, Fluid statics; pressure as a scalar, manometry, forces on submerged surfaces (NO moments NOR center of pressure), Description of flows; field approach, Euler Linear (NOT angular) Momentum balance, NavierStokes (NS) equation; elementary derivation; application; Poiseuille flow, Couette flow, Energy equationBernoulli equation, applications including flow mea nondimensionalization of NS equations and boundary conditions, simplifications for cases without free surfaces and without cavitation (scale factor approach should NOT be done); High Re flow: Prandtl's Boundary layerselementary results for flat plates. Separation, flow past immersed bodies (bluff, streamlined); physics of ballgames (qualitative) Heat Transfer: Introduction, rate law and conservation law, series and of critical thickness of insulation, Unsteady conduction; significance of Biot and Fourier numbers, Heissler charts; Low Bi case;
1094	HSS	MODAL LOGIC	PHI768A	The History of Modal notions; The Lewis systems; Strict Implication andEntaliment; The System E and Relevance Logic; Philosophical Problems.
1095	HSS	FUNDAMENTALS OF MODERN LINGUISTICS	ENG701A	Structure of language; statistical structure and information theory; phonetics,phonemics and the distinctive feature theory; grammatical structure; ICanalysis; phrase structure and transformational gramr
1096	CE	COMPUTATIONAL METHODS IN ENGINEERING	ESO208A	Introduction, Engineering Systems, Physical and Mathematical Modeling, Error Analysis Approximations and round off and Truncation errors, Roots of Equations single variable Method of Bisection, Meth Solution of Linear Simultaneous EquationsDirect MethodsGauss Elimination, GaussJordan, LU decomposition; Iterative MethodsGaussSeidel, Conjugate Gradient, Banded and Sparse systems Solution of Transforms, Regression Analysis for Multivariable, Eigen Valuesand Eigen Vectors Power method, Relaxation Method, Diagonalization method. Numerical Differentiation and IntegrationHighAccuracy Diff segments, Open Integration Formulas, Ordinary Differential Equations Finite Difference method, Method of Weighted Residuals, Analytical versus Numerical Methods, Initial Value and Boundary Value Pro EquationsElliptic and parabolic Equations, Explicit and Implicit Methods, Crank Nicholson Method, ADI method; Introduction to Finite Element Method, Applications.
1097	AE	SPACECRAFT GUIDANCE NAVIGATION AND CONTROL	AE345A	Attitude dynamics and stability ofthreeaxis stabilized, singlespin, dualspin, and multibodyspacecraft with a1ticulated antennas, sensors, and solar arrays.Design of control ofthreeaxis stabilized spacecraft threeaxis attitude maneuver controllers using reaction wheels and thrustersControl of spinning spacecraft in transfer orbit during delta_ v firing and in operational orbitsaround the Earth, and design of act control of dualspin spacecraftPrecision pointing and tracking controllers for tracking landmarks, moving objects, and othersatellites for crosslink communicationSolar array controllers for tracking the Sun with spacecraft dynamics andcontrol systemsAttitude determination with gyros, star trackers, sun sensors, and horizon sensors usingalgorithms such as TRIAD and QUEST (quaternion estimator); senso techniques will be related with the control of Indian communication, remotesensing, and other special purpose satellites (Ca1tosat, Edusat, telemedicine). Course Reference : * Hughes, P.C., Spacecraft Attitude Dynamics, John Wiley, 1986.* Sidi, M.J., Spacecraft Dynamics and Control, Cambridge University Press, 1997* Noton, M., Spacecraft Navigation and of Geosynchronous Spacecraft, Prentice Hall, 1986* Bryson, A.E., Control of Spacecraft and Aircraft, Princeton University Press, 1994* Pocha, J.J., An Introduction to Mission Design for Geostationary Sate Wie, B., Space Vehicle Dynamics and Control, AIAA Education Series, 1998
1098	BSBE	FUNCTIONAL GENOMICS	BSE653A	Introduction; The impact of genomics on biological research; Identification of a large set of genes involved in a biological process; Highthroughput expression analysis; Genomewide search for interacting Pharmacogenomics; Proteomics; Functional genomics in model organisms; High throughput phenotypic analysis; Recent developments in genomics.
1099	BSBE	FUNCTIONAL GENOMICS	BSE653	Introduction; The impact of genomics on biological research; Identification of a large set of genes involved in a biological process; Highthroughput expression analysis; Genomewide search for interacting Pharmacogenomics; Proteomics; Functional genomics in model organisms; High throughput phenotypic analysis; Recent developments in genomics.
1100	AE	MEASUREMENTS IN FLUID MECHANICS	AE611A	Introduction to measurement, Wind Tunnels and Water Tunnels, Flow visualization techniques,Measurement of Pressure and volume flow rate, Force Measurements, TemperatureMeasurements, Hotwire l systems, PIV Measurements, Integral opticalmeasurement techniques: Shadowgraph, Schlieren & Interferometers, LDV Measurements, LIFMeasurements. Measurement of Wall Shear Stress * Fluid Mechanics Measurement, by Richard J. Goldstein SpringerVerlIng. 1983* Experimental Methods for Engineers by J.P. Holman McGrawHill 2008* Measurement in Fluid Mechanics by Stavros Tavoul 1998.* Instrumentation, Measurements, and Experiments in Fluids, byE Rathakrishnan, CRCPress, 2007* The Laser Doppler Technique, by L. E. Drain, John Wiley & Sons 1980.* Hotwire anemometry, by Pe CambridgeAerospace Series, 2010.
1101	ME	MICROMACHINING	ME665A	1. Introduction to the Course & Classification of MMPs, Part1: Mechanical Type Advanced Micromachining Processes, 2. Abrasive Jet Micro Machining (AJMM), 3. Ultrasonic Micro Machining (USMM), 4. Finishing (AFF), 6. Chemo mechanical Polishing (CMP), 7. Magnetic Abrasive Finishing (MAF), 8. Magnetorheological Finishing (MRF), 9. Magnetorheological Abrasive Flow Finishing (MRAFF), 10. Magne Electric Discharge Micro Machining (EDMM), 13. Wire EDM, EDDG, ELID, 14. Laser Beam Micro Machining (LBMM), 15. Electron Beam Micro Machining (EBMM),16. Ion Beam Machining 1. Introduction to Micromachining, V. K. Jain (Ed.), Narosa publisher, 2010.2. Micromachining, J. A. McGeough,3. Micromanufacturing, V. K. Jain (Ed.), CRC press, 2012.4. Micromanufacturing & Nanotechr
1102	EE	ELECTRICAL ENGINEERING COMMUNICATION SKILLS	EE390A	Technical Communication, definition and attributes. Ethics in communication. Technical Writing. Report and Article composition. How to write a Technical Brochure. Writing summary and abstracts of tech Presentation. Technical Presentation and use of multimedia.
1103	ME	BIOMEMS AND MICROSYSTEMS TECHNOLOGY	ME774	Introduction to BioMEMS and Microsystems technology, Biochips/ biosensors and introduction to device fabrication, Introduction to Cell biology, DNA & Protein chemistry, Microfluidics, Biochip Sensors & andLabonchip devices, Introduction to MEMS Design. Microfluidics: Continuum mechanics at small scales, Basics of microfluidics, Gas flows, liquid flows, boundary conditions, low Reynold's no. flo dielectrophoresis, microfluid'cs for internal flow eontrol (micropumps and microvalves, cfevice building and characterization), 1. Fundamentals ofMicrofabrication (Second Edition), Marc J. Madou, CRCpress Taylor and Francis Group, 6000 Broken Sound Parkway NW, Suite 300,Boca Raton, FL334872724, 2002.2. BioMEMS Techn Sound ParkwayNW, Suite 300, Boca Raton, FL334872724, 2006.3. Biomolecular sensing, processing and analysis, Rashid Bashir, Steve T.Werely, Mauro Ferrari, Springer Science and Business Media LLC, 2
1104	EE	MICROWAVE IMAGING CHARACTERIZATION AND NONDESTRUCTIVE TESTING	EE686A	1. Introduction to electromagnetic theory and microwaves; review of Maxwell's equations,interaction of microwaves with the dielectric materials and the concept of effectivepermittivity, definition of n parameters required for themicrowave imaging and their practical equivalents. 2. Review of transmission line theory and the equivalent network representation of field quantities at microwave frequencies 3. Electromagnetic scattering theory: direct and inverse problems, the inverse problem from the mathematic point of view, basis of the electromagnetic scattering problem formulation for the microwave c testing, resonant methods, cavity perturbation approach, reflection methods, transmission reflection methods. 5. Theory of transmission reflection methods for determining the permittivity and permeabili dispersive and anisotropic media. 1.2.3.4.5.Jaleel Akhtar, Microwave Imaging: Reconstruction of One Dimensional Permittivity Profiles,Vdm Verlag Dr. Mueller, Germany, 2008.L.F. Chen, C.K. Ong, C.P. Neo, V.V. Vardhan and V.K. Verdhan, Mi Introduction to Electromagnetic Inverse Scattering, KluwerAcademic Publishers, The Netherlands, 1992.Matteo Pastorino, Microwave Imaging, John Wiley & Sons, 2010.Literature from various Journals re
1105	EE	MICROWAVE IMAGING CHARACTERIZATION AND NONDESTRUCTIVE TESTING	EE686	1. Introduction to electromagnetic theory and microwaves; review of Maxwell's equations,interaction of microwaves with the dielectric materials and the concept of effectivepermittivity, definition of n parameters required for themicrowave imaging and their practical equivalents. 2. Review of transmission line theory and the equivalent network representation of field quantities at microwave frequencies 3. Electromagnetic scattering theory: direct and inverse problems, the inverse problem from the mathematic point of view, basis of the electromagnetic scattering problem formulation for the microwave c testing, resonant methods, cavity perturbation approach, reflection methods, transmission reflection methods. 5. Theory of transmission reflection methods for determining the permittivity and permeabili dispersive and anisotropic media. 1.2.3.4.5.Jaleel Akhtar, Microwave Imaging: Reconstruction of One Dimensional Permittivity Profiles,Vdm Verlag Dr. Mueller, Germany, 2008.L.F. Chen, C.K. Ong, C.P. Neo, V.V. Vardhan and V.K. Verdhan, Mi Introduction to Electromagnetic Inverse Scattering, KluwerAcademic Publishers, The Netherlands, 1992.Matteo Pastorino, Microwave Imaging, John Wiley & Sons, 2010.Literature from various Journals re
1106	EE	PROJECT WORK	EE490	PROJECT WORK
1107	EE	ELECTRICAL ENGINEERING UNDER GRADUATE	EE393A	ELECTRICAL ENGINEERING UNDER GRADUATE PROJECT I

S.No.	Branch	Course Title	Courses ID	Content
		PROJECT I		
1108	ME	MECHANICAL ENGINEERING COMMUNICATION SKILLS	ME399A	MECHANICAL ENGINEERING COMMUNICATION SKILLS
1109	ME	THERMODYNAMICS	ES0201A	Definitions & concepts: SI Units; System; Property; Energy; ThermodynamicEquilibrium; Work, State Postulate; Zeroth Law of Thermodynamics; TemperatureScale, Thermodynamic Properties of Fluids: M Compressibilitychart; Thermodynamic Diagrams including Mollier diagram; Steam Tables, Firstlaw of Thermodynamics & its applications to Non flow processes, Applicationsof First Law of Thermodynam Thermodynamics toChemically Reacting Systems Second Law of Thermodynamics & its Applications,Thermodynamic Potentials, Maxwells Relations; Thermodynamic Relations andAvailability, Power Cy interaction &various modes of work, Heat, State Postulate, Zeroeth Law of Thermodynamics, Temperature Scale, IPTS.Thermodynamic Properties of Fluids: Pure substance. Phase, Simple compressiblesu Pressurevolume; Temperaturevolumeand Phase diagrams; Mollier diagram and Steam tables.
1110	HSS	MARXISM GANDHI AND EXISTENTIALISM	PHI442A	This course deals with the philosophical ideas of Karl Marx, Gandhi and someof the important existentialist thinkers, especially JeanPaul Sartre and MatrinHeideggar. Major emphasis is laid on the moral
1111	HSS	INTRODUCTION TO PHILOSOPHY	PHI141B	General Introduction to the Course; Nature of Knowledge; Sources of Knowledge;The Problem of Perception; Skepticism; Kinds of Metaphysics; MetaphysicalSystems; MindBody Dualism; Proofs for the E
1112	HSS	ENGLISH LANGUAGE AND COMMUNICATION SKILLS	ENG112	This course imparts training in the use of English language for communicativepurposes, and aims to develop reading comprehension, writing, listening, andspoken language skills of the student. The Lang
1113	MTH	MATHEMATICS - II	MTH102AA	Matrices: matrix operations (Addition, Scalar Multiplication, Multiplication,Transpose, Adjoint and their properties; Special types of matrices (Null, Identity,Diagonal, Triangular, Symmetric, SkewSymmetric, Determinants and their properties, Vector Space $R^n(R)$;Subspaces; Linear Dependence/Independence; Basis; Standard Basis of R^n ;Dimension; Coordinates with respect to a basis; Complementary Subspa Linear Transformation from R^n to R^m (motivation, $X \cdot AX$); Image of a basis identifies the linear transformation;Range Space and Rank; Null Space and Nullity; Matrix Representation of a lineartransformati matrices; Similar Matricesand linear operators; Invertible linear operators; Inverse of a nonsingularmatrix; Cramers method to solve the matrix equation $Ax = b$; Eigenvalues andeigenvectors of a linear oper eigenvectorsof Hermitian, skewHermitian, Unitary, and Normal matrices (including symmetric,skewsymmetric, and orthogonal matrices), Implication of diagonalaizability ofthe matrix $A + A^T$ in the real qua
1114	MTH	MATHEMATICS - II	MTH102A	Matrices: matrix operations (Addition, Scalar Multiplication, Multiplication,Transpose, Adjoint and their properties; Special types of matrices (Null, Identity,Diagonal, Triangular, Symmetric, SkewSymmetric, Determinants and their properties, Vector Space $R^n(R)$;Subspaces; Linear Dependence/Independence; Basis; Standard Basis of R^n ;Dimension; Coordinates with respect to a basis; Complementary Subspa Linear Transformation from R^n to R^m (motivation, $X \cdot AX$); Image of a basis identifies the linear transformation;Range Space and Rank; Null Space and Nullity; Matrix Representation of a lineartransformati matrices; Similar Matricesand linear operators; Invertible linear operators; Inverse of a nonsingularmatrix; Cramers method to solve the matrix equation $Ax = b$; Eigenvalues andeigenvectors of a linear oper eigenvectorsof Hermitian, skewHermitian, Unitary, and Normal matrices (including symmetric,skewsymmetric, and orthogonal matrices), Implication of diagonalaizability ofthe matrix $A + A^T$ in the real qua
1115	MSE	MULTI FUNCTIONAL OXIDES: THIN FILMS & DEVICE	MSE689	Fundamentals of oxides : crystal structure, defect chemistry, and properties;focus on various material systems methods of fabrication e.g. solid statechemistry. Oxide thin films. polycrystalline versus epit (fundamentals of glow discharge processes and film deposition RFand DC magnetron sputtering new approaches), laser ablation (basic science,applications, various approaches), science and technolog chemical vapor deposition atomic layer deposition; PVD visa vischemical processes; issues related to epitaxy and case studies. Characterizationmethods : Structural techniques uses of X ray diffraction, a Measurements. Devices types of devices, fabrication: fundamentalsand issues; Lithographic methods: conventional and next generation, FIB (fieldion) techniques, Nanofabrication: principles, processes a
1116	MTH	ALGEBRA	MTH751	Groups, Basic properties, Isomorphism theorems, Permutation groups, SylowTheorems, Structure theorem for finite abelian groups, Rings, Integral domains,Fields, division rings, Ideals, Maximal ideals, Eu theorem and Chinese Remainder theorem, Field Extension, Cramer'srule, Algebraic elements and extensions, Finite fields.Determinants and their properties, Systems of linear equations, Eigenvaluesa transformations, Inner product spaces.
1117	MTH	SPECTRAL METHODS FOR PARTIAL DIFFERENTIAL EQUATIONS	MTH696	Galerkin, Collocation & Tau methods, Spectral approximation, The Fourier system,Continuous & discrete Fourier expansion, Orthogonal polynomials in (1,1),Fundamentals of spectral methods for PDEs, Te
1118	MTH	COMPUTATIONAL FLUID DYNAMICS	MTH694	Conservation laws, Weak solutions & shocks, Monotone difference schemes, Totalvariation diminishing schemes, Godunovtype schemes, Essentially nonoscillatorymethods, Flux limiters.
1119	MTH	NON-PARAMETRIC INFERENCE	MTH683	Order statistics, Tests of goodness of fit, Sign & signed rank tests, WaldWolfowitz, KolmogorovSmirnov, Median & MannWhitney tests, Linear ranktests for the location problem & scale problem, Measures
1120	SDS	STATISTICAL DECISION THEORY	SDS681	Decision function, Risk function, Optimal decision rules, Admissibility &completeness, The minimax theorem, The complete class theorem, Sufficientstatistics. Invariant decision problems, Admissible & m
1121	MTH	INTEGRAL EQUATIONS	MTH653A	Volterra and Fredholm integral equations, Resolvent Kernels. Operator equations,Fredholm theory, HilbertSchmidt theory. Nonlinear integral equations, Singularintegral equations.
1122	MTH	INTEGRAL EQUATIONS	MTH653	Volterra and Fredholm integral equations, Resolvent Kernels. Operator equations,Fredholm theory, HilbertSchmidt theory. Nonlinear integral equations, Singularintegral equations.
1123	MTH	LOCALLY CONVEX SPACES	MTH639	Topological linear spaces, Equicontinuity, Function spaces, Convexity & convextopological spaces, HahnBanach theorem, Barrelled spaces, Principle of uniformboundedness, Bornological spaces, Duality
1124	MTH	APPROXIMATION THEORY	MTH631A	Best approximation in normed spaces. Tchebycheff systems. TchebycheffWeierstrass Jackson Bernstein ZygmundNikolaev etc. theorems. Fourierseries, Splines, Convolutions, Linear positive, Variation
1125	MTH	APPROXIMATION THEORY	MTH631	Best approximation in normed spaces. Tchebycheff systems. TchebycheffWeierstrass Jackson Bernstein ZygmundNikolaev etc. theorems. Fourierseries, Splines, Convolutions, Linear positive, Variation
1126	MTH	APPLIED HARMONIC ANALYSIS	MTH627	Basic Fourier Analysisa review Convolutions, Multipliers and Filters, PoissonSummation Formula, Shannon Sampling Discrete Fourier Transform, Fast FourierTransform, Discrete Wavelets, Continuous Wa
1127	MTH	MEASURE THEORY	MTH620A	Algebras and salgebras, Measures, Outer measures, Lebesgue measure in R^n ,Completeness and regularity. Measurable functions and their properties,Convergence in measure, Integral, Convergence theo dual. Product measures, Construction, Fubini theorem and itsapplications, Differentiation of measures.
1128	MTH	MEASURE THEORY	MTH620	Algebras and salgebras, Measures, Outer measures, Lebesgue measure in R^n ,Completeness and regularity. Measurable functions and their properties,Convergence in measure, Integral, Convergence theo dual. Product measures, Construction, Fubini theorem and itsapplications, Differentiation of measures.
1129	MTH	BIOMATHEMATICS	MTH606	Biofluid dynamics; Blood flow & arterial diseases; Transport in intestines& lungs; Diffusion processes in human systems; Mathematical study of nonlinearVolterra equations, Stochastic & deterministic mo
1130	MTH	FINITE ELEMENT METHOD	MTH522A	Introduction and motivation, Weak formulation of BVP and Galerkin approximation,Piecewise polynomial spaces and finite element method, Computer implementationof FEM, Results from Sobolev spaces spaces, Polynomial approximation theory in Sobolev spaces, Variationalproblem for second order elliptic operators and approximations, Mixed methods,Iterative techniques.
1131	MTH	FINITE ELEMENT METHOD	MTH522	Introduction and motivation, Weak formulation of BVP and Galerkin approximation,Piecewise polynomial spaces and finite element method, Computer implementationof FEM, Results from Sobolev spaces spaces, Polynomial approximation theory in Sobolev spaces, Variationalproblem for second order elliptic operators and approximations, Mixed methods,Iterative techniques.
1132	MTH	NUMERICAL LINEAR ALGEBRA	MTH520	Computer arithmetic. Vector and matrix norms. Condition number of a matrixand its applications. Singular value decomposition of a matrix and its applications.Linear leastsquares problem. Householder control.

S.No.	Branch	Course Title	Courses ID	Content
1133	MTH	MULTIVARIATE ANALYSIS	MTH514A	Multivariate normal distribution, assessing normality, Wishart and HotellingsT2; Comparisons of several multivariate means, MANOVA; multivariate linearregression models; principal components, factor a
1134	MTH	MULTIVARIATE ANALYSIS	MTH514	Multivariate normal distribution, assessing normality, Wishart and HotellingsT2; Comparisons of several multivariate means, MANOVA; multivariate linearregression models; principal components, factor a
1135	PHY	ATOMIC MOLECULAR AND OPTICAL PHYSICS	PHY524A	Atomic structure: Review of one-electron atoms, Fine structure and Hyperfine structure, Spectral consequences of fine structure, Stark and Zeeman shifts, Interaction of One-electron atoms with one-electron atoms, Line intensities and lifetime of the excited states, Line shapes and widths, Photoelectric effect, Atom-light Hamiltonian, Density matrix, Optical Bloch equations, Electromag interactions, Helium energy levels, Exchange interaction, Thomas-Fermi model, Hartree-Fock method, Coupled angular momentum, Molecular structure: Van der Waals and Covalance Bond, Rotational spectroscopy, Selection rules, Molecular symmetries and their consequences. Special Topic: (one of the topics will be discussed} Atomic Bose-Einstein condensate/Non-linear optics/Nanomaterials Elementary atomic structure (Ciaredon Press). B. H. Bransden and C. J. Joachain, Physics of Atoms and Molecules (Pearson) M. Karplus, Atoms and Molecules (Benjamin-Cumming Pub. Co.)
1136	PHY	INTRODUCTION TO ATOMIC AND NUCLEAR PHYSICS	PHY524	Atomic Physics: Review of atomic structure of H, atomic structureof two electron system, alkali system, HartreeFock method, LS coupling,molecular binding, LCAO, LCMO; molecular spectra (electronic, ro of nuclei, nuclear two body problem, nuclearforce and nuclear models, nuclear decay, nuclear reaction kinematicsand classification of nuclear reactions (compound nuclear, direct etc),heavy ion reactions, experimental toolsof pure and applied nuclear physics.
1137	ME	MATERIALS SELECTION IN MECHANICAL DESIGN	ME770A	The evolution of engineering materials; Materials and the design process;Functional requirements of engineering materials; Materials selection based onproperties alone; Materials selection based on pro
1138	ME	MATERIALS SELECTION IN MECHANICAL DESIGN	ME770	The evolution of engineering materials; Materials and the design process;Functional requirements of engineering materials; Materials selection based onproperties alone; Materials selection based on pro
1139	ME	EVOLUTIONARY ALGORITHMS IN SEARCH & OPTIMIZATION	ME767	Traditional optimization methods. Simple genetic algorithms reproduction,crossover and mutation. Analysis of GAoperators; Deception; Multimodel andmultiobjective optimization; Engineering application
1140	ME	ROBOT MOTION PLANNING	ME766A	Configuration spaces of mobile vehicles and manipulators, Geometric modellingand sensor based map building. Path planning and obstacle avoidance. Objectmanipulation and grasping. Design of user in
1141	ME	ROBOT MOTION PLANNING	ME766	Configuration spaces of mobile vehicles and manipulators, Geometric modellingand sensor based map building. Path planning and obstacle avoidance. Objectmanipulation and grasping. Design of user in
1142	ME	MANUFACTURING AUTOMATION	ME765A	Automation strategies, flow lines, automated assembly systems, transfersystems; Vibratory bowl feeders, nonvibratory feeders. Part orienting, feedtrack, part placing and part escapement systems; Progr
1143	ME	MANUFACTURING AUTOMATION	ME765	Automation strategies, flow lines, automated assembly systems, transfersystems; Vibratory bowl feeders, nonvibratory feeders. Part orienting, feedtrack, part placing and part escapement systems; Progr
1144	ME	INTRODUCTION TO ROBOTICS	ME762A	Types of Robots. Spatial transformation and kinematics of open chain linkages.Mobile robots, Actuators, sensors, programming and control. Applications motion planning, grasping and industrial automa
1145	IME	MANAGEMENT OF RISK IN FINANCIAL SYSTEM	MBA678A.	Concept of Risk and Risk Management., Different types of Risks like SystematicRisk, Interest Rate Risk, Liquidity Risk, Operational Risk, Regulatory Risk,Market Risk, Foreign Exchange Risk, Commodity Pr Methods of identifying and measuring different types of risks. Useof Risk Models. Methods of Risk control and Management, i.e., requirementof active Risk Management techniques through use of VaR m forward rate agreements etc.
1146	IME	MANAGEMENT OF RISK IN FINANCIAL SYSTEM	MBA678.	Concept of Risk and Risk Management., Different types of Risks like SystematicRisk, Interest Rate Risk, Liquidity Risk, Operational Risk, Regulatory Risk,Market Risk, Foreign Exchange Risk, Commodity Pr Methods of identifying and measuring different types of risks. Useof Risk Models. Methods of Risk control and Management, i.e., requirementof active Risk Management techniques through use of VaR m forward rate agreements etc.
1147	IME	PROJECT FINANCING & MANAGEMENT	MBA677A	Generation and Screening of Project Ideas, Project Appraisal and Evaluation,Financial Projections, Investment Criteria, Cost Benefit Analysis, Project Finance,Financing Infrastructure Projects, Sources of f Techniques for Project Management.
1148	IME	PROJECT FINANCING & MANAGEMENT	MBA677.	Generation and Screening of Project Ideas, Project Appraisal and Evaluation,Financial Projections, Investment Criteria, Cost Benefit Analysis, Project Finance,Financing Infrastructure Projects, Sources of f Techniques for Project Management.
1149	IME	SECURITY ANALYSIS DERIVATIVES & PORTFOLIO MANAGEMENT	MBA676A.	Financial Markets, Investment Alternatives, Risk and Return, Portfolio Theoryand Capital Asset Pricing, Capital Asset Pricing Theory and Arbitrate PricingTheory, Efficient Market Hypothesis, Security Anal Strategies, Derivatives, Options, Futures, Swaps, BlackScholesmodel, Value at risk, Estimating volatility and correlations, Hedging and PortfolioManagement.
1150	ME	ADHESION FRICTION AND LUBRICATION FOR MICROMACHINES	ME667	I. History of friction and lubrication studies, Origins of friction, Coulomb's theory, Archard's theory, Bowden and Tabor's theory, Hardness of metals and ductile materials, contact area calcul hydrogen bonding etc.), adhesion models/nanorheology, meniscus and surfce models, static friction, stickslip phenomenon, friction anisotropy, concept ofsuperlubricity, micronanoscale wearphenomenon regimes, load bearing equation, journal bearings, elastohydrodynamic lubrication, film thickness calculation, Boundary lubrication, characteristics of boundary lubricants, liquid lubricants, additives, confin (91ectures)3. Nanotribological measurements: nanoscratching, atomic force microscopy tests, friction force microscopy, surface force apparatus, Tabor and Winterton design for surface force measurem surfaces. (9 lectures)4. Effects of surface/material parameters on micro/nanoscale friction (Load effects, speed effects, surface energy, surface l roughness effects on adhesion and friction, humidity mol Tribology on the Small Scale: A Bottom Up Approach to Friction, Lubrication, and Wear, C. Mathew Mates, Oxford University Press, 2007Nanotribology and Materials in MEMS, Sujeet K. Sinha, N. Satyanarayan, Brian J. Briscoe (editors) (Imperial College Press)Biological Micro and Nanotribology: Nature's Solutions by M. Scherge, S. N. Gorb, Springer Verlag
1151	CE	RISK & RELIABILITY IN GEOTECHNICAL ENGINEERING	CE731	Introduction: Sources and types of uncertainties associated with geotechnical analysis,importance of probabilistic methods and reliabilitybased analysis in geotechnicalengineering Review of probability a analysis Fundamentals of reliability analysis: FirstOrder Second Moment (FOSM) method, FirstOrderReliability Method (FORM), SecondOrder Reliability Method (SORM), Monte Carlosimulation Application of reliability analysis to soilproperties, uncertainty in interpretation techniques Spatial variability of soil properties, scale of fluctuations, estimation of autocorrelationand autocovariance Probabilistic groundwater modeling, flow design of shallow anddeep foundations, settlement analysis Reliabilitybased liquefaction analysis, lateral spreading Development of fragility curves for geotechnical problems 1. Fenton, G. A and Griffiths, D. V. (2008). Risk Assessment in Geotechnical Engineering,Wiley, New York.2. Phoon, K. K. (Ed.) (2008). ReliabilityBased Design in Geotechnical Engineering:Computations and GeotechnicalEngineering, SpringerWienNewYork.4. Baecher, G. and Christian, J. (2003). Reliability and Statistics in GeotechnicalEngineering, Wiley, New York.
1152	HSS	APPRECIATING FAR-EASTERN CINEMA	ART401	This is a (theoretical) cinemastudy course, whichpresents a survey of the FarEastern films both as an art form forcreative expression and as a medium of mass communication. The coursewill broadly ori countriesof the FarEast Japan, Korea, Mainland China, HongKong and Taiwan. Thefilms included in this course, span the decades between the post SecondWorld War era to the 21st century and they are course content comprise of vast level oftheoretical and textual study. Berggreen, ShuLing Chen, and Rob Peaslee. "TransChinese imagination:film and crossStrait perception as a historical case study for contextualjournalism education." Asia Pacific Media Education Review, 17(1), 2006. "The history."InterAsia Cultural Studies. Volume 9.Number 2 (2008): 239 250.Fang, Karen. "Arresting Cinema: Surveillance and the CityState in theRepresentation of Hong Kong." New Formation 17-SEP-14
1153	CSE	DATA STRUCTURES AND ALGORITHMS	ESO211	Order Analysis: Objectives of time analysis of algorithms; Bigoh and Thetanotations, Data Structures: Arrays, Linked lists, Stacks (example: expressionevaluation), Binary search trees, RedBlack trees, Has Quicksort with average case analysis, Heaps and heapsort, Lowerbound on comparisonbased sorting and Counting sort, Radix sort, Btrees,Dynamic Programming: methodology and examples (Fibonacci Methodology, examples (lecture scheduling,process scheduling) and comparison with DP (more examples to come later ingraph algorithms), Graph Algorithms: Basics of graphs and their representations,

S.No.	Branch	Course Title	Courses ID	Content
				set and Fibonacci heap datastructures), Shortest Paths (Dijkstra, BellmanFord, FloydWarshall).
1154	EE	INTRODUCTION TO ELECTRONICS	ESC102N	Stored program concept (with simple computer simulator), machine language and instruction formats,assembly language for the simple computer. Integer representation, finite representation of real numbers. Unary, binary, ternary operations. Conditionals, ifthen, ifthenelse, nested conditionals, switchcase.Loops, for, while, repeat, loopinvariants, precondition, postcondition. Functions and return values,arguments, arrays, linked lists, dynamic data structures, stack, queue, graphs, trees related algorithms,memory and its management. Elementary complexity motivation, concrete complexity, big O notation.Linux tools, introduction to system programming. Solutions of systems of linear equations, integration, solution ofODEs. Course Reference : 1. Brian W Kernighan and Dennis M Ritchie, The C Programming Language,2nd Ed. ANSI C version,Pearson, 2006
1155	MME	DESIGN OF SINTERED PRODUCTS	MME674	Factors affecting designmaterials and geometry: Specific design of productslike permeable materials, structural parts, bearings and cutting tool materials:conditioning of metal powders to influence processing
1156	IME	STRATEGIC MARKETING- CONTEMPORARY ISSUES	MBA639A	Successful Marketing in highly competitive global markets of today needs breakthrough concepts, socially responsible and innovative execution. And mastering that blend entails participatory, immersive learning from foundations courses.Course Plan and Modules: 1) Market opportunity recognition and evaluation 2) Generating business models 3) Green and sustainable marketing scenarios 4) Contextual strategic marketing
1157	IME	INTRODUCTION TO GAME THEORY	IME636A	Description of Game Theory, Representation of games in extensive form, Normal form and Coalition form, Concept of preferences and utility, Introduction to solution concepts for normal form games, Description of oligopoly, extensive form games of perfect and imperfect information, refinements of Nash equilibrium, finite and infinite horizon, alternating bargaining models, games with incomplete information; Bayesian mechanism design, Properties of mechanism and implementation
1158	IME	INTRODUCTION TO GAME THEORY	IME636	Description of Game Theory, Representation of games in extensive form, Normal form and Coalition form, Concept of preferences and utility, Introduction to solution concepts for normal form games, Description of oligopoly, extensive form games of perfect and imperfect information, refinements of Nash equilibrium, finite and infinite horizon, alternating bargaining models, games with incomplete information; Bayesian mechanism design, Properties of mechanism and implementation
1159	MSE	MICROSCOPY AND MICROANALYSIS OF MATERIALS	MSE642A	Advanced Optical microscopy : Special microscopy techniques and applications: Bright field and dark field imaging; confocal microscopy; interference microscopy; polarized light microscopy; phase contrast microscopy : Basis of image contrast and various operating modes in SEM SE and BSE, Xray, EBIC, cathodoluminescence, voltage contrast mode, Magnetic contrast mode. (2) Environmental SEM, Low voltage SEM, image acquisition and analysis, applications (3) Fractography and failure analysis (2)Transmission electron microscope : Wave properties of electrons, lens defects, aberration corrected TEM and subAngstrom resolution, dynamic diffraction and anomalous absorption effects, image artifacts (3L) BF, OF, Weak beam OF images and applications (1 L) Electron Diffraction: SADP, Microdiffraction, CBED. Diffuse scattering and finestructure simulation of lattice and structural images, Interpretation of images and illustrative examples(2L) STEMHAADF imaging, information limit (1L) Lorentz microscopy and holography (1) Specimen preparation and analysis Course Reference : 1. Fundamental of light microscopy and electronic imaging, 0.8. Murphy, WileyLiss, 20012. Microstructural Characterization of Materials D. Brandon and W.D. Kaplan, John Wiley and Sons, 20003. Scanning Electron Microscopy & XRay Microanalysis, J.Goldstein et.al, Springer, 20004. Scanning Electron Microscopy & XRay Microanalysis, J.Goldstein et.al, Springer, 20005. Scanning Electron Microscopy & XRay Microanalysis, J.Goldstein et.al, Springer, 20006. Scanning Electron Microscopy & XRay Microanalysis, J.Goldstein et.al, Springer, 20007. Scanning Electron Microscopy & XRay Microanalysis, J.Goldstein et.al, Springer, 20008. Scanning Electron Microscopy & XRay Microanalysis, J.Goldstein et.al, Springer, 20009. 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S.No.	Branch	Course Title	Courses ID	Content
1172	IME	MANAGING SERVICE OPERATIONS	MBA671A.	Service as Product, Design of Service Systems, Location and Layout of ServiceFacilities, Service Engineering including Work Design, Human Factors, Automationand Communication, Productivity and Service Delivery Planning. Professional Service: Achieving differentiation through knowledge andrelationship, Service and Competitive Strategy; Service delivery systems andIT applications; IT enabled services; Service delivery systems andIT applications; IT enabled services; centersand related services, Cross Cultural issues; Pricing and Transfer Pricing ofConnected Services, Project Implementation, Learning, Innovation and KnowledgeManagement in the Service based business.
1173	IME	MANAGING SERVICE OPERATIONS	MBA671.	Service as Product, Design of Service Systems, Location and Layout of ServiceFacilities, Service Engineering including Work Design, Human Factors, Automationand Communication, Productivity and Service Delivery Planning. Professional Service: Achieving differentiation through knowledge andrelationship, Service and Competitive Strategy; Service delivery systems andIT applications; IT enabled services; centersand related services, Cross Cultural issues; Pricing and Transfer Pricing ofConnected Services, Project Implementation, Learning, Innovation and KnowledgeManagement in the Service based business.
1174	IME	PROJECT MANAGEMENT	MBA666A.	Characteristics of Projects, Project Economics, Screening and Selection, Evaluation, Structuring Organizational and Work Breakdown, Scheduling, Budgeting,Resource Management, Life Cycle Costing, Project Management, Agile Project Management, Privacy and Security. John Nicholas and Herman Steyn: Project Management for Engineering, Business and Technology, Routledge Taylor and Francis Group, Fourth Edition, Indian Special Edition.
1175	IME	PROJECT MANAGEMENT	MBA666.	Characteristics of Projects, Project Economics, Screening and Selection, Evaluation, Structuring Organizational and Work Breakdown, Scheduling, Budgeting,Resource Management, Life Cycle Costing, Project Management, Agile Project Management, Privacy and Security. John Nicholas and Herman Steyn: Project Management for Engineering, Business and Technology, Routledge Taylor and Francis Group, Fourth Edition, Indian Special Edition.
1176	IME	SIMULATION OF BUSINESS SYSTEMS	MBA643A.	Simulation Philosophy and Methodologies, Review of Basic Probability andStatistics, Random number Generation, Programming Considerations, Languagesand Data Structures, Verification and Validation of Simulation, Analysis by Simulation Using Spreadsheets, Simulation of Production System Inventories, Queues and Production Scheduling.
1177	IME	SIMULATION OF BUSINESS SYSTEMS	MBA643.	Simulation Philosophy and Methodologies, Review of Basic Probability andStatistics, Random number Generation, Programming Considerations, Languagesand Data Structures, Verification and Validation of Simulation, Analysis by Simulation Using Spreadsheets, Simulation of Production System Inventories, Queues and Production Scheduling.
1178	LT	INTRODUCTION TO COHERENT AND LASER OPTICS	LT631	Maxwell eqns. and electro magnetic waves, Wave properties of light andpropagation. Waveguides, Optical resonators. Ray Optics, Matrix methods,Optical Anisotropic materials. Interference and interferometry.
1179	LT	LASER SYSTEMS AND APPLICATIONS	LT611	Atomic, ionic, molecular, excimer and liquid laser systems and applications, Solidstate lasers. Short Pulse generation and measurement. Laser applications inmedicine and surgery, Materials processing, Communication.
1180	HSS	THEORIES OF RIGHTS	HSS701	The aim of the course is to understand the origins of the notions in theclassical period, its rejuvenation in the medieval and early modern periods,and contemporary debates surrounding different theories of rights. Themes include: Stoicism, Republicanism, Natural Law Discourse,Liberalism, Utilitarianism, Legal Positivism, Marxism, and Communitarianism.
1181	HSS	COMPARATIVE AESTHETICS	ENG752	The course explores the convergence and divergence of eastern and westernaesthetic practices, in an attempt to two. Its scope includes classical Greekand Sanskrit texts like Bharata s Natyasastra and Aesthetics. The concept of dhvani and postulates of New Criticism.
1182	HSS	CONTEMPORARY CRITICAL THEORIES & LITERARY PRAXES	ENG751	Any postgraduate course in English Literature is desirableContemporary critical theories particularly those of the post1960s, haveproblematised the writing, reading and receiving of literature. This course explores narratology, ecology, and notions ofculture and history. It aims to expose students to contemporary theories thatwould enhance their research into chosen areas of literature. Delving on relevantissues and texts, the course will explore the role of literature in the contemporary world. The course will also explore the role of literature in the contemporary world. The course will also explore the role of literature in the contemporary world. The course will also explore the role of literature in the contemporary world.
1183	HSS	INTRODUCTION TO PSYCHOLOGY	PSY151A	Psychological Perspectives and Approaches; Perception; Learning; Memory;Higher Cognitive Processes; Motivation and Emotion; Intelligence; Personality;Individual Differences.
1184	HSS	INTRODUCTION TO PSYCHOLOGY	PSY151	Psychological Perspectives and Approaches; Perception; Learning; Memory;Higher Cognitive Processes; Motivation and Emotion; Intelligence; Personality;Individual Differences.
1185	SDS	ECONOMETRIC THEORY	SDS785	Multiple linear model , estimation of parameters under spherical and nonsphericaldisturbances by least squares and maximum likelihood methods, testsof hypothesis, R2 and adjusted R2. Prediction, with confidence intervals. Specification error analysis related to explanatory variables,inclusion and deletion of explanatory variables. Idea of Steinrule estimation.Exactand stochastic linear restrictions, restricted and mixed regression. Heteroskedasticity, problem and test, estimation under Heteroskedasticity.Autocorrelation, DurbinWatson test. Errorsinvariables, inconsistencyof least squares method, methods of consistent estimation, nonlinear least squares and feasible generalized least squares estimators.Simultaneous equations model, structural and reduced forms, rank and orderconditions for identifiability, indirect least squares, two stage least squares and unit roots & cointegration.
1186	MTH	MATHEMATICAL METHODS	MTH752A	Calculus of Variations; Sturm Liouville Problem and Green's Function; PerturbationMethods and Similarity Analysis; Stability Theory.
1187	MTH	MATHEMATICAL METHODS	MTH752	Calculus of Variations; Sturm Liouville Problem and Green's Function; PerturbationMethods and Similarity Analysis; Stability Theory.
1188	MTH	BANACH ALGEBRAS C* ALGEBRAS AND SPECTRAL THEORY	MTH734	Elementary properties of Banach Algebras and examples; Ideals and quotients,the Spectrum, the Riesz Functional Calculus.Abelian Banach Algebras, C* Algebras;Representations of C* Algebras and the Gelfand-Naimark theorem; The Spectral theorem; Some applications.
1189	MTH	REPRESENTATION THEORY OF LINEAR LIE GROUPS	MTH733	Representation theory of Compact groups; Peter Weyl Theorem. Linear LieGroups; The Exponential map, Lie Algebra, Invariant Differential Operators.Representations of the group and its Lie Algebra. Four fundamental representations of the Euclidean Motion Group.
1190	MTH	MODAL LOGIC	MTH701A	Modal Propositional LogicSystems K, T, D, S4, S5, B; Automated Proof Methods,Decidability; Consistency, Frames, Canonical Models, Completeness; FiniteModels, Incompleteness.Algebraic semanticsLinear Modal Logic; Automated Proof Methods; Identity. someModal Systems and applicationsTemporal, dynamic and epistemic Logics,Topology via Modal Logic
1191	MTH	MODAL LOGIC	MTH701	Modal Propositional LogicSystems K, T, D, S4, S5, B; Automated Proof Methods,Decidability; Consistency, Frames, Canonical Models, Completeness; FiniteModels, Incompleteness.Algebraic semanticsLinear Modal Logic; Automated Proof Methods; Identity. someModal Systems and applicationsTemporal, dynamic and epistemic Logics,Topology via Modal Logic
1192	MTH	NUMERICAL SOLUTION OF PARTIAL DIFFERENTIAL EQUATIONS	MTH693A	Basic linear algebra vector and matrix norms and related theorems. Parabolicsystems in one and two space dimensions explicit and implicit formulae.Consistency, stability and convergence. Iterative methods for linear systems. Dirichlet, Neumann and mixed problems. Direct factorizationmethods and successive overrelaxation (S.O.R.). ADI and conjugate gradientmethods. Hyperbolic equations. First order hyperbolic systems in one space dimension. Galerkin method and applications.
1193	MTH	NUMERICAL SOLUTION OF PARTIAL DIFFERENTIAL EQUATIONS	MTH693	Basic linear algebra vector and matrix norms and related theorems. Parabolicsystems in one and two space dimensions explicit and implicit formulae.Consistency, stability and convergence. Iterative methods for linear systems. Dirichlet, Neumann and mixed problems. Direct factorizationmethods and successive overrelaxation (S.O.R.). ADI and conjugate gradientmethods. Hyperbolic equations. First order hyperbolic systems in one space dimension. Galerkin method and applications.
1194	MTH	TIME SERIES ANALYSIS:FORECASTING AND CONTROL	MTH685	Linear stationary processes, Autocovariance & spectral density functions &moving average processes, Linear nonstationary processes, Model estimation& identification, Forecasting, Transfer function models.

S.No.	Branch	Course Title	Courses ID	Content
1195	SDS	STATISTICAL SIMULATION DATA ANALYSIS AND MODEL BUILDING	SDS684	Introduction to simulation & MonteCarlo studies; Generation of random variables.Interactive computational & graphical techniques in model building; Data basedinference methods such as JackKnife, Boo
1196	MTH	ORDER STATISTICS	MTH682	Basic distribution theory, Moments of order statistics including recurrence relations, Bounds and approximations, Estimation of parameters, Life testing,Short cut procedures, Treatment of outliers, Asympt
1197	MTH	TRIBOLOGY	MTH664A	The fundamentals of lubrication, friction & wear. Boundary lubrication,Hydrodynamic lubrication, Elastohydrodynamic lubrication. Compressibility &thermal effects, Non Newtonian lubrication, Roughness
1198	MTH	TRIBOLOGY	MTH664	The fundamentals of lubrication, friction & wear. Boundary lubrication,Hydrodynamic lubrication, Elastohydrodynamic lubrication. Compressibility &thermal effects, Non Newtonian lubrication, Roughness
1199	MTH	GRAPH THEORY	MTH657	Basic definitions. Blocks. Ramsey Numbers. Degree sequences. Connectivity.Eulerian and Hamiltonian Graphs. Planar graphs and 5colour theorem. Chromaticnumbers. Enumeration. MaxFlow MinCut Th
1200	ME	MECHANICS OF BIOLOGICAL MEMBRANES	ME724A	1. Organisation of Animal cells : Structure and function of cell membrane. Roll of fluid lipid bilayers in cell functionality. Discussion on the experimental methods of study membranes.2. A brief review of d determination of the stable equilibirium shapes of red blood cells. Application of the theory to equilibrium shapes of phase separated fluid lipid bilayers.4. A discussion on the adhesion of vecicles and cel mechanical and thermodynamic model. Course Reference : 1. B Albert at zI.; Molecular biology of the cell. Garland Science, NY, 2002 (fourth edition).2. J. N. Israelachvili, Intermolecular and Surface Forces: With Applications to Colloidal and Biol Membranes, Handbook of Biological Physics Vol. 1, Elsevier, Amsterdam, 1995.4. S. A. Safran, Statistical Thermodynamics of Surfaces, Interfaces, and Membranes, Westview Press, 2003.
1201	EE	INTELLIGENT SYSTEMS & CONTROL	EE705	Norms of Signals, Vectors and Matrices,Positive Definite Functions, Positive Definite Matrices; Continuoustime Statespace Model, L TI Statespace Model, Nonlinear Statespace model, Equilibrium point an Stability Theory, Lyapunov stability of time invariant system, LaSalle's Invariance Theorem, Chetaev's Instability Theorem, Lyapunov stability of time varying system, Lyapunov's indirect me Nonlinear Statespace model, ARMAX and NARMAX Models, Lyapunov Stability for Discrete Time Systems; Modeling of Different Nonlinear Systems: Inertial Wheel Pendulum, Two Link Manipulator, Invert Design, Statefeedback linearizable systems.FeedForward Networks: Multilayered Neural Networks, Radial Basis Function Networks. Adaptive Learning Rate; Feedback Networks, Back Propagation Throug NetworksClassical sets, Fuzzy Sets, Concept of a fuzzy number, Operations on Fuzzy sets, Properties of Fuzzy Sets, Some Typical Membership Functions; Extension Principle of Fuzzy Sets, Crisp Relation Base and ApproximateReasonjng, Fuzzy Linguistic Variables, Linguistic modifier, Rulebase systems, Fuzzy Rulebase, Fuzzy Implication Relations, Fuzzy Compositional Rules, Inference mechanism comp Identification Using TS Fuzzy Models,The TS Model From InputOutput Data, The TS Fuzzy Model Using Linearization. 1. Alexander M. Meystel and James S. Albus , Intelligent Systems: Architecture, D Systems with LabVIEW(TM),Springer, 2009
1202	PHY	THERMAL PHYSICS	PHY211	Principles of thermodynamics (with applications to simple fluids), applicationsof thermodynamics: concept of thermodynamic state, extensive and intensivevariables; heat and work, internal energy functi temperature as conjugate pair of variables; secondlaw of thermodynamics, entropy maximum and energy minimum principles;thermodynamic potentials: enthalpy, Helmholtz potential, Gibbs potential;cor relations; firstorder phase transitionsand ClausiusClapeyron equation; concepts associated with critical and multicriticalphenomena, some chosen applications from surfaces and interfaces,chemical reac gases: equilibriumproperties pressure and equation of state; transport processes momentumtransport and viscosity, energy transport and thermal conductivity, chargetransport & electrical conductivity (thandisorder, illustration with simple examples; Maxwells demon; qualitativejustifications of laws of thermodynamics (without introducing ensembles),thermodynamics of irreversible processes: entropy p
1203	MTH	REGRESSION ANALYSIS	MTH416	Simple and multiple linear regression, Polynomial regression and orthogonalpolynomials, Test of signficance and confidence intervals for parameters.Residuals and their analysis for test of departure from observations, Power transformation of dependent and independentvariables. Problem of multicollinearity, ridge regression and principal componentregression, subset selection of explanatory variables, M Asymptotic properties of estimators. Generalised LinearModels (GLIM), Analysis of binary and grouped data using logistic and loglinearmodels.
1204	MTH	REAL AND COMPLEX ANALYSIS	MTH413A	Real and complex numbers; Open, closed and compact sets in Rn ; Limits andcontinuity; Differentiation and Integration; Sequences and series; Sequencesand series of functions; Complex integration.
1205	MTH	REAL AND COMPLEX ANALYSIS	MTH413	Real and complex numbers; Open, closed and compact sets in Rn ; Limits andcontinuity; Differentiation and Integration; Sequences and series; Sequencesand series of functions; Complex integration.
1206	MTH	APPLIED STOCHASTIC PROCESS	MTH412A	Definition and classification of general stochastic processes. Markov Chains:definition, transition probability matrices, classification of states, limitingproperties. Markov Chains with Discrete State Space: Markov Process with Continuous State Space: Introduction toBrownian motion. 22-MAR-2014
1207	MTH	APPLIED STOCHASTIC PROCESS	MTH412	Definition and classification of general stochastic processes. Markov Chains:definition, transition probability matrices, classification of states, limitingproperties. Markov Chains with Discrete State Space: Markov Process with Continuous State Space: Introduction toBrownian motion.
1208	MTH	PROBABILITY THEORY II	MTH411	Tight families of probability distributions, Convergence of probability distributionfunctions, Hellys theorem, HellyBray theorem, Skorohods fundamental theorem,Scheffes theorem; Weak convergence, Unif Expansion of characteristic functions, Polyas theorem, Bochnerstheorem. Moments and uniqueness of the probability distribution, FrechetShohat theorem. Central limit theorems: LindebergLevy, Lyapunov
1209	MTH	ANALYSIS II	MTH404A	Lebesgue measure on Rn : Introduction, outer measure, measurable sets,Lebesgue measure, regularity properties, a nonmeasurable set, measurablefunctions, Egoroffs theorem, Lusins theorem. Lebesgue convergence theorem, integral of nonnegative functions, FatousLemma, monotone convergence theorem, the general Lebesgue integral, Lebesgueconvergence theorem, change of variable formula. Differ The Minkowskis inequality and Hlders inequality,completeness of Lp, denseness results in Lp. Fourier series: Definition of Fourierseries, formulation of convergence problems, The L2 theory of Fourier ser
1210	MTH	ANALYSIS II	MTH404	Lebesgue measure on Rn : Introduction, outer measure, measurable sets,Lebesgue measure, regularity properties, a nonmeasurable set, measurablefunctions, Egoroffs theorem, Lusins theorem. Lebesgue convergence theorem, integral of nonnegative functions, FatousLemma, monotone convergence theorem, the general Lebesgue integral, Lebesgueconvergence theorem, change of variable formula. Differ The Minkowskis inequality and Hlders inequality,completeness of Lp, denseness results in Lp. Fourier series: Definition of Fourierseries, formulation of convergence problems, The L2 theory of Fourier ser
1211	MTH	PROBABILITY THEORY - I	MTH311A	Sets and set operations, Sample space, Sigma fields, Measurable spaces, Events.Measure spaces, Caratheodorys extension theorem, Construction of measures,Product spaces, Product measures. Probab simplefunctions, Random variables. Induced measures and probability distributionfunctions: discrete, continuous and absolutely continuous, one to onecorrespondence with induced probability measure, Fatous lemma, Monotone convergence and dominated convergencetheorems, Uniform integrability, Markov, Chebyshev, CauchySchwarz, Minkowski,Holder, Jensen and Lyapunov inequalities. Absolute co theorem, Convolution. Functions of random variables,Jacobian theorem.
1212	MTH	PROBABILITY THEORY - I	MTH311	Sets and set operations, Sample space, Sigma fields, Measurable spaces, Events.Measure spaces, Caratheodorys extension theorem, Construction of measures,Product spaces, Product measures. Probab simplefunctions, Random variables. Induced measures and probability distributionfunctions: discrete, continuous and absolutely continuous, one to onecorrespondence with induced probability measure, Fatous lemma, Monotone convergence and dominated convergencetheorems, Uniform integrability, Markov, Chebyshev, CauchySchwarz, Minkowski,Holder, Jensen and Lyapunov inequalities. Absolute co theorem, Convolution. Functions of random variables,Jacobian theorem.
1213	HSS	POSTWAR AMERICAN FICTION	ENG750	Postwar American fiction is characterized by a complex sensibility that is oftenpervasive in the novels of the period from 1945 through the Cold War of theseventies and eighties to the present. With Ameri give way to introspection and selfdebate concerning theproblematic of defining American identity and nationhood against a distinctmulticultural presence, and the unviability of "American Dream Interestingly, fiction of this period frames war both asa trope and a realistic concern. The protracted cultural wars which began withthe Civil Rights era of the 1960s infused new life into the literature of the immigrants, began to be heard. And thevariegated literary movements (postmodernism, humanism, and feminism toname a few) and fictional strategies (such as protest, fantasy, black humor)employed in reality into their imaginativeartistic vision. The fiction of this period is marked by thematic concerns suchas politics, paranoia, race, money, technology, sex, suburbia, urban decay,immigration, and spiritual
1214	HSS	POSTMODERN THEORY AND LITERATURE	ENG749	Any postgraduate course in Modern LiteratureAcquainting the students with the major critical concepts, selfreflexive textsand amorphous themes of Postmodernism, the course aims to explore theuseful introduction to the central conceptsof postmodernism by providing an approach to contemporary American, LatinAmerican, European and Indian (Writing in English) literature. Major topics fordiscussion a

S.No.	Branch	Course Title	Courses ID	Content
				the Author, Rhizome, Knowledgeand Power, Entropy, the Literature of Exhaustion, the Hyperreal and theSimulacrum, Cyberspace and Cyberpunk, Postmodern Ethics, Postmodernism andPopular Culture, P
1215	HSS	COGNITIVE LINGUISTICS	ENG748	The course explores languagecognition mappings in varied contexts such asunilingual, bilingual and sign language use, Crosslinguistic variations in basicconceptual domains such as time and apace and semantics of grammar type hierarchies and continuums,compositional structures and cocompositionality, noncompositional structures,constrains on possible grammars, perceptual processes and g
1216	HSS	FEMINIST THEORY AND LITERATURE	ENG743	Definitions: the Sex/Gender debate, Masculinity and femininity, patriarchy, thefeminism/gender studies debate.Gender and Identity: Processes of Identity formation, gender roles in differentsocial contexts external vs internalselfrepresentations, the media and representation.The Politics of Feminism: The reasons and implications of focusing on genderas a subject of inquiry, the relationship between theory
1217	ME	MANUFACTURING PROCESSES II	TA212	Engineering materials; Microstructureprocessing relationship; Solidificationprocesses: molding and casting; Joining processes: welding, brazing, and soldering; Deformationprocesses: hot and cold worki treatment; Surface modificationtechniques: carburization, CVD, PVD.Specialized Infrastructure requirement:Rolling Mill, Pressure Die Casting Unit
1218	CHM	PHYSICAL CHEMISTRY -II	CHM322A	Equilibrium Thermodynamics: Laws of thermodynamics,the equilibrium state, thermodynamic variables, conjugate quantities,thermodynamic potentialsStatistical Mechanics: Kinetic Theory of gases, Boltzmonatomic, diatomic and polyatomicgases, chemical equilibrium, quantum statisticsTransport Phenomena: Transport coefficients, thermal conductivity,diffusivity, viscosity, ionic conductivity, Limiting law activation energy,transitionstate theory, reactions as trajectories, molecular beam experiments,reactions in liquid phase, Kramers theory, Diffusion limited reactions 1. P.W. Atkins and Julio de Paula : Physical Chemistry2. I.N. Levine: Physical Chemistry3. D.A. McQuarrie and J. D. Simon : Physical Chemistry A Molecular Approach4. D. A. McQuarrie, Statistical Mechan Statistical Mechanics7. R.S. Berry, S.A. Rice and John Ross : Physical Chemistry
1219	MTH	SOBOLEV SPACES AND APPLICATIONS	MTH656	Elements of operator theory and Hilbert spaces; Introduction to the theoryof distributions. Sobolev Spaces : Imbedding and compactness theorems,Fractional spaces and elements of trace theory. Applica
1220	MTH	COMPLEX FUNCTION THEORY	MTH644	Fundamental theorems, Winding number & applications,Normal families, Riemannmapping theorem, Fundamentals of univalent functions & entire functions,PhragmenLindelf theorems, Gamma, Riemann
1221	MTH	SEVERAL COMPLEX VARIABLES	MTH640	Cauchy integral formula, Taylor series, Associated radii of convergence, Analyticfunctions, Reinhardt domain, Logarithmic convexity, Laurents expansion, Envelopeof holomorphy, Goldbergs growth param structure.
1222	MTH	TOPICS IN OPERATOR THEORY AND HARMONIC ANALYSIS	MTH637	Operators on Hilbert spaces: Compact operators, Schatten class and HilbertSchmidt operators, Spectral theorem. Fourier series, Smooth functions anddistributions. Hardy spaces, Carleson measures, H1 SO(3).
1223	MTH	NUMBER THEORY	MTH215	Divisibility, Primes, Congruences, Residue systems, Primitive roots; Quadraticreciprocity, Some arithmetic functions, Farey fractions, Continued fractions,Some Diophantine equations, Bertrands postulate
1224	MTH	LINEAR ALGEBRA	MTH201N	Fields and linear equations. Vector spaces. Linear transformations and projections,Determinants. Elementary canonical forms: diagonalization, triangulation,primary decomposition etc. Secondary decom spaces,Selfadjoint, Unitary and normal operators, Orthogonal projections. Bilinearforms, Symmetric, Skewsymmetric, Positive and semipositive forms etc.
1225	MTH	MATHEMATICS I (PREP)	MTH001	COURSE CONTENT: 3.1. ALGEBRA Unit-I: (4 Lectures) Arithmetic, geometric and harmonic progressions and their sums. Infinite geometric series. Sums of the squares and cubes of the first n natural n Orthogonal and symmetric matrices. Determinant of matrices with polynomial entries. Determinant of AT, A-1, adj(A), A2, -A, PTAP. Solution of third- and fourth-degree polynomial equations. Unit-III: (2 Lec sytems. Cramer rule. Unit-IV: (2 Lectures) Complex numbers. Basic inequalities of complex numbers. Few geometrical regions by the inequalities of complex variables. 3.2. PERMUTATION AND COMBINA formulae for nPr and nCr and their connections. Simple applications. 3.3. ANALYTICAL GEOMETRY OF TWO DIMENSIONS Unit-VI: (7 Lectures) Equations of the conic sections in the standard form, focus, of tangent and normal. Change of coordinates by Y = P X. Converting quadratic forms of two variables into standard forms. Polar coordinates system. 3.4. ANALYTICAL GEOMETRY OF THREE DIMENSION a point from a plane. Skew lines. Unit-IX: (4 Lectures) Equation of a sphere. Ellipsoid and hyperboloids. Section of a sphere and an ellipsoid by a plane. General quadratic equation of three variables and its The Elements of Coordinate Geometry, Arihant, 2010 [3] E. Kreyszig, Advanced Engineering Mathematics, John Wiley & Sons, 10th Edition, 2011
1226	MSE	THERMODYNAMICS & PHASE EQUILIBRIA	MSE201A	I. .Introduction Introduction, Thermodynamics terminology2. First law First law of thermodynamics and its applications3. Second Law Second law of thermodynamics and itsconsequences, Combined sta Thermodynamic Auxiliary functions: Helmholtz free energy,functions Gibbs free energy, Chemical potential,Maxwell's relations, GibbsHelmholtz equations.6. Third law Third law of thermodynamics7 with temperatureand pressure, ClausiusClapeyron equation, PTdiagram8. Solutions Thermodynamics_ of solutions: Raoult's andHenry's Law, activity of a component, Regularsolutions, GibbsDu dilutesolutions and interaction ____mtrameters.9. Reaction Equilibria Equilibrium constant, Reaction equilibria for (a)homogeneous reactions consisting of gasmixtures, (b) heterogeneous reactions consist RuleII. Electrochemical cells Them1odynamics of electrochemical cells:Relation between chemical and electrical drivingforces, Nemst equation, Concentration andformation cells, thenndynamics of aqu ofcomposition diagram common binary Free energycompositiondiagrams: Eutectic, Eutectoid, Peritectic etc.13. Phase diagrams Study of some common phase diagrams, such asFeC, CuZn, AlCu, Fe0Si0. I. Thermodynamics of Solids: Richard A. Swalin2. Introduction to Thermodynamics of Materials: David R. Gaskell3. Physical Chemistry of Metals: L.Darken and R.W.Gury4. Problems in Metallurgical Therm Transformation: Porter and Esterling.
1227	MME	MATERIALS DEGRADATION & ITS PREVENTION	MME480	Types of processes leading to degradation of materials, viz Oxidation Corrosion,Wear, Creep and fatigue review of basics of thermodynamics and kinetics relatedto oxidation and corrosion studies, Pourba atmospheric galvanic, intergranular, dealloying, crevice and pittingcorrosion, microbiological, stress corrosion cracking, hydrogen damage, radiationdamagte; Oxidation and hot corrosion of materials at high environment conditioning design modification cathodic and anodicprotection, metallic coating inorganic coating organic coating, inhibitors andpassivators wear resistant materials structural modification
1228	MME	PHYSICAL METALLURGY LAB	MME415	Laboratory techniques for studying phase transformations in materials,recrystallization and grain growth, eutectoid transformations in steels,hardenability, tempering of martensite; resistivity of metals, co transmission measurement on various metals.
1229	MME	HISTORY OF SCIENCE AND TECHNOLOGY OF METALLURGY	MME420	Mining, Smelting, Alloying (mainly copper tin), Metal forming : forging, castingOrigins of metallurgy in Balkans, Near and Middle East ; Metallurgy in PrehistoricWorld, Iron and Steel, Silver, Lead, Brass, Zinc Asia, Metallurgy of Greece and Rome, Metallurgy in Europe andthe Middle East, Metallurgy of the Americas and Africa.
1230	PHY	THERMAL PHYSICS	PHY210	Principles of thermodynamics (with applications to simple fluids), applicationsof thermodynamics: concept of thermodynamic state, extensive and intensivevariables; heat and work, internal energy functi temperature as conjugate pair of variables; secondlaw of thermodynamics, entropy maximum and energy minimum principles;thermodynamic potentials: enthalpy, Helmholtz potential, Gibbs potential;cor relations; firstorder phase transitionsand ClausiusClapeyron equation; concepts associated with critical and multicriticalphenomena, some chosen applications from surfaces and interfaces,chemical reac gases: equilibriumproperties pressure and equation of state; transport processes momentumtransport and viscosity, energy transport and thermal conductivity, chargetransport & electrical conductivity (thandisorder, illustration with simple examples; Maxwells demon; qualitativejustifications of laws of thermodynamics (without introducing ensembles),thermodynamics of irreversible processes: entropy p
1231	PHY	PHYSICS - II	PHY103N	Vector calculus; electrostatics; Gauss law and applications, electrostatic potentialand curl of E; work and energy in electrostatics, Laplaces equation and (first)LTPD[C]1020[0]CoreLTPD[C]0003[2]CoreLTPD andtorque on dipoles; polarization, bound charges, electric displacementand boundary conditions, linear dielectrics. force on dielectrics; motion ofcharges in electric & magnetic fields; magnetostatics: cur densities, magnetic field H, magnetic susceptibility,ferro, para and diamagnetism, boundary conditions on B and H, Faradayslaw, energy in magnetic field, displacement current, Maxwells equations inmedium momentum of plane e.m. waves.propagation through linear media and conductors. reflection and transmissionat normal incidence from dielectric and metal interfaces. magnetism as arelativistic phenolr uncertaintyprinciple, electron diffraction experiments, de Broglie hypothesis, Borninterpretation, Schrdinger equation and application to 1d box problem.
1232	PHY	PHYSICS-I (PREP)	PHY002	<p>COURSE CONTENT UNIT I: Mechanics [23 lectures] Newton's laws of motion for a single particle, Conservation of linear momentum and energy, Moment of force and Torque, Conservation of ang inertia, Parallel and perpendicular axis theorem. Binary collisions in the centre of mass coordinates and lab frame, Elastic and inelastic collisions. Kepler's laws of planetary motion, Escape velocity a Stefan's and Planck radiation law, Newton's law of cooling, Photoelectric effect, Wave particle duality, de Broglie hypothesis, Davisson Garmer experiment, Heisenberg uncertainty principle. UNI Distribution,Zeroth, first and second laws of thermodynamics. 4. READINGS 4.1 TEXT BOOKS: 1. 2. D. Kleppner and R. Kolenkow, An Introduction to Mechanics, Tata McGraw-Hill, 2007. D. Halliday and R. F

S.No.	Branch	Course Title	Courses ID	Content
				Modern Physics, McGraw-Hills Co., Inc., New York. Feynman Lectures on Physics, Volume I& II, Narosa Publishing House, 1998.</p>
1233	HSS	HUMAN COGNITIVE PROCESSES	PSY790	The course deals with scientific study of human mind and explores cognitiveprocesses involved in perception, memory, pattern recognition, psycholinguistics,and bilingualism. Recent research techniques
1234	HSS	MANAGEMENT OF ORGANIZATIONAL BEHAVIOUR	PSY787	The course focuses on a practical training to understand the human behaviour at workin order to predict the effectiveness and wellbeing in various types of organizations,and to enhance the understanding
1235	HSS	APPLIED SOCIAL PSYCHOLOGY	PSY784	This course will explain how social psychological theories can be applied to reallife. The prime objective is to disseminate findings from behavioural scienceresearch which have relevance for problems of s
1236	HSS	ADVANCED EXPERIMENTAL SOCIAL PSYCHOLOGY	PSY783	The course includes experimental paradigms in contemporary social psychologycovering areas such as attitude and attitude change, group processes, socialpower, reward allocation, prosocial behavior, s
1237	HSS	PSYCHOLOGY OF PERSONALITY	PSY780	Introduction to the field of personality. Trait and situational approaches topersonality are covered, along with a critical assessment of the major theoriesof personality.
1238	HSS	FUNDAMENTAL OF NEUROPSYCHOLOGY	PSY775	Development and evolution of brain : Brain organization and function; Cerebralasymmetry; Neuropsychology of higher order functions memory language,emotional processes, spatial behaviour ; Applied
1239	HSS	COMMONWEALTH LITERATURE	ENG735	The course covers literature in English in the Commonwealth countries of Africa,Australia, Canada, India, New Zealand, and the West Indies. An attempt willbe made to show that commonwealth writers, w
1240	HSS	INDIAN LITERATURE	ENG733	The course deals with major trends and developments in Indian literaturefrom ancient to modern times. The course will focus primarily on the followingtopics: the problem of definition;the growth of nationa
1241	HSS	AMERICAN RENAISSANCE	ENG719	The course will study the major themes and techniques in the literature of theAmerican Renaissance, with special attention to the writings of Melville,Hawthorne, Emerson, Thoreau and Poe.
1242	HSS	INDIVIDUAL PROJECTS	ENG716	This course is intended to provide the students an opportunity to take up ontheir own a shortterm fieldwork or library project, with constant guidancefrom the Instructor. Evaluation will depend on the final
1243	MTH	LINEAR PROGRAMMING AND EXTENSIONS	MTH306A	Linear Models: Formulation and Examples, Basic Polyhedral Theory Convexity,Extreme points, Supporting hyperplanes etc, Simplex Algorithm Algebraic andGeometrical approaches, Artificial variable tech
1244	MTH	LINEAR PROGRAMMING AND EXTENSIONS	MTH306	Linear Models: Formulation and Examples, Basic Polyhedral Theory Convexity,Extreme points, Supporting hyperplanes etc, Simplex Algorithm Algebraic andGeometrical approaches, Artificial variable tech
1245	MTH	TOPOLOGY	MTH438	Topological spaces, Basis for a topology, The order topology, Subspace topology,Closed sets. Countability axioms, Limit points, Convergence of nets in topologicalspaces, Continuous functions, The produ
1246	MTH	TOPOLOGY	MTH304	Topological spaces, Basis for a topology, The order topology, Subspace topology,Closed sets. Countability axioms, Limit points, Convergence of nets in topologicalspaces, Continuous functions, The produ
1247	MTH	BASIC STRUCTURE OF MATHEMATICS	MTH300	Finite and Infinite Sets: Finite sets, Countable sets, Uncountable sets. Groupsand Symmetry: Groups, Subgroups, Lagrange theorem, Normal subgroups,Quotient groups, Group actions, Homomorphisms, C
1248	MTH	MATHEMATICS - III	MTH203N	Introduction and Motivation to Differential Equations, First Order ODE $y' = f(x,y)$ geometrical Interpretation of solution, Equations reducible to separable form,Exact Equations, Integrating factor, Linear Equations, Improved Eulers Method, Elementary types of equations. $F(x,y,y'')$ 0; not solved for derivative, Second Order Linear differential equations:fundamental system of solutions and general solution of hom
1249	MTH	SET THEORY AND DISCRETE MATHEMATICS	MTH202	Permutations and combinations and basic definitions. Generating functions.Polyas enumeration theory. Recurrence relations. Principle of inclusion andexclusion. Balanced incomplete block design. Diffe
1250	MTH	A FIRST COURSE IN LINEAR ALGEBRA	MTH201A	Fields and linear equations. Vector spaces. Linear transformations and projections,Determinants. Elementary canonical forms: diagonalization, triangulation,primary decomposition etc. Secondary decom
1251	MTH	INTRODUCTION TO PROFESSION	MTH100	Mathematical thought process: Proofs by construction, existence, specialization,induction, contradiction, Abstraction, Sets: Russels paradox, Axiom of Choice,Counting, Infinity, Continuum Hypothesis, Nu
1252	MSE	INTRODUCTION TO MANUFACTURING PROCESSES	TA201N	Introduction to Manufacturing, Historical perspective; Importance ofmanufacturing; Classification of manufacturing processes, Engineering materials,Casting, Fundamentals of casting, Sand casting, Perr
1253	MSE	MANUFACTURING PROCESSES I	TA201T	Introduction to manufacturing; Evolution of manufacturing; Importance ofdesign in manufacturing; Conventional material removal processes: chip formation, tooldynamics, practical machining and finishi
1254	CSE	MATHEMATICS FOR COMPUTER SCIENCE -II	CS202A	1. Propositional logic syntax and semantics.2. Tautologies, axiom system and deduction.3. Proof of soundness and completeness.4. First order logic syntax and semantics.5. Structures, models, satisfact
1255	CSE	MATHEMATICS FOR COMPUTER SCIENCE -I	CS201A	1. Mathematical proofs, proofs by induction, by contradiction, proving the contrapositive.2. Basic counting techniques, pigeonhole principle, recurrence relations, generating functions, principle of inclusion
1256	CHM	INORGANIC CHEMISTRY LABORATORY	CHM344A	Estimation of iron in minute quantities by UVvis spectrophotometryPrinciples of colorimetric analysis: determination of iron content of an unknown sample.Preparation of hexamminenickel(II)chloride: est

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		FABRICATION TECHNOLOGY		
1279	EE	SOLAR PHOTOVOLTAIC TECHNOLOGIES	EE615	Review of Semiconductor properties Crystal structure of semiconductors, bandtheory, occupation statistics, electrical properties, optical properties, recombinationkinetics, avalanche process in semiconductors; Frequency multipliers, frequency amplifiers and oscillators, Resonant tunneling structures, RTDoscillators; Intervalley scattering, Gunn diodes, IMPATT diodes; TRAPATTs; Mixerdiodes; Step recovery diodes; Introduction to epitaxial growth techniques
1280	EE	STATISTICAL SIGNAL PROCESSING-I	EE602A	Power Spectrum EstimationParametric and Maximum Entropy Methods, Wiener,Kalman Filtering, LevinsonDurban Algorithms Least Square Method, AdaptiveFiltering, Nonstationary Signal Analysis, Wigner Distribution
1281	EE	STATISTICAL SIGNAL PROCESSING-II	EE602	Power Spectrum EstimationParametric and Maximum Entropy Methods, Wiener,Kalman Filtering, LevinsonDurban Algorithms Least Square Method, AdaptiveFiltering, Nonstationary Signal Analysis, Wigner Distribution
1282	HSS	QUALITATIVE RESEARCH : THEORY AND PRACTICE	PSY774	In the last few years, research in psychology has witnessed a remarkable shiftto a post positivist paradigm. A large number of qualitative methods haveemerged in consonance with this shift. For a researcher to understand the nature and understanding of the qualitative methods. This course isaimed at providing an extensive overview of qualitative research methods,methodological roots, major theoretical principles and issues in qualitative research, developmental psychology discursive psychology,counseling and psychotherapy.
1283	NET	NUCLEAR FUEL CYCLE	NT652	Introduction, nuclear fuels, uranium technology, xirconium process, babricationof fuel assemblies, PWR fuel, mixedoxide fuel, irradiated fuel, reprocessing,radioactivity, contamination, waste management, environmental impact
1284	NET	FAST REACTOR TECHNOLOGY	NT651	Introduction, Core design, Fuelelement design, Fuel management, Heat transportsystems, Steamgenerators, IHX design, Sodium pumps & piping, Instrumentation& controls, safety, extractive and physical chemistry of fast reactors, radiation environment, advanced NDT techniques, corrosion.
1285	NET	NON-DESTRUCTIVE EVALUATION	NT642A	Introduction, various NDE techniques ultrasonics, eddy current, magnetic fluxleakage, radiography, optical, tomographic extensions of classical NDE/NDT methodsRadon inversion, data collection mechanical testing
1286	ME	NON-DESTRUCTIVE EVALUATION	NT642	Introduction, various NDE techniques ultrasonics, eddy current, magnetic fluxleakage, radiography, optical, tomographic extensions of classical NDE/NDT methodsRadon inversion, data collection mechanical testing
1287	NET	INTRODUCTION TO COMPUTERIZED TOMOGRAPHY	NT641	Overview, medical imaging, nondestructive testing, radiographic techniques,various applications, data collection, design of CT scanners for materials testing,flow measurement, related instrumentation, Radon inversion, reconstruction transformmethods, serieexpansion methods, convolution algorithms, error estimates,direct theorems, inverse theorems.
1288	NET	NUCLEAR FUSION	NT633	Basic physics of fusion reactions, thermonuclear crosssections. Radiation losses:bremsstrahlung and cyclotron radiation. Energy balance: Lawson criterion,neutronnics in a fusion reactor. Plasma confinement: tokamak, laser fusion, Inertial confinement of plasma microexplosion and laser fusion.
1289	NET	NUCLEAR POWER ENGINEERING III	NT615	Health Physics: introduction, radiation protection, regulatory aspects, radiationbiology, operational radiation protection, radiation protection monitoring.Process Instrumentation and Control: basic concepts, PID control systems, safety systems. Materials: fabrication and properties of zircaloy, metallic fuels, ceramicfuels, applications.Nuclear Chemistry: role of chemistry in nuclear engineering, chemical processesin the nuclear fuel cycle, product separation
1290	NET	NUCLEAR POWER ENGINEERING II	NT614	Introduction to control theory. Point reactor kinetics with introduction to feedbackeffects. Nonlinear effects. Shielding. Introduction to reactor reliability and safety analysis. Radioactive waste disposal. Economics of nuclear power generation.
1291	NET	NUCLEAR POWER ENGINEERING I	NT611A	Types of nuclear reactors. Heat generation in fuel elements and temperaturedistributions. Heat removal, Reactor coolants. Single phase and two phase heattransfer. Boiling and flow regimes. Heat transfer correlations. Safety systems.
1292	ME	NUCLEAR POWER ENGINEERING I	NT611	Types of nuclear reactors. Heat generation in fuel elements and temperaturedistributions. Heat removal, Reactor coolants. Single phase and two phase heattransfer. Boiling and flow regimes. Heat transfer correlations. Safety systems.
1293	NET	NUCLEAR AND REACTOR PHYSICS	NT602A	Introduction to quantum mechanics, Schrodinger equation and its solution byseparation of variables. Potential well, quantum states. Nuclear charge, radiusand mass. Binding energy. Nuclear forces and types of nuclear interactions. Compoundnucleus. Energy dependence of neutron cross sections and Brecitwigner formula.The fission process.Neutron diffusion theory showing down length. Critical mass and size. Numericalcriticality calculations. Two group analysis. Age theory. Slowing down theory.
1294	NET	NUCLEAR AND REACTOR PHYSICS	NT602	Introduction to quantum mechanics, Schrodinger equation and its solution byseparation of variables. Potential well, quantum states. Nuclear charge, radiusand mass. Binding energy. Nuclear forces and types of nuclear interactions. Compoundnucleus. Energy dependence of neutron cross sections and Brecitwigner formula.The fission process.Neutron diffusion theory showing down length. Critical mass and size. Numericalcriticality calculations. Two group analysis. Age theory. Slowing down theory.
1295	ME	ENGINEERING ACOUSTICS AND ITS CONTROL	ME778A	Wave propagation in solids and fluids. Admittance and impedance concepts forinfinite and finite acoustic waveguides. Sound radiation from vibrating structures:Measurement of radiated power, elementarystress and strain, stress intensity factor, fatigue crack growth rate, fracture toughness of elastic interlayers, blocking masses, changes of material and crosssection.Active control of structureborne sound : Elements of frequency domaincontrol system analysis and synthesis, wave absorbing materials
1296	CHM	PHYSICAL CHEMISTRY I	CHM321A	1. Introduction: importance, historic background, quantum mechanics vs classical mechanics,waveparticle duality, uncertainty principle2. Schroedinger equation: wavefunction and interpretation, time dependent perturbation theory3. Molecular spectroscopy: electronic transitions, vibrational transitions, rotational transitions4. Angular Momentum: rigid rotor, orbital and spin angular momentum5. Hydrogen and hydrogen like atoms6. Quantum numbers and atomic spectra7. Pauli antisymmetry principle, Slater determinant, He atom, Li atom. 1. I. N. Levine, Quantum Chemistry2. J. P. Lowe and K. A. Peterson, Quantum Chemistry3. D. A. McQuarrie, Quantum Chemistry4. D. A. McQuarrie, J. D. Simon, Physical Chemistry: A molecular approach5. Atkins and de Paula, Physical Chemistry
1297	CHM	ORGANIC CHEMISTRY I	CHM303A	Stereochemistry (8)Dynamic stereochemistry: Conformation and Reactivity. Various chemo, regio and stereoselectivereactions.Reactive Intermediates : (8) Carbenes and carbenoids Radicals: Structure, reactivity and radical anions. Carbocations: Nonclassical carbocation. Sigma and pi participation.Mechanistic and Stereochemical Aspects of : (10)BaeyerVilliger, Claisen (including JohnsonClaisen, Ireland rearrangements);ene and metalloene reactions; (2+2), (3+2) and (4+2) cycloadditions; Barton reaction.Organometallic Chemistry : (5)Mechanism and stereochemistry of various reactions. Palladium based catalysed reactions.Enzymatic Reactions : (3)Mechanism and stereochemical aspects of hydrolases (including esterases and lipases), oxidoreductases.Green Chemistry : Concepts and applications (3) 1. March, J., Advanced Organic Chemistry, 4th ed, 1999.2. Nasipuri, D., Stereochemistry of Organic Compounds, 2nd ed., 1995.3. Solomons, T. W. G., Organic Chemistry 6th ed, 1996.4. Sykes, Peter, A guide to organic chemistry mechanisms, 20107. M. B. Smith, Organic Synthesis, 3rd Ed. 20108. Clayden, Greeves, Warren, and Wothers, Organic Chemistry, 1st ed, 2001
1298	CHM	BASIC ORGANIC CHEMISTRY -II	CHM202A	Oxidation: (5)With Cr and Mn compounds; with peracids and other peroxides; with periodic acid, Pb(OAc)4,Hg(OAc)2 and SeO2.Reduction: (6)Catalytic hydrogenation; metal hydride, dissolving metal and hydrazine reductions; Aldol condensation and Benzoin condensations, Darzens glycidic estersynthesis; Dieckmann reactions, Wittig reaction, DielsAlder and ene reactions, Reformatsky reaction.Acetoacetic ester and malonic ester synthesis. Acyl substitution reactions. Polynuclear Hydrocarbons: (2)Carbohydrate Chemistry: Introduction, Structural elucidation and some typical eactions of mono and disaccharides (5)Heterocyclic Chemistry: Furan, Pyrrole, Thiophene, Pyridine, Imidazole Reduction (to be solved inthe class and supplemented by home assignments). 1. Carruthers, W., Coldham, I. Some Modern Methods of Organic Synthesis, 2008.2. House, H. O., Modern Synthetic Reactions.3. March, J., Advanced Organic Chemistry, 4th ed, 1999.4. Clayden, Greeves, Warren, and Wothers, Organic Chemistry, 1st ed, 20015. Organic Mechanisms, 20107. M. B. Smith, Organic Synthesis, 3rd Ed. 2010
1299	CE	SPECIAL TOPICS IN STRUCTURAL DESIGN	CE471A	Elements of Prestressed concrete; Introduction to seismic design,Introduction to design of masonry structures; Case studies in design of RC Structures(RC water tank, Building frame, etc.), Case studies in design of Steel Structures(Steel moment-resisting frames, Steel trusses, Steel towers, etc.) 1 Pillai, S.U., and Menon, D., Reinforced Concrete Design, Tata McGraw Hill, 20092 Sinha, S.N., Reinforced Concrete Design, Tata McGraw Hill.3 Subramanian N. (2008). Design of Steel Structures, Oxford University Press International5 Krishna Raju N. (1995). Prestressed Concrete, Tata McGraw Hill6 Dayaratnam, P., "Brick and Reinforced Brick Structures", Oxford & IBH PublishingHouse, 1997
1300	CE	DESIGN OF STEEL STRUCTURES	CE371A	Steel structures, Limit states and design philosophy; partial safetyfactors and load combinations; Analysis and design methods; Design of tensionmembers based on net section including shear lag effect; Design of compression members based on Euler's buckling formula, Local buckling and buckling class, End restraints and effective length factor; Role of plate buckling, Plastic hinge, Classification of section: plastic,compact, semicompact, slender, Design strength of welded joints, Bolted connections, Moment-resisting steel frames, Design of composite steel-concrete beams, Design of steel deck slabs, Design of steel bracing systems, Design of steel towers, Design of steel bridges, Design of steel chimneys, Design of steel storage tanks, Design of steel silos, Design of steel piers, Design of steel piles, Design of steel foundations, Design of steel offshore structures, Design of steel marine structures, Design of steel industrial structures, Design of steel agricultural structures, Design of steel power plant structures, Design of steel petrochemical structures, Design of steel pharmaceutical structures, Design of steel food processing structures, Design of steel textile mill structures, Design of steel paper mill structures, Design of steel pulp mill structures, Design of steel rubber mill structures, Design of steel glass mill structures, Design of steel cement mill structures, Design of steel sugar mill structures, Design of steel oilseed mill structures, Design of steel distillery structures, Design of steel brewery structures, Design of steel winery structures, Design of steel dairy structures, Design of steel fish processing structures, Design of steel meat processing structures, Design of steel leather processing structures, Design of steel tanneries, Design of steel printing press structures, Design of steel bookbinding structures, Design of steel furniture manufacturing structures, Design of steel machinery manufacturing structures, Design of steel electrical equipment manufacturing structures, Design of steel electronics manufacturing structures, Design of steel pharmaceutical manufacturing structures, Design of steel food processing manufacturing structures, Design of steel textile manufacturing structures, Design of steel paper manufacturing structures, Design of steel pulp manufacturing structures, Design of steel rubber manufacturing structures, Design of steel glass manufacturing structures, Design of steel cement manufacturing structures, Design of steel sugar manufacturing structures, Design of steel oilseed manufacturing structures, Design of steel distillery manufacturing structures, Design of steel brewery manufacturing structures, Design of steel winery manufacturing structures, Design of steel dairy manufacturing structures, Design of steel fish processing manufacturing structures, Design of steel meat processing manufacturing structures, Design of steel leather processing manufacturing structures, Design of steel tanneries manufacturing structures, Design of steel printing press manufacturing structures, Design of steel bookbinding manufacturing structures, Design of steel furniture manufacturing manufacturing structures, Design of steel machinery manufacturing manufacturing structures, Design of steel electrical equipment manufacturing manufacturing structures, Design of steel electronics manufacturing manufacturing structures, Design of steel pharmaceutical manufacturing manufacturing structures, Design of steel food processing manufacturing manufacturing structures, Design of steel textile manufacturing manufacturing structures, Design of steel paper manufacturing manufacturing structures, Design of steel pulp manufacturing manufacturing structures, Design of steel rubber manufacturing manufacturing structures, Design of steel glass manufacturing manufacturing structures, Design of steel cement manufacturing manufacturing structures, Design of steel sugar manufacturing manufacturing structures, Design of steel oilseed manufacturing manufacturing structures, Design of steel distillery manufacturing manufacturing structures, Design of steel brewery manufacturing manufacturing structures, Design of steel winery manufacturing manufacturing structures, Design of steel dairy manufacturing manufacturing structures, Design of steel fish processing manufacturing manufacturing structures, Design of steel meat processing manufacturing manufacturing structures, Design of steel leather processing manufacturing manufacturing structures, Design of steel tanneries manufacturing manufacturing structures, Design of steel printing press manufacturing manufacturing structures, Design of steel bookbinding manufacturing manufacturing structures, Design of steel furniture manufacturing manufacturing manufacturing structures, Design of steel machinery manufacturing manufacturing manufacturing structures, Design of steel electrical equipment manufacturing manufacturing manufacturing structures, Design of steel electronics manufacturing manufacturing manufacturing structures, Design of steel pharmaceutical manufacturing manufacturing manufacturing structures, Design of steel food processing manufacturing manufacturing manufacturing structures, Design of steel textile manufacturing manufacturing manufacturing structures, Design of steel paper manufacturing manufacturing manufacturing structures, Design of steel pulp manufacturing manufacturing manufacturing structures, Design of steel rubber manufacturing manufacturing manufacturing structures, Design of steel glass manufacturing manufacturing manufacturing structures, Design of steel cement manufacturing manufacturing manufacturing structures, Design of steel sugar manufacturing manufacturing manufacturing structures, Design of steel oilseed manufacturing manufacturing manufacturing structures, Design of steel distillery manufacturing manufacturing manufacturing structures, Design of steel brewery manufacturing manufacturing manufacturing structures, Design of steel winery

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				oflaterally unsupported beams, Lateral torsional buckling, Effect of restraints andeffective length; Effect of axial load on flexure behaviour, Crosssection yielding andmember instability, PM interaction and action,Common simple and eccentric joints and frame connections, Column bases. Subramanian, N. (2008). Design of Steel Structures, OxfordUniversity Press.Bhavikatti, S. S. (20 10). Design of Steel Structures (by LimitState Method as Per IS: 8002007), IK International.Segui, W.T. (2007)
1301	EE	DIGITAL SWITCHING	EE629A	Network Architecture; time division multiplexing; digital switching; space & timedivision switching, cross point and memory requirements; blocking probabilities.traffic Analysis, models for circuit and pack
1302	EE	DIGITAL SWITCHING	EE629	Network Architecture; time division multiplexing; digital switching; space & timedivision switching, cross point and memory requirements; blocking probabilities.traffic Analysis, models for circuit and pack
1303	EE	TOPICS IN CRYPTOGRAPHY AND CODING	EE628	Cryptography and error control coding in communication and computing systems.Stream and block ciphers; DES; publickey cryptosystems; key management,authentication and digital signatures. Codes
1304	EE	SPEECH SIGNAL PROCESSING	EE627A	Spectral and nonspectral analysis techniques; Modelbased coding techniques;Noise reduction and echo cancellation; Synthetic and coded speech qualityassessment; Selection of recognition unit; Model
1305	EE	SPEECH SIGNAL PROCESSING	EE627	Spectral and nonspectral analysis techniques; Modelbased coding techniques;Noise reduction and echo cancellation; Synthetic and coded speech qualityassessment; Selection of recognition unit; Model
1306	EE	TOPICS IN STOCHASTIC PROCESSES	EE626A	Martingale convergence theorem, stopping times, sequential analysis. ErgodicTheory: Measure preserving transformations, stationary processes, mixingconditions, ergodic theorem, ShannonMillanBreim
1307	EE	TOPICS IN STOCHASTIC PROCESSES	EE626	Martingale convergence theorem, stopping times, sequential analysis. ErgodicTheory: Measure preserving transformations, stationary processes, mixingconditions, ergodic theorem, ShannonMillanBreim
1308	EE	DETECTION & ESTIMATION THEORY	EE623A	Classical Detection and Estimation Theory, Signal Representation, Detection ofsignals in Gaussian noise, Waveform estimation, Linear estimation problems,Wiener filtering, Kalman filtering.
1309	EE	DETECTION & ESTIMATION THEORY	EE623	Classical Detection and Estimation Theory, Signal Representation, Detection ofsignals in Gaussian noise, Waveform estimation, Linear estimation problems,Wiener filtering, Kalman filtering.
1310	CSE	FUNCTIONAL PROGRAMMING	CS653A	ML (CAML dialect); Xcalculus and combinators; abstraction and higher order functions; lazy and eager evaluation; types, polymorphism and type inference; Equations and pattern matching; SECD machin
1311	CSE	MOBILE COMPUTING	CS634A	Introduction: Mobile computing a vision for future, ubiquitous computing versusvirtual reality, software models for mobile computing. Data managementIssues. Distributed algorithms and mobility: structur the air: pull and push based data transfers, data disseminationby broadcast, treating air as cache, energy efficient indexing in air. Handoffmanagement: handoff detection, failures, channel assignments. L location management. Approximatequery processing: concept hierarchy, summary database, updates and viewmaintenance, approximate query processing. Mobile Transaction Models. MobileComputing voice and data serviceson 3G networks, battery problem and power dissipation, low energy processors.File system support for mobile computing: Coda and Bayou file systems.Adhoc network routing prot
1312	CSE	COMPUTER SYSTEMS SECURITY	CS628A	Introduction: need and basic goals for computer security, security threatsetc.Cryptographic building blocks: Symmetric and asymmetric key cryptography,cryptographic hash functions, digital signature so access control: models for access control, some confidentiality,integrity, and hybrid models of access control such as BellLa Padula, Biba,Chinese Wall etc., discretionary v/s mandatory access control.Ca which have security implicationssuch as buffer overflows, race conditions etc. Malicious code: Viruses, worms,Trojan horses; how they work and how to defend against them.Network Security: problems i VPN,SSL, S/MIME etc. Protocol vulnerabilities: examples of protocol vulnerabilitiessuch as in TCP/IP, denial of service attacks etc. Tools for network securitysuch as firewalls and intrusion detection syste
1313	CSE	COMPUTING LABORATORY II	CS252A	1. System administration: Students will learn to setup and manage a network server including a web server andan email server. They should also be familiar with various network protocols. Further, studer learn simpleloadbalancing tools.2. Security: While setting up different network and database servers, students will learn to manage the securityissues including different attacks. Public key infrastructure of maintaining sys terns, students willneed to use system vulnerability and intrusion testing tools as well.3. Compiler tools: Students will need to learn the various lowlevel tools used routinely in compilers various integrateddevelopment environments (IDEs) such as eclipse and debuggers such as gdb. They should also use tools foreffective tagging and browsing of source code. Finally, they will learn to use 1. Manuals.2. Online help that is available for different tools.
1314	EE	MATHEMATICAL METHODS IN SIGNAL PROCESSING	EE601A	Generalized inverses, regularization of illposed problems. Eigen and singularvalue decompositions, generalized problems. Interpolation and approximationby least squares and minimax error criteria. Optim
1315	EE	MATHEMATICAL METHODS IN SIGNAL PROCESSING	EE601	Generalized inverses, regularization of illposed problems. Eigen and singularvalue decompositions, generalized problems. Interpolation and approximationby least squares and minimax error criteria. Optim
1316	EE	MATHEMATICAL STRUCTURES OF SIGNALS & SYSTEMS	EE600A	Nature of definitions; Theory of measurement and scales; Symmetry, invarianceand groups; Groups in signals and systems; Algebraic and relational structuresof signal spaces and convolutional systems,
1317	EE	MATHEMATICAL STRUCTURES OF SIGNALS & SYSTEMS	EE600	Nature of definitions; Theory of measurement and scales; Symmetry, invarianceand groups; Groups in signals and systems; Algebraic and relational structuresof signal spaces and convolutional systems,
1318	EE	TRANSDUCERSAND INSTRUMENTATION	EE455	Measurement process; scales of measurement; configuration and functionaldescription of measurement systems; performance characteristics; sensingelements and transducers for measurement of mo recording; microprocessor based instrumentation.
1319	EE	ANTENNAS AND PROPAGATION	EE442A	Retarded potential, radiation from current element and dipole, radiationpatterns, impedance, reciprocity. Various types of antennas, interferometers andmultielement arrays, Antenna Measurements. Groun lonospheric Propagationrefractive index, critical frequencies, effectsof magnetic field.
1320	EE	SIGNALS SYSTEMS AND NETWORKS	EE200	Continuous and discrete time signals; Fourier series, Fourier, Laplace and Ztransform techniques; DFT. Sampling Theorem. LTI systems: I/O description,impulse response and system functions, pole/ zero analysis.
1321	IME	BUSINESS TO BUSINESS MARKETING	MBA637A.	Business to Business Marketing encompasses those management activities thatenable a supplier firm to understand, create, and deliver value to otherbusinesses, governments, and/or institutional custom past, the topical area applied largely to industrial manufacturingfirms. Today, business to business marketing provides practical frameworks,concepts, and tools for organizations as diverse as managem among many other leadingedge technology and servicecompanies.As business to business marketing expands its scope and stature, this courseaims at reinvigorating training in marketing beyond the tir multifunctional teams, strategic alliance environmentalsensitivity, interorganizational trust, organizational learning and adherence toethical principles. Furthermore, with the advent of relationship and netw share value with customer firms.Understanding of business buying and marketing behaviour within the contextof relationship/network theories is the central learning from this proposedelective.Given this organizational buying behaviour andbusinesstobusiness marketing.
1322	IME	CONSUMER BEHAVIOUR	MBA634A.	Consumers in the Market place: (a) An introduction to Consumer Behaviour,Consumers as individuals: (a) Perception, (b) Learning and Memory, (c) Motivation,Values and Involvement, (d) Attitudes, (e) Att The Purchase Situation, Postpurchase Evaluation and ProductDisposal, (c) Group Influence, Opinion Leadership, (d) Organizational andHousehold Decision Making, Consumers and Sub Cultures: (a) Incon Consumer Behaviour, (b)Lifestyles and Global Culture, (c) Sacred and Profane Consumption.

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1323	IME	CONSUMER BEHAVIOUR	MBA634.	Consumers in the Market place: (a) An introduction to Consumer Behaviour,Consumers as individuals: (a) Perception, (b) Learning and Memory, (c) Motivation,Values and Involvement, (d) Attitudes, (e) Attitudes and Attitude Change, (f) The Purchase Situation, Postpurchase Evaluation and ProductDisposal, (c) Group Influence, Opinion Leadership, (d) Organizational andHousehold Decision Making, Consumers and Sub Cultures: (a) Incorporating Consumer Behaviour, (b)Lifestyles and Global Culture, (c) Sacred and Profane Consumption.
1324	IME	MARKETING RESEARCH	MBA633A	Nature and scope of Marketing research: (a) The marketing research process,(b) Research design and Implementation. Data collection : (a) Secondary sourcesof marketing data, (b) Standardized sources, (c) Customized sources, (d) Measurement, (f) Experimentation, (g) Sampling fundamentals. Data Analysis: (a) Hypothesis testing: Basic concepts and tests of associations, (b)Correlation regression analysis, (c) Discriminant and Canonical Correlation results. Recent Trends: (a) Social media for market research (b) Bayesian methods for marketing research (c) Agent based modeling.
1325	IME	MARKETING RESEARCH	MBA633.	Nature and scope of Marketing research: (a) The marketing research process,(b) Research design and Implementation. Data collection : (a) Secondary sourcesof marketing data, (b) Standardized sources, (c) Customized sources, (d) Measurement, (f) Experimentation, (g) Sampling fundamentals. Data Analysis: (a) Hypothesis testing: Basic concepts and tests of associations, (b)Correlation regression analysis, (c) Discriminant and Canonical Correlation results. Recent Trends: (a) Social media for market research (b) Bayesian methods for marketing research (c) Agent based modeling.
1326	ME	ENGINEERING ACOUSTICS AND ITS CONTROL	ME778	Wave propagation in solids and fluids. Admittance and impedance concepts forinfinite and finite acoustic waveguides. Sound radiation from vibrating structures:Measurement of radiated power, elementarystress and strain, propagation of elastic interlayers, blocking masses, changes of material and crosssection.Active control of structureborne sound : Elements of frequency domaincontrol system analysis and synthesis, wave absorbing materials.
1327	LT	INTRODUCTION TO LASERS	LT601	Introduction to general lasers and their types, Brief intro. quantum physics,Schrodinger wave eqn., Atomic systems, emission and absorption processes,Population inversion, gain, optical cavities, three and four level systems, atomic, molecular, liquid, and excimer lasers, Optical pumping.
1328	CE	ENGINEERING HYDRAULICS	CE262A	Introduction. Review of the basic equations: continuity, momentum, and energy. Flow through closed conduits: Laminar flow, Turbulent flow, Pipes in Series and Parallel, Pipe Networks, Unsteady flow. Flow measurement: Pressure, Velocity and Discharge measurements. Forces on immersed bodies:Drag and Lift. Basics of Irrigation Engineering: Crop water requirements, Irrigation methods and systems.
1329	HSS	INTRODUCTION TO STATISTICAL INFERENCE	SOC723	Basic statistics; correlation and regression analysis; probability and probabilitydistributions; sampling designs; tests of significance; analysis of variance; nonparametric methods; recent developments in statistical inference.
1330	HSS	SOCIOLOGICAL THEORY	SOC721	Relation between theory and research. The nature and use of theory, basic concepts, fundamental perspectives of society, conceptualization of social system social structure and culture, Various perspectives on social theory: Phenomenology, Ethnomethodology & Structuration theory.
1331	HSS	RESEARCH METHODS	SOC720	Nature and types of scientific explanation, Values and objectivity in social scienceresearch, Various research designs: (i) Descriptive, (ii) Exploratory, (ii)Experimental (two and multigroup designs). Tools and techniques of research: scaling techniques (Differential scales, Summated scalesand Cumulative scales, etc), problems of measurement, Various types ofreliability and validity of measures, Qualitative research.
1332	HSS	SOCIETY AND SOCIAL PROBLEMS OF INDIA	SOC481A	Sociological perspectives on social problems; Social change; Interactionbetween structural and cultural components in the process of social change; Major socialproblems of India such as social and economic inequality, caste, tribalism, governance, communalism andterrorism; The concept of inclusive growth . Beteille, Andre (2000) Antinomies of Society: Essays on Ideologies & Institutions. New Delhi:Oxford University Press.Dumont, Louis (1970) Homo Hierarchicus. London: Paladin, Granada Publishing Ltd.Engel, Karl (1987) The Great Transformation. New York: Oxford University Press.Five Year Plan, 200712. New Delhi: Oxford University Press.Srinivas, M.N. (1966) Social Change in Modern India. Berkeley: California University Press.
1333	IME	MANAGEMENT OF TECHNOLOGY	MBA626.	Policy Technology Choice: Linkage; National Technology Policies; Technology,Competition and Industrial Structure; formulating the technology strategy,Technology Development and Acquisition process, Technology Transfer, Licensing and joint Ventures, Managing Technology Spillovers;Justification of new technology; management accounting and technology;Integration of New with Old technology, Assimilation of Technology.
1334	IME	CORPORATE INNOVATION & ENTREPRENEURSHIP	MBA624.	In the first module this course will take an applied approach to learn theimperatives of lateral thinking and accelerated innovation in large organizationsunder relentless pressure of discontinuity. It will explore the role of innovation in accelerating targeted innovation and new product concept to marketprocesses.In the second module this course will develop from theories of entrepreneurshipan applied approach for managing disruptive innovation. In the third module, otherentrepreneurial strategies, real time monitoring and adaptive control systemsfor small businesses and the role of clusters, community of practitioners forstrategic flexibility will be some of the emerging trends.
1335	IME	MANAGERIAL COMMUNICATION: STRATEGIES AND TACTICS	MBA621A.	Communication Theory & Practice: What is communication? Why is it needed?When is it needed? How is it done? Theory and practical essentials of communication Context,Purpose, Participants, Messages, Channels, Media, Feedback, Barriers to Communication, Managerial Communication? What are thedifferent levels of managerial communication? What is intrapersonal, interpersonal, group and mass communication fundamentals? When are they needed by managers? Communication System, The Media and Tools of Communication Climate, Low Structure: One toOne Communication, High Structure: One to One Communication, Meetings and Conferences,Interaction and Communication, Planning and Producing Effective Business Reports,Business and Managerial Communication Research.Module II Oral presentations by every student to class representing unique contexts; writtenreport on unique corporate situations, such as crisismanagement, conflict resolution, and negotiation. Team Learnings
1336	IME	MANAGERIAL COMMUNICATIONS	MBA621.	The Manager, Interpersonal Communication, Ongoing Communication Process andflow, Organizational Managerial Communication, Personal Language, use andCommunication System, The Media and Tools of Communication, Meetings and Conferences, Interactional Presentation, Keys to Functional Writings,Formats for Business letters and Memos, Exposure to eCommunication, Planningand Producing Effective Business Reports, Business and Managerial Communication class.
1337	ECO	DEVELOPMENT ECONOMICS	ECO332	Economic growth versus economic development, indicators of economic development, economic inequality, poverty, population growth and economic development, models of dual economy, land tenancy and agricultural development.
1338	ECO	MICROECONOMICS II	ECO311	Economic growth versus economic development, indicators of economic development, economic inequality, poverty, population growth and economic development, models of dual economy, land tenancy and agricultural development.
1339	ECO	MICRO ECONOMICS II	ECO303	Static games with complete information Nash Equilibrium; Dynamic games with complete information Subgame perfect Nash Equilibrium; Static games with incomplete information Bayesian Nash Equilibrium; Selection and Moral Hazard; Exchange Economy and definition of Competitive Equilibrium; Pareto efficiency and Pareto set; One consumer one producer economy; Two sector production economy; First and Second welfare theorems; and social welfare functions; Public goods and externalities; Property rights and Coase theorem; Social Choice Theory
1340	EE	SMART GRID TECHNOLOGIES APPLICATIONS	EE680A	Introduction, data flow and graphical programming techniques, Virtualinstrumentation (VI), advantages, VIs and SubVIs, Data acquisition methods,DAQ hardware, PC hardware; Structure, Operating system and software.
1341	EE	SMART GRID TECHNOLOGIES APPLICATIONS	EE680	Introduction, data flow and graphical programming techniques, Virtualinstrumentation (VI), advantages, VIs and SubVIs, Data acquisition methods,DAQ hardware, PC hardware; Structure, Operating system and software.
1342	EE	SMART GRID TECHNOLOGIES	EE679	Review of probability and stochastic processes, Markov chains, Little's theorem,modeling & analysis of M/M/ queues, Burke's Theorem, Reversibility, Methodof stages, Analysis of M/G/1 queues, Queues with priority, queueing networks, diffusionapproximations.
1343	EE	MIMO WIRELESS COMMUNICATION	EE677	Knowledge representation, statespace techniques, logic, semantic networks,frames, script. Production system, object oriented and ANN models. Applicationsin robotic vision and processing of document images. Sequential circuits, Logic families, Number systems, Arithmeticcircuits using SSI/MSI chips. Basic microprocessor architecture, Essentials ofa microcomputer system, Instruction sets, Machine cycles, Interrupts.
1344	EE	NEURAL NETWORKS	EE671	Theory of representation; Two computational paradigms; Multilayer networks;Autoassociative and heteroassociative nets; Learning in neural nets: Supervisedand unsupervised learning; Application of neural networks to pattern recognition.
1345	EE	SPECIAL TOPICS IN POWER ELECTRONICS	EE666A	PWM inverters, Multilevel inverters, Neutral point controlled inverters, Softswitching converters: DCD resonant link inverters, Hybrid resonant linkinverters, Quasi resonant link converters, Switched mode converters.
1346	EE	SPECIAL TOPICS IN POWER	EE666	PWM inverters, Multilevel inverters, Neutral point controlled inverters, Softswitching converters: DCD resonant link inverters, Hybrid resonant linkinverters, Quasi resonant link converters, Switched mode converters.

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		ELECTRONICS		
1347	EE	ADVANCED ELECTRIC DRIVES	EE665A	Closed loop control of solid state DC drives, Scalar and vector control of inductionmotor, Direct torque and flux control of induction motor, Self controlledsynchronous motor drive, Vector control of synchro
1348	EE	ADVANCED ELECTRIC DRIVES	EE665	Closed loop control of solid state DC drives, Scalar and vector control of inductionmotor, Direct torque and flux control of induction motor, Self controlledsynchronous motor drive, Vector control of synchro
1349	EE	SMART ANTENNAS FOR MOBILE COMMUNICATION	EE643A	Statistical signal processing concepts, Basics of mobile wireless communications.Radiofrequency signal modeling and channel characterization. Smart antennasand generalized array signal processing.
1350	EE	SMART ANTENNAS FOR MOBILE COMMUNICATION	EE643	Statistical signal processing concepts, Basics of mobile wireless communications.Radiofrequency signal modeling and channel characterization. Smart antennasand generalized array signal processing.
1351	EE	ANTENNAS ANALYSIS & SYNTHESIS	EE642A	Vector potential; antenna theorems and definitions; dipole, loop, slot radiators;aperture antennas; array theorems; pattern synthesis; self and mutual impedances;scanning antennas; signal processing ant
1352	EE	ANTENNAS ANALYSIS & SYNTHESIS	EE642	Vector potential; antenna theorems and definitions; dipole, loop, slot radiators;aperture antennas; array theorems; pattern synthesis; self and mutual impedances;scanning antennas; signal processing ant
1353	EE	COMPUTATIONAL ELECTROMAGNETICS	EE640A	Review of complex variables, conformal mappings, matrix calculus; SturmLiouville equation; Eigenvalue problem; Guiding structures; Scattering media;Greens function approach; Variational formulation, F
1354	EE	COMPUTATIONAL ELECTROMAGNETICS	EE640	Review of complex variables, conformal mappings, matrix calculus; SturmLiouville equation; Eigenvalue problem; Guiding structures; Scattering media;Greens function approach; Variational formulation, F
1355	EE	HVDC TRANSMISSION & FLEXIBLE A C TRANSMISSION SYSTEMS	EE635A	General aspects of DC transmission, converter circuits and their analysis, DClink controls, faults and abnormal operation and protection;Mechanism of activeand reactive power flow contro; Basic FACTS o
1356	EE	HVDC TRANSMISSION & FLEXIBLE A C TRANSMISSION SYSTEMS	EE635	General aspects of DC transmission, converter circuits and their analysis, DClink controls, faults and abnormal operation and protection;Mechanism of activeand reactive power flow contro; Basic FACTS o
1357	IME	MARKETING OF SERVICES	MBA635.	Service Businesses today are global from inception yet needs intricate localization. Managing on line and on demand multi tasking multiplexities makes Service Marketing challenging and a thriving area o contemporary research findings. Strategic Marketing of Services, Contemporary issues, understanding Customer Requirements, expectations and complexities of Customer Behaiour in the service domain. Positioning and Relationship Marketing. Service Delivery and Service Recovery, Service scapes, Service Quality, aligning Service Design and Standards, Service Pricing, Yield Management. KanoQFD and c number of case studies there will be an extended term project for Case Building.
1358	IME	E- MARKETING	MBA632.	Marketing Fundamentals (environment, competition, consumer behavioursegmentation, Targeting, and positioning, 4Ps product, price, promotion,place), Marketing strategy, Digital marketing Opportunities in sales and marketing, web services, ASP and other financialchoices, Realtime Analytic, Frontline Access, Miscellaneous Emerging opportunities.
1359	IME	ECONOMICS OF BUSINESS POLICY	MBA630A	Internationally, a lot of intergration is taking place between economic theory,particularly industrial organization theory and management sterategy theory. Onthe one hand, industrial organization theorists and management strategy theorists are coming to depend on industrialorganization theory to provide a general framework for organizing the otherwiseincoherent mass of facts available to them.In this context, industrial organization theoryand strategic management theory. It uses the business related tenents ofeconomics (old and new) to develop a coherent analytical basis for the formulationand evaluation of its internalorganization. The course emphasizes practical managerial applications of topicsfrom industrial economics and strategy: economics of scale and scope, industryanalysis market structure compe marketsand executive remuneration.
1360	IME	ECONOMICS OF BUSINESS POLICY	MBA630.	Internationally, a lot of intergration is taking place between economic theory,particularly industrial organization theory and management sterategy theory. Onthe one hand, industrial organization theorists and management strategy theorists are coming to depend on industrialorganization theory to provide a general framework for organizing the otherwiseincoherent mass of facts available to them.In this context, industrial organization theoryand strategic management theory. It uses the business related tenents ofeconomics (old and new) to develop a coherent analytical basis for the formulationand evaluation of its internalorganization. The course emphasizes practical managerial applications of topicsfrom industrial economics and strategy: economics of scale and scope, industryanalysis market structure compe marketsand executive remuneration.
1361	IME	INTERNATIONAL BUSINESS MANAGEMENT	MBA628A.	This course is relevant to all executives who plan or operationalise businessstrategies across multiple countries for international marketing, internationalsourcing, or international ownership. It focuses on the production and marketing of appropriate products and services,and key aspects in operationalising the strategy through organization structure,human resource, international coordination and leadership.While working on the strategy, shall invite frequentattention to businesses that originate or operate in India/Asia. We will also beconscious of Governance relationships of investors from developed countries thataffect their businesses and the theme in ourdiscussions. And, we shall be conscious of how organizations may, throughbusiness without borders, stretch their capacities, and develop new competencesand relationships.Much of the course is for the understanding.
1362	IME	INTERNATIONAL BUSINESS MANAGEMENT	MBA628.	This course is relevant to all executives who plan or operationalise businessstrategies across multiple countries for international marketing, internationalsourcing, or international ownership. It focuses on the production and marketing of appropriate products and services,and key aspects in operationalising the strategy through organization structure,human resource, international coordination and leadership.While working on the strategy, shall invite frequentattention to businesses that originate or operate in India/Asia. We will also beconscious of Governance relationships of investors from developed countries thataffect their businesses and the theme in ourdiscussions. And, we shall be conscious of how organizations may, throughbusiness without borders, stretch their capacities, and develop new competencesand relationships.Much of the course is for the understanding.
1363	HSS	PHILOSOPHICAL AESTHETICS	PHI450A	The Aesthetic attitude; Aesthetic Experience; Art and Aesthetic; Defining Artand its Problems; Art and Emotion; Literary Aesthetics; Art, Society and Morality;Philosophy of Literature.
1364	HSS	MORAL THINKING	PHI447A	Introduction to Ethical Theories(Consequentialist and Nonconsequentialist theories, Hedonism, Utilitarianism, Deontological theories, Ethical Rules, Situation Ethics, Virtue Ethics).Metaethical Theories (Ethical Foundations of an Adequate Approach to an Adequate Theory; the Moral point of view; Why be Moral?). Ethics in the Indian tradition. Applied Ethics: Issues and Dilemmas. Text (excerpts):1. Frankena, W.K. Ethics. New Delhi: Prentice Hall ofindia,19992. Hospers, John. An Introduction to Philosophical Analysis. New Delhi: Allied Publishers,19673. LaFollette, Hugh, ed. Ethics in the Indian Tradition. University Press, 19405. Pojman, Louis (Ed.) Ethical Theory: Classical and Contemporary Readings, Belmont:Wadsworth, 19986. Cahn, Steven M. & Peter Markie (Eds.) Ethics: History. Theory and Contemporary Issues. Belmont:Wadsworth, 19987. Green, Jonathon and Nicholas Karolides. The Encyclopedia on Censorship. New York:Info base Publishers, 2005.Jansen, Sue Curry. Censorship: The Knot that binds Power and Knowledge. Oxford:Oxford University Press, 2003.
1365	HSS	LITERATURE AND CENSORSHIP	ENG439A	Censorship is defined as the restrictions that are imposed on the writerand writings by centres of power like the state, religious institutions or other socialorganizations. These external restrictions can lead to selfcensorship and thoughtsopenly. The course attempts to understand why writing is perceived as a threat toestablished institutions like the state or organized religion, and the impact of suchrestrictions on writing. Besides this, the course also attempts to understand the efficacy of thisrestrictive measure will be evaluated with the help oftexts that have created controversiesfor going against accepted notions of morality or religious beliefs. Texts: Green, Jonathon and Nicholas Karolides. The Encyclopedia on Censorship. New York:Info base Publishers, 2005.Jansen, Sue Curry. Censorship: The Knot that binds Power and Knowledge. Oxford:Oxford University Press, 2003. Republic.Chicago: University of Chicago Press, 2003.

S.No.	Branch	Course Title	Courses ID	Content
1366	HSS	POSTCOLONIAL LITERATURE	ENG438A	This course will unpack the many meanings of the term ‘postcolonial’ through a study of literature that has emerged from the colonial encounter in various parts of the world.Focused mainly on literature from Africa, the Caribbean, and India, in addition to British literature itself, to understand how such literature deals with and/or resists the experience and legacy of colonialism. In addition, the course will explore the role of literature within current global relations between the North and the South. 1) Loomba, Ania. Colonialism/Postcolonialism. New York: Routledge, 1998.2) The Empire Writes Back: Theory and Practice in PostColonial Literatures. Edited byBill Ashcroft, Gareth Griffiths, Helen Tiffin. New York: Routledge, 1993.
1367	ECO	MACROECONOMICS I	ECO221A	Circular Flow of Income, Output and Expenditure, National IncomeAccounting Methods, Classical Model, Keynesian Cross Model, .Theories of Consumption andInvestment, Demand for Money, ISLM Model and its application to Macroeconomics.
1368	HSS	SOCIAL AND BEHAVIOURAL ASPECTS OF HEALTH	HSS403	From biomedical model of health to sociopsychological model of health;Illness cognition; Social processes, illness and coping; Methodological issues in study of health;Stigma and testing and treatment of mental health.
1369	HSS	SOCIAL & BEHAVIOURAL ASPECTS OF HEALTH	HSS403A	From biomedical model of health to sociopsychological model of health; Illness cognition; Social processes, illness and coping; Methodological issues in study of health; Stigma and testing and treatment of mental health.
1370	ECO	MACROECONOMICS	ECO202	Circular Flow of Income, Output and Expenditure, National IncomeAccounting Methods, Classical Model, Keynesian Cross Model, .Theories of Consumption andInvestment, Demand for Money, ISLM Model and its application to Macroeconomics.
1371	ECO	MICROECONOMICS I	ECO201A	Theory of consumer behavior; Theory of demand and supply; Production theory;Costs of production; Markets Perfect competition, Monopoly oligopoly; Theoryof distribution; Pricing.
1372	DES	INTRODUCTION TO CRITICAL ART APPRECIATION	DES629	Contemporary Art Theory & Practice, Art and its implications Society & Culture,Role of Art and Artist in Socio Cultural Framework, Understanding Visual CultureHistorical and Global Perspective, Methodological issues in study of art. Relative studies in diverse cultural expressions.
1373	DES	DESIGN CULTURE AND SOCIETY	DES628	Understanding the Dynamics of Indian Society : Social Units and Institutions,Cultural Adaptations, Exploring Culture : Attributes, Cultural Practices, CulturalGrowth and Cultural Integration , Methods of cultural studies. Culture : Proletarian Design Innovations, Role of Design withinSocietal Structure and Cultural Framework and Understanding Users, UserExperience Design : UE Research Techniques, Trend Mapping : Research triangulationhow observation, interrogation and intuitionadd the key and core vital layer to understanding the data assessed, theknowledge acquired and the insights delivered; Quantitative sampling; Visual brand directions.
1374	DES	MANAGEMENT OF DESIGN AND INNOVATION	DES627	Identification of Opportunities and the creative mind, Problem Based ideation,Creative Problem solving, Market Innovation and Brands, Strategy and Organizationfor the Creative Business, Networks and Communication for the Creative Business over the Life Cycle.
1375	DES	INTERACTION DESIGN	DES626	Interaction Design defines behavioral pattern and builds the performance levelof technical, environmental, biological and organizational systems i.e. software,products, mobile devices, environments, services, etc. The primary concern ofthis topic in design. Students are expected to collect information through userresearch following the various user requirements and consider the evaluation process of design in terms of usability and emotionalfactors and propose solutions for the ease of use in physical or virtual productsor a system. Interactive objects, spaces and environments. Communication Technogy (ICT) potentialities as term projects.
1376	DES	FORM AND STYLE	DES625	Theory : Form envelops and assists function and in the process creates a networkof values that is termed as Style. Studies in form and style concerns theconceptualization, exploration, and development of form. Visualization like Fantasy. Metaphors, Cultural connotations andBionics in the context of form making.From analysis Analysis of principles of form in relation with society study of evolution of forms in product design. Motion andform, Light space analysisFor Visual communication form will be explored in the context of conceptcommunicationand the application of information theory to visual structures.This will include ergonomic and market behaviorstowards forms.Studio: Form development assignments in both Product and visual communicationswill deal with the evolution of form in relation with exploration and functional studies, and usability issues to market trendsas related to the study of Form. The study of the design process will help inunderstanding the shifts between analytical and creative phases in differentstages of design. The course is envisaged in collaboration with assignments from the industrywherever possible in which the primary work will take place in the Lab andthrough assignments.
1377	DES	ELEMENTS AND PRINCIPAL OF DESIGN	DES624	The course intends to develop the following areas(i) Elements of Design value, color, form, shape, line and texture. Each element is to be examined theoretically along with studio exercises andevaluated through practical work. Movement, and balance on the basis of Design Elements. Thecourse proposes to develop through understanding of the elements andprinciples of design and their corelationship.The course intends in developing a design process. Studentsare expected to develop knowledge and practical skill through theoreticaland practical training. Students are required to develop products, giveseminars and submit research paper on chosen topics.
1378	EE	ELECTRIC POWER SYSTEM OPERATION AND MGMT. UNDER RESTRUCTURED ENVIRONMENT	EE633A	Fundamentals of deregulation: Privatization and deregulation, Motivations forRestructuring the Power industry; Restructuring models and Trading Arrangements:Components of restructured systems, India (Bilateral, Regional, and multilateral),Open Access Transmission Systems; Different models of deregulation: U K Model,California model, Australian and New Zealand models, Deregulation in Asiaincluding India, Bidding strategies and Ancillary services; Wheeling charges and pricing: Wheeling methodologies,pricing strategies.
1379	EE	ELECTRIC POWER SYSTEM OPERATION AND MGMT. UNDER RESTRUCTURED ENVIRONMENT	EE633	Fundamentals of deregulation: Privatization and deregulation, Motivations forRestructuring the Power industry; Restructuring models and Trading Arrangements:Components of restructured systems, India (Bilateral, Regional, and multilateral),Open Access Transmission Systems; Different models of deregulation: U K Model,California model, Australian and New Zealand models, Deregulation in Asiaincluding India, Bidding strategies and Ancillary services; Wheeling charges and pricing: Wheeling methodologies,pricing strategies.
1380	HSS	POPULATION ECONOMY AND SOCIETY	SOC479A	Relationship between demographic trends and the socioeconomic context;Demography and population studies; World population growth; Population of India; Populationtheories; Data and methods in population studies. Course Reference : Davis, Kingsley (1968) The Population of India and Pakistan. New York: Russell and Russell.Sharma, A. K. (2011) Population and Society. New Delhi: Concept Publishing Company Pvt.Ltd. Bureau of Census.Smith David, and Keyfitz, Nathan (1977) Mathematical Demography. Berlin: SpringerVerlag.Smith, David P. (1992) Formal Demography. New York: Plenum Press.Sorokin, Pitirim (1978) Consequences of Population Trends. Vol. 1 NewYork: United Nations.United Nations (1974) Methods of Projections of Urban and Rural Population. New York:United Nations.United Nations (1983) Indirect Estimation of Population. New York: United Nations.
1381	HSS	URBAN HABITAT AND SOCIAL INTERACTION	SOC477A	From rural to urban: debates around the concepts of gemeinschaftand gesellschaft; Origin of city in history; Theories of urban sociology (Classical andContemporary); Technology and urban life; Poetics and aesthetics ofurban sociology. Course Reference : Rao, M. S. A., ed. (1974) Urban Sociology in India: Reader and Source Book. NewDelhi: Orient Longman.Gottdiener, Mark and Ray Hutchison (2006) The New Urban Sociology. Boulder: Lynne Rienner Publishers.David et al (1977) Being Urban: A Sociology of City Life. Lexington, Mass: D. C.Heath and Company.Lin Jan and Mele Christopher, ed. (2005) The Urban Sociology Reader. London:Routledge.Palen, J. John. (1993) Politics in TwentiethCenturyUrbanism. New York: Routledge.
1382	CSE	COMPUTER VISION AND IMAGE PROCESSING	CS676A	Human and Computer Vision, Image Representation and Modelling, Line andEdge detection, labeling, Image Segmentation. Pattern Recognition: Statistical,Structural, Neural and Hybrid Techniques. Training and Testing. Analysis, Robotic Vision. Role of Knowledge.
1383	CSE	COMPUTER VISION AND IMAGE PROCESSING	CS676	Human and Computer Vision, Image Representation and Modelling, Line andEdge detection, labeling, Image Segmentation. Pattern Recognition: Statistical,Structural, Neural and Hybrid Techniques. Training and Testing. Analysis, Robotic Vision. Role of Knowledge.
1384	CSE	INTRODUCTION TO NATURAL LANGUAGE PROCESSING	CS671A	A computational framework for natural language. A framework such as LFG,GPSP or Panini in some depth. Partial description of English or an Indianlanguage in the frame work, lexicon, algorithms and data structures and applications like machine translation, database interface.
1385	CSE	INTRODUCTION TO	CS671	A computational framework for natural language. A framework such as LFG,GPSP or Panini in some depth. Partial description of English or an Indianlanguage in the frame work, lexicon, algorithms and data structures and applications like machine translation, database interface.

S.No.	Branch	Course Title	Courses ID	Content
		NATURAL LANGUAGE PROCESSING		applications like machine translation, database interface.
1386	CSE	SOFTWARE ARCHITECTURE	CS654A	In this course we study, typical software system structures (architectural styles),techniques for designing and implementing these structures, models forcharacterizing and reasoning about architectures, Framework; Architectural Styles, Design Patterns; ArchitectureDescription Languages; Productline architectures; Component based development.
1387	CSE	SOFTWARE ARCHITECTURE	CS654	In this course we study, typical software system structures (architectural styles),techniques for designing and implementing these structures, models forcharacterizing and reasoning about architectures, Framework; Architectural Styles, Design Patterns; ArchitectureDescription Languages; Productline architectures; Component based development.
1388	CSE	FUNCTIONAL PROGRAMMING	CS653	ML (CAML dialect); Xcalculus and combinators; abstraction and higher order functions; lazy and eager evaluation; types, polymorphism and type inference; Equations and pattern matching; SECD machine
1389	CSE	RANDOMIZED ALGORITHMS	CS648A	Review of discrete probability; Notion of randomized algorithms, motivatingexamples; Markov, Chebyshev inequalities, Chernoff bounds; Probabilistic method;Hashing, fingerprinting; Random walks and Monte Carlo methods; Verifying checkableproofs; some number theoretic problems; Approximate counting.
1390	CSE	RANDOMIZED ALGORITHMS	CS648	Review of discrete probability; Notion of randomized algorithms, motivatingexamples; Markov, Chebyshev inequalities, Chernoff bounds; Probabilistic method;Hashing, fingerprinting; Random walks and Monte Carlo methods; Verifying checkableproofs; some number theoretic problems; Approximate counting.
1391	CSE	ADVANCED TOPICS IN ALGORITHMS & DATA STRCUTURES	CS647	The course intends to deal with advanced aspects of algorithm: design andanalysis including data structures, analysis and lower bound proofs, amortizedcomplexity of algorithms. Fibonacci heaps and splay trees. Shortest path problem. Network flows preflowpush algorithms,max flow algorithm, and scaling algorithms. Matching, blossoms, MicaliVaziranialgorithm. Lower bound theory for parallel computations.
1392	CSE	PARALLEL ALGORITHMS	CS646A	Complexity measure for parallel algorithms. Parallel combinatorial algorithms:permutations with and without repetitions, combinations, derangements. Parallelsearching algorithms: maximum/minimum, tree traversal algorithms, parallel algorithms for connectivity problems,parallel algorithms for path problems.
1393	CSE	PARALLEL ALGORITHMS	CS646	Complexity measure for parallel algorithms. Parallel combinatorial algorithms:permutations with and without repetitions, combinations, derangements. Parallelsearching algorithms: maximum/minimum, tree traversal algorithms, parallel algorithms for connectivity problems,parallel algorithms for path problems.
1394	CSE	TOPICS IN DESIGN AND ANALYSIS OF ALGORITHMS	CS645	Introduction. Disjoint set UnionFind algorithms. Redblack trees. Selectionalgorithms and application to convex hull. Planar graph separators. Priorityqueues. Fusion trees and their applications to integer sorting. Polynomial time algorithms for matching.
1395	CSE	FINITE AUTOMATA ON INFINITE INPUTS	CS644A	Finite automata on infinite words and trees: Complementatation, determinizationand algorithms for checking emptiness.Connections with logic: Finite automata and monadic second order (MSO)logic on wordtrees. Applications: Decision procedures for temporal logics. Modelling, verificationand synthesis of systems. Effective theory of infinite games.
1396	CSE	FINITE AUTOMATA ON INFINITE INPUTS	CS644	Finite automata on infinite words and trees: Complementatation, determinizationand algorithms for checking emptiness.Connections with logic: Finite automata and monadic second order (MSO)logic on wordtrees. Applications: Decision procedures for temporal logics. Modelling, verificationand synthesis of systems. Effective theory of infinite games.
1397	ECO	ECONOMETRICS II	ECO351	Non Linear Regression Models, Non Parametric Regression Models, Measurement Errors, Stochastic Regressor, Multinomial and Conditional Logit Models, Ordered Logit Models, Nested Logit Models, Random Effects Dynamic Panel Data Models, Time Series Models including ARCH,GARCH and Multivariate GARCH Models, DEA and Stochastic Frontier Models, Total Factor Productivity Estimation, Levinsohn Petrin Approach, Models to Trade, Finance, Microeconomics, Development Economics, Growth, Macroeconomics, Industry, among others.
1398	ECO	ENVIRONMENTAL ECONOMICS AND POLICY	ECO341	Classical Linear Regression Model (CLRM), Assumptions of the CLRM and its violation, Hypothesis Testing, Gauss Markov Theorem, Properties of the Estimator, Multiple Regression Model, Heteroscedasticity, Simultaneous Equations, Introduction to Time Series, Discrete Choice, Limited Dependent Variable and Panel Data Models, Estimation ProceduresOrdinary Least Square(OLS), Generalized Least Squares(GLS), IV(Instrumental Variables), Econometric Software: Eviews and Stata
1399	ECO	INTRODUCTION TO MATHEMETICAL ECONOMICS	ECO261	Basic Mathematical Logic; Set Theory; Review of Functions of a Single Variable and Its Application in Economics: Choice, Utility and Aggregation; Functions of Several Variables : Level Curves, Homogeneous Functions, Optimization with Equality and Inequality Constraints; The Envelope Theorem and Duality; Dynamics and Dynamic Optimization.
1400	ECO	MICROECONOMICS I	ECO211	Theory of consumer behavior; Theory of demand and supply; Production theory;Costs of production; Markets Perfect competition, Monopoly oligopoly; Theoryof distribution; Pricing.
1401	ECO	INTRODUCTION TO PROFESSION	ECO100	For whom the bell tolls; What is economics?; Basic concepts of supply anddemand; Central Problems of very economic society; Development, freedom andopportunities; Dilemma of development; Linkages between business and industry; The role of economist in societal development; Therole of economist in governance
1402	DES	SPECIAL STUDIES/PROJECT COURSES IN DESIGN	DES698	Cognitive Design; Design Management; Human factors in Ergonomics Design;Usability & Usercentric Design; Axiomatic Design; Human Computer InterfaceDesign (HCI), etc.
1403	DES	TOPICS IN DESIGN	DES689	Lectures and Workshops on Various Topics in Design like Ergonomics, GraphicDesign and Typography, Design Management, Visual Image Design, Compositionand Media Art, Aesthetics and Forms, Role of Design in Society, Design Exercises,Studio Projects, Field Trips.
1404	DES	DESIGN PROJECT II	DES682	Manufacturability Studies, Embedded products, Product styling, Package Designing,Information design, HCI, GUI, Animation, Film appreciation, Print Making.
1405	DES	CREATIVE VISUALIZATION	DES621A	Theory: Art movements and design : Principles of style in art, product design,architecture and graphic design : Photography : Design and performing art :Art and perception : Methodology of criticism and analysis (drawing, lights, stage craft, script writing, editing, productdevelopment) to emphasize visualization powers in an attempt to build a personalvision.
1406	DES	CREATIVE VISUALIZATION	DES621	Theory: Art movements and design : Principles of style in art, product design,architecture and graphic design : Photography : Design and performing art :Art and perception : Methodology of criticism and analysis (drawing, lights, stage craft, script writing, editing, productdevelopment) to emphasize visualization powers in an attempt to build a personalvision.
1407	CSE	CIRCUIT COMPLEXITY THEORY	CS642	The course aims at a comprehensive overview of results on the circuit complexityclasses and their relationship with the Turing based classes. The topics to be covered in the course are as follows: The class P, the class NC, proof of the relationship RL is a subset of SC. The class NC1 and its characterizations. The class TC0 and its characterizations. The class ACC and its characterizations. The class AC0 and its characterizations.
1408	DES	TOPICS IN MOTION PICTURES	DES623	Topics in motion pictures require a combined practical and theoretical approachfor realizing the creative media in its totality. The proposed course would offerstudents the opportunity to acquire a range of knowledge and skills related to film and television media; Approach and methods in these forms of media productions; Overview of the digital media arts field with an emphasis on technological developments, historical directions, keyconcepts and methodologies through seminar lectures, research presentations,practical exercises and a final project. Classes would be supplemented withviewing a range of produced works and discussing them in relation to the modernmethods and explore both technical and creative approaches to the medium.The course will also include short workshops supported by specialized professionals in the related fields.
1409	DES	DESIGN PRACTICE II	DES603	Construction of Forms, Geometrical Transformations; Surface Modeling;Representation of 3D Shapes; Solid Modeling; Simulation in CAED, Rapid Prototyping & Tooling; Strength and Stiffness of Structural Members; Design of ElectroMechanical Systems; Mechatronics; Design of EmbeddedSystems; Intelligent Product Design.Laboratory: 2D3D Modeling, Operation of Rapid Tooling and Prototyping Instruments, Development of CAD Models.
1410	CSE	ADVANCED GRAPH ALGORITHMS	CS743A	Review of important sequential graph algorithms. Introduction to parallel modelsfor computation. General techniques for fast parallel computations on vectorsand lists and their applications to design of efficient algorithms for depth first search of directed and undirected graphs.NC algorithms for ST numbering and open ear decomposition. Parallel algorithmsfor graph optimization problems. Algorithms for graph coloring and testing.
1411	CSE	ADVANCED GRAPH	CS743	Review of important sequential graph algorithms. Introduction to parallel modelsfor computation. General techniques for fast parallel computations on vectorsand lists and their applications to design of efficient algorithms for depth first search of directed and undirected graphs.NC algorithms for ST numbering and open ear decomposition. Parallel algorithmsfor graph optimization problems. Algorithms for graph coloring and testing.

S.No.	Branch	Course Title	Courses ID	Content
		ALGORITHMS		algorithms for depth first search of directed and undirected graphs.NC algorithms for ST numbering and open ear decomposition. Parallel algorithmsfor graph optimization problems. Algorithms for graph testing.
1412	CSE	ADVANCED COMPILER OPTIMIZATIONS	CS738	Introduction to Advanced topics, Compiler Algorithms Notation, Symbol tablestructure, Intermediate representation, Run time support, Producing codegenerators automatically, Control flow analysis, Data optimizations, Redundancy elimination, Loop optimizations, procedureoptimizations, Register allocation, Code scheduling, control flow and low leveloptimizations, Inter procedural Analysis and optimizati
1413	CSE	TOPICS IN INTERNET TECHNOLOGIES	CS727	Today the Internet is being used for myriad of applications electronic publishing,electronic commerce, distance education, collaborative working, etc. This courseintends to investigate the underlying princ DHTML, VRML, SGML, XML and other markup schemes; Images compression, formats; Audio compression, formats; Content Delivery protocols HTTP and variants, Internet servers, proxy servers; Search Component technologies, Javabeans, CORBA;Security, Electronic payment systems, Firewalls, Encryption, Watermarks;Performance, Benchmarking the Web.
1414	CSE	VLSI TESTING AND FAULT TOLERANCE	CS720	The course is primarily intended to familiarize students with the problemof testing large and complex electronic circuits. Various techniques to solvethis problem and concepts of design for easy testabilit will be covered in detail. The problem of reducedyield and reliability of circuits in presence of faults will be discussed andtechniques to improve the yield and reliability of these circuits by introducingfaultt software
1415	CSE	SPECIAL TOPICS IN COMPUTER SCIENCE	CS697	Special and advanced topics in different areas of Computer Science andEngineering will be covered under this course.
1416	CSE	COMPUTATIONAL NUMBER THEORY AND ALGEBRA	CS681	Elementary operations: the complexity of basic operations like additions,multiplications for integers and polynomials. Polynomials: The complexity offactorization, irreducibility testing, ideal membership of short vector in an integer lattice. Motivating example;polynomial factorization. Integers: The complexity of factorization, primalitytesting, discrete log computation etc for integers. Motivating examples: RS Motivating examples: Elliptic curve cryptosystemsand integer factoring.
1417	CSE	CATEGORY THEORY	CS680	Introduction: Basic definition and diagram; sources and sinks; monicity andepicity; isomorphisms of objects and morphisms; duality; Universal Structures:Initial terminal and zero; Category of sources and hierarchy; extension of categories; factorization; chains and exactness;Morphism algebra: Biproduct; semiadditive category; Additive category; Functors:Natural transformation; categories on natural tran and colimit; Hreflection morphism and adjoint functor; Representablefunctors Category in context of another category; Application to Logic (Topoi);application to Programming Languages.
1418	EE	OPTO-ELECTRONICS	EE416	LEDs, semiconductor lasers, modulation of laser sources. Avalanche and PINphotodetectors and their characteristics. Solar cells. Optical fibers and theircharacteristics. Integrated optics. Fiber optic com
1419	EE	COMMUNICATION SYSTEMS	EE321A	Information measures. Source coding. ISI & channel equalization, partialresponse signalling. Mary modulation systems, error probability calculations.PLLs and FM threshold extension. Error control codin
1420	EE	COMMUNICATION SYSTEMS	EE321	Information measures. Source coding. ISI & channel equalization, partialresponse signalling. Mary modulation systems, error probability calculations.PLLs and FM threshold extension. Error control codin
1421	EE	SIGNALS SYSTEMS & NETWORKS	EE200A	Continuous and discrete time signals; Fourier series, Fourier, Laplace and Ztransform techniques; DFT. Sampling Theorem. LTI systems: I/O description,impulse response and system functions, pole/ zero analysis.
1422	CE	ENGINEERING GRAPHICS	TA101N	Introduction to sketching; Principal views, principles ofdimensioning Introduction to computeraidedgraphics Missing view, sectional view and assembly drawings Overview of pictorial representation, and intersection oflines and planes Intersections of solids and development of lateral surfaces
1423	HSS	INTRODUCTION TO INDIAN SOCIETY	SOC173	Indian Sociology: An Introduction, Social Roots of Indian Society:Vedic heritage,Brahminic, Islamic, British, Indian Social Structure:Rural Context, Urban Context,Indian Social Institutions and Organisations India:Reformist,Nationalist, Agrarian, Backward Caste, Processes of Social Change in India,Westernization, Sanskritisation, Contemporary Social Problems. Secularism,Common civil code, Reservation pol
1424	HSS	INTRODUCTORY SOCIOLOGY	SOC171B	What is Sociology? Explain how the new science of societycame about in the nineteenth century inWestern Europe and how it wasdifferent from earlier social thought. Discuss the intellectual and socialco theEnlightenment. The socioeconomic roots: the TwinRevolutions The Industrial Revolution The French RevolutionTheoretical thinking in sociology(A) Classical sociological theorists: Karl Marx Max Weber Symbolic InteractionismSociety, Community and Self 21ecturesNorms, values, roles, custom, statusDeviance Theories of subcultureThe concept of anomieSociology of Organizations Different sociologica and Ethnicity Gender Caste DisabilityPoverty Definition and measurement Extent of poverty Theories of povertyPower and Politics Debating basic concepts in politicalsociology Authoritarianism and democ Religion Sociological theories of religion Religious organizations Secularization debates
1425	ECO	LAW AND ECONOMICS	ECO751	Introduction; Economic Analysis of Property Law: An Economic Theory of Property,and Intellectual Property Rights; Economic Analysis of Contrat Law: EconomicTheory of Contract, and Economics of Ren Economics:Legislative and Executive Participation and Discretion: Economic anaylsis ofjudicial review, and general Applications to Indian Institutes: Economic theoryof Corporation Law: The Theory of the Problemsof Corporation Law, Economic Analysis of Labour Law: Unions and Productivity,"WorkersProtective" Legislation, Issues in Employment discremination on groundsof Race Sex, and Age
1426	ECO	MONETARY ECONOMICS	ECO407	Demand for money; Supply of money; Monetary aggregates; Neutrality of money;Money in a growth model; Money in an OLG model; Central bank goals, targets,and instruments; Time consistent versus di
1427	ECO	CONTRACT ECONOMICS	ECO405	Introduction; Economics of contracts (adverse selection and moral hazard);Hidden action, monitoring, and control; Efficient contracts; Transfer prices;Contingent claims; Principal agent models; Incomplete
1428	ECO	HISTORY OF ECONOMIC THOUGHT	ECO404	Introduction; Social Choice : How much environmental protection; Market failure:Public bad and externatities; Internalisation of externalities; Regulations :Emission Fees & marketable permits; Environmen pollution Sustainable development; Economics of natural resources,Natural resorces accounting;The environmental issues and policies in India.
1429	EE	CONTROL TECHNIQUES IN POWER ELECTRONICS	EE662A	State space modeling and simulation of linear systems, Discrete time models,conventional controllers using small signal models, Fuzzy control, Variablestructure control, Hysteresis controllers, Output an
1430	EE	CONTROL TECHNIQUES IN POWER ELECTRONICS	EE662	State space modeling and simulation of linear systems, Discrete time models,conventional controllers using small signal models, Fuzzy control, Variablestructure control, Hysteresis controllers, Output an
1431	EE	POWER ELECTRONICS APPLICATIONS IN POWER SYSTEMS	EE661A	Basics of flexible AC transmission systems, Controlled rectifier and energystorage plants, Tap changers and phase shifters, Thyristor controlled VARcompensation and series compensation, Modern (sync Power quality conditioners, Power electronics in power generation.
1432	EE	POWER ELECTRONICS APPLICATIONS IN POWER SYSTEMS	EE661	Basics of flexible AC transmission systems, Controlled rectifier and energystorage plants, Tap changers and phase shifters, Thyristor controlled VARcompensation and series compensation, Modern (sync Power quality conditioners, Power electronics in power generation.
1433	EE	FUZZY SET LOGIC & SYSTEMS & APPLICATIONS	EE658A	Introduction, Uncertainty, Imprecision and Vagueness, Fuzzy systems, Briefhistory of Fuzzy logic, Foundation of Fuzzy Theory, Fuzzy Sets and Systems,Fuzzy Systems in Commercial Products, Research to crisp conversions, Fuzzy arithmetic, Numbers, Vectors and the extensionprinciple, Classical logic and Fuzzy logic, Mathematical background of FuzzySystems, Classical (Crisp) vs, Fuzzy sets, Represent set operations, Sand T Norms, Properties of Fuzzy sets,Sets as Points in Hypercube, Cartesian Product, Crisp and Fuzzy Relations,Examples, Liguistic variables and hedges, Membership function design. I Approximate Reasoning, Forms of Fuzzy Implication,Fuzzy Inference Engines, Graphical Techniques of Inference, Fuzzyifications/DeFuzzification, Fuzzy System Design and its Elements, Design options. I Universal Approximators, Additive Fuzzy Systems(standard additive model).
1434	EE	FUZZY SET LOGIC & SYSTEMS & APPLICATIONS	EE658	Introduction, Uncertainty, Imprecision and Vagueness, Fuzzy systems, Briefhistory of Fuzzy logic, Foundation of Fuzzy Theory, Fuzzy Sets and Systems,Fuzzy Systems in Commercial Products, Research to crisp conversions, Fuzzy arithmetic, Numbers, Vectors and the extensionprinciple, Classical logic and Fuzzy logic, Mathematical background of FuzzySystems, Classical (Crisp) vs, Fuzzy sets, Represent set operations, Sand T Norms, Properties of Fuzzy sets,Sets as Points in Hypercube, Cartesian Product, Crisp and Fuzzy Relations,Examples, Liguistic variables and hedges, Membership function design. I Approximate Reasoning, Forms of Fuzzy Implication,Fuzzy Inference Engines, Graphical Techniques of Inference, Fuzzyifications/DeFuzzification, Fuzzy System Design and its Elements, Design options. I Universal Approximators, Additive Fuzzy Systems(standard additive model).

S.No.	Branch	Course Title	Courses ID	Content
1435	EE	THE FINITE ELEMENT METHOD FOR ELECTRIC AND MAGNETIC FIELDS	EE649A	Introduction : Review of Electromagnetic Theory. Introduction to the Finite Element Method using electrostatic fields :Galerkin s method of weighted residuals, Minimum energy principle,Calculation of capacitance functions, isoparmetricelements, meshing, solvers, post processing. finite Element Modeling : Conductive media, steady currents ; Magnetostaticfields, permanent Magnest, scalar and vector potentials ; E
1436	EE	THE FINITE ELEMENT METHOD FOR ELECTRIC AND MAGNETIC FIELDS	EE649	Introduction : Review of Electromagnetic Theory. Introduction to the Finite Element Method using electrostatic fields :Galerkin s method of weighted residuals, Minimum energy principle,Calculation of capacitance functions, isoparmetricelements, meshing, solvers, post processing. finite Element Modeling : Conductive media, steady currents ; Magnetostaticfields, permanent Magnest, scalar and vector potentials ; E
1437	ECO	TOPICS IN APPLIED ECONOMETRICS	ECO761	This course provides a methodology of emprical research in applying econometrictools for policy oriented research in social sciences. Emphasis will be given tovarious ways of handling problems faced in
1438	ECO	DEVELOPMENT ECONOMICS	ECO735	The international trends in economic development, various development theoriesand operational strategies towards capital formation, international capital flows,foreign trade, agriculture, industry, HRD, te neweconomic policies and emerging new international structures. Besides, theshidents will undertake some empirical project as group assign many.
1439	ECO	INDUSTRIAL ORGANISATION AND POLICY	ECO734	Scope of Industrial organization; Industrial Efficiency; Basic Framework forthe Study of Industrial Organization; Recent Approaches to Industrial Organization;Market Structure and Its Elements; Market Co
1440	ECO	ECONOMETRICS	ECO732A	Problems of statistical inference in economics constraints on the parameterspace, random coefficient models, distributed lags, decision models, and systemsof equations. New functional forms kernel e
1441	MSP	ENGINEERING POLYMERS	MS614	Classification and structure of polymers, Glass transition, Linear viscoelasticity,Stress relaxation and dynamic experiments, Mechanical models, Superpositionprinciples, Effect of structure on mechanical
1442	MSP	MATERIALS FOR ENERGY CONVERSION & STORAGE	MS611	Characteristics of solar radiation, Basic features of solar cells, Various junctionconfigurations, p n homojunction, Schottky barrier, Heterojunction, Photoelectrochemicalcells, Desired material properties, P Thermal oxidation, Chemical displacement, Plasmadeposition, Energy storage devices.
1443	MSP	ELECTRONIC MATERIALS	MS606	Classification, Crystal growth techniques, Wafer processing, Doping methods,Formation of oxide layer, CVD, MOCVD and MBE, Metallic contacts and interconnects,Lithography, Processing integration.Pho
1444	MSP	MATERIALS ENGINEERING	MS605	Solidification, Powder processing, Crystal growth, Heat treatment andmicrostructures, Non destructive evaluation, Processing of glasses and polymers,Novel processing methods, Thin films, Surface phen
1445	MSP	ELECTRICAL AND DIELECTRIC MATERIALS	MS602	Metallic conduction, Energy bands, Brillouin zones, Temperature dependence ofmetallic conductivity, Impurity contributions, Semiconductor materials, Dopingeffects, Law of mass action, Electrical resistivity Semiconductortechnology, Point defects, Diffusion phenomenon, Ionic conduction, Temperatureand (aliovalent) impurity effects, Superionic conductors and devices, Di, piezoandferroelectric materials, Me
1446	MME	THIN FILM PHYSICS & APPLICATIONS	MME685	Surface science; experimental techniques to study surfaces; kinetics of surfaceprocesses impingement of atoms, scattering, adsorption, sticking coefficient;Film nucleation and growth mechanisms, critica chemical vapour deposition, atomic layer epitaxy, liquid phaseepitaxy, solgel technique etc, Electrical, optical, magnetic and mechanicalproperties of thin films and their applications.
1447	MME	SELECTION AND DESIGNING WITH ENGINEERING MATERIALS	MME667	Overview of the design process: concepts and stages of engineering design anddesign alternatives to develop materials with tailored properties; Performanceindices of materials; function, objective and c for thermal, mechanical, thermomechanical applications,damage tolerant designs for structural applications; Basic concepts of materialsscience: processingstructurepropertyperformance correlation; over defect structure of different material classes, Briefintroduction to the manufacturing processes for metals, polymers, ceramics,glasses and composite materials; design for manufacturability, Ashbys mate
1448	MME	MODERN TRENDS IN METAL FORMING PROCEESS	MME655	Limitation of conventional metal forming methods: Powder rolling and its variousvariants, spray rolling, direct strip process: Powder, spray, rotary and isothermalforging: Hydrostatic and powder extrusion;
1449	MME	FUNDAMENTALS OF STEREOLOGY & APPLICATIONS TO MICROSTRUTURAL ANALYSIS	MME650	Concepts and language of stereology; geometrical probability; fundamentaloperations in stereology; averaging with respect to orientation; basic stereologicalparameters on true 2D sections and thick sections andtransmission electron micrographs; numerical density and size distribution ofparticles and grains of various shapes and sizes; stereological analysis ofanisotropic microstructures; fractal description of polycrystalline and other structures and fracture surfaces.
1450	MME	X-RAY CRYSTALLOGRAPHY - I	MME646	Elemental compound and alloy crystals, modes of bonding, crystal types, densityof packing, atomic stacking, inter atomic voids, coordination polyhedra,Paulings rules, symmetry elements, space and point
1451	MME	ADVANCES IN IRON AND STEEL MAKING	MME630	Recent trends in iron and steel making: Gassolid and slagmetal reaction: Spongeiron making: Continuous steel making: Continuous casting: Vacuum degassingand electros slag remelting: Advances in aggr impurity control: Emphasis on application of physical chemistry and transportphenomena.
1452	MME	HEAT AND MASS TRANSFER	MME626	Review of the basic concepts in heat,mass and momentum transfer: Advancedtopics in convective heat and heat transfer: Radiative heat transmission.Simultaneous heat and mass transfer: Selected topics
1453	MME	ELECTROCHEMICAL TECHNOLOGY IN MATERIALS PROCESSING	MME613	Thermodynamic of electrolyte, electrochemical potential, conduction of ions insolution, overpotential, absorption, phase formation: Economics of an electrolyticprocess, principles of cell design, Electrochemical electroplating,anodizing, pickling, electrophoretic painting, electrochemical treatment of minerals,batteries and fuel cells, water treatment and environmental protection.
1454	IME	SOCIO-POLITICAL AND ETHICAL ASPECTS OF BUSINESS	MBA617A	Industrial revolution and industrialization, Political economy of underdevelopment,Sociology of development, Indian rural and urban society, Influence of religionand karma, Multiplicity of languages, culture government,Directive principles of state policy, Welfare state and Civil society, Socialstratification, Environmental issues and legislation, and social movements,Corporate social responsibility and business negotiable instruments act, Income tax Act, Environmental legislation.
1455	IME	SOCIAL POLITICAL AND LEGAL ENVIRON OF BUSINESS	MBA617.	Industrial revolution and industrialization, Political economy of underdevelopment,Sociology of development, Indian rural and urban society, Influence of religionand karma, Multiplicity of languages, culture government,Directive principles of state policy, Welfare state and Civil society, Socialstratification, Environmental issues and legislation, and social movements,Corporate social responsibility and business negotiable instruments act, Income tax Act, Environmental legislation.
1456	IME	HUMAN RESOURCE MANAGEMENT	MBA616A.	Meaning of Work and Humans as Resource, Human Resource Planning andSelection, Motivation and Compensation Management, Performance Appraisal,Career Management, Training and HRD, Group D Empowerment and Participation, Technology & HRM, Japanese HRM.
1457	IME	HUMAN RESOURCE MANAGEMENT	MBA616.	Meaning of Work and Humans as Resource, Human Resource Planning andSelection, Motivation and Compensation Management, Performance Appraisal,Career Management, Training and HRD, Group D Empowerment and Participation, Technology & HRM, Japanese HRM.
1458	IME	INVESTMENT VALUATION AND REAL OPTIONS	MBA610.	Investment Valuation Estimating Cost of Equity and Cost of Capital, Option PricingTheory, Option Pricing Applications in Valuation Real Options in ManagerialDecision Making Binomial Tree Method for Val Options,Appraising Projects with Real Options.
1459	IME	FINANCIAL MANAGEMENT	MBA607A	Fund and Cost Flow Analysis, Working capital management, Determination ofcapital structure of the firm, Cost of Capital, Capital asset pricing models,Leverages, Investment Analysis, Portfolio Management articles.
1460	IME	FINANCIAL MANAGEMENT	MBA607	Fund and Cost Flow Analysis, Working capital management, Determination ofcapital structure of the firm, Cost of Capital, Capital asset pricing models,Leverages, Investment Analysis, Portfolio Management articles.

S.No.	Branch	Course Title	Courses ID	Content
1461	IME	SOFTWARE PROJECT MANAGEMENT	IME671	This course will cover the techniques for managing software projects. It is intended to give the students both knowledge about, and practical experience in, the design and development of production quality software. The course follows: Software Process; Software Configuration Management, CMM Levels, Software Project Planning and Costing; Requirements Engineering; Software Project Design; Testing; Software Metrics; Quality Assurance; Software Project Management Applications and solution methods.
1462	IME	OPERATIONS RESEARCH FOR MANAGEMENT	IME605A	Introduction, Mathematical Modeling, Linear programming Formulation, solution procedures, Duality, Sensitivity, Applications, Network methods Max Flow, Min cost, Shortest path, Dynamic programming and Simulation Applications and solution methods.
1463	IME	INTRODUCTION TO COMPUTING	IME603A	Computing Computer Organization, Data Representation, Data Structures such as Arrays, Stacks, Queues and Trees, Algorithms for Searching and Sorting, Complexity, File Processing, Structured Programming and Database Systems.
1464	HSS	EDUCATION AND SOCIAL CHANGE	SOC748	General introduction to the place of learning in society. Learning, education and training. Changing meanings of education across time and society. A brief historical perspective on education in India. Social functionalist perspectives and structural conflict perspectives on education. Class, conflict, legitimation processes, reproduction of society. Anarchist perspectives. New Sociology of Education. Symbolic interactionist perspectives on education. Status politics and education. Caste, class, gender and education in India. Indian thinkers on education. Current debates on the place of education in India.
1465	HSS	SOCIAL MOVEMENTS : OLD AND NEW	SOC746	The aim of the course is to define the various types of social movements, and understand their role in striving for social change or transformation. It will examine theories of social movements in the context of various actors, will be studied with special reference to India.
1466	MTH	BASES IN LOCALLY CONVEX SPACES AND KOETHE SEQUENCE SPACES	MTH634	Preliminaries, Elements of basis theory, Types of bases, Summability (summation of infinite series), Koethe sequence spaces, Bases in OTVS, Isomorphism theorems.
1467	MTH	TOPICS IN TOPOLOGY	MTH628	Classification of 2-dimensional surfaces; Fundamental group; Knots and covering spaces; Braids and links; Simplicial homology groups and applications; Degree and Lefschetz Number; Borsuk-Ulam Theorem.
1468	MTH	DIFFERENTIABLE MANIFOLDS AND LIE GROUPS	MTH624A	Differentiable manifolds; Tangent space. Vector fields; Frobenius theorem; Relation between Lie subalgebras & Lie subgroups; Cartan's theorem on closed subgroups; One parameter subgroups; Exponential map.
1469	MTH	DIFFERENTIABLE MANIFOLDS AND LIE GROUPS	MTH624	Differentiable manifolds; Tangent space. Vector fields; Frobenius theorem; Relation between Lie subalgebras & Lie subgroups; Cartan's theorem on closed subgroups; One parameter subgroups; Exponential map.
1470	MTH	FOURIER ANALYSIS	MTH621A	Fourier series; Norm and pointwise convergence, Approximate identities, Plancherel theorem, Conjugation, Maximal functions, Classical Hardy spaces, F. and M. Riesz theorem, Interpolation of linear operators and applications.
1471	MTH	FOURIER ANALYSIS	MTH621	Fourier series; Norm and pointwise convergence, Approximate identities, Plancherel theorem, Conjugation, Maximal functions, Classical Hardy spaces, F. and M. Riesz theorem, Interpolation of linear operators and applications.
1472	MTH	ALGEBRA II	MTH611A	Fields: definition and examples. Ring of polynomials over a field. Field extensions. Algebraic and transcendental elements, Algebraic extensions. Splitting field of a polynomial. Algebraic closure of a field, UFDs. Fundamental theorem of Galois. Solvability by radicals. Solutions of cubic and quartic polynomials, Insolubility of quintic and higher degree polynomials. Geometric constructions. Cyclotomic extensions. Frobenius properties. Free modules, submodules and quotient modules, isomorphism theorems. Localization. Direct sum and direct products. Noetherian and Artinian rings and modules, structure of Artinian rings, Hilbert's basis theorem.
1473	MTH	APPLIED MATRIX THEORY	MTH610	Review of basic lin. alg. canonical factorization. Q-Forms. Courant-Fischer min-max & related theorems. Perron-Frobenius theory. Matrix stability. Inequalities, $\ A^{-1}\ \geq 1/\lambda_{\min}(A)$. Direct, iterative, projection and Krylov subspace methods.
1474	MTH	MATHEMATICAL MODELLING	MTH603	Elementary mathematical models; Role of mathematics in problem solving; Concepts of mathematical modelling; System approach; formulation, Analysis of models; Sensitivity analysis, Simulation approach.
1475	MTH	ALGORITHMS	MTH524	Preliminaries: Introduction to algorithms; Analyzing algorithms: space and time complexity; growth of functions; summations; recurrences; sets, etc. Greedy Algorithms: General characteristics; Graphs: minimum spanning tree, quicksort. Dynamic Programming: Elements of dynamic programming; The principle of optimality; The knapsack problem; Shortest paths; Chained matrix multiplication. Graph Algorithms: Depth-first search, Breadth-first search, FFT; Efficient FFT implementation. Number Theoretic Algorithms: Greatest common divisor; Modular arithmetic; Solving modular linear equations. Introduction to cryptography. Computational Geometry: Line segment intersection, Heuristic and Approximate Algorithms: Heuristic algorithms; Approximate algorithms; NP-hard approximation problems.
1476	SDS	NON-PARAMETRIC INFERENCE	SDS516	Order statistics, Run tests, Goodness of fit tests, rank order statistics, sign test and signed rank test. general two sample problems, Mann-Whitney test, Linear rank tests for location and scale problem, k-sample problems.
1477	SDS	NON-PARAMETRIC INFERENCE	MTH516	Order statistics, Run tests, Goodness of fit tests, rank order statistics, sign test and signed rank test. general two sample problems, Mann-Whitney test, Linear rank tests for location and scale problem, k-sample problems.
1478	CE	WATER SUPPLY AND WASTEWATER DISPOSAL SYSTEMS	CE412A	Introduction and Scope; Analysis and Design of Water Treatment Systems; Design of Water Distribution Networks; Analysis and Design of wastewater Collection Systems; Analysis and Design of Wastewater Treatment Engineering. Authors: H. Peavy, D. Row and G. Tchobanoglous. Publisher: Tata McGraw-Hill. Water Supply and Pollution Control. Authors: Warren Viessman Jr. and Mark J. Hammer. 7th Edition (1990). Publisher: Tata McGraw-Hill (Indian Edition Available). Additional class notes and reference material will be provided during lectures, either in softcopy or hardcopy form.
1479	AE	ROCKET PROPULSION	AE441A	Rocket propulsion – Introduction, Single and multi-stage rockets. Performance of chemical rockets- Principles of combustion, Estimation of adiabatic flame temperature, Thrust coefficient, Characteristic velocity.
1480	ECO	ECONOMETRICS	ECO732	Problems of statistical inference in economics – constraints on the parameter space, random coefficient models, distributed lags, decision models, and systems of equations. New functional forms – kernel estimators.
1481	ECO	INPUT-OUTPUT TECHNIQUES	ECO514	Basic input-output system; Projection of input-output coefficients; Non-survey and partial survey methods; Input demand functions; Capital coefficients and dynamic considerations in input-output analysis; Input-output models in the field of energy and environment.
1482	ECO	INTERNATIONAL ECONOMICS	ECO434	Global trade in goods and services; Why study international trade and finance; Growth and trade; Basic theory of international trade; Theory of comparative advantage; Implications of Heckscher-Ohlin theory; import tariff; Non-tariff import barriers; Arguments for and against protection; Regional trading blocks; Trade policies for development; International factor movements; Exchange rate and open economy; Inflation; Balance of payments; International monetary system; Benefits and costs of the Globalization Process.
1483	ECO	ECONOMICS OF REGULATION AND IPR	ECO422	Introduction; Rationale for regulation and antitrust; Regulatory practices, rules vs. implementation agencies; Regulatory capture; Discriminatory pricing; Controlling franchises; Public enterprise regulation; Government controls; Product quality, safety, and health issues; Patents; Copyrights; Trademark and Service Mark; Industrial Designs Registration; Protection of Layout of Designs Integrated Circuit; Protection of New Plant Varieties; Biotechnology and IPR, TRIPS and Access to Medicines, etc.).
1484	ECO	INDIAN ECONOMIC PROBLEMS	ECO413	Social accounting matrix; Leontief's open and closed IO models; Multipliers and linkages in the IO model; Organization of basic data for IO models; Identification of the key sectors; Role of project in development; Profitability analysis; Methods of measurement of benefits and costs of projects; Case studies.
1485	CSE	DATA STREAMING: ALGORITHMS & SYSTEMS	CS719A	Motivating applications: network monitoring, sensor networks, need for highly efficient processing of high speed and high volume data streams, Space and time efficient randomized algorithms as a candidate for streaming algorithms, expectation, variance, Markov and Chebyshev's inequality, Chernoff and Hoeffding (CH) tail inequalities, hash functions, limited independence, CH bounds for limited independence. Finding frequent items in streaming data, histograms over data streams, Transforms over data streams, wavelets, Fourier and DCT clustering over data streams, Applications to graphs.
1486	CSE	DATA STREAMING:	CS719	Motivating applications: network monitoring, sensor networks, need for highly efficient processing of high speed and high volume data streams, Space and time efficient randomized algorithms as a candidate for streaming algorithms.

S.No.	Branch	Course Title	Courses ID	Content
1508	IME	PROJECT I	MBA699.	In this course, each student will take up a management project or management topic under the guidance of a specific faculty. Towards the end of the semester, the student will present a final report of the
1509	IME	MANAGEMENT SEMINARS	MBA698A.	Management Seminars
1510	IME	ENERGY AND CARBON MARKETS: ECONOMICS POLICY AND REGULATION	MBA681A.	Energy and Economic Development: National and International Perspective, Structure of Energy Demand and Supply,, Energy Value Chain and Energy Accounting, India Energy Scenario 2047, Economics of Applications in General Algebraic Modelling System (GAMS), International Markets for Energy: Oil, Coal, Natural Gas and Uranium, Indian Energy Markets: Oil & Gas, Coal and Electricity, Private Investment Sectors, Pricing in Energy Markets: Electricity, Coal, Oil and Natural Gas, Functioning of Power Exchange and Commodity Exchanges (Energy), Cross Border Energy Cooperation, Energy and Environment, Carbon Markets and Carbon Finance, National Action Plan on Climate Change, JNN National Solar Mission, Renewable Energy: Technology, Economics and Policy, Market for Renewable Energy Certificate on relevant topic. Recommended Test: 1. Environmental and Natural Resource Economics, Tom Tietenberg, AddisonWesley, 2000. 2. International Energy Markets: Understanding Pricing, Policies, and Profits, Carol Dahl, Pe 9781593701789, 2009.4. Economics of Regulation and Antitrust Viscusi, Vernon & Harrington, MIT Press, 2000.5. Integrated Energy Policy, Report of the Expert Committee, Govt. of India, Planning Commission Bank and Asian Development Bank.7. Relevant reports of respective Ministries.8. Other relevant journal articles, reports, policy documents, regulations of regulatory commissions, case studies and class r
1511	MSE	PRINCIPLES OF METAL EXTRACTION AND REFINING	MSE320	History and importance of metal extraction; Introduction of mineral dressing:Comminution, Tabling, Jigging and flotation; Metallurgical fuels and the energyscenario; Pyrometallurgical operations roasting
1512	MSE	MECHANICAL BEHAVIOUR OF MATERIALS	MSE310	Stress tensor and stress transformation equations, Principal stresses; Straintensor and strain transformation equations; Isotropic and anisotropic elasticity,elastic strain energy; Yield criteria and constitutive Dislocation theory edge, screw and mixed dislocations, resistanceto dislocation motion and elastic properties of dislocations, dislocation interactions,multiplication and dissociation; Strengthening mechanisms under complex stressstates, prediction of long time properties; Fracturetoughness and fatigue Griffiths crack theory, energy release rate analysis,modes of loading, stress analysis of cracks, fracture toughness of complex stressstates, environmental assisted crackingand fatigue; Some case studies related to design.
1513	MSE	MATERIALS CHARACTERIZATION	MSE250	Chemical bonding, fundamentals of crystallography, reciprocal lattice, structuresin metals, inorganic compounds, polymers, silicates and glasses, stereographicprojections; Production, characterization, and applications; Optical principles of microscopy resolution, magnification,depth of focus; electron diffraction, imaging (various contrasts), determinationof crystal structure, burgers vector, electron beamspectroscopy Microscopy and, Electron Probe Micro Analyser; Principles of QuantitativeMicroscopy: volume density, surface density, length density, numerical density,particle and grain size; Overview of other character
1514	IME	SEMINAR	IME698A	SEMINAR
1515	CSE	COMPUTATIONAL COMPLEXITY	CS640A	Introduction to Computational Complexity. Complexity Classes. P and NPcompleteness. Hierarchy Theorems. Space Complexity. NL completeness. Savitchs Theorem. ImmermanSzelepsćenyi Theorem. Lower bounds for small sized circuits. Uniformity. Circuitclasses NC and AC. Randomized Computations. RP, BPP, ZPP. Relationship between BPP and otherclasses. Randomized space complexity. Interactive Proofs. Various problems. Hardness of Approximation. NP ⊆ PCP(poly(n), 1) Communication Complexity. Definition and lower bound techniques. Circuit Lower Bounds. Lower bounds on AC0. Counting Complexity. The class #P. Codes. Derandomization. Average case complexity.
1516	CSE	INTRODUCTION TO SOFTWARE ENGINEERING	CS455A	Introduction industrial strength software, problem of software development,problem of scale, basic processbased approach, etc. Software Process Models concept of processes, ETVX model for process management, the sub phases in the phase, analysis techniques (structuredanalysis), specification, validation, function point analysis, Project planning effort, schedule, quality, project monitoring, and basic CMM model coupling and cohesion, Coding style, structuredprogramming, verification concepts. Testing testing purpose, levels of testing,black box testing, white box testing, different test case generation approaches. 1. Ian Somerville, Software Engineering, 9th Ed., Pearson, 2010.2. C Ghezzi, M Jazayeri, D Mandrioli, FUndamentals of Software Engineering, 2nd Ed., PrenticeHall, 2002.3. RN Taylor, N Medvidovic, EM Dassen, Software Engineering, 2nd Ed., PrenticeHall, 2006.4. Software Engineering: Checking, ACM Computing Surveys, 41{4}, Article 21, 2009.5. AV Aho, MS Lam, R Sethi, JD Ullman, Compiler Design: Principles, Techniques and Tools, 2nd Ed., PrenticeHall,2006
1517	ES	M TECH THESIS	ES699	M. Tech. Thesis
1518	MTH	APPLIED STOCHASTIC PROCESSES	SE359	Definition and classification of general stochastic processes. Markov Chains:definition, transition probability matrices, classification of states, limitingproperties. Markov Chains with Discrete State Space: Definition and classification. Markov Process with Continuous State Space: Introduction toBrownian motion.
1519	EE	M TECH THESIS	EE699	M. Tech. Thesis
1520	MTH	REGRESSION ANALYSIS	SE358	Generalized Inverses, Cochrans theorem, Gauss Markov Setup, Least squaresestimators with restriction on parameters, Test of Hypothesis of linear parametericfunction, ANOVA, power of tests, Confidence intervals, Linearity, Ridge Regression and Principal Component Analysis, SubsetSelection, Nonlinear Regression.
1521	DES	M DES THESIS	DES699	M. Des. Thesis
1522	MTH	MATHEMATICAL LOGIC	SE354	Formal theories, consequence and deduction. Classical Propositional Calculus:Syntax, truth, validity, Adequacy of connectives, normal forms, applications tocircuit design, Axiomatic treatment, deduction theorem. Boolean algebras, Computerassisted formal proofs: tableaux,resolution. Classical first order theories: Syntax, satisfaction, truth validity,Axiomatic treatment, Equality, Examples of firstorder theories : Peano arithmetic, Consistency, completeness, LowenheimSkolemtheorems, compactness, Firstorder theories with equality, Decidability, Computerassistedformal proofs: tableaux, resolution. Godels incompleteness theorem
1523	CSE	M TECH THESIS	CS699	M. Tech. Thesis
1524	BSBE	M.TECH THESIS	BSE699	M. Tech. Thesis
1525	ES	PH D THESIS	ES799	Ph. D. Thesis
1526	CHM	SYMMETRY IN CHEMISTRY	SE335	Introduction to symmetry: Symmetry elements, symmetry operations and symbols, multiple symmetry operations, symmetry operations on molecules; Symmetry and group theory: Definition of groups. Molecular symmetry characters and character tables, reducible and irreducible representations; Symmetry and molecular structure calculations: Introduction to quantum mechanics, Symmetry of wave functions of atoms and molecules, atomic orbitals of conjugated molecules (Huckel MOs) and of transition metal complexes; Symmetry and spectroscopy: Transition between stationary states, Microwave, Infrared and Raman spectra, The symmetry of vibrational modes of some diatomic and polyatomic molecules, selection rules of electronictransition, vibronic analysis. Some important special effects such as the exclusion rule for centrosymmetric molecules, Fermi resonance, WoodwardHoffmann rule of electrocyclic reactions;
1527	HSS	PHD THESIS	PSY799	Ph. D. Thesis
1528	HSS	PHD THESIS	SOC799	Ph. D. Thesis
1529	CHM	BIOSYSTEMS	SE334	Buffers (their use in study of biomolecules), pH, pKa of aminoacids, D and L amino acid nomenclature ; Proteins: protein sequencing by chemical and mass & NMR spectroscopic methods, Use of spectroscopy to study proteins, 310, Betahelices, parallel and antiparallel betasheets, gamma 1 turns, betaturns), circular Dichroism of proteins, supersecondary structural motifs, tertiary (motifs and domains: some Important motifs like alpha helix, beta sheet, Protein Engineering; Biophysical techniques to purify and study proteins. Dialysis, salting out and precipitation by organic solvents, Ion exchange, gel filtration, reversed phase, affinity chromatography, ultrafiltration, nucleic acids (chemical, dideoxy and fluorescence), Prokaryotic Transcription, translation, genetic code, genomes, genes, over expression of recombinant proteins, mutagenesis (random and site directed) and inhibition of enzyme activity, uncompetitive, noncompetitive and irreversible inhibition of enzymes. Effect of pH, temperature on enzymes activity; Metabolism: Photosynthesis, Calvins cycle, Glycolysis, Krebs cycle, electron transport, oxidative phosphorylation.
1530	CHM	INDUSTRIAL ORGANIC CHEMISTRY	SE333	Various aspects of the energy and raw material supply: Coal, oil, natural gas,nuclear, and biomass as energy sources; Basic products of industrial synthesis:synthesis gas, methanol, formaldehyde, halogenated hydrocarbons, butanes, higher olefins, unbranched higher olefins and their use inmetathesis reactions, Acetylene: Significance and manufacturing process foracetylene, manufacture through calcium carbide, thermal cracking, applications of butadiene, Synthesis using carbonmonoxide: Hydroformylation, industrial operations, utilization of oxo products,carbonylation of olefins;Oxidation products of ethylene: Ethylene oxide, propylene oxide, propanol, butanols, amyl alcohols, aldol synthesis,polyhydric alcohols, neopentyl glycol Vinylhalogen and oxygen compounds: Vinylchloride, vinylidene chloride, vinylacetate, vinyl ethers; Polyamides: Adipic acid, aliphatic polyamides, allylchloride, acrylonitrile.; Aromatics: Sourceof feedstocks, coking of hard coal, isolation, special separation techniques,condensed aromatics, naphthalene, anthracene, hydrodealkylation.

S.No.	Branch	Course Title	Courses ID	Content
1556	ME	DYNAMICS OF ROTATING MACHINERY	ME757	Rotorbearing interaction. Flexural vibration, critical speeds of shafts, Effectsof anisotropic bearings, unbalanced response of an assymetric shaft. Gyroscopiceffects. Aerodynamic effects. Equivalent disc
1557	ME	VIBRATION CONTROL	ME756A	Factors affecting level of vibration, vibration reduction at the source, vibrationcontrol by structural design, selection of materials. Vibration control by additivedamping; Dynamic vibration absorbers, vibrati
1558	ME	VIBRATION CONTROL	ME756	Factors affecting level of vibration, vibration reduction at the source, vibrationcontrol by structural design, selection of materials. Vibration control by additivedamping; Dynamic vibration absorbers, vibrati
1559	ME	OPTIMIZATION METHODS IN ENGINEERING DESIGN	ME752A	Classical optimization methods, unconstrained minimization; Univariate, conjugatedirection, gradient and variable metric methods, constrained minimization,Feasible direction and projections. Integer and
1560	ME	OPTIMIZATION METHODS IN ENGINEERING DESIGN	ME752	Classical optimization methods, unconstrained minimization; Univariate, conjugatedirection, gradient and variable metric methods, constrained minimization,Feasible direction and projections. Integer and
1561	ME	COMPUTER AIDED ENGINEERING DESIGN	ME751A	Methodology of interactive, graphical, engineering design; Discretization,optimization, simulation in CAED. Design of curves and surfaces. Applicationsin conveyor systems, sheet metal working, tool design
1562	ME	COMPUTER AIDED ENGINEERING DESIGN	ME751	Methodology of interactive, graphical, engineering design; Discretization,optimization, simulation in CAED. Design of curves and surfaces. Applicationsin conveyor systems, sheet metal working, tool design
1563	ME	BOILING & CONDENSATION	ME742	Pool boiling: Nukiyama Experiment, theory of vapour bubble formation, Mechanismof CHF, various models and correlations. Flow Boiling, Homogeneous, andhetrogeneous models, Boiling enhancement te
1564	ME	COMBUSTION ENGINES & AIR POLLUTION	ME741	Thermochemistry, Chemical Equilibrium, Kinetics; Laminar and Tarbulent Flamepropagation in SI Engines, Unburned and Burned Mixture States, Flame Quenching;Fuel Injection, Spray Atomization, Penetr
1565	ME	MODELLING OF MECHANICAL PROPERTIES OF MATERIALS	ME729	Introduction, Homogenisation, Ductile Materials, Dislocations, single crystalplasticity, Size effects on mechanical properties, Thermodynamics of constitutivemodelling, Examples of constitutive models, M
1566	ME	FRACTURE AND FATIGUE	ME728A	Fracture: Energy release rate, crack tip stresses and deformation fields, plasticzone, Elastoplastic fracture through Jintegral and CTOD, Dynamic fracture,Testing for Fracture. Toughness. Fatigue: Crack nu
1567	ME	EXPERIMENTAL STRESS ANALYSIS	ME671	Strain Gauge, strain rosettes and transducer applications. Photoelasticity,materials and their selection. Introduction to 3D photoelasticity. Brittle coatingmethods, Moire method of strain analysis and nond
1568	ME	METAL FORMING	ME663	Fundmentals of plasticity, yield and flow, anisotropy, instability, limit analysis,slipline field theory. Applications to forging, wire and tube drawing, deep drawing,extrusion and rolling. High velocity forming.
1569	ME	MECHANICS OF GROUND VEHICLES	ME654	Simple dynamical models of ground vehicles, mechanics of pneumatic tires,mechanics of vehicleterrain interaction, performance characteristics of roadvehicles, Handling characteristics. Directional sta
1570	ME	PRINCIPLES OF DYNAMICS	ME652	Review of kinematics and kinetics of a particle and a rigid body in plane motion.Eulers equations; Methods of analytical dynamics, Lagranges equations; Hamiltonsprinciple; Dynamics in phase space and
1571	ME	EXPERIMENTAL METHODS IN THERMAL SCIENCES	ME649A	Probes and transducers; Calibration; Turbulence measurement via statisticalmeasures; Single and multipoint correlations; Signal conditioning; Opticalmethods, Interferometry, Schlieren, shadowgraph, LC
1572	ME	EXPERIMENTAL METHODS IN THERMAL SCIENCES	ME649	Probes and transducers; Calibration; Turbulence measurement via statisticalmeasures; Single and multipoint correlations; Signal conditioning; Opticalmethods, Interferometry, Schlieren, shadowgraph, LC
1573	CHM	BASIC PHYSICAL CHEMISTRY	SE331	States of matter and properties of gases: Description of states of matter, perfectgas equation of state, The Maxwell distribution of speeds, diffusion and effusion,real gases, intermolecular interactions, cri
1574	PHY	STATISTICAL MECHANICS	SE316A	Review of Thermodynamics 2. Basic Principles and Applications of StatisticalMechanics 3. Ideal Quantum Gases 4. Interacting Systems5. Theories of PhaseTransitions 6. Computer Simulations 7. Eleme
1575	STA	PH.D THESIS	STA799	Ph. D. Thesis
1576	PHY	STATISTICAL MECHANICS	SE316	Review of Thermodynamics 2. Basic Principles and Applications of StatisticalMechanics 3. Ideal Quantum Gases 4. Interacting Systems5. Theories of PhaseTransitions 6. Computer Simulations 7. Eleme
1577	ME	PHD THESIS	ME799	Ph. D. Thesis
1578	IME	PHD THESIS	IME799	Ph. D. Thesis
1579	PHY	MODERN OPTICS	SE313	Review of Maxwells and electromagnetic wave equations; Wave propagationin anisotropic media; Polarized light; Diffraction from circular aperture andconcept of resolution; Fourier transforms and Fourie
1580	PHY	ORDER AND CHAOS	SE312A	Dynamical Systems; Nonlinear dynamics of one dimensional flows, Fixed point,linear stability analysis, bifurcations; Nonlinear dynamics of two dimensionalflo
1581	PHY	ORDER AND CHAOS	SE312	Dynamical Systems; Nonlinear dynamics of one dimensional flows, Fixed point,linear stability analysis, bifurcations; Nonlinear dynamics of two dimensionalflo
1582	PHY	PHYSICS OF UNIVERSE	SE311	Astronomical observations and instruments, Photometry; Stellar spectra andstructure; Stellar evolution, nucleosynthesis and formation of elements; Variablestars; Compact stars (white dwarfs. Neutron s
1583	PHY	PRINCIPLES OF LASERS & THEIR APPLICATIONS	SE309	Gaussian Optics, Optical Resonators and their Mode Structure; Atomic levels,Absorption, Spontaneous and Stimulated emission, Einstein coefficients; RateEquations, Population Inversion, Gain media, 3 a
1584	PHY	ENERGY	SE308	Indian and global energy resources, current energy exploitation, energy demand,energy planning, renewable energy sources, wind energy, energy from water,solar energy, energy from mineral oils, nuclear
1585	PHY	QUANTUM PHYSICS	SE301	Origin of quantum theory and related experiments, WaveParticle duality forphotons and material particles, Wave function and its born interpretation,Relation with measurement of dynamical variables, dfu
1586	CSE	COMPUTER ARCHITECTURE	CS422A	Introduction: Overview of Computer Architecture, Performance evaluation ofprocessors, Pipelining, Superpipelines, Advanced pipelines, static and dynamicscheduling, Instruction level parallelism, loop un

S.No.	Branch	Course Title	Courses ID	Content
				1. JL Hennessy, DA Patterson, Computer Architecture: A Quantitative Approach, 4th Ed., Morgan Kaufmann/ElsevierIndia, 2006.
1587	ME	MACHINE LEARNING FOR ENGINEERS	ME698A	Multiobjective optimization, Robust design techniques (variation reductiontechniques), Optimal control, stochastic programming. Role of Optimization inCAD: Why optimization? OptimizationGeometric m analysis softwares, Graphics interfacing, Choice ofHardware platform. Application to engineering design problems, comparison withexisting solutions.
1588	ME	MICROSCALE THERMAL ENGINEERING	ME689	MicroMechanical systems (MEMS), Micro Channels, Heat pipes, jets, valves,Heat Sinks, Solar cells, Bearings, Pumps, Heat pipes, Jets, valves, Heat sinks,Solar Cells, Bearings, Pumps, Flow Sensors and ac CharacteristicNondimensional parameters, Microscale Heat conduction, Heat transport in thinfilms and at solidsolid interfaces, Convective diffusion phenomena, Enzymesubstratereactions, channel flow electroosmosis,Electroosmotic pumps, Surface tension driven flows, Coating flows,Thermocapillary flows, Molecular dynamics simulations.
1589	ME	COMPUTER AIDED DESIGN OF THERMAL SYSTEMS	ME648	Simulation of thermal processes, application to casting, extrusion, heat treatment,thermal design of heat exchangers, electronic circuitry. Optimization searchmethod and geometric programming, control
1590	ME	INTRODUCTION TO TURBULENT FLOW	ME647A	Measure of turbulence, diffusivity, length scales. Reynolds equation. Mixinglength models. Homogeneous, isotropic turbulence, correlation and energyspectrum functions, integral micro scales. Grid turbu
1591	ME	INTRODUCTION TO TURBULENT FLOWS	ME647	Measure of turbulence, diffusivity, length scales. Reynolds equation. Mixinglength models. Homogeneous, isotropic turbulence, correlation and energyspectrum functions, integral micro scales. Grid turbu
1592	ME	ADVANCED COMPUTATIONAL FLUID DYNAMICS	ME634A	Discretisation procedure in Finitedifference and Finitevolume. NavierStokes,Energy equations. Staggered rectilinear grids. Explicit methods : MAC, SMAC.Implicit Methods, SIMPLE and SIMPLER. Matrix m methods. Finitedifference NavierStokes solution on nonorthogonalgrids, transformation. Collocated grids. Finitevolume methods onnonorthogonal grids. Turbulence modelling, ke modelling
1593	ME	ADVANCED COMPUTATIONAL FLUID DYNAMICS	ME634	Discretisation procedure in Finitedifference and Finitevolume. NavierStokes,Energy equations. Staggered rectilinear grids. Explicit methods : MAC, SMAC.Implicit Methods, SIMPLE and SIMPLER. Matrix m methods. Finitedifference NavierStokes solution on nonorthogonalgrids, transformation. Collocated grids. Finitevolume methods onnonorthogonal grids. Turbulence modelling, ke modelling
1594	ME	NONLINEAR VIBRATION	ME627A	Phase space, singular points, limit cycle; Analytical methods, perturbationtechniques, equivalent linearization; Duffings equation, jump phenomenon,Van der Pol equation. Stability criterion; Floquets theor
1595	ME	NON-LINEAR VIBRATION	ME627	Phase space, singular points, limit cycle; Analytical methods, perturbationtechniques, equivalent linearization; Duffings equation, jump phenomenon,Van der Pol equation. Stability criterion; Floquets theor
1596	ME	VIBRATION OF CONTINUOUS SYSTEMS	ME626A	Vibration of discrete systems with single and multi degree of freedom. Hamiltonsprinciple, Langranges equations. Longitudinal vibration of bars, lateralvibration of straight and curved beams, vibration of
1597	ME	VIBRATION OF CONTINUOUS SYSTEM	ME626	Vibration of discrete systems with single and multi degree of freedom. Hamiltonsprinciple, Langranges equations. Longitudinal vibration of bars, lateralvibration of straight and curved beams, vibration of
1598	ME	AUTOMATION AND CONTROL	ME453	Modeling, Analysis, and Simulation of Dynamic System; Mechanical, electronic,electrohydraulic and electomechanical systems; Stepper and Servomotors; useof MATLAB; StateSpace, laplace and frequer sensitivity; PID, Phase lag and Phase lead compensation; Sampled datasystems and Digital controllers; DA/AD converters; Microprocessors; Sensors andactuators; interfacing with computers.
1599	ME	VIBRATION & CONTROL	ME354A	Introduction to modeling of dynamical systems. Single Degree of FreedomSystems Free undamped vibration, Free damped vibration, Forced vibration, Transmissibility,Convolution method, Mechanisms o Multi Degree of Freedom Systems(undamped and proportional damping) Matrix methods, Modal analysis. Approximate Methods.Vibration of continuous systems (free vibration only).Introduction to contr Bode plots. Controller performance and types.Steady state errors and constants. Types of feedback control systems Derivative errorcompensation, Integral error compensation, Proportional error comper 1. Theory of Vibrations. W. T. Thomson, Prentice Hall.2. Control Systems Engineering. N. S. Nise, John Wiley & Sons.3. Vibration Problems in Engineering. W. Weaver, S. P. Timoshenko and D. H. Young, John Franklin,). D. Powell, and A. EmamiNaeini,Prentice Hall.6. Modern Control Engineering. K. Ogata, Prentice Hall.
1600	ME	DYNAMICS AND VIBRATION OF MACHINERY	ME353	Three dimensional motion of rigid bodies kinematics and kinetics. Gyrodynamics.Vibrations of single, two and multiple degrees of freedom systems, free andforced vibrations; Time and frequency domain
1601	IME	BUSINESS PROCESS MANAGEMENT	MBA647A.	Introduction to Business Process Management (BPM); History; Importance of improving business processes; Drivers and triggers of BPM; Stakeholders; Importance of organizational strategy and process BPM solution; Importance of a structured approach to implementing BPM; The BPM implementation framework:Organizational strategy phase, process architecture phase, Launch pad phase, Understanding Project management; People change management; Leadership.BPM maturity; Embedding BPM within the organization; Methods, tools and techniques of business process modelling, analysis and design Multiple instance patterns, State based patterns, cancellation patterns; Business Process Languages.Best practices in BPM; BPM in eBusiness, eCommerce and eGovernment; BPM case studies.
1602	IME	BUSINESS PROCESS MANAGEMENT	MBA647.	Introduction to Business Process Management (BPM); History; Importance of improving business processes; Drivers and triggers of BPM; Stakeholders; Importance of organizational strategy and process BPM solution; Importance of a structured approach to implementing BPM; The BPM implementation framework:Organizational strategy phase, process architecture phase, Launch pad phase, Understanding Project management; People change management; Leadership.BPM maturity; Embedding BPM within the organization; Methods, tools and techniques of business process modelling, analysis and design Multiple instance patterns, State based patterns, cancellation patterns; Business Process Languages.Best practices in BPM; BPM in eBusiness, eCommerce and eGovernment; BPM case studies.
1603	IME	INTELLECTUAL PROPERTY MANAGEMENT VALUE CREATION AND VALUE CAPTURE	MBA640A.	ModuleI: Intellectual Property Management. Market Capitalization, Intellectual Capital (IC), Components of Intellectual Capital, Tangible and Intangible Assets of Firms, Goodwill, Linkage between IC, Corporate the need for Intellectual Property Management, Various Types of Intellectual Property trademarks, Copyrights, Patents, Trade Secrets, and Industrial Design, International IP Treaties/Agreements on IP Rights Management, Intellectual Property Strategies in Indian Context Universities, CSIR and Commercial Firms ModuleII: The Dynamics of Value Creation and Value Capture. ModuleIII: Patent Mapping.
1604	IME	INTELLECTUAL PROPERTY MANAGEMENT VALUE CREATION AND VALUE CAPTURE	MBA640.	ModuleI: Intellectual Property Management. Market Capitalization, Intellectual Capital (IC), Components of Intellectual Capital, Tangible and Intangible Assets of Firms, Goodwill, Linkage between IC, Corporate the need for Intellectual Property Management, Various Types of Intellectual Property trademarks, Copyrights, Patents, Trade Secrets, and Industrial Design, International IP Treaties/Agreements on IP Rights Management, Intellectual Property Strategies in Indian Context Universities, CSIR and Commercial Firms ModuleII: The Dynamics of Value Creation and Value Capture. ModuleIII: Patent Mapping.
1605	ME	INTRODUCTION TO CONTINUUM MECHANICS	ME622A	Analysis of stress and strain; Equilibrium, Compatibility and constitutiveequations; Plane problems; Stress functions; Applications; Complex potentialsin two dimensional and axisymmetric problems; Vari
1606	ME	INTRODUCTION TO CONTINUUM MECHANICS	ME622	Analysis of stress and strain; Equilibrium, Compatibility and constitutiveequations; Plane problems; Stress functions; Applications; Complex potentialsin two dimensional and axisymmetric problems; Vari
1607	ME	ADVANCED THEORY OF TURBOMACHINERY	ME617	The equations of motion in rotating coordinate system, effects of Coriolis andCentrifugal forces, energy equation, classification of turbomachines; twodimensionalcascade theory and experimental results secondary flows and loss estimation; offdesign performance; radialand mixed flow machines; multistage axial compressors and turbines; predictionof stage performance and stacking; rotating stall and s
1608	ME	MECHANICAL ENGINEERING LAB	ME471N	Experimentation in automation and control, Solid Mechanics, Heat transfer,Energy conversion.
1609	ME	MECHANICAL ENGG. RESEARCH	ME374	To learn about a topic in depth through independent study under the guidanceof a faculty member from the department. Based on a Original Research Project/design project/experimental project.

S.No.	Branch	Course Title	Courses ID	Content
1610	ME	MANUFACTURING SYSTEMS	ME371	Experimentation in machine dynamics, Materials, Manufacturing Sciences andFluid mechanics.
1611	ME	INTERNAL COMBUSTION ENGINES	ME359A	Classification, Construction, Valve arrangements. Fuels, Fuel air cycle. Combustion,Effect of engine variables, Combustion chambers, Carburation and fuel injection,Knocking. Engine cooling, Friction and
1612	ME	INTERNAL COMBUSTION ENGINES	ME359	Classification, Construction, Valve arrangements. Fuels, Fuel air cycle. Combustion,Effect of engine variables, Combustion chambers, Carburation and fuel injection,Knocking. Engine cooling, Friction and
1613	ME	MICROSCALE THERMAL SCIENCES	SE381	Introduction: Micro ElectroMechanical Systems (MEMS), Micro channels, Heatpipes, Jets, Valves, Heat sinks, Solar cells, Bearings, Pumps, Flow sensors andactuators, Fins, Drug delivery systems; Transp CharacteristicNondimensional parameters; Microscale Heat Conduction: Microscale energytransport in solids, Heat transport in thin films and at solidsolid interfaces,Heat conduction in semiconductor de Channelflow with soluble or rapidly reacting walls, Flow past reacting flat plate; Solutionsof Electrolytes: Electric double layer of Debye sheath, Electrokinetic phenomena,Electro osmosis, Electroosmotic Thermocapillary pump, Diffusocapillary flows, Marangoni convection and instability;Modeling: Continuum model, Molecular model, Introduction to molecular dynamicssimulations and direct simulation M
1614	NET	M TECH THESIS	NT699	M. Tech. Thesis
1615	MSP	M TECH THESIS	MS699	M. Tech. Thesis
1616	MTH	MATHEMATICAL MODELING	SE364	Elementary mathematical models. Role of mathematics in problem solving.Concepts of mathematical modeling. System approach. Formulation. Analysisof models. Sensitivity analysis & parameter estim
1617	ME	M TECH THESIS	ME699	M. Tech. Thesis
1618	MTH	FINITE ELEMENT METHOD	SE363	Introduction and motivation, Weak formulation of BVP and Galerkin approximation,Piecewise polynomial spaces and finite element method, Computer implementationof FEM, Results from Sobolev spaces spaces, Polynomial approximation theory in Sobolev spaces, Variationalproblem for second order elliptic operators and approximations, Mixed methods,Iterative techniques.
1619	MTH	STATISTICAL SIMULATION & DATA ANALYSIS	SE361	Simulation of random variables from discrete, continuous, multivariate distributionsand stochastic processes, MonteCarlo methods. Regression analysis, scatterplot, residual analysis. Computer Intensive representation of multivariate data, Cluster analysis, Principal componentanalysis for dimension reduction.
1620	LT	M TECH THESIS	LT699	M. Tech. Thesis
1621	MTH	MATHEMATICAL METHODS	SE360	Multiple Integral Theorems and their Applications: Greens theorem, Stokestheorem and Gauss divergence theorem. Integral Transforms: Fourier, Fouriersine/cosine and Hankel Transforms with their inver Perturbation theory, Regular perturbation theory, Singular perturbationtheory, Asymptotic matching. Calculus of Variation: Introduction, Variationalproblem with functionals containing first order derivative problem with moving boundaries. Boundaries with constraints. Higherorder necessary conditions, Weistrass function, Legendres and Jacobis condition.Existence of solutions of variational problems. Ray
1622	CSE	COMPUTING LABORATORY-I	CS251A	1. Basic operating system commands. Students are expected to know the basic shell (e.g., bash) commands and should be able to understand the options and functioning of a command by reading the m emacs. How ever, they should be able to utilize the multiple features of the editors (such as automatic indentation, ontextsensitive colouring, letypesensitive autowrap, etc.) and not use them simply as a ty cvs, svn, git, dares) work. They_should be able to check in, check out, resolve errors and put tags on a snapshot. On all subsequent assignments, they must use a form of version .control.4. Scripting and a familiar with the different command syntax in other shells as well. Also, they will need to know the various functions (e.g., seq, for) that a shell provides. The choice of the scripting language(per! or python know how to format properly the equations, gures, tables, theorems, etc. using different packages and options. For bibliography management, they should use bibtex, and must use it within the latex docum (they includegnuplot, xfig, etc.). 6. Hardware. Students will work handson to learn how to install hard drives, RAM, etc., and in general, assemble a computer from its different parts. available for different tools.
1623	IME	INNOVATION FOR SUSTAINABLE BUSINESS ADVANTAGE	MBA654A.	ModuleI: Innovation Concepts and Principles. Historical Perspectives, Innovation Myths and Realities, Challenges, Triggers and Sources for Creativity and Innovation, Innovation by individuals, communities and Unstructured Innovation, Link between Corporate Vision, Strategy and Innovation, Components of Strategic Innovation, Organizational Architecture for Strategic Innovation, The Role of Government Po Twelve Principles for Breaking Innovation Barriers, Innovation Principles for Sustainable Competitive Advantage and Generation of Wealth and ValueModuleII: Innovation Approaches and Frameworks. Innovation Levels of Innovation: Business Model Innovation, Process Innovation, Market Innovation and Product/Service Innovation; Innovation Frameworks: HansenBirkshaw, TracyWiersema, SawhneyWolcott; Re Model; Jugad Innovation, Essential Principles and Practice of Jugad Innovation, Jugad Innovations Future for Emerging Markets, Illustrative commercial examples for InnovationModuleIII: Innovation Proj
1624	CSE	PARALLEL PROGRAMMING	CS433A	1. Introduction: Why parallel computing; Ubiquity of parallel hardware/multicores; Processes and threads; Programming models: shared memory and message passing; Speedup and efficiency; Amdahls L architectures; cache hierarchy and coherence; sequential consistency.3. Introduction to parallel software: Steps involved in developing a parallel program; Depen dence analysis; Domain deco:rriposition; comm'tmication, false sharing, computationtocommunication ratio as a guiding metric for decomposition,hot spots and staggered communication.4. Shared memory parallel programming: Synchron models; Highlevel language memory models (such Java and/or C++); Memory fences. Developing parallel programs with UNIX fork model: !PC with shared memory and message pass ing; UNIX semapho with POSIX thread library: Thread creation; Thread join; Mutex; Condition variables. Example case studies (see note below for some details). Developing parallel programs with OpenMP directives: Parallel Reduction. Example case studies (see note below for some details).5. Message passing programming: Distributed memory model; Introduction to message passing interface (MPI); Synchronization as Se distribution, Scatter, Gather; MPI derived data types. Example case studies (see note below for some details). Notes:1. The example case studies should be chosen to cover a wi parallel prex, parallel search, graph algorithms,parallel ranking, reduction, algorithms using tree, fan, pipe paradigms, matrix computa tion, equation solvers,nbody simulation, ray tracing, etc.2. The instruct are encouraged to offer large semesterlong programming projects.
1625	ECO	BAYESIAN DATA ANALYSIS	ECO543A	Course Description:This is an introductory level course in Bayesian methods aimed to introduce students to the Bayesian framework (originating from Bayes rule), that formally allows to incorporate nonsa techniques that will be useful in academia as well as industry. The course is targeted for advanced undergraduate students or any individual seeking to learn Bayesian techniques.The course will be heavil linear regression. Consequently, students who have taken the course Econometric Methods (ECO341A) will have an advantage. However, students who are willing to take this course but have not taken eco linear regression.Softwares: Bayesian methods are highly computational, hence proficiency in coding is required or one should be willing to put in extra hours to code programs. The official software for th
1626	ME	ADHESION FRICTION AND LUBRICATION FOR MICROMACHINES	ME667A	I. History of friction and lubrication studies, Origins of friction, Coulomb's theory, Archard's theory, Bowden and Tabor's theory, Hardness of metals and ductile materials, contact area calcul hydrogen bonding etc.), adhesion models/nanorheology, meniscus and surfce models, static friction, stickslip phenomenon, friction anisotropy, concept ofsuperlubricity, micronanoscale wearphenomeno regimes, load bearing equation, journal bearings, elastohydrodynamic lubrication, film thickness calculation, Boundary lubrication, characteristics of boundary lubricants, liquid lubricants, additives, confin (91ectures)3. Nanotribological measurements: nanoscratching, atomic force microscopy tests, friction force microscopy, surface force apparatus, Tabor and Winterton design for surface force measurem surfaces. (9 lectures)4. Effects of surface/material parameters on micro/nanoscale friction (Load effects, speed effects, surface energy, surface l roughness effects on adhesion and friction, humidity mol Tribology on the Small Scale: A Bottom Up Approach to Friction, Lubrication, and Wear, C. Mathew Mates, Oxford University Press, 2007Nanotribology and Materials in MEMS, Sujeet K. Sinha, N. Satyanara Brian J. Briscoe (editors) (Imperial College Press)Biological Micro and Nanotribology: Nature's Solutions by M. Scherge, S. N. Gorb, Springer Verlag
1627	IME	COMMERCIAL BANKING RISK MODELING AND RISK MANAGEMENT	MBA679.	The course will deal with the theory, tools and techniques necessary for efficient modeling and management of risk in financial services with emphasis on commercial banking. At the macro level emphas operation of a commercial bank. At the micro level various facets of risk management which include interest rate, credit and market risk are covered in sufficient depth. The course also covers related topi The course does not have a specific text book and consists of prescribed readings provided by the instructor.
1628	IME	STRATEGIC MARKETING- CONTEMPORARY ISSUES	MBA639.	Successful Marketing in highly competitive global markets of today needs breakthrough concepts, socially responsible and innovative execution. And mastering that blend entails participatory, immersive from foundations courses.Course Plan and Modules: 1) Market opportunity recognition and evaluation 2) Generating business models 3) Green and sustainable marketing scenarios 4) Contextual strategi
1629	MME	MATERIALS CHARACTERIZATION	MME250	Chemical bonding, fundamentals of crystallography, reciprocal lattice, structuresin metals, inorganic compounds, polymers, silicates and glasses, stereographicprojections; Production, characterization, a applications; Optical principles of microscopy resolution, magnification,depth of focus; electron diffraction, imaging (various contrasts), determinationof crystal structure, burgers vector, electron beamspe

S.No.	Branch	Course Title	Courses ID	Content
1630	MME	METALLURGICAL KINETICS	MME210	Microscopy and, Electron Probe Micro Analyser; Principles of QuantitativeMicroscopy: volume density, surface density, length density, numerical density,particle and grain size; Overview of other character
1631	MME	THERMODYNAMICS OF MATERIALS	MME200	Thermodynamics vs. kinetics, homogeneous and heterogeneous reactions;Chemical Reaction Controlrate equation, reaction rate constant, reaction order,nonelementary reactions; Solid State Diffusion Fi fluid flow and its relevance to mass transfer, generalmass transport equation, concept of mass transfer coefficient, models of masstransfer film theory and Higbie's penetration theory; Internal Mass Trans adsorption isotherms; Langmuir, BET, adsorptionas the rate limiting step; gasification of C by CO2, dissolution of N2 in moltensteel, porous solids, specific surface area and pore size distribution; ReactorD time distribution; models of industrialreactors; Electrochemical Kineticsconcept of polarization, activation over potential,ButlerVolmer and Tafels equation, applications in electrodeposition and corrosion,Co
1632	MME	INTRODUCTION TO PROFESSION	MME100	Heterogeneous and homogeneous systems, extensive and intensive properties,simple equilibrium; First Law of thermodynamics, constant volume and constantpressure processes; Spontaneous processes of thermodynamics, criterion for equilibrium, Entropy and disorder;most probable microstate, configurational entropy and thermal entropy; auxiliaryfunctions, Maxwells relations, GibbsHelmholtz equation; thermodynamic properties of mixtures of ideal andimperfect gases; reactions in gas mixtures; reactions of pure condensed phaseswith gas mixtures standard Gibbs energy of reactions, Ellingham diagram quasichemical model of solution, activity andalternative standard states; reaction equilibrium in condensed system, Gibbsphase rule, binary systems involving compound formation, solubility of gasesin m equation, concentration andformation cells, Pourbaix diagrams; thermodynamics of Point Defects.
1633	BSBE	MODERN INSTRUMENTAL METHODS IN BIOLOGICAL SCIENCES	BSE621A	Historical aspects of various materials, including some landmarks; Naturalresources of materials; Cost, economics, energy, environmental and politicalissues relating to materials industry and application growth of automotive, aerospace and railway sectors, electrical, electronicand telecommunication equipment/systems, energy sector, military hardware,structural and general engineering applications, bio properties and processing.
1634	BSBE	MODERN INSTRUMENTAL METHODS IN BIOLOGICAL SCIENCES	BSE621	Applications of spectroscopic and other techniques to the study of biomolecules:UVVis spectroscopy, Circular dichroism, Fluorescence, NMR, Mass, IR and Ramanspectroscopy, XRay diffraction. Cellular techniquesto purify and study proteins. Dialysis, salting out and precipitaion by organicsolvents, Ion exchange, gel filtration, reversed phase, affinity chromatography,ultracentrifugation, gel electrophoresis electrophoretic analysis), detection (staining, blotting and immunodetection)and purification (various chromatograpy, HPLC, immunoprecipitation)of proteins, and specialized applications (in vitro synthes
1635	IME	RESEARCH METHODOLOGY	IME700	Applications of spectroscopic and other techniques to the study of biomolecules:UVVis spectroscopy, Circular dichroism, Fluorescence, NMR, Mass, IR and Ramanspectroscopy, XRay diffraction. Cellular techniquesto purify and study proteins. Dialysis, salting out and precipitaion by organicsolvents, Ion exchange, gel filtration, reversed phase, affinity chromatography,ultracentrifugation, gel electrophoresis electrophoretic analysis), detection (staining, blotting and immunodetection)and purification (various chromatograpy, HPLC, immunoprecipitation)of proteins, and specialized applications (in vitro synthes
1636	MTH	OPERATIONS RESEARCH - I	SE356	Introduction to Social Science Research Perspective, Different Approaches to Social Research, Approaches to Theory Building, Sampling , Measurement Issues & Scale construction, Research Design , Qu and Presentation, Research Evaluation and Critique, Issues in Current Research Practice.
1637	MTH	BASIC STRUCTURES OF MATHEMATICS	SE353	Linear Models: Formulation and Examples, Basic Polyhedral Theory Convexity,Extreme points, Supporting hyperplanes etc, Simplex Algorithm Algebraic andGeometrical approaches, Artificial variable tech BoundedVariable L.P.P. Transportation Problems: Models and Algorithms, Network Flows:Shortest path Problem, MaxFlow problem and Mincost Flow problem, DynamicProgramming: Principle of optimal
1638	MTH	DISCRETE MATHEMATICS	SE352	Finite and Infinite Sets: Finite sets, Countable sets, Uncountable sets. Groupsand Symmetry: Groups, Subgroups, Lagrange theorem, Normal subgroups,Quotient groups, Group actions, Homomorphisms, C Closed sets, Sequences, Continuity, Complete metric spaces,Contraction principle and applications, Connectedness and compactness. Fractals:Metric space of fractals and its completeness, Iterated func by identification: Torus, mobius strip, Klein bottle.
1639	MTH	LINEAR ALGEBRA	SE351	Permutations and combinations and basic definitions. Generating functions.Polyas enumeration theory. Recurrence relations. Principle of inclusion andexclusion. Balanced incomplete block design. Diffe
1640	PHY	INTRODUCTION TO ATOMIC & NUCLEAR PHYSICS	SE317	Fields and linear equations. Vector spaces. Linear transformations and projections,relations, etc. Determinants. Elementary canonical forms: diagonalization,triangulation, primary decomposition etc. Seco spaces, Selfadjoint, Unitary and normal operators, Orthogonal projections.Bilinear forms, Symmetric, Skewsymmetric, Positive and semipositive formsetc.
1641	CSE	PARALLEL PROGRAMMING	CS433	Quantum mechanics of one and two electron atoms, Many electron atoms, Centralfield approximation, ThomasFermi approximation, Molecular binding, LCAO,LCMO and VB methods, Hydrogen molecules, H
1642	PHY	MODERN PHYSICS LABORATORY	PHY315	1. Introduction: Why parallel computing; Ubiquity of parallel hardware/multicores; Processes and threads; Programming models: shared memory and message passing; Speedup and efficiency; Amdahls L architectures; cache hierarchy and coherence; sequential consistency.3. Introduction to parallel software: Steps involved in developing a parallel program; Depen dence analysis; Domain deco:rriposition; comm'tmication, false sharing, computationtocommunication ratio as a guiding metric for decomposition,hot spots and staggered communication.4. Shared memory parallel programming: Synchron models; Highlevel language memory models (such Java and/or C++); Memory fences. Developing parallel programs with UNIX fork model: !PC with shared memory and message pass ing; UNIX semapho with POSIX thread library: Thread creation; Thread join; Mutex; Condition variables. Example case studies (see note below for some details). Developing parallel programs with OpenMP directives: Parallel Reduction. Example case studies (see note below for some details).5. Message passing programming: Distributed memory model; Introduction to message passing interface (MPI); Synchronization as Se distribution, Scatter, Gather; MPI derived data types. Example case studies (see note below for some details). Notes:1. The example case studies should be chosen to cover a wi parallel prex, parallel search, graph algorithms,parallel ranking, reduction, algorithms using tree, fan, pipe paradigms, matrix computa tion, equation solvers,nbody simulation, ray tracing, etc.2. The instruct are encouraged to offer large semesterlong programming projects.
1643	PHY	NATURAL NANO MACHINES	PHY314	Modern experimental techniques with a view to demonstrate the basic conceptsin physics through experiments. this course has three components: a) onelecture per week: observation, measurements, qu the discovery of electron, quantization of charge, e/m ratios,Millikans oil drop experiment, SternGerlach experiment, Rutherford scattering,Davisson Germer experiment, discovering atomic nature through femtoseconds to light years. b) laboratory work(twice a week): a current list of experiments is available with the Department.c) small project/open ended experiments: These experiments will be chosenby and using central facilities.
1644	IME	INNOVATION FOR SUSTAINABLE BUSINESS ADVANTAGE	MBA654.	Examples of nanomachines in living cells; differences between macroscopic andnano machines; world of nanometer and picoNewton; stochastic dynamics ofnanomachines; experimental, computational ratchetmechanism; mechanochemistry of nanomachines; energetics and efficiency ofnanomachines; intracellular cargo transporters; nanosize unzippers; nanosizeengines for polymerization of macrom rotary motors of ATP synthesizer; molecular sensors hair cells; nanopistonsand cell crawling.
1645	IME	SUMMER PROJECT	MBA697.	ModuleI: Innovation Concepts and Principles. Historical Perspectives, Innovation Myths and Realities, Challenges, Triggers and Sources for Creativity and Innovation, Innovation by individuals, communities and Unstructured Innovation, Link between Corporate Vision, Strategy and Innovation, Components of Strategic Innovation, Organizational Architecture for Strategic Innovation, The Role of Government Po Twelve Principles for Breaking Innovation Barriers, Innovation Principles for Sustainable Competitive Advantage and Generation of Wealth and ValueModuleII: Innovation Approaches and Frameworks. Ino Levels of Innovation: Business Model Innovation, Process Innovation, Market Innovation and Product/Service Innovation; Innovation Frameworks: HansenBirkinshaw, TracyWiersema, SawhneyWolcott; Re Model; Jugad Innovation, Essential Principles and Practice of Jugad Innovation, Jugad Innovations Future for Emerging Markets, Illustrative commercial examples for Innovation ModuleIII: Innovation Proj
1646	CE	COMPUTATIONAL METHODS IN ENGINEERING	ESO218	During the summer after first two semesters, each student will take up a summerproject in an industrial or service organization for 810 weeks. During this period,the student will work under the guidance o presentationduring the third semester.
1647	PHY	COMPUTATIONAL PHYSICS	PHY473A	Introduction, Engineering Systems, Physical and Mathematical Modeling, ErrorAnalysis Approximations and round off and Truncation errors, Roots ofEquations single variable Method of Bisection, Metho of Linear Simultaneous EquationsDirect MethodsGauss Elimination,GaussJordan, LU decomposition; Iterative MethodsGaussSeidel, ConjugateGradient, Banded and Sparse systems, Solution of Nonlinear Regression Analysis for Multivariable, Eigen Valuesand Eigen Vectors Power method, Relaxation Method, Diagonalization method.Numerical Differentiation and IntegrationHighAccuracy Differentiation Fo Open Integration Formulas, OrdinaryDifferential Equations Finite Difference method, Method of Weighted Residuals,Analytical versus Numerical Methods, Initial Value and Boundary Value ProblemsEulers and parabolic Equations,Explicit and Implicit Methods, Crank Nicholson Method, ADI method; Introductioonto Finite Element Method, Applications.

S.No.	Branch	Course Title	Courses ID	Content
1648	PHY	COMPUTATIONAL PHYSICS	PHY473	Introduction to computers, FORTRAN/C; finite difference calculus, interpolationand extrapolation, roots of equations, solution of simultaneous linear algebraicequation, least squares curve fitting, numeric
1649	PHY	MATHEMATICAL METHODS II	PHY422A	Functions of a complex variable, ordinary differential equations, special functions,differential operations and SturmLiouville theory, partial differential equations,Greens functions.
1650	PHY	MATHEMATICAL METHODS II	PHY422	Functions of a complex variable, ordinary differential equations, special functions,differential operations and SturmLiouville theory, partial differential equations,Greens functions.
1651	PHY	STATISTICAL MECHANICS	PHY412A	1. Review of Thermodynamics, Probability theory, Random Walk, Brownianmotion, Diffusion Equation, idea of Langevin and FokkerPlanck Equations [8]2. Basic principles of Equilibrium Classical Statistical Mechanics, Matrix, Ideal Quantum Gases andtheir properties, BoseEinstein Condensation, Free Electron gas [10]4 a)Ising model of Magnetism, Transfer Matrix method, Mean field theory [4]b) Phase Transitions, Curie temperature, S. K.Ma, Chaudhury and Stauffer. 1. A.C. Melissiuos and J. Napolitano, Experiments in Modern Physics, znct ed. (Academic Press,Amsterdam, 2003).2. Resource Files on Experiments maintained in the Laboratory.3. P. R. Beviugton, Data F
1652	PHY	STATISTICAL MECHANICS	PHY412	1. Review of Thermodynamics, Probability theory, Random Walk, Brownianmotion, Diffusion Equation, idea of Langevin and FokkerPlanck Equations [8]2. Basic principles of Equilibrium Classical Statistical Mechanics, Matrix, Ideal Quantum Gases andtheir properties, BoseEinstein Condensation, Free Electron gas [10]4 a)Ising model of Magnetism, Transfer Matrix method, Mean field theory [4]b) Phase Transitions, Curie temperature, S. K.Ma, Chaudhury and Stauffer. 1. A.C. Melissiuos and J. Napolitano, Experiments in Modern Physics, znct ed. (Academic Press,Amsterdam, 2003).2. Resource Files on Experiments maintained in the Laboratory.3. P. R. Beviugton, Data F
1653	HSS	TWENTIETH CENTURY PHILOSOPHY II	PHI765	An Analytical Study of the Logic of Ordinary Language; Problems of Meaning,Reference and Truth; the AnalyticSynthetic Distinction; the SchemeContentDistinction; Antifoundationalism and Neopragmatism
1654	ECO	ADVANCED MICROECONOMICS	ECO750	Discussions on macroeconomic theory within the ISLM framework, add the supplyconsiderations, and then review some of the new clasical and new keynesiancontributions; budget financing, interactions
1655	IME	ADVANCED STATISTICAL METHODS FOR BUSINESS ANALYTICS	IME692A	The course is designed to train students on understanding research problems and situations requiring multivariate approaches, selecting appropriate multivariate techniques of data analysis, interpreting results and dealing with multiple noninterdependence techniques (such as Factor Analysis, Cluster Analysis, Multidimensional Scaling), multiple dependence techniques (such as Multiple Regression Analysis, Discriminant Analysis, etc.)
1656	MSE	MATERIALS SCIENCE TECHNOLOGIES FOR APPLICATIONS IN LIFE SCIENCES	MSE693A	Introduction to integrating nanotechnology and materials science with life sciences : Introduction to various size regimes in life science and materials science Importance of integration of materials science with life sciences of important molecules in cell (1) Protein structure, organization, functions with emphasis on antibodies and enzymes, regulation of enzyme activity, protein phosphorylation DNA: structure and function of DNA microfluidic devices and Labonchip devices Materials for MEMS Photolithography: (single crystal silicon, mask, oxide formation, resist application, baking, exposure, positive and negative resist, developing, etching, micromachining, 3D structure with sacrificial layer, LIGA Deposition: physical and chemical vapour deposition Soft fabrication: application of polymers in bioMEMS, microcontact printing, microtransfer methods, response to implanted device, in vivo and in vitro tests for biocompatibility Overview of immune system (innntate and adaptive immunity, cell mediated and humoral immunity), B cells, T cells, MHC 1. B. Alberts eta/., Essential Cell Biology. (Garland Publishing Inc., New York, ed. Third, 2009).2. S. S. Saliterman, Fundamentals of bioMEMS and medical microdevices. (WileyInterscience,Bellingham, 2005).3. S. S. R. Kumar, Ed., Nanomaterials. C. S. S. R. Kumar, Ed., Nanotechnologiesfor the life sciences (WileyVCH, Weinheim, 2006), vol. 1, pp. 366.5. C. S. S. R. Kumar, Nanomaterials for biosensors. C. S. S. R. Kumar, Ed., Nanotechnologies for the life sciences. C. S. S. R. kumar, Ed.,Nanotechnologies for the life sciences (WileyVCH, Weinheim, 2005), vol. 3.7. J. M. Anderson, Annu Rev of Matl Res, 31, 81 (2001)
1657	MSE	MATERIALS SCIENCE TECHNOLOGIES FOR APPLICATIONS IN LIFE SCIENCES	MSE693	Introduction to integrating nanotechnology and materials science with life sciences : Introduction to various size regimes in life science and materials science Importance of integration of materials science with life sciences of important molecules in cell (1) Protein structure, organization, functions with emphasis on antibodies and enzymes, regulation of enzyme activity, protein phosphorylation DNA: structure and function of DNA microfluidic devices and Labonchip devices Materials for MEMS Photolithography: (single crystal silicon, mask, oxide formation, resist application, baking, exposure, positive and negative resist, developing, etching, micromachining, 3D structure with sacrificial layer, LIGA Deposition: physical and chemical vapour deposition Soft fabrication: application of polymers in bioMEMS, microcontact printing, microtransfer methods, response to implanted device, in vivo and in vitro tests for biocompatibility Overview of immune system (innntate and adaptive immunity, cell mediated and humoral immunity), B cells, T cells, MHC 1. B. Alberts eta/., Essential Cell Biology. (Garland Publishing Inc., New York, ed. Third, 2009).2. S. S. Saliterman, Fundamentals of bioMEMS and medical microdevices. (WileyInterscience,Bellingham, 2005).3. S. S. R. Kumar, Ed., Nanomaterials. C. S. S. R. Kumar, Ed., Nanotechnologiesfor the life sciences (WileyVCH, Weinheim, 2006), vol. 1, pp. 366.5. C. S. S. R. Kumar, Nanomaterials for biosensors. C. S. S. R. Kumar, Ed., Nanotechnologies for the life sciences. C. S. S. R. kumar, Ed.,Nanotechnologies for the life sciences (WileyVCH, Weinheim, 2005), vol. 3.7. J. M. Anderson, Annu Rev of Matl Res, 31, 81 (2001)
1658	HSS	EXISTENTIALISM	PHI782	The course seeks to go into the sources of modern existentialist movementin the thoughts of Hegel, Husserl and Kierkegaard. A detailed study of Sartresphilosophy will be undertaken in the light of his ea
1659	HSS	SOCIAL AND POLITICAL PHILOSPHY	PHI774	Nature and Method of Social Philosophy; Nature of Society; Theories of Originsof Society; Place and Role of Social Institutions; Social Values and Ends ofPolitical Authority; Sources of Justification and Legitimacy
1660	HSS	INDIAN PHILOSOPHY I	PHI769	A discussion of Epistemological, Metaphysical and Value problems raised in theClassical Systems of Indian Philosophy. Depending on the students interest,aptitude and progress, either some texts will be studied or all.
1661	HSS	MODAL LOGIC	PHI768	The History of Modal notions; The Lewis systems; Strict Implication andEntaliment; The System E and Relevance Logic; Philosophical Problems.
1662	HSS	MORAL JUDGEMENT	PHI757	The details of the studies in this course will be designed for advanced studentsdepending on their fields of research.
1663	PHY	MEASUREMENT TECHNIQUES	PHY692A	Typical experiments in various areas is physics; vacuum techniques; transducers:temperature, pressure, charge particles, photons, etc; electronic noise; surveyof analog and digital I/Cs; signal processing
1664	PHY	MEASUREMENT TECHNIQUES	PHY692	Typical experiments in various areas is physics; vacuum techniques; transducers:temperature, pressure, charge particles, photons, etc; electronic noise; surveyof analog and digital I/Cs; signal processing
1665	PHY	PARTICLE PHYSICS	PHY680	Natural units; evidence for 4 fundamental interactions, leptons and hadrons,historical introduction to particle zoo, relativistic kinematics, Lorentzinvariantphase space, calculation of 2 and 3body phase space integrals, helicity representations and Young tableaux, the GellMann eightfold way,prediction of Λ , quark model, construction of hadronic wave functions, magneticmoment of the neutron, statistics of baryons & concept of confinement, gauge theory, local U(1) gauge theory andMaxwell equations, YangMills theories, SU(2) and SU(3) gauge theories,construction of SU(2)xU(1) gauge theory, gauge boson selfinteractions,spontaneous breaking of symmetry, construction ofthe GlashowSalamWeinberg model (outline only).
1666	PHY	GENERAL RELATIVITY & COSMOLOGY	PHY660	Machs principle. Riemanniangeometry. energymomentum tensor and Einsteinsequations. Schwarzschild metric and singularities of space time. postNewtonianapproximations. spherically symmetric solutions
1667	PHY	ELECTRONICS	PHY647	Survey of network theorems and network analysis, basic differential amplifiercircuit, op amp characteristics and applications, simple analog computer, analogintegrated circuits, PLL, etc., digital electronic logic gates, flipflops, multichannelanalyzer etc., introduction to microprocessors.
1668	PHY	COHERENT OPTICS	PHY646	Fourier transforms, diffraction theory, coherence theory, twodimensional systemstheory, optical processing, holography, holographic interferometry and itsapplications; astronomical correlation interferometry
1669	PHY	QUANTUM ELECTRONICS	PHY644	Semiclassical theory of lasers, single and multimode operation, gas laser theory,ring and Zeeman lasers, coherence in lasers. nonlinear optical phenomena,Feynman diagrams in multiphoton problems.
1670	PHY	CONDENSED MATTER PHENOMENA IN LOW DIMENSIONAL MATERIALS	PHY642	Characteristic length scales for quantum phenomena; scaling as a heuristic tool;scientific and technological significance of nanostructures and mesoscopicstructures. brief introduction to quantum view of condensed matter, electronic properties: science and technology realizing low dimensionalstructures; MBE, MOCVD, LangmuirBlodgett films, novel processes; electronicproperties of heterostructures, quantum wells, quantum dots, quantum tunneling, hot electrons, conductance and transmission of nanostructures;principles of application of electronic devices. quantum structures:optical properties: optical process in low dimensional semiconductors

S.No.	Branch	Course Title	Courses ID	Content
				megnetotransport: transport in magnetic field, semiclassicaldescription, quantum approach, AharonovBohm effect, Shubnikov deHaas effect; introduction to quantum Hall effect.
1671	PHY	NUCLEAR TECHNIQUES IN SOLID STATE STUDIES	PHY638	Different solid state physics/materials science aspects which can bestudied using nuclear techniques. Rutherford backscattering, channeling, elasticrecoil detection analysis, positron annihilation, Mssbauer spectroscopy.
1672	PHY	LOW TEMPERATURE PHYSICS	PHY634	Production of low temperatures; cryostat design and experimental techniquesapplied to low temperature; thermometry; specific heat, transport phenomena,thermal, electrical and magnetic properties; superconductivity.
1673	PHY	PHYSICS AND TECHNOLOGY OF THIN FILMS	PHY629	Introduction to thin films, nucleation theories and growth processes, PVD andCVD processes, epitaxial growth, microstructure, electronic transport, opticalproperties of thin films, size effects, physics and applications.
1674	PHY	PHYSICS OF SOFT MATTER AND FLUIDS	PHY628	Tightbinding band structure; shallow impurities, deep impurities, densityfunctional theory, manybody theory of impurities, quantized Hall effect,metastability.
1675	HSS	PROBLEMS IN PHILOSOPHICAL AESTHETICS	PHI752	Theories of Art as Mimesis, Expression and Form; Aesthetic Experience; Artas a Cultural System; Art and Morality; the Philosophy of Literature: Truth,Meaning, Interpretation and Evaluation; Literature and Aesthetics.
1676	HSS	PHILOSOPHICAL AESTHETICS	PHI450	The Aesthetic attitude; Aesthetic Experience; Art and Aesthetic; Defining Artand its Problems; Art and Emotion; Literary Aesthetics; Art, Society and Morality;Philosophy of Literature.
1677	HSS	INDIVIDUAL VS AUTHORITY	PHI448A	The distinction between Authority, Power and Sovereignty; Political forms ofAuthority; The individuals Rights and his Legitimate Autonomy; Encroachmenton the Individuals Legitimate Sphere: its Sources and Remedies.
1678	HSS	INDIVIDUAL VS AUTHORITY	PHI448	The distinction between Authority, Power and Sovereignty; Political forms ofAuthority; The individuals Rights and his Legitimate Autonomy; Encroachmenton the Individuals Legitimate Sphere: its Sources and Remedies.
1679	PHY	QUANTUM PROCESSES IN LOW DIMENSIONAL SEMICONDUCTORS	PHY312A	Characteristic length scales for quantum phenomena; scaling as a heuristic tool;scientific and technological significance of nanostructures and mesoscopicstructures. brief introduction to quantum view of electronic properties: science and technology realizing low dimensionalstructures; MBE, MOCVD, LangmuirBlodgett films, novel processes; electronicproperties of heterostructures, quantum wells, quantum dots, quantum tunneling, hot electrons, conductance and transmission of nanostructures;principles of application of electronic devices. quantum structures:optical properties: optical process in low dimensional semiconductors. megnetotransport: transport in magnetic field, semiclassicaldescription, quantum approach, AharonovBohm effect, Shubnikov deHaas effect; introduction to quantum Hall effect.
1680	PHY	QUANTUM PROCESSES IN LOW DIMENSIONAL SEMICONDUCTORS	PHY312	Characteristic length scales for quantum phenomena; scaling as a heuristic tool;scientific and technological significance of nanostructures and mesoscopicstructures. brief introduction to quantum view of electronic properties: science and technology realizing low dimensionalstructures; MBE, MOCVD, LangmuirBlodgett films, novel processes; electronicproperties of heterostructures, quantum wells, quantum dots, quantum tunneling, hot electrons, conductance and transmission of nanostructures;principles of application of electronic devices. quantum structures:optical properties: optical process in low dimensional semiconductors. megnetotransport: transport in magnetic field, semiclassicaldescription, quantum approach, AharonovBohm effect, Shubnikov deHaas effect; introduction to quantum Hall effect.
1681	PHY	MODERN OPTICS (ITS PHYSICS & ENGINEERING)	PHY307	Review of Maxwells and electromagnetic wave equations, wave propagation inanisotropic media, polarized light, diffraction from circular aperture and conceptof resolution, Fourier transforms and Fourier optics, optical fibres, optical communication sources (LED, lasers etc.) and detectors,and optical, electro and magnetooptic effects, lasermatter interaction.
1682	PHY	ORDER AND CHAOS IN NATURE	PHY306A	Dynamical systems, importance of nonlinearity, nonlinear dynamics of flows (in1, 2 and 3 dimensions) and maps (in 1, 2 dimensions) in phase space(equilibrium, periodicity, bifurcation, catastrophe, determinism, renormalization), measurement of chaos (Poincar section, Lyapunovindex, entropy), fractal geometry and fractal dimension, examples fromphysicalsciences, engineering and biology.
1683	PHY	ORDER AND CHAOS	PHY404	Dynamical systems, importance of nonlinearity, nonlinear dynamics of flows (in1, 2 and 3 dimensions) and maps (in 1, 2 dimensions) in phase space(equilibrium, periodicity, bifurcation, catastrophe, determinism, renormalization), measurement of chaos (Poincar section, Lyapunovindex, entropy), fractal geometry and fractal dimension, examples fromphysicalsciences, engineering and biology.
1684	PHY	PHYSICS OF UNIVERSE	PHY305	Astronomical observations and instruments, photometry, stellar spectraand structure; stellar evolution, nucleosynthesis and formation of elements,variable stars, compact stars, star clusters and binary systems.
1685	PHY	PRINCIPLES OF LASERS & THEIR APPLICATIONS	PHY303A	Gaussian optics, optical resonators and their mode structure, atomiclevels, absorption, spontaneous and stimulated emission, Einstein coefficients,rate equations, population inversion, gain media, 3 and 4 level systems, optical communication, materials processing, holography,medical applications.
1686	PHY	PRINCIPLES OF LASERS & THEIR APPLICATIONS	PHY303	Gaussian optics, optical resonators and their mode structure, atomiclevels, absorption, spontaneous and stimulated emission, Einstein coefficients,rate equations, population inversion, gain media, 3 and 4 level systems, optical communication, materials processing, holography,medical applications.
1687	PHY	ENERGY	PHY301A	Indian and global energy resources, current energy exploitation, energy demand,energy planning, renewable energy sources, wind energy, energy from water,solar energy, energy from mineral oils, nuclear energy.
1688	PHY	ENERGY	PHY301	Indian and global energy resources, current energy exploitation, energy demand,energy planning, renewable energy sources, wind energy, energy from water,solar energy, energy from mineral oils, nuclear energy.
1689	PHY	RELATIVITY	PHY226B	Special Relativity: empirical evidence for the constancy of c, frames ofreference; Lorentz transformations; relativity of simultaneity; twin and otherparadoxes; Spacetime diagrams; Transformation laws for physical quantities; collisions; Maxwell's equationsin covariant form.
1690	ECO	MONEY AND BANKING	ECO333	Functions of Money and Interest Rates; Money Supply; Money Market; Regulatoryand promotional institutions; Banking institutions; and Term structure of InterestRates.
1691	ECO	ECONOMETRIC PRACTICE	ECO312	Economic theory and econometrics; Types of models; Sources of data;Microeconometric models; Models of the firm and industry; Applications todevelopment economics; Macroeconometric models; Monetarism.
1692	ECO	ENVIRONMENTAL ECONOMICS LEGISLATION AND SOCIAL IMPACT	ECO747	With a view to establish sustainable development and to overcome the dilemmaof development, the course proposes to focus on the broad aspects ofenvironmental economics, environmental legislations and policy. The Government under the leadership of U.N. and World Bank will be discussedalong with a number of case studies. The legal aspects of various enviromentalprojects will be discussed. Besides the students will be encouraged to do projects on environmental issues.
1693	ECO	ADVANCED MONETARY THEORY	ECO745A	The course begins with the role of money in static macroeconomic models,neoclassical growth models and consumption loan models, then focusses uponsupply of money and monetary aggregate demand.
1694	ECO	ADVANCED MONETARY THEORY	ECO745	The course begins with the role of money in static macroeconomic models,neoclassical growth models and consumption loan models, then focusses uponsupply of money and monetary aggregate demand.
1695	ECO	SEMINAR ON SELECTED TOPICS IN ADVANCED ECONOMIC ANALYSIS	ECO740	The course will cover some frontier topics in advanced economic analysis fora detailed discussion in seminars.
1696	ECO	INTER-INDUSTRY ECONOMICS	ECO738A	The course will highlight the basic inputoutput models and their extensions.Various examples would be drawn from the empirical analysis of the inputoutputframework in India and abroad.
1697	ECO	INTER-INDUSTRY ECONOMICS	ECO738	The course will highlight the basic inputoutput models and their extensions.Various examples would be drawn from the empirical analysis of the inputoutputframework in India and abroad.
1698	ECO	ADVANCES IN MICROECONOMIC THEORY	ECO737	Developments in monopolistic competition (nonprice decisions in particular), themicroeconomic theory of modern organizations (discretionary managerialbehaviour, the organizational dimension) as emerging issues.

S.No.	Branch	Course Title	Courses ID	Content
1699	ECO	PUBLIC ECONOMICS	ECO535	Scope of public economics; Equity, social welfare and taxation; Taxation, incomesupport and social insurance; Taxation and individuals; Market failure andgovernment intervention; Optimal provision of pu
1700	ECO	MULTINATIONAL ENTERPRISES	ECO527	Introduction; Foreign direct versus portfolio investment; Global and regionaltrends in direct investment; Horizontal and vertical integration; Ownershiplocationinternalization framework; Endogenous mark Overseas investment and firm exports; Vertical multinationals,fragmentation and outsourcing; Licensing versus direct investment: models ofinternalization of the multinational enterprise; Contracts, intelle trade; International taxation and transfer prices; Dodomestic firms benefit from direct foreign investment?; MNEs and host countryrelations
1701	ECO	ECONOMIC ANALYSIS OF LAW	ECO424A	Introduction to economic analysis of law; Economic theory of property law;Contracts and warranties; and Economic theory of tort law in addition, Thecourse focuses on any are of the following madule(s) analysis of labor law; Economic analysis of competition law;Project assignment
1702	ECO	ECONOMIC ANALYSIS OF LAW	ECO424	Introduction to economic analysis of law; Economic theory of property law;Contracts and warranties; and Economic theory of tort law in addition, Thecourse focuses on any are of the following madule(s) analysis of labor law; Economic analysis of competition law;Project assignment
1703	ECO	FINANCIAL ECONOMICS	ECO423A	Basic accounting principles; Basics of financial markets; Return, risk and riskaversion; How securities are traded?; Mutual funds and the institutionalenvironment; Portfolio selection; The capital asset pric prices and yields; The term structure of interest rates; Managingbond portfolio; Security analysis; Options and other derivatives; Option valuation;Futures and forward markets; Portfolio management tech
1704	ECO	FINANCIAL ECONOMICS	ECO423	Basic accounting principles; Basics of financial markets; Return, risk and riskaversion; How securities are traded?; Mutual funds and the institutionalenvironment; Portfolio selection; The capital asset pric prices and yields; The term structure of interest rates; Managingbond portfolio; Security analysis; Options and other derivatives; Option valuation;Futures and forward markets; Portfolio management tech
1705	ECO	MACROECONOMICS I	ECO231	Global economic issues and challenges; From global to national issues; Economicinstitutions: markets, government, and social groups; Indian economic issuesand policy; Indian agriculture and rural econ towardsinternational capital flow
1706	IME	ADVANCED STATISTICAL METHODS FOR BUSINESS ANALYTICS	IME692	The course is designed to train students on understanding research problems and situations requiring multivariate approaches, selecting appropriate multivariate techniques of data analysis, interpreting dealing with multiple noninterdependence techniques (such as Factor Analysis, Cluster Analysis, Multidimensional Scaling), multiple dependence techniques (such as Multiple Regression Analysis, Discriminant Analysis)
1707	EE	INTELLIGENT SYSTEMS & CONTROL	EE705A	Norms of Signals, Vectors and Matrices,Positive Definite Functions, Positive Definite Matrices; Continuoustime Statespace Model, L TI Statespace Model, Nonlinear Statespace model, Equilibrium point and Stability Theory, Lyapunov stability of time invariant system, LaSalle's Invariance Theorem, Chetaev's Instability Theorem, Lyapunov stability of time varying system, Lyapunov's indirect method, Nonlinear Statespace model, ARMAX and NARMAX Models, Lyapunov Stability for Discrete Time Systems; Modeling of Different Nonlinear Systems: Inertial Wheel Pendulum, Two Link Manipulator, Inverted Pendulum, Design, Statefeedback linearizable systems.FeedForward Networks: Multilayered Neural Networks, Radial Basis Function Networks. Adaptive Learning Rate; Feedback Networks, Back Propagation Through Time, Recurrent Neural NetworksClassical sets, Fuzzy Sets, Concept of a fuzzy number, Operations on Fuzzy sets, Properties of Fuzzy Sets, Some Typical Membership Functions; Extension Principle of Fuzzy Sets, Crisp Relation Composition, Fuzzy Base and ApproximateReasonjng, Fuzzy Linguistic Variables, Linguistic modifier, Rulebase systems, Fuzzy Rulebase, Fuzzy Implication Relations, Fuzzy Compositional Rules, Inference mechanism composition, Fuzzy Identification Using TS Fuzzy Models,The TS Model From InputOutput Data, The TS Fuzzy Model Using Linearization. 1. Alexander M. Meystel and James S. Albus , Intelligent Systems: Architecture, Design and Applications, John Wiley & Sons, 1997. 2. J. R. Koopman, Adaptive Systems with LabVIEW(TM),Springer, 2009
1708	MSE	MATERIALS SCIENCE TECHNOLOGIES FOR APPLICATIONS IN LIFE SCIENCES	MSE693N	Introduction to integrating nanotechnology and materials science with life sciences Introduction to various size regimes in life science and materials science. Importance of integration of materials science with life sciences. Important molecules in cell (1) Protein structure, organization, functions with emphasis on antibodies and enzymes, regulation of enzyme activity, protein phosphorylation DNA: structure and function of DNA, microfluidic devices and Labonchip devices Materials for MEMS Photolithography: (single crystal silicon, mask, oxide formation, resist application, baking, exposure, positive and negative resist, development, etching, micromachining, 3D structure with sacrificial layer, LIGA Deposition: physical and chemical vapour deposition Soft fabrication: application of polymers in bioMEMS, microcontact printing, microtransfer printing, response to implanted device, in vivo and in vitro tests for biocompatibility Overview of immune system (innntate and adaptive immunity, cell mediated and humoral immunity), B cells, T cells, MHC Self assembly, (organothiol molecules on gold substrate) Techniques for assembly: microcontact printing, dip pen nanolithography Layer by layer self assembly: methods, materials applications Application of multilayered systems. 1. B. Alberts eta/., Essential Cell Biology. (Garland Publishing Inc., New York, ed. Third, 2009).2. S. S. Saliterman, Fundamentals of bioMEMS and medical microdevices. (WileyInterscience,Bellingham, 2003).3. Nanomaterials. C. S. S. R. Kumar, Ed., Nanotechnologies for the life sciences (WileyVCH, Weinheim, 2006), vol. 1, pp. 366.5. C. S. S. R. Kumar, Nanomaterials for biosensors. C. S. S. R. Kumar, Ed., Nanotechnologies for the life sciences. C. S. S. R. kumar, Ed., Nanotechnologies for the life sciences (WileyVCH, Weinheim, 2005), vol. 3.7. J. M. Anderson, Annu Rev of Matl Res, 31, 81 (2001)
1709	ME	MICRO-MECHANICS	ME725	1. Introduction: Defects in continua, their properties and characterization (1 lecture).2. Review of linear elastostatics. Defects. Eigenstrains. Static Green's functions. Superposition principle. (2 lectures)3. Physical interpretation, invariants, principal values. (6 lectures)5. Linear elasticity with defects. Holes, inclusions and dislocations. Interaction of defects. Equivalent macroscopic properties. (6 lectures)6. Thermoelasticity (6 lectures)7. Elastodynamics of defecte
1710	PHY	COMPUTATIONAL SIMULATION METHODS IN PHYSICS	PHY627	FORTTRAN/C programming, structured programming, errors, numerical analysis,differentiation, integration, solution of differential equations, solution ofSchrddinger equation, simulations of planetary motion, molecular dynamics simulation of Ar, density functional theory, CarParrinello simulation, Monte Carlo simulation, simulation of Ising model, quantum MonteCarlo simulation, genetic algorithms.
1711	AE	INTRODUCTION TO PROFESSION	AE100	History of aviation, spaceflight. Aerodynamic shape, generation of forces.Aerodynamics of airfoils. Atmosphere. Performance, stability & control. Structural layout. Power plants. Instruments & navigation. Flight, structural, propulsion and high speed aerodynamiclaboratories together with demonstrative experiments.
1712	IME	OPERATIONS MANAGEMENT	IME642A	Overview of Manufacturing Planning and Control; Forecasting; SmoothingMethods, Time Series Analysis, Decomposition Methods. Autoregressive and BoxJenkins Models. Qualitative Models; Aggregate Measures, Single Machine Models, Flow Shop and Job Shops, DynamicScheduling, Evaluation of Heuristics and Dispatching Rules.
1713	IME	OPERATIONS MANAGEMENT	IME642	Overview of Manufacturing Planning and Control; Forecasting; SmoothingMethods, Time Series Analysis, Decomposition Methods. Autoregressive and BoxJenkins Models. Qualitative Models; Aggregate Measures, Single Machine Models, Flow Shop and Job Shops, DynamicScheduling, Evaluation of Heuristics and Dispatching Rules.
1714	IME	DESIGN OF PRODUCTION SYSTEMS	IME641A	Production systems : concepts and integrated view, Policy Decisions, Capacityplanning, Product development, Plant location, Plant layout, Materials handling,Assembly line balancing, Work design, Method of layout
1715	IME	MANAGEMENT DECISION ANALYSIS	IME634A	Multiobjective decisions, Decisions under uncertainty, Statistical Decision Trees,Applications from Quality Control and Production Control.
1716	IME	MANAGEMENT DECISION ANALYSIS	IME634	Multiobjective decisions, Decisions under uncertainty, Statistical Decision Trees,Applications from Quality Control and Production Control.
1717	IME	INTRODUCTION TO STOCHASTIC PROCESSES AND THEIR APPLICATIONS	IME625A	a) Introduction to stochastic process, Random walks, Markov chains, Markovprocesses, poisson processb) Application of Stochastic processes in (i) Queueing Theory, (ii) Scheduling,(iii) Manufacturing (ii) Queueing Theory, (iii) Scheduling, (iv) Manufacturing
1718	IME	INTRODUCTION TO STOCHASTIC PROCESSES AND THEIR APPLICATIONS	IME625	a) Introduction to stochastic process, Random walks, Markov chains, Markovprocesses, poisson processb) Application of Stochastic processes in (i) Queueing Theory, (ii) Scheduling,(iii) Manufacturing (ii) Queueing Theory, (iii) Scheduling, (iv) Manufacturing
1719	IME	COMPUTER AIDED DECISION	IME624A	System Analysis: Information System Analysis and Design, Decision SupportSystem, Database Management Systems, Query Languages, Emerging Areas likecommunication network distributed systems

S.No.	Branch	Course Title	Courses ID	Content
		SYSTEMS		Structures, Statistical Considerations, Validation, Simulation Languages,Applications.
1720	IME	COMPUTER AIDED DECISION SYSTEMS	IME624	System Analysis: Information System Analysis and Design, Decision SupportSystem, Database Management Systems, Query Languages, Emerging Areas likecommunication network distributed systems Structures, Statistical Considerations, Validation, Simulation Languages,Applications.
1721	IME	FINANCIAL ENGINEERING	IME611A	Module 1 : Basic Elements of Financial Systems and Financial Management Fundamentals of Financial Systems and Domain Knowledge of FinacialManagementModule 2 : Mathematical Background Intr Measures Numerical procedures : Binomial & trinomial trees, Monte Carlo simulation;finite difference methodsModule 3 : Options and Futures Markets Forward and futures contracts : Basic definition, Diff Rate Futures. Swaps : Currency Swaps, Interest Rate Swaps Options : Definitions, Payoff Diagrams, General Arbitrage Relationships,The Binomial Method, Applications to Hedging & Speculating, Delta He Financial Risk Management Introduction : Different types of risk ; approaches to risk management;history of bank regulation. Greek letters : Definitions and how they are used.
1722	HSS	SOCIAL THEORY IN LATE TWENTIETH CENTURY	SOC745	Canons in sociological theory, breaking with modernity, post positivist and poststructural social thought, post modernism in social theory: present images andfuture possibilities, Writings of J Baudrillard
1723	HSS	SOCIAL INEQUALITY AND STRATIFICATION	SOC734	Critical evaluation of concepts and theories of social inequality and stratification;aspects of inequality in contemporary societies; methodology of stratification;stratification in India.
1724	HSS	SOCIOLOGY OF DEVELOPMENT	SOC732	Sociological perspective on development problems with special reference to India,theories/models of development, Modernization Theory (New modernization studies) Dependence Theory (New depende Developmentindicators. The sustainable nature of Development, Development reconsideredvoices of Dissent.
1725	CE	RISK & RELIABILITY IN GEOTECHNICAL ENGINEERING	CE731A	Introduction: Sources and types of uncertainties associated with geotechnical analysis,importance of probabilistic methods and reliabilitybased analysis in geotechnicalengineering Review of probability analysis Fundamentals of reliability analysis: FirstOrder Second Moment (FOSM) method, FirstOrderReliability Method (FORM), SecondOrder Reliability Method (SORM), Monte Carlosimulation Application soilproperties, uncertainty in interpretation techniques Spatial variability of soil properties, scale of fluctuations, estimation of autocorrelationand autocovariance Probabilistic groundwater modeling, flow design of shallow anddeep foundations, settlement analysis Reliabilitybased liquefaction analysis, lateral spreading Development of fragility curves for geotechnical problems 1. Fenton, G. A and Griffiths, D. V. (2008). Risk Assessment in Geotechnical Engineering,Wiley, New York.2. Phoon, K. K. (Ed.) (2008). ReliabilityBased Design in Geotechnical Engineering:Computations and GeotechnicalEngineering, SpringerWienNewYork.4. Baecher, G. and Christian, J. (2003). Reliability and Statistics in GeotechnicalEngineering, Wiley, New York.
1726	MSE	PRACTICAL TRANSMISSION ELECTRON MICROSCOPY & NANOANALYSIS OF MATERIALS	MSE665A	Specimen preparation for TEM analysis : Metallic selfsupporting samplesElectropolishingCeramics and electronic devices: Crosssectional specimen preparation Ion Milling Soft materials, polymers, biolo and use of Plasma CleanersTEM: Instrumental details and requirements : Physics of Different Electron Sources:Lenses, Ape11ures, and Resolution: Electron Detection Display and Image RecordingPumps and Images; STEM Imaging System: Alignment and Stigmatation Lens Rotation Centers, Correction of Astigmatism in the Imaging Lenses, Calibrating the Imaging System; Magnification Calibration, Camera pattern analysis Mathematical Definition ofthe Reciprocal Lattice; Laue Equations and their Relation to Bragg's Law, Ewald Sphere of Reflection; The Excitation Error Experimental SAD Techniques, In Precession Diffraction, Double Diffraction, Orientation ofthe Specimen, Orientation RelationshipsAmplitude contrast imaging and image interpretations : Image artifacts in TEM Systematic Crystal defect a Defects and Particles Springer, 20092. Sample preparation handbook for TEM: Methodology, J. Ayache et al., Springer, 20103. Sample preparation handbook for TEM: Techniques, J. Ayache et al., Springer, 20104. Aberrationcor materials science, D. Shindo and T. Oikawa, Springer, 20026. Handbook of Microscopy:Applications in materials science, solidstate phsysics and chemistry. EditedBy: S. Amelinckx, D. van Dyck, J. van Lan Kaplan, John Wiley and Sons,2008
1727	MSE	PRACTICAL TRANSMISSION ELECTRON MICROSCOPY & NANOANALYSIS OF MATERIALS	MSE665	Specimen preparation for TEM analysis : Metallic selfsupporting samplesElectropolishingCeramics and electronic devices: Crosssectional specimen preparation Ion Milling Soft materials, polymers, biolo and use of Plasma CleanersTEM: Instrumental details and requirements : Physics of Different Electron Sources:Lenses, Ape11ures, and Resolution: Electron Detection Display and Image RecordingPump and Images; STEM Imaging System: Alignment and Stigmatation Lens Rotation Centers, Correction of Astigmatism in the Imaging Lenses, Calibrating the Imaging System; Magnification Calibration, Camera pattern analysis Mathematical Definition ofthe Reciprocal Lattice; Laue Equations and their Relation to Bragg's Law, Ewald Sphere of Reflection; The Excitation Error Experimental SAD Techniques, In Precession Diffraction, Double Diffraction, Orientation ofthe Specimen, Orientation RelationshipsAmplitude contrast imaging and image interpretations : Image artifacts in TEM Systematic Crystal defect a Defects and Particles 20092. Sample preparation handbook for TEM: Methodology, J. Ayache et al., Springer, 20103. Sample preparation handbook for TEM: Techniques, J. Ayache et al., Springer, 20104. Aberrationcorrected an science, D. Shindo and T. Oikawa, Springer, 20026. Handbook of Microscopy:Applications in materials science, solidstate phsysics and chemistry. EditedBy: S. Amelinckx, D. van Dyck, J. van Landuyt, ad G Wiley and Sons,2008
1728	ME	MECHANICS OF BIOLOGICAL MEMBRANES	ME724	1. Organisation of Animal cells : Structure and function of cell membrane. Roll of fluid lipid bilayers in cell functionality. Discussion on the experimental methods of study membranes.2. A brief review of d determination of the stable equilibrium shapes of red blood cells. Application of the theory to equilibrium shapes of phase separated fluid lipid bilayers.4. A discussion on the adhesion of vecicles and cel mechanical and thermodynamic model. 1. B Albert at zl.; Molecular biology of the cell. Garland Science, NY, 2002 (fourth edition).2. J. N. Israelachvili, Intermolecular and Surface Forces: With Applications to Colloidal and Biological Systems. Aca of Biological Physics Vol. 1, Elsevier, Amsterdam, 1995.4. S. A. Safran, Statistical Thermodynamics of Surfaces, Interfaces, and Membranes, Westview Press, 2003.
1729	ECO	HETEROGENOUS FIRMS AND INTERNATIONAL TRADE	ECO524A	Static representative agent model, Dynamic representative model, Bellmao equation!Pontryagin's Maximum Principle, Asset pricing, Dynamic endowment economy, Overlapping generations model, F 1. Cooley, T. 1995. Frontiers of Business Cycle Research, Princeton University Press.2. Krueger, Dirk, Lecture Notes (available on his website).3. Ljungqvist, L. and Sargent, Thomas J. 2004. Recursive Macr Edition, New York: McGrawHill.5. Williamson, Stephen D., Lectures in Macroeconomics (available on his website).
1730	ECO	MONETARY ECONOMICS	ECO522A	Monetary Policy in the Basic Macro Model, Demand for Money, Targets versus Instruments, Role of Expectations and Monetary Policy, Time versus Discretionary Policy, Credibility of the Central Bank, The I. Monetary Economics by J Handa2. Handbook of Monetary Economics, edited by Benjamin Friedman and F H Hahn, Vols 1&23. Journal articles to be suggested in class.
1731	ECO	ECONOMIC ANALYSIS OF LAW	ECO504A	
1732	ECO	INDUSTRIAL ECONOMICS	ECO411A	Introduction: StructureConductPerformance (SCP) paradigm, Chicago School view, modem developments and relevance of industrial policy; Monopoly Power and Industrial Concentration; Oligopolistic In exit, barriers to entry, predatory pricing and limit pricing; Empirical Studies of SCP Paradigm
1733	ECO	ECONOMETRICS II	ECO342A	Non Linear Regression Models, Non Parametric Regression Models, Measurement Errors, Stochastic Regressor, Multinomial and Conditional Logit Models, Ordered Logit Models, Nested Logit Models, Rare Dynamic Panel Data Models, Time Series Models including ARCH,GARCH and Multivariate GARCH Models, DEA and Stochastic Frontier Models, Total Factor Productivity Estimation, Levinsohn Petrin App Models to Trade, Finance, Microeconomics, Development Economics, Growth, Macroeconomics, Industry, among others.
1734	ECO	ECONOMETRICS I	ECO341A	Classical Linear Regression Model (CLRM), Assumptions of the CLRM and its violation, Hypothesis Testing, Gauss Markov Theorem, Properties of the Estimator, Multiple Regression Model, Heteroscedast Equations, Introduction to Time Series, Discrete Choice, Limited Dependent Variable and Panel Data Models, Estimation ProceduresOrdinary Least Square(OLS), Generalized Least Squares(GLS), IV(Instrum Econometric Software: EvIEWS and Stata
1735	ECO	DEVELOPMENT ECONOMICS	ECO311A	Economic growth versus economic development, indicators of economic development, economic inequality, poverty, population growth and economic development, models of dual economy, land tenancy development.
1736	ECO	MICROECONOMICS II	ECO301A	Static games with complete information Nash Equilibrium; Dynamic games with complete information Subgame perfect Nash Equilibrium; Static games with incomplete information Bayesian Nash Equilib Selection and Moral Hazard; Exchange Economy and definition of Competitive Equilibrium; Pareto efficiency and Pareto set; One consumer one producer economy; Two sector production economy; First

S.No.	Branch	Course Title	Courses ID	Content
				and social welfare functions; Public goods and externalities; Property rights and Coase theorem; Social Choice Theory
1737	ECO	INTRODUCTION TO MATHEMATICAL ECONOMICS	ECO261A	Basic Mathematical Logic; Set Theory; Review of Functions of a Single Variable and Its Application in Economics: Choice, Utility and Aggregation; Functions of Several Variables: Level Curves, Homogeneity
1738	HSS	INTRODUCTION TO PSYCHOLOGY	PSY151B	Psychological Perspectives and Approaches; Perception; Learning; Memory; Higher Cognitive Processes; Motivation and Emotion; Intelligence; Personality; Individual Differences.
1739	ME	NATURE AND PROPERTIES OF MATERIALS	ME222A	History of engineering materials, Engineering materials, Materials property chart, Crystal structure, Imperfections of solids, Mechanism of strengthening in metals, Hall-Petch effect, X-ray diffraction, Fracture
1740	ME	FLUID MECHANICS	ME231A	Reynolds Transport Theorem; Integral form of continuity, momentum and energy; Eulerian and Lagrangian viewpoints; Constitutive relations; Navier-Stokes equations: Exact solutions; Potential flow; Boundary layers; channels; compressible flows.
1741	ME	FLUID MECHANICS	ME231	Reynolds Transport Theorem; Integral form of continuity, momentum and energy; Eulerian and Lagrangian viewpoints; Constitutive relations; Navier-Stokes equations: Exact solutions; Potential flow; Boundary layers; channels; compressible flows.
1742	SDS	STATISTICAL INFERENCE	MTH755	Population and samples; Parametric and nonparametric models; Exponential and location-scale families; Sufficiency and minimal sufficiency; Complete statistics; Unbiased and UMVU estimation; Asymptotic theory of maximum likelihood; Asymptotically efficient estimation; Variance estimation; The jackknife; The bootstrap; The NP lemma; MLR; UMP tests for one and two-sided hypotheses; Unbiased and similar; U-tests; Bayes tests; Pivotal quantities; Inverting acceptance regions of tests; The Bayesian confidence interval; Prediction sets; Length of confidence intervals; UMA and UMAU confidence sets; Invariant confidence
1743	MTH	ALGEBRA	MTH751A	Groups, Basic properties, Isomorphism theorems, Permutation groups, Sylow Theorems, Structure theorem for finite abelian groups, Rings, Integral domains, Fields, division rings, Ideals, Maximal ideals, Euclidean domain and Chinese Remainder theorem, Field Extension, Cramer's rule, Algebraic elements and extensions, Finite fields. Determinants and their properties, Systems of linear equations, Eigenvalues and eigenvectors, Linear transformations, Inner product spaces.
1744	ECO	INDIAN ECONOMIC PROBLEMS	ECO413A	India in a Global Perspective (broad comparison with developed, developing and emerging economies), Trends in Aggregate Economic Activities (trends in Indian national income and related variables, price index methods etc), Industry (industrial policy resolutions, growth, industrial bottlenecks, structural reforms etc), and Infrastructure (education, poverty alleviation programmes, energy, transport etc)
1745	ECO	INTERNATIONAL ECONOMICS AND FINANCE	ECO412A	World Trade and Empirical Facts of International Trade with its Explanations, Neoclassical Trade Theories, Alternative and New Trade Theories, Trade Policy Instruments and their Impact on Welfare, Trade Movements, Past and Present Arrangements of the International Monetary System including discussion on the European Monetary System and Optimum Currency Area, Regional Trade Agreements, Trade and BOP.
1746	ECO	MACROECONOMICS-II	ECO321A	Microeconomic foundations of Macroeconomics, Representative agent model, Application of representative agent model in asset pricing, consumption theories etc., Long-run economic growth models, Monetary cycle models.
1747	PHY	MAGNETISM IN MATERIALS	PHY624	Magnetism in atoms and ions; crystal field; diamagnetism and paramagnetism, ferro and antiferromagnetism; complex orders; experimental techniques; molecular fields and exchange interaction; direct interaction
1748	PHY	CONDENSED MATTER II	PHY622	Fermi liquid, second quantization, interaction picture, electron-electron interaction; plasmons; electron-phonon interactions; polarons, advanced methods of band structure calculations. Cooperative phenomena. Ginzburg-Landau theories.
1749	PHY	ELECTRONIC STRUCTURE OF MATERIALS	PHY621	One electron model, Born-Oppenheimer approximation, Hartree & Hartree-Fock approximation, density functional theory, local density approximation, beyond LDA. electrons in periodic solids, Bloch's theorem, pseudopotential method, KKR method, LMTO method, the full potential methods. applications to different types of solids; electron in disordered solids, mean-field theories, coherent potential approximation, methods and its applications to clusters and amorphous semiconductors, applications of electronic structure methods to materials design.
1750	PHY	COMPUTATIONAL PHYSICS	PHY617A	Examples of subcellular nanomachines of life; difference between macroscopic and nanomachines: world of nanometer and piconewton, stochastic dynamics at low Reynolds number; experimental, computational, optical tweezers, and AFM; Langevin and Fokker-Planck equations for Brownian rectifiers; power stroke vs. Brownian ratchet mechanisms for directed movements. mechanochemistry of subcellular machines. cytoskeleton associated nanomachines: intracellular motor transport; nucleotide-based machines DNA/RNA helicase/polymerase, ribosome G-proteins switches and latches; membrane-bound machines, transmembrane proteins, crawling of cells.
1751	PHY	NON-EQUILIBRIUM STATISTICAL MECHANICS	PHY615	Linear response theory, Fokker-Planck and Langevin equations, master equation; nucleation and spinodal decomposition, critical dynamics, Boltzmann equation.
1752	PHY	CLASSICAL ELECTRODYNAMICS II	PHY614A	Path integral method of formulating quantum mechanics and its application to elementary quantum systems, formal scattering theory; Lippmann-Schwinger formulation, scattering of particles with spin, stationary scattering
1753	PHY	ADVANCED STATISTICAL MECHANICS	PHY613	Equilibrium statistical mechanics, phase transitions, critical phenomena, superfluidity, superconductivity, nonequilibrium statistical mechanics, Langevin equations, Fokker-Planck equations, ergodic hypothesis
1754	PHY	ATOMIC, MOLECULAR & OPTICAL PHYSICS (AMO)	PHY612A	Elements of finite groups. representation theory. applications to physical systems: crystal symmetries. continuous groups. Lie algebras and their elementary applications. global properties of groups.
1755	PHY	INTRODUCTION TO NUCLEAR AND PARTICLE PHYSICS	PHY611A	Second quantization; interaction picture; S-matrix; diagrammatic methods; many-particle Green's functions; basic techniques in many-body physics; additional topics (at the discretion of the Instructor).
1756	PHY	REVIEW OF QUANTUM PHYSICS II	PHY604	Problem-oriented survey of statistical mechanics, deuteron problem, nuclear scattering, alpha and beta decay, elementary particle phenomenology, crystal structure, symmetry, periodic potential, bands, metals
1757	PHY	REVIEW OF QUANTUM PHYSICS I	PHY602	Problem-oriented review of basic quantum mechanics: Schrödinger equation, simple potential problems, quantum dynamics, angular momentum, perturbation theory, scattering, applications to atoms and molecules
1758	PHY	THEORETICAL PROJECT II	PHY571	Advanced research oriented theoretical study in continuation of project work undertaken in PHY 570, or study of another research oriented topic in Theoretical Physics with an aim to bring the student in contact with a concrete research area of current interest. Statement of the problem, relevance and context, steps involved, tools employed, proposed work plan and results obtained.
1759	PHY	THEORETICAL PROJECT I	PHY570	Study of a research oriented topic in Theoretical Physics with an aim to bring the student in contact with a concrete research area of current interest. Solving a small problem in this area is required, detailing the results obtained.
1760	PHY	M.SC. PROJECT IV	PHY568	Experimental project in a research laboratory: 4. Preparation of report and interpretation of results.
1761	PHY	M.SC. PROJECT III	PHY566	Experimental project in a research laboratory: 3. Data acquisition and analysis.
1762	PHY	M.SC. PROJECT II	PHY565	Experimental project in a research laboratory: 2. Development and testing of experimental setup.

S.No.	Branch	Course Title	Courses ID	Content
1763	PHY	M.SC. PROJECT I	PHY563	Experimental project in a research laboratory: 1. Literature survey andpreparation for the project.
1764	PHY	CLASSICAL ELECTRODYNAMICS II	PHY553B	Special relativity, Minkowski space and four vectors, concept of fourvelocity,four acceleration and higher rank tensors, relativistic formulation ofelectrodynamics, Maxwell equations in covariant form, gauge potentials, radiation from an accelerated charge,Larmor formula, bremsstrahlung and synchrotron radiation, multipole radiation,dispersion theory, radiative reaction, radiative damping, scattering byfree ch
1765	PHY	CLASSICAL ELECTRODYNAMICS II	PHY553	Special relativity, Minkowski space and four vectors, concept of fourvelocity,four acceleration and higher rank tensors, relativistic formulation ofelectrodynamics, Maxwell equations in covariant form, gauge potentials, radiation from an accelerated charge,Larmor formula, bremsstrahlung and synchrotron radiation, multipole radiation,dispersion theory, radiative reaction, radiative damping, scattering byfree ch
1766	ME	MICROMECHANICS	ME725A	1. Introduction: Defects in continua, their properties and characterization (1 lecture).2. Review of linear elastostatics. Defects. Eigenstrains. Static Green's functions. Superposition principle. (2 lectures)3. Physical interpretation, invariants, principal values. (6 lectures)5. Linear elasticity with defects. Holes, inclusions and dislocations. Interaction of defects. Equivalent macroscopic properties. (6 lectures)6. Linear Thermoelasticity (6 lectures)7. Elastodynamics of defecte
1767	MSE	ELECTRONIC DEVICES AND CHARACTERIZATION	MSE628A	Electronic Device related characteristics of a semiconductor material Review: n and p type semiconductors, wafers, carrier mobility, conductivity, equilibrium carrier statistics, generationrecombination processes, Photo Spectroscopy (SIMS) {0.75}{b} Resistivity: Fourpoint probe {0.75} {c} Charge carrier type, density, mobility: Hall effect {1}{d} Bandgap: UVVisible spectroscopy {0.75} {e} Absorption coefficient {0.5}2 Semiconductor Junctions: {a} Schottky and Ohmic contacts, thermionicemission, Tunnelling, Schottky diodes {2} {b} Contact resistance: Two terminal; Four terminal technique {0. 5} 5 {c} Barrier height: From 1V, CV, Photocapacitance profiling (1) {e} Bandoffset for a semiconductor semiconductor junction using CV technique {0.5} 4 Metalinsulatorsemiconductor junctions : {a} MOS capacitor, quantitative analysis of a flat band device, C trapped and interfacetrapped charges {1} {d} MOSFET {5} {e} FinFET MOSFET: Architecture for creating simple Boolean logic, memory Dopant density profiling using CV (already discussed in section 3 above)3 Semiconductor Material and Device Characterization, Dieter K. Schroder, January 2006,WileyIEEE Press(2) MetalSemiconductor Contacts (Electrical & Electronic Engineering Monographs), E. H.Rhoderick and R. H. Williams, Winfried Monch (Nov 30,I990)Springer; I edition (November 30, 1990)(4) Optical Techniques for SolidState Materials Characterization, Rohit P. Prasankumar and Antoinette J. Taylor, CRC Press; I edition (June 1, 1996)2nd edition(April 12, I 996)(6) Advanced Semiconductor Fundamentals (2nd Edition) by Robert F. Pierret, PrenticeHall; 2 edition (August 19, 2002)(7) Semiconductor Devices: Physics and Technology, Simon M. Sze, MingK wei Lee,Wiley
1768	MSE	ELECTRONIC DEVICES AND CHARACTERIZATION	MSE628	Electronic Device related characteristics of a semiconductor material Review: n and p type semiconductors, wafers, carrier mobility, conductivity, equilibrium carrier statistics, generationrecombination processes, Photo Spectroscopy (SIMS) {0.75}{b} Resistivity: Fourpoint probe {0.75} {c} Charge carrier type, density, mobility: Hall effect {1}{d} Bandgap: UVVisible spectroscopy {0.75} {e} Absorption coefficient {0.5}2 Semiconductor Junctions: {a} Schottky and Ohmic contacts, thermionicemission, Tunnelling, Schottky diodes {2} {b} Contact resistance: Two terminal; Four terminal technique {0. 5} 5 {c} Barrier height: From 1V, CV, Photocapacitance profiling (1) {e} Bandoffset for a semiconductor semiconductor junction using CV technique {0.5} 4 Metalinsulatorsemiconductor junctions : {a} MOS capacitor, quantitative analysis of a flat band device, C trapped and interfacetrapped charges {1} {d} MOSFET {5} {e} FinFET MOSFET: Architecture for creating simple Boolean logic, memory Dopant density profiling using CV (already discussed in section 3 above)3 Semiconductor Material and Device Characterization, Dieter K. Schroder, January 2006,WileyIEEE Press(2) MetalSemiconductor Contacts (Electrical & Electronic Engineering Monographs), E. H.Rhoderick and R. H. Williams, Winfried Monch (Nov 30,I990)Springer; I edition (November 30, 1990)(4) Optical Techniques for SolidState Materials Characterization, Rohit P. Prasankumar and Antoinette J. Taylor, CRC Press; I edition (June 1, 1996)2nd edition(April 12, I 996)(6) Advanced Semiconductor Fundamentals (2nd Edition) by Robert F. Pierret, PrenticeHall; 2 edition (August 19, 2002)(7) Semiconductor Devices: Physics and Technology, Simon M. Sze, MingK wei Lee,Wiley
1769	CHM	INORGANIC CHEMSTRY II	CHM442	Coordination Chemistry: Bonding, Spectra, Magnetism, Structure and ReactionMechanism, Supramolecular Chemistry, Molecular Magnetism, OrganometallicChemistry, Inorganic Chemistry of Biological s
1770	CHM	PHYSICAL CHEMISTRY II	CHM422	Thermodynamics, Statistical Thermodynamics, Chemical kinetics.
1771	CHE	FUNDM. OF COLLOID & INTERFACE SCI. & TECH.	CHE688A	Capillarity, interfacial thermodynamics, surfactants, stability of muliphasesytems, foam, emulsion, multiphase reactors, wetting and adhesion, catalystsintering/redispersion; Stability and coagulation of f
1772	CHE	BIOCHEMICAL ENGINEERING	CHE362A	Cell Structure and Cell Types, Chemicals of Life (RNA, DNA, enzymes etc.),Kinetics of Enzyme Reactions, Applied Enzyme Catalysis, Metabolic Stoichiometricand Energetics, Molecular Genetics and Contr
1773	CE	INSTRUMENTATION LABORATORY AND FIELD PRACTICES IN GEOINFORMATICS	CE673	Use of automatic and digital levels, electronic theodolites, total stations, planetabling; Control surveys using GPS, Total station and triangulation methods(adjustment and computations of coordinates); Ca
1774	CE	INTRODUCTION TO REMOTE SENSING	CE671	Remote sensing system; Physics of remote sensing, EMR characteristics andinteraction in atmosphere and with ground objects; Sensor types characteristics:types of resolution, FOV, IFOV, PSF; RS satellite applications; Georeferencing, resampling methods; Atmospheric errors andremoval; Satellite orbits and characteristics; Applications of optical and microwaveremote sensing techniques in Civil Engineering.
1775	CE	GEOTECHNICAL INVESTIGATIONS FOR CIVIL ENGINEERING PROJECTS	CE638N	Subsurface exploration planning, drilling and sampling techniques, field andlaboratory tests, instrumentation and monitoring of field data, report preparation.
1776	CE	STABILITY OF STRUCTURES	CE622N	Criteria for design of structures: stability, strength, and stiffness; Classicalconcept of stability; Stability of discrete systems: linear and nonlinear behaviour;Stability of continuous systems: stability of columns; Buckling of members: member buckling versus global buckling, slenderness ratio of framemembers; Stability of beams: lateraltorsion buckling; Stability of plates: axialflexural buckling, shear flexural buckling, buckling under co
1777	CE	ENGINEERING MECHANICS	CE621N	Stress analysis: forces and moments, theory of stress, principal stresses andstress invariants, compatibility equations, equilibrium equations; Strain:deformation and velocity gradients, Lagrangian and Eulerian conditions; Fundamental physical principles: conservation of mass,linear momentum, angular momentum, and energy, second law of thermodynamics;Constitutive theory: St. Venants principal, linear elastic, viscoelasticity, yield criteria; Applications: Airy stress function,twodimensional elastostatics problems, torsion.
1778	CE	ADVANCED HYDRAULICS	CE611	Basics: dimensional analysis, equations of continuity, motion, and energy,irrotational flow, drag and lift of immersed bodies; Pipe flow: laminar flow,turbulent flow, boundary layer theory, wall turbulent shear stress,hydraulic jump, rapidly varied flow, spatially varied flow,unsteady flow.
1779	CE	ADVANCED MATHEMATICS FOR CIVIL ENGINEERS	CE602N	Linear differential equations; Fourier integrals and transforms; Partial differentialequations; Numerical methods in general; Numerical methods for differentialequations; Linear algebra; Numerical method
1780	AE	FUNDAMENTAL OF AEROSPACE PROPULSION - I	AE650A	Introduction to propulsion, conservation equations, basic thermodynamics,dynamics and thermodynamics of 1 D flows, 1 D isentropic flows, normal andoblique shocks, compressible flows, Rayleigh flow, rocket propulsion, thrust equation, solid rockets, liquid rockets,hybrid rockets, gas turbine cycles.
1781	AE	MOLECULAR GAS DYNAMICS	AE747A	The molecular model; binary elastic collisions: basic kinetic theory; referencestates & boundary conditions; collisionless flow; transition regime flows. Directsimulation Monte Carlo method. One dimension
1782	AE	THEORY OF VIBRATIONS	AE678A	Introd. Hamiltons principles, Lagranges eqn, Eigenvalue problem (EVP), discrete& continuous system. Boundary value problem formulation. General EVP, positivedefinite system, self adjoint property. Vibrations of continuous vibration, approximate methods. Response to excitation.
1783	AE	AEROELASTICITY	AE676A	Influence coefficients and function. Formulation of static and dynamic aeroelasticequations. Static aeroelasticity; divergence, aileron reversal & controleffectiveness, solutions by matrix and energy methods. Dynamic aeroelasticity,flutter calculation, panel flutter.
1784	PHY	HIGH ENERGY PHYSICS II	PHY781A	Current topics in Particle Physics and quantum field theory.

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S.No.	Branch	Course Title	Courses ID	Content
1827	CSE	THEORY OF COMPUTATION	CS340A	Scope and motivation for theory of computation; informal introduction tocomputability and complexity; set membership problem as idealization ofcomputing problems; alphabets, strings, languages, automata, regular languages, equivalence with finite automata; pumping lemma; someapplications of FAs (e.g., text pattern matching); decision properties of regularlanguages; MyhillNerode theorem; minimization of DFAs, Grammars and normal forms of CFGs, pumping lemma for contextfreelanguages; decision and closure properties; some applications (e.g., YACC,markup languages, XML and document type definition, etc.), Why consider LR model; universal TM, Recursive and r.e.languages; notion of undecidability; undecidability of halting problem; reducibility;other undecidable problems; Rices theorem; separation of r.e. and recursivelanguages; Cook-Levin theorem, automated theorem proving,Motivation for examining feasibility/infeasibility distinction; definition of timeand space complexity classes; P and NP, and their importance; polynomialtime reducibility; definiteness of complexity classes; notion of randomized algorithms; probabilistic TMs; classes RP, BPP, and ZPP; relationshipsto P and NP; proof of inclusion of BPP in P/poly.
1828	PHY	RESEARCH	PHY799	To be registered by Ph.D. students from Semester I itself, and by M.ScPh.D.(DualDegree) students from Semester V onwards.
1829	PHY	CLASSICAL MECHANICS	SE314A	Review of the Newtonian mechanics; Lagrangian Mechanics, Generalizedcoordinates, constraints, Principle of virtual work, Lagranges equation, Calculusof variations; Central forces, Collisions, Scattering of particles, phase trajectories, canonical transformations, Poisson brackets, HamiltonJacobi theory; Rigid body dynamics; Nonlinear Dynamics.
1830	PHY	CLASSICAL MECHANICS	SE314	Review of the Newtonian mechanics; Lagrangian Mechanics, Generalizedcoordinates, constraints, Principle of virtual work, Lagranges equation, Calculusof variations; Central forces, Collisions, Scattering of particles, phase trajectories, canonical transformations, Poisson brackets, HamiltonJacobi theory; Rigid body dynamics; Nonlinear Dynamics.
1831	PHY	SPECIAL & GENERAL RELATIVITY	PHY407A	Special Relativity: empirical evidence for the constancy of c, frames of reference,Lorentz transformations; relativity of simultaneity; twin and other paradoxes,transformation laws for velocity, momentum, energy, mass, etc.; relativistic Doppler effect, form, representations of the Lorentz group and SL(2,C).Introduction to General Relativity: principle of equivalence; Machs principle,Riemannian geometry; Christoffel symbols, the curvature and stressenergy tensor; tests, basic cosmology,FRW metric; cosmological expansion; cosmic microwave background; heliumabundance; anisotropies in the CMBR.
1832	PHY	SPECIAL & GENERAL RELATIVITY	PHY407	Special Relativity: empirical evidence for the constancy of c, frames of reference,Lorentz transformations; relativity of simultaneity; twin and other paradoxes,transformation laws for velocity, momentum, energy, mass, etc.; relativistic Doppler effect, form, representations of the Lorentz group and SL(2,C).Introduction to General Relativity: principle of equivalence; Machs principle,Riemannian geometry; Christoffel symbols, the curvature and stressenergy tensor; tests, basic cosmology,FRW metric; cosmological expansion; cosmic microwave background; heliumabundance; anisotropies in the CMBR.
1833	PHY	INTRODUCTION TO THE DEPARTMENT	PHY400	The course will expose the students to research areas being pursued in theDepartment, and issues relevant to research as a profession. Faculty membersfrom different subdisciplines would deliver lectures and seminars to be given.
1834	PHY	MODERN PHYSICS LABORATORY	PHY315A	Modern experimental techniques with a view to demonstrate the basic conceptsin physics through experiments. this course has three components: a) onelecture per week: observation, measurements, quantitative analysis, etc. b) the discovery of electron, quantization of charge, e/m ratios,Millikans oil drop experiment, SternGerlach experiment, Rutherford scattering,Davisson Germer experiment, discovering atomic nature through electron diffraction, femtoseconds to light years. b) laboratory work(twice a week): a current list of experiments is available with the Department.c) small project/open ended experiments: These experiments will be chosenby the students and using central facilities.
1835	PHY	PHYSICS OF NON-EQUILIBRIUM PHENOMENA	SE305A	INTRODUCTION: Examples of nonequilibrium phenomena(i) Glass transition; (ii)Nucleation; (iii) Phase separation; Experimental probes: Dynamic scattering;inelastic neutron scattering, THEORETICAL TOPOLOGY: (i) random walks,translational Brownian motion; (b) FokkerPlanck equationdiffusion and drift; Illustration with (i) translational Brownian motion, (ii) rotational Brownianmotion. Master equationloss and gain of probabilities; crossingapplications in (i) chemical reactions (ii) rockmagnetism. Enhancing signals with the help of noise applications ofstochastic resonance in (a) nonlinear optics, (b) solid state devices, (c) neuroscience extensionsapplications in (a)condensation and (b) crystallization. UNSTABLE STATES: AllenCahn scenario ofinterfacial dynamics and domain growthapplications to domain growth inquenched magnets; polymer mixtures.Theory of phase separation controlled by topological defects application to liquidcrystals. Theory of coarsening of Cellular Patterns applications to soap froths(e.g., shaving foams). NON-EQUILIBRIUM PHENOMENA: Traffic; Drivesurfaces applications in molecular beam epitaxy (MBE).
1836	PHY	PHYSICS OF NON-EQUILIBRIUM PHENOMENA	SE305	INTRODUCTION: Examples of nonequilibrium phenomena(i) Glass transition; (ii)Nucleation; (iii) Phase separation; Experimental probes: Dynamic scattering;inelastic neutron scattering, THEORETICAL TOPOLOGY: (i) random walks,translational Brownian motion; (b) FokkerPlanck equationdiffusion and drift; Illustration with (i) translational Brownian motion, (ii) rotational Brownianmotion. Master equationloss and gain of probabilities; crossingapplications in (i) chemical reactions (ii) rockmagnetism. Enhancing signals with the help of noise applications ofstochastic resonance in (a) nonlinear optics, (b) solid state devices, (c) neuroscience extensionsapplications in (a)condensation and (b) crystallization. UNSTABLE STATES: AllenCahn scenario ofinterfacial dynamics and domain growthapplications to domain growth inquenched magnets; polymer mixtures.Theory of phase separation controlled by topological defects application to liquidcrystals. Theory of coarsening of Cellular Patterns applications to soap froths(e.g., shaving foams). NON-EQUILIBRIUM PHENOMENA: Traffic; Drivesurfaces applications in molecular beam epitaxy (MBE).
1837	PHY	QUANTUM FIELD THEORY	PHY681A	Lorentz and Poincar groups; relativistic wave equations; Lagrangian formalismfor fields; symmetry transformations and Noethers theorem; quantizationof fields; divergences and renormalization; YangMills theory and fundamental interactions.
1838	PHY	QUANTUM FIELD THEORY	PHY681	Lorentz and Poincar groups; relativistic wave equations; Lagrangian formalismfor fields; symmetry transformations and Noethers theorem; quantizationof fields; divergences and renormalization; YangMills theory and fundamental interactions.
1839	PHY	REVIEW OF CLASSICAL PHYSICS II	PHY603	Problemoriented review of electromagnetism, optics and thermodynamics:electric fields, potentials, Gausss law, dielectrics, magnetic fields, Ampreslaw, Faradays law, Maxwells equations, electromagnetism and optics.
1840	PHY	CONDENSED MATTER PHYSICS I	PHY543A	Free electron theory; heat capacity; transport properties; Hall effect;elementary concepts of quantum Hall effect, quantization of conductance in a metallic nanowireStructure and scattering; crystalline solids; tightbinding model; application to graphene and nanotubes,semiclassical dynamics; notion of an electron in a DCelectric field; effective mass, holes, crystal binding; types of solids; van der Waals sinteractions; harmonicapproximations, vibrations of mono and diatomic lattices, lattice heatcapacity, Einstein and Debye modelsSemiconductors; intrinsic & extrinsic semiconductors, laws of mass action, electron & hole transport, semiconductor equation, BCS theory, GinzburgLandau theory, flux quantization,Magnetism: exchange interaction, diamagnetism, paramagnetism,ferromagnetism & antiferromagnetism, Hund's rules, Pauli paramagnetism, etc. 1. Introduction to Solid State Physics by C. Kittel2. Solid State Physics by N. W. Ashcroft and N. D. Mermin3. Solid State Physics by H Ibach and Hans Luth
1841	PHY	EXPERIMENTAL PHYSICS II	PHY462	Experiments in General Physics, Optics, Nuclear Physics and Condensed MatterPhysics (List of current experiments available with the Department in the formof a manual).
1842	PHY	ELECTRONICS	PHY441A	Review of network theorems and network analysisOperation Amplifier and negative feedbackOpAmp limitations and applicationsCircuits with OpAmps and diodesInterfacing BJT with OpAmpsMOSFETsDigital logic design and programming with microcontroller applications "The art of electronics" by P. Horowitz and W. Hill"Student Manual for The Art of Electronics" by T. C. Hayes and P. Horowitz"OpAmps and linear integrated circuits" by R.A. Gray"The art of electronics" by P. Horowitz and W. Hill"Student Manual for The Art of Electronics" by T. C. Hayes and P. Horowitz"OpAmps and linear integrated circuits" by R.A. Gray"The 8051 Microcontroller: architecture, programming and applications" by K. J. Ayala
1843	PHY	MATHEMATICAL METHODS I	PHY421A	Vector analysis; curvilinear coordinates; matrices and vector spaces, tensors,function spaces; Hilbert spaces; orthogonal expansions; operators ininfinite dimensional spaces, Fourier series and Fourier transforms, Lie algebras, applications.
1844	PHY	MATHEMATICAL METHODS I	PHY421	Vector analysis; curvilinear coordinates; matrices and vector spaces, tensors,function spaces; Hilbert spaces; orthogonal expansions; operators ininfinite dimensional spaces, Fourier series and Fourier transforms, Lie algebras, applications.
1845	CE	EARTHQUAKE ANALYSIS AND DESIGN OF STRUCTURES	CE629A	Characteristics of earthquakes; Earthquake response of structures; Concept ofearthquake resistant design; Code provisions of design of buildings; Design ofliquid storage tanks; Liquefaction; Nonengineered structures.
1846	BSBE	HUMAN MOLECULAR GENETICS	BSE636A	Simple Mendelian traits; Loss of function mutations; Gain of function mutations;Gene interactions; Dynamic mutations; Genetics of neoplasia; Genomic imprintingand human disease; X inactivation and DNA methylation; Genetic disorders;Pharmacogenetics and biochemical genetics; Animal models in human genetics:Methods used for diagnosis and detection of gene mutations; Gene Therapy.

S.No.	Branch	Course Title	Courses ID	Content
1847	HSS	METHODOLOGY: ART CRITICISM & APPRECIATION	ART701A	To familiarize students with variety of methodological approach in art criticismand analysis The course proposes to develop visual skills and analytical skillsin writing about various forms of art in using c media. The methodology of criticism is base on some of the followingauthors John Dewey (Criticism and Perception), Barkan, Fidmans, Jack Hobbs,and Solomns Phenomenological Model of Criticism, and 2dimension and 3dimension visual art.
1848	HSS	METHODOLOGY: ART CRITICISM & APPRECIATION	ART701	To familiarize students with variety of methodological approach in art criticismand analysis The course proposes to develop visual skills and analytical skillsin writing about various forms of art in using c media. The methodology of criticism is base on some of the followingauthors John Dewey (Criticism and Perception), Barkan, Fidmans, Jack Hobbs,and Solomns Phenomenological Model of Criticism, and 2dimension and 3dimension visual art.
1849	HSS	ARCHITECTURE AND ENVIRONMENTAL DESIGN	ART104A	Manmade Environment and Natural Environment Primary Function of EnclosedSpaceshelter building, material consideration, Introduction to Architecture whatis architecture, difference between building and
1850	AE	INTRO TO VIRTUAL INSTRUMENTATION	AE698A	Introduction to VI, typical applications, functional systems, graphical programming,data flow techniques, advantages of VI techniques. VI programming techniques;VIs and subVIs, loops and charts, arrays, interfacenodes and DLL links. Sensors, transducers and signal conditioning; commontransducers for displacement, temperature, load, pressure, flow etc. Singleended, floating and differential inputs, group interrupts, DMA, operating system, PCI buses. Bus based instrumentation;instrumentation buses, GPIB, RS232C.
1851	AE	HIGH TEMPERATURE GAS DYNAMICS	AE690A	Nature of high temperature flows, perfect and real gas, Gibbs free energy andentropy production, microscopic description of gases, thermodynamic properties,equilibrium properties kinetic theory, inviscid
1852	AE	AEROSPACE STRUCTURAL ANALYSIS II	AE687A	General loads on aircraft, load factor, Vn diagram, effect of gust loading. Energyprinciples, potential and complementary potential; deflection analysis,indeterminate structures. Analysis of plates, Kirchhoff
1853	AE	AIRCRAFT MATERIALS AND PROCESSES	AE684A	Definition of various terms used for classification of materials. Mechanicalproperties. Testing of aircraft materials. Classification of alloys of aluminium,steel, titanium etc. High temperature problems; aer
1854	AE	COMPOSITE MATERIALS	AE681A	Introduction, Definition, classification, behaviors of unidirectional composites:prediction of strength, stiffness, factors influencing strength & stiffness, failuremodes, analysis of lamina; constitutive classifi
1855	AE	INTRODUCTION TO FINITE ELEMENT METHODS	AE675A	Discussion on mathematical models, reliability of computer aided engineeringanalysis. Model problem of linear elastostatics in onedimension, principle ofminimum potential energy, beam bending problem of twodimensional code. Convergence analysis in two dimensions.Characterizational of solution smoothness, rate of convergence in energy norm,a posteriori error estimation. Direct computation of stres
1856	AE	SOLID MECHANICS	AE672A	Introduction. Material behaviour of idealized bodies, mathematical preliminaries,tensor analysis, partial derivatives, etc. Analysis of stress and strain measures.Laws of conservation, eqn. of motion, conse
1857	AE	FLIGHT STABILITY AND CONTROL	AE648A	Linearized equations of aircraft motion for small perturbations in stability axes.Stability analysis of linearized equations of motion. Airplane longitudinal motion.Airplane lateral motion. Airplane handling q
1858	AE	SPACE DYNAMICS-I	AE641A	Introduction, performance of single and multistage rockets, central force motion,twobody problem, ballistic trajectories, trajectory transfer, rendezvous andinterception, Eulers eqns, satellite attitude dynam
1859	AE	TURBULENCE	AE621A	Origin, examples & character of turb, Reynolds stress, energy relations, closureproblem, phenomenology, eddy viscosity. Staistics. spectra, spacetime correlations,macro & microscales, stat. theory of turb
1860	MTH	ANALYSIS	MTH753	Metric spaces, Open and closed sets, Compactness and connectedness,Completeness, Continuous functions (several variables and on metric spaces),uniform continuity C(X), X, compact metric space, Uniform Lebesgue Integration, Lpspaces.Complex Analysis: Analytic functions, Harmonic conjugates, Cauchy theoremsand consequences, Power series, Zeros of analytic functions, Maximum modulustheorem, Schwarz bases, Riesz Lemma,The space L2 as a Hilbert space.
1861	MTH	FOURIER ANALYSIS-I & DISTRIBUTION THEORY	MTH736A	Introduction, Test function spaces, Calculus with distributions, supports ofdistributions, Structure theorems, convolutions, Fourier transforms, L1, L2 theoryof Fourier Transform, Tempered distributions, Poisson equations.
1862	MTH	FOURIER ANALYSIS-I & DISTRIBUTION THEORY	MTH736	Introduction, Test function spaces, Calculus with distributions, supports ofdistributions, Structure theorems, convolutions, Fourier transforms, L1, L2 theoryof Fourier Transform, Tempered distributions, Poisson equations.
1863	MTH	NONLINEAR DYNAMICAL SYSTEMS	MTH658A	Picard's theorem, Boundedness of solutions, Omega limit points of boundedtrajectories.LaSalle's invariance principle; Stability via Lyapunov's indirect method,Converse Lyapunov function, Brokett's theorem,Applications to control system; Stable and unstable manifolds of equilibria, Stablemanifold theorem, HartmanGrobman theorem, Examples and applications,Center manifold theore
1864	MTH	NONLINEAR DYNAMICAL SYSTEMS	MTH658	Picard's theorem, Boundedness of solutions, Omega limit points of boundedtrajectories.LaSalle's invariance principle; Stability via Lyapunov's indirect method,Converse Lyapunov function, Brokett's theorem,Applications to control system; Stable and unstable manifolds of equilibria, Stablemanifold theorem, HartmanGrobman theorem, Examples and applications,Center manifold theore
1865	MTH	ALGEBRAIC TOPOLOGY	MTH649A	Homotopy, Path homotopy. The fundamental group. Covering spaces. Thefundamental group of the circle, S1, sphere, S2, Surfaces 2dimensional,Punctured plane etc. Techniques of calculation. The special fixed point theorem for the disc etc. Triangulations.Simplicial complexes. Barycentric subdivision. Simplicial mappings, The simplicialapproximation theorem. Simplicial homology groups; Calculations for co
1866	MTH	ALGEBRAIC TOPOLOGY	MTH649	Homotopy, Path homotopy. The fundamental group. Covering spaces. Thefundamental group of the circle, S1, sphere, S2, Surfaces 2dimensional,Punctured plane etc. Techniques of calculation. The special fixed point theorem for the disc etc. Triangulations.Simplicial complexes. Barycentric subdivision. Simplicial mappings, The simplicialapproximation theorem. Simplicial homology groups; Calculations for co
1867	MTH	DIFFERENTIAL GEOMETRY	MTH648A	Theory of Space CurvesThe SerretFrenet formulas. Gauss Theory of SurfacesFirst and second fundamental form, Examples, Weingarten map, Principalcurvatures, Gaussian curvature, Examples. Computations Gauss theorem Egregium, Hilberts theorem on the positivity ofcurvature at a point on a compact surface in R3. Geodesics, Equations ofgeodesics, Examples. Jacobi fields, Conjugate points etc. Riemannian Immersions, Embeddings. Metric tensor,Riemannian connection and curvature.
1868	MTH	DIFFERENTIAL GEOMETRY	MTH648	Theory of Space CurvesThe SerretFrenet formulas. Gauss Theory of SurfacesFirst and second fundamental form, Examples, Weingarten map, Principalcurvatures, Gaussian curvature, Examples. Computations Gauss theorem Egregium, Hilberts theorem on the positivity ofcurvature at a point on a compact surface in R3. Geodesics, Equations ofgeodesics, Examples. Jacobi fields, Conjugate points etc. Riemannian Immersions, Embeddings. Metric tensor,Riemannian connection and curvature.
1869	MTH	COMPLEX ANALYTIC DYNAMICS AND FRACTALS	MTH647A	Chordal & spherical metrics, Normal families. Iteration of polynomials andrational functions, Periodic points & orbits, Julia & Fatous sets and theircharacterizations, Dynamics of Julia and Fatous sets for c
1870	MTH	COMPLEX ANALYTIC DYNAMICS AND FRACTALS	MTH647	Chordal & spherical metrics, Normal families. Iteration of polynomials andrational functions, Periodic points & orbits, Julia & Fatous sets and theircharacterizations, Dynamics of Julia and Fatous sets for c
1871	ME	ENERGY SYSTEMS-II	ME401A	Introduction: General Theory and Classification of Turbomachines; Similarity andDimensional Analysis; Twodimensional Cascade Theory; Axial and Radial Flow Machines: Turbines,Compressors and Fans; Pelton, Francis and Kaplan Turbines; Pump and Cavitation. I. S. L. Dixon and C. A. Hall, Fluid Mechanics and Thermodynamics of Turbomachinery, Elsevier,Sixth Edition, 2010.2. H. Cohen, G. F. C. Rogers and H. I. H. Saravanamuttoo, Gas Turbine Theory, Addison W

S.No.	Branch	Course Title	Courses ID	Content
				2003.4. B.Lakshminarayana, Fluid Dynamics and Heat Transfer ofTurbomachinery, John Wiley and Sons,Inc, 1996.
1872	MTH	INTRODUCTION TO COMMUTATIVE ALGEBRA	MTH612A	Commutative rings, ideals, prime and maximal ideals, Noetherian Artinian rings,Primary decomposition and Noetherian rings, Modules over commutative rings,Exact sequences, the Hom and tensorfunct
1873	MTH	INTRODUCTION TO COMMUTATIVE ALGEBRA	MTH612	Commutative rings, ideals, prime and maximal ideals, Noetherian Artinian rings,Primary decomposition and Noetherian rings, Modules over commutative rings,Exact sequences, the Hom and tensorfunct
1874	MTH	ALGEBRA II	MTH611	Fields: definition and examples. Ring of polynomials over a field. Fieldextensions. Algebraic and transcendental elements, Algebraic extensions. Splittingfield of a polynomial. Algebraic closure of a field, U Fundamental theorem of Galois. Solvability by radicals Solutionsof cubic and quartic polynomials, Insolvability of quintic and higher degreepolynomials. Geometric constructions. Cyclotomic extensions. F properties. Free modules, submodules andquotient modules, isomorphism theorems. Localization. Direct sum and directproducts. Noetherian and Artinian rings and modules, structure of Artinian rings,Hi
1875	MTH	FLUID MECHANICS	MTH523A	Review and General Properties of Navier Stokes Equations; Some Exact solutionsof NS equations; Introduction to boundary layer theory; Introduction to turbulentflow; Introduction to compressible flow; A
1876	MTH	TIME SERIES ANALYSIS	MTH517A	Linear stationary processes, AR, MA, ARMA and ARIMA; identification, estimationof the models; forecasting time series regression; Fourier analysis, spectralrepresentation of a stochastic process, proper
1877	MTH	TIME SERIES ANALYSIS	MTH517	Linear stationary processes, AR, MA, ARMA and ARIMA; identification, estimationof the models; forecasting time series regression; Fourier analysis, spectralrepresentation of a stochastic process, proper
1878	SDS	INFERENCE - II	SDS515	Group families, the principle of equivariance, location family, scale family,location scale family. Minimum risk equivariance estimators, risk functions,admissibility, prior distribution, posterior distribution, g their relations. Review of convergence in probability andconvergence in distributions. Consistency results of the mle's, and the mme's.Asymptotic relative efficiency. Consistent and Asymptotic moments and MLE methods in one parameter xponential familyand multiparameter exponential family. Sequential Probability Ratio Tests andits applications in different practical problems. Invariant test a for goodness of fit. Large sample tests and confidenceintervals based on CAN estimators. Consistency of large sample tests andasymptotic powers of large sample tests.
1879	SDS	INFERENCE - II	MTH515	Group families, the principle of equivariance, location family, scale family,location scale family. Minimum risk equivariance estimators, risk functions,admissibility, prior distribution, posterior distribution, g their relations. Review of convergence in probability andconvergence in distributions. Consistency results of the mle's, and the mme's.Asymptotic relative efficiency. Consistent and Asymptotic moments and MLE methods in one parameter xponential familyand multiparameter exponential family. Sequential Probability Ratio Tests andits applications in different practical problems. Invariant test a for goodness of fit. Large sample tests and confidenceintervals based on CAN estimators. Consistency of large sample tests andasymptotic powers of large sample tests.
1880	MTH	MATHEMATICAL METHODS	MTH428A	Multiple Integral Theorems and their Applications: Greens theorem, Stokestheorem and Gauss divergence theorem. Integral Transforms: Fourier, Fouriersine/cosine and Hankel Transforms with their inver Perturbation theory, Regular perturbation theory, Singular perturbationtheory, Asymptotic matching. Calculus of Variation: Introduction, Variationalproblem with functionals containing first order derivatives problem with moving boundaries. Boundaries with constraints.Higher order necessary conditions, Weiretrass function, Legendres and Jacobiscondition. Existence of solutions of variational problems. Ray algebras.
1881	MTH	MATHEMATICAL METHODS	MTH428	Multiple Integral Theorems and their Applications: Greens theorem, Stokestheorem and Gauss divergence theorem. Integral Transforms: Fourier, Fouriersine/cosine and Hankel Transforms with their inver Perturbation theory, Regular perturbation theory, Singular perturbationtheory, Asymptotic matching. Calculus of Variation: Introduction, Variationalproblem with functionals containing first order derivatives problem with moving boundaries. Boundaries with constraints.Higher order necessary conditions, Weiretrass function, Legendres and Jacobiscondition. Existence of solutions of variational problems. Ray algebras.
1882	MTH	SAMPLING THEORY	MTH417A	Principles of sample surveys; Simple, Stratified and unequal probability samplingwith and without replacement; ratio, product and regression method of estimation;systematic sampling; cluster and subsa
1883	MTH	SAMPLING THEORY	MTH417	Principles of sample surveys; Simple, Stratified and unequal probability samplingwith and without replacement; ratio, product and regression method of estimation;systematic sampling; cluster and subsa
1884	MTH	MATRIX THEORY AND LINEAR ESTIMATION	MTH415A	Review of finite dimensional vector spaces (Null space and nullity), Lineardependence and independence, Matrix algebra, Rank of a Matrix, Inverse ofa nonsingular matrix. Hermite canonical forms, Genera Idempotent matrices. Real quadratic forms, reduction of pair of realsymmetric matrices, Singular value decomposition. extrema of a quadraticforms, Vector and matrix differentiation. Least squares theory and more than one hypothesis, ANOVA table,Confidence interval and regions, Power of Ftest. Multiple comparisons andsimultaneous confidence intervals.
1885	MTH	MATRIX THEORY AND LINEAR ESTIMATION	MTH415	Review of finite dimensional vector spaces (Null space and nullity), Lineardependence and independence, Matrix algebra, Rank of a Matrix, Inverse ofa nonsingular matrix. Hermite canonical forms, Genera Idempotent matrices. Real quadratic forms, reduction of pair of realsymmetric matrices, Singular value decomposition. extrema of a quadraticforms, Vector and matrix differentiation. Least squares theory and more than one hypothesis, ANOVA table,Confidence interval and regions, Power of Ftest. Multiple comparisons andsimultaneous confidence intervals.
1886	MTH	COMPUTER PROGRAMMING AND DATA STRUCTURES	MTH409	Fortran 77: Integer and real operations, logic and complex operations, Controlstatements, Do statement, arrays subroutines and functions. Introduction todata structures in C Programming Language; Arra circular linked lists, Application of Linked Lists; Polynomial addition,sparse matrices, Trees: binary trees, redblack trees, Hash tables. somediscussion about data structures in F90F95 with examples.
1887	MTH	FUNCTIONAL ANALYSIS	MTH405A	Fundamentals of normed linear spaces: Normed linear spaces, Riesz lemma,characterization of finite dimensional spaces, Banach spaces. Bounded linearmaps on a normed linear spaces: Examples, line Geometric and extension forms and their applications. Three maintheorems on Banach spaces: Uniform boundedness principle, divergence ofFourier series, closed graph theorem, projection, open mappi convergence, Banach Alaoglu theorem, adjoint of anoperator. Hilbert spaces : Inner product spaces, orthonormal set, GramSchmidtorthonormalization, Bessels inequality, Orthonormal basis, Separable Hi theorem, Riesz representation theorem.Bounded operators on Hilbert spaces: Adjoint, normal, unitary, self adjointoperators, compact operators, eigen values, eigen vectors, Banach algebras.Spectral theo operators.
1888	MTH	FUNCTIONAL ANALYSIS	MTH405	Fundamentals of normed linear spaces: Normed linear spaces, Riesz lemma,characterization of finite dimensional spaces, Banach spaces. Bounded linearmaps on a normed linear spaces: Examples, line Geometric and extension forms and their applications. Three maintheorems on Banach spaces: Uniform boundedness principle, divergence ofFourier series, closed graph theorem, projection, open mappi convergence, Banach Alaoglu theorem, adjoint of anoperator. Hilbert spaces : Inner product spaces, orthonormal set, GramSchmidtorthonormalization, Bessels inequality, Orthonormal basis, Separable Hi theorem, Riesz representation theorem.Bounded operators on Hilbert spaces: Adjoint, normal, unitary, self adjointoperators, compact operators, eigen values, eigen vectors, Banach algebras.Spectral theo operators.
1889	MTH	A FIRST COURSE IN LINEAR ALGEBRA	MTH201	Fields and linear equations. Vector spaces. Linear transformations and projections,Determinants. Elementary canonical forms: diagonalization, triangulation,primary decomposition etc. Secondary decom spaces,Selfadjoint, Unitary and normal operators, Orthogonal projections. Bilinearforms, Symmetric, Skewsymmetric, Positive and semipositive forms etc.
1890	MSP	INTRODUCTION TO NANOMATERIALS AND NANOTECHNOLOGY	MS617	Effects of confinement and finite sizezero, one and two dimensional nanostructures(concepts of surface and interfacial energies), Intermolecular andinterfacial forces in organic, polymeric, biological and mesoscale thermodynamics, Gibbs treatment of interfaces,mesoscale fluid dynamics, thin soft films, mesoscale phenomena in soft matterand applications: adhesion, wetting, nucleation, Nanofabrication cluster fabrication, LangmuirBlodget growth, Nanolithography,Scanning probe lithography, Micro contact printing, Synthesis of nanoparticlesand films: solgel, hydrothermal, freeze drying, intercalation, att of nanocrystlline materials, Carbonnanotubes, short and long term applications and perspectives, Demonstrationof some techniques in preparation and characterization of nanomaterials.
1891	MSP	CHARACTERIZATION OF MATERIALS	MS604	Crystallography, Reciprocal lattice, Diffraction methods, Electron microscopy,Metallography, Thermal analysis, Chemical analysis, Spectroscopic techniques,Laboratory sessions.
1892	MSP	MECHANICAL PROPERTIES OF MATERIALS	MS603	Stress and strain tensors, Elastic constants, Effect of structure on elasticbehaviour, Elastic stress distributions, Viscosity and viscoelasticity in polymers,Yielding criteria, Dislocations and plastic deformat ceramics and composites, Fatigue, Mechanical testing, Strengthand engineering design with brittle solids, Heat treatment, Powder processing.
1893	MSP	STRUCTURAL & MAGNETIC PROPERTIES OF MATERIALS	MS601	Crystal structure, Bonding of atoms, Crystal chemistry, Equilibrium thermodynamics,Phase equilibria, Phase transformations, Dia, Para, Ferro, Ferri and Antiferromagnetism,Magnetic domains, Anisotropy

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1894	MSE	DIFFUSION IN SOLIDS	MSE648A	Diffusion equations and mathematical solutions: Phenomenological diffusiontheories: Atomic theory of diffusion, theoretical and experimental investigationof diffusion phenomena: Diffusion in ionic solids
1895	MSE	DIFFUSION IN SOLIDS	MSE648	Diffusion equations and mathematical solutions: Phenomenological diffusiontheories: Atomic theory of diffusion, theoretical and experimental investigationof diffusion phenomena: Diffusion in ionic solids
1896	MSE	ELECTROCHEMICAL TECHNOLOGY IN MATERIALS PROCESSING	MSE613A	Thermodynamic of electrolyte, electrochemical potential, conduction of ions insolution, overpotential, absorption, phase formation: Economics of an electrolyticprocess, principles of cell design, Electroch electroplating,anodizing, pickling, electrophoretic painting, electrochemical treatment of minerals,batteries and fuel cells, water treatment and environmental protection.
1897	MSE	ELECTROCHEMICAL TECHNOLOGY IN MATERIALS PROCESSING	MSE613	Thermodynamic of electrolyte, electrochemical potential, conduction of ions insolution, overpotential, absorption, phase formation: Economics of an electrolyticprocess, principles of cell design, Electroch electroplating,anodizing, pickling, electrophoretic painting, electrochemical treatment of minerals,batteries and fuel cells, water treatment and environmental protection.
1898	MSE	ELECTRONIC & MAGNETIC PROPERTIES OF MATERIALS	MSE410	DC conductivity of metals, Hall effect and magnetoresistance, AC conductivityof metals, thermal conductivity and specific heat of metals, Thermopower ofmetals; Review of quantum mechanics and free electron model, Fermi energy, Fermi zone, Free electron band diagrams, potential in a crystal, electrondynamics and concept of holes, conductivity in relation to band structure, bandstructures of metals and semiconductors; empirical estimation of carrier concentration in semiconductors; Degenerate and non degeneratesemiconductors, intrinsic and extrinsic semiconductors, determination of dopantlevels and mobility measurements; Ionic conduction review of defect equilibrium, extrinsic defects on conduction, applications insensors and batteries; Dielectric Materials Dielectric constant and polarization,linear dielectric materials, capacitors and insulators, polarization mechanismOptical Materials electronholerecombination, solidstate LEDs, lasers and IR detectors, band gap engineering;Light interaction with materials transparency, translucency and opacity, refractionand refractive index, birefringence, optical activity, optical activity, optical activity, optical spin, permanentmagnetic moment of atoms, diamagnetism, paramagnetism and Pauliparamagnetism,ferro, antiferro and ferri magnetism, Fe, Co and Ni and alloyadditions, ferrites, magnetic hysterisis, soft magnetic materials
1899	PHY	PHYSICS OF NON-EQUILIBRIUM PHENOMENA	PHY311A	Introduction: examples of nonequilibrium phenomena(i) glass transition;(ii) nucleation; (iii) phase separation; experimental probes: dynamicscattering; inelastic neutron scattering, theoretical tools: two al Brownian motion; (b) FokkerPlanck equationdiffusionand drift; illustration with (i) translational Brownian motion, (ii)rotational Brownian motion; master equation loss and gain of probabilities;concept of chemical reactions(ii) rock magnetism. enhancing signals with the help of noise applicationsof stochastic resonance in (a) nonlinear optics, (b) solid state devices, (c)neuroscience, (d) molecular motors and (e) quantum transport (a) condensation and (b) crystallization. unstable states: AllenCahn scenarioof interfacial dynamics and domain growth applications to domain growth inquenched magnets; LifshitzSlyozov arguments for patterns applications to soap froths (e.g., shaving foams); nonequilibriumsteadystates in driven system: drivensystems of interacting particles applications to vehicular traffic; drivensurfaces applications to catalysis
1900	PHY	PHYSICS OF NON-EQUILIBRIUM PHENOMENA	PHY311	Introduction: examples of nonequilibrium phenomena(i) glass transition;(ii) nucleation; (iii) phase separation; experimental probes: dynamicscattering; inelastic neutron scattering, theoretical tools: two al Brownian motion; (b) FokkerPlanck equationdiffusionand drift; illustration with (i) translational Brownian motion, (ii)rotational Brownian motion; master equation loss and gain of probabilities;concept of chemical reactions(ii) rock magnetism. enhancing signals with the help of noise applicationsof stochastic resonance in (a) nonlinear optics, (b) solid state devices, (c)neuroscience, (d) molecular motors and (e) quantum transport (a) condensation and (b) crystallization. unstable states: AllenCahn scenarioof interfacial dynamics and domain growth applications to domain growth inquenched magnets; LifshitzSlyozov arguments for patterns applications to soap froths (e.g., shaving foams); nonequilibriumsteadystates in driven system: drivensystems of interacting particles applications to vehicular traffic; drivensurfaces applications to catalysis
1901	PHY	OPTICS LAB	PHY218	Experiments based on Fresnels equations, study of optical surfaces, Fraunhoferand Fresnel diffraction, interferometers, modulation transfer function,fibre optics, spatial filtering, characteristics of HeNe a
1902	PHY	INTRODUCTION TO PROFESSION	PHY100	Frontiers of physics at various scales, unifying themes and tools ofphysics, significant discoveries, which shaped our current understanding ofthe physical World. physicsinduced technologies.The course covers the CondensedMatter laboratories of the Physics Department.
1903	NET	NUCLEAR MEASUREMENTS LABORATORY	NT621A	Biological effects of radiation; Radiation monitoring; GM Counter characteristics,counting statistics. Scintillation detectors and gamma spectrometry. Multichannelanalysis. Semiconductor detectors for a range of applications. Ratiomeasurements. Neutron diffusion length and age measurements. Experimentsusing Van de Graaff. Radioisotope applications, Computer simulation studies.
1904	NET	NUCLEAR MEASUREMENTS LABORATORY	NT621	Biological effects of radiation; Radiation monitoring; GM Counter characteristics,counting statistics. Scintillation detectors and gamma spectrometry. Multichannelanalysis. Semiconductor detectors for a range of applications. Ratiomeasurements. Neutron diffusion length and age measurements. Experimentsusing Van de Graaff. Radioisotope applications, Computer simulation studies.
1905	ME	SMART MATERIALS AND STRUCTURE	ME771A	Materials for both actuation and sensing: Piezoelectric Materials, MagnetostrictiveMaterials, Materials for actuation: Shape Memory alloys Magnetic shape memorymaterial, Electro/Magneto rheological fluids; state space basedapproach: HellingerReissner Principle, Variational Formulation, Finite ElementsModelling of Vibration of smart Laminates; state space based analysis & designof smart controllers, Concepts of Control Selfsensingactuators; Placement of Smart Actuators/Sensors Vibration damping.
1906	ME	SMART MATERIALS AND STRUCTURE	ME771	Materials for both actuation and sensing: Piezoelectric Materials, MagnetostrictiveMaterials, Materials for actuation: Shape Memory alloys Magnetic shape memorymaterial, Electro/Magneto rheological fluids; state space basedapproach: HellingerReissner Principle, Variational Formulation, Finite ElementsModelling of Vibration of smart Laminates; state space based analysis & designof smart controllers, Concepts of Control Selfsensingactuators; Placement of Smart Actuators/Sensors Vibration damping.
1907	ME	ROBOT MANIPULATORS: DYNAMICS AND CONTROL	ME763	Review of robot manipulators. Manipulator kinematics, dynamics and control.Singularity and workspace analysis. Introduction to manipulator design.
1908	ME	COMPUTER AIDED MANUFACTURING	ME761A	Direct numerical control (DNC) and computer numerical control (CNC): adapativecontrol of manufacturing processes. Manufacturing system concepts. Computerprocesses monitoring and control, offline simulation
1909	ME	COMPUTER AIDED MANUFACTURING	ME761	Direct numerical control (DNC) and computer numerical control (CNC): adapativecontrol of manufacturing processes. Manufacturing system concepts. Computerprocesses monitoring and control, offline simulation
1910	ME	WAVE PROPAGATION IN SOLIDS	ME723A	Waves in infinite and semiinfinite elastic media. Reflection and refraction atplane interface. Dispersion of waves in bounded solids. Waves in rods and plates.Solution of transient problems. Rayleigh wave
1911	ME	WAVE PROPAGATION IN SOLIDS	ME723	Waves in infinite and semiinfinite elastic media. Reflection and refraction atplane interface. Dispersion of waves in bounded solids. Waves in rods and plates.Solution of transient problems. Rayleigh wave
1912	ME	ALTERNATIVE FUELS & ADVANCE IN IC ENGINES	ME690A	Combustion and Fuels, Combustion process in SI and CI engines, Petroleum basedliquid fuels and refining, Liquid alternative Fuels, Advantages, potential,problems associated with utilization, Vegetable oil, Biodiesel, Liquefied petroleum Gas, Dimethyl ether, Hythane, Multifuelengines, Modern developments in IC Engines, EGR, MPFI, GDI, HCCI, Turbochargedengines, Optical Measurement techniques, Fuel atomization and spray characteristics, engines, sources and Nature of various types ofpollutants: Pollution monitoring instruments and techniques, Control measures,emission legislations.
1913	ME	ALTERNATIVE FUELS & ADVANCE IN IC ENGINES	ME690	Combustion and Fuels, Combustion process in SI and CI engines, Petroleum basedliquid fuels and refining, Liquid alternative Fuels, Advantages, potential,problems associated with utilization, Vegetable oil, Biodiesel, Liquefied petroleum Gas, Dimethyl ether, Hythane, Multifuelengines, Modern developments in IC Engines, EGR, MPFI, GDI, HCCI, Turbochargedengines, Optical Measurement techniques, Fuel atomization and spray characteristics, engines, sources and Nature of various types ofpollutants: Pollution monitoring instruments and techniques, Control measures,emission legislations.
1914	ME	APPLIED NUMERICAL METHODS	ME685A	1. Concepts of Algorithms and Programming; Revision of computer languages such as MATLAB, Fortran, and C++. 2. Introduction to Mathematical Modelling. 3. Taylor and Fourier series expansion. 4. Root finding, linear algebraic systems; nonlinear algebraic equations. 8. igenvalues and eigenvectors. 9. Solution of simultaneous nonlinear algebraic systems. 10. Numerical integration, Simpsons rule, Gaussian quadrature convergence and error studies. 13. Linear PDEs by finite differences. Programming projects based on mathematical modelling followed by an application of the numerical methods given above.
1915	ME	APPLIED NUMERICAL METHODS	ME685	1. Numerical Methods for Engineers; Steven C. Chapra and Raymond P. Canale, 7th2. Introduction to Numerical Analysis, S.S. Sastry; Prentice Hall of India, 2012. edition, McGrawHill, 2014.3. Numerical Methods for Engineers, M.L. James, G.M. Smith & J.C. Wolford, Harper Collins College Division; 4thedition, 1993.

S.No.	Branch	Course Title	Courses ID	Content
1916	ME	MACHINING SCIENCE II	ME662	General classification of unconventional machining, chemical machining, electricdischarge machining, Abrasive Jet and Ultrasonic Machining, electron beammachining, laser beam machining, ion beam machining, withmodifications.
1917	MTH	MS PROJECT II	MTH698A	Fundamentals of parallel computing; Parallel techniques and algorithms; Parallelalgorithms for linear algebric equations; Design of parallel algorithms for eigenvalue problem; Parallel issues of factorization methods;Conjugate gradient method; Parallel methods for ordinary and partial differentialequations.
1918	MTH	MS PROJECT II	MTH698	Fundamentals of parallel computing; Parallel techniques and algorithms; Parallelalgorithms for linear algebric equations; Design of parallel algorithms for eigenvalue problem; Parallel issues of factorization methods;Conjugate gradient method; Parallel methods for ordinary and partial differentialequations.
1919	MTH	NUMERICAL SOLUTION OF ORDINARY DIFFERENTIAL EQUATIONS	MTH692A	Introduction. RungeKutta methods derivation, error bounds and error estimates.Weak stability theory for RungeKutta methods. Order and convergence of thegeneral explicit onestep methods. Linear multistep methods. PredictorCorrector methods. Stiff systems. 22-MAR-2014
1920	MTH	NUMERICAL SOLUTION OF ORDINARY DIFFERENTIAL EQUATIONS	MTH692	Introduction. RungeKutta methods derivation, error bounds and error estimates.Weak stability theory for RungeKutta methods. Order and convergence of thegeneral explicit onestep methods. Linear multistep methods. PredictorCorrector methods. Stiff systems.
1921	MTH	NUMERICAL LINEAR ALGEBRA	MTH691A	Triangular form, Matrix norms, Conditioning of linear systems, Direct methods(Gauss, Cholesky, Householder), Iterative methods (Jacobi, GaussSeidel,Relaxation) for solving linear systems, Computing of preconditioning.
1922	MTH	NUMERICAL LINEAR ALGEBRA	MTH691	Triangular form, Matrix norms, Conditioning of linear systems, Direct methods(Gauss, Cholesky, Householder), Iterative methods (Jacobi, GaussSeidel,Relaxation) for solving linear systems, Computing of preconditioning.
1923	SDS	NON-LINEAR REGRESSION	SDS686	Estimation methods, Commonly encountered problems in estimation, Statisticalinference, Multiresponse nonlinear model, Asymptotic theory, Computationalmethods.
1924	MTH	NON-LINEAR REGRESSION	MTH686	Estimation methods, Commonly encountered problems in estimation, Statisticalinference, Multiresponse nonlinear model, Asymptotic theory, Computationalmethods.
1925	ME	MACHINING SCIENCE I	ME661	Mechanics of chip formation, chip curl. Bluntness and cutting forces. Thermalaspects of machining. Tool wear, tool life and economics of machining. Mechanicsof grinding, forces and specific energy, ten
1926	ME	NUMERICAL CONTROL OF MACHINE TOOLS	ME658	Basic principle of numerical control, Classification of NC systems,. NC partprogrammingmanual and computer aided. Drives, feedback devices, Countingdevices used in NC system. Interpolators for Manu
1927	ME	CONDUCTION AND RADIATION	ME641	Conduction: Steady and unsteady problems and their solutions in cartesian,cylindrical and spherical coordinates. Separation of variables. Duhamels theorem.Laplace transform. Problems involving chang spectral surfaces, Shape factors. Applications to cavities and enclosures.Integral equations approach. Radiation from gases, vapours and flames.
1928	ME	VISCOUS FLOW THEORY	ME631A	Introduction: Fluid Properties, Definition of Continuum, Examples of Viscous Flow Phenomena, Laminar and Turbulent Flow, Vector and Tensor notation, Lagrangian/Eulerian Methods,Streamline, Path line, Vorticity and Circulation.Fundamental Equations of Viscous Flow: Conservation of Mass, Momentum and Energy,Finite Volume Approach, Derivation of Continuity Equation: conservative and non conservative NS equations. Exact Solutions: Parallel Flow in a Straight Channel, Couette Flow, Lubrication Theory, HagenPoiseuille Flow,Unsteady Parallel Flow, Stokes Problems, Similarity Solution and Creeping Flow, Momentum and Energy Thickness, Order of Magnitude Analysis, Shape Factor, MomentumIntegral Approach, Boundary Layer Separation, Effect of Pressure Gradient, Boundary Layer Control by Suction and Holstenand Bohlen Method (Modified PohlhausenMethod), WaltzsQuadratureFormula and Example Problems.
1929	ME	VISCOUS FLOW THEORY	ME631	Stressdeformation relations, NavierStokes equation, exact solutions, twodimensional and axisymmetric boundary layers, Separation, Compressible boundarylayers, Elements of stability theory, Turbulent f
1930	ME	COMPUTATIONAL FLUID DYNAMICS AND HEAT TRANSFER	ME630A	ODE, matrix methods, root finding. Classification of PDE, finite differences,Steady and unsteady conduction, explicit and implicit method, advectiondiffusionproblems, upwinding, boundarylayers, NavierSt
1931	ME	COMPUTATIONAL FLUID DYNAMICS AND HEAT TRANSFER	ME630	ODE, matrix methods, root finding. Classification of PDE, finite differences,Steady and unsteady conduction, explicit and implicit method, advectiondiffusionproblems, upwinding, boundarylayers, NavierSt
1932	ME	APPLIED DYNAMICS AND VIBRATIONS	ME625	Review of single degree of freedom systems; Generalised coordinates, constraints,virtual work; Lagranges equation; Continuous systems; strings, beams;RaleighRitz and Galerkins methods; Dynamics of
1933	ME	FINITE ELEMENT METHODS IN ENGINEERING MECHANICS	ME623A	Introduction to ID FEM. Problems in structural mechanics using two dimensionalelements; Plane stress, plane strain, axisymmetric analysis; Three dimensionalstress analysis; Shell analysis; Solution of h
1934	ME	FINITE ELEMENT METHODS IN ENGINEERING MECHANICS	ME623	Introduction to ID FEM. Problems in structural mechanics using two dimensionalelements; Plane stress, plane strain, axisymmetric analysis; Three dimensionalstress analysis; Shell analysis; Solution of h
1935	SDS	PROBABILITY THEORY	SDS754	Algebras and sigma algebras; Measurable spaces; Methods of introducingprobability measures on measurable space; Random variables; Lebesgue integral;Expectation; Conditional probabilities and con variables;Fubini's theorem; Various kinds of convergence of sequence of random variables;Convergence of probability measures; Central limit theorem; delta method;Infinitely divisible and stable dis andtheir basic properties.
1936	SDS	PROBABILITY THEORY	MTH754	Algebras and sigma algebras; Measurable spaces; Methods of introducingprobability measures on measurable space; Random variables; Lebesgue integral;Expectation; Conditional probabilities and con variables;Fubini's theorem; Various kinds of convergence of sequence of random variables;Convergence of probability measures; Central limit theorem; delta method;Infinitely divisible and stable dis andtheir basic properties.
1937	MTH	ANALYSIS	MTH753A	Metric spaces, Open and closed sets, Compactness and connectedness,Completeness, Continuous functions (several variables and on metric spaces),uniform continuity C(X), X, compact metric space, Uniform Lebesgue Integration, Lpspaces.Complex Analysis: Analytic functions, Harmonic conjugates, Cauchy theoremsand consequences, Power series, Zeros of analytic functions, Maximum modulustheorem, Szegobases, Riesz Lemma,The space L2 as a Hilbert space.
1938	MTH	INTRODUCTION TO HYPERBOLIC GEOMETRY	MTH633A	Models of Hyperbolic Space: Upper Half Space Model & Disc Model; Isometries of HyperbolicSpace; Geodesics; Slimness of Triangles and Exponential Divergence of Geodesics in HyperbolicSpace; Isoperimetric Isometries and Fundamental groups; Properly DiscontinuousGroup actions; Fundamental Domains; Hyperbolic Surfaces. * Lectures on Hyperbolic Geometry; Riccardo Benedetti and Carlo Petronio;SpringerVerlag.* Kleinian Groups; Bernard Maskit; SpringerVerlag.* Metric Spaces of NonPositive Curvature; Martin R. Bridson, A group theory;Brian H. Bowditch, preprint.
1939	HSS	LITERATURE & ADAPTATION	ENG446A	This course will examine the processes and politics of adaptation as texts travel across genres, media, and cultures. Focusing primarily on literary texts, graphic narrativesand films, it will investigate the creative process, it will delve into topics such as the semiotics of translation, the 'fidelity' debate, the relationship between adaptation and appropriation, ideology and intention in the creation of adaptations.1) Hutcheon, Linda. A Theory of Adaptation. New York: Routledge, 2006.2) Sanders, Julie. Adaptation and Appropriation. New York: Routledge, 2006.

S.No.	Branch	Course Title	Courses ID	Content
1940	ECO	APPLIED GAME THEORY	ECO502A	Introduction: Optimal contracts under uncertainty, hidden information (adverse selection), or hidden action (moral hazard); Bilateral contracting: Hiddeninformation screening and signaling, hidden action private information; moral hazard in teams, tournaments and organizations; Incomplete contracts: Institution design, implementation theory, bilateral and multilateral contracts 1. Patrick Bolton and Mathias Dewatripont, Contract Theory (2005), The MIT Press.2. T.V. S. Rammohan Rao, Contract Economics (2004), New Age International.
1941	HSS	ADVANCED COURSE IN SOCIAL COGNITION	PSY798	The course aims at providing an understanding of the concepts and issues insocial cognition research. The course contents include historical roots of socialcognition; social cognitive principles a cost socialrepresentations; rituals and rhetoric; knowledge and social process; socialcognition and the study of stereotyping, prejudice and discrimination, socialcognition and discourse; social sensibility and
1942	HSS	LANGUAGES OF SOUTH ASIA	ENG448A	South Asia as a linguistic region 5Indo Aryan languages 5Dravidian languages 4TibetoBurman languages 3Munda languages 3Languages of the Andamans 2Convergence features 4Language contact . 4 Conflicts 2. Abbi, An vita. 1997. Languages of Tribal and Indigenous Peoples of India: TheEthnic Space. MLBDAbbi, Anvita. 2006. Endangered Languages of the Andaman Islands. Lin comEuropa.
1943	ECO	ENVIRONMENTAL ECONOMICS AND POLICY	ECO501A	Price mechanism and its limitations to mitigate externalities, economic instruments to promote sustainable development, missing market methods to set market prices for environmental goods and services 1. Shyam Divan & Armin Rosencranz, Environmental Law and Policy in India, OxfordUniversity Press2. R Rajagopalan, Environmental Studies, Oxford University Press3. Charles D. Kolstad, Environmental Economics, Oxford University Press4. Resource and Environmental Economics, Roger Perman, Yue Ma, JamesMcGilvray, and Michael Common6. Environmental Economics for three Huggers and other Skeptics, William K. Jaeger7. Renewable Economic Theory of Exhaustible Resource: R Ranganathan10. Environmental Economics: Hartwick & Olewaler
1944	ECO	ADVANCED MACRO ECONOMICS	ECO408	Introduction; The Solow growth model; Infinitehorizon and overlappinggenerationmodels; New growth theory; Realbusinesscycle theory; Traditional Keynesiantheories of fluctuations; Incomplete nominal
1945	ECO	INDUSTRIAL ECONOMICS	ECO321	Microeconomic foundations of Macroeconomics, Representative agent model, Application of representative agent model in asset pricing, consumption theories etc., Longrun economic growth models, Monetarism, business cycle models.
1946	EE	DIGITAL COMMUNICATION NETWORKS	EE673A	OSI model, queueing theory, physical layer, error detection and correction,data link layer, ARQ strategies, framing, media access layer, modelling andanalysis of important media access control protocols, F
1947	EE	DIGITAL COMMUNICATION NETWORKS	EE673	OSI model, queueing theory, physical layer, error detection and correction,data link layer, ARQ strategies, framing, media access layer, modelling andanalysis of important media access control protocols, F
1948	EE	FUNDAMENTALS OF ELECTRIC DRIVES	EE664A	Motor load dynamics, starting, braking & speed control of dc and ac motors.DC drives: converter and chopper control. AC Drives: Operation of inductionand synchronous motors from voltage and current i
1949	EE	FUNDAMENTALS OF ELECTRIC DRIVES	EE664	Motor load dynamics, starting, braking & speed control of dc and ac motors.DC drives: converter and chopper control. AC Drives: Operation of inductionand synchronous motors from voltage and current i
1950	EE	BASICS OF POWER ELECTRONIC CONVERTERS	EE660A	Power semiconductor devices, BJT, MOSFET, IGBT, GTO and MCT: ACDCConverters; Forced communication; synchronous link converters, DCAC converters,buck, boost, buckboost, cuk, flyback configurati
1951	EE	BASICS OF POWER ELECTRONIC CONVERTERS	EE660	Power semiconductor devices, BJT, MOSFET, IGBT, GTO and MCT: ACDCConverters; Forced communication; synchronous link converters, DCAC converters,buck, boost, buckboost, cuk, flyback configurati
1952	EE	ROBUST CONTROL SYSTEMS	EE654A	Linear Quadratic Regulators: return ratio & difference, sensitivity function.Kalmans optimality condition. Gain/phase margins, robustness to time delayand nonlinearity. Characterization of sensitivity. Khar
1953	EE	ROBUST CONTROL SYSTEMS	EE654	Linear Quadratic Regulators: return ratio & difference, sensitivity function.Kalmans optimality condition. Gain/phase margins, robustness to time delayand nonlinearity. Characterization of sensitivity. Khar
1954	EE	DIGITAL CONTROL	EE653A	Discretetime signals and systems, Ztransform, pulse transfer functions.Compensator design by root locus, error coefficients and frequency response.Statespace models of discrete time systems, controll
1955	CSE	COMPUTATIONAL GEOMETRY	CS663A	Historical perspective: complexity notions in classical geometry. Towardscomputational geometry, geometric preliminaries, models of computation.Geometric searching: point location problems, location statement and lower bounds. Grahams scan, Jarvismarch, quick hull technique, convex hulls in more than one dimension, extensionand applications. Proximity: divide and conquer approach, locus appro
1956	CSE	COMPUTATIONAL GEOMETRY	CS663	Historical perspective: complexity notions in classical geometry. Towardscomputational geometry, geometric preliminaries, models of computation.Geometric searching: point location problems, location statement and lower bounds. Grahams scan, Jarvismarch, quick hull technique, convex hulls in more than one dimension, extensionand applications. Proximity: divide and conquer approach, locus appro
1957	CSE	TOPICS IN DISTRIBUTED SYSTEMS	CS632A	Local area networks, concurrency control and recovery, distributed languagesand communication primitives, file servers, case studies of distributed systems.
1958	CSE	MATHEMATICS FOR COMPUTER SCIENCE	CS601A	Programming utilities, lab exercise for developing large system and applicationprograms.
1959	CSE	COMPUTER NETWORKS	CS425A	Introduction: Advantages of computer networks, LAN vs. WAN, ISO/OSI sevenlayerarchitecture, networks topologies, Physical Layer: transmission media,data encoding, Data Link Layer: Framing, Error det Other examples of MAC protocols, Network Layer: Internetworking Tunneling, Encapsulation, Fragmentation. Internet Protocol (IP)Header structure, addresses, options, etc. Routing Algorithms and Routin Protocol header, services, connectionmanagement, congestion control, sliding window, timers. User Datagram Protocol.Domain Name Service, Unix network programming, socket abstraction. clientserver Course Reference : 1. AS Tanenbaum, DJ Wetherall, Computer Networks, 5th Ed., PrenticeHall, 2010.2. LL Peterson, BS Davie, Computer Networks: A Systems Approach, 5th Ed., MorganKauffman, 2011.3. Cryptography and Network Security, Principles and Practice, 5th Ed., PrenticeHall, 2010
1960	CSE	INTRODUCTION TO COMPUTER GRAPHICS	CS360A	Introduction to Picture Synthesis and Analysis. Conceptual Framework of anInteractive Graphical Simulation System, Graphics hardware. Basic RasterGraphics Algorithms. Introduction to Simple Raster, Techniques, Geometric Modeling in 3D. Viewing in 3D. Concept of Synthetic Camera. Dialogue Design. Graphics User Interfaces. WindowingSystems Graphi cal Modeling of Discrete events. Simulation of Role of Virtual Reality in Graphical Simulation.
1961	CE	LABORATORY COURSE IN INFRASTRUCTURE ENGINEERING AND MANAGEMENT (GI+TE PART)	CE642A	Geological hazards and environmental impact; Landslides: cause, classification,zonation and protection; Earthquakes: historical seismicity, classification, interplateand intraplate earthquakes, effect on gr extentof flood hazard, flood hazard zoning, flood control and protection; Landsubsidence; Snow avalanches; Rock bursts; Mapping, monitoring and managementof geological hazards.
1962	CE	STRUCTURAL DYNAMICS	CE620A	Loading: nature of dynamic loading, harmonic, random, types of dynamic loading;Continuous systems: rods (axial vibrations), beams (shear, axial and axialshearflexuralvibrations); Discrete mass systems analysis); Introduction of random vibration: stochastic processes, stochasticanalysis of linear dynamical systems to Gaussian inputs, SDOF, MDOF.
1963	CE	ADVANCED MATHEMATICS	CE602A	Linear differential equations; Fourier integrals and transforms; Partial differentialequations; Numerical methods in general; Numerical methods for differentialequations; Linear algebra; Numerical method

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		FOR CIVIL ENGINEERS		
1964	CE	GEOINFORMATICS	CE331A	Preview of Geotechnical Problems in Civil Engineering and InfrastructureDevelopment, Description of soil, Engineering geology of soils and theirformation, earthquakes and their effects, Stresses within a and tests for its determination, different heads,piping, quicksand condition, seepage, flownets, flow through dams, filters,Compressibility and consolidation characteristics, maximum past stress, OCR,deter Strength and direct and triaxialshear tests, Mohr Coulomb strength criterion, drained, consolidated undrainedand undrained tests, strength of loose and dense sands, NC and OC soils, dilation,pore pressure unit weight, fieldcompaction control, etc. Introduction to Geosynthetics, classification, functions,properties physical, mechanical, hydraulic, environmental, etc. Stability ofslopes, limit equilibrium methods,
1965	EE	DIGITAL CONTROL	EE653	Discretetime signals and systems, Ztransform, pulse transfer functions.Compensator design by root locus, error coefficients and frequency response.Statespace models of discrete time systems, controll
1966	EE	BASICS OF MODERN CONTROL SYSTEMS	EE650A	Vector spaces, Linear systems, similarity transformations, Canonical forms,Controllability, Observability, Realisability etc. Minimal realization, Digital systems,Nonlinear systems, Phaseplane analysis, Poin
1967	EE	BASICS OF MODERN CONTROL SYSTEMS	EE650	Vector spaces, Linear systems, similarity transformations, Canonical forms,Controllability, Observability, Realisability etc. Minimal realization, Digital systems,Nonlinear systems, Phaseplane analysis, Poin
1968	EE	MICROWAVE MEASUREMENTS AND DESIGN	EE647A	Experiments in basic microwave measurements; passive and active circuitcharacterization using network analyser, spectrum analyser and noise figuremeter; PC based automated microwave measureme
1969	EE	MICROWAVE MEASUREMENTS AND DESIGN	EE647	Experiments in basic microwave measurements; passive and active circuitcharacterization using network analyser, spectrum analyser and noise figuremeter; PC based automated microwave measureme
1970	EE	PHOTONIC NETWORKS & SWITCHING	EE646A	Optical communications: Introduction to basic optical communications anddevices. Optical multiplexing techniques Wavelength division multiplexing,Optical frequency division multiplexing, time division 802.3, DQDB, FCS, HIPPI etc. Multiple access optical networks,Topologies, Single channel networks, Multichannel networks, FTFR, FTTR, TTFRand TTTR, Single hop networks, Multihop networks, Multiacc subscriber access networks. Design issues. Optical switching:Motivation, Spatial light modulator, Relational and nonrelational switchingdevices, Fundamental limits on optical switching elements, Switchin
1971	EE	PHOTONIC NETWORKS & SWITCHING	EE646	Optical communications: Introduction to basic optical communications anddevices. Optical multiplexing techniques Wavelength division multiplexing,Optical frequency division multiplexing, time division 802.3, DQDB, FCS, HIPPI etc. Multiple access optical networks,Topologies, Single channel networks, Multichannel networks, FTFR, FTTR, TTFRand TTTR, Single hop networks, Multihop networks, Multiacc subscriber access networks. Design issues. Optical switching:Motivation, Spatial light modulator, Relational and nonrelational switchingdevices, Fundamental limits on optical switching elements, Switchin
1972	EE	MONOLITHIC MICROWAVE ICS	EE645A	Scattering parameters of nports, Conductor and dielectric losses in planartransmission lines, coupled lines, multiconductor lines, discontinuities, GaAsMESFET fabrication devices, High electron mobility t
1973	EE	MONOLITHIC MICROWAVE ICS	EE645	Scattering parameters of nports, Conductor and dielectric losses in planartransmission lines, coupled lines, multiconductor lines, discontinuities, GaAsMESFET fabrication devices, High electron mobility t
1974	EE	ADVANCED ENGINEERING ELECTROMAGNETICS	EE641A	Transmission line theory; Greens function and integral transform techniques;Wave propagation and polarization parameters; reflection and transmissionacross an interface; waveguides, cavity resonators
1975	EE	ADVANCED ENGINEERING ELECTROMAGNETICS	EE641	Transmission line theory; Greens function and integral transform techniques;Wave propagation and polarization parameters; reflection and transmissionacross an interface; waveguides, cavity resonators
1976	EE	ELECTRICAL INSULATION IN POWER APPARATUS AND SYSTEMS	EE634A	Properties of dielectrics and breakdown mechanisms ; composites and novelmaterials; insulators for outdoor applications. Issues in design of insulators and insulator systems. Overvoltages and insulation Testing and Evaluation : Procedures and standards, ageing studies. On line and off line condition monitoring of substation equipment. Advances in measurement and diagnostic technologies : partial disc
1977	EE	SIMULATON OF MODERN POWER SYSTEMS	EE630A	Modern power systems operation and control, Power system deregulation; staticand dynamic modeling; Load flow and stability studies; Electromagneticphenomenon; Insulation and partial discharge.
1978	EE	SIMULATON OF MODERN POWER SYSTEMS	EE630	Modern power systems operation and control, Power system deregulation; staticand dynamic modeling; Load flow and stability studies; Electromagneticphenomenon; Insulation and partial discharge.
1979	EE	INFORMATION AND CODING THEORY	EE624A	Entropy and mutual information, rate distortion function, source coding, variablelength coding, discrete memoryless channels, capacity cost functions, channelcoding, linear block codes, cyclic codes. Cor
1980	EE	INFORMATION AND CODING THEORY	EE624	Entropy and mutual information, rate distortion function, source coding, variablelength coding, discrete memoryless channels, capacity cost functions, channelcoding, linear block codes, cyclic codes. Cor
1981	EE	REPRESENTATION AND ANALYSIS OF RANDOM SIGNALS	EE621A	Review of probability, random variables, random processes; representation ofnarrow band signals. Transmission of signals through LTI systems; Estimationand detection with random sequences; BAYES, sequences, linear stochastic dynamical systems.
1982	EE	REPRESENTATION AND ANALYSIS OF RANDOM SIGNALS	EE621	Review of probability, random variables, random processes; representation ofnarrow band signals. Transmission of signals through LTI systems; Estimationand detection with random sequences; BAYES, sequences, linear stochastic dynamical systems.
1983	EE	WAVELET TRANSFORMS FOR SIGNAL & IMAGE PROCESSING	EE607A	Basics of functional Analysis; Basics of Fourier Analysis; Spectral Theory; TimeFrequency representations; Nonstationary Processes; Continuous WaveletTransforms; Discrete TimeFrequency Transforms
1984	EE	INTRODUCTION TO SIGNAL ANALYSIS	EE605A	Discrete and Continuous time signals and systems, LTI systems, Convolution,Difference equations. Frequency domain representation: Fourier transform andits properties. Random discrete signals. Sampling systems of equations. Over and Underdetermined systems. Row andColumn spaces, Null spaces. Least square and minimum norm solutions. Inverseand pseudo inverse, Symmetry transformations. Eigen transmission, matched filtering, ISI, equalization. Coherent and noncoherentdetection.
1985	EE	LINEAR ALGEBRA FOR ENGINEERING	EE605	Discrete and Continuous time signals and systems, LTI systems, Convolution,Difference equations. Frequency domain representation: Fourier transform andits properties. Random discrete signals. Sampling systems of equations. Over and Underdetermined systems. Row andColumn spaces, Null spaces. Least square and minimum norm solutions. Inverseand pseudo inverse, Symmetry transformations. Eigen transmission, matched filtering, ISI, equalization. Coherent and noncoherentdetection.
1986	EE	IMAGE PROCESSING	EE604A	Human visual system and image perception, monochrome & colour visionmodels, colour representation ; image sampling & quantization; 2D systems,image transforms; image coding; stochastic models

S.No.	Branch	Course Title	Courses ID	Content
				techniques.
1987	EE	IMAGE PROCESSING	EE604	Human visual system and image perception, monochrome & colour visionmodels, colour representation ; image sampling & quantization; 2D systems;image transforms; image coding; stochastic models
1988	DES	DESIGN PROJECT I	DES681	Introduction to Graphics, Introduction to Graphics Software, Introduction toStyle, Introduction to Multimedia Application, Introduction to CAD Application,User Consumer Interaction Study, Media Communi
1989	DES	DESIGN PRACTICE I	DES602	Stages of a Product and Concurrent Engineering; Problem Formulation, Specificationsand Constraints; Creating Forms; Configuration Optimisation; Coupled, Decoupledand Uncoupled designs; Product of Making,Addressing Failures and Courage to Create; Interpersonal Skills; Robust Design,Incubation; Economic Considerations; Micro and Macro Designs; Introduction toElectronics; Laboratory on Problem andJournals Review.
1990	DES	DESIGN THEORY	DES601	Design Philosophy, Art (aesthetics) in ID, History of Design, Human Experiencein Design, Design Elements, Design Principles, Theory of Colour, Colour Aesthetics,Subject of Colour, Design Paradigm, Art D Texturein 2D and 3D surface; Colour Aesthetics; Product Analysis and Ergonomics,Product Design and Developments.
1991	EE	ELECTRICAL INSULATION IN POWER APPARATUS AND SYSTEMS	EE634	Properties of dielectrics and breakdown mechanisms ; composites and novelmaterials; insulators for outdoor applications. Issues in design of insulators and insulator systems. Overvoltages and insulation Testing and Evaluation : Procedures and standards, ageing studies. On line and off line condition monitoring of substation equipment. Advances in measurement and diagnostic technologies : partial disc
1992	EE	SEMICONDUCTOR DEVICE MODELLING	EE616A	Models for metalsemiconductor contacts and heterojunctions. MOSFET quantum theory of 2DEG, gradual channel approximation, charge control models,BSIM model, secondorder effects. MESFETShock
1993	EE	SEMICONDUCTOR DEVICE MODELLING	EE616	Models for metalsemiconductor contacts and heterojunctions. MOSFET quantum theory of 2DEG, gradual channel approximation, charge control models,BSIM model, secondorder effects. MESFETShock
1994	EE	SOLID STATE DEVICES I	EE614A	Basic semiconductor physics. Diodes (PN junction, Schottky, contact), JunctionTransistors (BJT, HBT), Field Effect Transistors (JEFT, MESFET, MOSFET, HEMT).Other semiconductor devices.
1995	EE	SOLID STATE DEVICES I	EE614	Basic semiconductor physics. Diodes (PN junction, Schottky, contact), JunctionTransistors (BJT, HBT), Field Effect Transistors (JEFT, MESFET, MOSFET, HEMT).Other semiconductor devices.
1996	EE	FIBER OPTIC SYSTEMS I	EE612A	Review of semiconductor physics radiative recombination. LEDs, optical cavity,DH and other lasers. PIN and APD detectors, detector noise. Optical fibers ray and mode theories, multimode and singlemod
1997	EE	ANALOG CIRCUITS FOR SIGNAL PROCESSING	EE612	Review of semiconductor physics radiative recombination. LEDs, optical cavity,DH and other lasers. PIN and APD detectors, detector noise. Optical fibers ray and mode theories, multimode and singlemod
1998	EE	ANALOG/DIGITAL VLSI CIRCUITS	EE610A	Analog MOS circuits, opamps, frequency and transient responses, stability andcompensation. Analog switches, sampleandhold circuits, switchedcapacitorcircuits. MOS inverters and gate circuits, interfac array, standard cell, and PLA based designs. Digital toAnalog andAnalogtoDigital converters.
1999	EE	ANALOG IC DESIGN - I	EE610	Analog MOS circuits, opamps, frequency and transient responses, stability andcompensation. Analog switches, sampleandhold circuits, switchedcapacitorcircuits. MOS inverters and gate circuits, interfac array, standard cell, and PLA based designs. Digital toAnalog andAnalogtoDigital converters.
2000	EE	ELECTRICAL ENGINEERING LAB I	EE380A	Experiments from various areas of Electrical Engineering with emphasis on electronicdevices, circuits, control systems and machines.This course has three labs: Electronic Circuits Lab. (7 experiments),C
2001	CSE	MODERN CRYPTOLOGY	CS641A	Basics of finite fields. Private and Publickey cryptography, existing cryptosystemsand their security. Cryptanalysis of existing systems. Zeroknowledge protocols,Oneway functions. Advanced protocols fo
2002	CE	TRAFFIC ENGINEERING	CE683A	Microscopic and macroscopic traffic parameters, traffic flow models, carfollowingmodels, capacity and level of service analysis, design of traffic facilitieslike unsignalized and signalized intersections, inte
2003	CE	CONSTITUTIVE MODELING OF FRICTIONAL MATERIALS	CE637A	Role of constitutive modeling; Importance of laboratory testing with relationto constitutive modeling; Elasticity: linear, quasilinear, anisotropic; Plasticitybasics: yield criteria, flow rule, plastic potential, hard Critical state soil mechanics: critical state concept,camclay models, simulation of single element test using camclay, consolidation,drained and undrained triaxial test; Stressdilatancy theory; Work harden hypoelasticityplasticity, disturbed state concept.
2004	CE	EXPERIMENTAL METHODS IN STRUCTURL ENGINEERING	CE623A	Similitude and Structural Models: Dimensional analysis, Buckingham's Pi theorem, Scale factors and dynamicsimilitudeUses and Applications of Models: Types of model investigation, Indirect and d and unce1tainty in experiment, Measurement systems, Accuracy in models and reliability of resultsTest Planning, Design and Implementation: Testing sequence and experimental plan, Loading systems, c various types of sensors for displacement, velocity, acceleration, pressure, loads, strains, etc, fullfield measurements, .Data Acquisition System and Data Processing: Analog systems, digital systems using Experiments to illustrate buckling of stiUctural members; loaddefb1mation behaviour of beams, columns, joints, and flames under various loads; mode shapes, natural frequency, damping factors from fre
2005	IME	INFRASTRUCTURE REGULATION POLICY & FINANCE	MBA675.	Role of Infrastructure in Economic Development, Natural Monopoly and Economicsof Infrastructure Regulation, Rate of Return Regulation, Performance BasedRegulation, Pricing for Infrastructure Sector, Privatisation in Infrastructure Sector, Reform Acts, Competition in InfrastructureSector (Bulk Power, Telecom, Transportation), Issues in Infrastructure Finance,Modes of Project Financing, Risks in Infrastr
2006	IME	SUPPLY CHAIN MANAGEMENT	MBA664A.	Strategic Framework for Supply Chain, Materials Management Functions,Forecasting and market Analysis, Purchasing and Procurement, Physical Supply,Managing Inventories, MRP and Capacity Plannin and Distribution, IT and Role of EBusiness, Financial Evaluation.
2007	IME	MANUFACTURING STRATEGY	MBA622.	Product and factory life cycle, strategic dimensions of technology, characteristicsof job shops and flow shops, learning curve effects, economies of scale,resolution of conflicts between manufacturing an department with manufacturing, marketing, service and purchasing.Concept of aligning of manufacturing and the corporate strategy.
2008	IME	ORGANIZATION STRUCTURE AND BEHAVIOUR	MBA611A.	Introduction to Organizations, Work motivation and Organization Goals, Organizations and Markets, Organization Structures and Systems, Strategy, Structure & Technology, Organization Environment and Evolution, Employee empowerment, Learning Organizations and organization Effectiveness, Service Organizations, Organizations as Networks. State ofart research papers and case studies will be used fo
2009	IME	ORGANIZATION STRUCTURE AND BEHAVIOUR	MBA611.	Introduction to Organizations, Organization Goals, Organizations and Markets,Organization Structures and Systems, Strategy, Structure & Technology, OrganizationEnvironment and Culture, Various Design Organizationsand organization Effectiveness, Service Organizations, Organizations as Networks.State ofart research papers and case studies will be used for the selected topics.
2010	IME	ECONOMIC ANALYSIS FOR MANAGEMENT	MBA606A.	Basic concepts in business economics, Economics of Market, Utility theory, Determination of Price, Production Function, Theories of Competition, Theory of Supply and Demand, Micro Level Firm Behaviour and domestic product. Keynesian theory of income determination, Monetary approach, Inflation, balance of Payments, Structure of Indian economy, Indian economic growth and development, Aggregate S projects)
2011	IME	ECONOMIC ANALYSIS FOR MANAGEMENT	MBA606.	Basic concepts in business economics, Economics of Market, Utility theory,Determination of Price, Production Function, Theories of Competition, Theoryof Supply and Demand, Micro Level Firm Behaviour and domestic product. Keynesian theory of income determination,Monetary approach, Inflation, balance of Payments, Structure of Indian economy,Indian economic growth and development.
2012	IME	ACCOUNTING & FINANCE	MBA601A.	1) Introduction: Forms of Organization & Corporate Reporting, Reporting Fundamentals & Financial Reporting. 2) Financial Accounting: Accounting Principles, Accounting Mechanics, Accounting Standar SystemsTools & Techniques, Activity Based Costing (ABC). 4) Corporate finance: The Time Value of Money, Valuation, Capital Budgeting, Capital Structure & Dividend Policy, Working Capital Management. 1. Williams et al., Financial and Managerial Accounting, Tata McGraw Hill. 2. Fundamentals of Financial management, J. Van Horne & J M Wachowicz. 3. Principles of Corporate Finance Brealey, Myers &
2013	IME	ACCOUNTING & FINANCE	MBA601.	Balance sheet, profit and loss concepts, accounting principles and mechanics,Inventory Valuation and Depreciation accounting, Ratio and Fund flow analysis.Introduction to cost Accounting. Various met responsibilityaccounting. Use of costing systems in decision making. Extensive case studieswill employed in this course.

S.No.	Branch	Course Title	Courses ID	Content
2014	IME	PROBABILITY AND STATISTICS	IME602	Probability: Sample space, events and probability; Conditional probability and independence; Random variable (RV); Expectation, variance and higher moments; Some standard probability distributions; Finite Population and sample; Sampling distributions; Properties of point estimators; Maximum likelihood method; Interval estimation; Hypothesis testing; Some standard hypothesis tests; Analysis of variance; Probability Theory and Statistical Inference, John Wiley & Sons.2)R.V. Hogg, A.T. Craig, and J.W. McKean, Introduction to Mathematical Statistics, Pearson Education. 1)V.K. Rohatgi and A.K.Md.E. Saleh, An Introduction to Probability and Statistics, Wiley India Private Limited.2)R.J. Larsen and M.L. Marx, An Introduction to Mathematical Statistics and Its Applications, Prentice Hall, Inc.3)G. Port, and S. Stone, Introduction to Statistical Theory, Houghton Mifflin.5)W. Feller, An Introduction to Probability Theory and its Applications (Vol. 1 & 2), Wiley India Private Limited.6)A.M. Goon, M.K. Gupta and B. Dasgupta, Fundamentals of Statistics (Vol. 1 & 2), World Press Private Limited.8)J.P. Romano and E.L. Lehmann, Testing Statistical Hypotheses, Springer (SIE).9)N.R. Draper and H. Smith, Applied Regression Analysis, Wiley.
2015	CSE	OPERATING SYSTEMS	CS330A	Introduction: Role of operating System; System Calls; Processes and ThreadsConcepts of Processes, Threads; Process Control Block. CPU Scheduling: Examplesfrom contemporary OSES (UNIX and NT).Process Synchronization Critical Section Problem; Hardware Mechanism for synchronization; Semaphores and Mutex objects; Classical Problems (Producer-Consumer, dining philosophers etc.); Monitoring and avoidance mechanisms.Virtual Memory: Address binding process (compilation and linking); Dynamic Linking; Segmentation; Paging Protection; Demand Paging; Page Replacement policies Thrashing and its prevention; File organization in directories; File attributes; Operations on files; Directory attributes and operations on directories; Directory organizations;File and directory protections.File system implementation Concepts of file systems; caching;Examples of file systems from one or more of DOS, BSD, Linux, HPFS and NTFS;Device Drivers:Storage management Disk scheduling; Disk Management; Swap and swap management.Security and Virus Protection Trojan Horses etc.
2016	IME	SUPPLY CHAIN MANAGEMENT	MBA664.	Strategic Framework for Supply Chain, Materials Management Functions,Forecasting and market Analysis, Purchasing and Procurement, Physical Supply,Managing Inventories, MRP and Capacity Planning, Inventory Control and Distribution, IT and Role of EBusiness, Financial Evaluation.
2017	IME	MARKETING MANAGEMENT	MBA631A.	Marketing Environment, Company analysis (strength, weaknesses, opportunities and threats), the concept of marketing mix., four Ps of marketing, and the concept of marketing strategy. The concept of marketing research, Behaviour and Marketing Research, International marketing, Marketing economy and public policy issues, Emarketing. In this course concepts will be elaborated by the use of cases and research papers, Branding, Student Project (Field research & Data Interpretation based term projects)
2018	IME	MARKETING MANAGEMENT	MBA631.	Marketing Environment, Company analysis (strength, weaknesses, opportunities and threats), the concept of marketing mix., four Ps of marketing, and the concept of marketing strategy. The concept of marketing research, Behaviour and Marketing Research. International marketing. Marketing economy and public policy issues. Emarketing. In this course concepts will be elaborated by the use of cases and research papers.
2019	IME	STRATEGIC MANAGEMENT	MBA623A.	General Management Function, Introduction to the corporate strategy, concept of organizational purpose, environmental scanning and formulation of objectives, strategy for growth such as concentric growth, diversification, organizational structure and designing control systems to support the implementation of the strategy. Role of implementation issues in strategy formulation. Impact of organizational culture, structure, systems on strategy.
2020	IME	STRATEGIC MANAGEMENT	MBA623.	General Management Function, Introduction to the corporate strategy, concept of organizational purpose, environmental scanning and formulation of objectives, strategy for growth such as concentric growth, diversification, organizational structure and designing control systems to support the implementation of the strategy. Role of implementation issues in strategy formulation. Impact of organizational culture, structure, systems on strategy.
2021	IME	MANUFACTURING STRATEGY	MBA622A.	Product and factory life cycle, strategic dimensions of technology, characteristics of job shops and flow shops, learning curve effects, economies of scale, resolution of conflicts between manufacturing and marketing, department with manufacturing, marketing, service and purchasing. Concept of aligning of manufacturing and the corporate strategy.
2022	IME	INDUSTRIAL PROJECT	IME697	A 68 week industrial project for M Tech students during the period intervening the II and III semesters on a problem of practical relevance completed in an industrial or service organization. The student will be assigned to an officer/executive of the host organization. On completion of the summer project the student will submit a written report and give a seminar to the IME Department.
2023	IME	FINANCIAL ENGINEERING	IME611	Module 1 : Basic Elements of Financial Systems and Financial Management Fundamentals of Financial Systems and Domain Knowledge of Financial ManagementModule 2 : Mathematical Background Introduction to Probability Measures Numerical procedures : Binomial & trinomial trees, Monte Carlo simulation; finite difference methodsModule 3 : Options and Futures Markets Forward and futures contracts : Basic definition, Diffusion Processes, Rate Futures. Swaps : Currency Swaps, Interest Rate Swaps Options : Definitions, Payoff Diagrams, General Arbitrage Relationships, The Binomial Method, Applications to Hedging & Speculating, Delta Hedging, Financial Risk Management Introduction : Different types of risk ; approaches to risk management; history of bank regulation. Greek letters : Definitions and how they are used.
2024	IME	OPERATIONS RESEARCH FOR MANAGEMENT	IME605	Introduction, Mathematical Modeling, Linear programming Formulation, solution procedures, Duality, Sensitivity, Applications, Network methods Max Flow, Min cost, Shortest path, Dynamic programming Shortest Path Problem, Programming Applications and solution methods.
2025	IME	INTRODUCTION TO COMPUTING	IME603	Computing Computer Organization, Data Representation, Data Structures such as Arrays, Stacks, Queues and Trees, Algorithms for Searching and Sorting, Complexity, File Processing, Structured Programming.
2026	HSS	INDIAN SOCIETY AND CULTURE	SOC473A	Approaches to study Indian Society. Social stratification: caste, jati and varna among Hindus, non-Hindus. Marriage and Family among Hindus, Muslims and Christians; polyandry and polygyny; regulations on marriage; Sanskritization and Westernization; parochialization and universalization; structural, dialectical, cognitive historical, and indological approaches. Origin and evolution of cultural change in culture traditions and processes of change: peasant religious and sectarian movements; dalit and backward caste movements; tribal, labour movements. New Social Movement: Ecological Movement. Social Movements: Dumont, Louis. Homo Hierarchicus: The Caste System and its Weidenfeld and Nicholson, 1970. Ghurye, G. S. Caste and Race in India. Mumbai: Popular Prakashan, 1969, 5th edition. Gupta, Dipankar, editor. Social Movements in India. Oxford University Press, 1966. Rao, M. S. A. Social Movements and Social Transformation: A Study of Two Backward Classes Movements in India. New Delhi: Macmillan, 1979. Shah, Ghanshyam. Social Movements in India. Jaipur: Rawat Publications, 1986 (reprint). Srinivas, M. N. Social Change in Modern India. New Delhi: Orient Longman, 1995 (reprint 2002).
2027	MSE	PHASE TRANSFORMATIONS IN MATERIALS	MSE340	Thermodynamic order of transformations; Theory of nucleation Kinetics of homogeneous, transient and heterogeneous nucleation; Theory of Thermally Activated Growth: Interface controlled growth, Diffusion controlled growth, Massive transformation; Transformation Kinetics Johnson-Mehl equation, Avrami model, Transformation kinetics in diffusion controlled transformations, Isothermal and continuous cooling transformation of phase particles; Martensitic transformation Nature of martensitic transformations, Bain distortion, Nucleation, and growth of martensite, Athermal, isothermal and burst transformations, Thermomechanical effects on transformation energy; Solidification Nature and growth of solid-liquid interfaces, Rapid solidification, Glass transition, metallic glasses; Heat Treatment IT and CCT Diagrams in steels, quench hardening and tempering of non-ferrous metals and cast irons, heat treatment of Ni-base superalloys and Ti alloys, Thermomechanical treatments.
2028	IME	SECURITY ANALYSIS DERIVATIVES & PORTFOLIO MANAGEMENT	MBA676.	Financial Markets, Investment Alternatives, Risk and Return, Portfolio Theory and Capital Asset Pricing, Capital Asset Pricing Theory and Arbitrage Pricing Theory, Efficient Market Hypothesis, Security Analysis, Portfolio Strategies, Derivatives, Options, Futures, Swaps, Black-Scholes model, Value at risk, Estimating volatility and correlations, Hedging and Portfolio Management.
2029	IME	QUANTITATIVE METHODS FOR DECISION MAKING	MBA651A.	Introduction to decision analysis and process, Simplex Method, Duality Theory, Network Problems, Elementary probability theory, Decision Sciences, EVPI, random variables, probability distributions, conditional probability, statistics, Sampling techniques, Estimation, Hypothesis testing and Regression (linear and nonlinear), Student Project (Field research & Data Interpretation based term projects)
2030	IME	QUANTITATIVE METHODS FOR DECISION MAKING	MBA651.	Introduction to decision analysis and process. Elementary probability theory, conditional probability, Bayesian decision analysis, EVPI, moment generating functions, the central limit theorem, Descriptive and inferential statistics.
2031	IME	ENTERPRISE INTEGRATION WITH IT	MBA646A.	Need for integration, Evolution of ERP, Components of ERP, Enterprise evaluation, Business process mapping, Business Process Reengineering, Understanding and evaluating ERP packages, Technology evaluation, Implementation on SAP system, Case studies
2032	IME	ENTERPRISE INTEGRATION WITH IT	MBA646.	Need for integration, Evolution of ERP, Components of ERP, Enterprise evaluation, Business process mapping, Business Process Reengineering, Understanding and evaluating ERP packages, Technology evaluation, Implementation on SAP system, Case studies
2033	IME	COMPUTING FOR MANAGEMENT	MBA641A.	Computers and Management Function, Introduction to an appropriate high level language, Introduction to Data Structures, Computer Organization, System Configuration, Introduction to data base management.
2034	IME	COMPUTING FOR	MBA641.	Computers and Management Function, Introduction to an appropriate high level language, Introduction to Data Structures, Computer Organization, System Configuration, Introduction to data base management.

S.No.	Branch	Course Title	Courses ID	Content
MANAGEMENT				
2035	IME	BUSINESS TO BUSINESS MARKETING	MBA637.	Business to Business Marketing encompasses those management activities thatenable a supplier firm to understand, create, and deliver value to otherbusinesses, governments, and/or institutional custom... past, the topical area applied largely to industrial manufacturingfirms. Today, business to business marketing provides practical frameworks,concepts, and tools for organizations as diverse as managemen... among many other leadingedge technology and servicecompanies.As business to business marketing expands its scope and stature, this courseaims at reinvigorating training in marketing beyond the tir... multifunctional teams, strategic alliance environmentalsensitivity, interorganizational trust, organizational learning and adherence toethical principles. Furthermore, with the advent of relationship and netw... share value with customer firms.Understanding of business buying and marketing behaviour within the contextof relationship/network theories is the central learning from this proposedelective.Given this... organizational buying behaviour andbusinesstobusiness marketing.
2036	PHY	CLASSICAL MECHANICS I	PHY401A	1. Review of Newton's laws of motion, Galilean transformations, Frames of reference and pseudoforces. Symmetries in Newton's laws, Lagrangian formulation, Configuration space. Calculus of variatio... oscillations and normal modes, Anharmonic oscillators, Resonances in harmonic and anharmonic oscillaors*, Parametric resonance. Secular (regular) perturbation theory*, Lindtstedt-Poincare methodR... pendulum, Motion in central force field, System of particles, Charged particle in an Elecromagnetic field, Lagrangian formulation of relativistic mechanics, etc. 2.Â Fixed points and linear stability a... (no chaos in 2D autonomous flow).Â Legendre transformation, Hamiltonian formulation, Phase plane, IntegralÂ invariants, Symplectic area conservation, (Generalized) Liouville'sÂ theorem, Poincare recur... transformations,Â Poisson brackets, Active view versus passive view of canonicalÂ transformations.Â Principle of varying action and Hamilton-Jacobi theory, Optico-mechanicalÂ analogy, Action-angle va... integrability and chaos in HamiltonianÂ systems. N.B.: The topics in boldface* are essential for Classical Mechanics II. Reference books: 1. J. V. Jose & E. J. Saletan, Classical Dy... University Press (1982). 3. L. D. Landau & E. M. Lifshitz, Mechanics, Butterworth-Heinemann (1976). 4. H. Goldstein, Classical Mechanics, Addison-Wesley (1 980). 5. J. L. McCauley, Class... Publications (2000). 7. S. H. Strogatz, Nonlinear Dynamics and Chaos, Westview Press (2001).
2037	PHY	CLASSICAL MECHANICS	PHY401	Review of Newtonian mcdnnics: Basicassumption; constraints, principle of virtual work, D'Alembert's cprinciple,generalized coordinates [3]Lagrangian mechanics: Calculu of variations, Princip... phase space&. phase trajectories, LiuviuUe's theorem [3]smalt oscillations, normal modes, anharmonic and nonlinear oscillators, Nonlinear dynamicsChaos (Hamiltonian systems in dissipative syste... Waves in continuous media, NavierStokes Eqn., Pressure waves [3]canonical transformations, Poisson brackets, Hamitton Jacobi theory, Actionangle variables(recommend: central force discussed here) [3]... L. D. Landau and E. M. Lisfsitz,.Mechanics, Courses of Theoretical Physics, Oxford: Pergamon, 1976H. Goldstein, C. P. Poole, and J. L. Safko, Classical Mechanics, Addison Wesley, 2001 (Indian Ed. availab... 1994 .. (Indian Ed. available)H. Strogatz, Nonlinear Dynamics and Chaos, Levant, 2007 (Indian Ed.)
2038	MTH	SET THEORY & LOGIC	MTH302A	Formal theories, Consequence and deduction. Propositional Calculus:Syntax, Semantics, Applications; Axiomatic approach, Soundness, Consistency, Completeness.Other proof techniques: Sequent calcul... to the classof Boolean algebras. Classical rst order theories: Syntax, Semantics; Axiomatic approach,Soundness; Sequent calculus, Tableaux. Equality, examples of rst order theorieswith equality. Consiste... 1. R. Cori and D. Lascar, Mathematical Logic, Oxford, 2001.2. A. Margaris, First Order Mathematical Logic, Dover, 1990.3. J. GoubaltLarrecq and J. Mackie, Proof Theory and Automated Deduction, Kluwer,1...
2039	MTH	SET THEORY & LOGIC	MTH302	Formal theories, Consequence and deduction. Propositional Calculus:Syntax, Semantics, Applications; Axiomatic approach, Soundness, Consistency, Completeness.Other proof techniques: Sequent calcul... to the classof Boolean algebras. Classical rst order theories: Syntax, Semantics; Axiomatic approach,Soundness; Sequent calculus, Tableaux. Equality, examples of rst order theorieswith equality. Consiste... 1. R. Cori and D. Lascar, Mathematical Logic, Oxford, 2001.2. A. Margaris, First Order Mathematical Logic, Dover, 1990.3. J. GoubaltLarrecq and J. Mackie, Proof Theory and Automated Deduction, Kluwer,1...
2040	MTH	ANALYSIS - I	MTH408	Real Number system: Completeness property. Countable and Uncountable.Metric Spaces: Metric spaces, Examples: Ip, C[a; b]; Limit, Open sets, Convergenceof a sequence, Closed sets, Continuity. Compl... Totallybounded, Characterizations of compactness, Finite intersection property, Continuous functionson compact sets, Uniform continuity. Connectedness: Characterizations of connectedness,Continuou... theorem of calculus, Set of measurezero, Cantor set, Characterization of integrable functions. Convergence of sequence andseries of functions: Pointwise and uniform convergence of functions, Series of... Weierstrass approximation theorem. 1. N.L. Carothers, Real Analysis. 2. R. R. Goldberg, Methods of Real Analysis. 3. W. Rudin, Principles of Mathematical Analysis.
2041	MTH	ANALYSIS - I	MTH301	Real Number system: Completeness property. Countable and Uncountable.Metric Spaces: Metric spaces, Examples: Ip, C[a; b]; Limit, Open sets, Convergenceof a sequence, Closed sets, Continuity. Compl... Totallybounded, Characterizations of compactness, Finite intersection property, Continuous functionson compact sets, Uniform continuity. Connectedness: Characterizations of connectedness,Continuou... theorem of calculus, Set of measurezero, Cantor set, Characterization of integrable functions. Convergence of sequence andseries of functions: Pointwise and uniform convergence of functions, Series of... Weierstrass approximation theorem. 1. N.L. Carothers, Real Analysis.2. R. R. Goldberg, Methods of Real Analysis.3. W. Rudin, Principles of Mathematical Analysis.
2042	PHY	CLASSICAL ELECTRODYNAMICS I	PHY552	Electrostatics Laplace and' POisson equatlOhs' ciil.d "'their SOh.f. 'CToil.S,Uniqueness theorem, multipole expansion (911 lectures)Magnetostatics (12 lectures)Boundaryvalue p... medium, Poynting'stheorem,momentum and angular momentum of electromagnetic fields (810 lectures)Electromagnetic radiation, retarded potentials, Lorentz and CoulombgaugeI 911 lectures)rela... Classical Electrodynamics, John D. Jackson
2043	PHY	CONDENSED MATTER PHYSICS I	PHY543	Free electron theory; heat capacity; transport properties; Hall effect;elementary concepts of quantum Hall effect, quantization of conductance in a metallic nanowireStructure and scattering; crystalline so... model; tightbinding model;application to graphene and nanotubes,semiclassical dynamics; notion of an electron in a DElectric field; effective mass, holes, crystal binding; types of solids; van der Waals s... harmonicapproximations, vibrations of mono and diatomic lattices, lattice heatcapacity, Einstein and Debye modelsSemiconductors; intrinsic & extrinsic semiconductors, laws of mass action, electron & h... equation, BCS theory, GinzburgLandau theory, flux quantization,Magnetism: exchange interaction, diamagnetism, paramagnetism,ferromagnetism & antiferromagnetism, Hund's rules, Pauli paramag... 1. Introduction to Solid State Physics by C. Kittel2. Solid State Physics by N. W. Ashcroft and N. D. Mermin3. Solid State Physics by H Ibach and Hans Luth
2044	PHY	OPTICS	PHY224A	Review of Maxwell's equations, wave equation and solutions for plane andspherical wavesLinear dispersion theoryDerivation and discussion of Fresnel's equationsPolarization states of polariz... Michelsonand FabryPerot, multilayers for high and antireflectionDiffraction Kirchhoff integral, Fraunhofer diffraction, diffraction gratingCoherence spatial, temporal coherence, measurement techniques, n... 1.M.Born and E.Wolf, Principles of Optics (Cambridge Univ Press)2.J. B. Peatross and M. Ware: Physics of Light and OpticsThis book is freely available at http://optics.byu.edu/BYUOpticsBook.pdf3.F.L. Pe... McGrawHill)5.E.Hecht, Optics (AddisonWesley)6. K.K.Sharma, Optics: Principles and Applications (Academic Press)
2045	PHY	OPTICS	PHY224	Review of Maxwell's equations, wave equation and solutions for plane andspherical wavesLinear dispersion theoryDerivation and discussion of Fresnel's equationsPolarization states of polariz... Michelsonand FabryPerot, multilayers for high and antireflectionDiffraction Kirchhoff integral, Fraunhofer diffraction, diffraction gratingCoherence spatial, temporal coherence, measurement techniques, n... 1.M.Born and E.Wolf, Principles of Optics (Cambridge Univ Press)2.J. B. Peatross and M. Ware: Physics of Light and OpticsThis book is freely available at http://optics.byu.edu/BYUOpticsBook.pdf3.F.L. Pe... McGrawHill)5.E.Hecht, Optics (AddisonWesley)6. K.K.Sharma, Optics: Principles and Applications (Academic Press)
2046	PHY	QUANTUM PHYSICS	PHY204	Foundations of quantum mechanics Black body radiation, photoelectric effect,Compton effect, de Broglie hypothesis and its experimental verification 5 lecturesTimeindependent and timedependent Schrod... principle 5 lecturesSolution of stationarystate Schrodinger equation for particle in a box, particle in a finitewell, reflection and transmission across a step potential, application to phenomena likealphadeca... hydrogen,Discussion of excited state, Explanation of the periodic table by introduction of electronspin and Pauli's exclusion principle, StemGerlach experiment, two level systems 5lecturesFree partic... principle for approximate solutions and simple applications, groundstateenergy of helium atom 3 lecturesInteraction of light with matter Einstein's phenomenological theory, lifetime of a state,LASER... F eynman lectures volume IIIMax Jammer Conceptual development of quantum mechanicsB.L. van der Waerden Sources of quantum mechanicsE. Shroedinger Papers on wave mechanicsR. Shankar Pri...
2047	MTH	PARTIAL DIFFERENTIAL EQUATIONS	MTH424A	Introduction to PDEs, First order quasilinear and nonlinear equations;Higher order equations and classications; Solution of wave equations, Duhamel's principleand applications; Existence and uniqu... the Laplace equation; Heatequation, Maximum principle for the heat equation, Uniqueness of solutions of IVPs forheat conduction equation. 1. Robert C. McOwen: Partial Dierential Equations, Pearson Education Inc.2. Alen Jerey: Applied Partial Dierential Equations, Academic Press3. Ervin Kreyszig: Advanced Engineering Mathematics, John W...
2048	MTH	PARTIAL DIFFERENTIAL EQUATIONS	MTH424	Introduction to PDEs, First order quasilinear and nonlinear equations;Higher order equations and classications; Solution of wave equations, Duhamel's principleand applications; Existence and uniqu... the Laplace equation; Heatequation, Maximum principle for the heat equation, Uniqueness of solutions of IVPs forheat conduction equation.

S.No.	Branch	Course Title	Courses ID	Content
				1. Robert C. McOwen: Partial Dierential Equations, Pearson Education Inc.2. Alen Jerrey: Applied Partial Dierential Equations, Academic Press3. Ervin Kreyszig: Advanced Engineering Mathematics, John W
2049	PHY	ELECTRONICS	PHY441	Review of network theorems and network analysisOperation Amplifier and negative feedbackOpAmp limitations and applicationsCircuits with OpAmps and diodesInterfacing BJT with OpAmpsMOSFETsD programming with microcontroller applications " by P. Horowitz and W. Hill"Student Manual for The Art of Electronics" by T. C. Hayes and P. Horowitz"OpAmps and linear integrated circuits" by R.A. G Brown"The 8051 Microcontroller: architecture, programming and applications" by K. J. Ayala
2050	MTH	INTRODUCTION TO CONTINUUM MACHANICS	MTH423A	Fundamental concepts; Introduction to Cartesian tensors; Stress tensorsand equilibrium equations; Theory of strain and rate of deformation tensor; Conservationlaws and basic equations; Linear Elasticity inviscid ow, Incompressibleviscous ow, Introduction to boundary layer theory 1. Introduction to Continuum Mechanics{M. Lai, D. Rubin, E. Krempl2. Continuum Mechanics for Engineers{G. T. Mase and G. E. Mase3. Elementary uid mechanics{D. J. Acheson4. Fluid Mechanics{P K Ku
2051	MTH	PRINCIPLES OF NUMERICAL COMPUTATIONS	MTH437	Root nding problem: Methods and analysis; Interpolation: Methodsand analysis; Approximation: Least squares and minimax approximation; Numerical differentiation;Numerical integration: Methods and decomposition. 1. Elementary Numerical Analysis, an algorithmic approach, S.D. Conte and Carl DeBoor2. An Introduction to Numerical Analysis, Kendall E Atkinson3. Numerical Methods for Scientic and Engineering Cor
2052	ME	ME PROJECT-1	ME451A	Project work involving the analysis, synthesis, material/component selection and detaileddesign of a mechanical system including the preparation of working drawings. The systemmay be integrated with areas (or a combination there of) Fluidmechanics and Thermal sciences, Solid Mechanics and Design, Manufacturing Science andRobotics and/or any other related mechanical system(s).
2053	ME	SECONDARY MANUFACTURING PROCESSES	ME361	Introduction to manufacturing processes and system concept and itsevolution; Metal casting: Solidification Mechanism, Gating and Riser Design, Defects andProduct Design; Metal Forming: Fundamentals Machining: ToolSpecifications, Orthogonal and Oblique cutting, Tool wear and Tool Life, Economics ofMachining; Shaping processes for Plastics and Tool Design; joining Processes; UnConventionalMater technologies; Metrology and Selection of Manufacturing Processes. 1. Ghosh, A., Mallik, A.K., Manufacturing Science (2nd edition}, EastWest Press, 2010.2. Lal, G.K., Introduction to Machining Science (2nd edition}, New Age Internationalpublishers, 2009.3. Groover, M.P, Fundamentals of Manufacturing Engineering and Technology, PearsonEducation5. Galyer, j.F.W., Shotbolt, C.R., Metrology for Engineers, ELBS
2054	ME	THEORY OF MECHANISMS & MACHINES	ME352	Kinematic pairs, diagrams and inversion. Mobility and range of movement.Displacement, velocity and acceleration analysis of planar linkages .. Dimensional synthesis formotion, path and function genera mechanisms, Cam profile synthesis.Gears and gear trains. Theory of Mechanisms and Machines by Ghosh and Mallik (EWP).
2055	ME	HEAT & MASS TRANSFER	ME341A	Introduction. Onedimensional and Twodimensional Steady and Transient Conduction.Forced Convection over a flat plate and inside tubes. Natural Convection over a vertical flatplate. Mass Transfer. Boiling 1. Fundamentals of Heat and Mass Transfer by Frank P. Incropera and David P. Dewitt (4th ed.,John Wiley & Sons, New York, 1998)2. Heat Transfer by J.P.Holman (9th ed., Tata McGrawHill Edition, New De Transfer by Yunus A. Cengel (3rd ed., Tata McGrawHill Edition, New Delhi,2007)5. Solar Energy by S. P.Sukhatme (2nd ed., Tata McGrawHill, New Delhi, 1996)
2056	ME	HEAT TRANSFER	ME341	Introduction. Onedimensional and Twodimensional Steady and Transient Conduction.Forced Convection over a flat plate and inside tubes. Natural Convection over a vertical flatplate. Mass Transfer. Boiling 1. Fundamentals of Heat and Mass Transfer by Frank P. Incropera and David P. Dewitt (4th ed.,John Wiley & Sons, New York, 1998)2. Heat Transfer by J.P.Holman (9th ed., Tata McGrawHill Edition, New De Transfer by Yunus A. Cengel (3rd ed., Tata McGrawHill Edition, New Delhi,2007)5. Solar Energy by S. P.Sukhatme (2nd ed., Tata McGrawHill, New Delhi, 1996)
2057	MTH	NUMERICAL ANALYSIS AND SCIENTIFIC COMPUTING-I	MTH308	Root nding problem: Methods and analysis; Interpolation: Methodsand analysis; Approximation: Least squares and minimax approximation; Numerical differentiation;Numerical integration: Methods and decomposition. 1. Elementary Numerical Analysis, an algorithmic approach, S.D. Conte and Carl DeBoor2. An Introduction to Numerical Analysis, Kendall E Atkinson3. Numerical Methods for Scientic and Engineering Cor
2058	MTH	SEVERAL VARIABLE CALCULUS & DEFFERENTIAL GEOMETRY	MTH419	Diferentiation: Denition and examples, Mean value inequality, Tangentplanes to level sets of functions; Implicit mapping theorem, Inverse mapping theoremand applications; Taylor's theorem and app inequality forplane curves; Space curves, FrenetSerret formula for space curves; Local existence theoremcurves. Surfaces: Denition and examples; Tangent planes, Maps between surfaces;First fundamen Egremium. 1. Spivak, Calculus on manifolds, Springer2. Kumaresan, Dierential geometry and Lie groups, TRIM Series3. M P do Carmo, Dierential geometry of curves and surfaces, Prentice Hall4. A Pressley, Elementa
2059	MTH	SEVERAL VARIABLE CALCULUS & DEFFERENTIAL GEOMETRY	MTH305	Diferentiation: Denition and examples, Mean value inequality, Tangentplanes to level sets of functions; Implicit mapping theorem, Inverse mapping theoremand applications; Taylor's theorem and app inequality forplane curves; Space curves, FrenetSerret formula for space curves; Local existence theoremcurves. Surfaces: Denition and examples; Tangent planes, Maps between surfaces;First fundamen Egremium. 1. Spivak, Calculus on manifolds, Springer2. Kumaresan, Dierential geometry and Lie groups, TRIM Series3. M P do Carmo, Dierential geometry of curves and surfaces, Prentice Hall4. A Pressley, Elementa
2060	ME	MANUFACTURING SYSTEMS	ME461	Introduction to manufacturing, Manufacturing system concept.Manufacturing automation, FMS, CIMS, Flow lines and assembly systems, Automated storage /retrieval systems, AGV. Introduction to CAD/ application inmanufacturing. Process planning and Computer Aide Process planning. Group Technology,Opitz System and GT benefits. Material Management and Inventory control, MRP and MRP II.Just i charts, sampling. Total Quality Management. Manufacturingsystem simulation. 1. Computer Integrated Design and Manufacturing by Nanua Singh, john Wiley.2. Computer Aided Manufacturing by Chang, Wysk, Wang, Prentice Hall.3. Computer Aided Manufacturing by Rao, Tewari, Ku
2061	ME	ENERGY SYSTEMS - II	ME401	Introduction: General Theory and Classification of Turbomachines; Similarity andDimensional Analysis; Twodimensional Cascade Theory; Axial and Radial Flow Machines: Turbines,Compressors and Fans Pelton, Francis and Kaplan Turbines; Pump and Cavitation. I. S. L. Dixon and C. A. Hall, Fluid Mechanics and Thermodynamics of Turbomachinery, Elsevier,Sixth Edition, 2010.2. H. Cohen, G. F. C. Rogers and H. I. H. Saravanamuttoo, Gas Turbine Theory, Addison W 2003.4. B.Lakshminarayana, Fluid Dynamics and Heat Transfer ofTurbomachinery, John Wiley and Sons,Inc, 1996.
2062	ME	MANUFACTURING SCIENCE & TECHNOLOGY	ME361A	Introduction to manufacturing processes and system concept and itsevolution; Metal casting: Solidification Mechanism, Gating and Riser Design, Defects andProduct Design; Metal Forming: Fundamentals Machining: ToolSpecifications, Orthogonal and Oblique cutting, Tool wear and Tool Life, Economics ofMachining; Shaping processes for Plastics and Tool Design; joining Processes; UnConventionalMater technologies; Metrology and Selection of Manufacturing Processes. 1. Ghosh, A., Mallik, A.K., Manufacturing Science (2nd edition}, EastWest Press, 2010.2. Lal, G.K., Introduction to Machining Science (2nd edition}, New Age Internationalpublishers, 2009.3. Groover, M.P, Fundamentals of Manufacturing Engineering and Technology, PearsonEducation5. Galyer, j.F.W., Shotbolt, C.R., Metrology for Engineers, ELBS
2063	ME	DESIGN OF MACHINE ELEMENTS	ME351A	Introduction to design of systems and machine elements; Modes of failure,strength, stiffness and stability; Failure theories; Fatigue failure; Probabilistic approach todesign; Design of Bolted and Welded jo Design of shafts. Lab sessions:Detailed design of the above machine elements starting functional specifications to final sizing;Design of a subsystem involving multiple machine elements. Introduction to 1. Mechanical Engineering Design by).E. Shigley, C.R. Mischke & R.G. Budynas, McGraw Hill2. Machine elements in Mechanical Design by R.L. Mott, Prentice Hall3. Mechanical Design by P. Childs, Elsevier Norton
2064	ME	DESIGN OF MACHINE ELEMENTS	ME351	Introduction to design of systems and machine elements; Modes of failure,strength, stiffness and stability; Failure theories; Fatigue failure; Probabilistic approach todesign; Design of Bolted and Welded jo Design of shafts. Lab sessions:Detailed design of the above machine elements starting functional specifications to final sizing;Design of a subsystem involving multiple machine elements. Introduction to 1. Mechanical Engineering Design by).E. Shigley, C.R. Mischke & R.G. Budynas, McGraw Hill2. Machine elements in Mechanical Design by R.L. Mott, Prentice Hall3. Mechanical Design by P. Childs, Elsevier Norton
2065	CHM	CHEMICAL BINDING	CHM621A	Intoroduction, The origin of quantum numbers, some constant potentialproblems, commutation relationships, the atomic selfconsistentfield (SCF),Hartree and Hartree Fock methods. Screening effects Sta features of the chemical bond Chemical Binding.

S.No.	Branch	Course Title	Courses ID	Content
2066	CHM	MATHEMATICS FOR CHEMISTRY	CHM521	Preparations of various organic compounds employing different reactions willbe carried out, with a view to give the student sufficient training in syntheticorganic chemistry.
2067	CHM	ORGANIC PREPARATION LAB	CHM503	Preparations of various organic compounds employing different reactions willbe carried out, with a view to give the student sufficient training in syntheticorganic chemistry.
2068	CHM	INORGANIC CHEMISTRY I	CHM441	Principles of modern inorganic chemistry, discussion of the chemistry of nontransitionelements.
2069	CHM	ORGANIC CHEMISTRY I	CHM401	Stereochemistry, Dynamic stereochemistry, Mechanistic and Stereochemicalaspects, Reactive Intermediates: Carbenes, Nitrenes, Radicals, Carbocations.
2070	CHE	SPECIAL TOPICS	CHE679A	Part 1: Introduction to granular materials, Issues and challenges inexperimental studies; Discrete element method for granular simulations;Hard and soft particle models; Contact force modelling, Algorithmic insights obtained from DEM.Part 2: Continuum models (Balance laws for mass, momentum and energy);Static properties of granular piles (Role of friction, Reynolds dilatancy,Pressure distribution in cylinders, Force chains, Force transmission in granular media);Inelasticgrains in simple geometries, Hydrodynamic description of rapid granularflows, Heuristic theory and introduction to Kinetic theory of inelasticgases.Part 3: Dense granular flow rheology, inertialnumber, Surface flows of granular materials, Flow overinclined plane and in rotating cylinders and heaps, depth average equationsfor surface flows; Mixing and segregation of granular mixtures, Savage'skinetic theory of granular media.
2071	CHE	MATHEMATICAL METHODS IN CHEMICAL ENGINEERING	CHE641A	Modelling, Vector Spaces, Matrices, Linear Operators, Initial Value Problem,Partial Differential Equation, Sturm-Liouville Theory, Separation of Variables,Green's Functions, Transform Techniques, Nonlinear Systems.
2072	CHE	PRINCIPLES OF HETEROGENEOUS CATALYSIS	CHE633A	Adsorption; Energetics; Isotherms and Rates: Experimental aspects of adsorptionand allied phenomena on catalyst surfaces; Pore structure and surface areaestimation and their significance; Important catalytic reactions.
2073	CHE	CHEMICAL REACTION ENGINEERING	CHE631A	Behaviour of chemical reactions; Behaviour of chemical reactors: ideal and nonidealflow, nonisothermal reactor performance, reactor stability; Heterogeneousreactions: interphase and intraphase heat and mass transfer.
2074	CHE	NEW SEPARATION PROCESSES	CHE618A	Separation Factors for Rate Governed Separation Processes, MembraneCharacterization; Reverse Osmosis: Models of Solvent and Solute Transport,Concentration Polarization; Ultrafiltration: Types of Transmembrane Processes; Diafiltration: Process Design; Dialysis: Solute Transport analysisof dialyzer operation, Mode of Dialysis; Electrodialysis: Types of Electro dialysis Ion Transport Fundamentals, Concept of Limiting Current Density.
2075	CHE	TRANSPORT PHENOMENA	CHE611A	Kinematics, Transport theorem, constitutive relations, Equations of motion andtheir solutions, Boundary layer theory, Turbulence; Energy equation and its exactsolutions, Continuity equation for multicomponent systems.
2076	CHE	SUMMER IN-PLANT TRAINING	CHE494A	A written report and an oral presentation/interview after successful completionof an 8week industrial inplant training with a chemical industry.
2077	CHE	UNIT OPERATIONS AND PROCESS CONTROL LABORATORY	CHE492A	Mass Transfer, Heat transfer, chemical reactors, Computer aided data acquisitionand online analysis, Process dynamics and control.
2078	CHE	CHEMICAL ENGINEERING DESIGN	CHE453A	Introduction to Design Process Development, process alternatives, ProcessFlowsheeting and simulation using ASPEN PLUS, Conceptual Process Synthesis,Conceptual design of reactors, pressure vessel design, Heat Exchanger Network Analysis, Scaleup & pilot plant studies, HAZOPand Safety in design, Batch process Design for sequential processing usingSUPERPRO, Continuous plant design (group term project).
2079	BSBE	STRUCTURAL BASIC OF PROTEIN FUNCTION	BSE632A	Conformational studies of biomolecules, protein secondary structure, 3Dstructuresof proteins, DNA structures, membrane proteins. X-ray and NMR techniques,electron crystallography, other spectroscopic techniques. Computational studies to understand the structurefunction relationshipsof proteins, molecular mechanics and dynamics, force fields, biomolecularsimulation programs. Structural genomics, highthroughput screening.
2080	BSBE	NEUROBIOLOGY	BSE629A	Cell and molecular biology of neurons. Membrane potential, local signaling andgeneration of action potential. Mechanism of synaptic transmission. Sensory perception : Vision, hearing, taste, touch and smell. Motor system and the Cerebral cortex learning and memory, sleep and dream.
2081	BSBE	BIOMATERIALS	BSE613A	Introduction to Materials Science: Bulk and surface properties of materials; Polymeric materials; synthesis, characterization, and fabrication methods Inert, biodegradable, hydrogels, Natural, Genetically engineered Biomaterials: Protein structure, interaction of proteins with synthetic material; characterization of cell/material interactions; inflammatory responses; acute inflammation, chronic inflammation, foreign body response.
2082	AE	DYNAMICS AND VIBRATION	AE688A	Rigid body dynamics: Newton's second law, impulse and momentum, moment of a force and angular momentum, work and energy, system of particles, rigid bodies, Euler's equations. Analytical mechanics: Lagrange's theory: frequency response, transform methods, transfer function, transition matrix, Eigen value problem, Modal analysis. Lumped parameter systems: single degree of freedom system, two degrees of freedom systems, vibrations of slender members.
2083	AE	HELICOPTER THEORY : DYNAMICS AND AEROELASTICITY	AE686A	Historical development, configurations of helicopters, rotor system, flight control mechanism, momentum theory and blade element theory in hover, vertical flight and forward flight. Idealization of rotor blades. Flap pitch, lag pitch and flap lag coupling, simple model of rotor fuselage dynamics, longitudinal and lateral stability and control of helicopters. Course Reference : 1. Hopton, J. R. Helicopter Dynamics, Butterworths, London, 1971. 2. Hopton, J. R. Helicopter Dynamics, Butterworths, London, 1971. 3. Hopton, J. R. Helicopter Dynamics, Butterworths, London, 1971.
2084	AE	ROCKET ENGINE DESIGN	AE662A	Introduction to rocket propulsion, Types of rocket engines, Elements of combustion, Chemical propellants and their burning characteristics, Aerothermodynamic design analysis of solid propellant rocket engines.
2085	AE	AEROSPACE STRUCTURAL ANALYSIS - I	AE670A	Free body diagram, equilibrium equations, examples from three dimensional truss problems; bending moment, shear force. Introduction to theory of elasticity, stress, strain, stress-strain relations, constitutive equations, temperature effects, nonhomogeneous materials, modulus weighted sectional properties, thin walled sections. Deflection of beams. Torsion of circular and noncircular sections, thin walled sections, single and double shear. Bending and torsion. Plane strain and plane stress problems in elasticity. Euler's buckling of columns.
2086	AE	FLIGHT DYNAMICS	AE647A	Fundamentals of vectors. Transformation of coordinates. Particle kinematics. Rigid body kinematics. Force equations in a moving frame. Moment equations in a moving frame. Atmospheric flight Dynamics.
2087	AE	FINITE ELEMENT METHODS FOR FLUID DYNAMICS	AE618A	Fundamental concepts; strong form, weak form, Galerkin's approximation; matrix equations, element and global point of view; numerical integration Gaussian quadrature; temporal discretization generalized traction boundary conditions.
2088	AE	B TECH PROJECT	AE471A	Registration for project with the selection of topic & getting started on the design, fabrication work, algorithm etc.
2089	AE	AERODYNAMICS-I	AE610A	Basic fluid mechanics Navier-Stokes equation, vorticity kinematics, Basic potential flows viscous flows including boundary layer theory, turbulence (introduction)
2090	AE	AIRCRAFT PROPULSION	AE652A	Gas turbine engines, performance analysis, subsonic and supersonic diffusers, centrifugal and axial compressors, stage dynamics, compressor stall, axial turbines, compressor turbine matching, gas turbine engine design.
2091	AE	MATHEMATICS FOR AEROSPACE ENGG.	AE602A	differential equations, homogeneous linear equations of second order, nonhomogeneous linear equations of second order, free and forced oscillation problems, problems with variable coefficients, systems of partial differential equations, classification of second order equations, wave propagation in one dimension, parabolic equations, higher dimensional problems, Laplace equation, series solutions, transform methods.
2092	PHY	QUANTUM MECHANICS I	PHY431A	Origins of Quantum Theory, Schrodinger Equation, Application to One Dimensional Problems, WKB Approximation, Central Potentials, Quantum Harmonic Oscillator, Hydrogen Atom, Hilbert Space Formalism, Identical Particles, Spin and Statistics, Pauli Exclusion Principle, Variational Method, Applications to Helium Atom and Hydrogen Molecule Ion.
2093	PHY	QUANTUM MECHANICS II	PHY432A	Bound State Perturbation Theory, Time Dependent Perturbation Theory, Semiclassical Treatment of Radiation, Scattering Theory, Relativistic Wave Equations, Foundational Issues in Quantum Mechanics, Quantum Field Theory (1) Powell and Craseman (2) Feynman Lectures vol. III (3) Merzbacher (4) Gasiorowicz (5) Schiff (6) Sakurai (7) Landau-Lifshitz (8) Cohen-Tannoudji (9) Griffiths
2094	PHY	QUANTUM MECHANICS I	PHY431	Origins of Quantum Theory, Schrodinger Equation, Application to One Dimensional Problems, WKB Approximation, Central Potentials, Quantum Harmonic Oscillator, Hydrogen Atom, Hilbert Space Formalism, Identical Particles, Spin and Statistics, Pauli Exclusion Principle, Variational Method, Applications to Helium Atom and Hydrogen Molecule Ion.

S.No.	Branch	Course Title	Courses ID	Content
				Identical Particles,Spin and Statistics, Pauli Exclusion Principle, Variational Method, Applications to Helium Atomand Hydrogen Molecule Ion.
2095	MTH	THEORY OF COMPUTATION	MTH401A	Regular languages, Deterministic and nondeterministic nite automata,Closure properties, Languages that are and are not regular, State minimizationin deterministic nite automata. Contextfree languages, computability, ChurchTuring thesis, Halting problem, Some undecidableproblems. Computational complexity, Classes P and NP, NPcompleteness, Examples ofNPcomplete problems. 1. H.R. Lewis and C.H. Papadimitriou: Elements of the Theory of Computation, PrenticeHall, 1998.2. J.E. Hopcroft, R. Motwani, J.D. Ullman, Introduction to Automata Theory, Languages,and Computation, P
2096	MTH	THEORY OF COMPUTATION	MTH401	Regular languages, Deterministic and nondeterministic nite automata,Closure properties, Languages that are and are not regular, State minimizationin deterministic nite automata. Contextfree languages, computability, ChurchTuring thesis, Halting problem, Some undecidableproblems. Computational complexity, Classes P and NP, NPcompleteness, Examples ofNPcomplete problems. 1. H.R. Lewis and C.H. Papadimitriou: Elements of the Theory of Computation, PrenticeHall, 1998.2. J.E. Hopcroft, R. Motwani, J.D. Ullman, Introduction to Automata Theory, Languages,and Computation, P
2097	MTH	INTRODUCTION TO CONTINUUM MECHANICS	MTH423	Fundamental concepts; Introduction to Cartesian tensors; Stress tensorsand equilibrium equations; Theory of strain and rate of deformation tensor; Conservationlaws and basic equations; Linear Elasticity inviscid ow, Incompressibleviscous ow, Introduction to boundary layer theory 1. Introduction to Continuum Mechanics{M. Lai, D. Rubin, E. Kremp12. Continuum Mechanics for Engineers{G. T. Mase and G. E. Mase3. Elementary uid mechanics{D. J. Acheson4. Fluid Mechanics{P K Ku
2098	MTH	COMPLEX ANALYSIS	MTH403A	Topology on C, Convergence and continuity. CauchyRiemann equation,Elementary Functions. Power series: Convergence, Exponential, Trigonometric functions.Integration along curves, CauchyGoursat Th fundamentaltheorem of Algebra, Identity theorem, Morera's theorem. Zeros and poles, Residuetheorem, Evaluation of some integrals. Riemann theorem on removable singularities,Essential singularities Maximum modulus principle, Cauchy'stheorem for simply connected domain, Analyticity of complex logarithm. Harmonic functions,Poisson integral formula, Characterization of harmonic functions on half plane. Montel theorem, Riemann mapping theorem. 1. Stein and Shakarchi: Complex Analysis, Princeton Lect. in Analysis2. Gamelin : Complex Analysis, Springer
2099	MTH	COMPLEX ANALYSIS	MTH403	Topology on C, Convergence and continuity. CauchyRiemann equation,Elementary Functions. Power series: Convergence, Exponential, Trigonometric functions.Integration along curves, CauchyGoursat Th fundamentaltheorem of Algebra, Identity theorem, Morera's theorem. Zeros and poles, Residuetheorem, Evaluation of some integrals. Riemann theorem on removable singularities,Essential singularities Maximum modulus principle, Cauchy'stheorem for simply connected domain, Analyticity of complex logarithm. Harmonic functions,Poisson integral formula, Characterization of harmonic functions on half plane. Montel theorem, Riemann mapping theorem. 1. Stein and Shakarchi: Complex Analysis, Princeton Lect. in Analysis2. Gamelin : Complex Analysis, Springer
2100	MTH	ORDINARY DIFFERENTIAL EQUATIONS	MTH421A	Introduction to ODE; Existence and uniqueness of solution; Continuityand dierentiability of solution w.r.t. initial condition and parameters; Generaltheory of linear dierential equations; Methods of solving nonlinear dierential equations; Stability theory for system of linear dierential equations; StrumLiouville boundary value problems, Oscillation theory; Green's function. 1. Martin Brown, Dierential Equations and Their Applications, Springer, 1992.2. S. L. Ross, Introduction to Ordinary Dierential Equations, Wiley, 1980.3. Deo, Lakshmikantham, Raghavendra, Textbook of Ordinary World Scientic,2011.
2101	MTH	ORDINARY DIFFERENTIAL EQUATIONS	MTH421	Introduction to ODE; Existence and uniqueness of solution; Continuityand dierentiability of solution w.r.t. initial condition and parameters; Generaltheory of linear dierential equations; Methods of solving nonlinear dierential equations; Stability theory for system of linear dierential equations; StrumLiouville boundary value problems, Oscillation theory; Green's function. 1. Martin Brown, Dierential Equations and Their Applications, Springer, 1992.2. S. L. Ross, Introduction to Ordinary Dierential Equations, Wiley, 1980.3. Deo, Lakshmikantham, Raghavendra, Textbook of Ordinary World Scientic,2011.
2102	MTH	ABSTRACT ALGEBRA	MTH204A	Some set theoretic notions: Relations, Functions, Partitions, Divisionalgorithm. Various binary operations and examples. Groups and their properties,Subgroups, Cyclic groups and its subgroups, Group of subgroupand Quotient group, Homomorphism, Isomorphism theorems. Symmetric group,Cyclic decomposition of a permutation, Alternating group. Group action, Class equation,Cauchy's theorem, Sylow's Ideals and Quotient ring,Homomorphism, Isomorphism theorems. Polynomial ring, Unique factorization domain,Principal Ideal domain, Euclidean domain, Gaussian ring. 1. Contemporary Abstract Algebra, Joseph A Gallian (Narosa Publishing House, NewDelhi, 1998).2. Algebra, Michael Artin (Prentice Hall of India, New Delhi, 1994).3. Abstract Algebra, John B Fraleigh (Narosa NewDelhi, 1999).
2103	MTH	ABSTRACT ALGEBRA	MTH204	Some set theoretic notions: Relations, Functions, Partitions, Divisionalgorithm. Various binary operations and examples. Groups and their properties,Subgroups, Cyclic groups and its subgroups, Group of subgroupand Quotient group, Homomorphism, Isomorphism theorems. Symmetric group,Cyclic decomposition of a permutation, Alternating group. Group action, Class equation,Cauchy's theorem, Sylow's Ideals and Quotient ring,Homomorphism, Isomorphism theorems. Polynomial ring, Unique factorization domain,Principal Ideal domain, Euclidean domain, Gaussian ring. 1. Contemporary Abstract Algebra, Joseph A Gallian (Narosa Publishing House, NewDelhi, 1998).2. Algebra, Michael Artin (Prentice Hall of India, New Delhi, 1994).3. Abstract Algebra, John B Fraleigh (Narosa NewDelhi, 1999).
2104	HSS	TOPICS IN INDIAN PHILOSOPHY	PHI451A	Introduction and Historical Sketch; Comparative Ontology: Object of knowledge (prameya);Irreducible ontological categories (padarthas) in different schools; Different theories of error(khyativada); Theories of cognitions: Definition and number (indifferent schools); Logic: Nyaya Logic and theory of Inference; Logical fallacies; Nyayadoctrine on formal representation of cognitions. Datta, D. M. Six Ways of Knowing. Calcutta: Calcutta University. 1972Chatterjee, S. and D. Datta. An Introduction to Indian Philosophy. Calcutta: Rupa & Co. 2007's;Chatterjee, S. C. Nyaya theory of knowledge Publishers. 2008Dasgupta, S. N. A History of Indian Philosophy. Cambridge University Press. 1922Potter, K. (ed). Encyclopedia of Indian Philosophy. New Delhi: Motilal Banarasidass
2105	HSS	TOPICS IN INDIAN PHILOSOPHY	PHI451	Introduction and Historical Sketch; Comparative Ontology: Object of knowledge (prameya);Irreducible ontological categories (padarthas) in different schools; Different theories of error(khyativada); Theories of cognitions: Definition and number (indifferent schools); Logic: Nyaya Logic and theory of Inference; Logical fallacies; Nyayadoctrine on formal representation of cognitions. Datta, D. M. Six Ways of Knowing. Calcutta: Calcutta University. 1972Chatterjee, S. and D. Datta. An Introduction to Indian Philosophy. Calcutta: Rupa & Co. 2007's;Chatterjee, S. C. Nyaya theory of knowledge Publishers. 2008Dasgupta, S. N. A History of Indian Philosophy. Cambridge University Press. 1922Potter, K. (ed). Encyclopedia of Indian Philosophy. New Delhi: Motilal Banarasidass
2106	HSS	LITERATURE AND IDEOLOGY	ENG432A	This course will engage with the concept of Ideology, and itsrelationship with literature. Beginning with a definition of Ideology and its various subcategories,the course will go on to examine how ideology operates in the many different kinds of relationships that a literary text may have withideology, ranging from propagation and elaboration, to refutation and critique. We willalso attempt to understand how ideology creates and Eagleton, Terry. Criticism and Ideology. New York: Verso, 1978.Hawkes, David. Ideology. London: Routledge, 2003.Williams, Raymond. Keywords: A Vocabulary of Culture and Society. New York:Oxford University Press, 1978.
2107	ECO	ENVIRONMENTAL ECONOMICS AND POLICY	ECO501	(Basic Mathematical Logic; Set Theory; Review of Functions of a Single Variable and Its Application in Economics: Choice, Utility and Aggregation; Functions of Several Variables: Level Curves, Homogeneous Optimization with Equality andInequality Constraints; The Envelope Theorem and Duality; Dynamics and Dynamic Optimization. 1. Shyam Divan & Armin Rosencranz, Environmental Law and Policy in India, OxfordUniversity Press2. R Rajagopalan, Environmental Studies, Oxford University Press3. Charles D. Kolstad, Environmental Economics Resource and Environmental Economics, Roger Perman, Yue Ma, JamesMcGilvray, and Michael Common6. Environmental Economics for three Huggers and other Skeptics, William K. Jaeger7. Renewable Resource Economic Theory of Exhaustible Resource: R Ranganathan10. Environmental Economics: Hartwick & Olewaler
2108	EE	ELECTROMAGNETIC THEORY	EE340	1. Overview of Static Electric and Magnetic Fields, Steady Electric Currents;2. TimeVarying Electromagnetic Fields, Maxwell's Equations, Boundary Conditions;3. Plane Electromagnetic Waves, Propagation Transmission Lines: TEM waves, Transmission Line Equations, Wave Propagation alongFinite Transmission Lines, Transients on Lines, The Smith Chart;6. Waveguides, Waves in Guided Media, Parallel Plate Elementary Types of Antennas. I. Field and Wave Electromagnetics David K. Cheng, Second Edition, PearsonEducation, 2008.2. Engineering Electromagnetics W A Haytt & J A Buck, Seventh Edition, Tata McGrawHill, 2006.3. Electromagnetics Hill4. Principles ofElectromagnetics Mathhew N.O. Sadiku, Fourth Edition, OxfordUniversity Press.5. Electromagnetics with Applications Kraus and Fleisch, Fifth Edition, McGraw Hill,1999.
2109	EE	DIGITAL ELECTRONICS	EE370A	Digital and Analogue, Boolean and Binary Metric for Digital Circuits and Systems Transistors, Inverters, CMOS Inverter Inverter Parameters and design CombinatinalCircuits, Logic Gates, NAND, NOR, XOR, Multiplexers, Demultiplexers, Shift Registers, Counter, State Machine, PipeliningArrayBased Logic Implementations Memories, Organisation, Cell Designs, Peripherals Digital Systems, CPU Register Transfers and Data Paths Sequencing and Control Instruction Sets Introduction to VHDL and Verilog at appropriate places in the course. Software to be used for solvinghomework assignment problems, including some design problems.

S.No.	Branch	Course Title	Courses ID	Content
2110	EE	DIGITAL ELECTRONICS	EE370	Digital and Analogue, Boolean and Binary Metric for Digital Circuits and Systems Transistors, Inverters, CMOS Inverter Inverter Parameters and design CombinatinalCircuits, Logic Gates, NAND, NOR, XOR, Multiplexers, Demultiplexers, Shift Registers, Counter, State Machine, PipeliningArrayBased Logic Implementations Memories, Organisation, Cell Designs, Peripherals Digital Systems, CPU Register Transfers and Data Paths Sequencing and Control Instruction Sets Introduction to VHDL and Verilog at appropriate places in the course. Software to be used for solvinghomework assignment problems, including some design problems.
2111	ME	ADVANCE MECHANICS OF SOLIDS	ME321A	Introduction to Cartesian tensors; Strains: Concept of strain, derivation ofsmall strain tensor and compatibility; Stress: Derivation of Cauchy relations and,equilibrium and symmetry equations, principal stress and principal strain;symmetry;Boundary Value Problems: Definition of the bvp in linear elasticity including concepts ofuniqueness and superposition; 2d plane stress and plane strain problems, introduction to governing equations of thermoelasticity, torsion of noncircularcross sections, contact problems in 2d, problems on wedges and crack tip fields);3d problems by potential methods; Energy methods and problems. Timoshenko and Goodier, Theory of Elasticity, McGraw Hill Publishing Company, 1970.Bower, Applied Mechanics of Solids, CRC Press, 2009.Saad, Elasticity: Theory Application and Numerics, Academic Press, 2009.
2112	ME	INTRODUCTION TO ELASTICITY	ME321	Introduction to Cartesian tensors; Strains: Concept of strain, derivation ofsmall strain tensor and compatibility; Stress: Derivation of Cauchy relations and,equilibrium and symmetry equations, principal stress and principal strain;symmetry;Boundary Value Problems: Definition of the bvp in linear elasticity including concepts ofuniqueness and superposition; 2d plane stress and plane strain problems, introduction to governing equations of thermoelasticity, torsion of noncircularcross sections, contact problems in 2d, problems on wedges and crack tip fields);3d problems by potential methods; Energy methods and problems. Timoshenko and Goodier, Theory of Elasticity, McGraw Hill Publishing Company, 1970.Bower, Applied Mechanics of Solids, CRC Press, 2009.Saad, Elasticity: Theory Application and Numerics, Academic Press, 2009.
2113	HSS	SOCIAL COGNITION	PSY468A	The major emphasis of the course will be to provide an overview of theindividual level social cognitive processes. In addition to this, students will also be exposed toconcepts pertaining to collective and social cognitive processes. The course will also aim to understand the culturalunderpinning of human social cognition. Overall, the course will foster an understanding ofcognitive and social processes of human sense making in the social world. Both, individual and social cognitive processes will be covered in the course. Augoustinos, M. and Walker, I. Social Cognition: An Integrated Introduction. London: SagePublications. 1995.Brewer, M. B. and Hewstone, M (eds.). Social Cognition. Blackwell Publishing: Cornwall, U.K.2003. Tajfel, H. and Turner, J. Social Identity Theory and Social Movement. Cambridge University Press: Cambridge, U.K. 2004. people. MIT Press: Cambridge. !999.Postmes, T. and Jetten, J (eds.). Individuality and the Group: Advances in Social Identity. SagePublications: London. 2006.In addition to above texts, reading assignments will be given.
2114	HSS	GLOBAL COMMUNICATION	ENG451	The focus of the course is on the use of English language in national/multinational corporations andcommunication mediated through mobile, telephone, email, internet and other advanced technologies.3. MediaCommunication 3. Ethnographic Perceptions 4. Global Imbalances in Informational and Cultural Exchange 5.Benefits of Intercultural Communication 6. Intercultural Competence 7. Interpersonal Communication 8. Cross Cultural Communication 9. Intercultural Communication 10. Intercultural Communication 11.Qualities of Effective Report Writing 12. Writing Effective Emails 13. Communication Information ThroughVisuals 14. Channels of Nonverbal Communication 15. Competence in Interviewing Contexts 16. Competence in Interviewing Contexts 17. Competence in Interviewing Contexts 18. Competence in Interviewing Contexts 19. Competence in Interviewing Contexts 20. Competence in Interviewing Contexts
2115	HSS	THE INDO-ANGLIAN NOVEL	ENG431	The genre of the novel is seen as a colonial legacy as it developed in Indiasince the arrival of the British. The Indian English novel bears the colonial imprint more as it iswritten in the coloniser's language. However, the IndianEnglish novel has acquired quite a reputation in world literature today and has writers who have remarkable felicity of style and catholicity of subject matter. The course will trace the development of this progress. Novels that have made a mark nationally and internationallywill be used to exemplify this. Mehrotra, Arvind Krishna. History of Indian literature in English. New York: ColumbiaUniversity Press, 2003.Mukherjee, Meenakshi. The Twice Born Fiction: Themes and Techniques of the Indian Novel in English. New York: Oxford University Press, 2003.
2116	EE	ELECTRICAL ENGINEERING LABORATORY -II	EE381A	Experiments from various areas of Electrical Engineering with emphasis on digitalelectronics, communication, machines, drives and power systems, and electromagnetics.This course has three labs: Electrical Engineering Laboratory -II
2117	EE	ELECTROMAGNETIC THEORY	EE340A	1. Overview of Static Electric and Magnetic Fields, Steady Electric Currents;2. TimeVarying Electromagnetic Fields, Maxwell's Equations, Boundary Conditions;3. Plane Electromagnetic Waves, Propagation of Plane Waves, Transmission Lines: TEM waves, Transmission Line Equations, Wave Propagation alongFinite Transmission Lines, Transients on Lines, The Smith Chart;6. Waveguides, Waves in Guided Media, Parallel Plate Waveguides, Rectangular Waveguides, Cylindrical Waveguides, Spherical Waveguides, Antennas. Elementary Types of Antennas. I. Field and Wave Electromagnetics David K. Cheng, Second Edition, PearsonEducation, 2008.2. Engineering Electromagnetics W A Haytt & J A Buck, Seventh Edition, Tata McGrawHill, 2006.3. Electromagnetics W A Haytt & J A Buck, Seventh Edition, Tata McGrawHill, 2006.4. Principles ofElectromagnetics Mathew N.O. Sadiku, Fourth Edition, OxfordUniversity Press.5. Electromagnetics with Applications Kraus and Fleisch, Fifth Edition, McGraw Hill,1999.
2118	HSS	SOCIAL COGNITION	PSY468	The major emphasis of the course will be to provide an overview of theindividual level social cognitive processes. In addition to this, students will also be exposed toconcepts pertaining to collective and social cognitive processes. The course will also aim to understand the culturalunderpinning of human social cognition. Overall, the course will foster an understanding ofcognitive and social processes of human sense making in the social world. Both, individual and social cognitive processes will be covered in the course. Augoustinos, M. and Walker, I. Social Cognition: An Integrated Introduction. London: SagePublications. 1995.Brewer, M. B. and Hewstone, M (eds.). Social Cognition. Blackwell Publishing: Cornwall, U.K.2003. Tajfel, H. and Turner, J. Social Identity Theory and Social Movement. Cambridge University Press: Cambridge, U.K. 2004. people. MIT Press: Cambridge. !999.Postmes, T. and Jetten, J (eds.). Individuality and the Group: Advances in Social Identity. SagePublications: London. 2006.In addition to above texts, reading assignments will be given.
2119	HSS	ORGANIZATIONAL AND ADMINISTRATIVE PSYCHOLOGY	PSY458A	Organizations and the systems concept, Chronological sequence ofdevelopment of thought in Organizational Behavior area, A road map for understandingorganizational behavior: The elementary aspects of organizational behavior: Satisfaction, Motivation andperformance, Communication, Conflict and Negotiation, Leadership, Power, Control,Organizational Citizenship, and Antiorganization behaviors, Organizational Change andOrganizational Development. Significance of individual and group effectiveness throughawareness of self and others. This last portion requires experiential learning. I. Katz, D., & Kahn, R.L. (1978). The social psychology of organizations. (2nd ed.). NewYork: John Wiley & Sons.2. Robbins, S.P. Organizational behavior. New Delhi: PrenticeHall of India PrivateLimited. (13th edition) India Pvt Ltd.
2120	HSS	ORGANIZATIONAL AND ADMINISTRATIVE PSYCHOLOGY	PSY458	Organizations and the systems concept, Chronological sequence ofdevelopment of thought in Organizational Behavior area, A road map for understandingorganizational behavior: The elementary aspects of organizational behavior: Satisfaction, Motivation andperformance, Communication, Conflict and Negotiation, Leadership, Power, Control,Organizational Citizenship, and Antiorganization behaviors, Organizational Change andOrganizational Development. Significance of individual and group effectiveness throughawareness of self and others. This last portion requires experiential learning. I. Katz, D., & Kahn, R.L. (1978). The social psychology of organizations. (2nd ed.). NewYork: John Wiley & Sons.2. Robbins, S.P. Organizational behavior. New Delhi: PrenticeHall of India PrivateLimited. (13th edition) India Pvt Ltd.
2121	HSS	INTERPERSONAL DYNAMICS	PSY457	General introduction to the scope of interpersonal dynamics, and the methods used. Major theoretical approaches (Social exchange theory, FRIEDMAN, InterpersonalCircumplex). Interpersonal judgment and interpersonal communication (style, Transactional Analysis). Interpersonal influence and power. Interpersonal conflict and itsmanagement. The social self (selfmonitoring and selfdisclosure). Shyness and loneliness as personality traits (intelligence). Hinde: Towards Understanding Relationships.Berne : Games People Play.Duck & Gilmour : Personal Relationships.No single textbook is available that covers all the topics included in the course. Therefore, a variety of readings will be assigned from books that deal with topics relevant to the course.
2122	HSS	SOCIAL PSYCHOLOGY	PSY454A	Social psychology as a branch of psychology its historical background, Majorparadigms of contemporary social psychology, Methods adopted in social psychology,Aggression and violence, Person perception, Social perception, Compliance, Obedience to authority,Distributive and procedural justice, Group dynamics and intergroup relations, Issues of gender,poverty and marginalization Baron, R. A., Branscombe, N. R., & Byrne, D. (2009). Social Psychology (12th ed.). Boston,MA: Pearson/Allyn and Bacon.Gergen, Kenneth J. (1999). An invitation to social construction. Thousand Oaks, CA: Sage.Oxford, UK and Cambridge, MT: Blackwell.Misra, G. & Dalal, A. K. (Eds.) (2006). New directions in Indian psychology (Vol. 1): SocialPsychology. New Delhi: Sage.Strickland, L. H., Aboud, F. E., & Gergen, K. J. (Eds.) (2006). New directions in Indian psychology (Vol. 1): SocialPsychology. New Delhi: Sage.
2123	HSS	SOCIAL PSYCHOLOGY	PSY454	Social psychology as a branch of psychology its historical background, Majorparadigms of contemporary social psychology, Methods adopted in social psychology,Aggression and violence, Person perception, Social perception, Compliance, Obedience to authority,Distributive and procedural justice, Group dynamics and intergroup relations, Issues of gender,poverty and marginalization Baron, R. A., Branscombe, N. R., & Byrne, D. (2009). Social Psychology (12th ed.). Boston,MA: Pearson/Allyn and Bacon.Gergen, Kenneth J. (1999). An invitation to social construction. Thousand Oaks, CA: Sage.Oxford, UK and Cambridge, MT: Blackwell.Misra, G. & Dalal, A. K. (Eds.) (2006). New directions in Indian psychology (Vol. 1): SocialPsychology. New Delhi: Sage.Strickland, L. H., Aboud, F. E., & Gergen, K. J. (Eds.) (2006). New directions in Indian psychology (Vol. 1): SocialPsychology. New Delhi: Sage.
2124	HSS	PSYCHOLOGY OF ADJUSTMENT	PSY451A	The course content, along with lecture wise breakup, is given below.Examinations: Choice of format of the examinations is left to the instructor.Laboratory content: As of now, the time table does not provide the understanding of the course content. I. Wayne Weiten, Dana S Dunn, and Elizabeth Yost Hammer (2011). PsychologyApplied to Modern Life: Adjustment in the 21st Century. Wadsworth publishing(10th edition).2. Robert C. Carson, James Neill (1984). Job stress and burnout. Sage.

S.No.	Branch	Course Title	Courses ID	Content
2125	HSS	PSYCHOLOGY OF ADJUSTMENT	PSY451	The course content, along with lecture wise breakup, is given below.Examinations: Choice of format of the examinations is left to the instructor.Laboratory content: As of now, the time table does not provide the understanding of the course content. 1. Wayne Weiten, Dana S Dunn, and Elizabeth Yost Hammer (2011). Psychology Applied to Modern Life: Adjustment in the 21st Century. Wadsworth publishing (10th edition). 2. Robert C. Carson, James Neale (1984). Job stress and burnout. Sage.
2126	HSS	PHILOSOPHY OF SCIENCE	PHI446A	On what Philosophy of Science is; Logic and Empiricism; Theory and Observation; Evidence, Confirmation and Falsificationism; Induction and Probability; Scientific Revolution versus Normal Science; Science, dispositions and causes and Conditionals, Causality and Indeterminism, Bayesian and Modern theories of Evidence; Science and Values * Peter Godfrey-Smith (2003), Theory and Reality: An Introduction to Philosophy of Science, University of Chicago Press. [Standard Text for this course] James Ladyman (2002), Understanding Philosophy of Science, Cambridge University Press. Chalmers, A. F. (1982). What Is This Thing Called Science? Open University Press, Milton Keynes, second edition. Thomas Kuhn (1996) The Structure of Scientific Revolutions, University of Chicago Press. (2nd edition, 1996), Philosophy of natural science, Prentice Hall foundations of philosophy series. Englewood Cliffs, N.J.; Prentice Hall. Gillies, D. A. (1993), Philosophy of Science in the Twentieth Century: Four Central Themes, University Press, USA.
2127	HSS	PRINCIPLES OF COMMUNICATION DESIGN	ART411A	History of communication, Psychology of perception, Signs and symbols, Elements of visual display, Layout design, History of typography, Information structuring, Claymation, Visual branding, Case studies on composition, typography, illustration techniques, story boarding, claymation, logo design and advertisement campaigns. 1. Art and Visual perception Rudolph Arneheim 2. Manufacturing Consent: The Political Economy of the Mass Media: Noam Chomsky 3. Design and Form Johannes Itten 4. The Visual Display of Quantitative Information Edward Tufte 5. Evidence and Narrative: Edward Tufte 6. Principles of Three Dimensional design Stephen Luecking 7. Principles of 2D design Wucius Wong 8. Principles of Form and Design Wucius Wong 9. Pedagogical sketches
2128	HSS	LITERATURE AND CENSORSHIP	ENG439	Censorship is defined as the restrictions that are imposed on the writer and writings by centres of power like the state, religious institutions or other social organizations. These external restrictions can lead to self-censorship and thought control. The course attempts to understand why writing is perceived as a threat to established institutions like the state or organized religion, and the impact of such restrictions on writing. Besides, it also explores the efficacy of this restrictive measure will be evaluated with the help of texts that have created controversies for going against accepted notions of morality or religious beliefs. Green, Jonathon and Nicholas Karolides. The Encyclopedia on Censorship. New York: Info base Publishers, 2005. Jansen, Sue Curry. Censorship: The Knot that binds Power and Knowledge. Oxford: Oxford University Press, 2003.
2129	EE	DIGITAL SIGNAL PROCESSING	EE301A	Sampling and Reconstruction of continuous time signals, Characterization and properties of discrete time signals and systems, Computation of the discrete time Fourier transform and its properties, Complex representation of discrete time signals, Properties, Transform analysis of linear time invariant systems, Implementation of structures for discrete time systems, Digital filter design techniques, Homomorphic filtering, Applications of DSP in speech processing. [1] Discrete Time Signal Processing (Second Edition), Alan V. Oppenheim, Ronald W. Schaffer, and John R. Buck, Pearson Education India
2130	EE	DIGITAL SIGNAL PROCESSING	EE301	Sampling and Reconstruction of continuous time signals, Characterization and properties of discrete time signals and systems, Computation of the discrete time Fourier transform and its properties, Complex representation of discrete time signals, Properties, Transform analysis of linear time invariant systems, Implementation of structures for discrete time systems, Digital filter design techniques, Homomorphic filtering, Applications of DSP in speech processing. [1] Discrete Time Signal Processing (Second Edition), Alan V. Oppenheim, Ronald W. Schaffer, and John R. Buck, Pearson Education India
2131	HSS	MODERN ART	ART402A	Art movements from 1840s to 1960s will be discussed in class from an aesthetic and socioeconomic perspective. The study of movements will also be assisted with an exploration of philosophy of art where movements like Abstract expressionism, Pop art, Minimalism, Introduction to Post modernism. Philosophy of art Formalism (Clive Bell and Roger Fry), Dream in psycho analysis (Freud), Existentialism (Sartre, Camus), Modernism. 1. Story of Modern Art, Cheney, Sheldon 2. Art in theory: Charles Harrison & Paul Wood 3. Abstract art: Anna Moszynska 4. Shape of the pocket: John Berger 5. Marg (Journal) 6. Art India (Journal)
2132	HSS	MODERN ART	ART402	Art movements from 1840s to 1960s will be discussed in class from an aesthetic and socioeconomic perspective. The study of movements will also be assisted with an exploration of philosophy of art where movements like Abstract expressionism, Pop art, Minimalism, Introduction to Post modernism. Philosophy of art Formalism (Clive Bell and Roger Fry), Dream in psycho analysis (Freud), Existentialism (Sartre, Camus), Modernism. 1. Story of Modern Art, Cheney, Sheldon 2. Art in theory: Charles Harrison & Paul Wood 3. Abstract art: Anna Moszynska 4. Shape of the pocket: John Berger 5. Marg (Journal) 6. Art India (Journal)
2133	HSS	LANGUAGES OF SOUTH ASIA	ENG448	South Asia as a linguistic region 5 Indo Aryan languages 5 Dravidian languages 4 Tibeto Burman languages 3 Munda languages 3 Languages of the Andamans 2 Convergence features 4 Language contact 4. Conflicts 2. Abbi, Anvita. 1997. Languages of Tribal and Indigenous Peoples of India: The Ethnic Space. MLBD Abbi, Anvita. 2006. Endangered Languages of the Andaman Islands. Lincom Europa.
2134	HSS	TOPICS IN LITERARY MOVEMENTS	ENG435	The literature of any language is usually categorized in terms of movements, or discernible trends in style and theme at a particular point in the course of its development. These trends in literature constitute Realism, Naturalism, Modernism and Postmodernism. These are but generalized categories that actually include a wide variety of topics which have been dealt with in diverse ways by different writers. The course will focus on particular movements. It will also take into account the changing perspectives on writing and writers that usually change according to the movements. Each topic will be illustrated using a seminal text. Peck, John and Martin Coyle. A Brief History of English Literature. New York: Palgrave Macmillan, 2002. Sanders, Andrew. Short Oxford History of English Literature. Oxford: Oxford University Press, 2004.
2135	HSS	MODERN BRITISH AND AMERICAN NOVEL	ENG434A	1. Introduction to the Novel 2. Novel in the Anglo American literary tradition 3. Major concerns in 19th & 20th century British, American Novel 4. Romance & the Novel in America 5. Hawthorne's The Scarlet Letter 6. Feminism and Woolf 10. Woolf's To the Lighthouse 11. Forster's A Passage to India 12. American Modernism 13. Fitzgerald's Tender is the Night 14. Hemingway's A Farewell to Arms 15. Summary 1. Margaret Drabble. The Oxford Companion to English Literature. 2. The Cambridge Companion to American Literature.
2136	HSS	MODERN BRITISH AND AMERICAN NOVEL	ENG434	1. Introduction to the Novel 2. Novel in the Anglo American literary tradition 3. Major concerns in 19th & 20th century British, American Novel 4. Romance & the Novel in America 5. Hawthorne's The Scarlet Letter 6. Feminism and Woolf 10. Woolf's To the Lighthouse 11. Forster's A Passage to India 12. American Modernism 13. Fitzgerald's Tender is the Night 14. Hemingway's A Farewell to Arms 15. Summary 1. Margaret Drabble. The Oxford Companion to English Literature. 2. The Cambridge Companion to American Literature.
2137	EE	POWER SYSTEMS	EE330A	1. Introduction to power system structure, types of components. 2. Power Calculations, in balanced three phase Circuits, Method of Symmetrical Components for unbalanced three phase Circuits 3. Different Models for short and long transmission lines, steady state operation, shunt and series compensation, transients on transmission lines. 6. Network models of power system using admittance and impedance 7. Newton Raphson methods 8. Analysis of faulted power systems using sequence networks. 9. Basic stability analysis, derivation of swing equation, equal area criteria, calculation of critical clearing angle, number of generators in a plant. 1. Power System Analysis John J. Grainger & William D. Stevenson Jr, McGraw Hill 1994. 2. Power System Analysis and design J. Duncan Glover, M. Sarma 2nd edition, PWS Publishing Co Boston 2001. 3. Power System Analysis
2138	EE	POWER SYSTEMS	EE330	1. Introduction to power system structure, types of components. 2. Power Calculations, in balanced three phase Circuits, Method of Symmetrical Components for unbalanced three phase Circuits 3. Different Models for short and long transmission lines, steady state operation, shunt and series compensation, transients on transmission lines. 6. Network models of power system using admittance and impedance 7. Newton Raphson methods 8. Analysis of faulted power systems using sequence networks. 9. Basic stability analysis, derivation of swing equation, equal area criteria, calculation of critical clearing angle, number of generators in a plant. 1. Power System Analysis John J. Grainger & William D. Stevenson Jr, McGraw Hill 1994. 2. Power System Analysis and design J. Duncan Glover, M. Sarma 2nd edition, PWS Publishing Co Boston 2001. 3. Power System Analysis
2139	HSS	LITERATURE & ADAPTATION	ENG446	This course will examine the processes and politics of adaptation as text travel across genres, media, and cultures. Focusing primarily on literary texts, graphic narratives and films, it will investigate the dynamic process, it will delve into topics such as the semiotics of translation, the 'fidelity' debate, the relationship between adaptation and appropriation, ideology and intention in the creation of adaptations. 1) Hutcheon, Linda. A Theory of Adaptation. New York: Routledge, 2006. 2) Sanders, Julie. Adaptation and Appropriation. New York: Routledge, 2006.
2140	HSS	LITERATURE & THE INDIVIDUAL	ENG445A	The precarious role of an individual in a dynamic society: totalitarian, democratic, technological, globalized, etc. has been an intense and perennial subject of concern in literature. Many literary works deal with individual survival, pangs of isolation and alienation, and attempts for salvaging sanity from a maddeningly disordered world. These works, through fables, fantasies, realistic as well as futuristic representations of their societies. Apart from a utopian prophecy of deferred but ultimate triumph of humanity, these literary works often inspire, and instill in the readers enduring ideas, ideals, and values. The works for study include: Narayan, Saul Bellow, Somerset Maugham and Upamanyu Chatterjee to semi philosophical and popular texts of Ayn Rand, Khalil Gibran, Paulo Coelho, Richard Bach and Robert Pirsig.

S.No.	Branch	Course Title	Courses ID	Content
2141	HSS	LITERATURE & THE INDIVIDUAL	ENG445	The precarious role of an individual in a dynamic societytotalitarian, democratic,technological, globalized, etchas been an intense and perennial subject of concern in literature.Many literary works deal with individual survival, pangs of isolation andalienation, and attempts for salvaging sanity from a maddeningly disordered world. Theseworks, through fables, fantasies, realistic as well as futuristic representations of theirsocieties. Apart from a utopian prophecy of deferred but ultimate triumph of humanity, theseliterary works often inspire, and instill in the readers enduring ideas, ideals, and values. Theworks for study range from classical texts of Greek antiquity to modern Indian literature by Rabindranath Narayan, Saul Bellow, Somerset Maugham and UpamanyuChatterjee to semiphilosophical and popular texts of Ayn Rand, Khalil Gibran, Paulo Coelho,Richard Bach and Robert Pirsig.
2142	HSS	TOPICS IN LITERARY GENRES	ENG440A	This course will explore the concept of literary genre and itsimplications for both the production and reception of literary works. Delving beyond thebroad divisions of poetry, prose, and drama, this course will examine the characteristics of various genres such as epic, novel, short story, play, memoir, Nair or romance/gothic/science fiction/fantasy, etc. It will situate these genreswithin a historical context and explore their regional variations by studying Western textsalongside nonWestern ones from the same period. The course will also explore the evolution of various literarygenres and the extent to which genres are contained within culturaltemporal boundaries. No specific textbook. Reference material willbe decided on the basis of the specific group of genres under study during a particularsemester, and may include the following:1) Frow, John. Genre. New York: Routledge, 2009. 2) Fowler, E. M. Language and Literature. Oxford: Clarendon Press, 1968. 3) Genette, Gerard. Fictional Function. Ithaca, NY: Cornell University Press, 1988. 4) Genette, Gerard. Narrative Discourse. Ithaca, NY: Cornell University Press, 1988. 5) Genette, Gerard. Palimpsest. Ithaca, NY: Cornell University Press, 1997. 6) Genette, Gerard. Paratext. Cambridge: Cambridge University Press, 2004. 7) Genette, Gerard. Poetics. Ithaca, NY: Cornell University Press, 1988. 8) Genette, Gerard. The Novel. Ithaca, NY: Cornell University Press, 1988. 9) Genette, Gerard. The Novel's History. Ithaca, NY: Cornell University Press, 1988. 10) Genette, Gerard. The Novel's Theory. Ithaca, NY: Cornell University Press, 1988. 11) Genette, Gerard. The Novel's Practice. Ithaca, NY: Cornell University Press, 1988. 12) Genette, Gerard. The Novel's Poetics. Ithaca, NY: Cornell University Press, 1988. 13) Genette, Gerard. The Novel's Politics. Ithaca, NY: Cornell University Press, 1988. 14) Genette, Gerard. The Novel's Economics. Ithaca, NY: Cornell University Press, 1988. 15) Genette, Gerard. The Novel's Sociology. Ithaca, NY: Cornell University Press, 1988. 16) Genette, Gerard. The Novel's Psychology. Ithaca, NY: Cornell University Press, 1988. 17) Genette, Gerard. The Novel's Philosophy. Ithaca, NY: Cornell University Press, 1988. 18) Genette, Gerard. The Novel's Religion. Ithaca, NY: Cornell University Press, 1988. 19) Genette, Gerard. The Novel's Law. Ithaca, NY: Cornell University Press, 1988. 20) Genette, Gerard. The Novel's Medicine. Ithaca, NY: Cornell University Press, 1988. 21) Genette, Gerard. The Novel's Education. Ithaca, NY: Cornell University Press, 1988. 22) Genette, Gerard. The Novel's Military. Ithaca, NY: Cornell University Press, 1988. 23) Genette, Gerard. The Novel's Art. Ithaca, NY: Cornell University Press, 1988. 24) Genette, Gerard. The Novel's Science. Ithaca, NY: Cornell University Press, 1988. 25) Genette, Gerard. The Novel's Technology. Ithaca, NY: Cornell University Press, 1988. 26) Genette, Gerard. The Novel's Culture. Ithaca, NY: Cornell University Press, 1988. 27) Genette, Gerard. The Novel's Society. Ithaca, NY: Cornell University Press, 1988. 28) Genette, Gerard. The Novel's Community. Ithaca, NY: Cornell University Press, 1988. 29) Genette, Gerard. The Novel's Nation. Ithaca, NY: Cornell University Press, 1988. 30) Genette, Gerard. The Novel's World. Ithaca, NY: Cornell University Press, 1988.
2143	HSS	TOPICS IN LITERARY GENRES	ENG440	This course will explore the concept of literary genre and itsimplications for both the production and reception of literary works. Delving beyond thebroad divisions of poetry, prose, and drama, this course will examine the characteristics of various genres such as epic, novel, short story, play, memoir, Nair or romance/gothic/science fiction/fantasy, etc. It will situate these genreswithin a historical context and explore their regional variations by studying Western textsalongside nonWestern ones from the same period. The course will also explore the evolution of various literarygenres and the extent to which genres are contained within culturaltemporal boundaries. No specific textbook. Reference material willbe decided on the basis of the specific group of genres under study during a particularsemester, and may include the following:1) Frow, John. Genre. New York: Routledge, 2009. 2) Fowler, E. M. Language and Literature. Oxford: Clarendon Press, 1968. 3) Genette, Gerard. Fictional Function. Ithaca, NY: Cornell University Press, 1988. 4) Genette, Gerard. Narrative Discourse. Ithaca, NY: Cornell University Press, 1988. 5) Genette, Gerard. Palimpsest. Ithaca, NY: Cornell University Press, 1997. 6) Genette, Gerard. Paratext. Cambridge: Cambridge University Press, 2004. 7) Genette, Gerard. Poetics. Ithaca, NY: Cornell University Press, 1988. 8) Genette, Gerard. The Novel. Ithaca, NY: Cornell University Press, 1988. 9) Genette, Gerard. The Novel's History. Ithaca, NY: Cornell University Press, 1988. 10) Genette, Gerard. The Novel's Theory. Ithaca, NY: Cornell University Press, 1988. 11) Genette, Gerard. The Novel's Practice. Ithaca, NY: Cornell University Press, 1988. 12) Genette, Gerard. The Novel's Poetics. Ithaca, NY: Cornell University Press, 1988. 13) Genette, Gerard. The Novel's Politics. Ithaca, NY: Cornell University Press, 1988. 14) Genette, Gerard. The Novel's Economics. Ithaca, NY: Cornell University Press, 1988. 15) Genette, Gerard. The Novel's Sociology. Ithaca, NY: Cornell University Press, 1988. 16) Genette, Gerard. The Novel's Psychology. Ithaca, NY: Cornell University Press, 1988. 17) Genette, Gerard. The Novel's Philosophy. Ithaca, NY: Cornell University Press, 1988. 18) Genette, Gerard. The Novel's Religion. Ithaca, NY: Cornell University Press, 1988. 19) Genette, Gerard. The Novel's Law. Ithaca, NY: Cornell University Press, 1988. 20) Genette, Gerard. The Novel's Medicine. Ithaca, NY: Cornell University Press, 1988. 21) Genette, Gerard. The Novel's Education. Ithaca, NY: Cornell University Press, 1988. 22) Genette, Gerard. The Novel's Military. Ithaca, NY: Cornell University Press, 1988. 23) Genette, Gerard. The Novel's Art. Ithaca, NY: Cornell University Press, 1988. 24) Genette, Gerard. The Novel's Science. Ithaca, NY: Cornell University Press, 1988. 25) Genette, Gerard. The Novel's Technology. Ithaca, NY: Cornell University Press, 1988. 26) Genette, Gerard. The Novel's Culture. Ithaca, NY: Cornell University Press, 1988. 27) Genette, Gerard. The Novel's Society. Ithaca, NY: Cornell University Press, 1988. 28) Genette, Gerard. The Novel's Community. Ithaca, NY: Cornell University Press, 1988. 29) Genette, Gerard. The Novel's Nation. Ithaca, NY: Cornell University Press, 1988. 30) Genette, Gerard. The Novel's World. Ithaca, NY: Cornell University Press, 1988.
2144	EE	POWER ELECTRONICS	EE360A	1. Power Electronics components: Switching characteristics of power devices e.g. diodes, thyristors, GTOs, MOS transistors, IGBTs etc; snubber circnits, calculation of switching losses, resonantZVS and ZCS converters, single phase and three phase rectifier configurations, various modesof operation, a.c .regulators, and reactive power compensators.3. Inverters: Basic Single phase and three phase VSI confignrations, modulation techniques, dynamic analysis of Buck, Boost, BuckBoost and Cukconverters, switchmode power supplies.
2145	EE	POWER ELECTRONICS	EE360	1. Power Electronics components: Switching characteristics of power devices e.g. diodes, thyristors, GTOs, MOS transistors, IGBTs etc; snubber circnits, calculation of switching losses, resonantZVS and ZCS converters, single phase and three phase rectifier configurations, various modesof operation, a.c .regulators, and reactive power compensators.3. Inverters: Basic Single phase and three phase VSI confignrations, modulation techniques, dynamic analysis of Buck, Boost, BuckBoost and Cukconverters, switchmode power supplies.
2146	HSS	PHILOSOPHICAL LOGIC	PHI455A	Brief overview of first order logic, Problem of Logical Consequence, Deviant Logics: ManyvaluedLogic I: Three valued Logics, Many Valued Logic2: Many valued logic and Degrees oftruth, Para consistent Logic, Problem of Counterfactuals; LogicalPluralism 1. James Graham Priest, An Introduction to NonClassical Logic, Cambridge: CambridgeUniversity Press 2001 (standard Text for the course),22. Colin McGinn, Logical Properties, Oxford University Press, 2002.3. Peter Smith, Formal Logic, Cambridge University Press, 2003.4. Peter Smith, Modal Logic. Cambridge [England]:Cambridge University Press, 2001.9. Lewis, David K. Counterfactuals. Cambridge, Harvard University Press, 1973.10. S. Read, Thinking about Logic, Oxford University Press, 2002.11. S. Read, Thinking about Logic, Oxford University Press, 2002.12. S. Read, Thinking about Logic, Oxford University Press, 2002.13. S. Read, Thinking about Logic, Oxford University Press, 2002.14. S. Read, Thinking about Logic, Oxford University Press, 2002.15. S. Read, Thinking about Logic, Oxford University Press, 2002.16. S. Read, Thinking about Logic, Oxford University Press, 2002.17. S. Read, Thinking about Logic, Oxford University Press, 2002.18. S. Read, Thinking about Logic, Oxford University Press, 2002.19. S. Read, Thinking about Logic, Oxford University Press, 2002.20. S. Read, Thinking about Logic, Oxford University Press, 2002.21. S. Read, Thinking about Logic, Oxford University Press, 2002.22. S. Read, Thinking about Logic, Oxford University Press, 2002.23. S. Read, Thinking about Logic, Oxford University Press, 2002.24. S. Read, Thinking about Logic, Oxford University Press, 2002.25. S. Read, Thinking about Logic, Oxford University Press, 2002.26. S. Read, Thinking about Logic, Oxford University Press, 2002.27. S. Read, Thinking about Logic, Oxford University Press, 2002.28. S. Read, Thinking about Logic, Oxford University Press, 2002.29. S. Read, Thinking about Logic, Oxford University Press, 2002.30. S. Read, Thinking about Logic, Oxford University Press, 2002.
2147	HSS	PHILOSOPHICAL LOGIC	PHI455	Brief overview of first order logic, Problem of Logical Consequence, Deviant Logics: ManyvaluedLogic I: Three valued Logics, Many Valued Logic2: Many valued logic and Degrees oftruth, Para consistent Logic, Problem of Counterfactuals; LogicalPluralism 1. James Graham Priest, An Introduction to NonClassical Logic, Cambridge: CambridgeUniversity Press 2001 (standard Text for the course),22. Colin McGinn, Logical Properties, Oxford University Press, 2002.3. Peter Smith, Formal Logic, Cambridge University Press, 2003.4. Peter Smith, Modal Logic. Cambridge [England]:Cambridge University Press, 2001.9. Lewis, David K. Counterfactuals. Cambridge, Harvard University Press, 1973.10. S. Read, Thinking about Logic, Oxford University Press, 2002.11. S. Read, Thinking about Logic, Oxford University Press, 2002.12. S. Read, Thinking about Logic, Oxford University Press, 2002.13. S. Read, Thinking about Logic, Oxford University Press, 2002.14. S. Read, Thinking about Logic, Oxford University Press, 2002.15. S. Read, Thinking about Logic, Oxford University Press, 2002.16. S. Read, Thinking about Logic, Oxford University Press, 2002.17. S. Read, Thinking about Logic, Oxford University Press, 2002.18. S. Read, Thinking about Logic, Oxford University Press, 2002.19. S. Read, Thinking about Logic, Oxford University Press, 2002.20. S. Read, Thinking about Logic, Oxford University Press, 2002.21. S. Read, Thinking about Logic, Oxford University Press, 2002.22. S. Read, Thinking about Logic, Oxford University Press, 2002.23. S. Read, Thinking about Logic, Oxford University Press, 2002.24. S. Read, Thinking about Logic, Oxford University Press, 2002.25. S. Read, Thinking about Logic, Oxford University Press, 2002.26. S. Read, Thinking about Logic, Oxford University Press, 2002.27. S. Read, Thinking about Logic, Oxford University Press, 2002.28. S. Read, Thinking about Logic, Oxford University Press, 2002.29. S. Read, Thinking about Logic, Oxford University Press, 2002.30. S. Read, Thinking about Logic, Oxford University Press, 2002.
2148	HSS	MORAL THINKING	PHI447	Introduction to Ethical Theories(Consequentialist and Nonconsequentialist theories, Hedonism,Utilitarianism, Deontological theories, Ethical Rules, Situation Ethics, Virtue Ethics).MetaethicalTheories (Ethical Foundations, the Moral point of view; Why be Moral?). Ethics in the Indian tradition. Applied Ethics: Issues andDilemmas. 1. Frankena, W.K. Ethics. New Delhi: Prentice Hall ofindia,19992. Hospers, John. An Introduction to Philosophical Analysis. New Delhi: Allied Publishers,19673. LaFollette, Hugh, ed. Ethics in Practice: An Applied Ethics Reader. New York: Oxford University Press, 19944. Macneil, I. R. Ethics and Economics. New York: Oxford University Press, 19805. Parfit, D. Reasons and Persons. Oxford: Oxford University Press, 19846. Rawls, J. A Theory of Justice. Oxford: Oxford University Press, 19717. Scanlon, T. M. What We Owe to Each Other. Oxford: Oxford University Press, 19988. Singer, P. Practical Ethics. Cambridge: Cambridge University Press, 19939. Toulmin, S. E. Ethics and the Philosophy of Language. Cambridge: Cambridge University Press, 199010. Veatch, R. L. The Foundations of Bioethics. New York: Oxford University Press, 199011. Wiggins, J. The Invention of Solitude. New York: Oxford University Press, 199612. Williams, G. Ethics and the Limits of Reason. Cambridge: Cambridge University Press, 198013. Williams, G. The Unity of Virtue. Cambridge: Cambridge University Press, 198014. Williams, G. What We Owe to Each Other. Cambridge: Cambridge University Press, 198015. Williams, G. The Unity of Virtue. Cambridge: Cambridge University Press, 198016. Williams, G. The Unity of Virtue. Cambridge: Cambridge University Press, 198017. Williams, G. The Unity of Virtue. Cambridge: Cambridge University Press, 198018. Williams, G. The Unity of Virtue. Cambridge: Cambridge University Press, 198019. Williams, G. The Unity of Virtue. Cambridge: Cambridge University Press, 198020. Williams, G. The Unity of Virtue. Cambridge: Cambridge University Press, 198021. Williams, G. The Unity of Virtue. Cambridge: Cambridge University Press, 198022. Williams, G. The Unity of Virtue. Cambridge: Cambridge University Press, 198023. Williams, G. The Unity of Virtue. Cambridge: Cambridge University Press, 198024. Williams, G. The Unity of Virtue. Cambridge: Cambridge University Press, 198025. Williams, G. The Unity of Virtue. Cambridge: Cambridge University Press, 198026. Williams, G. The Unity of Virtue. Cambridge: Cambridge University Press, 198027. Williams, G. The Unity of Virtue. Cambridge: Cambridge University Press, 198028. Williams, G. The Unity of Virtue. Cambridge: Cambridge University Press, 198029. Williams, G. The Unity of Virtue. Cambridge: Cambridge University Press, 198030. Williams, G. The Unity of Virtue. Cambridge: Cambridge University Press, 1980
2149	HSS	PHILOSOPHY OF SCIENCE	PHI446	On what Philosophy of Science is; Logic and Empiricism; Theory and Observation; Evidence,Confirmation and Falsificationism; Induction and Probability; Scientific Revolution versus NormalScience; Scientific revolutions and causes and Conditionals,Causality and Indeterminism, Bayesian and Modern theories of Evidence; Science and Values * Peter GodfreySmith(2003), Theory and Reality: An Introduction to Philosophy of Science,University of Chicago Press. [Standard Text for this course] James Ladyman(2002), Understanding Philosophy of Science, Routledge, London, UK. (2002), Understanding Philosophy of Science, Routledge, London, UK. (2002), Understanding Philosophy of Science, Routledge, London, UK. (2002), Understanding Philosophy of Science, Routledge, London, UK. (2002), Understanding Philosophy of Science, Routledge, London, UK. (2002), Understanding Philosophy of Science, Routledge, London, UK. 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2157	HSS	LITERATURE AND IDEOLOGY	ENG432	This course will engage with the concept of Ideology, and itsrelationship with literature. Beginning with a definition of Ideology and its various subcategories,the course will go on to examine how ideology the many different kinds of relationships that a literary text may have withideology, ranging from propagation and elaboration, to refutation and critique. We willalso attempt to understand how ideology cre Eagleton, Terry. Criticism and Ideology. New York: Verso, 1978.Hawkes, David. Ideology. London: Routledge, 2003.Williams, Raymond. Keywords: A Vocabulary of Culture and Society. New York:Oxford Uni
2158	EE	MICROELECTRONICS II	EE311A	1. Semiconductors and Crystal Structures2.. Basic Semiconductor PhysicsEE3113. Excess Carriers and Transport Processes in Semiconductors4. Junction Diode5. Bipolar Junction Transistor (BJT)6. MO Diode, Varactor Diode, Light Emitting Diode (LED),Photodetector, and Solar Cell I. Solid State Electronic Devices, B.G. Streetman arid .S. Banerjee, Prentice Hall, 2000.2. Device Electronics for Integrated Circuits, R.S. Muller and T.I. Kamins, John Wiley, 2003.3. Introduction to Semicond 990.
2159	EE	MICROELECTRONICS II	EE311	1. Semiconductors and Crystal Structures2.. Basic Semiconductor PhysicsEE3113. Excess Carriers and Transport Processes in Semiconductors4. Junction Diode5. Bipolar Junction Transistor (BJT)6. MO Diode, Varactor Diode, Light Emitting Diode (LED),Photodetector, and Solar Cell I. Solid State Electronic Devices, B.G. Streetman arid .S. Banerjee, Prentice Hall, 2000.2. Device Electronics for Integrated Circuits, R.S. Muller and T.I. Kamins, John Wiley, 2003.3. Introduction to Semicond 990.
2160	HSS	INTRODUCTION TO LOGIC	PHI142	Introduction: Logic as the science of distinguishing valid arguments from invalid ones. Fallacies.Aristotelian Logic and its limitationsPropositional Logic: Propositional Calculus. Logical Operators. Transla ofInvalidity. Semantic Tableaux MethodPredicate Logic: Predicate Calculus. Quantifiers and Properties. Translations. Bondage andFreedom. Rules of Deduction (Generalization and Instantiation) and Cons MethodAxiomatic System: Introduction to the System of Principia Mathematica. Godel'sIncompleteness Theorem Texts:Copi, Irvin and Carl Cohen, Introduction to Logic, Prentice Hall, New Jeresy, 101h edn., 1998Hausman, Alan, Howard Kahane, and Paul Tidman, Logic and Philosophy: A ModernIntroduction, Wadswor Donais and Richard Montague, Logic: Techniques of Formal Reasoning, Harcourt, Brace& World, Inc., 1964References:Crossley, John, et al, What is Mathematical Logic, Oxford University Press, Oxford, 19 course in Mathematical Logic, Oxford University Press, Oxford, 2004, pp1115
2161	HSS	INTRODUCTION TO PHILOSOPHY	PHI141	General introduction to philosophy: Nature of philosophy; its relations with, and differencesfrom, science, religion, art, and culture. Metaphysics: Causality; determinism and free willMaterialism and Ideali knowledge; The Gettier problem. Theories of perception.Descartes and foundationalism. Coherentism. Naturalized epistemology.Ethics: Utilitarianism. Kant. Virtue ethics.Applied Ethics: Some problems o Chisholm, Roderick. Theory of Knowledge. Prentice Hall, Englewood Cliffs NJ, 1989.Horner, Chris and Emrys Westacott. Thinking through Philosophy. Cambridge University Press,Cambridge, 2000.Russell Cambridge University Press, Cambridge, 1993.
2162	CHE	FLUID MECHANICS AND RATE PROCESSES	ESO212	I. FLUID MECHANICS (34 lectures)1. Introduction to transport phenomena 2. Fluid statics; pressure as a scalar, manometry, forces on submerged surfaces by integration of pressure forces3. Description c mass; integral and differential approaches 5. Linear momentum balance: stress; deformation; Newtons law of viscosity; NavierStokes equation;6. Applications of NavierStokes equation for simple 1D prob (Pitot tube, orifice meters)8. Pipe flows; friction factor; Reynolds experiment; losses in fittings 9. Similitude and modeling using nondimensionalization of Navier Stokes equations and boundary conditions Low Re flows: flow past circular cylinders; stream functions; Stokes flow; drag coefficient correlations11. High Re flow: Prandtls approximation; basic inviscid flow; need for boundary layer; MagnusRobin ballgames(qualitative)II. HEAT TRANSFER1. Introduction: Fouriers law; unsteady conduction equation; boundary conditions2. Convection: heat transfer coefficient and correlations III. MASS TRANSFER1. (a) V. Gupta and S.K. Gupta, Fluid Mechanics and its Applications, New Age Intl., NewDelhi, 1984.(b) V. Gupta, Elements of Heat and Mass Transfer, New Age Intl., New Delhi, 1995.(c) R. W. Fox, P. J. Pritchard Mechanics, Springer, 1972.(e) J. P.Holman, Heat Transfer, 10th Ed., McGraw Hill, 2009.
2163	EE	INTRODUCTION TO ELECTRICAL ENGINEERING	ESO203	1. Introduction, SinglePhase Circuits, Power Calculations, Analysis of ThreePhase Circuits,Mutually Coupled Circuits2. Transformers: Magnetic Circuits, Equivalent Circuit and Performance3. DirectCurrent Construction, Equivalent Circuit, Torquespeed characteristics, SpeedControl, Starting, Applications 5. Synchronous Machines: Construction, Equivalent Circuit, Generator & Motor Operation, /,Power Angle controls,lLabomtmy Work: Nine experiments performed in the laboratory work cover the following.Power Circuit Measurements, SinglePhase Transfonners, D.C Motor & Generator,SinglePhase Induction M 1. Electric Machines and Power Electronics P.C.Sen, Second Edition, Wiley,'97 2. Basic Electrical Engineering D.P.Kothari & I.J.Nagrath, Tata McGraw Hill 3. Electric Machines 2nd edition, D.P.Kothari &
2164	CHM	NUCLEAR MAGNETIC RESONANCE SPECTROSCOPY	CHM689A	Introduction to NMR Spectroscopy.(1)Angular momentum, matrix representation of angular momentum operators. (6)Density matrices, pure and mixed states, density operator and calculation of expectat experiments.(5)Construction of hamiltonian matrix for multispin systems and the solution of AB and ABX spinsystems.(5)Product operator formalism and vector diagrams.(5)Analysis of multipulse experi structure determination. (2) 1. M.Goldman, Quantum Description of High Resolution NMR in Liquids, Clarendon Press,New York(1988).2. J.A.Pople, W.G.Schneider and H.J.Bernstein, High Resolution NMR, McGraw Hill, NewYork(195 C.P.Slichter, Principles of Magnetic Resonance, SpringerVerlag, Berlin(1990).
2165	CSE	COMPUTING LABORATORY-I	CS251	1. Basic operating system commands. Students are expected to know the basic shell (e.g., bash) commands and should be able to understand the options and functioning of a command by reading the m emacs. How ever, they should be able to utilize the multiple features of the editors (such as automatic indentation, ontextsensitive colouring, letypesensitive autowrap, etc.) and not use them simply as a ty cvs, svn, git, dares) work. They_should be able to check in, check out, resolve errors and put tags on a snapshot. On all subsequent assignments, they must use a form of version .control.4. Scripting and a familiar with the different command syntax in other shells as well. Also, they will need to know the various functions (e.g., seq, for) that a shell provides. The choice of the scripting language(per! or python know how to format properly the equations, gures, tables, theorems, etc. using different packages and options. For bibliography management, they should use bibtex, and must use it within the latex docu (they includegnuplot, xfig, etc.). 6. Hardware. Students will work handson to learn how to install hard drives, RAM, etc., and in general, assemble a computer from its different parts. available for different tools.
2166	MTH	SPECTRAL THEORY FOR SELF-ADJOINT OPERATORS	MTH632	Unbounded Operators, Matrix representation, Selfadjointness Criterion, Quadratic Forms, Differential Operators, Selfadjoint Extensions, Functional Calculus, Spectra of Selfadjoint Operators, Semianalytic Uncertainty Principle of Bargmann, Canonical Commutation Relations, Schrodinger representations, Schrodinger Operators, Selfadjointness, A Theorem of Kato, Spectral Theory for Schrodinger Operators N. Akhiezer, I. Glazman, Theory of Linear Operators in Hilbert Space II, Dover, 1961.* J. Blank, P. Exner, M. Havlivcek, Hilbert Space Operators in Quantum Physics, Springer, 2008.* T. Kato, Perturbation The M. Reed and B. Simon, Methods of Modern Mathematical Physics II, Academic Press, 1975.
2167	CHM	CHEMISTRY OF DRUG DESIGN AND METABOLISM	CHM698A	Physicochemical Principles of Drug Action; Partition Coefficients; ReceptorEffector Theories;Role of Second Messengers in Drug Action; Methods of Receptor Isolation,Characterization and Modeling (5)F Drugs and Regulatory Implications; TheoreticalApproaches: QSAR, Topliss Tree, MSA, CoMFA (2)Neuroactive Drugs: Neurons and Neurotransmitters; Brainrelated Disorders andChemotherapy; Drugs Inter Antimalarial, Antiviral, and Cardiovascular Drugs; Emerging Trends in DrugDesign: Inhibitors of DNA Topoisomerase and Protein Farnesylation & Prenylation; GeneBased Medicines (8)Biopharmaceuticals Conventional and GeneticDrugs; TissueSpecific Delivery of Antitumor Agents (8)Drug Administration, Distribution, Metabolism and Elimination (ADME); Pathways of DrugMetabolism: Enzymology and Mo Metabolism; Toxicological Aspects of Metabolism: MetabolicActivation of Environmental Carcinogens and DNA Damage; Drug Pharmacokinetics andFinal Body Clearance (2) 1. Medicinal Chemistry: A Biochemical Approach, Thomas Nogrady2. Principles of Medicinal Chemistry, William O. Foye3. The Pharmacological Basis of Therapeutics: Goodman and Gilman4. Introduction
2168	CHM	CHEMISTRY OF DRUG DESIGN AND METABOLISM	CHM698	Physicochemical Principles of Drug Action; Partition Coefficients; ReceptorEffector Theories;Role of Second Messengers in Drug Action; Methods of Receptor Isolation,Characterization and Modeling (5)F Drugs and Regulatory Implications; TheoreticalApproaches: QSAR, Topliss Tree, MSA, CoMFA (2)Neuroactive Drugs: Neurons and Neurotransmitters; Brainrelated Disorders andChemotherapy; Drugs Inter Antimalarial, Antiviral, and Cardiovascular Drugs; Emerging Trends in DrugDesign: Inhibitors of DNA Topoisomerase and Protein Farnesylation & Prenylation; GeneBased Medicines (8)Biopharmaceuticals Conventional and GeneticDrugs; TissueSpecific Delivery of Antitumor Agents (8)Drug Administration, Distribution, Metabolism and Elimination (ADME); Pathways of DrugMetabolism: Enzymology and Mo Metabolism; Toxicological Aspects of Metabolism: MetabolicActivation of Environmental Carcinogens and DNA Damage; Drug Pharmacokinetics andFinal Body Clearance (2) 1. Medicinal Chemistry: A Biochemical Approach, Thomas Nogrady2. Principles of Medicinal Chemistry, William O. Foye3. The Pharmacological Basis of Therapeutics: Goodman and Gilman4. Introduction
2169	CHM	QUANTUM COMPUTING	CHM696A	Thermodynamics of computing, Shannon Theory, elementary information theory (2)Basics of computers, ChurchTuring hypothesis, basics of computing complexity (4)Basic of quantum mechanics, Feynr approaches (8)Twolevel Systems, Coherence, Superposition Principle, Density Matrix, Entanglement, RelaxationProcesses.(6)Quantum gates and circuits, Theory of Quantum Information and Computation (8)Quantum Complexity, Quantum Turing Machine. (6)Physical implementations of Quantum Computation, Light polarization, NMR, Cavity QED, IonTraps,Lasermatter interaction, Coherent Control. (8)

S.No.	Branch	Course Title	Courses ID	Content
				1. Stuart Russell, Peter Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009. Can also use 2nd Ed., Pearson Education International, 2003.2. Nils Nilsson, Artificial Intelligence: A New Approach, Morgan Kaufmann, 1998.3. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.4. Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.5. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.6. Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.7. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.8. Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.9. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.10. Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.11. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.12. Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.13. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.14. Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.15. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.16. Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.17. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.18. Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.19. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.20. Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.21. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.22. Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.23. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.24. Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.25. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.26. Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.27. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.28. Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.29. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.30. Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.31. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.32. Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.33. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.34. Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.35. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.36. Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.37. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.38. Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.39. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.40. Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.41. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.42. Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.43. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.44. Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.45. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.46. Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.47. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.48. Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.49. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.50. 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Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.61. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.62. Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.63. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.64. Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.65. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.66. Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.67. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.68. Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.69. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.70. 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Russell and Norvig, Artificial Intelligence: A Modern Approach, 3rd Ed., Prentice Hall, 2009.
2183	HSS	INTRODUCTORY SOCIOLOGY	SOC171	What is Sociology? Explain how the new science of society came about in the nineteenth century in Western Europe and how it was different from earlier social thought. Discuss the intellectual and social context of the Enlightenment. The socioeconomic roots: the Twin Revolutions The Industrial Revolution The French Revolution Theoretical thinking in sociology (A) Classical sociological theorists: Karl Marx Max Weber Emile Durkheim (B) Symbolic Interactionism Society, Community and Self (C) Modernization Theory (D) Social Stratification Norms, values, roles, custom, status Deviance Theories of subculture The concept of anomie Sociology of Organizations Different sociological approaches to the study of organizations and Ethnicity Gender Caste Disability Poverty Definition and measurement Extent of poverty Theories of poverty Power and Politics Debating basic concepts in political sociology Authoritarianism and democracy Religion Sociological theories of religion Religious organizations Secularization debates
2184	HSS	INTRODUCTION TO PHILOSOPHY	PHI141A	General introduction to philosophy: Nature of philosophy; its relations with, and differences from, science, religion, art, and culture. Metaphysics: Causality; determinism and free will Materialism and Idealism Knowledge; The Gettier problem. Theories of perception. Descartes and foundationalism. Coherentism. Naturalized epistemology. Ethics: Utilitarianism. Kant. Virtue ethics. Applied Ethics: Some problems of business ethics. Chisholm, Roderick. Theory of Knowledge. Prentice Hall, Englewood Cliffs NJ, 1989. Horner, Chris and Emrys Westacott. Thinking through Philosophy. Cambridge University Press, Cambridge, 2000. Russell and Norvig, Introduction to the Theory of Computation, 3rd Ed., Morgan Kaufmann, 2001.
2185	ME	DESIGN FOR MANUFACTURE & ASSEMBLY	ME701	
2186	PSE	RESEARCH IN PHOTONICS AND LASERS	PSE606	Biophotonics, nanobiophotonics, Optical communication, Quantum cryptography Nonlinear optics Tomography Quantum dots, photonic crystals Laser manufacturing and materials processing Laser instrumentation and imaging.
2187	PSE	PHOTONICS LAB TECHNIQUES	PSE605	* Electrooptic effect using LiNbO3 crystal* Acoustooptic modulator* Study of effects of loss, dispersion, amplifier noise on 10Gbps links* 40Gbps QAM modulation and coherent demodulation* Nonlinear optical effects in crystal* Fresnel and Fraunhofer Diffraction* HeNe laser beam parameters* Laser diode characteristics: LI characteristic, beam profile measurement, modes and spectrum using FP cavity* Michelson interferometry* Holography
2188	PSE	PHOTONICS SYSTEMS AND APPLICATIONS	PSE604	Principles and Applications of Solid State Laser Systems Laser diode Structures, Mechanism of photon emission in semiconductor laser, Tunable semiconductor diode laser, Rare earth doped lasers, Nd:YAG Lasers, Quantum cascade Laser. Principles and Applications of Liquid and Gas Laser Systems Dye laser, Tunable Lasers, Frequency stabilization, Tuning Techniques, Ar+ lasers, HeNe laser, CO2 lasers. * Kjell J. Gasvik, Optical Metrology, 3rd Edition, John Wiley and Sons, 2002 * O. Svelto and D C Hanna, Principles of Lasers, 4th Ed., 1998 * R W Boyd, Nonlinear optics, Academic Press, 2nd Ed, 2003 * M. H. Nayfeh, Nonlinear Dynamics, Wiley, 1991.
2189	PSE	NUMERICAL METHODS IN OPTICS	PSE603	Linear algebra: matrices, matrix inversion; QR, Singular value decomposition, systems of equations, eigenvalues, eigenvectors, orthonormalization, condition number Laplace and Fourier transforms Vector calculus: gradient, divergence, curl, Gauss theorem, Stokes theorem, Helmholtz theorem, direct methods for linear algebraic equations; generalized inverses, least squares Numerical differentiation and integration; Numerical solution of 1st and second order ODEs, Runge-Kutta method, stability, error analysis. * E. Kreyszig, Advanced Engineering mathematics, 8th edition, McGraw-Hill, New York, 2000. * M.D. Greenberg, Advanced Engineering Mathematics, Prentice Hall, New Jersey, International edition, 1998. * I. F. G. B. Greenberg, Advanced Engineering Mathematics, Prentice Hall, New Jersey, International edition, 1998. * I. F. G. B. Greenberg, Advanced Engineering Mathematics, Prentice Hall, New Jersey, International edition, 1998.
2190	PSE	PRINCIPLES OF LASERS AND DETECTORS	PSE602	Introduction to light sources, Lasers, principle of lasing Optical cavities, longitudinal, transverse modes, Stability Interaction of radiation with matter, Spontaneous emission Absorption and stimulated emission (rate equation based Lasing model) Three and four level lasers CW and pulsed lasers, Q-switching and mode locking * Principles of Lasers, O. Svelto and D. C. Hanna, 5th edition, 2010 * Laser Physics, Peter W. Milonni and Joseph H. Eberly, Wiley, 2nd edition, 2010 * Lasers, Anthony E. Siegman, University Science Books; 1st edition, 1986; 2nd edition, 2004
2191	PSE	INTRODUCTION TO PHOTONICS	PSE601	Photonics deals with light generation, amplification, guiding, manipulation, and detection for harvesting information. This course introduces some of the fundamental aspects of photonics excluding general optics. Topics include: Gaussian Beams, Absorption and Dispersion Spatial and Temporal Coherence Boundary conditions, Fresnel's equations and coefficients, Brewster and critical angles, * E. Hecht, Optics, 4th ed., Pearson Education (2001). * G. R. Fowles, Introduction to Modern Optics, Dover (1989). * Smith, F.G. & King, T.A. Optics and Photonics: An introduction, John Wiley and Sons (2007).
2192	DES	INTERACTION DESIGN FOR PHYSICAL PRODUCTS	DES606	
2193	DES	PRODUCT DESIGN AND PROTOTYPING	DES605	
2194	DES	DESIGN THINKING & RESEARCH METHODS	DES604	
2195	MTH	VECTOR BUNDLES AND CHARACTERISTIC CLASSES	MTH761	Smooth Manifolds, Vector bundles, Constructing New vector bundles Out of Old. Grassmann Manifolds and Universal bundles, The classification of vector bundles. Characteristic classes for vector bundles. The Chern theorem, The construction of Stiefel Whitney classes, Chern, Pontryagin, and Euler classes. J. P. May. A concise course in Algebraic Topology. Chicago Lectures in Mathematics. University of Chicago Press, Chicago, IL, 1999. x+243 pp. John W. Milnor and James D. Stasheff. Characteristic classes and the cobordism problem. 1974. vii+331 pp.
2196	AE	DEFORMATION AND FRACTURE	AE704	
2197	ECO	PUBLIC ECONOMICS AND PUBLIC POLICY	ECO731	Unbounded Operators, Matrix representation, Selfadjointness Criterion, Quadratic Forms, Differential Operators, Selfadjoint Extensions, Functional Calculus, Spectra of Selfadjoint Operators, Semianalytic Functions, Uncertainty Principle of Bargmann, Canonical Commutation Relations, Schrodinger representations, Schrodinger Operators, Selfadjointness, A Theorem of Kato, Spectral Theory for Schrodinger Operators * N. Akhiezer, I. Glazman, Theory of Linear Operators in Hilbert Space II, Dover, 1961. * J. Blank, P. Exner, M. Havlicek, Hilbert Space Operators in Quantum Physics, Springer, 2008. * T. Kato, Perturbation Theory for Linear Operators, Springer, 1978. * M. Reed and B. Simon, Methods of Modern Mathematical Physics II, Academic Press, 1975.
2198	CHE	INTRODUCTION TO HYDRODYNAMIC STABILITY	CHE614	
2199	HSS	INTRODUCTION TO INDIAN PHILOSOPHY	PHI140A	1. Introduction: The Dialectical Style of Indian Philosophy The Khandana Mandana style. Stating the real and imaginary objections. Refutation of Objections. Establishing the position. 2. Historical Sketch: The development of Indian schools (at least in Nyaya, Bauddha, Mimamsa and Advaita). Genesis and categorization of valid cognition in different schools. 4. Content of Valid cognition (prameya): That which is there Indian Metaphysics (Adhityasamutpada, Arambhavada, Parinamavada, Vivartavada and Pratityasamutpada) 6. Logic: Inference (anumana) and its features in Nyaya. Logical Blockers or Hetvabhasas in Nyaya 7. Ethics: Theory of Karma * Six Ways of Knowing, D M Datta, Calcutta University, 1972. * An Introduction to Indian Philosophy, S Chatterjee and D Datta, Rupa & Co., 2007 (paperback) * Outlines of Indian Philosophy, M Hiriyanna, Motilal Banarasi Das, 1992. * Philosophical Traditions of India, P T Raju, Motilal Banarasi Das, 1992. * A History of Indian Philosophy, S N Dasgupta, Cambridge University Press, 1922. * Encyclopedia of Indian Philosophy, Vol 12, ed. Kar
2200	CSE	PRINCIPLES OF PROGRAMMING LANGUAGES	CS350	Brief history of development of programming languages, Introduction imperative programming, functional programming, logic programming and object oriented programming, Values and types, Notion of type scope block structure, static and dynamic scoping, Abstraction procedural and function abstractions, Type systems monomorphic type systems. Introduction to polymorphism, Types of polymorphism over monomorphic types, introduction to polymorphic type inference, Functional programming, Logic Programming, Object oriented programming. 1. Martn Abadi, Luca Cardelli, A Theory of Objects, Springer 1996. 2. Luca Cardelli, P Wegner, On Understanding Types, Data Abstraction and Polymorphism, ACM Computing Surveys, 17(4), pp 471, 1985. 3. Luca Cardelli, P Wegner, On Understanding Types, Data Abstraction and Polymorphism, ACM Computing Surveys, 17(4), pp 471, 1985. 4. Benjamin C. Pierce, Types and programming languages, MIT Press, 2002. 5. L. C. Paulson, ML for the Working Programmer, 2nd Ed., Cambridge University Press, 1996. 6. Benjamin C. Pierce, Types and programming languages, MIT Press, 2002. 7. Benjamin C. Pierce, Types and programming languages, MIT Press, 2002. 8. Benjamin C. Pierce, Types and programming languages, MIT Press, 2002. 9. Michael L Scott, Programming Language Pragmatics, 3rd Ed., Morgan Kaufmann, 2009

S.No.	Branch	Course Title	Courses ID	Content
2201	MTH	INTRODUCTION TO HYPERBOLIC GEOMETRY	MTH633	Models of Hyperbolic Space: Upper Half Space Model & Disc Model; Isometries of HyperbolicSpace; Geodesics; Slimness of Triangles and Exponential Divergence of Geodesics in HyperbolicSpace; Isoperimetric Inequalities; Isometries and Fundamental groups; Properly DiscontinuousGroup actions; Fundamental Domains; Hyperbolic Surfaces. * Lectures on Hyperbolic Geometry; Riccardo Benedetti and Carlo Petronio;SpringerVerlag.* Kleinian Groups; Bernard Maskit; SpringerVerlag.* Metric Spaces of NonPositive Curvature; Martin R. Bridson, A. Haefliger, Cambridge University Press, 1999.* Group theory;Brian H. Bowditch, preprint.
2202	HSS	INTRODUCTION TO LOGIC	PHI142A	Introduction: Logic as the science of distinguishing valid arguments from invalid ones. Fallacies.Aristotelian Logic and its limitationsPropositional Logic: Propositional Calculus. Logical Operators. Translations. Truth Tables. Validity and Invalidity. Semantic Tableaux MethodPredicate Logic: Predicate Calculus. Quantifiers and Properties. Translations. Bondage andFreedom. Rules of Deduction (Generalization and Instantiation) and Consistency. The Completeness Theorem. The Compactness Theorem. The Skolem MethodAxiomatic System: Introduction to the System of Principia Mathematica. Godel'sIncompleteness Theorem Texts:Copi, Irvin and Carl Cohen, Introduction to Logic, Prentice Hall, New Jeresy, 101h edn., 1998Hausman, Alan, Howard Kahane, and Paul Tidman, Logic and Philosophy: A ModernIntroduction, Wadsworth, 1998Donais and Richard Montague, Logic: Techniques of Formal Reasoning, Harcourt, Brace& World, Inc., 1964References:Crossley, John, et al, What is Mathematical Logic, Oxford University Press, Oxford, 1998 course in Mathematical Logic, Oxford University Press, Oxford, 2004, pp1115
2203	HSS	INTRODUCTION TO INDIAN PHILOSOPHY	PHI140	1. Introduction: The Dialectical Style ofIndian Philosophy TheKhandanaMandana style. Stating the real and imaginary objections.Refutation of Objections. Establishing the position.2. Historical Sketch: The six schools (at leastin Nyaya, Bauddha, Mimamsa and Advaita). Genesis and categorization ofvalid cognition in different schools.4. Content of Valid cognition (prameya): That which is thereIndianMetaphysics (Adhityasamutpada,Arambhavada, Parinamavada, Vivartavada and Pratityasamutpada)6. Logic: Inference (anumana) and its features in Nyaya. Logical Blockersor Hetvabhasas in Nyaya7. Ethics: Theory and Practice * Six Ways of Knowing, D M Datta, Calcutta University, 1972.* An Introduction to Indian Philosophy, S Chatterjee and D Datta, Rupa & Co., 2007(paperback)* Outlines ofIndian Philosophy, M Hiriyanna, Motilal Banarsidass, 1992.* Philosophical Traditions ofIndia, P T Raju, Motilal Banarasidass, 1992.* A History of Indian Philosophy, S N Dasgupta, Cambridge University Press, 1922.* Encyclopedia ofIndian Philosophy,Voll 12, ed. Kar
2204	MME	NATURE AND PROPERTIES OF MATERIALS	ESO214	Importance of structure, visavis, property with examples, periodic patterns in 2D/3D,symmetry, point and space groups, Bravais Lattices. Crystallographic planes and directions.Glasses, polymers, concepts of optical microscopy). Evolution of microstructure,phase diagrams, diffusion and phase transformation, application to microstructure design.Mechaual properties: stressstrain curves, elastic modulus, plasticity, fatigue andcreep properties of materials. Electrical and magnetic properties of materials, Distinctionbetween conductors, semiconductors and insulators, Brief summary of Free electron theory andBand Theory a brief introduction to ferroelectrics, Magnetic materials andstructure relations e.g. texture. 1. V. Raghavan, Materials Science and Engineering: A first course.2. L. H. VanVlack, Elements of Materials Science3. W. D. Callister, Jr., Materials Science and Engineering: An futroduction4. M.F. Ashby and C. Humphrey, Materials Selection: Engineering Materials 2: An introduction to microstructuresprocessing and design.6. Wulff series (Vol.I4), Science and Properties of Materials7. WF Smith, Principles of Materials Science and Engineering
2205	HSS	INTRODUCTION TO THE ART OF VIDEO MAKING	ART105	TheoryCinematography, Production Process, Writing for Screen, Basics Picture Making, CameraAngles, Direction Time and Space continuity, Cuts, Art of Editing The 5C's of Cinematography, MascelliDirecting the Documentary, Michail RabigerA History ofthe Cinema, Eric RhodeWriting for Screen and Television, TobiasSingle camera video production, Robert L. Stone Short Film & Video, Irving & ReaThe Art of Cinema, B.D.GargaA compauion to Film Theory, Miller & Starn
2206	CHM	STATISTICAL MECHANICS & ITS APPL. TO CHEMISTRY	CHM650	Course description: Thermodynamics, Statistical Mechanics of noninteracting systems,interacting systems, and nonequilibrium systemsCourse Details1. Equilibrium Thermodynamics: Thermodynamic Equilibrium criteria, phase equilibria (4 lectures)2. Ensembles in Statistical Mechanics: Ensemble postulate and ergodicity, microcanonical,canonical and grand canonical ensembles, quantum and classicalpartition functions, quantum correlations,collective modes, occupation numbers, collections of fermions, bosons,photons, classical deal gas of spinless particles, molecular partition functions, idealparamagnets. (6 lectures)3. Correlation function, neutron scatteringexperiments, Virial equation, Meyer cluster diagrams (6 lectures)5. Interacting Systems 2: Computer Simulations Ensemble averages, ergodicity,random numbers, Monte Carlo Transitions in Lattice models Lattice gas,Ising Model, order parameter, Mean Field theory, Renormalization group theory(6 lectures)7. Nonequilibrium Statistical Mechanics (6 lectures) Linear Response theory 1. D.A. McQuarrie, Statistical Mechanics2. David Chandler, Introduction to Modern Statistical Mechanics3. K.L. Huang, Statistical Mechanics4. B. Widom, Statistical Mechanics: A concise introduction for chemists
2207	CHM	THE CHEMISTRY OF METAL-CARBON BOND: STRUCTURE REACTIVITY & APPLICATIONS	CHM648	Introduction to Organometallics: Bonding, Types of Ligands, and some basics concepts likeisoelectronic and isolobal analogyCharacterization techniques of Organometallic compounds (NMR and IR spectroscopy)lithium and magnesium: synthesis, structures, fluxionality andreactivityChemistry of Aluminum: Aluminum alkyls. Use of aluminum alkyls in polymerization of olefins.Organometallic chemistry of transition metals: phosphines / metal nitrosyls / metal isocyanide: structures,reactivity and bonding.Metal carbenes, metallocenes, Fischer carbenes, Schrock, carbenes, complexes with Nheterocycliccarbenes (NHCs), boranes Organometallic Chemistry: Oxidative addition, reductive elimination, insertion,elimination, and migrationApplications of organometallics in organic synthesisCC bond coupling reactions (Heck, sangoshira, Suzuki) Most of the material for this course will be accessed from primary literature viz., Journalarticles. Some text books that will be followed are as follows:Elschenbroich, C.; Salzer, A.; Organometallics: A Concise Introduction to Concepts, Syntheses andApplications, First Edition, Universities Press, 2010Douglas, B.; McDaniel, D.; and Alexander, J. Concepts and Models of Inorganic Chemsitry 3rdEdn. John Wiley, New York. 1993.Fischer, G. and sons 2005.
2208	HSS	INTRODUCTION TO ART APPRECIATION & CRITICISM	ART102AA	What is Art, What is art appreciation, History of Art Criticism, Methodology of Critical ArtAppreciation, , Elements and Principles of Visual Art, Characteristics of art of Prehistoric,Ancient, middle age, Dark Age and Modern Art methods and materials.Practice: Still life, Nature Study, Composition, Symmetry, Narrative Continuity, 2D and 3D Exercises Jaason's Basic History of Western Art by Penelope CaviesArt of Twentieth Century by Loredana ParmesaoiIndiao Art A Concise History by Roy C. CravenEarly India From the Origins to AD 1300 by R. S. Sharma Eastern Art (Chinese aod Japaoese Art) Skira, Sharman Lee PublishersMoving Focus: Essays on Indian Art by K.G. SubramaoyaoThe Making of a New 'India' Art by Tapati Guha ThakurtaIndiao Art
2209	HSS	APPLICATION OF PSYCHOLOGY TO LIFE	PSY152A	Nature and Scope of Applied Psychology, Historical Perspective, Areas ofApplied Psychology, Roles and Skills of Applied Psychologists, Ethical Issues, Clinical andCounseling Psychology, Community Psychology, Legal Psychology
2210	HSS	INTRODUCTORY SOCIOLOGY	SOC171A	What is Sociology? Explain how the new science of societycame about in the nineteenth century inWestern Europe and how it wasdifferent from earlier social thought. Discuss the intellectual and social context of theEnlightenment. The socioeconomic roots: the TwinRevolutions The Industrial Revolution The French RevolutionTheoretical thinking in sociology(A) Classical sociological theorists: Karl Marx Max Weber Emile Durkheim Symbolic InteractionismSociety, Community and Self 21ecturesNorms, values, roles, custom, statusDeviance Theories of subcultureThe concept of anomieSociology of Organizations Different sociological perspectives on Society and Ethnicity Gender Caste DisabilityPoverty Definition and measurement Extent of poverty Theories of povertyPower and Politics Debating basic concepts in politicalsociology Authoritarianism and democracy Religion Sociological theories of religion Religious organizations Secularization debates
2211	CHM	COMPUTER PROGRAMMING FOR CHEMISTRY	CHM684A	Fortran and Computer Programming: Elements of Fortran programming, constants, variablesand operators, control statements, I/O operations, functions and subprograms, common,equivalence, arrays, storage management, Round off and truncation errors, pitfalls and debugging (18)Numerical Methods: Roots of equations Bisection and NewtonRaphson methods, Systemof linear equations, Gaussian, GaussJordan elimination, Linear regressionanalysis, eigenvalues and eigenvectors, Numerical differentiation and integration, Differentialequations (15)Applications to chemistry : statistical thermodynamics, chemical kinetics, Curve fitting (4) 1) Michael Boillot, Understanding Fortran77, wesspublishing company, New York(1987).2) Fortran95/2003 for scientists and engineers, S.J.Chapman, McGrawHill (2008).3) (Ed) D.F. De Tar, Computer programming, Benjamin Inc, New York (1965).5) S.C.Chapra and R.P.Canale, Numerical Methods for Engineers, Tata McGraw Hill, NewDelhi (2003).
2212	CHM	COMPUTER PROGRAMMING FOR CHEMISTRY	CHM684	Fortran and Computer Programming: Elements of Fortran programming, constants, variablesand operators, control statements, I/O operations, functions and subprograms, common,equivalence, arrays, storage management, Round off and truncation errors, pitfalls and debugging (18)Numerical Methods: Roots of equations Bisection and NewtonRaphson methods, Systemof linear equations, Gaussian, GaussJordan elimination, Linear regressionanalysis, eigenvalues and eigenvectors, Numerical differentiation and integration, Differentialequations (15)Applications to chemistry : statistical thermodynamics, chemical kinetics, Curve fitting (4) 1) Michael Boillot, Understanding Fortran77, wesspublishing company, New York(1987).2) Fortran95/2003 for scientists and engineers, S.J.Chapman, McGrawHill (2008).3) (Ed) D.F. De Tar, Computer programming, Benjamin Inc, New York (1965).5) S.C.Chapra and R.P.Canale, Numerical Methods for Engineers, Tata McGraw Hill, NewDelhi (2003).
2213	HSS	LANGUAGE AND SOCIETY	ENG124A	Problems of definition; Multilingual communities; Language variation; Language andidentity; Standardization; Language, culture and cognition; Language and social control; Methodologicalissues. Clyne, Michael, 2003. Dynamics of Language Contact, Cambridge University Press.Downes, William, 1998. Language and Society, Cambridge University Press.Romaine, S., 2000. Language in Society: An Introduction

S.No.	Branch	Course Title	Courses ID	Content
2228	CHM	ADVANCED MAIN GROUP CHEMISTRY	CHM668	A. Topics for SelfStudy1. VSEPR Theory and prediction of molecular geometry2. Symmetry. Symmetry elements and symmetry operations. Point groups. Introduction to character tables. Uses of character tables in predicting molecular geometry and reactivity, Carboranes and metallo carboranes. Low valent boron compounds, organoboron compounds and organic synthesis. Use of ¹¹ B NMR in structure elucidation of organoboron compounds and reactivity. Organomagnesium and organosodium compounds. Chemistry of Aluminum. Aluminum Alkyls. Unusual organometallic compounds of aluminum. Including low oxidation state Al compounds. 15 compounds and the recent advances Multiple bonding in main group elements: Compounds involving silicon, phosphorus, bismuth etc. Synthesis, structure and reactivity. Recent literature on multiple bonding in cyclophosphazanes. Polysilanes and Polyphosphazenes Noble gas compounds Most of the material for this course will be accessed from primary literature viz., Journal articles. Some text books that will be followed are as follows: Inorganic Chemistry Principles of Structure and Reactivity, Chemsitry. 3rd Edn. Douglas, B.; McDaniel, D.; and Alexander, J. John Wiley, New York. 1993 Chemistry of the Elements. 2nd Edn. Greenwood, N. N.; and Earnshaw, A. Pergamon, Oxford, 1989 Elschenbroich, C. Springer Verlag, Heidelberg, 2005
2229	CHM	THE CHEMISTRY OF METAL-CARBON BOND: STRUCTURE REACTIVITY & APPLICATIONS	CHM648A	Introduction to Organometallics: Bonding, Types of Ligands, and some basic concepts like isoelectronic and isolobal analogy Characterization techniques of Organometallic compounds (NMR and IR spectroscopy) Lithium and magnesium: synthesis, structures, fluxionality and reactivity Chemistry of Aluminum: Aluminum alkyls. Use of aluminum alkyls in polymerization of olefins. Organometallic chemistry of transition metals: phosphines / metal nitrosyls / metal isocyanide: structures, reactivity and bonding. Metal carbenes, metal carbynes, Fischer carbenes, Schrock, carbenes, complexes with N-heterocyclic carbenes (NHCs), and others. Organometallic Chemistry: Oxidative addition, reductive elimination, insertion, elimination, and migration Applications of organometallics in organic synthesis C-C bond coupling reactions (Heck, Saito, etc.) Most of the material for this course will be accessed from primary literature viz., Journal articles. Some text books that will be followed are as follows: Elschenbroich, C.; Salzer, A.; Organometallics: A Concise Introduction to Concepts, Syntheses and Applications, First Edition, Universities Press, 2010 Douglas, B.; McDaniel, D.; and Alexander, J. Concepts and Models of Inorganic Chemistry 3rd Edn. John Wiley, New York. 1993. Fieser, R. M. and sons 2005.
2230	CHM	BIO-INORGANIC CHEMISTRY	CHM646A	1. Mineral Origin of life. Archaeal, Eucarial and Bacterial domain. (3 Lectures) 2. Transition metal ions in biology. Metalloproteins. Electron carriers, oxygen carriers, enzymes. Biogeochemical cycles of transition metals. Proteins, Cytochrome P450, Nitrophorin, NO synthase, peroxidase, catalase, Ferritin, cytochrome c oxidase, ceruloplasmin, blue copper proteins, di and trinuclear copper proteins. Other enzymes like, hydrogenase, metalloproteins like sulfite oxidase, xanthine oxidase, nitrate reductase, DMSO reductase, tungsten containing formate dehydrogenase and tungsten bearing hyperthermophilic and thermophilic enzymes. Zn enzymes like carbonic dehydratase, water splitting. (15 Lectures) 4. Active site analogue reaction models and structural models of these enzymes. (5 Lectures) 5. Environmental chemistry, auto exhaust, arsenic and other heavy metal pollution and anticancer drugs. 1. Principles of Bioinorganic Chemistry, S. J. Lippard and J. M. Berg, University Science Books, Mill Valley, 1994. 2. Bioinorganic Chemistry: Inorganic Elements in the Chemistry of Life, W. Kaim and B. Schwabe, Wiley, New York, 1999.
2231	HSS	ENGLISH LANGUAGE AND COMMUNICATION SKILLS	ENG112C	Reading Comprehension Skills Discovering structure; identifying themes and subthemes; understanding and interpreting facts; distinguishing facts from opinions and specific from general statements; summarizing and paraphrasing; prewriting activities through drafting and revision; Developing a composition using techniques such as definition, classification, analogy, etc; Descriptive, narrative, and argumentative techniques in writing essays; emphasis on Sentence structure, Diction, Agreement, Tense, and Point of view. (b). Strategies: Economy, emphasis, clarity, concreteness, unity and coherence (7 Hrs). Spoken Language Skills Descriptive, narrative, and argumentative essays.
2232	CHM	MODERN PHYSICAL METHODS IN CHEMISTRY	CHM664A	1. Symmetry and Group Theory: Group multiplication table, elements of a symmetry group, symmetry group classification, characters, group representation, The Great Orthogonality theorem, basis of representation, selection rules, transitions. (15) 2. Interaction of light with matter, Einstein coefficients, relationship of Einstein coefficients with transition moments (no derivation) and transition probabilities, oscillator strength. Beer-Lambert law, energy levels, potential energy curves, MO and term symbols, Franck-Condon principle, symmetry and selection rules, spin and parity forbidden transitions, vibronic interaction. (8) 3. Simple harmonic motion, anharmonic oscillator, vibrational coordinates. Force constants, selection rules, (F and G matrix if time permits). Applications: Organic molecules, functional groups versus frequency approach, Fermi resonance, frequency shifts because of anharmonicity. Mass Spectrometry, Magnetic Resonance, Photoelectron Spectroscopy (4) 4. Chemical Applications of Group Theory F.A. Cotton 2. Theory and applications of UV spectroscopy. H.H. Jaffe and M. Orchin. 3. Quantum mechanics, Cohen-Tannoudji, Diu and Lalo (2005) 4. Molecular Spectroscopy, P. W. M. Andrews, Wiley, New York, 1991.
2233	ECO	INTRODUCTION TO ECONOMICS	ECO101AA	Definition and central problems of Economics, Microeconomics vs. Macroeconomics, Normative vs. Positive Economics, Stock vs. Flow variables; Aggregate economic variables: Gross domestic product (GDP), Exchange rate; Laws of demand and supply, Elasticity, Market equilibrium, Consumer and producer surplus, Price controls; Utility theory: Laws of diminishing marginal and equi-marginal utility, Indifference curves, Production of variable proportion, Short-run cost functions, Isoquant and isocost, Optimizing conditions, Long-run cost functions; Types of market: Perfect Competition, Monopoly, Oligopoly, Monopolistic Competition. 1. Economics, by Paul A. Samuelson, any edition. 2. Macroeconomics, by N. G. Mankiw, 6th edition. 3. Microeconomics, by R. S. Pindyck, D. L. Rubinfeld and P. L. Mehta, 6th edition. 4. Intermediate Microeconomics, by R. R. Barlow, any edition.
2234	CHM	MODERN PHYSICAL METHODS IN CHEMISTRY	CHM664	1. Symmetry and Group Theory: Group multiplication table, elements of a symmetry group, symmetry group classification, characters, group representation, The Great Orthogonality theorem, basis of representation, selection rules, transitions. (15) 2. Interaction of light with matter, Einstein coefficients, relationship of Einstein coefficients with transition moments (no derivation) and transition probabilities, oscillator strength. Beer-Lambert law, energy levels, potential energy curves, MO and term symbols, Franck-Condon principle, symmetry and selection rules, spin and parity forbidden transitions, vibronic interaction. (8) 3. Simple harmonic motion, anharmonic oscillator, vibrational coordinates. Force constants, selection rules, (F and G matrix if time permits). Applications: Organic molecules, functional groups versus frequency approach, Fermi resonance, frequency shifts because of anharmonicity. Mass Spectrometry, Magnetic Resonance, Photoelectron Spectroscopy (4) 4. Chemical Applications of Group Theory F.A. Cotton 2. Theory and applications of UV spectroscopy. H.H. Jaffe and M. Orchin. 3. Quantum mechanics, Cohen-Tannoudji, Diu and Lalo (2005) 4. Molecular Spectroscopy, P. W. M. Andrews, Wiley, New York, 1991.
2235	CHM	CHEMISTRY OF NATURAL PRODUCTS	CHM662	Biosynthetic aspects and Synthesis of selected natural products of biological and structural importance: Discussions on synthetic methods, strategies towards these natural products mostly in chiral form: Total synthesis of structures ranging from 3-membered to macrocycles, and complex natural products such as Taxol, rapamycin, leishmanin, etc. K. C. Nicolaou, Classics in Total Synthesis Vols III, Wiley-VCH, 1996; 2003; 2011 T. Hudlicky and J. W. Reed, The way of synthesis, Wiley-VCH, 2007 E. J. Corey and X. M. Cheng, The logic of chemical synthesis, Wiley, New York, 1990 chemistry Vols 19, Elsevier, 1999
2236	CHM	NUCLEAR MAGNETIC RESONANCE SPECTROSCOPY	CHM689	Introduction to NMR Spectroscopy. (1) Angular momentum, matrix representation of angular momentum operators. (6) Density matrices, pure and mixed states, density operator and calculation of expectation values. (5) Construction of hamiltonian matrix for multi-spin systems and the solution of AB and ABX spin systems. (5) Product operator formalism and vector diagrams. (5) Analysis of multipulse experiments for structure determination. (2) 1. M. Goldman, Quantum Description of High Resolution NMR in Liquids, Clarendon Press, New York (1988). 2. J. A. Pople, W. G. Schneider and H. J. Bernstein, High Resolution NMR, McGraw Hill, New York (1958). 3. C. P. Slichter, Principles of Magnetic Resonance, Springer Verlag, Berlin (1990).
2237	CE	MECHANICS OF SOLIDS	ESO202	Free body diagram with examples on modelling of typical supports and joints, Conditions for equilibrium in 3D and 2D, Friction: limiting and non-limiting cases; Force-displacement relationship and geometrical analysis of thin-walled pressure vessels; Concept of stress at a point, Plane stress case: transformation of stresses at a point, principal stresses and Mohr's circle, Displacement field, Concept of strain at a point, Comparison of experimental results on 1D material behaviour, Concepts of elasticity, plasticity, strain hardening, failure (fracture/yielding), Idealization of 1D stress-strain curve, Generalized Hooke's law (without anisotropy), bending moment, and twisting moment diagrams) of slender members (singularity functions not to be used); Torsion of circular shafts and thin-walled tubes (plastic analysis and rectangular shafts not to be discussed); Cases of combined stresses, Concept of strain energy, Yield criteria; Deflection due to bending, Integration of the moment-curvature relationship, Strain energy and complementary strain energy for simple structural elements (those under axial load, shear force, bending moment, and torsion), Castigliano's theorems for deflection analysis and indeterminacy, Plastic behaviour not to be covered) * Crandall, S.H., Dahl, N.C., and Lardner, T. J., An Introduction to the Mechanics of Solids, McGraw-Hill, Second Ed. with 51 Units, 1978. * Beer, F.P., Johnston, Jr., E.R., Engineering Mechanics, Vol. 1: Statics, John Wiley, Second Ed. with 51 Units, 1980. * Popov, E.P., Engineering Mechanics of Solids, Prentice-Hall, First Ed., 1990
2238	CHM	MOLECULE RADIATION INTERACTION	CHM685A	1. Classical Electromagnetism (5) Fields, Maxwell's equations, gauges and optics. 2. Quantization of Electromagnetic Fields (10) Photons, polarizations, Stokes parameters, nontrivial role of the vector potential. 3. Spectroscopy (10) Born-Oppenheimer limit, time-dependent viewpoints, nonadiabatic effects. 5. Advanced topics (7) Beyond dipole approximation, attosecond spectroscopy. 1. Sakurai, Advanced Quantum Mechanics, 1967. 2. Cohen-Tannoudji & Dupont-Roc, Atom-Photon Interactions, 2004. 3. Landau & Lifshitz, Classical Theory of Fields, 1951. 4. Cohen-Tannoudji, Dupont-Roc & Grynberg, Photons and Atoms, Wiley, New York, 1989.
2239	CHM	SUPRAMOLECULAR CHEMISTRY	CHM654A	Introduction to the meaning of supramolecular chemistry, phenomenon of molecular recognition and their quantification (2 Lectures) Building blocks of supramolecular chemistry: acyclic receptors for neutral and charged species, cyclodextrins (12 Lectures) Sensors and information processing, electro-optic phenomena, molecular machines (12 Lectures) Amphiphilic molecules and their aggregation, Langmuir-Blodgett, molecular recognition, etc.

S.No.	Branch	Course Title	Courses ID	Content
				catalysis and otherapplications. (12 Lectures)Future scopes (1 Lecture) Supramolecular Chemistry: Concepts and Perspectives, J.M. Lehn, VCH, Weinheim, 1995Principles and Methods in Supramolecular Chemistry, H. J. Schneider and A. Yatsimirsky,Wiley, New York, 2000Sup
2240	CHM	CRYSTAL AND MOLECULAR STRUCTURE DETERMINATION	CHM651A	Generation of Xrays, monochromators, safety (1 Lecture)Concept of direct and reciprocal lattices, Braggs law of Xray diffraction in direct andreciprocal lattice, crystal systems, point groups, Bravais lattices groups (8 Lectures)Argand diagram, intensity data collection and quantitative aspects of Xray diffraction,temperature factor and scaling of data (10 Lectures)The phase problem, direct method of solving critical evaluation (5 Lectures) Xray Structure Determination A Practical Guide by G. H. Stout and H. L. Jensen, MacMillan,N.Y. 1968.Contemporary Crystallography, M. J. Buerger, McGrawHill, N.Y. 1970Crystal Structure Analysis A Primer by R. W. L. Palmer, KluwerAcademic/Plenum, N.Y. 2003
2241	CHM	GENERAL CHEMISTRY	CHM102A	Atomic Structure (4 Lectures). Chemical Bonding (5 Lectures). Molecular Spcctmscopy (5 Lectures). Coordination complexes of transition metal ions (3 Lectures). Organometallic Chemistry (3 Lectures). Chemistry of New Materials: For example, Fullerenes, nanotubes and graphene (1 Lecture) 1. Atkins & Julio de Paula: Physical Chemist1y, 8th Edition, Freeman & Co.2. Atkins & Shriver: Inorganic Chemistry, 4th Edition (2008).3. Clayden, Greaves, Warren &Wothers: Organic Chemistry, Oxtbrd Univ
2242	CHM	SUPRAMOLECULAR CHEMISTRY	CHM654	Introductionthe meaning of supramolecular chemistry, phenomenon of molecular recognitionand their quantification (2 Lectures)Building blocks of supramolecular chemistry acyclic receptors for neutral and charged molecules cyclodextrins (12 Lectures)Sensors and information processing, electrooptic phenomena, molecular machines (12Lectures)Amphiphilic molecules and their aggregation, LangmuirBlodgettry, molecular recognition catalysis and otherapplications. (12 Lectures)Future scopes (1 Lecture) Supramolecular Chemistry: Concepts and Perspectives, J.M. Lehn, VCH, Weinheim, 1995Principles and Methods in Supramolecular Chemistry, H. J. Schneider and A. Yatsimirsky,Wiley, New York, 2000Sup
2243	CHM	CHEMISTRY LABORATORY	CHM101A	Chemical analysis with relevance to everyday life. Determination of amount of Ca2+ in milk by complexometric titration2. Estimation of iodine in common iodized salt by iodometry3. Estimation of phosphorus by permanganometric titratton5. Extraction of DNA from green peas or onions and its identification6. Extraction of caffeine, an alkaloid from tea leaves7. How many pigments are there in the green portion of tea leaves?8. Chemical CompoundsI. Preparation and characterization of aspirin, a common medicine2. DielsAlder reaction a versatile organic reaction to form CC bonds: a reaction betweenanthracene and maleic anhydride3. Organometallic compounds: bridges between inorganic and organic compounds. AcetylationofF errocene5. An environmentrelated synthesis: Preparation of potash alum fromscrap aluminumPhotochemistry6. Reaction on Physical Chemistry ConceptsI. Partition of solutes in mixture of solvents: Acetic acid inwater/nbutanol2. Kinetics of reactions: An example using the iodidehydrogen peroxide clock reaction3. Determination of rate constant A. J. Elias:" A Collection ofInteresting General Chemistry Experiments", UniversitiesPress, Hyderabad, 2009
2244	CHM	CRYSTAL AND MOLECULAR STRUCTURE DETERMINATION	CHM651	Generation of Xrays, monochromators, safety (1 Lecture)Concept of direct and reciprocal lattices, Braggs law of Xray diffraction in direct andreciprocal lattice, crystal systems, point groups, Bravais lattices groups (8 Lectures)Argand diagram, intensity data collection and quantitative aspects of Xray diffraction,temperature factor and scaling of data (10 Lectures)The phase problem, direct method of solving critical evaluation (5 Lectures) Xray Structure Determination A Practical Guide by G. H. Stout and H. L. Jensen, MacMillan,N.Y. 1968.Contemporary Crystallography, M. J. Buerger, McGrawHill, N.Y. 1970Crystal Structure Analysis A Primer by R. W. L. Palmer, KluwerAcademic/Plenum, N.Y. 2003
2245	CHM	STATISTICAL MECHANICS & ITS APPL. TO CHEMISTRY	CHM650A	Course description: Thermodynamics, Statistical Mechanics of noninteracting systems,interacting systems, and nonequilibrium systemsCourse Details1. Equilibrium Thermodynamics: Thermodynamic Equilibrium criteria, phase equilibria (4 lectures)2. Ensembles in Statistical Mechanics: Ensemble postulate and ergodicity, microcanonical,canonical and grand canonical ensembles, quantum and classicalpartition functions, quantum correlations,collective modes, occupation numbers, collections of fermions, bosons,photons, classical deal gas of spinless particles, molecular partition functions, idealparamagnets. (6 lectures)3. Correlation function, neutron scatteringexperiments, Virial equation, Meyer cluster diagrams (6 lectures)5. Interacting Systems 2: Computer Simulations Ensemble averages, ergodicity,random numbers, Monte Carlo Transitions in Lattice models Lattice gas,Ising Model, order parameter, Mean Field theory, Renormalization group theory(6 lectures)7. Nonequilibrium Statistical Mechanics (6 lectures) Linear Response theory8. 1. D.A. McQuarrie, Statistical Mechanics2. David Chandler, Introduction to Modern Statistical Mechanics3. K.L. Huang, Statistical Mechanics4. B. Widom, Statistical Mechanics: A concise introduction for chemists
2246	HSS	INTRODUCTION TO THE ART OF VIDEO MAKING	ART105A	TheoryCinematography, Production Process, Writing for Screen, Basics Picture Making, CameraAngles, Direction Time and Space continuity, Cuts, Art of Editing The 5C's of Cinematography, MascelliDirecting the Documentary, Michail RabigerA History ofthe Cinema, Eric RhodeWriting for Screen and Television, TobiasSingle camera video production, Robert L. StoneShort Film & Video, Irving & ReaThe Art of Cinema, B.D.GargaA compauion to Film Theory, Miller & Starn
2247	HSS	HUMAN RIGHTS THEORY AND PRACTICE	SOC486A	1 Introduction: Magna Carta, English Bill of Rights, American/French Declaration, Universal Declaration ofHuman Rights: Background, Content and Relevance2 Theories/Justification/ Perspectives on Human Rights3 The Universality of Rights; Rights Vs Duties; Individual Vs Group Rights; Civil and Political Rights Vs social, Economic and cultural rights4 The notion of rights in various religious traditions (Hindu, Muslim, Buddhist, Jain)5 The roles ofGandhi, Ambedkar and Nehru)6 Intergovernmental Organization: The United Nations (study of specific UN agencies related to human rights);Regional Instruments7 International NGO Amnesty International8 Fundamental Rights Vs DirectivePrinciples of State_Policy) and Emergency 9 Case Studies of Selected National NGOs10 Case Studies of Selected Regional NGOs 11 National Human Rights Commission 12 Humanitarian Law; International Refugee Law14 Some Issues in Human Rights: Violence and Terrorism; Women's Rights, Child Rights, Dalit Rights,Minority Rights, Tribal Rights, Capital Punishment, etc.15 Basu, D. D. Introduction to the Constitution of India. New Delhi: Prentice Hall,2008, 18th edition.Desai, A. R., editor. Violation of Democratic Rights in India, 3 volumes.Mumbai: Popular Prakashan, 1986199 Sakiko and A. K. Shiva Kumar, editors. Readings in HumanDevelopment: Concepts, Measures and Policies for a DevelopmentParadigm. New Delhi: Oxford University Press, 2003.Mohanty, Manoranjan, Paik, S. World.New Delhi: Sage Publications, 1998.Nanda, Ved P., James R. Scarritt and George W Shepherd, Jr., editors.Global Human Rights: Public Policies, Comparative Measures andNGO Strategies. Boulder: Lynne Rienner Publishers, 1998
2248	HSS	INTRODUCTION TO INDIAN SOCIETY	SOC173A	The question of identificationPostive Discrimination:Features, Legal provisionsFamily and Kinship:Family as a functional unit; Types of family; Kinship usageHindu marriage:Aims, Age, Preference and residence; Marriage; marriage; Mahr; Divorce;RecentchanqesMarriage among Christians:Historical background and chanqesReligious Traditions:Hinduism, Buddhism, Jainism, Sikhism, IslamSecularism; Reliqious violenceSocial Movements: So
2249	CHM	MOLECULAR SPECTROSCOPY	CHM637A	1. Group theory (6)Review of point groups, permutation inversion and molecular symmetry groups.2. Interaction of radiation and matter (5)Qualitative aspects, Einstein A, B coefficients, absorptionemission coefficients3. Vibrational Spectroscopy (8)Normal modes, selection rules, Fermi and Coriolis perturbations, polyatomic molecules.5. Electronic Spectroscopy (6)Introduction, symmetry aspects, vibronic effects.6. Advanced Topics in Spectroscopy: WilsonHowardWatson Hamiltonian, timedependent viewpoint. 1. Bernath, Spectra of Atoms and Molecules, 1995.2. Bunker & Jensen, Molecular Symmetry & Spectroscopy, 1998.3. Papousek & Aliev, Molecular VibrationalRotational Spectra, 1982.4. Hollas, Modern Spectroscopy, 1996
2250	CHM	PHYSICAL PHOTOCHEMISTRY	CHM636A	1. Theory of electronic absorption spectra, Beers law, Absorption crossection,Solvatochromism (4)2. Radiative and nonradiative transitions, Vibrational relaxation, Internal conversion, Intersystem crossing3. Dynamic stokes shift, Dynamic and static quenching,SternVolmer equation, Forster resonance energy transfer and theory, Fluorescenceanisotropy and Perrin equation, Excited state proton/electron transfer4. Laser: cavity dumping, Qswitchching, modelocking (4)5. Spectroscopic techniques, Spectrophotometer, Spectrofluorimeter, Time correlatedsingle photon counting, Transient absorption, Streak camera, Fluorescence lifetime measurements (3) 1. Nicholas J. Turro, Modern Molecular Photochemistry2. K. K. RohatgiMukherjee, Fundamentals of Photochemistry3. J. R. Lakowicz, Principles of Fluorescence Spectroscopy4. W. T. Silvast, Laser Fundamentals, 1992
2251	CHM	SOLID STATE CHEMISTRY	CHM626	Chemical crystallography (10)Introduction, Space lattice, Crystal point groups, space group (working knowledge),Stereographic projections, Packing in solids, Crystal structures of representative systems, Crystal field splitting (03)Crystal classifications, Madelung constant and Lattice energy.Characterization techniques (working knowledge) for solids (07)Xray diffraction, Electron microscopy (SEM, TEM, AFM), Thermal techniques (04)Magnetic momentsVSM /SQUID, Electrical resistivity V Two / Fourprobe methods and thermal conductivity, Optical band gap, XPES, XAS).Electronic properties and Band theory of solids (04)Free electron model of metals (05)Diffusion (05)Point defects, Dislocations, Extended defects, Clusters and aggregates, Color centres, Nonstoichiometryof compounds, Diffusion mechanisms, Ficks law, Kirkenall effect.Phase transitions (06)Liquid crystals, Structureproperty relations (magnetic,electrical, superconductivity, optical and thermal).Preparative techniques (07)Powder synthesis by conventional and modern chemical methods, Reaction kinetics Special methods for single crystal growth and thin films depositions. Wiley & Sons, 1989.* L Smart and E. Moore, Solid state chemistry, Chapman and Hall , 1992.* A K. Cheetham and P. Day, Solid state chemistry compounds, Clarendon Press,Oxford 1992.* C N. R. Rao and J. J. Van Wazer, Solid state chemistry property relations, SpringerVerlag, 1975. P. A. Cox,Electronic structure and chemistry of solids, Oxford Univ. Press 1987.
2252	CE	ENGINEERING GRAPHICS	TA101A	Introduction to sketching; Principal views, principles ofdimensioning Introduction to computeraidedgraphics Missing view, sectional view and assembly drawings Overview of pictorial representation, and isometric drawing

S.No.	Branch	Course Title	Courses ID	Content
				intersection of lines and planes Intersections of solids and development of lateral surfaces
2253	ME	MANUFACTURING PROCESSES II	TA202A	Engineering materials; Microstructureprocessing relationship; Solidificationprocesses: molding and casting; Joining processes: welding, brazing, and soldering; Deformationprocesses: hot and cold working; Surface modificationtechniques: carburization, CVD, PVD.Specialized Infrastructure requirement:Rolling Mill, Pressure Die Casting Unit
2254	HSS	SOCIETY AND SOCIAL PROBLEMS OF INDIA	SOC481	Sociological perspectives on social problems; Social change; Interactionbetween structural and cultural components in the process of social change; Major socialproblems of India such as social and economic inequality, governance, communalism andterrorism; The concept of inclusive growth . Beteille, Andre (2000) Antinomies of Society: Essays on Ideologies & Institutions. New Delhi:Oxford University Press.Dumont, Louis (1970) Homo Hierarchicus. London: Paladin, Granada Publishing Ltd.Engel, Karl (1987) The Dialectic of Enlightenment. New York: Oxford University Press.Five Year Plan, 200712. New Delhi: Oxford University Press.Srinivas, M.N. (1966) Social Change in Modern India. Berkeley: California University Press.
2255	BSBE	INTRODUCTION TO BIOLOGY	LIF101A	Part I Principles of Cellular LifeEffects of water's polarity: importance of cohesion of water molecules for life; moderation of Earth's temperature; Suitability of water as a solvent for life.Carbon and the molecular biology of lifeMonomers.Carbohydrates structure and functionProteins structure and functionLipids structure and functionNucleic acids structure and functionIntroduction to metabolism: pathways; energy transformation and regulation of metabolismB. Cell: Structural and functional unit of life (5 lectures)Structure: how we study cells; bird's eye view of cell structure; subcellular structures; organelles; cytoskeleton; cell surface and cell-cell communication; cell cyclePart II Principles of Inheritance: Information processing in living systemsIntroduction to heredity: inheritance of chromosomes; comparison of asexual and sexual reproduction; mechanisms of inheritance in humanChromosomal basis of inheritance: Relating Mendel's principles to chromosomes; Sex chromosomes and sex-linked inheritance; errors and exceptions in chromosomal inheritanceDNA replication and processing of RNA; synthesis of proteinOrganization and control of prokaryotic genomesOrganization and control of eukaryotic genomes: chromatin structure; control of gene expressionDNA technology and its development: single cell to multicellular organism;
2256	HSS	POPULATION ECONOMY AND SOCIETY	SOC479	Relationship between demographic trends and the socioeconomic context;Demography and population studies; World population growth; Population of India; Population theories; Data and methods in population studies.Davis, Kingsley (1968) The Population of India and Pakistan. New York: Russell and Russell.Sharma, A. K. (2011) Population and Society. New Delhi: Concept Publishing Company Pvt.Ltd.Shryock, Henry S. (1977) Demography. New York: Census.Smith David, and Keyfitz, Nathan (1977) Mathematical Demography. Berlin: SpringerVerlag.Smith, David P. (1992) Formal Demography. New York: Plenum Press.Sorokin, Pitirim (1978) Contemporary Demography. New York: Population Trends. Vol. 1 New York: United Nations.United Nations (1974) Methods of Projections of Urban and Rural Population. New York:United Nations.United Nations (1983) Indirect Techniques for Demographic Estimation.
2257	HSS	INDUSTRIAL SOCIOLOGY	SOC474	01 Industrial Sociology nature and scope, Evolution of Industry as a productive system,02 Formal organization: Bureaucracy and its functions, 03 Roles of Executives in bureaucratic organizations, Functions of Informal organizations, Importance of informal organizations, Group dynamics,04 Industrial Relations: Trifold role: management (Employers Associations/ Federations), labour (national/ International organizations), and work in industry, Labour problems, Social structure of Trade unions, Trade Unionism (in India) and labour policies in India (through five year plans),07 Industrial conflicts, and conflict resolution mechanisms,08 Unionism vrs. professionalism08 Participatory Management Issues (Indian as well as Global experiences), 1. Monappa Arun,2006, Industrial Relations, Tata McGrawhill publishing Co.(Reprint)2. Stephen Edgell, 2006, The Sociology of work, Sage publications.3. Schnieder E V, 1972, Industrial Sociology, The social structure of industry, Holt Rinehart and Winston5. Ivar E Berg, 1981, Sociological Perspectives on Labour markets, Academic Press.6. Forsyth, D.R. 2010. Group Dynamics, 5th Edition. Belmont, CA: Thomson Higher Education7. Form,1968, Industrial Sociology, London: Harper & Row,9. Parker Stanley Robert, 1981,The Sociology of Industry, Allen and Unwin
2258	CHM	SOLID STATE CHEMISTRY	CHM626A	Chemical crystallography (10)Introduction, Space lattice, Crystal point groups, space group (working knowledge),Stereographic projections, Packing in solids, Crystal structures of representative systems, (03)Crystal classifications, Madelung constant and Lattice energy.Characterization techniques (working knowledge) for solids (07)X-ray diffraction, Electron microscopy (SEM, TEM, AFM), Thermal techniques (08)Magnetic momentsVSM /SQUID, Electrical resistivity Van der Pauw / Four-probe methods and thermal conductivity, Optical band gap, XPS, XAS).Electronic properties and Band theory of solids (04)Free electron theory (05)Diffusion (05)Point defects, Dislocations, Extended defects, Clusters and aggregates, Color centres, Nonstoichiometry of compounds, Diffusion mechanisms, Fick's law, Kirkendall effect.Phase transitions (06)Liquid crystals, Structure-property relations (magnetic, electrical, superconductivity, optical and thermal).Preparative techniques (07)Powder synthesis by conventional and modern chemical methods, Reaction kinetics, Special methods for single crystal growth and thin films depositions. Wiley & Sons, 1989.* L Smart and E. Moore, Solid state chemistry, Chapman and Hall, 1992.* A K. Cheetham and P. Day, Solid state chemistry compounds, Clarendon Press,Oxford 1992.* C N. R. Rao and J. Van Wazer, Inorganic Chemistry, property relations, SpringerVerlag, 1975. P. A. Cox,Electronic structure and chemistry of solids, Oxford Univ. Press 1987.
2259	CHM	CHEMICAL BINDING	CHM621	Review of basic principles of quantum theory and atomic structure. Introduction to chemical bonding and weak intermolecular forces. (4)Electronic structure of many electron atoms and variation principles (5)Molecular orbital(MO) theory of ground and excited states of H ₂ ⁺ , homo and heteronuclear diatomic molecules, electronic term symbols, valence bond(VB) theory of diatomic molecules,comparison of VB and MO wavefunctions for diatomic molecules, configuration interaction(CI) wave functions.(6)Electronic structures of polyatomic molecules. SCFMO treatment of closed shell systems.Basis functions. SCFMO treatment of open shell systems. Equilibrium geometry, molecular vibrational frequencies. Koopmans' theorem; and Brillouin's theorems. Brief introduction to electron correlation.Miller-Plesset (MP) perturbation theory and CI calculations. Density Functional Theory (DFT), Semiempirical methods (extended Hückel and CNDO), Molecular mechanics, Topological characteristics of electron density.(8) 1. I.N. Levine, Quantum Chemistry, Fifth edition, Pearson Education (2000)2. F.L. Pilar, Elementary Quantum Chemistry, Second edition, McGraw-Hill (1990)3. J.P. Lowe, Quantum Chemistry, Second edition, McGraw-Hill (1997)5 A. Szabo and N. S. Ostlund, Modern Quantum Chemistry, Dover (1996).
2260	CHE	PROCESS CONTROL	CHE381A	Introduction: to analog and digital feedback control; measurement devicesLaplace transforms: of simple functions, derivatives, integrals, zero order holdZ transforms: of simple functions, improper sampling, state, systems with time delaysDynamic behavior of first order systems: and their responses to simple, non-sinusoidal inputsDynamic behavior of second order systems: and their responses, underdamped, overdamped, critically damped, time delays, inverse responses; Padé approximationsFrequency responses: Bode and Nyquist diagrams: first, second order systems, time delays, pure P, I and D controllers D.R. Coughanowr, Process Systems Analysis and Control, 2nd ed., McGraw Hill, New York, 1991.
2261	CE	FOUNDATION DESIGN	CE352A	Ideal and nonideal VLE (TxyPxy plots, Azeotrope): Computation of 2+2VLE data (Temperaturecomposition, Pressurecomposition plots) using (i)ideal mixture assumption and (ii) using various activity coefficient models. Benzene-ethanol, Furfural-water, benzene-cyclohexane mixtures.Two-film model for mass transfer between gas and liquid (e.g. CO ₂ absorption using K ₂ CO ₃ solution, NH ₃ absorption in dilute acid solution) :Stoichiometry, mass balance, column through the two-film model. Formulation of the steady-state problem in terms of differential equations for gas and liquid phase species using Fick's law, the gas-liquid equilibrium relations, and rate of reaction. Introduction to chemical engineering computing by B. A. Finlayson; Indian edition by John Wiley* Chemical Process Modelling and Computer Simulation by A. K. Jana; PHI Learning, 2008* Luyben W.L.,Heurleghem, 2003
2262	CHM	CHEMISTRY OF ORGANOMETALLIC COMPOUNDS	CHM616A	Introduction (2)History of organometallic chemistry; Werner complexes; Coordination number and geometry;Crystal field theory and ligand field theory; Bonding and molecular orbitals.Ligands (10)Bonding in transition metal complexesLimitationLewis Base Ligands: Halide donors, Oxygen ligands, Sulphur ligands, Nitrogen ligands: R ₃ N, R ₂ N, RN ₂ ; Tris(pyrazolyl)borate, A Few Biologically Important N-Ligands: imidazole, purine, porphyrineCyclophosphorane, arenes, metallocenes and sandwich complexes/bonded carbon ligands: carbenes, carbynesBimetallic complexes and Metal Clusters (3)Metal-metal bond in bimetallic complexes; Cluster chemistry; Reductive elimination; Intramolecular insertions/eliminations; Nucleophilic/Electrophilic attacks on coordinated ligands.Homogeneous catalysis and Catalysts (10)Introduction; Alkene isomerization; Hydrogenation; Metathesis; C-H activation and functionalization; Oxidation of olefins; Metal Clusters and catalysisPhysical methods in Organometallic chemistry (6) ¹ H NMR spectroscopy; ¹³ C NMR spectroscopy; ³¹ P NMR spectroscopy (5)Introduction, Coenzyme B ₁₂ , Nitrogen fixation; Nickel enzyme; and catalyst innovations in organic synthesis John Wiley & Sons, Ltd, New York, 20002. M. B. Smith Organic Synthesis Wavefunction, Inc 20003. H.J. Schmalz, T. Wirth Organic Synthesis Highlights, 20034. The Chemistry of the Transition Metals" by Robert Crabtree(3rd Edition, Wiley).6. The Principles and Applications of Transition Metal Chemistry, by Collman, Hegedus, Norton and Finke (2nd eds, University Science Books, 1993)
2263	HSS	SOCIOLOGY OF DEVELOPMENT	SOC470A	Objectives of the course is to portray the process of development and underdevelopment from a sociological perspective. Being essentially a theory-based course, it would analyse the process of development from a sociological perspective.1 Defining the concept of Development: Sociocultural factors, 2 The modernization perspective of development (the Structural-Functional view): Its intellectual heritage classical modernisation's methodology and its limitations,3 The Modernisation perspective: New modernization studies. 4 The Dependence perspective of development (the Neo-Marxist view): Its intellectual heritage, the classical dependence's limitations.5 The Dependence Perspective of development: New dependence studies.6 The World System Perspective: Its intellectual heritage, Contributions of Immanuel Wallerstein, the theory of world cycles, world system perspective.8 Global System Interdependence, New Structural analysis, agent-centered analyses, a new substantive focusDynamics of complex change.9 Globalization (neo Liberalism), Development and Globalization * Alvin Y So, 2005, Social Change and Development,, Sage publications, Thousand Oaks* P W Preston, 2001, Development Theory and Introduction, Blackwell Publishers.* David Lehmann, 1989, Development and Modernization, Sage Publications, Thousand Oaks
2264	HSS	SOCIOLOGY OF	SOC470	Objectives of the course is to portray the process of development and underdevelopment from a sociological perspective. Being essentially a theory-based course, it would analyse the process of development from a sociological perspective.

S.No.	Branch	Course Title	Courses ID	Content
		DEVELOPMENT		sociology.1 Defining the concept of Development: Sociocultural factors, 2 The modernization perspective of development (the Structural Functional view): Its intellectual heritage classical modernisations methodology and itslimitations,3 The Modernisation perspective: New modernization studies. 4 The Dependence perspective of development (the NeeMarxist view): Its intellectual heritage, the classical d limitations.5 The Dependence Perspective of development: New dependence studies.6 The World System Perspective: Its intellectual heritage, Contributions of Immanuel Wallerstein, the theory of world c world system perspective.8 Global System Interdependence, New Structural analysis, agent centered analyses, a newsubstantive focusDynamics of complexchange.9 Globalization (neo Liberalism), Deve * Alvin Y So, 2005, Social Change and Development,, Sage publications, Thousand Oaks* P W Preston, 2001, Development Theory and Introduction, Blackwell Publishers.* David Lehmann, 1989, Developm
2265	CHM	CHEMISTRY OF ORGANOMETALLIC COMPOUNDS	CHM616	Introduction (2)History of organometallic chemistry; Werner complexes; Coordination number and geometry;Crystal field theory and ligand field theory; Bonding and molecular orbitals.Ligands (10)Bondin limitationLewis Base Ligands: Halide donors, Oxygen ligands, Sulphur ligands, Nitrogen ligands: R3N,R2N, RN2; Tris(pyrazolyl)borate, A Few Biologically Important NLigands: imidazole,porphine, porphyrineC arenes, arenes, metalacylesMetallocene and sandwich complexes/bonded carbon ligands: carbenes, carbinesBimetallic complexes and Metal Clusters (3)Metalmetal bond in bimetallic complexes; Cluste Reductive elimination; Intramolecularinsertions/eliminations; Nucleophillic/Electrophillic attacks on coordinated ligands.Homogeneous catalysis and Catalysts (10)Introduction; Alkene isomerization; Hydro Metathesis; CH activation andfunctionalization; Oxidation of olefins; Metal Clusters and catalysisPhysical methods in Organometallic chemistry (6)1H NMR spectroscopy; 13C NMR spectroscopy; 31P NM (5)Introduction, Coenzyme B12, Nitrogen fixation; Nickel enzyme; and catalyst innovations in organic synthesis JohnWiley & Sons, Ltd, New York, 20002. M. B. Smith Organic Synthesis Wavefunction, Inc 20003. HJ. Schmalz, T. Wirth Organic Synthesis Highlights, 20034. the Transition Metals" by Robert Crabtree(3rd Edition, Wiley).6. The Principles and Applications of Transition Metal Chemistry, by Collman, Hegedus,Norton and Finke (2nd eds, University Science Boc
2266	HSS	INDIAN ART AND CIVILIZATION	ART101A	Indian Art from Ancient Times: Rock Paintings of Bhimbetka and Indus Valley Civilization;Early Indian Art (1" 10'h Century A.D.): Buddhist Art from Gandhara and Kushan School,Buddhist Arts c 14'h Century A.D.): Chola Art of Deccan Temples ofKhajuraho, Temple of Konark, Paintings of Lepakshi; 15th to 19th Century Art: MoghulMiniatures, Jain Miniature, Mysore and Tanjore Schools of Art Art20th 21" Century: Individual Artists *Indian Art A Concise History by Roy C. Craven*Early India From the Origins to AD 1300 by Romila Thapar*Moving Focus: Essays on Indian Art by K.G. Subramanyan*TheMakingofaNew'Indian'
2267	CHM	FRONTIERS IN ORGANIC CHEMISTRY	CHM612A	Asymmetric Synthesis : Including organo and metal based catalysis (13)Synthesis using Organometallic Chemistry : Transition and main group elements basedreactions involving region, stereo, and enan etc. (5)Green Chemistry, Glycobiology, Synthetic aspects using Domino reactions, (12)Principles of atom economy with examples, Templated and solid supportedOrganic Synthesis. M. B. Smith Organic Synthesis Wavefunction, Inc 2000HJ. Schmalz, T. Wirth Organic Synthesis Highlights, 2003J. Tsuji, Transition metal reagents and catalyst innovations in organic synthesis JohnWiley&
2268	CHM	FRONTIERS IN ORGANIC CHEMISTRY	CHM612	Asymmetric Synthesis : Including organo and metal based catalysis (13)Synthesis using Organometallic Chemistry : Transition and main group elements basedreactions involving region, stereo, and enan etc. (5)Green Chemistry, Glycobiology, Synthetic aspects using Domino reactions, (12)Principles of atom economy with examples, Templated and solid supportedOrganic Synthesis. M. B. Smith Organic Synthesis Wavefunction, Inc 2000HJ. Schmalz, T. Wirth Organic Synthesis Highlights, 2003J. Tsuji, Transition metal reagents and catalyst innovations in organic synthesis JohnWiley&
2269	CHM	BIO-INORGANIC CHEMISTRY	CHM646	1. Mineral Origin of life. Archaeal, Eucarial and Bacterial domain. (3 Lectures)2. Transition metal ions in biology. Metallobiomolecules. Electron carriers, oxygencarriers, enzymes. Biogeoinorganic chemistr proteins, Cytochrome P450, Nitrophorin, NOsynthase,peroxidase, catalase, Ferritin, cytochromeC oxidase, cerulplasmin, bluecopper proteins, di and tricopper proteins. Other enzymes like,hydrogenase, me enzymeslike sulfite oxidase, xanthine oxidase, nitrate reductase, DMSO reductase,tungsten containing formate dehydrogenase and tungsten bearinghyperthermophilic and thermophilic enzymes. Zn enzy water splitting. (15 Lectures)4. Active site analogue reaction models and structural models of theseenzymes. (5 Lectures)5. Environmental chemistry, auto exhaust, arsenic and other heavymetal pollution anticancer drugs. 1. Principles of Bioinorganic Chemistry, S. J. Lippard and J. M. Berg,University Science Books, Mill Valley, 1994.2. Bioinorganic Chemistry: Inorganic Elements in the Chemistry of Life, W. Kaim and B.Schw
2270	CHM	PHYSICAL PHOTOCHEMISTRY	CHM636	1. Theory of electronic absorption spectra, Beers law, Absorption crosssection,Solvatochromism (4)2. Radiative and nonradiative transitions, Vibrational relaxation, Internal conversion, Intersystem crossing Dynamic stokes shift, Dynamic and static quenching,SternVolmer equation, Forster resonance energy transfer and theory, Fluorescenceanisotropy and Perrin equation, Excited state proton/electron trans laser: cavity dumping, Qswitching, modelocking (4)5. Spectroscopic techniques, Spectrophotometer, Spectrofluorimeter, Time correlatedsingle photon counting, Transient absorption, Streak camera, Fluor (3) 1. Nicholas J. Turro, Modern Molecular Photochemistry2. K. K. RohatgiMukherjee, Fundamentals of Photochemistry3. J. R. Lakowicz, Principles of Fluorescence Spectroscopy4. W. T. Silvast, Laser Fundam
2271	CHM	ENZYME REACTIONS MECHANISM AND ENZYME KINETICS	CHM632	Enzyme kinetics of single and multiple substrate systems including Enzyme assays andinhibition (15 lectures)Cooperativity and multienzyme systems (4 lectures)Enzyme structure and identification of a and mechanisms of some enzymes likeSerine proteases, polymerases, ribonucleases, lysozyme and ribonucleotide reductases(radical enzyme) (15 lectures)Mechanism based enzyme inhibition and drug 1. Allan Ferhst, Structure and Mechanism in Protein Science: A Guide to Enzyme Catalysisand Protein Folding2. N. C. Price and E. Stevens, Fundamentals of Enzymology: The Cell and Molecular Biologyof 2nd Edition.
2272	HSS	URBAN HABITAT AND SOCIAL INTERACTION	SOC477	From rural to urban: debates around the concepts of gemeinschaftand gesellschaft; Origin of city in history; Theories of urban sociology (Classical andContemporary); Technology and urban life; Poetics a ofurban sociology. Rao, M. S. A., ed. (1974) Urban Sociology in India: Reader and Source Book. NewDelhi: Orient Longman.Gottdiener, Mark and Ray Hutchison (2006) The New Urban Sociology. Boulder: WestView Press.Got Being Urban: A Sociology of City Life. Lexington, Mass: D. C.Heatb and Company.Lin Jan and Mele Christopher, ed. (2005) The Urban Sociology Reader. London:Routledge.Palen, J. John. (2008) The Urban TwentiethCenturyUrbanism. New York: Routledge.
2273	CHE	UNIT OPERATION LABORATORY -I	CHE391A	Safety, ethical guideline, error analysis, data presentation Brief introduction to topics of experiments, relevance in industry Fluid flow, fluid pru1icle system Thermodynamics Heat and mass transfer Demon hydraulic conveying, lbelt conveyingAnalysis and testing of crude oil and petroleum products: 1Determination of octane number, cetane number, pour point,cloud point, smoke point * Unit Operation Laboratory Manual.* "Unit Operations of Chemical Engineering" (7th Edition), by W.L. McCabe, J.C. Smith,and P. Harriott, McGraw Hill Publishing, 2005 (ISBN 0072848235).* Cou
2274	CHM	ORGANIC PHOTOCHEMISTRY	CHM614	An overview of basic concepts of photochemistryEnergy transfer; theoretical aspects of organic photochemistry; reaction mechanisms;photoreduction and photosubstitution reactions; photocycloaddition functional groups.Singlet oxygen: generation and reactions; photoinduced electron transfer{basic concepts,illustrative examples of application to organic synthesis; photochemistry in organizedmedia.Na Modern Molecular Photochemistry by N. J. TurroOrganic Photochemistry by J. M. Coxan and B. HaltonEssentials of Molecular Photochemistry by A. Gilbert and J. BaggotOrganic Photochemistry and Pho
2275	MSE	MANUFACTURING PROCESSES I	TA201A	Introduction to manufacturing:Evolution of manufacturing, Engineering materials classification, Evolution of manufacturing and its classifications. Engineering Materials: Structure of materials, Types of r casting processes, Patterns, Core making, Gating system, Solidificationof pure metals and alloys, shrinkage, gas solubility, Riser design, Investment casting, Casting defects. Joining processes: Fusion wel and Soldering, Solid state welding processes, Thermit welding. Deformation processes: Engineering stressstrain curve, Effect of temperature on the workability, Extrusion (direct and indirect)Rolling class metallurgy: Introduction, Powder production, Compaction, and Sintering, Engineering stressstrain curve. Plastic injection molding: Flow forming of plastic components. Heat Treatment: Special techniques
2276	ECO	INTRODUCTION TO ECONOMICS	ECO101A	Definition and central problems of Economics, Microeconomics vs.Macroeconomics, Normative vs. Positive Economics, Stock vs. Flow variables; Aggregateeconomic variables: Gross domestic product (G Exchange rate;Laws of demand and supply, Elasticity, Market equilibrium, Consumer and producer surplus,Price controls; Utility theory: Laws of diminishing marginal and equimarginal utility,Indifference c of variable proportion, Shortrun cost functions,Isoquant and isocost, Optimizing conditions, Longrun cost functions; Types of market:Perfect Competition, Monopoly, Oligopoly, Monopolistic Competition. 1. Economics, by Paul A. Samuelson, any edition.2. Macroeconomics, by N. G. Mankiw, 6th edition.3. Microeconomics, by R. S. Pindyck, D. L. Rubinfeld and P. L. Mehta, 6th edition.4. Intermediate Microecc Microeconomic Analysis, by R. R. Barthwal, any edition.
2277	PHY	PHYSICS-II	PHY103A	Vector calculus; Electrostatic with full use of vector calculus calculation of electric fields, electrostatic potential and Laplaces equation and uniqueness of its solution; Method of images; Energy in electro and torque on a dipole in an electric field; Electrostatics in a medium, Displacement vector and boundary conditions, linear dielectrics, force on a dielctric; Magnetostatics with full use of vector calculus, in field Magnetostatics in a medium, magnetization, bound currents, magnetic field H, boundary condition on B and H, magnetic susceptibility, ferro, para and diamagnetism.; Faradays law, energy in magneti approximation; Maxwells equations in vacuum and conducting and nonconducting medium, Energy in electromagnetic field, Poynting vector, plane electromagnetic waves; Reflection and refraction of ele

S.No.	Branch	Course Title	Courses ID	Content
2278	PHY	PHYSICS-I	PHY102A	Transformation of Scalars and Vectors under Rotation, Form Invariance of Newtons Second Law, Forces in Nature; Solving Newtons Equations of Motion in Polar Coordinates for problems including Conservative and NonConservative Forces; Central Forces, Energy Equation and Diagrams, Elliptical, Parabolic, and Hyperbolic Orbits, Harmonic Oscillator with Damping, Forced Oscillations and Resonance, inclusion of Newtons Law of Gravitation, Forces, Weather Systems, Foucault Pendulum; Angular Momentum and Torque, Rigid Body Dynamics, Moment of Inertia Tensor, Principal Axes, TorqueFree Precession, Gyroscopes, Eulers Equations; Special Relativity; EnergyMass Relation
2279	EE	INTRODUCTION TO ELECTRONICS	ESC201A	Circuit analysis techniques (nodal, mesh, superposition, Thevenins, andNortons theorems); Transient analysis of capacitive and inductive circuits; Sinusoidal steadystateanalysis of circuits containing resistors, capacitors, inductors, diodes, MOSFETs and amplifiers; ICfabrication; Operational amplifier circuits and waveform generators; Number system, logic gates,logic minimization, combinational circuits; Field programmable gate arrays (FPGAs)
2280	HSS	INDIAN SOCIETY AND CULTURE	SOC473	Approaches to study Indian Society. Social stratification: caste, jati and varna among Hindus, nonHindus. Marriage and Family among Hindus, Muslims and Christians; polyandry and polygyny; regulations on marriage, dowry, divorce, Sanskritization and Westernization; parochialization and universalization; structural, dialectical, cognitive historical,and indological approaches. Ortilogenetic processes of cultural change in culture traditions. Processes of change: peasant religious and sectarian movements; dalit and backward caste movements; tribal, labour movements. New Social Movement: Ecological Movement. Dumont, Louis. Homo Hierarchius: The Caste System and its ImplicationsLondon: Weidenfeld and Nicholson, 1970.Ghurye, G. S. Caste and Race in India. Mumbai: Popular Prakashan, 1969, 5thedition.Gupta, R. S. India. Calcutta: Oxford University Press,1966.Rao, M. S. A. Social Movements and Social Transformation: A Study of TwoBackward Classes Movements in India. New Delhi: Macmillan, 1979.Shah, Ghanshyam. Indian Tradition. Jaipur: Rawat Publications,1986 (reprint).Srinivas, M. N. Social Change in Modern India. New Delhi: Orient Longman,1995 (reprint 2002).
2281	SPEC	EVENING EXERCISE	PE102	There are two components in this course, namely: 1. Physical exercise2. Personality Development Activities.Activities under physical exercise: Warming up exercises: jogging, running, stretching, simpleexercises like: 1. Strength 2. Speed 3. Endurance 4. Flexibility 5. CoordinateActivities. Intramural activities: Group sports.Personality development activities: There are five streams in this component: 1. Games and sports, the students choose one of the following 12 sports: 1.Athletics 2. Badminton 3. Cricket 4. Hockey 5. Table Tennis 6. Tennis 7. Football. 8.Swimming 9. Basket ball10. Volleyball11. Weightlifting12. Badminton
2282	CHM	ADVANCED ORGANIC CHEMISTRY II	CHM602A	Principles of retrosynthetic analysis: Linear and convergent synthesis, Synthesis 7under steric control, Regio and stereoselective synthesis, Basic synthetic methods.Methodologies for the construction of 3- and 4-membered heterocyclic rings. Application In organic synthesis.10 Corey and Cheng, The Logic of Chemical Synthesis, Wiley, 1989.Nicolaou and Sorensen, Classics in Total Synthesis, 1996.Nicolaou and Snyder, Classics in Total Synthesis II, 2003.Carey and Sundberg, Advanced Organic Chemistry, 5th edition, Wiley, 2000.
2283	BSBE	BIOINFORMATICS & COMPUTATIONAL BIOLOGY	BSE322A	Introduction to bioinformatics, biological databases and their growth, Concept of homology and definition of associated terms, pairwise sequence alignment,dotmatrix plot, dynamic programming algorithm (Needleman-Wunsch algorithm), gap penalty, statisticalsignificance of alignment Multiple sequence alignment, Sum of pairs method, CLUSTAL W, Genetic AlgorithmPattern finding in protein and DNA sequencing, Gibbs Sampler, Hidden Markov Model, distance method(neighborjoining), maximumlikelihood methodGene prediction in prokaryotes and eukaryotes, homology and ab initio methodsGenome analysis and annotation, comparative genomics*Bioinformatics: Sequence and Genome Analysis by David W. Mount, Cold Spring Harbor Laboratory Press (2001)* Developing Bioinformatics Computer Skills by C. Gibas and P. Jambeck, 2003; Reilly (2003) and G. Mitchison, Cambridge University Press (1998)
2284	BSBE	SCIENTIFIC & PROFESSIONAL COMMUNICATION	BSE301A	Essential elements of written communication: discussion of a topic, identification of thekey elements, clarity of the rationale in a scientific and technical work, elaboration oftechnical details, key elements of a presentation. How to underscore thesignificance of a project/finding, and its larger meaning, conclusionsEssential elements of verbal communication: what communication skills interestsaudience, how to navigate through a presentation. How to convey a scientific message for thelay audience, making new ideas understandable, how to engage with the audience.
2285	CHE	MASS TRANSFER & ITS APPLICATIONS	CHE313A	Introduction: Definition of rate of reaction, types of reactors, industrial reactions and reactorsBasic Concepts in Chemical Kinetics: reaction rate constant, reaction order, elementary reactions and molecular theory of reaction rates. Variable volume reaction systems, differential and integral methods ofkinetic analysis, half-lives, least-squares analysis.Noulementary Homogeneous Reactions: reaction mechanisms, pseudosteady state approximation, multiple reactor systemsDesign for Multiple Reactions in Isothermal Batch Reactors. H. Scott Fogler, Elements of Chemical Reaction Engineering, 4th edition, 2006,PrenticeHall of India, New Delhi.
2286	SPEC	MORNING EXERCISE	PE101A	There are two components in this course, namely: 1. Physical exercise2. Personality Development Activities.Activities under physical exercise: First Phase of physical fitness test. Warming up exercises: jogging, running, stretching, simpleexercises like: 1. Strength 2. Speed 3.Endurance 4. Flexibility 5. Coordinate Activities. Intramural Athletics Meet. Final phase ofphysical fitness test.Personality development activities: There are five streams in this component: 1. Games and sports, the students have to opt for one of theabove five streams. In Games and sports, the students choose one of the following 12 sports: 1.Athletics 2. Badminton 3. Cricket 4. Hockey 5. Table Tennis 6. Tennis 7. Football. 8.Swimming 9. Basket ball10. Volleyball11. Weightlifting12. Badminton
2287	SPEC	MORNING EXERCISE	PE101	There are two components in this course, namely: 1. Physical exercise2. Personality Development Activities.Activities under physical exercise: First Phase of physical fitness test. Warming up exercises: jogging, running, stretching, simpleexercises like: 1. Strength 2. Speed 3.Endurance 4. Flexibility 5. Coordinate Activities. Intramural Athletics Meet. Final phase ofphysical fitness test.Personality development activities: There are five streams in this component: 1. Games and sports, the students have to opt for one of theabove five streams. In Games and sports, the students choose one of the following 12 sports: 1.Athletics 2. Badminton 3. Cricket 4. Hockey 5. Table Tennis 6. Tennis 7. Football. 8.Swimming 9. Basket ball10. Volleyball11. Weightlifting12. Badminton
2288	CHM	BIOSYSTEMS	CHM481	Buffers (their use in study of biomolecules), pH, pKa of amino acids, D and L amino acidnomenclature. (1)Proteins: protein sequencing by chemical and mass & NMR spectroscopic methods, Use ofspectroscopy to study protein structure (primary, secondary, tertiary, quaternary), helices, parallel and antiparallel sheets, turns, turns), circular dichroism of secondary structures, tertiary (motifs and domains : someimportant motifs like Rossmann fold, helix turn helix, 4 helix bundles, beta-sheets, structures, Method of replication, sequencing of nucleic acids(chemical, dideoxy and fluorescence), Transcription, Translation, genetic code, genomes,genes, over expression of recombinant proteins, mutation, Glycoproteins, proteoglycans, Membranes and lipids, bacterial cell wallsynthesis and mechanism of some important antibiotics like penicillin, antibiotic resistance. (4)Metabolism: Photosynthesis, Calvin cycle, Glycolysis, Citric Acid Cycle, Reaction order, competitive, uncompetitive,noncompetitive and irreversible inhibition of enzymes. Effect of pH, temperatureon enzyme activity. (4)Biophysical techniques to purify and study proteins. Dialysis, Gel permeation chromatography,ultracentrifugation, gel electrophoresis. (3) L. Stryer,Proteins by T.E. Creighton,Genes VII by B. Lewin,Introduction to protein structure by Branden and Tooze,Enzyme structure and Mechanism by Alan Fersht.
2289	BSBE	BIOCHEMICAL ENGINEERING	BSE222A	energetics of microbial metabolism; transport phenomena; enzyme catalyzed reactions andprocesses; bioreactor design and applications; sterilization; instrumentation and control.Bioseparations and Biochemical Engineering: Filtration, precipitation; affinity precipitation;column chromatography; plate theory and principles of chromatography; different types ofchromatography, polishingcrystallization, drying, separation case studies. Processmonitoring and process economics.Environmental Bioprocesses: Interaction of mixed microbial population; aerobic and anaerobicprocesses; applications; biological wastewater treatment, bioremediation case studies. *Blanch, H. W. and Clark, D. S. Biochemical Engineering. Marcel Dekker, Inc.* Bailey, J. E. and Ollis, D. F. Biochemical Engineering Fundamentals. McGrawHill, Inc.* Belter, P. A., Cussler, E. M. and Hammel, R. T. Bioprocess Engineering Principles. Elsevier, 1988. Mohamed. A. Downstream Processing of Proteins: Methods and Protocols. Humana Press* Shuler, M. L. and Kargi, F. Bioprocess Engineering Basic Concepts* Amersham Biosciences literature notes on chromatography
2290	CHM	ORGANIC CHEMISTRY II	CHM402A	Oxidation: (6)Oxidation involving organosulfur (such as Swern) and organoselenium compounds; DessMartin, IBXand related hypervalent iodine based oxidations, Ag2CO3/celite Prevost, photosensitized oxidation of alkenes, atunfunctionalised carbons, Fleming-Tamara oxidation, and microbial oxidations.Reduction: (6)Using silanes, Al and B based reagents (e.g. DIBAL, L-selectride, K-selectride, RedAl etc.), low valentTi species, Jacobsen's epoxidation, Corey's oxazaborolidinecatalyzed reduction, Noyori's BINAP reduction, SAMP, RAMP, Evans oxazoline and Oppolzer'sultams,Aldol reaction (in brief: only principles using models).Clemmensen reduction and Si enolates via imines Michael additions (cuprates etc.) Via allyl boron, allyl tin, allyl and vinyl silanes Metal catalyzed Cyclopropanation reactions (including Simmons-Smith reaction) Ringclosing, ring opening reactions synthesis of natural products. 1. Carruthers, W., Coldham, I. Some Modern Methods of Organic Synthesis, 2008.2. Smith, M. B., Organic Synthesis, 2nd ed., 2002.3. Carreira, E. M.; Kvaerno, L. Classics in stereoselective synthesis, 20094. Smith, M. B. Organic Synthesis, 2nd ed., 2002.5. Tsuji, J., Transition metal reagents and catalysts, 2000
2291	AE	TURBULENT COMBUSTION	AE754A	Introduction, Governing equations, Statistical description of turbulence, Turbulent scales and correlations, Reynoldsaveraged equations, Mixing, Flows with premixed and nonpremixed reactants, Numerical simulation of turbulent flows Introduction Turbulent scales Spatial and temporal correlations Reynoldsaveraged equations Wallbounded shear flows Free shear flows Statistical descriptionTurbulence modeling and Mixing Introduction to premixed and nonpremixed mixtures Moment methods Wellstirred reactor Conserved scale methodsNumerical and experimental methods for Reacting flows Combustion Measurement techniques

S.No.	Branch	Course Title	Courses ID	Content
				* A first course in Turbulence, Tennekes and Lumley* An introduction to combustion, Stephen Turns* Turbulent Combustion, N. Peters* Combustion Theory, F. Williams* Theoretical and Numerical Combustion
2292	AE	TURBULENT COMBUSTION	AE754	Introduction, Governing equations, Statistical description of turbulence, Turbulent scales and correlations, Reynolds averaged equations, Mixing, Flows with premixed and nonpremixed reactants, Numerical simulation of turbulent flows, Introduction to premixed and nonpremixed mixtures Moment methods Well stirred reactor Conserved scale methods Numerical and experimental methods for Reacting flows Combustion measurement techniques * A first course in Turbulence, Tennekes and Lumley* An introduction to combustion, Stephen Turns* Turbulent Combustion, N. Peters* Combustion Theory, F. Williams* Theoretical and Numerical Combustion
2293	BSBE	BIOCHEMISTRY & BIOCHEMICAL ENGINEERING LAB	BSE223A	Fundamentals of Biochemistry: This section will deal with the experiments which can be used to determine the biomolecules both quantitatively and qualitatively. Experiment related to Spectrophotometry using bovine serum albumin (BSA). Quantifying the unknown protein using spectrophotometric measurements by UV adsorption, Lowry method, dye binding method, Bicinchoninic acid (BCA) method. Analysis of the presence of reducing sugars using Benedict's test. Kinetics, Starch conversion to glucose by salivary amylase. Paranitrophenyl 13D-glucopyranoside (pNPG) assay for 13glucosidase. Analysis of proteins by SDS-PAGE. Biochemical Engineering Bioseparation Process: Design of a microbial mass culture. Downstream processing; harvesting of cell mass and extraction of protein/enzyme (adsorption, filtration), cell sonication, precipitation, column chromatography, column regeneration and product recovery. Ethanol Fermentation. Bioreactor operation (25 lts), tangential/cross flow filtration for cell harvesting, computer simulation, mass transfer phenomenon and cell growth kinetics. * Introductory Practical Biochemistry by S. K. Sawhney and Randhir Singh, Publisher: Alpha Science International, Ltd (August 1, 2005). * Protein Purification Applications Practical Approach by Simon Roe O'Neil, Publisher: Horwood Publishing, 2004. * Biochemical Engineering Design by R. W. Smith, 2nd edition, 1995, McGraw-Hill, New York. * Protein Purification Applications Practical Approach by Simon Roe O'Neil, 2nd edition, 2004, Horwood Publishing, Chichester. * Biochemical Engineering Design by R. W. Smith, 2nd edition, 1995, McGraw-Hill, New York.
2294	CHM	BASIC PHYSICAL CHEMISTRY II	CHM222	1. Zeroth Law of Thermodynamics: Equilibrium, State Functions, Temperature, Equations of State 2. First Law of Thermodynamics: Work, Heat, Internal Energy, Heat Capacity, Concept of Enthalpy 3. Second Law of Thermodynamics: Spontaneous Change, Entropy 4. Third Law of Thermodynamics: Concept of the absolute zero temperature 5. Free Energy and Standard States: Free energies and Thermodynamic potentials, Le Chatelier's Principle, Thermodynamics, Equilibrium Constant 6. Equilibrium Thermodynamics: Chemical Potential of Mixtures, Phase Equilibrium, Phase Rule, Clapeyron Equation, Phase Diagram 7. Real Gases: Equations of State, Virial Equation, van der Waals Equation, Compressibility Factor, Real Gas Behavior, Solutions, Electrolytes, Ionic activity and the Debye-Hückel Theory, the Nernst Equation, Colligative properties, Multicomponent phase diagrams 9. Kinetic Theory of Gases and Transport Processes 10. Reaction Kinetics 1. P. W. Atkins and Julio de Paula, Physical Chemistry 2. I. N. Levine, Physical Chemistry 3. R. J. Silbey, R. A. Alberty, and M. G. Bawendi, Physical Chemistry
2295	CHE	HEAT TRANSFER & ITS APPLICATIONS	CHE312A	Introduction to heat transfer, scope, variables and general applications Modes of heat transfer (conduction, radiation, convection); Heat conduction: Fourier law, thermal conductivity in gases, liquids and solids; heat transfer through composite walls; heat transfer through fins; heat transfer through cylindrical and spherical surfaces: General differential energy balance equation, dimensionless form, simplified forms, boundary conditions, similarity and scaling analyses Steady state conduction in stagnant fluids and solids, fins, and heat exchangers: 1D and 2D numerical solutions and Matlab, etc. Convective heat transfer (laminar): heat transfer in ducts, flat plate, falling film, natural convection, correlations Convective heat transfer (turbulent): heat transfer in ducts, flat plate, natural convection, correlations * Frank P. Incropera and David P. Dewitt, Fundamentals of Heat and Mass Transfer, Sixth ed. Wiley India * J.P. Holman, Heat Transfer, McGraw Hill; latest International or Indian edition. * D. Q. Kern, Process Heat Transfer, McGraw Hill; latest International or Indian edition.
2296	BSBE	MOLECULAR BIOLOGY LAB	BSE312A	Microbiology: Introduction to sterilization techniques and certain equipment used for sterilization or maintenance of a sterile environment e.g. the autoclave and the laminar flow hood. Learning the basic techniques for microbial culture etc. Measurement of pH and introduction to the pH meter. Learning how to make liquid and solid medium for microbial culture and the various techniques for growing microbial cultures. Identification of bacteria using various subtypes e.g. Gram staining. Collection of soil samples and water samples from different locations on campus for identification and quantification of bacteria in these samples through Gram staining and colony counting. Microscopy and the observation of bacterial cells under a compound microscope.
2297	BSBE	MOLECULAR CELL BIOLOGY	BSE311A	Molecular Genetic Techniques and Genomics: Genetic analyses of mutations to identify and study genes; DNA cloning and characterization; Genome wide analyses of gene structure and gene expression. Structure of Genes and Chromosomes: Chromosomal organization genes and non coding DNA; Mobile DNA; Structural organization of eukaryotic chromosomes; organelle DNAs Transcriptional Control of Gene Expression: Regulation of transcription by activators and repressors of transcription; mechanism of transcription activation and repression. Posttranscriptional Gene Control: Processing of eukaryotic pre-mRNA; transport across nuclear envelope; cell signalling: Signalling molecules and cell surface receptors; intracellular signal transduction; G protein coupled receptors Membrane trafficking: Translocation of secretory proteins across the ER membrane. Biochemical and genetics studies on cell cycle; mechanisms regulating mitotic events; meiosis a special type of cell division. * Molecular Cell Biology, by Lodish et al (5th edition or recent), W.H. Freeman and Company, New York * Molecular Biology of the Cell, by Alberts et al (4th edition or later), Garland Science
2298	CHM	PHYSICAL ORGANIC CHEMISTRY	CHM611	Pericyclic Reactions: (7) Conservation of orbital symmetry, and Woodward and Hoffmann rules. Cycloadditions, Electrocyclizations, Sigmatropic rearrangements, and Chelotropic reactions. Orbital overlap and molecular orbital functions. Reactions at sp³, sp², and sp carbons. Examples in synthesis and biological processes. Felkin-Ahn model, Houk model, Cieplak model, EFOE model, and Cation complexation model as applied to 1,3-diene reactions and further reactions in reference to Baldwin rules for ring closure) Chemical Equilibria and Chemical Reactivity: (4) Correlation of reactivity with structure, Hammett equation, substituent constants and reaction rates. Elucidation of organic reaction mechanisms. Electron Transfer Reactions: (3) Theoretical basis, Examples of photoinduced and chemically induced electron transfer reactions (PET and CET). Organic Photochemistry: fluorescence, phosphorescence, energy transfer and electron transfer, and properties of excited states, Representative photochemical reactions of carbonyl compounds, olefins, and aromatic compounds. Miscellaneous: Kinetic control of reactions. Mechanism and Theory in Organic Chemistry. 3. Deslongchamps, P., Stereoelectronic Effects in Organic Chemistry.
2299	SPEC	EVENING EXERCISE	PE102A	There are two components in this course, namely: 1. Physical exercise 2. Personality Development Activities. Activities under physical exercise: Warming up exercises: jogging, running, stretching, simple exercises like: 1. Strength 2. Speed 3. Endurance 4. Flexibility 5. Coordinate Activities. Intramural activities: Group sports. Personality development activities: There are five streams in this component: 1. Games and sports, the students choose one of the following 12 sports: 1. Athletics 2. Badminton 3. Cricket 4. Hockey 5. Table Tennis 6. Tennis 7. Football. 8. Swimming 9. Basketball 10. Volleyball 11. Weightlifting
2300	CHE	CHEMICAL REACTION ENGINEERING	CHE331A	Introduction: Definition of rate of reaction, types of reactors, industrial reactions and reactors Basic Concepts in Chemical Kinetics: reaction rate constant, reaction order, elementary reactions and molecular theory of reaction rates; variable volume reaction systems, differential and integral methods of kinetic analysis, half-lives, least squares analysis. Nine elementary Homogeneous Reactions: reaction mechanisms, pseudosteady state approximation, reactor; multiple reactor systems Design for Multiple Reactions in Isothermal Reactors: mass balances, selectivity and yield, parallel reactions, series reactions, complex reactions, best operating conditions. Nonisothermal reactors, nonadiabatic reactors, multiple reactors with interstage cooling, multiple steady states. Nonideal flow: residence time distribution, tank in series model, dispersion model, applications to design H. Scott Fogler, Elements of Chemical Reaction Engineering, 4th edition, 2006, Prentice Hall of India, New Delhi.
2301	CHM	ADVANCED ORGANIC CHEMISTRY II	CHM602	Principles of retrosynthetic analysis: Linear and convergent synthesis, Synthesis under steric control, Regio and stereoselective synthesis, Basic synthetic methods. Methodologies for the construction of 3- and 4-membered heterocyclic rings. Application in organic synthesis. 10 Corey and Cheng, The Logic of Chemical Synthesis, Wiley, 1989. Nicolaou and Sorensen, Classics in Total Synthesis, 1996. Nicolaou and Snyder, Classics in Total Synthesis II, 2003. Carey and Sundberg, Advanced Organic Chemistry, 5th edition, 1991, McGraw-Hill, New York.
2302	CE	HYDRAULIC AND HYDROLOGIC DESIGN	CE462A	Synthetic design storms & Estimation of peak discharge, Urban storm drainage design, Culvert design, Detention storage design, Watershed modeling, Flood frequency analysis and hydrologic design under uncertainty concepts in channel design, Design of canal headworks, distribution works, and cross drainage works, Design of gravity dams, spillways, and energy dissipators. 1. Hydrologic Analysis and Design by Richard H. McCuen, Prentice Hall, New Jersey, USA. 2. Applied Hydrology by V. T. Chow, David Maidment, and Larry Mays, Tata McGraw Hill, New Delhi, India. 3. Hydraulic Engineering by R. C. Chow, McGraw-Hill, New York.
2303	CE	ENGINEERING HYDROLOGY	CE361A	Introduction: Hydrologic cycle, water budget, world water quantities Precipitation and Abstractions: Forms of precipitation, data analysis, rain gauge networks; Infiltration process, infiltration indices and Horton's infiltration equation; evaporation and evapotranspiration; Transpiration Runoff and Hydrographs: Rainfall runoff relations, time area concept, flow duration curve, mass curve, flow hydrograph, Unit Hydrograph (UH), its analysis and synthesis Log-Pearson Type III distributions, Rational method, risk, reliability, and safety factor; Hydrologic storage routing Groundwater Hydrology: Types of aquifers and properties, Darcy's law, steady flow in a confined aquifer. 1. Engineering Hydrology by K Subramanya, Tata McGraw Hill, New Delhi. 2. Applied Hydrology by V. T. Chow, David Maidment, and Larry Mays, Tata McGraw Hill, New Delhi, India.
2304	CHM	ORGANIC CHEMISTRY II	CHM402	Oxidation: (6) Oxidation involving organosulfur (such as Swern) and organoselenium compounds; Dess-Martin, IBX and related hypervalent iodine based oxidations, Ag₂CO₃/celite Prevost, photosensitized oxidation of alkenes, atunfunctionalised carbons, Fleming-Tamam oxidation, and microbial oxidations. Reduction: (6) Using silanes, Al and B based reagents (e.g. DIBAL, LiAlH₄, K-selectride, Red-Al etc.), low valent Ti species, Jacobsen's epoxidation, Corey's oxazaborolidine catalyzed reduction, Noyori's BINAP reduction, SAMP, RAMP, Evans oxazoline and Oppolzer's sultams, Aldol reaction (in brief: only principles using models). C-C bond formation and Si enolates via imines Michael additions (cuprates etc.) Via allyl boron, allyl tin, allyl and vinyl silanes Metal catalyzed Cyclopropanation reactions (including Simmons-Smith reaction) Ring closing, ring opening synthesis of natural products.

S.No.	Branch	Course Title	Courses ID	Content
				1. P. W. Atkins and Julio de Paula, Physical Chemistry2. I. N. Levine, Physical Chemistry3. R. J. Silbey, R. A. Alberty, and M. G. Bawendi, Physical Chemistry
2318	BSBE	STRUCTURAL BIOLOGY	BSE321A	Part I Principles of Protein Structure from primary sequence to three dimensional structure.Elementary ideas of bonding and structure. The building blocks. Motifs of Protein structure.Prediction, design a functions ofproteins, Recognition, Complementarity and Active Sites. Flexibility and Protein Function,Location and nature of Binding Sites , Functional Properties of, Structural Proteins, Catalysis:Overview, Control of Protein Function. Mechanisms ofRegulation.Part III Determination of 3DStructures using Xray crystallography an overview of themethod. Laboratory experiments protein preparation for crystal fordata collection. Xray diffraction data collection and processing a demonstration session. *Protein Structure and Function, By Gregory A Petsko and Dagmar Ringe, New Science Press.* Introduction to Protein Structure, Branden & Tooze, Garland Publishing.* Outline of Crystallography for Biolo
2319	AE	MEASUREMENTS IN FLUID MECHANICS	AE611	Introduction to measurement, Wind Tunnels and Water Tunnels, Flow visualization techniques,Measurement of Pressure and volume flow rate, Force Measurements, TemperatureMeasurements, Hotwire L systems, PIV Measurements, Integral opticalmeasurement techniques: Shadowgraph, Schlieren & Interferometers, LDV Measurements, LIFMeasurements. Measurement of Wall Shear Stress * Fluid Mechanics Measurement, by Richard J. Goldstein SpringerVerlIng. 1983* Experimental Methods for Engineers by J.P. Holman McGrawHill 2008* Measurement in Fluid Mechanics by Stavros Tavoularis 1998.* Instrumentation, Measurements, and Experiments in Fluids, byE Rathakrishnan, CRCPress, 2007* The Laser Doppler Technique, by L. E. Drain, John Wiley & Sons 1980.* Hotwire anemometry, by Pe CambridgeAerospace Series, 2010.
2320	AE	EXPERIMENTS IN AEROSPACE ENGINEERING - III	AE451	List of Experiments:Low Speed Aerodynamics Lab:1. Turbulence measurement'2.0 Boundary Layer. measurement3. Aerodynamic characterization of a model aircraftHigh Speed Aerodynamics Lab:1. Experiments in photoelasticity2. Experiments in vibration3. Dynamic characterization of elastomeric materials4. Inertia measurementPropulsion Lab:I. Characterization of intake2. Experiments in compress continuous combustion unit * Measurement Systems Application and Design, E. O. Doebelin*Data Reduction and Error Analysis for Physical Sciences, P. R. Bevington and D. K.Robinson.* Experimental Stress Analysis, James W. Dally
2321	ME	HAMILTONIAN MECHANICS AND SYMPLECTIC ALGORITHMS	ME726A	Part 1a Some relevant definitions and results in the theory of differentiable manifolds, smooth vector fields,differential forms,exterior) calcu lus (differentiation and integration using differential Hamiltonian mechanics (Lagrange's vs Hamilton's Equations) , Canonical Transformation , Legendre T:raus formal.ion , Symplectic Tlansfonna!.ions, Some defi.nit.ions and :result.s in the t. he Partial Differential Equation. Integrable systems (simple examples).Part 2a Some basic notions of numerical algorithms (order conditions etc). Examples of Numerical methods, Symplectic Integra 2b Symplectic RungeKutta IVlethods, Generating Function for Symplectic Rm1geKutta Methods and Symplectic Ivlethods Based on it. Variational Integrators. Introduction to Hamiltonian Perturbat geometric numerical integration with some applications to mechanical systems.
2322	EE	OPTIMIZATION FOR BIG DATA	EE606A	DSP Architecture: Von Neumann vs. Harvard architecture, VelociTI architecture Memory management of TI DSP processors C2X, C6X. Peripheral overview of TI C2X, C6X processor. Code Composer Stu processor. Optimization of assembly and C code. Pipelining. General Extension Language (GEL). Assembler, Linker. Interrupts RTDX. DSP/BIOSList of Experiments Introduction to Code Composer Stu and its Applications Audio Codec and its Applications Real Time Data Exchange Using Matlab and Labview FIR filtering by interfacing Matlab with Code Composer Studio Introduction to Interrupts Digital control of a singlephase inverter with passive load PLL for a threephase ac system PLL for a singlephase system Current control of a threephase STA TCOM Speed control of an induction motor by V/f me Rulph Chassaing, "Digital Signal Processing And Applications With The C6713 And C6416 DSK", WileyInterscience, 2004. Thad B. Welch, Wright, H.G. Cameron, and Michael G. Morrow, Tretter, "Communication System Design Using DSP Algorithms:With Laboratory Experiments for the TMS320C6701 and TMS320C6711", Springer, 2003. Nasser Kehtarnavaz, "Real Chassaing, "DSP Applications Using C And The TMS320C6x DSK", Wiley Interscience, 2002. Shehrzad Qureshi, "Embedded Image Processing On The TMS320C6000 DSP: Ex
2323	AE	FLIGHT MECHANICS	AE321	Standard atmosphere Definition of altitude, relation betweengeopotential and geometric altitudes, pressure, temperature, density altitudes 1Airfoil nomenclature, Airfoil data, infinite vs finite wings,critical r '3Airplane drag estimation for subsonic and supersonic flightregime for fuselage, wings, tail and other components of aircraft 3Flaps mechanism of high lift, estimation ofCL, CD, CLICD, for different level and unaccelerated flight, thrust available andmaximum velocity, power required for level and unacceleratedflight, power available and maximum velocity (reciprocatenginepropeller combination, jet to climb, range andendurancepropeller driven airplane, range and endurance jetairplane, take off and landing performance, turning flight andthe Vn diagram, accelerated rate of climb (energy method),spec of fixed wing Unmanned aerial Vehicles2 I. Introduction to Flight: J.D. Anderson, McGraw Hill International Editions.2. Miele, A., "Flight MechanicsTheory of Flight Paths, Vol.I",AddisonWesley, Reading, MA.3. Tewari, A., "Atmospheri Sons, Inc
2324	AE	COMPRESSIBLE AERODYNAMICS	AE311	Review of thermodynamics Governing equations of compressible flowIsentropic flow. Area Mach number relation .Speed of sound, Mach cone, Flow regimes in terms of MachnumberStationary and moving and expansion wavesConvergingdiverging nozzle, supersonic wind tunnel1D unsteady flow: Riemann problemMethod of characteristicsSmall perturbations applied to, subsonic & supersonic airfoils,slend interactionTransonic small perturbation (TSP) equationsTransonic full potential equationsRayleigh & Fanno flowExperimenta I techniquesIntroduction to hypersonics I) Elements of gasdynamics: Leipmann and Roshko, John Wiley and Sons.2) The dynamics and thermodynamics of compressible flows: A. H Shapiro, John Wileyand Sons.
2325	AE	ELEMENTS OF VIBRATION	AE231	Introduction: Particle dynamics 4Rigid body dynamics: Planar and threedimensional 12Theory of vibrations: Singledegreeoffreedom, multidegreeoffreedomsystems; free and forced vibrations; modal anal virtual work 2Hamilton's Principle; Lagrange's equation 4 1. Engineering Mechanics: Dynamics (Fifth Edition): J.L. Meriam and L.G. Kraige,John Wiley and Sons.2. Elements of vibration analysis: L. Meirovitch, McGrawHill International Editions3. Theory of vibratio
2326	ME	DIFFERENCE EQUATIONS FOR ENGINEERS	ME682A	1. Introduction and Some Applications of Difference Equations in Engineering (1 lecture)2. Preliminaries in algebra and analysis (3 lectures)3. Analogies between differential and difference equations (Calculus. Summation (3 lectures)6. Interpolation, Extrapolation (1 lecture)7. Generating functions (2 lectures)8. Linear difference equations, First order equations (2 lectures)9. Higher Order Difference equations with variable coefficients (2 lectures)12. Method of undetermined coefficients and variation of parameters (2 lectures)13. Limiting behavior of solutions (1 lecture)14. Systems of Linear differ theory and Fourier analysis (3 lectures)17. Nonlinear difference equations and boundary value problems (1 lecture)18. Asymptotic methods (1 lecture)19. Stability theory and relevance to dynamical s Difference Equations. Symmetry (2 lectures)22. Difference Equations with Continuous Time, Differentialdifference equations (1 lecture)23. Discrete Mechanics (1 lecture)24. Open problems (1 lecture) 1. An Intoduction to Difference Equations, S. N. Elaydi (Springer) (textbook)2. Difference Equation, W. G. Kelley and A. C. Peterson (Academic Press)3. Finite Difference Equations. H. Levy and F. Lessma
2327	CHE	MECHANICS OF SOFT MATERIALS	CHE678A	1. Fundamental Equations: The strain tensor; The stress tensor; Thermodynamics ofdeformation; Hooke's law; Homogeneous deformations; Equilibrium of an elasticmedium bounded by a plane; St of rods; Bending and torsion of rods. Equation of equilibrium for a bent plate; The energy of a bent plate; Application of bending plate geometry for solving problems related to Adhesion. (10 hours)3. Non elasticity. Solutions for incompressible materials. Cavitation in crosslinked networks. (10 hours)4. Mechanics of cell wall: Elasticity of cellular filaments; soft networks in cell;biomembranes, membran 1. "Theory of Elasticity, 3rd edition" by Landau and Lifshitz. Course of theoreticalphysics, vol7.2. "A treatise on the mathematical theory of elasticity" by A. E. H. Love.3. "Contact
2328	AE	THEORY OF COMBUSTION	AE753A	Introduction to combustion : Types of flames, role of chemical kineticsChemical Kinetics : Formulation of chemical kinetics equations, reactionmechanisms, steady state approximation, Arrhenius Law : Fo branching explosions, Chemical equilibrium Conservationequations for reacting flows : Shvab Zeldovich formulation. Laminar premixedcombustion : Flame speed Thermal theory (Mallard and Le Chatellie limits micro combustion. Detonation andDeflagration : Chapman Hugoniot relations, Chapman Jouguet points Liminarnon premixed combustion: Burke Schuman Analysis, Phenomenological AnalysisIgnit stirred reactor. Turbulent nonpremixedcombustion : Conserved scalar approach, two variable approach,flamelet model, direct closure.
2329	CHE	ANALYTICAL & MATERIAL CHARACTERIZATION TECHNIQUES OF ENGINEERS	CHE661A	1. Introduction: Various terms in measurement. Accuracy, precision, selectivity, limit of detection, sensitivity, response time, reliability, signal/noise and sources of noise/errors (2 lectures).2. Characterization of materials. Xray photoelectron spectroscopy (XPS), mass spectrometry (MS) and other spectroscopy techniques at atomic and molecular levels (12lectures).3. Microscopy and diffraction: Optical mic practical aspects of the microscopy and diffraction techniques will be introduced. Imaging using scanning tunneling microscope (STM) will also be considered (12 lectures).4. Thermal analysis: Applica
2330	AE	OPTIMAL SPACE FLIGHT CONTROL	AE777	1. Control Systems (Basic definitions, notation, tracking systems.)2. Guidance and navigation(Basic concepts, linear regulation and tracking, proportional navigation, crossproduct steering). 3. Optimal control and the minimum principle, HamiltonJacobiBellman formulation, endpoint constraints, EulerLagrange formulation, twopoint boundaryvalue solution techniques, optimal terminal control with interior const and Hinfinity robust optimal control.) 4. Optimal guidance and control of rocket flight (Terminal guidance of interceptors, nonplanar tracking systems (3DPN),Goddards problem, 2PBVP solutions for gravit

S.No.	Branch	Course Title	Courses ID	Content
				control (Introduction to orbital mechanics, HillClohesseyWiltshire model, autonomous rendezvous and docking, minimum energy transfer, Lamberts problem, optimal guidance of reentry vehicles, nonplanar trajectories). 1. Tewari, A., Advanced Control of Aircraft, Spacecraft, and Rockets, John Wiley & Sons, Chichester, 2011. References: 1. Tewari, A., Atmospheric and Space Flight Dynamics, Birkhuser, Boston, 2006. 2. Bryson, A. E., Optimal Control of Dynamical Systems, Wiley, New York, 1975. 3. Bryson, A. E., Introduction to Optimal Control, Wiley, New York, 2007.
2331	CHE	NUCLEAR CHEMICAL ENGINEERING	CHE617	Lectures Chemical Engineering Aspects of Nuclear Power 1 Uranium, Zirconium & Thorium: Concentration, Purification and Production 4 Extraction of Nuclear Metals: Principles, Theory & Equipment; Solvent Extraction, Thorex, TTA chelation, Precipitation, Ion Exchange & Fluoride Distillation with simulation a 10 Separation of Light Isotopes: Principles; Distillation & Electrolysis of Water, Electrolysis & Steam Hydrogen/Deuterium 6 Nuclear Hydrogen Production : Sulfur Iodine Cycle, Hybrid Sulfur Route, High Temperature Electrolysis; with simulation b Nuclear heat for coal liquefaction & gasification; with simulation c Nuclear Heat for Chemical Processes Books Recommended: Text: "Nuclear Chemical Engineering" by Benedict, Pigford & Han, 2nd edition, McGraw Hill 1981. General Reading: "Solvent Extraction Principles and Practice" by Treybatt, McGraw Hill 1969. "Nuclear Reactor Engineering" by W. L. Marshall, McGraw Hill 1970. "Relevant research papers from AIChE J, Ind. Eng. Chem. Res., Nuc. Eng. & Des. SIMULATOR: ASPEN PLUS based simulations for a) Separation Flowsheets b) SI Cycle, Hybrid Sulfur Route c) Coal Liquefaction
2332	CHE	FUNDAMENTALS OF CHEMICAL ENGINEERING -II	CHE602A	Module on Thermodynamics: 1. Review of 1st and 2nd Law of Thermodynamics (including applications) (2 lectures) 2. PVT relations of pure fluids (cubic EOS, generalized correlations, compressibility factor, Maxwell's relations, residual properties. Two phase systems: Clapeyron equation, Phase diagrams. Generalized property correlations for gases) (3 lectures) 4. Multicomponent Mixtures: Partial molar properties, The ideal solution and excess molar properties. Activity coefficients and models (Margules, NRTL, Wilson, UNIQUAC) (3 lectures) 5. Vapor Liquid Equilibrium: multicomponent phase envelopes, generalized correlations (4 lectures) 6. Thermodynamic properties and VLE from cubic EOS (2 lectures) 7. Other types of phase equilibria: LLE, VLE, SLE (1 lecture) 8. Heat of adsorption, multicomponent adsorption equilibrium, reaction engineering (1 lecture; Chap. 1 of HSF) 11. Steady and unsteady state operation in idealized reactors (3 lectures; Chap. 4 of HSF) 12. Multiple reactions (2 lectures; Chap. 6 of HSF) 13. Nonelementary reaction kinetics and diffusion effects on heterogeneous reactions (2 lectures; Chap. 11 of HSF) 16. Diffusion and reaction in porous catalysts (4 lectures; Chap. 12 of HSF) 17. Design of fixed bed reactors (2 lectures) (HSF) Elements of Chemical Reaction Engineering, Prentice Hall (India) 2. S. I. Sandler (SIS) Chemical and Biological Engineering Thermodynamics, John Wiley (Indian Edition).
2333	AE	AUTOMATIC CONTROL OF AIRCRAFT ROCKETS AND SPACECRAFT	AE649A	1. Introduction to Automatic Control Systems. (Plant Models, Control Algorithms, Sensors, Actuators, Control Systems Classification.) 2. Introduction to Rigid Body Dynamics and Flight Models. 3. Linear Systems Analysis (Performance, and Robustness). 4. Single Variable Linear Control (Proportional, Integral, Derivative Control, Rate and Rate Integrating Gyros, Single Input Regulation by Pole Placement, Linear Observers, SISO Autopilots for Aircraft, Rockets, and Entry Vehicles, Planar Tracking Systems for Rockets and Entry Vehicles, Pitch Stabilization of Gravity Gradient Spacecraft, Spacecraft Single Axis Maneuvers). 5. Multivariable and Multivariable Tracking, Longitudinal Autopilots for Aircraft, Bank to Turn Missiles, and Entry Vehicles, Lateral Directional Autopilots for Atmospheric Flight Vehicles, Attitude Stabilization of Spacecraft, Roll Autopilots, Gyroscopes, Variable Speed Control Moment Gyroscopes, Thrust Vectoring Attitude Control of Rockets). 6. Terminal Time Weighted Linear Optimal Control (Time Varying Tracking Systems, Guidance and Control Systems. Simulink, John Wiley & Sons, Chichester, 2002. 1. , 2. Tewari, A, Atmospheric and Space Flight Dynamics, Springer (Birkhauser), Boston, 2006.
2334	BSBE	BIOCHEMISTRY	BSE221A	Enzymes: Concepts, kinetics, catalytic strategies and regulation: Free energy as a thermodynamic function, formation of transition states, The Michaelis-Menten model, enzyme inhibitors, coenzymes, protein structure and interconnectivities. Glycolysis and Gluconeogenesis: Energy conversion pathways in organisms, control of glycolytic pathway, synthesis of glucose from noncarbohydrate precursors, reciprocal regulation, glyoxylate cycle. Oxidative phosphorylation and electron transport chain: Mitochondrial membrane, electron transfer, proton pumps and physical link to citric acid cycle, regulation of cellular metabolism, glycogen breakdown and synthesis. Fatty acid Metabolism: Triacylglycerols as energy stores, stages of processing, pathways of synthesis and degradation, acetyl Coenzyme A Synthesis of the molecules of cholesterol, synthesis of lipids and steroids, regulation of cholesterol biosynthesis Protein Folding and turnover: Stability, pathways of folding, chaperones, proteasomes, amino acid degradation, urea formation. *Biochemistry by Jermy M Berg, John L Tymoczko and Lubert Stryer, Publisher: W. H. Freeman; Seventh Edition (December 24, 2010). *Principles of Biochemistry by Albert Lehninger, David L Nelson, Michael Cox, Publisher: W. H. Freeman; Sixth Edition (October 1, 2008).
2335	AE	MECHANICS OF HIGHLY DEFORMABLE STRUCTURES	AE692	Definition of large deformation of structures as large deflection, large rotation with small strains. Linear and nonlinear structural responses. Theory of elastica with exact and numerical solutions. Elements of stress, work conjugate stress and strain measures. Governing equations in strong and weak form. Hyperelastic material constitutive law. Incremental numerical solution of large deformation boundary value problems. Total Lagrangian method. Corotational scheme. * James F. Doyle. Nonlinear analysis of thin walled structures. Springer Verlag. 2001. * M. A. Crisfield. Nonlinear finite element analysis of solids and structures. Vols. I and II. John Wiley & Sons. 1991. * M. A. Crisfield. Nonlinear finite element analysis of solids and structures. Vols. I and II. John Wiley & Sons. 1991. 25-JUL-2013
2336	BSBE	ORGAN SYSTEM PHYSIOLOGY AND ANATOMY	BSE211A	Animal Tissues & Organ Systems: Concept of self assembly, energy and evolution, Basic structure of the cell, Organization of the cells to form different tissue systems, Assembly of different tissue systems into organs. Concept of pH, Intracellular & extracellular fluid, Thermoregulation Transport Phenomenon: Energy dependent and energy independent transport of molecules in a system Nervous System: Organizational aspects of the different sensory system of the body, Signal reception from the outside environment by these different sensory modalities and relaying the information to the nervous system Endocrine System: Outline of the endocrine system in carrying out physiological functions Circulatory System and Blood: Network of the blood vessel in the body and their fine structural modifications, Composition of the circulating fluid: Blood, Different blood components Controls system of the heart Immune System and Lymphatic System: Basic concept of immunity, Classification of immune system, Basic functioning of different components of immune system, Basic concepts of the immune system System: Anatomy of the musculoskeletal system, Tissue organization of the bone, cartilage and muscle, Role of the musculoskeletal system in movement Respiratory System: Structure of the lungs, Process of ventilation, digestive system, Functioning of the individual organs Biology Eight Edition, N. A. Campbell & J. B. Reece (2008)
2337	AE	INCOMPRESSIBLE AERODYNAMICS	AE211	Introduction to aerodynamics Atmosphere (ISA) and its stability Continuum hypothesis, dynamic similarity, Aero foil nomenclature, forces and moments Incompressible irrotational flow, Complex potential, Joukowski airfoil, Kutta-Joukowski theorem Conformal Mapping and Joukowski airfoil. Kelvin's circulation theorem Thin Airfoil theory Helmholtz theorems, Finite wing theory Computational methods; Panel and vortex lattice methods boundary layer equations, Similarity solutions. Integral approach Introduction to transition & turbulence. Turbulent boundary layer 1) Aerodynamics for Engineering Students by E.L. Houghton, P.W. Carpenter 2) Physical Fluid Dynamics by D.J. Tritton, Oxford Science Publication. 3) An Introduction to Fluid Dynamics by G.K. Batchelor, Cambridge University Press. 4) Aerodynamics by P.W. Carpenter, E.L. Houghton, Oxford Science Publication.
2338	AE	AEROSPACE PROPULSION	AE341	Introduction 2 Principle of Propulsion Air breathing and Rocket Propulsion *Reading assignment and Home work on Basic Fluid Mechanics, Thermodynamics and Compressible Flows should be given Aero Engines (thrust, efficiency, range) Cycle Analysis of Airbreathing Jet Engines (Ideal and Actual Cycles) 12 Ramjet Turbojet Turbofan Turbo prop Turbo shaft Air Intakes 3 Rocket Propulsion 6 Introduction to Solid Rocket Motors 1. Flame Temperature Thrust Coefficient Characteristic Velocity Types of Nozzles and Efficiencies Gas Turbine Combustors and Afterburners 2 1. Mechanics and Thermodynamics of Propulsion, P. Hill and C. Peterson, 2. Gas Turbine Theory, H. Cohen, G. F. C. Rogers, H. I. H. Saravanamuttoo 3. Jet Propulsion, N. Cumpsty 4. Rocket Propulsion Elements, G. B. Smith, McGraw Hill 5. Rocket Propulsion, D.P. Mishra, Prentice Hall of India, New Delhi, revised edition, 2010. 7. Gas Turbine Propulsion, D.P. Mishra, Annamaya Publisher, New Delhi 2011
2339	AE	AIRCRAFT CONTROL SYSTEMS	AE322	Definition of stability and control static stability and dynamic /stability. Moments on an airplane. Definition of pitch angle, flight path angle and angle of attack. Criteria for longitudinal and static stability. 2A total moment about CG of aircraft. Equations of longitudinal static stability. Calculation of elevator angle to trim in stick fixed vs stick free longitudinal static stability, estimation of neutral point (both stick fixed and stick free). 12 Directional static stability, lateral static stability, estimation of static margin, estimation of trim condition 6 Equations of airplane motion 3 Concept of stability and control derivatives 5 Longitudinal and lateral stability augmentation systems 3 1. Flight Stability and Automatic Control: R. Nelson, McGraw Hill Education 2. Introduction to Flight: J.D. Anderson, McGraw Hill International Editions 3. Elkin, B., "Dynamics of Flight: Stability and Control", McGraw Hill Education 4. Introduction to Aircraft Control and Stability: R. V. Schmitz, McGraw Hill Education
2340	AE	EXPERIMENTS IN STRUCTURES	AE331	Loads on an aircraft 2 Elements of Linear Theory of Elasticity 4 Idealization of Aerospace Structure ' 1 Stress Resultant 2 Extension bending of nonhomogeneous Euler-Bernoulli 4 Bending shear stress (open and closed box beams. 5 Shear center for open and closed section 4 Tapered beams 2 Beam Column Euler Buckling 5 Principle of virtual work 4 1, Theory and analysis of flight structures: R.M. Rivello. No. of lectures 2, Aircraft Structures for Engineering Students (Fourth edition): T.H.G. Megson, Elsevier Aerospace Engineering Series. 3. Analysis of Aircraft Structures: R.M. Rivello, McGraw Hill Education
2341	CHE	CFD OF MULTIPHASE REACTORS	CHE644	Mathematical Modeling: conservation equations; dimensionless Lectures numbers; boundary conditions; Numerical solution of model equations: Finite volume method; periodic flows; pressure velocity coupling; pressure correction equation based solution method; under relaxation factors; Algorithms SIMPLE/R, QUICK/Superbee; convergence and error analysis; 3 Multiphase Flow: Approaches: Eulerian & Volume of fluid (VOF); dispersed multiphase flows; flow through porous media; dense and dilute suspensions; solids pressure; granular temperature; granular flows. 5 Reactive flow processes: reactive mixing micro, macro & turbulent; RANS based models: species transport; homogeneous and heterogeneous reaction types: stirred tank, bubble column, fixed bed, fluidized bed.; Reactor Internals: impellers, baffles, spargers, heating & cooling coils, distributors & mixing grids.; population balance models: coalescence & breakage; temperature, particle size distribution, dense & dilute suspensions. 5

S.No.	Branch	Course Title	Courses ID	Content
2355	AE	EXPERIMENTS IN AEROSPACE ENGINEERING-I	AE251A	ContentPrinciples of measurement Introduction Description of MeasuringInstruments Performance Characteristics ofInstruments, Calibration, Accuracy,Precision, Bias, Dynamic responseVirtual Instrumen clusters, file I/O Data acquisition: ADC, DAC, DIO,serial and GPIB communication Motion control systemSensors Strain Gage Motion Force, Torque, Power Pressure and Sound Temperature and Heat Flux R Estimates of Mean and Errors Curve fitsAdvanced Optical Measurements (PIV,LDV,etc.) *. Measurement Systems Application and Design, E. O. Doebelin*. Data Reduction and Error Analysis for Physical Sciences, P. R. Bevington and D. K.Robinson.*. Experimental Stress Analysis, James W. Dr
2356	AE	SPACECRAFT GUIDANCE NAVIGATION AND CONTROL	AE645	Attitude dynamics and stability ofthreearxis stabilized, singlespin, dualspin, and multibodyspacecraft with a1ticulated antennas, sensors, and solar arrays.Design of control ofthreearxis stabilized spacecraft threeaxis attitude maneuver controllers using reaction wheels and thrustersControl of spinning spacecraft in transfer orbit during delta_ v firing and in operational orbitsaround the Earth, and design of act control of dualspin spacecraftPrecision pointing and tracking controllers for tracking landmarks, moving objects, and othersatellites for crosslink communicationSolar array controllers for tracking the Sun with spacecraft dynamics andcontrol systemsAttitude determination with gyros, star trackers, sun sensors, and horizon sensors usingalgorithms such as TRIAD and QUEST (quaternion estimator); senso techniques will be related with the control of Indian communication, remotesensing, and other special purpose satellites (Ca1tosat, Edusat, telemedicine). * Hughes, P.C., Spacecraft Attitude Dynamics, John Wiley, 1986.* Sidi, M.J., Spacecraft Dynamics and Control, Cambridge University Press, 1997* Noton, M., Spacecraft Navigation and Guidance, Springer Geosynchronous Spacecraft, Prentice Hall, 1986* Bryson, A.E., Control of Spacecraft and Aircraft, Princeton University Press, 1994* Pocha, J.J., An Introduction to Mission Design for Geostationary Satelli Wie, B., Space Vehicle Dynamics and Control, AIAA Education Series, 1998
2357	AE	AIRCRAFT DESIGN-II	AE462A	Concepts of structural design; Vn diagram; airworthiness requirementsStress resultants for swept and unswept wings; application of modifiedbeam theoryMethods for wing stress analysis; yielding based preliminary layout of wing; margin of safety;advanced analysis (using FEM based commercial/opensource software)for full wing.TotalLaboratory component of the course:TopicConcepts of structural des modifiedbeam theory; methods for wing stress analysis; yielding based design;buckling (of columns, panels and stiffened panels) based design of thinstructures; ribsspacing; sizing and preliminary layout *. Analysis and design of Flight Vehicle Structures: E.F. Bruhn*. Airframe Structural Design: M. Niu*. Aircraft Design: A Conceptual Approach, D. Raymer (4th Edition), AIAA Press,2006.
2358	AE	AIRCRAFT DESIGN-II	AE462	Concepts of structural design; Vn diagram; airworthiness requirementsStress resultants for swept and unswept wings; application of modifiedbeam theoryMethods for wing stress analysis; yielding based preliminary layout of wing; margin of safety;advanced analysis (using FEM based commercial/opensource software)for full wing.TotalLaboratory component of the course:TopicConcepts of structural des modifiedbeam theory; methods for wing stress analysis; yielding based design;buckling (of columns, panels and stiffened panels) based design of thinstructures; ribsspacing; sizing and preliminary layout *. Analysis and design of Flight Vehicle Structures: E.F. Bruhn*. Airframe Structural Design: M. Niu*. Aircraft Design: A Conceptual Approach, D. Raymer (4th Edition), AIAA Press,2006.
2359	AE	AIRCRAFT DESIGN-I	AE461	Conceptual design based on preliminary mission requirementsSurvey of existing vehicular configurations (in similar category); lofting(preliminary layout sketches); preliminary weight estimationSelection aerodynamic characteristics and performance evaluationDesign of tail areas and control surfacesEstimation of spanwise load distributions on wing and tailTOTAL LABORATORY OF THE COURSEConcept lofting (preliminarylayout sketches); preliminary weight estimation; selection of wingloading; thrust loading; wing section and planform; fuselage layout andweight balance; estimation of aerodynamic cha distributions on wing and tail. Aircraft Design: A Conceptual Approach, D. Raymer (4th Ed.), AIAA Press, 2006.
2360	MSE	POWDER METALLURGY	MSE659A	Powder Production (Chemical Methods, Electrolytic Methods, Atomization, Mechanical Methods),Powder Characterization (Chemical Composition and Structure, Particle Size and Surface Topography,Pyrr Approach to Sintering, NonIsothermal Sintering, Microstructural Evolution, LiquidPhase Sintering, Stages of Liquid Phase Sintering, Supersolidus Sintering, Activated Sintering, PressureAssistedSintering, I. R.M. German, Powder Metallurgy Science, 2"d ed. John Wiley, 1999.2. A. Upadhyaya, G.S. Upadhyaya, Powder Metallurgy: Science, Technology and Materials, 20 II.3. ASM Handbook, Volume 7: Pow
2361	MSE	Deformation Processing	MSE657	Slip planes and systems in various crystal systems; Elasticity and Plasticity; Deformation processesincluding Rolling ;Forging, Extrusion; Drawing and deep drawing etc.; Deformation of plastics and polym I. Hosford, W. F., and Cadell, R. M., 2007, Metal Forming: Mechanics and Metallurgy, CambrideUniversity Press, Cambridge.2. George Dieter, 1986, Mechanical Metallurgy, McGraw Hill
2362	MSE	SCIENCE AND TECHNOLOGY OF THIN FILMS AND DEVICE FABRICATION	MSE604A	1. Introduction Introduction to device processing steps 4 Examples of various devices (with emphasis on solar cellsand MOSFETs) Need for miniaturizationThin Film Deposition2. Basics of thin film Brief growth control3. PVD Processes Evaporation (Thermal and ebeam)5 Principles of glow discharge and various sputteringprocesses4. Chemical Growth Fundamentals of CVD growthProcesses 4 Modern analysis 1 Optical analysis Morphology analysisDevice Fabrication6. Substrate seiection Selection of substratesand preparation , Single crystal growth (Silicon)3 Role of substrate surface and contamin Photoresists (Negative, Positive etc) Exposure Development 5 Masking Variants of lithography with emphasis on Photo and ebeam Precautionary steps8. Material Removal Wet (Chemical) and dry (Plasma composition in devicesand heat treatment Oxidation and heat treatment10. Metallization and Adhesion and morphology issuesInterconnects Introduction to electromigration visavis metallization: 2Impa particular kind offor a particular case device e.g. Si Solar Cell, MOSFET, 111V solar cell devices 4(take the case of adevice) Gandhi Si Processing (Volume I and II) Lou and Mayer Introduction to Microfabrication by Sami Fransilla (Wiley) Fundamentals ofMicrofabrication: The Science of Miniaturization by Marc Madou (CRC Pre Press) Pulsed Laser deposition of thin films by Chrisey and Hubler (Wiley Interscience)
2363	AE	TRANSITION AND TURBULENCE	AE625	Elements of viscous flows and thin shear layer approximation. Different typesof TSL flows. Instabilities in flows. RayleighTaylor, KelvinHelmholtz mechanisms.Thin shear layer instabilities: for parallel and boundary layers, wakes, jets and free shear layers. Primary andsecondary instabilities and relationship of instability theories to transition.Receptivity of shear layers for 2 and 3D flows. Bypass transition in other kinematic tools of turbulence. Role of stretchingand dispersion in small scale turbulence. Route to turbulence: Chaos via nonlinearity,instabilities and bifurcation. Coherent structures in turbulence:U decomposition (POD). DNS, LES and other closureschemes of turbulence * Drazin. P.G. & Reid W.H.: Hydrodynamic Stability 1981, (CUP).* Davidson. P.A.: Turbulence (2003) (OUP)* Lmdahl. M.T. & MolloChristensen: Turbulence & Random Processes in Fluid Mechanics 1992 (CU (CUP)* Sagaut. P.: Large Eddy Simulation for Incompressible Flows. 2000 (Springer)* Sengupta T.K.: Foundation of CFD, 2004 (Univ. Press)
2364	ME	COMPLIANT MECHANISMS	ME851A	Genesis of compliant mechanisms; glimpse of applications of compliant mechanisms; mobilityanalysis; large deformation analysis: continuum and discrete perspectives: finite element analysis,pseudo ri based on linkage synthesis methods, selectionbased methods, and structural optimizationbased methods; kinetoelastic maps and understanding limits of compliant mechanisms; static balancing of com
2365	ME	BASIC CONTROL SYSTEM FOR MECHANICAL ENGINEERS	ME850A	1. Elementary review of dynamic systems. Equations of motion. Numerical solution of ODEs. Linearization.Stability. (5 lectures)2. Laplace transforms and inverse Laplace transforms. Block diagrams. Tra Nonminimum phase systems and their transient responses. Steady state responses. (7 lectures)3. Root locus plots. Nyquist plots. Bode plots. Implications for transient responses. (5 lectures)4. Compens motivation and sample problems. (3 lectures)6. Discrete time systems. Stability. (2 lectures)7. State space. Standard form for an LTI system. General solution. Controllability and observability. Pole placem lectures)9. Introduction to timedelayed control. (2 lectures)10. Simulations of nonlinear systems with linearization based controllers. Case studies from the literature as time permits. (41ectures)In addition 1. K. Ogata. Modern Control Engineering. (PHI)2. F. Golnaraghi and B. C. Kuo. Automatic Control Systems. (Wiley)
2366	CE	GROUND IMPROVEMENT TECHNIQUES	CE634N	Engineering properties of soft, weak and compressible deposits, principles of treatmentloading(static and Dynamic), Accelerated flow, Reinforcement, Drainage and fillers,Injections, Thermal, electrical and Methods and Case histories Moseley and Kirsch: Designing with Geosynthetics (2"d Edition), Routledge, (2004), ISBN: 0415274559
2367	MSE	POWDER METALLURGY	MSE659	Powder Production (Chemical Methods, Electrolytic Methods, Atomization, Mechanical Methods),Powder Characterization (Chemical Composition and Structure, Particle Size and Surface Topography,Pyrr Approach to Sintering, NonIsothermal Sintering, Microstructural Evolution, LiquidPhase Sintering, Stages of Liquid Phase Sintering, Supersolidus Sintering, Activated Sintering, PressureAssistedSintering, I. R.M. German, Powder Metallurgy Science, 2"d ed. John Wiley, 1999.2. A. Upadhyaya, G.S. Upadhyaya, Powder Metallurgy: Science, Technology and Materials, 20 II.3. ASM Handbook, Volume 7: Pow
2368	AE	AIRBREATHING PROPULSION	AE341A	Introduction- Principle of propulsion • Air-breathing and rocket propulsion • Reading assignment and home work on basic fluid mechanics, thermodynamics and compressible flows.</ • Performance of gas turbine engines (Thrust, efficient, range). Cycle analysis of air-breathing jet engines (ideal and actual cycles) • Ramjet • Scramjet • Turbojet • Scramjet • Turbojet • Turbofan • Turboprop • Turboshift. Introduction to turbo-machinery • Types of turbomachinery • Conservation of angular momentum and losses • Compressor characteristics • Rotating stall and surge. Axial compressors • Principle of operation • Stage dynamics • Multi staging • Radial ec • Efficiency and losses • Blade cooling • Compressor-turbine matching. Gas turbine combustors and after-burners.

S.No.	Branch	Course Title	Courses ID	Content
2369	AE	EXPERIMENTS IN FLIGHT MECHANICS	AE421A	Introduction to fliaht testing and instrumentation 1Techniques and data reduction methods, Error analysis 2Calibration of flight and special flight test instruments 1Evaluation of cruise and climb performance of airplane dvnamic modes and stall characteristics1Introduction to flight testing and instrumentation 3Techniaues and data reduction methods, Error analysis 6Calibration of flight and special flight test instruments 3 stability and control characteristics 3Observations of airplane dynamic modes and stall characteristics 3
2370	AE	AIRCRAFT CONTROL SYSTEMS	AE322A	Definition of stability and control static stability and dynamicstability. Moments on an airplane. Definition of pitch angle, flightpath angle and angle of attack. Criteria for longitudinal and staticstability . Aerodynamic tototal moment about CG of aircraft. Equations of longitudinal staticstability. Calculation of elevator angle to trim in stick fixed vsstick free longitudinal static stability, estimation of neutral point(both stick fixed and stick free) Directional static stability, lateral static stability, estimation ofstatic margin, estimation of trim condition Equations of airplane motion Concept of stability and control derivatives Longitudinal and lateral control system *. Flight Stability and Automatic Control: R. Nelson, McGraw Hill Education*. Introduction to Flight: J.D. Anderson, McGraw Hill International Editions*. Etkin, B., "Dynamics of Flight: Stability and Control
2371	AE	COMPRESSIBLE AERODYNAMICS	AE311A	Review of thermodynamics (2HRS) Governing equations of compressible flow (2HRS) Isentropic flow, Area Mach number relation (3 HRS) Speed of sound, Mach cone, Flow regimes in terms of Machnumber, shock wave expansion (3 HRS) Reflection, intersection of shocks and expansion waves (2 HRS) Convergingdiverging nozzle, supersonic wind tunnel (2 HRS) 1D unsteady flow: Riemann problem (2 HRS) Method of characteristics rules and area rule (2 HRS) Curved shock and Croccos Theorem (1 HRS) ShockBoundary layer interaction (1 HRS) Transonic small perturbation (TSP) equations (2 HRS) Transonic full potential equation *Elements of gasdynamics: Leipmann and Roshko, John Wiley and Sons.*The dynamics and thermodynamics of compressible flows: A. H Shapiro, John Wileyand Sons.
2372	AE	INCOMPRESSIBLE AERODYNAMICS	AE211A	Introduction to aerodynamics 1Atmosphere (ISA) and its stability 2Continuum hypothesis, dynamic similarity, 2Aerofoil nomenclature, forces and moments 1Incompressible irrotational flow, Complex potential, Joukowski theorem (2 HRS)Conformal Mapping and Joukowski airfoil, Kelvins circulation theorem (3 HRS)Thin Airfoil theory (3 HRS)Helmholtz theorems, Finite wing theory (5 HRS)Computational methods for aerodynamics Introduction to NS equations (2 HRS)Prandtl boundary layer equations, Similarity solutions. Integral approach (4 HRS)Introduction to transition & turbulence. Turbulent boundary layer (4 HRS) *Aerodynamics for Engineering Students by E.L. Houghton, P.W. Carpenter* Physical Fluid Dynamics by D.J. Tritton, Oxford Science Publication.* An Introduction to Fluid Dynamics by G.K. Batchelor, Cambridge University Press
2373	AE	INTRODUCTION TO AEROSPACE STRUCTURES	AE331A	Loads on an aircraft (2 HR)Elements of Linear Theory of Elasticity (4 HR)Idealization of Aerospace Structure (1 HR)Stress Resultant (2 HR)Extensionbending of nonhomogenous EulerBernoulli (4 HR) Bending of thin walled beams: Single celled and multi celled box beams. (5HR) Shear center for open and closed section (4 HR) Tapered beams (2 HRS) BeamColumn Euler Buckling (5 HRS)Principle of virtual work (4 HR) * Theory and analysis of flight structures: R.M. Rivello.* Aircraft Structures for Engineering Students (Fourth edition): T.H.G. Megson,Elsevier Aerospace Engineering Series.* Analysis of Aircraft Structures
2374	MSE	MATERIALS FAILURE: ANALYSIS AND PREVENTION	MSE676	1. Introduction to Failure analysis and prevention: Concepts, root causes analysis, primary rootcauses, design deficiencies, material defects, manufacturing/installation defects, categoriesof failure, failure prevention (2 hours)2. Fracture mechanics: Ductile fracture of metallic materials and their interpretations, factorsaffecting ductilebrittle relationships. (3 hours)3. Fracture modes, Ductile fracture of metallic materials and their interpretations, factorsaffecting ductilebrittle relationships. (3 hours)4. Failure characteristics of Ceramics and Plastics (2 hours)5. Failure characteristics of Ceramics and Plastics (2 hours)6. Failure characteristics of Ceramics and Plastics (2 hours)7. Wear Failures and Prevention (2 hours)8. Corrosion related failure mechanisms: macroscopic and microscopic characteristics, statistical aspects of fatigue,Fatigue failure prediction and life assessment. (3 hours)9. Wear Failures and Prevention (2 hours)10. Corrosion related failure mechanisms: macroscopic and microscopic characteristics, statistical aspects of fatigue,Fatigue failure prediction and life assessment. (3 hours)11. Structural life assessment methods, Nondestructive techniques. (2 hours)12. Fractography,, (4 hours)13. Illustrative "Case studies of engineering failure " due to: improper processing practice,improper treatment procedure, improper design, unanticipated service conditions, etc. (8 hours) components fail , D.J Wulpi, ASM International, The MaterialsInformation Society, 1 999.3. A.J. McEvily, Metal Failures: Mechanisms, Analysis, Prevention, Jolm Wiley and Sons,2002.4. Practical engineering failure prevention, Volume II , ASM Handbook, The MaterialsInformation Society, 2002.6. Failure analysis of engineering structures: Methodology and ase Histories, V.Ramachandran, A.C. Raghuram, R.V. Krishnamoorti, Prentice Hall
2375	ME	APPROXIMATE METHODS IN ENGINEERING MATHEMATICS	ME781A	This course is an introduction to approximate mathematical methods in engineering. These methods provide powerful tools with which to approximately analyze omplicated/intractable engineer ing/science problems in physics in an accessible manner. Finally, these approximate methods play a symbiotic role to computational analyses; these latter techniques typically falter near singularities, which is where approximation is most useful.1. Convergent/divergen power series; Asymptotic series; Error estimates; Subdominance; Stokes"phenomenon.2. Asymptotic analysis of integrals (11 lectures): Integration by parts; Laplace"s method; Introduction to perturbation theory (4 lectures): Algebraic equations; Iteration and expansion; Method of dominant balance; Regular and singular perturbation problems; J;lescaling; Nonintegral power series; Application.3. Summation of series (3 lectures): Improvement of convergence; Divergent series; Pade approximation; Application.4. Boundary layer theory (7 lectures): Introduction to boundary layer theory; Asymptotic limits; Application.5. vVKB Theory (4 lectures): Formal procedure; Patched asymptotic approximation; Turning point problems. Application6. Multiplescale analysis (6 lectures): Resonance and secular terms. Application.7. Bender, C. M. and S. O. Orszag, 1999. Advanced Mathematical Methods for Scientists andEngineers, SpringerVerlag: New York, USA.2. Hinch, E. J., 1991. Perturbation Methods, Cambridge U. Press: Cambridge
2376	ME	GRANULAR MATERIALS	ME773A	1. Introduction to granular materials: Illustrative examples.2. Revision of continuum mechanics.3. :Materials with yield strength: I lohrCoulolllb and DruckerPrager materials.4. Statics: Coulomb"s law, J and K. Micromechanical behavior: Particleparticle interaction. Meanfield theory.7. Other features: Closepacking, Crystallization, Force chains, Shear banding.8. Slow dense flow: Plasticity theory.9. Application: Flow of granular materials.10. Inclined cbntes.12. Rapid granular flow: Rough inelastic particles.13. Mixing and segregation: Brazilnut effect, Rotating drum and Avalanches.14. Effect of air: Porous beds. Barchau d 111H, ,, .. ud Homglas
2377	AE	FLIGHT MECHANICS	AE321A	Standard atmosphere Definition of altitude, relation betweengeopotential and geometric altitudes, pressure, temperature, densityaltitudesAirfoil nomenclature, Airfoil data, infinite vs finite wings, criticalMach number Airplane drag estimation for subsonic and supersonic flight regimefor fuselage, wings, tail and other components of aircraft Flaps mechanism of high lift, estimation of CL, CD, CL/CD, fordifferent flaps at different angles of attack and unaccelerated flight, thrust available and maximumvelocity, power required for level and unaccelerated flight, poweravailable and maximum velocity (reciprocating enginepropellercombination, jet engine) climb, range and endurancepropeller driven airplane, rangeand endurance jet airplane, take off and landing performance, turningflight and the Vn diagram, accelerated rate of climb (energy method), specific fuel consumption fixed wing Unmannedaerial Vehicles * Introduction to Flight: J.D. Anderson, McGraw Hill International Editions.* Miele, A., "Flight MechanicsTheory of Flight Paths, Vol.I",AddisonWesley, Reading, MA.* Tewari, A., "Atmospheric and Space Flight
2378	CE	DISCOUNTINUED	CE631N	Soil composition and soil structure, Steady State flow, 2D and 3D seepage, Transientflow; Compressibility and Rate of consolidation, One, two, and three dimensionalconsolidation theories; Shear strength
2379	CHE	PROCESS ENGG. PRINCIPLES IN MICRO. FABRICATION	CHE659A	This course provides an insight into the process engineering principles inmicroelectronic fabrication. The transport processes, reaction kinetics and reactor designaspects of the following topics will be covered:1. Processing, epitaxy,lithography, plasma processing (deposition and etch), electrochemical deposition andchemical mechanical planarization. Also, the cleanliness and purity in processenvironment (e.g. fine particle control).2. 1. Stanley Middleman, Arthur Hochberg, "Process Engineering Analysis inSemiconductor Device Fabrication", McGraw Hill, 1st Ed, 1993. 2. Stephen A. Campbell, "The Science and Engineering of Microelectronic Fabrication", McGraw Hill, 2nd Ed, 2003. 3. Peter VanZant, "Microchip Fabrication", McGraw Hill, 5th Ed, 2004.(Students likely to take the course: Chemical Engineering, Electrical Engineering,Metallurgical and Materials Engineering, Mechanical Engineering, Manufacturing Engineering, and Industrial Engineering)
2380	CE	EXPERIMENTAL METHODS IN STRUCTURAL ENGINEERNIG	CE623N	Similitude and Structural Models: Dimensional analysis, Buckingham"s Pi theorem, Scale factors and dynamicsimilitudeUses and Applications of Models: Types of model investigation, Indirect and direct methods Error and unce1tainty in experiment, Measurement systems, Accuracy in models and reliability of resultsTest Planning, Design and Implementation: Testing sequence and experimental plan, Loading systems, Calibration, various types of sensors for displacement, velocity, acceleration, pressure, loads, strains, etc, fullfield measurements, .Data Acquisition System and Data Processing: Analog systems, digital systems Exercises: Experiments to illustrate buckling of stiUctural members; loaddeff1mation behaviour of beams, columns, joints, and flames under various loads; mode shapes, natural frequency, damping factor Dally, J. W. and Riley, W. F. (1978). Experimental Stress Analysis, McGraw HillHarris H. G. and Sabnis, G. M. (1999). Structural Modeling and Experimental Techniques, CRC Press 1999Nachtigal, C.L. (1999). Structural Testing, Prentice Hall I Fairmont PressHolman, J. P. (200 I). Experimental Methods for Engineers, McGraw Hill
2381	CHE	THE STRUCTURE & RHEOLOGY OF COMPLEX FLUIDS	CHE613A	Introduction to Complex Fluids (CF) and Soft Condensed Matter (SCM) (2 hours)Basic Forces, Energies and Timescales in CF and SCM. (5 hours)Excluded volume interactions, Van Der Waals interactions (5 hours)Different types of flow fields, linear and nonlinear Viscoelasticity, Kinematics and stress, Boltzmann Superposition Principle.Various Characterization Techniques (6 hours)Microscopy, Light scattering, Rheology Elementary molecular theories, linear and nonlinear rheology. Rheology of entangled polymers: Reptation theories, Transient network formulations Colloidal dispersions: (6 hours)Rheological behavior of suspensions: (6 hours)Introduction to glass transition. Rheology of amorphous polymers, very concentrated suspensions, emulsions, foams, colloidal gels. Relaxation behavior, yield stres ,thixotropic behavior, thixotropic behavior Ronald G. Larson, The Structure and Rheology of Complex Fluids, Oxford University Press(1998).Richard A.L. Jones, Soft Condensed Matter, Oxford University Press (2002).Ch. W. Macosko, Rheology: Principles and Applications, Elsevier (1989). Montgomery T. Shaw and William J. MacKnight, Introduction to Polymer Viscoelasticity,WileyInterscience; (2005)
2382	CHE	FUNDAMENTALS OF	CHE601A	Mathematics modulue:1. Linear algebra: Basics (2lectures; Chap. 3 ofMLB)2. Linear algebra: Numerical methods (3 lectures; Chap. 1 of SKG)3. Nonlinear algebraic equations: Numerical solution, including Newton's method

S.No.	Branch	Course Title	Courses ID	Content
		CHEMICAL ENGINEERING -I		(2lectures; Chap. 8 ofMLB)5. Numerical solution of ODEs Initial value problems (2 lectures, Chap. 5 of SKG)6. Numerical solution of ODEs Boundary value problems (2 lectures, ChaP. 6 of SKG)7. Probab (!lecture; Chap. 1 ofBSL)9. Shell balances for 1D momentum transport (2lectures; Chap. 2 ofBSL)10. NavierStokes equations (2lectures; Chap. 3 ofBSL)11. Unsteady and 2D momentum transfer (3 lectures; Chap. 10 ofBSL)14. Temperature distributions with more than one independent variable (3 lectures; Chap. 12 ofBSL)15. Introduction to mass transfer (!lecture; Chap. 17 ofBSL).16. Diffusion and mass tran Chap. 20 of BSL) 1. M. L. Boas (MLB), Mathematical Methods in the Physical Sciences (Second Edition),John Wiley & Sons (Indian edition).2. S. K. Gupta (SKG) Numerical Methods for Engineers, New Age International.3. Edition).
2383	EE	ORGANIC ELECTRONICS	EE611A	Lectures: General Overview of Organic Semiconductors and Electronics; Introduction to some of the basics of Molecular Quantum Mechanics; Optical and Electrical Properties of Organic Semiconductor Materials; Organic Semiconductor Processing; OLED passive and active matrix displays, OTFT circuits;Organic Solar Cell physics and processing; Research opportunities in organic electronics and the associated technologies.Labs: Fabrication of Organic LEDs and Organic Solar Cells. The purpose of this course is (i) to give the student an introduction to the world of organi electronics and (ii) to help students appreciate practical organic electronic devices and th associated technologies. It was offered as EE698W by Dr. S. Sundar Kumar Iyer in Spring 2007 Summer courses have also been offered on this topic in the summers of 2005, 2006 and 2007 bhe faculty members in the EE department.
2384	CHE	INTRODUCTION TO NANOSCIENCE AND TECHNOLOGY	CHE674A	1. Overview nanoscience: Important concepts such as size, quantum effect, andMoores law. Characteristic length scales determining the behavior of physical andbiological systems; fundamental phenomena in nanoscience: quantum dots,lanthanidebased nanoparticles, polymer nanoparticles, carbon nanotubes and theirproperties (11 lectures).2. Synthesis and organization of nanomaterials: Chemical Routes for Synthesis ofNanomaterials: Sol-gel, reverse micelles, Solvothermal synthesis;Thermolysis routes, Microwave heating synthesis; Sonochemical synthesis; selfassembly;lithography; microfluidics; and chemical vapor deposition; surfacemodification techniques (X-ray scattering, Xray scattering), Scanning probe methods (STM and AFM) and other techniques(Optical Spectroscopy, Chromatography, Surface Plasmon Resonance and LightScattering) (1 0 lectures).4. Application of nanomaterials in various fields. Discussion on research papers based on the above syllabus. This will be done inbetween the above topics (8 lectures). 1. Christof M. Niemeyer, Chad A. Mirkin , Nanobiotechnology: Concepts, applicationsand perpectives, WileyInterscience (2004)2. Geoffery A. Ozin, Andre C. Arsenault, Nanochemistry: A chemical approach to nanotechnology, WileyInterscience (2004)
2385	CHE	AN INTRODUCTION TO SYSTEMS BIOLOGY	CHE684A	Introduction to transcription Networks; The Concept ofNetwork Motif; Autoregulation loops:Positive and Negative autoregulation; FeedForward loops; Global structure of Transcription Networks and Temporal Dynamics of Gene Networks; Circuits; Kinetic Proofreading; Optimal gene circuit design; Demand rule for gene regulation; Input function of a Gene: MichaelisMenten and Hill equations, Multidimensional input functions .Duration: One semester. 1. An Introduction to Systems Biology by Uri Alon. (Chapman & Hall/CRC, UK, 2007)2. Mathematical Models in Biology by EdelsteinKeshet, L. (Cambridge University Press,2005)3. Systems Biology: A Lectorial Course in Mathematical principles of immunology and virology by Martin A. Nowak And Robert May (Oxford University press,USA,2001)
2386	CHE	INTRODUCTION TO MOLECULAR SIMULATIONS	CHE622A	Theory methods, and application of molecular simulation. Elementary statisticalmechanics. Molecular modeling. Basic Monte Carlo and molecular dynamicstechniques and ensemble averaging. Evaluation of simulation results for complex fluids and solids. Commercial simulation software.
2387	EE	DIGITAL VIDEO SIGNAL PROCESSING	EE608A	Representation of digital video including modeling of video image formation, spatiotemporal sampling over lattices, conversion of signals sampled on different lattices and sampling rate conversion of video signals. Motion field Models, optical flow methods; blockbased methods, pelrecursive methods and Bayesian methods.Video filtering: Motion compensated filtering, including spatiotemporal spectrum, filtering and motion estimation. Video restoration, multiframe restoration; Standards conversion including practical Up/Down conversion Methods.Video compression: Basic concepts and techniques of video coding, Interframe Compression, Video Coding Standards, Codecs, Scalable Video coding. 1. Image and Video Processing, A.I. Bovik, Elosvier Academic Press, 2005.2. Digital Pictures, A.N. Netravali, B.G. Haskell, Plenum Press, 1997.3. Digital Video Transcoding, A. Vetro, K. Sreenivas, Prentice Hall Signal Processing Series, 1995.5. Wireless Video Communications, L. Hanzo, P.J. Cherriman, J. Streit, IEEE Series onDigital and Mobile Communication, 2001.
2388	EE	COMPUTATIONAL ASPECTS OF TOMOGRAPHIC IMAGING : MODELS TO INVERSIONS	EE659A	1. Motivations and overview of of tomography, limited data settings, approximate tomography, multimodal tomography.2. Typical Models : Maxwells equations, Helmholtz equation, eikonal equation, radiative transfer equation, element schemesand the method of moments (boundary element method).4. Linear tomography: Straight path tomography, Born and Rytov approximationsin diffraction tomography, algebraic reconstruction method of adjoints.7. Approximate tomography: Shape based tomography and topological derivatives.8. Introduction to stochastic reconstruction schemes, maximum likelihood andBayesian methods, Bayesian methods, electromagnetic wave tomography, elastography,multimodal tomography.
2389	MME	Advanced Structural Ceramics	MME672	1. Fundamentals of Material Properties and the importance of Ceramic materials (2L)2. Glass and glassceramics (3L)3. Processing and properties of different ceramic monoliths: (12L) Fundamental Sintering of Ceramics: (12L) sintering4. Mechanical behavior of Structural ceramics: (12L) Brittleness of ceramicsConcept of fracture toughness and different toughness measurement techniquesElastic modulusStrength measurement of ceramic composites: (6L)Examples of toughened particle reinforced compositesWhisker reinforced compositesFiber reinforced composites6. Recent advances in Structural Ceramics: (3L)Functionally graded ceramics: (6L) o W.D. Kingery, H.K. Bowen, D.R. Uhlmann, Introduction to Ceramics, 2nd ed. (John Wiley and Sons,NY, 1976). o R. W. Davidge; Mechanical behavior of ceramics; Cambridge University Press, 1989. o W.D. Kingery; Physical Ceramics, John Wiley & Sons. o J.B. Wachtman, Mechanical Properties of Ceramics. (John Wiley and Sons, NY, 1996). o B.R. Lawn, Fracture of Brittle Solids., 2nd ed. (Cambridge University Press, Cambridge University, UK,1998) o R.W. Hertzberg, Deformation and Fracture Mechanics of Engineering Materials., 3rd ed. (John Wiley & Sons, NY, 1983).
2390	MME	NON EQUILIBRIUM PROCESSING OF MATERIALS	MME603	Introduction to nonequilibrium processing; Thermodynamics and kinetics of metastable phase formation; Rapid solidification: Under cooling, Phase diagram of metastable states, Methods of rapid solidification of mechanical alloying, Mechanism of alloying, Energy criteria for mechanical alloying, Synthesis of nonequilibrium phases,Application of mechanical alloying; Metallic glass: Understanding of glass formation of metallic glass, application. Special nonequilibrium processing and phase transformations 1.Nonequilibrium processing of Materials Edited by C. Suryanarayana, Pergamon, 1999.Phase Transformation in Metals and Alloys D.A. Porter and K.E. Sterling, Van Nostrand Reinhold,Wokingham, UK (1991)2. Solidification W. Kurz and D.J. Fisher, 3rd edn, Trans Tech Publications, Switzerland,19894.Elements of rapid solidification Edited by M.A.Otooni, Springer 1998.5.Mechanical alloying and milling C. Suryanarayana, Science and Engineering Research Series, 11 (1994) 355408.7.Metallic glasses: production, properties and applications Edited by T.R. Anantharaman, Trans Tech publications 1984.
2391	ME	APPROXIMATE METHODS IN ENGINEERING MATHEMATICS	ME781	This course is an introduction to approximate mathematical methods in engineering. These methods provide powerful tools with which to approximately analyze complicated/intractable engineering/scientific problems in physics in an accessible manner. Finally, these approximate methods play a symbiotic role to computational analyses; these latter techniques typically falter near singularities, which is where approximate methods excel.1. Convergent/divergent power series; Asymptotic series; Error estimates; Subdominance; Stokes phenomenon.2. Asymptotic analysis of integrals (11 lectures): Integration by parts; Laplace method; Stationary phase method; Introduction to perturbation theory (4 lectures): Algebraic equations; Iteration and expansion; Method of dominant balance; Regular and singular perturbation problems; J;lescaling; Nonintegral power series; Application.5. Summation of series (3 lectures): Improvement of convergence; Divergent series; Pade approximation; Application.6. Boundary layer theory (7 lectures): Introduction to boundary layer theory; Asymptotic expansion; Limits; Application.7. vKVB Theory (4 lectures): Formal procedure; Patched asymptotic approximation; Turning point problems. Application8. Multiple scale analysis (6 lectures): Resonance and secular terms; Application.9. Bender, C. M. and S. O. Orszag, 1999. Advanced Mathematical Methods for Scientists and Engineers, SpringerVerlag: New York, USA.2. Hinch, E. J., 1991. Perturbation Methods, Cambridge University Press: Cambridge, UK.
2392	ME	FUNDAMENTALS OF CASTING & SOLIDIFICATION	ME664A	1 Introduction to Casting Problem: Principles of solidification, introduction to fluid flow and microstructure in a casting process.2 Solidification Transport Phenomena in Casting: Solidification of pure metals; Directional solidification.Mathematical treatment of solidification transport phenomena involved in a casting process (mass and heat transfer, fluid dynamics, mushy zone); Mathematical analysis of microstructural, mold filling, shrinkage and other flow and heat transfer related defects), their causes and remedies; Understanding the role of transport phenomena in the formation of these defects.4 Directional casting; Directional solidification, Micro casting.Manipulation and control of casting structure, properties and defects through advanced casting processes Inoculation practices, rheocasting, thixocasting, etc.5 Heat Transfer2 Fluid Flow and Solidification: Mathematical treatment of solidification rates exact and approximate methods;mold casting heat transfer.Riser design, feeding distance, mold filling and their relationship to casting.3 Solidification1. Science and Engineering of Casting Solidification, Doru M Stefanescu, 2nd ed.2. Solidification ProcessJung, M C Fleming, McGraw Hill.3. Principles of metal casting, R W Heine, C R Loper and Rosenthal, ASM International Technology, S Kalpakjian.6. Fundamentals of Modern Manufacturing, M P Groover.7. Fundamentals of Manufacturing Processes, G K Lal and S K Choudhury.8. Metals HandbookMetal Casting, ASM.
2393	MSE	TRANSMISSION ELECTRON MICROSCOPY & NANO-ANALYSIS OF MATERIALS	MSE653	1 Introduction :History, Interaction of electrons with matter, Different kinds ofTEMs 22 Electron scattering and diffraction: Terminology of scattering Interaction cross section; concept of mean free path; Elastic and inelastic scattering in TEM: Elastic scattering mechanisms; scattering at isolated atoms, atomic scattering factor, the structure factor, simple diffraction concepts. Inelastic processes occur in the form of plasmon excitation and phonon scattering.3 Aspects of diffractionpattern formation; reciprocal lattice, vector g, Ewald sphere of reflection; excitation error; diffraction from longperiod superlattices, small volumes, wedged shaped specimens, planar and curved surfaces.4 I. Transmission Electron Microscopy B.D. Williams & C.B.Carter, Springer, 20092. Aberration corrected analytical transmission electron microscopy, Ric Brydson, Wiley, 20113. Analytical electron microscopy: Principles and Applications Marc De Graef, Cambridge University Press, 2005. Science of Microscopy, P.W. Hawkes and J.C.H. Spence, Springer, 20076. In situ electron microscopy at high resolution, Editor: Florian Banhart, World Scientific, 2008.

S.No.	Branch	Course Title	Courses ID	Content
				ofMicroscopy:Applications in materials science, solidstate phsysics and chemistry. EditedBy: S. Amelinckx, D. van Dyck, J. van Landuyt, ad G. van Tendeloo, VCH, Weinheim, 19979. Microstructural Charac
2394	CE	COMPUTER METHODS IN HYDRAULICS AND HYDROLOGY	CE613A	Basic: Introduction to computer programming and computation with Matlab. (02 lectures)Open channel flow : Estimation of normal and critical depth; uniform flow computations; computation ofwater suhydraulic structures; unsteady flow SaintVenant equation, kinematicwave routing, diffusion routing, overland flow; steady and unsteady modelling using HECRAS. (07 lectures)Closed conduit flow: Steady hydrology: Estimation of Unit hydrographs; lumped and distributed flow routing;hydrologic statistics parameter estimation, time series analysis, frequency analysis, geostatistics;hydrologic modelling using Richards'equation, GreenAmpt infiltration model; introduction to MODFLOW. (05 lectures)Application of soft computing methods and GIS in Hydraulic and Hydrologic modelling. (03 lectures)Laborat1. Chow V.T., Maidment D.R. and Mays L.W. (1988), Applied Hydrology, McGrawHill2. Chaudhry M.H. (2007), OpenChannel Flow, 2"d Edition, Springer3. Deb K. (1995), Optimization for engineering desGeographical InformationSystems, 3'd Edition, Prentice Hall5. Mathews J.H. and Fink K.D. (2006), Numerical Methods using Matlab, 4th Edition, Prentice Hall6. Press etal., Numerical Recipes (C, C++ Edition, John Wiley & Sons8. Wasserman P.O. (1989), Neural computing, theory and practice, Van Nostrand Reinhold, New York
2395	ME	LIQUID-VAPOUR PHASE CHANGE TECHNOLOGY	ME639A	1st Term (14 hours) o 2nd Term (14 hours) 3rd Term (14 hours) Introduction to twophase flow and heat transfer technology Liquidvapor phase change phenomena , Interfacial tension, Wetting p Twophase flow fundamentals, Flow patterns and map representation, Development of homogeneous, separated flow and drift flux models, Flooding mechanismso Boiling Fundamentals, Homogeneo Enhancement techniqueso Condensation fundamentals, External and internal condensation , Film condensation theory , Dropwise condensation theory , Enhancement techniqueso Experimen techniques, Contact angle/Surface tension measurement, Conventional thermometry , Limitations, Data reduction, Application examples Application of twophase tlow and heat transfer, Th thermal management, Latent heat storage devices, Gravity assisted thermosyphons/ Vapor chambers, Conventional heat pipes, Micro heat pipes, Pulsating heat pipes, Capillary pumped loops/ Lo instabilities, microscale boiling and condensation , atomistic nucleation models, microgravity boiling, microfabrication/ integration techniques, contact resistance, surface roughness, Leidenfrost p Liquid Vapor Phase Change Phenomena by Van P. Carey (Taylor & Francis)(ii) Convective Boiling And Condensation by Collier John (Oxford Engineering Science) (iii) Heat Transfer Characteristics in Engineering Science)(v) One Dimensional TwoPhase Flow G. B. Wallis (McGraw Hill)(vi) Heat Pipe Science and Technology by Amir Faghri (Taylor and Francis)(vii) Heat Pipe Technology and Applic Hill)(ix) Heat Transfer Incropera and Dewitt (John Wiley and Sons)
2396	ME	ENGINE MANAGEMENT	ME691A	1. Introduction (1)2. Diesel Engine management: cylinder charge control systems, basic principles of diesel fuel injection, mixture distribution, fuel injection parameters, various designs and overview of di injection pumps, distributor fuel injection pump systems, helix and port controlled distributor injection pumps, overview of discrete cylinder systems,single plunger fuel injection pumps, unit injector syste diesel control (EDC), electronic control unit (ECU). (25)3. Gasoline Engine Management: cylinder charge control systems, fuels supply, manifold fuel injection, gasoline direct injection, operation of gasoline sensors, engine speed sensors, hall effect phase sensors, hot film air mass sensors,piezoelectric knock sensor, micromechanical pressure sensor, high pressure sensor, twostep lambda sensor,Electronic 1. Internal Combustion engine fundamentals: J B Heywood, McGraw Hill Publications.2. Gasoline Engine Management: Robert Bosch GMBH3. Diesel Engine Management: Robert Bosch GMBH4. The Inter Pollution: E FObert, Intext Educational Publishers, NY.6. Advanced Engine Technology: Heinz Heisler ISBN 0340568224, SAE Publications.
2397	MME	DISPLAY MATERIALS & TECHNOLOGIES	MME601	Contents: Overview of display industry, information capacity of displays, introduction to different flat panel display technologies, applications and comparison; Characterization and performance of display speed, memory and storage, degradation, resolution, addressability,physiological factors, and measurement instrumentation; Basic concepts of cathode ray tube (CRT); Luminescence and luminescent ma energy in luminescent materials,chemistry and preparation of luminescent materials, and emission properties of the prepared materials; Technical discussion of display technologies: Light Emitting Diode FieldEmission Displays (FEDs): Vacuum Fluorescent Displays (VFDs): Liquid Crystal Displays (LCDs):
2398	CE	CIVIL ENGINEERING MATERIALS	CE242A	Earth System: Lithosphere, Hydrosphere, Cryosphere and atmosphere and theiirinteractionsSolid Earth : Shape, size, interior of the solid earth. Geological materials: rocks,soils, minerals (clay mineralogy), consequences,Geomorphological features, structures (folds, faults)Earthquakes: Causes, classification, magnitude, intensity, Historical earthquakes,Seismic hazards zoning, strong ground motion, earthqu and quality of ground waterGeology of India : Physiographic and fectonic divisions; Minral resources.Geophysical mapping: seismic, resistivity, radar, geotomography, loggingRemote sensing, GIS and GPS forsite selections for Dam, tunnels, waste/radioactive disposal sites
2399	AE	TRANSITION AND TURBULENCE	AE625A	Elements of viscous flows and thin shear layer approximation. Different typesof TSL flows. Instabilities in flows. RayleighTaylor, KelvinHelmholtz mechanisms.Thin shear layer instabilities: for parallel and boundary layers, wakes, jets and free shear layers. Primary andsecondary instabilities and relationship of instability theories to transition.Receptivity of shear layers for 2 and 3D flows. Bypass transition in other kinematic tools of turbulence. Role of stretchingand dispersion in small scale turbulence. Route to turbulence: Chaos via nonlinearity,instabilities and bifurcation. Coherent structures in turbulence:U decomposition (POD). DNS, LES and other closureschemes of turbulence * Drazin. P.G. & Reid W.H.: Hydrodynamic Stability 1981, (CUP)* Davidson. P.A.: Turbulence (2003) (OUP)* Lmdahl. M.T. & MolloChristcnsen: Turbulence & Random Processes in Fluid Mechanics 1992 (CU (CUP)* Sagaut. P.: Large Eddy Simulation for Incompressible Flows. 2000 (Springer)* Sengupta T.K.: Foundation of CFD, 2004 (Univ. Press)
2400	MSE	SCIENCE AND TECHNOLOGY OF THIN FILMS AND DEVICE FABRICATION	MSE604	1. Introduction Introduction to device processing steps 4 Examples of various devices (with emphasis on solar cellsand MOSFETs) Need for miniaturizationThin Film Deposition2. Basics of thin film Brief growth control3. PVD Processes Evaporation (Thermal and ebeam)5 Principles of glow discharge and various sputteringprocesses4. Chemical Growth Fundamentals of CVD growthProcesses 4 Modern analysis 1 Optical analysis Morphology analysisDevice Fabrication6. Substrate seiection Selection of substratesand preparation , Single crystal growth (Silicon)3 Role of substrate surface and contamin Photoresists (Negative, Positive etc) Exposure Development 5 Masking Variants of lithography with emphasis on Photo and ebeam Precautionary steps8. Material Removal Wet (Chemical) and dry (Plasn composition in devicesand heat treatment Oxidation and heat treatment10. Metallization and Adhesion and morphology issuesInterconnects Introduction to electromigration visavis metallization: 2Impa particular kind offor a particular case device e.g. Si Solar Cell, MOSFET, 111V solar cell devices 4(take the case of adevice) Gandhi Si Processing (Volume I and II) Lou and Mayer Introduction to Microfabrication by Sami Fransilla (Wiley) Fundamentals ofMicrofabrication: The Science of Miniaturization by Marc Madou (CRC Pre Press) Pulsed Laser deposition of thin films by Chrisey and Hubler (Wiley Interscience)
2401	ME	ENERGY STORAGE SYSTEMS	ME686	contents: Review of semiconductor physics, carrier statistics, generationrecombination and carrier transport. Devices: PN junctions, Schottky barrier diodes, MOS capacitors, field effect transistors.Planar transient enhanced diffusion), ionimplantation, deposition (chemical and physical vapour techniques), etching; Lithography; Device and process integration, withMOSFET as an example. I. Nanomaterials, Nanotechnologies and Design: an Introduction to Engineers and Architects, D.Michael Ashby, Paulo Ferreira. Daniel L. Schodek , ButterworthHeinemann, 2009.2. Handbook ofNanoph Publishers, 2003.3. Encyclopedia ofNanoscience and Nanotechnology, Ed.: Hari Singh Nalwa, AmericanScientific Publishers, 2004.4. Handbook ojNanoceramics and their Based Nanodevices (Vol. 2) I
2402	MSE	Nanotribology and Nanomechanics	MSE637	Definition of micro/nanotribology Origin oftribology Measurement techniques Role oftribology in MEMS/NEMS Nanotribology: Measurement techniques and Concepts AFM/FFM Role of Surface r viscoelastici!YSurface maging, Friction and Adhesion Atomic scale imaging Friction: macro and microscale (Tomlinsonthermal/geometric effects) Surface roughness Nanoscale friction Wear mapping Lu Insitu local deformation characterization NanomachiningIndentation Pico/Nano indentation Localised surface elasticity/viscoelasticity Loadisplacement curve, indenter geometry,stiffuessBoundary Lub acoustic/imaging methods Defect NucleationScaling Effects Length scale effect on hardness, yield strength Roughness and contact parameters Model, Adhesion friction, 2body and 3body deformation, n controlling adhesion,friction and wear) Nanotribology and nanomechanics of magnetic storagedevices/ NEMS/MEMS Biomaterial/ Biological samples I. Introduction To Micromechanics And Nanomechanics, Shaofan Li and Gang Wang World Scientific Publishing Company (2008)2. Fundamentals oftribology, Ramsey Gohar, Homer Rahnejat Imperial Coll will be supplemented with handouts, and journal publications.
2403	MME	Nanomaterials: Processing and Properties	MME688	1. Fundamental Properties of various primary material classes (Metals, ceramics and Polymers) (8L)2. Definition and Classification ofNanomaterials (2L)3. Size dependent properties and various characte Mechanochemical synthesis to produce nanosized precursor powders4.2. Various routes to produce Nanometallic alloys (Rapid solidification)4.3 Challenges in processing bulk ceramic nanomaterials4.4 important bulk nanomaterials (10 L)5.1 Mechanical Properties5.2 Thermal Properties5.3 Tribological Properties5.4 Biological Properties (Biomedical applications)6. Applications of bulk nanomaterials (3L o Nanomaterials Synthesis, Properties and Applications, Edited by A. S. Edelstein and R. Cammarata, Institute of Physics Publishing, London, 1998 (paper back Edition).o Physics and Chemistry ofNanost Nanotechnology, Edited by H. S. Nalwa Vols. 15,Academic Press (2000).
2404	MME	MATERIALS FOR BIOMEDICAL APPLICATIONS	MME668	Introduction to basic concepts of Materials Science; Salient properties of important material classes; Property requirement of biomaterials; Concept of biocompatibility; cellmaterial interactions and foreign CoCrMo alloys; Bioinert, Bioactive and bioresorbable ceramics; Processing and properties of different bioceramic materials with emphasize on hydroxyapatite; Synthesis of biocompatible coatings on stru

S.No.	Branch	Course Title	Courses ID	Content
				developing new materials for bioimplant applications. 1. Biomaterials Science: An introduction to Materials in Medicine, Edited by Ratnes, Hoffman, Schoet and Lemons,Second Edition: Elsevier Academic Press, 2004.2. Comprehensive str Biomaterials Science and Biocompatibility, Frederick H. Silver and David L.Christiansen, Piscataway, Springer, New Jersey.4. Biological Performance of Materials: Fundamentals of Biocompatibility, Edited by J.M. Davis, IRL Press, Oxford University Press, New York, 1994.
2405	ME	BASIC CONTROL SYSTEMS FOR MECHANICAL ENGINEERS	ME850	1. Elementary review of dynamic systems. Equations of motion. Numerical solution of ODEs. Linearization.Stability. (5 lectures)2. Laplace transforms and inverse Laplace transforms. Block diagrams. Transfer functions. Nonminimum phase systems and their transient responses. Steady state responses. (7 lectures)3. Root locus plots. Nyquist plots. Bode plots. Implications for transient responses. (5 lectures)4. Compensation techniques. Root locus motivation and sample problems. (3 lectures)6. Discrete time systems. Stability. (2 lectures)7. State space. Standard form for an LTI system. General solution. Controllability and observability. Pole placement. (2 lectures)9. Introduction to timedelayed control. (2 lectures)10. Simulations of nonlinear systems with linearization based controllers. Case studies from the literature as time permits. (4 lectures)In addition to the above, students will be required to read the following books: 1. K. Ogata. Modern Control Engineering. (PHI)2. F. Golnaraghi and B. C. Kuo. Automatic Control Systems. (Wiley)
2406	EE	WIRELESS COMMUNICATIONS	EE670A	Introduction to Wireless Communications, multipatchannel models, capacity of wireless channels, performanc of digital modulation techniques over wireless channels, combining techniques, multicarrier systems. 1. Wireless Communications: Principles and Practice byTS Rappaport, PrenticeHall, second edition, 20022. Principles of Mobile Communications by GL Stuber,Kluwer Academic, second edition, 2001.3. Wireless Communications: Principles and Practice by J. G. Proakis, Prentice Hall, 1998.4. Wireless Communications: Principles and Practice by S. Haykin and M. Moher, Prentice Hall, 2004.
2407	AE	INTRODUCTION TO AEROSPACE ENGINEERING	AE201A	Fixed wing vehicles: History of Aviation, introduction to fixed wing vehicles,configuration and layout, propulsion, lift generation mechanism, balance of forces and moments, control mechanisms (10 hr)Rotary wing vehicles: History, configuration and layout, propulsion, lift generation mechanism, balance of forces and moments, control mechanisms (6 hr)Space Vehicles: History, configuration and layout, propulsion, lift generation mechanism, balance of forces and moments, navigation. (4 hr) Demonstration of flight simulation software. Introduction to Flight: J.D. Anderson, Jr., McGrawHill International Editions.
2408	MSE	TRANSMISSION ELECTRON MICROSCOPY & NANO-ANALYSIS OF MATERIALS	MSE653A	1 Introduction :History, Interaction of electrons with matter, Different kinds ofTEMs 22 Electron scattering and diffraction: Terminology of scattering Interaction cross section; concept of mean free path; elastic and inelastic scattering in TEM: Elastic scattering mechanisms; scattering at isolated atoms, atomicscattering factor, the structure factor, simple diffraction concepts. Inelastic processes occur in the tern: plasmon excitation, aspects of diffractionpattern formation; reciprocal lattice, vector g, Ewald sphere of reflection; excitation error; diffraction from longperiod superlattices, sniall volumes, wedgeshaped specimens, planar interfaces.
2409	MME	DiiTraction and Spectroscopy Techniques for Surface	MME695	Importance of Smfacc Characterization. Present status sensitivity and resolution achievable.Diffraction Techniques: Revicv.,: of basic diffraction theory; Various Small Angle Xray Scattering techniques, Electron diffraction, neutron sources: Small angle neutron scattering; Illustrative analysis using diffraction techniques Spectroscopy: Basic principles of Spectroscopy. Principles of XPS, Instrumentation, XPSpatterns, Quantitative analysis, e1Tect, Quantitative analysis, Depth profiling,Applications; Static and Dynamic SIMS, Common modes of analysis, quantitative and Qualitative analysis; Case studies on the spectroscopic analysis of surface and bulk materials. 1. Encyclopedia ofMaterials Characterisation, C. R. Brunelle. C. A. Evans and S. Vilson,ButterworthHennmann, 1992. Boston.2. Characterisation ofMaterials Volume 2. Editor:E.N. Kaufmann, WileyInterscience, 1998.3. Springer (2003).4. Materials Charocterization. published in 1986 as Volume I 0 of the 9th Edition MetalsHandbook, ASM International. 1986 (Fifth printing 1998).
2410	ME	ENGINE MANAGEMENT	ME691	1. Introduction (1)2. Diesel Engine management: cylinder charge control systems, basic principles of diesel fuel injection, mixture distribution, fuel injection parameters, various designs and overview of diesel engine fuel injection pumps, distributor fuel injection pump systems, helix and port controlled distributor injection pumps, overview of discrete cylinder systems,single plunger fuel injection pumps, unit injector systems, electronic diesel control (EDC), electronic control unit (ECU). (25)3. Gasoline Engine Management: cylinder charge control systems, fuels supply, manifold fuel injection, gasoline direct injection, operation of gasoline engine sensors, engine speed sensors, hall effect phase sensors, hot film air mass sensors,piezoelectric knock sensor, micromechanical pressure sensor, high pressure sensor, twostep lambda sensor,Electronic Control Unit (ECU). 1. Internal Combustion engine fundamentals: J B Heywood, McGraw Hill Publications.2. Gasoline Engine Management: Robert Bosch GMBH3. Diesel Engine Management: Robert Bosch GMBH4. The Internal Combustion Engine: John Heywood, McGraw Hill Publications.5. Pollution: E FObert, Intext Educational Publishers, NY.6. Advanced Engine Technology: Heinz Heisler ISBN 0340568224, SAE Publications.
2411	MME	MATERIALS SCIENCE TECHNOLOGIES FOR APPLICATIONS IN LIFE SCIENCES	MME693	Introduction to basic concepts of DNA and proteins: structure, their role in life sciencesand their application as nanotechnology tools; Microfabrication techniques forapplications in life sciences including SelfAssembled Monolayers (SAMs); Biofunctionalization of nanomaterials; Drug deliverytechniques using nanomaterials; Characterization techniques used in life sciences:Fluorescence microscopy, Quantitative analysis, etc. Sciences including:Sensing, BioMEMS devices, Drug delivery, and Biomedical devices. 1. B. Alberts etal. , Essential Cell Biology. (Garland Publishing Inc. , New York, ed.Third, 2009).2. C. S. S. R. Kumar, Biofimclionalization qfNanomaterials. C. S. S. R. Kumar, Ed.,Nanotechnologies for the life sciences (WileyVCH, Weinheim, 2006), vol. 8.
2412	MSE	ENERGY MATERIALS AND TECHNOLOGIES	MSE624	
2413	MSE	SOLID STATE IONICS	MSE664	Perfect Structure, Defects in Elemental Solid and Ionic Compound, Defect Classes,Point Defects, KrogerVink Notation for Point Defects, Point Defect Formation & Equilibrium, Law of Massaction, Thermodynamic Properties of Intrinsic and Extrinsic Defects, Intrinsic and Extrinsic Defect Associates and Effect of Impurities on the Concentration of Defect Complexes and Associate. Basic Concepts of Diffusion, Tracer Diffusion, Self Diffusion, Chemical Diffusion, Ambipolar Diffusion, NernstEinstein Relationship, and ConductivityDiffusion Relationship. Defect Equilibria in Pure and Stoichiometric Compounds with Schottky Defects,Frenkel Defect Pairs and Intrinsic Ionization of Electron Deficient Oxide, Metal Oxide with Excess Oxygen. Brouwer Diagrams for YSZ, Undoped and Doped CeO2, TiO2 and BaTiO3 Electrical Characterization Techniques such as AC Electrochemical Impedance Spectroscopy, Tafel Method. OpenCircuit Potential, Efficiency, Nernst Equation Analysis, Activation Losses (Tafel Equation), Ohmic Losses,Concentration Losses. Description of Operation, Configurations, Cell Components, Materials, Solid Oxide Fuel Cells, Gas Sensors and Batteries. Publisher: Oxford University Press,ISBNIO: 01951101452 A.J. Moulson & J. M. Herbert, &#34;Electroceramics: Materials, Properties, Applications&#34;, Publisher:Springer3 M. W. Barsoum, &#34;Fundamentals of Electroceramics&#34;, Publisher: John Wiley and Sons.5 Robert Huggins, &#34;Use of defect equilibrium diagrams to understand minority species transport in solid electrolytes&#34;, Edited by P. J. Gellings & H. J. M.Bouwmeester, Publisher: CRC Press7 &#34;High Temperature Solid Oxide Fuel Cells, Fundamental, Design and Applications &#34;, Edited by Subhan Ullah, Publisher: Elsevier
2414	MME	GRAIN BOUNDARY ENGINEERING	MME680	1. Grain boundary structure: Geometrical aspects, Degrees of freedom, Principles governing grain shape and size and their orientation. Theoretical formulations: Structural units model, Plane matching model, Grain Boundary Boundary types, GB mobility and boundarysolute interactions. GB structure and Properties: mechanical strength, wear, creep, magnetic, electrical etc., Simulation and Modelling of grain boundary additions, Magnetic Field etc. GB descriptors: Connectivity, density, junction distribution, Character Distribution (8 hours)4. Boundary Characterization techniques: X-ray diffraction, neutron diffraction methods. Microtexture analysis: Automated EBSD,Kikuchi pattern, Hough&#39;s transform, SEMOIM based, TEM based, Schemes for representation of Data and Microstructure. Resistance,Superconductivity, Electronic ceramics
2415	MME	SOLIDIFICATION PROCESSING	MME670	Introduction; Thermodynamics of solidification; Nucleation and growth ; Pure metalsolidification: GibbsThomson effect ; Alloy Solidification: Mathematical Analysis ofredistribution of solute during solidification; Multiphase solidification: eutectic, peritectic and monotectic; Modelling of solidification;Case studies. 1. Solidification Processing; Fleming, M.C., McGrawHill, N.Y., 19742. Solidification of Casting; Ruddle, R.W., Institute of Metals, 19573. Solidification and Casting, Davies, G.J., John Wiley and Sons, 19734. Solidification by Kurz, W. and Fisher, D.J., TransTech Publications,Switzerland, 19896. Theory of Solidification; Davis, Stephen, H. Cambridge University Press, 2001
2416	ME	RAPID PROTOTYPING & TOOLING TECH.	ME779	Importance and overView ofRapid Prototyping ario Manufacturing; Process Chain; Solid modeling;Data. formats, conversion, checking, repairing and transmission; Part Slicing and Build orientation;Classification of rapid prototyping processes; principles and chara.cteristics qf current and developing rapid prototyping, tooling and manufacturing processes; Use of rapid prototypes and tools in product design, development and service; Selection of materials for rapid prototyping; Chua, C.K., Leong, K.F., and Lim, C.S., 2003, Rapid Prototyping Principles and Applications,World Scientific. (second edition).Jacobs, P.F., 1992, Rapid Prototyping and Manufacturing Fundamentals of Stereolithography and Processes for Rapid Prototyping, KLuwer. /Gebhardt, A., 2003, Rapid Prototyping; Hanser. &#39;Pham, D.T., and Demov, S.S., 2002, Rapid Prototyping and Manufacturing, SpringerVerlag.Pique, A., and Chrisey, D.B., 2002, DirectWrite Techno ogies for Rapid PrototypingApplications: Sensors, Electronics and Integrated Power Sources, Academic Press.Vcnuinod, R., 2002, Rapid Prototyping and Manufacturing, SpringerVerlag.
2417	AE	ROCKET ENGINE DESIGN	AE662	Introduction to rocket propulsion, Types of rocket engines, Elements of combustion,Chemical propellants and their burning characteristics, Aerothermodynamic designanalysis of solid propellant rocket engine and liquid propellant rocket engine. 1. Sutton G.P and Ross D. M. &#34;Rocket Propulsion Elements&#34;, John Wiley & Sons,New York, 1985.2. Barrere M, Jaumotte A and Vandenkerckhove I. &#34;Rocket Propulsion&#34;, ElsevierPublishing Company, 1998.3. ofPropulsion&#34;, Addison Wesiey Publishing Company, 1965.4. Oater G.C, &#34;Aerothermodynamics of Gas Turbine and Rocket Propulsion&#34;, 3rdEdition, AIAA education Seria, 1998.5. Mishra D K, &#34;Rocket Propulsion&#34;, Tata Mc Graw Hill, 2002.

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				Fundamentals Of Solid Propellant, combustion"progress in Astronautics and Aeronautics, Vol. 90, AIAA New York.
2418	ME	GRANULAR MATERIALS	ME773	1. Introduction to granular materials: Illustrative examples.2. Revision of continuum mechanics.3. :Materials with yield strength: I lohrCoulolllb and DruckerPrager materials.4. Statics: Coulomb's, J a Micromechanical behavior: Particleparticle interaction. Meanfield theory.7. Other features: Closepacking, Crystallization, Force chains, Shear banding.8. Slow dense flow: Plasticity theory.9. Application: FI Inclined cbntes.12. Rapid granular flow: Rough inelastic particles.13. Mixing and segregation: Brazilnut effect, Rotating drum and Avalanches.14. Effect of air: Porous beds. Barchau d 111H, ,; .. ud Homgla
2419	ME	Combustion and Reacting Flow	ME744	(approximate number of lectures in brackets):1. Thermodynamic idealization of reacting, and hightemperature systems: classical and statistical thermodynamics review [6]2. Chemical kinetics, hete of multicomponent mass transfer, radiation in participating media [5]4. Coupling of thermodynamics, kinetics, and transport for idealized reactors [3]5. Approximate modeling of reacting system premixed and diffusion flames, chemical vapor deposition, hydrocarbon reforming processes, porous media reactor, membrane reactor [8]6. Thermal decomposition, ignition, and detonation [2] (8. Environmental impact; pollutant formation and control; exergybased analysis [3]9. Experimental techniques, theories of spectroscopy and mass spectrometry [3]10. Futuristic technologies and adv combustion, oxyfuel combustion, chemical looping combustion, carbon nanotube synthesis, burner design [3] I. Fundamentals of Combustion, D. P. Mishra (PrenticeHall India, 2008)2. Combustion, Glassman, Yetter (Associated Press, 2008)3. Irreversible phenomena, K. Terao. (Springer, 2007)4. Combustion, W Systems, A. K. Oppenheim (Springer, 2006)7. Chemical kinetics and reaction dynamics, S. K. Upadhyay (Springer, 2006)8. Chemically Reacting Flow, Kee, Coltrin, Glarborg (WileyInterscience, 2003)9. (Benjamin/Cummings 1985)
2420	EE	NONLINEAR FIBER OPTICS	EE639	1. Introduction: Linear vs Nonlinear effects in optical fiber, Important nonlinear effects and their impact on optical communication, applications of optical nonlinearities to signal processing2. Electroma transmission of waves, structure of an optical fiber, wave equation in cylindrical coordinates, fiber modes, characteristics of LPm mode3. Dispersion in optical fibers: Chromatic dispersion in singlemode second order models, effect on communication system , mitigation of PMD4. FourWave mixing (FWM): Mathematical description, phase matching, fiber parametric am plifiers using FWM, squeezed modulation: SelfPhase modulation, impact on communication system, modulation instability, CrossPhase modulation, impact on communication system6. Nonlinear Schrodinger equation (NLSE): Nonlin soliton solutions,numerical .solution of the NLSE
2421	ME	FUEL CELLS	ME743	(approximate number oflectures in brackets)1. Fundamentals: Review of classical thermodynamics; introductory electrochemistry;principles of chemical and electrochemical kinetics; tra fuel cell systems; analyses of fuel cell kinetics; quantification of fuel cell performance [14].3. Computational/experimental techniques: Conservation and rate equations; approximate analytical treatment mechanics of fuel cell systems; measurement of fuel cell performance; lab visits; introduction to electrochemical impedance spectroscopy [10].4. Special topics: Direct methanol fuel cell; n [4]. 1. Fuel Cell Systems Explained, J. Larminie and A. Dicks (John Wiley & Sons, 2003, USA)2. Fuel Cell Fundamentals, R. O'Hayre, SW. Cha, W. Colella, F. B. Prinz (John Wiley and Sons, 2005, USA)3. Srinivasan (Springer, 2006, USA)5. Principles of Fuel Cells, X. Li (CRC Press, 2005, USA)6. Fuel Cells: Principles and Applications, B. Viswanathan and M.A. Scibioh (Universities Press, 2006, India)7. Pl Cells: Fundamental, Design and Applications, S. C.Singhal, K. Kendall (Elsevier Science, 2004, USA)9. Transport Phenomena in Fuel cells, Ed. B. Sunden and M. Faghri (WIT Press, 2005, UK)10. Fundam
2422	ME	HAMILTONIAN MECHANICS AND SYMPLECTIC ALGORITHMS	ME726	Part 1a Some relevant definitions and results in the theory of differentiable manifolds, smooth vector fields, differential forms,exterior) calcu lus (differentiation and integration using differential Hamiltonian mechanics (Lagrange's vs Hamilton's Equations) , Canonical Transformation , Legendre T:raus forma!ion , Symplectic Tlansfonna!ions, Some defi.nit.ions and :result.s in the t. he Partial Differential Equation. Integrable systems (simple examples).Part 2a Some basic notions of numerical algorithms (order conditions etc). Examples of Numerical methods, Symplectic Integra 2b Symplectic RungeKutta IVlethods, Generating Function for Symplectic Rm1geKutta Methods and Symplectic Ivlethods Based on it. Variational Integrators. Introduction to Hamiltonian Perturbat geometric numerical integration with some applications to mechanical systems. 1. Arnold, V. I., 1989. IVlathematical Methods of Classical Mechanics. Springer. Second edition. [Te].tbook: for part 1 only sections 18, 3241, 4448, for part 2 only sections 1317, 19].2. Leimkuhler, 2, 47,9]. 3. Hairer, E., Lubich, C., vVa1mer, G., 2006. Geometric Numerical Integration: StructurePreserving Algorithms for Ordinary Differential Equations
2423	ME	DIFFERENCE EQUATIONS FOR ENGINEERS	ME682	1. Introduction and Some Applications of Difference Equations in Engineering (1 lecture)2. Preliminaries in algebra and analysis (3 lectures)3. Analogies between differential and difference equations (Calculus. Summation (3 lectures)6. Interpolation, Extrapolation (1 lecture)7. Generating functions (2 lectures)8. Linear difference equations, First order equations (2 lectures)9. Higher Order Difference equations with variable coefficients (2 lectures)12. Method of undetermined coefficients and variation of parameters (2 lectures)13. Limiting behavior of solutions (1 lecture)14. Systems of Linear diffe theory and Fourier analysis (3 lectures)17. Nonlinear difference equations and boundary value problems (1 lecture)18. Asymptotic methods (1 lecture)19. Stability theory and relevance to dynamical s Difference Equations. Symmetry (2 lectures)22. Difference Equations with Continuous Time, Differentialdifference equations (1 lecture)23. Discrete Mechanics (1 lecture)24. Open problems (1 lecture) 1. An Intoduction to Difference Equations, S. N. Elaydi (Springer) (textbook)2. Difference Equation, W. G. Kelley and A. C. Peterson (Academic Press)3. Finite Difference Equations. H. Levy and F. Lessma
2424	CE	STABILITY OF STRUCTURES	CE622	Criteria for Design of Structures: Stability, Strength, and StiffnessClassical Concept of StabilihjStabilihJ of Discrete Systems: Linear and nonlinear behaviourStabilihJ of Continuous Systems: Stability of Co Frames:member buckling versus global buckling, slenderness ratio of frame members. Stabilityof Beams: lateraltorsion buckling. Stability of Plates: axialflexural buckling, shearflexural buckling, buckling
2425	ME	FUNDAMENTALS OF CASTING & SOLIDIFICATION	ME664	1 Introduction to Casting Problem: Principles of solidification, introduction to fluid flow and micrstntcture in a casting process.2 Solidification Transnort Phenomena in Casting: Solidification of pur grO:vth; Directional solidification.Mathematical treatment of soliditication transp01t phenomena involved in a casting process (mass and heat transfer, fluid dynamics, mushy zone); Mathematical analy microstructural, mold filling, shrinkage and other flow and heat transfer related defects), their causes and remedies; Understanding the role of transport phenomena in the formation of these defects.4 casting; Directional solidification, Micro casting.Manipulation and control of casting structure, properties and defects through advanced casting processes Inoculation practices, rheocasting, thixocasting Transfer2 Fluid Flow and Solidification: Mathematical treatment of solidification rates exact and approximate methods;mold casting heat transfer.Riser design, feeding distance, mold filling and their rel 1. Science and Engineering of Casting Solidification, Doru M Stefanescu, 2nd ed.2. Solidification ProcessJng, M C Fleming, McGraw Hill.3. Principles of metal casting, R W Heine, C R Loper and Rosentl Technology, S Kalpakjian.6. Fundamentals of Modern Manufacturing, M P Groover.7. Fundamentals of Manufacturing Processes, G K Lal and S K Choudhury.8. Metals HandbookMetal Casting, ASM.
2426	ME	THERMODYNAMICS OF CONTINUOUS MEDIA	ME720	1. Foundations of continuum thermodynamics1.1 Geometry of continuous bodies: kinematics; singular surfaces; strain compatibility (3)1.2 Balance of mass and momentum; the concept of stress (1 entropy; the ClausiusDuhem inequal ity (4)1.5 Constitutive theory1.5.1 Material frame indifference and material symmetry (1)I1.5.2 Equilibrium response (1)1.5.3 Dynamic response: linear irreversible Thermodynamic equilibrium; stability (4)2. Rigidconductors2.1 Constitutive equations; Fourier 's law (2)2.2 Boundary value problems (2)3. Thermoelasticity3.1 Constitutive equations; linear therm Thermoplasticity4.1 Constitutive equations: simple phenomenological models with work hardening (3)4.2 Onedimensional problems: adiabatic shear bands (4)5. Thermodynamics of surfaces5.1 Gibl CalmHilliard type of theories for diffusive interfaces (2)5.4 Problems (can be from different sources, for e.g. adiabatic shock waves, phase transformations, grain growth, membrane mechanics) (4) (i) The Mechanics and Thermodynamics of Continuous :tvledia, M. Silhavy, Springer , 1997.(ii) An Introduction to Thermomechanics, H. Ziegler, NorthHolland, 1983.(iii) The Mechanics and Thermodyna and J. H. Weiner , Dover , 1997.(v) The Physics and Mathematics of Adiabatic Shear Bands, T. W. Wright , Cambridge , 2002. (vi) Statistical Thermodynamics of Surfaces, Interfaces, and Membranes, S
2427	EE	COMPUTATIONAL ASPECTS OF TOMOGRAPHIC IMAGING : MODELS TO INVERSIONS	EE659	1. Motivations and overview of of tomography, limited data settings, approximatetomography, multimodal tomography.2. Typical Models : Maxwells equations, Helmholtz equation, eikonal equation,radia element schemesand the method of moments (boundary element method).4. Linear tomography: Straight path tomography, Born and Rytov approximationsin diffraction tomography, algebraic reconstru method of adjoints.7. Approximate tomography: Shape based tomography and topological derivatives.8. Introduction to stochastic reconstruction schemes, maximum likelihood andBayesian methods, electromagnetic wave tomography, elastography,multimodal tomography.
2428	CHE	INTRODUCTION TO TRANSPORT PHENOMENA	CHE602M	Module on Thermodynamics:1. Review of 1st and 2nd Law of Thermodynamics (including applications) (2lectures)2. PVT relations of pure fluids (cubic EOS, generalized correlations, compressibility fac Maxwell's relations, residual properties. Two phase systems: Clapeyron equation, Phasediagrams. Generalized property correlations for gases) (3 lectures)4. Multicomponent Mixtures: Partial mo The ideal solution andexcess molar properties. Activity coefficients and models (Margules, NRTL, Wilson, UNIQUAC) (3 lectures)5. Vapor Liquid Equilibrium: multicomponent phase envelopes, ga (4lectures)6. Thermodynamic properties and VLE from cubic EOS (2 lectures)7. Other types of phase equilibria: LLE, VLLE, SLE (1 lecture)8. Heat of adsorption, multicomponent adsorption equilibrium reaction engineering (1 lecture; Chap. 1 ofHSF)11. Steady and unsteady state operation in idealized reactors (3 lectures; Chap. 4 of HSF)12. Multiple reactions (2 lectures; Chap. 6 ofHSF)13. Nonelementar diffusion effects on heterogeneous reactions (2lectures; Chap. 11 ofHSF)16. Diffusion and reaction in porous catalysts (4lectures; Chap. 12 ofHSF)17. Design of fixed bed reactors (2 lectures)

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				(HSF) Elements of Chemical Reaction Engineering, Prentice Hall (India)2. S. I. Sandler (SIS) Chemical and Biological Engineering Thermodynamics, John Wiley(Indian Edition).
2429	CE	GROUND IMPROVEMENT TECHNIQUES	CE634	Engineering properties of soft, weak and compressible deposits, principles of treatmentloading(static and Dynamic), Accelerated flow, Reinforcement, Drainage and fillers,Injections, Thermal, electrical and Methods and Case histories Moseley and Kirsch: Designing with Geosynthetics (2"d Edition), Routledge, (2004), ISBN: 0415274559
2430	CE	REINFORCED EARTH STRUCTURES	CE633	Reinforcing materials, Advantage of RE, behaviour of Reinforced earth walls, Soilreinforcement interaction internal and external stability condition, field application of RE.Randomly reinforced earth and an geogrids, geonets, andgeomernbranes. KoemerR. M.:Designing with Geosynthetics (4 u, Edition),Prentice Hall, New Jersey, (1997)
2431	CE	FOUNDATION ANALYSIS AND DESIGN	CE632	Settlement and bearing capacity; shallow spread footings, mats, and deep foundations.Foundation models, Contact pressure distribution for footings, Rafts, Piles, RetainingStructures; Soilstructure interme Bowles, J. E.: Foundation Analysis and Design (5nd Edition),The McGraw Hill Companies Inc. (2001)
2432	CE	ECOHYDROLOGY	CE619	Introduction: Origin and scope of ecohydrology. (03 Lectures)Ecohydrological processes: Interactions between physical, chemical and biological processesat basin scale soil water dynamics, land surface Landscapeconnectivity morphological, ecological and hydrological connections. (12 Lectures)Techniques in ecohydrological measurements: Measuring energy and water fluxes inatmosphere, soil and ve soil respiration and soil heat flux;vegetation leaf area index, stomatal conductance and transpiration. (08 Lectures)Ecohydrological modelling: Governing equations; mathematical models stochastic and ecohydrological modelling. (10 Lectures)Applications of ecohydrology: Use of ecohydrogical principles in paleohydrology and climatechange studies; ecohydrological approach for sustainable managem 1. D'Odorico P., Porporato A. 2006. Dry/and Ecohydrology, Springer, Dordrecht, TheNetherlands.2. Eagleson P.S. 2002. Ecohydrology: Darwinian expression ofvegetationform andfmction, Cambridge casestudies: An approach to the sustainable management of water resources, CABInternational, Cromwell Press, Trowbridge, UK.4. Rodriguezlturbe 1., Porporato A. 2005. Ecohydrology of watercontrolled Sadler J.P. 2007. Hydroecology and Ecohydrology: past,present andfitture, John Wiley & Sons, Chichester, UK.
2433	ME	LIQUID VAPOUR PHASE CHANGE PHENOMENA	ME639	1st Term (14 hours) o 2nd Term (14 hours) 3rd Term (14 hours) Introduction to twophase flow and heat transfer technology Liquidvapor phase change phenomena , Interfacial tension, Wetting p Twophase flow fundamentals, Flow patterns and map representation, Development of homogeneous, separated flow and drift flux models, Flooding mechanismso Boiling Fundamentals, Homogeneo Enhancement techniqueso Condensation fundamentals, External and internal condensation , Film condensation theory , Dropwise condensation theory , Enhancement techniqueso Experimen techniques, Contact angle/Surface tension measurement, Conventional thermometry , Limitations, Data reduction, Application examples Application of twophase tlow and heat transfer, Th thermal management, Latent heat storage devices, Gravity assisted thermosyphons/ Vapor chambers, Conventional heat pipes, Micro heat pipes, Pulsating heat pipes, Capillary pumped loops/ L instabilities, microscale boiling and condensation , atomistic nucleation models, microgravity boiling, microfabrication/ integration techniques, contact resistance, surface roughness, Leidenfrost p Liquid Vapor Phase Change Phenomena by Van P. Carey (Taylor & Francis)(ii) Convective Boiling And Condensation by Collier John (Oxford Engineering Science) (iii) Heat Transfer Characteristics in Engineering Science)(v) One Dimensional TwoPhase Flow G. B. Wallis (McGraw Hill)(vi) Heat Pipe Science and Technology by Amir Faghri (Taylor and Francis)(vii) Heat Pipe Technology and Applic Hill)(ix) Heat Transfer Incropera and Dewitt (John Wiley and Sons)
2434	EE	CONVEX OPTIMIZATION IN SIGNAL PROCESSING AND COMMUNICATIONS	EE609	
2435	EE	DIGITAL VIDEO SIGNAL PROCESSING	EE608	Representation of digital video including modeling of video image formation, spatiatemporal sampling over lattices, conversion of signals sampled on different lattices and sampling rate conversion o Motion field Models, optical flow methods; blockbased methods, pelrecursive methods and Bayesian methods.Video filtering: Motion compensated filtering, including spatiatemporal spectrum, filterin restoration, multiframe restoration; Standards conversion including practical Up/Down conversion Methods.Video compression: Basic concepts and techniques of video coding, Interframe Compression Codecs, Scalable Video coding. 1. "Image and Video Processing", A.I. Bovik, Elosovier Academic Press, 2005.2. "Digital Pictures", A.N. Netravali, B.G. Haskell, Plenum Press, 1997.3. "Digital Video Transco Prentice Hall Signal Processing Series, 1995.5. "Wireless Video Communications", L. Hanzo, P.J. Cherriman, J. Streit, IEEE Series onDigital and Mobile Communication, 200 1 .
2436	EE	WIRELESS COMMUNICATIONS	EE670	Introduction to Wireless Communications, multipatchannel models, capacity of wireless channels, performanc of digital modulation techniques over wireless channels, combining techniques, multicarri 1. Wireless Communications: Principles and Practice byTS Rappaport, PrenticeHall, second edition, 20022. Principles of Mobile Communications by GL Stuber,Kluwer Academic, second edition, 2001.3. W UniversityPress, 1998.5. Modern Wireless Communications by S Haykin and MMoher, PrenticeHall, 2004.
2437	CHE	SCIENCE & TECHNOLOGY OF DRUGS	CHE612	1. Physical Chemical properties and biological activity2. .Molecular orbifal theory in drug design3. Structural features and pharmacological activity4. Drug metabolism and procedure of tailoring drugs CNS stimulants and their synthesis8. Chemotherapy and oncology. Synthesis of drugs, mechanism of action and procedure treatment.9. Drugs affecting sugar and diabetes mellitus. Synthe and AIDS and procedure treatment12. Psychotic and related drugs 1. Harrison's Principles of Internal Medicine, 15th Ed, McGraw Hill, N.Y., 2001.2. Goodman and Gillman's The Pharmacological Basis of Therapeutics, gth Ed, McGraw Hill, N.Y., 1992.3.V
2438	EE	CIRCUIT DESIGN FOR FREQUENCY AND PHASE SYNTHESIS	EE606	DSP Architecture: Von Neumann vs. Harvard architecture, VelociTI architecture Memory management of TI DSP processors C2X, C6X. Peripheral overview of TI C2X, C6X processor. Code Composer Stu processor. Optimization of assembly and C code. Pipelining. General Extension Language (GEL). Assembler, Linker. Interrupts RTDX. DSP/BIOSList of Experiments Introduction to Code Composer Stu and its Applications Audio Codec and its Applications Real Time Data Exchange Using Matlab and Labview FIR filtering by interfacing Matlab with Code Composer Studio Introduction to Interrupts Digital control of a singlephase inverter with passive load PLL for a threephase ac system PLL for a singlephase system Current control of a threephase STA TCOM Speed control of an induction motor by V/f me Rulph Chassaing, "Digital Signal Processing And Applications With The C6713 And C6416 DSK", WileyInterscience, 2004. Thad B. Welch, Wright, H.G. Cameron, and Michael G. Morrow, Tretter, "Communication System Design Using DSP Algorithms:With Laboratory Experiments for the TMS320C6701 and TMS320C6711", Springer, 2003. Nasser Kehtarnavaz, "Rea Chassaing, "DSP Applications Using C And The TMS320C6x DSK", Wiley Interscience, 2002. Shehrzad Qureshi, "Embedded Image Processing On The TMS320C6000 DSP: Ex
2439	CHE	INTRODUCTION TO NANOSCIENCE AND TECHNOLOGY	CHE674	1. Overview nanoscience: Important concepts such as size, quantum effect, andMoores law. Characteristic length scales determining the behavior of physical andbiological systems; fundamental phenom dots,lanthanidebased nanoparticles, polymer nanoparticles, carbon nanotubes and theirproperties (11 lectures).2. Synthesis and organization of nanomaterials: Chemical Routes for Synthesis ofNanomate reverse micelles, Solvothermal synthesis;Thermolysis routes, Microwave heating synthesis; Sonochemical synthesis; selfassembly;lithography; microfluidics; and chemical vapor deposition; surfacemodi andXray scattering), Scanning probe methods (STM and AFM) and other techniques(Optical Spectroscopy, Chromatography, Surface Plasmon Resonance and LightScattering) (1 0 lectures).4. Application Discussion on research papers based on the above syllabus. This will be done inbetween the above topics (8 lectures). 1. Christof M. Niemeyer, Chad A. Mirkin , Nanobiotechnology: Concepts, applicationsand perspectives, WileyInterscience (2004)2. Geoffery A. Ozin, Andre C. Arsenault, Nanochemistry: A chemical approach
2440	CE	ENGINEERING MECHANICS	CE621	Stress analysis: forces and moments, theory of stress, principal stresses andstress invariants, compatibility equations, equilibrium equations; Strain:deformation and velocity gradients, Lagrangian and Eu conditions; Fundamental physical principles: conservation of mass,linear momentum, angular momentum, and energy, second law of thermodynamics;Constitutive theory: St. Venants principal, linear elas viscoelasticity, yield criteria; Applications: Airy stress function,twodimensional elastostatics problems, torsion.
2441	CE	FINITE ELEMENT METHODS FOR CIVIL ENGINEERING APPLICATIONS	CE723	1. Problem formulation, numerical and closed form solutions, weak form, collocation, least square, Galerkin technique, derivation of finite element equations, stiffness matrices, global assembly, co and possible sources of errors.(2. Formulation of one dimensional truss and beam elements. Application to 2D trusses and frames.3. Formulation of 2D problems involving plane stre pavements4. Formulation of plate bending elements. Bending of plate, Von Karman nonlinear plate theory and formulation.5. Formulation of thin shell elements. Applications to dome, water tank etc. embankment, tunnel, etc.7. Nonlinear static and dynamic problems; geometric and material nonlinearity, Pdelta effectsin tall buildings, elastoplastic analysis as encountered in structures and g infinite elements and crack tip elements.9. CE applications such as 3D elastic problems, consolidation, seepage, transpott and propagation through heterogeneous media.10. Finite element fo

S.No.	Branch	Course Title	Courses ID	Content
2454	CE	COMPUTER METHODS IN HYDRAULICS AND HYDROLOGY	CE613	Basic: Introduction to computer programming and computation with Matlab. (02 lectures)Open channel flow : Estimation of normal and critical depth; uniform flow computations; computation ofwater surface profiles;hydraulic structures; unsteady flow SaintVenant equation, kinematicwave routing, diffusion routing, overland flow; steady and unsteady modelling using HECRAS. (07 lectures)Closed conduit flow: Steady flow: Estimation of Unit hydrographs; lumped and distributed flow routing;hydrologic statistics parameter estimation, time series analysis, frequency analysis, geostatistics;hydrologic modelling using unit hydrograph, Muskhelishvili's equation, GreenAmpt infiltration model; introduction to MODFLOW. (05 lectures)Application of soft computing methods and GIS in Hydraulic and Hydrologic modelling. (03 lectures)Laboratory exercises: 1. Chow V.T., Maidment D.R. and Mays L.W. (1988), Applied Hydrology, McGrawHill2. Chaudhry M.H. (2007), OpenChannel Flow, 2nd Edition, Springer3. Deb K. (1995), Optimization for engineering design, Wiley4. Geographical Information Systems, 3rd Edition, Prentice Hall5. Mathews J.H. and Fink K.D. (2006), Numerical Methods using Matlab, 4th Edition, Prentice Hall6. Press eta!., Numerical Recipes (C, C++), Cambridge University Press7. Edition, John Wiley & Sons8. Wasserman P.O. (1989), Neural computing, theory and practice, Van Nostrand Reinhold, New York
2455	CHE	MECHANICS OF SOFT MATERIALS	CHE678	1. Fundamental Equations: The strain tensor; The stress tensor; Thermodynamics of deformation; Hooke's law; Homogeneous deformations; Equilibrium of an elastic medium bounded by a plane; Stress analysis of rods; Bending and torsion of rods. Equation of equilibrium for a bent plate; The energy of a bent plate; Application of bending plate geometry for solving problems related to Adhesion. (10 hours)3. Nonlinear elasticity. Solutions for incompressible materials. Cavitation in crosslinked networks. (10 hours)4. Mechanics of cell wall: Elasticity of cellular filaments; soft networks in cell;biomembranes, membranes. 1. "Theory of Elasticity, 3rd edition"; by Landau and Lifshitz. Course of theoretical physics, vol7.2. "A treatise on the mathematical theory of elasticity"; by A. E. H. Love.3. "Contact Mechanics"; by J. K. Knowles.4. "Mechanics of Materials"; by C. D. Coker.5. "Mechanics of Materials"; by C. D. Coker.6. "Mechanics of Materials"; by C. D. Coker.7. "Mechanics of Materials"; by C. D. Coker.8. "Mechanics of Materials"; by C. D. Coker.9. "Mechanics of Materials"; by C. D. Coker.10. "Mechanics of Materials"; by C. D. Coker.
2456	CHE	PETROLEUM REFINERY ENGINEERING	CHE662	Petroleum refining in India; refinery tests and crude oil evaluation ; crude distillation column design; Delayed Coking; Catalytic cracking; Catalytic reforming;Catalytic isomerization; Alkylation; Polymerization
2457	CHE	PROCESS ENGG. PRINCIPLES IN MICRO. FABRICATION	CHE659	This course provides an insight into the process engineering principles in microelectronic fabrication. The transport processes, reaction kinetics and reactor design aspects of the following topics will be covered: 1. processing, epitaxy,lithography, plasma processing (deposition and etch), electrochemical deposition and chemical mechanical planarization. Also, the cleanliness and purity in process environment (e.g. fine particles, organic vapors, etc.) 1. Stanley Middleman, Arthur Hochberg, "Process Engineering Analysis in Semiconductor Device Fabrication"; McGraw Hill, 1st Ed, 1993. 2. Stephen A. Campbell, "The Microelectronics Fabrication Handbook"; McGraw Hill, 2nd Ed, 1999. 3. Peter VanZant, "Microchip Fabrication"; McGraw Hill, 5th Ed, 2004.(Students likely to take the course: Chemical Engineering, Electrical Engineering, Metallurgical and Materials Engineering, Mechanical Engineering, etc.)
2458	CHE	NUMERICAL METHODS IN CHEMICAL ENGINEERING	CHE642	Systems of Linear and NonLinear Algebraic Equations: Successive Substitution,Newton Raphson, Eigenvalues and Eigenvectors of Matrices, Interpolation,Solutions of ODEs (IVP): Runge Kutta, Multistep Methods, Shooting Methods.
2459	CHE	PRINCIPLES OF HETEROGENEOUS CATALYSIS	CHE633	Adsorption; Energetics; Isotherms and Rates: Experimental aspects of adsorption and allied phenomena on catalyst surfaces; Pore structure and surface area estimation and their significance; Important catalytic reactions
2460	CHE	CHEMICAL REACTION ENGINEERING	CHE631	Behaviour of chemical reactions; Behaviour of chemical reactors: ideal and nonideal flow, nonisothermal reactor performance, reactor stability; Heterogeneous reactions: interphase and intraphase heat and mass transfer
2461	CHE	THERMODYNAMICS OF FLUIDS AND FLUID MIXTURES	CHE623	Classical thermodynamics of phase equilibria: Thermodynamic properties from volumetric data: Nature of intermolecular forces, Theory of corresponding states;Fugacities in gas mixtures and liquid solutions
2462	AE	PRINCIPLES OF ACOUSTICS	AE752	a. Wave theory of sound: Plane waves, Harmonic waves and complex Algebra, Speed of sound, Energy, Intensity and Power, Spherical wavesb. Quantitative measurement of sound: Frequency content and Propagation of Plane waves: Reflection from a rigid surface, Propagation in a tube,Radiation due to waves on the wall, Oblique reflection and transmission at a planar interface. Radiation from Vibrating Bodies
2463	AE	AUTOMATIC CONTROL OF AIRCRAFT ROCKETS AND SPACECRAFT	AE649	1. Introduction to Automatic Control Systems.(Plant Models, Control Algorithms, Sensors, Actuators, Control Systems Classification.)2. Introduction to Rigid Body Dynamics and Flight Models.3. Linear System Analysis (Performance, and Robustness).4. Single Variable Linear Control(Proportional, Integral, Derivative Control, Rate and Rate Integrating Gyros,Single Input Regulation by Pole Placement, Linear Observers, SISO Autopilots for Aircraft, Rockets, and Entry Vehicles, Planar Tracking Systems for Rockets and Entry Vehicles, Pitch Stabilization of Gravity Gradient Spacecraft, Spacecraft Single Axis Maneuvers).5. Multivariable and Multivariable Tracking, Longitudinal Autopilots for Aircraft, Bank to turn Missiles, and Entry Vehicles, Lateral Directional Autopilots for Atmospheric Flight Vehicles, Attitude Stabilization of Spacecraft, Roll Autopilots, Gyroscopes ,Variable Speed Control Moment Gyroscopes, Thrust Vectoring Attitude Control of Rockets).6. Terminal Time Weighted Linear Optimal Control(Time Varying Tracking Systems, Guidance and Control Systems. & Sons,Chichester, 2002.1 ., 2. Tewari, A, Atmospheric and Space Flight Dynamics, Springer (Birkhauser), Boston,2006.
2464	MSP	MECHANICAL PROPERTIES OF MATERIALS	MS603A	Stress and strain tensors, Elastic constants, Effect of structure on elastic behaviour,Elastic stress distributions, Viscosity and viscoelasticity in polymers,Yielding criteria, Dislocations and plastic deformation in metals, ceramics and composites, Fatigue, Mechanical testing, Strength and engineering design with brittle solids, Heat treatment, Powder processing.
2465	CHM	ORGANIC CHEMISTRY: FUNDAMENTALS AND APPLICATIONS	CSO201A	
2466	CHE	THERMODYNAMICS	ESO201	Introduction & scope of thermodynamics; system, thermodynamic state & equilibrium, work & heat,zeroth law, temperature scale, properties of pure substances, tables of thermodynamic properties, I law of thermodynamics, (iv) chemical reactions; II law: heat engines and refrigerators, Kelvin Planck and Clausius statements, Reversible and irreversible processes, Carnot cycle and efficiency,Thermodynamic temperature scale; potentials; Clayperon equation; Phase rule, Power and refrigeration cycles: Rankine (ideal & reheat), Otto, Diesel and Brayton cycles; Vapor compression cycle,Air standard refrigeration cycle. * Sonntag, Borgnakke, Van Wylen, Fundamentals of Thermodynamics, 6th Edition, Wiley India. * Y V C Rao, An Introduction to Thermodynamics, Universities Press, 2005. * Cengel and Boles, Thermodynamics: An Engineering Approach, 5th Edition, McGraw Hill, 2006.
2467	CHE	FUNDM. OF COLLOID & INTERFACE SCI. & TECH.	CHE688	Capillarity, interfacial thermodynamics, surfactants, stability of multiphase systems, foam, emulsion, multiphase reactors, wetting and adhesion, catalysts sintering/redispersion; Stability and coagulation of colloids
2468	CHE	ADVANCED PROCESS DYNAMICS & CONTROL	CHE681	Process Identification Techniques for SISO & MIMO systems off line & on line;Generalized Predictive Control (GPC); Model Predictive Control (MPC): Dynamic Matrix Control (DMC), Internal Model Control (IMC), MATLAB & SIMULINK software.
2469	AE	INSTRUMENTATION MEASUREMENTS AND EXPERIMENTS IN FLUIDS	AE696	Need and objective, fundamentals of fluid mechanics, wind tunnels, visualization,HWA, pressure and noise measurements, temperature, wall shear stress, flow measurements, geophysical flows, spin up and spin down
2470	AE	HIGH TEMPERATURE GAS DYNAMICS	AE690	Nature of high temperature flows, perfect and real gas, Gibbs free energy and entropy production, microscopic description of gases, thermodynamic properties, equilibrium properties kinetic theory, inviscid flow
2471	CE	FLUID MECHANICS LABORATORY	CE612	Verification of momentum equation; Friction loss in pipes; Rainfall runoff relationship; Flow over sharp crested weir; Flow in pipe networks; Bernoulli theorem; Fall velocity of objects; Point velocity measurement; abstraction; Hydrogen bubble flow visualization; Hydraulic jump; Flow past a cylinder
2472	AE	HELICOPTER THEORY : DYNAMICS AND AEROELASTICITY	AE686	Historical development, configurations of helicopters, rotor system, flight control mechanism, momentum theory and blade element theory in hover, vertical flight and forward flight. Idealization of rotor blades. Flap pitch, lag pitch and flap lag coupling, simple model of rotor fuselage dynamics, longitudinal and lateral stability and control of helicopters.
2473	AE	DETERMINISTIC & RANDOM	AE685	Free and forced vibration of discrete multi degree of freedom systems with and without viscous damping; impulse and frequency response methods. Continuous systems; natural modes, free & forced vibration

S.No.	Branch	Course Title	Courses ID	Content
		VIBRATION		processes,response of system to random excitations
2474	AE	AIRCRAFT MATERIALS AND PROCESSES	AE684	Definition of various terms used for classification of materials. Mechanicalproperties. Testing of aircraft materials. Classification of alloys of aluminium,steel, titanium etc. High temperature problems; aer
2475	AE	RANDOM VIBRATIONS	AE683	Intro. to probability theory, random process. Excitation response relations forstationary random processsingple and multidegree of freedom system withlinear and nonlinear characteristics, continuous sy
2476	ME	MODELING OF MULTIPHYSICS SYSTEMS	ME698N	
2477	CHE	M. TECH. SEMINAR	CHE701	M. Tech. Seminar
2478	CHE	M TECH THESIS	CHE699	M. Tech. Thesis
2479	CE	ANALYTICAL AND NUMERICAL METHODS IN GEOMECHANICS	CE639	Finite difference, finite element and other analytical methods of solution to(i) Elasticity and stability problems in Geomechanics, (ii) Analysis of responseof soil media to applied loads, (iii) Limiting equilibri
2480	ECO	QUANTITATIVE ANALYSIS OF INTERNATIONAL TRADE	ECO752	
2481	CE	M. TECH THESIS	CE699	M. Tech. Thesis
2482	PHY	TRANSPORT IN MESOSCOPIC SYSTEMS	PHY632	
2483	PHY	HIGH ENERGY PHYSICS II	PHY781	Current topics in Particle Physics and quantum field theory.
2484	CE	TRAFFIC ENGINEERING	CE683	Microscopic and macroscopic traffic parameters, traffic flow models, carfollowingmodels, capacity and level of service analysis, design of traffic facilitieslike unsignalized and signalized intersections, int
2485	CHM	PRINCIPLES OF PHYSICAL CHEMISTRY	CHM629	Atomic and Molecular structure, Molecular Spectroscopy, Concepts of StatisticalThermodynamics, Electrochemistry, Chemical Kinetics, Photochemistry
2486	CSE	DATA MINING	CS685	
2487	CHM	PRINCIPLES OF INORGANIC CHEMISTRY	CHM649	Principles of modern inorganic chemistry discussion of the chemistry of nontransition elements, Coordination Chemistry, organometallic chemistry, Inorganicchemistry of biological systems.
2488	CHM	CHEMISTRY LAB (PREP)	CHM001	COURSE TOPICS: 1. Calibration of Burette 2. To determine the density of a given sample of liquid. 3. To determine the surface tension of benzene. 4. 5. To prepare of standard oxalic acid solution and u READINGS: 1. Laboratory Manuals (Department of Chemistry, IIT (BHU), Varanasi).
2489	CE	FOUNDATION DYNAMICS	CE635	Dynamics of elastic systems; Single and multidegrees of freedom systems;Empirical and semiempirical approaches to the theory of soil dynamics; Elastictheories of soil dynamics; Wave propagation; Dy
2490	AE	TURBULENCE	AE621	Origin, examples & character of turb, Reynolds stress, energy relations, closureproblem, phenomenology, eddy viscosity. Staistics. spectra, spacetime correlations,macro & microscales, stat. theory of turb Experimentaltechniques.
2491	AE	FINITE ELEMENT METHODS FOR FLUID DYNAMICS	AE618	Fundamental concepts; strong form, weak form, Galerkins approximation; matrixeqns, element and global point of view; numerical integration Guassianquadrature; temporal discretization generalized tr performance computing.
2492	AE	BOUNDARY LAYER INSTABILITY AND TRANSITION	AE617	NavierStokes eqn. and its various forms, thin shear layer approxn. Varioustypes of flows. Instabilities in laminar flows. Relationship of instability theorywith transition to turbulence. Transition prediction. F
2493	AE	FLIGHT STABILITY AND CONTROL	AE648	Linearized equations of aircraft motion for small perturbations in stability axes.Stability analysis of linearized equations of motion. Airplane longitudinal motion.Airplane lateral motion. Airplane handling q
2494	AE	FLIGHT DYNAMICS	AE647	Fundamentals of vectors. Transformation of coordinates. Particle kinematics.Rigid body kinematics. Force equations in a moving frame. Moment equationsin a moving frame. Atmospheric flight Dynamic
2495	MSE	SELECTION AND DESIGN OF ENGINEERING MATERIALS	MSE422	
2496	MSE	IRON AND STEEL MAKING	MSE421	
2497	MSE	DISPLAY MATERIALS & TECHNOLOGIES	MSE601	
2498	PHY	OPTICAL IMAGING	PHY690P	
2499	PHY	TYPE II SUPERCONDUCTORS VORTICES & APPLICATIONS	PHY690S	
2500	MSE	MANUFACTURING PROCESS	MSE470	
2501	ECO	LABOUR ECONOMICS	ECO753	
2502	ECO	PROJECT-I	ECO598	PROJECTI
2503	IME	COMPUTER INTEGRATED MANUFACTURING SYSTEMS	VLM605.	
2504	IME	AUTOMATION & ROBOTICS IN MANUFACTURING	VLM604.	
2505	IME	ADVANCE MATERIALS	VLM603.	
2506	AE	MATHEMATICS FOR AEROSPACE ENGG.	AE602	Matrices, determinants, vector spaces, linear transformation, eigensystems,linear equations, introduction to ordinary differential equations, homogeneouslinear equations of second order, nonhomogeneouslinear equations of second order, partial differentialequations, Fourier series, Fourier transform, Laplace transform,introduction to differencing methods; basic concepts of partial differentialequations, classification of second order equations, wave propaga

S.No.	Branch	Course Title	Courses ID	Content
				methods, elements of complex variables.
2507	AE	INTRO TO AEROSPACE ENGG.	AE601	History of aviation. History of spaceflight. Earths atmosphere and gravitationalfield. Anatomy of Flight vehicles. Bluff bodies v/s streamlined body, airfoil. Liftgeneration, significance of L/D ratio. Aerodynamic diagrams. Instruments and navigational aids. Exposure to flight testing.
2508	AE	AEROSPACE STRUCTURAL ANALYSIS - I	AE670	Free body diagram, equilibrium equations, examples from three dimensionaltruss problems; bending moment, shear force. Introduction to theory of elasticity,stress, strain, stressstrain relations, constitutive temperatureeffects, nonhomogeneous materials, modulus weighted sectional properties, thinwalled sections. Deflection of beams.Torsion of circular and noncircular sections,thin walled sections, single a bending andtorsion. Plane strain and plane stress problems in elasticity. Eulers bucklingof columns.
2509	CE	GEOTECHNICAL EARTHQUAKE ENGINEERING	CE636	Introduction; Seismic Hazards: Mitigation of Seismic Hazards, seismology andearthquakes, strong ground motion, seismic hazard analysis; Wave propagationin unbounded media: in semiinfinite bodies, i conditions on ground motion; Liquefaction: evaluation of liquefactionhazards, effects of liquefaction; Case studies.
2510	ME	PROJECT II	ME452	Fabrication of a prototype and appropriate modification in the design (if necessary) to meetthe qualitative and quantitative performance parameters as envisaged in the Project I
2511	ME	PROJECT - I	ME451	Project work involving the analysis, synthesis, material/component selection and detaileddesign of a mechanical system including the preparation of working drawings. The systemmay be integrated with areas (or a combination there of) Fluidmechanics and Thermal sciences, Solid Mechanics and Design, Manufacturing Science andRobotics and/or any other related mechanical system(s).
2512	AE	PROJECT-I	AE471	Registration for project with the selection of topic & getting started on thedesign, fabrication work, algorithm etc.
2513	CHM	INORGANIC CHEMISTRY LAB	CHM443	
2514	MTH	ERGODIC THEORY	MTH729	
2515	ME	SUMMER INDUSTRIAL TRAINING	ME300	Six weeks training after the sixth semester in the Industry.
2516	IME	M TECH THESIS	IME699	M. Tech. Thesis
2517	EE	POWER MANAGEMENT CIRCUITS	EE698	
2518	MSP	PHD THESIS	MS799	Ph. D. Thesis
2519	MTH	ANALYTICAL TECHNIQUES FOR PDE	MTH663	
2520	ME	ADVANCED THERMODYNAMICS	ME698L	
2521	ME	BUBBLES AND DROPLETS	ME698K	
2522	ME	COMPRESSIBLE FLOW	ME698J	
2523	EE	ANALOG CIRCUITS FOR SIGNAL PROCESSING	EE698W	
2524	CSE	SKETCHING AND SAMPLING FOR BIG DATA ANALYSIS (NUMERICAL LINEAR ALGEBRA)	CS698C	
2525	ME	INTRODUCTION TO PROFESSION	ME100	
2526	EE	INTRODUCTION TO PROFESSION	EE100	
2527	MSE	UNDER GRADUATE PROJECT III	MSE497A	UG PROJECT (UGPIII)
2528	PHY	EFFECTIVE FIELD THEORY	PHY690U	
2529	PHY	BACHELOR OF SCIENCE PROJECT -IV	PHY558A	
2530	MTH	INTRODUCTION TO HOMOGENIZATION	MTH719	
2531	CHM	SURFACES INTERFACES THIN FILMS & RELATED ANALYTICAL TECHNIQUES	CHM683A	
2532	MTH	FLUID MECHANICS	MTH523	Review and General Properties of Navier Stokes Equations; Some Exact solutionsof NS equations; Introduction to boundary layer theory; Introduction to turbulentflow; Introduction to compressible flow; A
2533	MME	IRON AND STEELMAKING	MME330	
2534	AE	SHEAR FLOWS	AE312	Dynamics & thermodynamics of 1D flow, isentropic flow,1D wave: normal shock,central expansion. Supersonics, oblique shock, 2D steady flows: reflection,intersection; PrandtlMeyer flow. Method of chara rule. Hypersonics; similitude, high temp and rarefaction.
2535	EE	PROGRAMMING FOR SIGNAL PROCESSING	EE698K	
2536	CSE	PRACTICAL CYBER SECURITY FOR CYBER PRACTITIONERS	CS698S	

S.No.	Branch	Course Title	Courses ID	Content
2537	PHY	PHYSICS OF TURBULENCE	PHY672	
2538	EE	NUMERICAL MODELING AND SPICE SIMULATION OF SEMICONDUCTOR DEVICES	EE698P	
2539	EE	SMART GRID TECHNOLOGY APPLICATIONS	EE698B	
2540	CSE	SPECIAL TOPICS IN NATURAL LANGUAGE PROCESSING	CS698O	
2541	AE	FUNDAMENTAL OF AEROSPACE PROPULSION - I	AE650	ntroduction to propulsion, conservation equations, basic thermodynamics,dynamics and thermodynamics of 1 D flows, 1 D isentropic flows, normal andoblique shocks, compressible flows, Rayleigh flow, rocket propulsion, thrust equation, solid rockets, liquid rockets,hybrid rockets, gas turbine cycles.
2542	EE	5G WIRELESS TECHNOLOGIES	EE698Q	
2543	CHE	GRADUATE SEMINAR	CHE802	Ph. D. Seminar
2544	CHE	M TECH SEMINAR	CHE702	M. Tech. Seminar
2545	AE	SOLID MECHANICS	AE672	Introduction. Material behaviour of idealized bodies, mathematical preliminaries,tensor analysis, partial derivatives, etc. Analysis of stress and strain measures.Laws of conservation, eqn. of motion, conse
2546	CE	PHD THESIS	CE799	Ph. D. Thesis
2547	AE	PHD THESIS	AE799	Units : As arranged
2548	AE	M TECH THESIS	AE699	Units : As arranged
2549	LT	M TECH LAB TECHNIQUES	LT680	
2550	IME	DATA MINING AND KNOWLEDGE DISCOVERY	IME672	
2551	AE	SPACE DYNAMICS-I	AE641	Introduction, performance of single and multistage rockets, central force motion,twobody problem, ballistic trajectories, trajectory transfer, rendezvous andinterception, Eulers eqns, satellite attitude dynam
2552	CHM	GRADUATE SEMINAR	CHM801	Graduate Seminar
2553	CHM	GENERAL SEMINAR2	CHM800	General Seminar
2554	CHM	PROJECT	CHM700A	PROJECT
2555	CHM	INTRODUCTION TO PROFESSION	CHM100	
2556	CE	MACHINE PROCESSING OF REMOTELY SENSED DATA	CE672	Image processing system; Preprocessing of remotely sensed data; Radiometricand Geometric distortions and corrections; Image enhancement; Imagetransformations; Pattern recognition.
2557	CE	GLOBAL CLIMATE CHANGE	CE646	Introduction to global climate; Global climatic models; Methods of reconstructingclimate; Quaternary climates, sea level changes, glacial/interglacial cycles;Geological records of climate change, sediment migrationhistory, response of vegetation to climatic reversals; Prequaternary climates,evolution of climate through geological time.
2558	CE	QUALITY AND SAFETY IN CONSTRUCTION	CE645M	Introduction to physical and structural geology; Landforms and drainage patterns;Elements of photogeology; Stereoscopy; Elementary photogrammetry; Photographicsystems, types of cameras, films and geology,landuse, land wastage, hydrogeology, mineral exploration and change detection.
2559	BSBE	BIOMECHANICS	BSE314	Introduction to biomechanics. Biosensors, actuators and control. Analysis ofbiosystem as a flexible structure. Aerodynamics, hydrodynamics and locomotion.Biostatics and Biodynamics I Mechanics of f Archimedes principles, Pascal Law, Bernoullis theory and the livingworld, Viscosity and turbulence, Human circulatory system.
2560	AE	MOLECULAR GAS DYNAMICS	AE747	The molecular model; binary elastic collisions: basic kinetic theory; referencestates & boundary conditions; collisionless flow; transition regime flows. Directsimulation Monte Carlo method. One dimension
2561	ECO	PHD THESIS	EC0799	Ph. D. Thesis
2562	HSS	PHD THESIS	PHI799	Ph. D. Thesis
2563	CSE	SPECIAL TOPICS IN COMPUTER SCIENCE	CS797	
2564	AE	DYNAMICS AND VIBRATION	AE688	Rigid body dynamics: Newtons second law, impulse and momentum, momentof a force and angular momentum, work and energy, system of particles, rigidbodies, Eulers equations. Analytical mechanics: theory: frequency response, transform methods, transfer function,transition matrix, Eigen value problem, Modal analysis. Lumped parametersystems: single degree of freedom system, two degrees of free vibrations of slender members.
2565	CSE	APPLICATIONS OF MARKOV CHAINS IN COMBINATORIAL OPTIMIZATION AND IN EVOLUTIONARY DYNAMICS	CS698P	APPLICATIONS OF MARKOV CHAINS IN CO AND EVOLUTIONARY DYNAMICS
2566	PHY	COSMOLOGY	PHY690I	
2567	CSE	TOPICS IN COMPUTER ARCHITECTURE AND OPERATING SYSTEMS	CS698E	
2568	EE	INTRODUCTION TO COMPUTATIONAL BIOMEDICAL IMAGING	EE698J	

S.No.	Branch	Course Title	Courses ID	Content
2569	CSE	INTRODUCTION TO ALGORITHMS AND LOGICS IN GAME THEORY	CS684	
2570	DES	PSYCHOLOGICAL PRINCIPLES AND DESIGN	DES631	
2571	CSE	DESIGNING VERIFIABLY SECURE SYSTEMS	CS698K	
2572	AE	ROCKET AND MISSILE STRUCTURES	AE673	Mission analysis, design approaches, analytical techniques, rocket grain analysis,structural types and optimization, honeycomb and sandwich construction,structural materials, aeroelasticity of cylindrical structure.
2573	HSS	THE WRITERS VISION OF THE FUTURE	ENG436	This course identifies and defines visionary literature through a broadrange of disciplines and/or literary genres. Open to engaging with texts from differentdomains, from philosophy to economics to science and technology, and to the future asprophecy. In the process, the course examines the ways in which the visions articulatedby such texts intersect with their and our realities, their purpose and effectiveness, as well as their relevance for society. No specific textbook or reference material. Thespecific texts chosen for the course in a particular semester will determine the appropriatereference materials.
2574	HSS	INTRODUCTION TO LINGUISTICS	ENG122	Introducing Language; Natural Language and Artificial Languages; NaturalLanguage and Animal Communication Systems; Evolution of Language; Acquisition ofLanguage; Ancient Indian Linguistics; Modern Linguistics. 1. Hirschberg, S. and T. Hirschberg, Reflections on language, OUP, 1999.2. Radford, R., Linguistics: An Introduction, CUP, 1999.3. Jackendoff, Ray, Foundations of Language, OUP, 2000.*4. Bergmann, Anouska, Introduction to Linguistics, OUP, 2000.
2575	HSS	APPLICATION OF PSYCHOLOGY TO LIFE	PSY152	Nature and Scope of Applied Psychology, Historical Perspective, Areas ofApplied Psychology, Roles and Skills of Applied Psychologists, Ethical Issues, Clinical andCounseling Psychology, Community Psychology, Legal Psychology
2576	PHY	M.SC. REVIEW PROJECT III	PHY502	M.Sc. Review Project III
2577	BSBE	BIOLOGY LAB	BSE641	
2578	EE	CONVEX OPTIMIZATION FOR ML AND WIRELESS COMMUNICATIONS	EE698X	
2579	CSE	HARDWARE SECURITY OF INTERNET OF THINGS	CS698U	
2580	EE	ELECTROMAGNETIC INTERFERENCE AND COMPATIBILITY TECHNIQUES	EE644	
2581	ME	MECHANICS OF ELECTROMAGNETIC CONTINUA	ME698B	
2582	CSE	FUNDAMENTALS OF DATA ENGINEERING - I	CS698B	
2583	BSBE	PHD THESIS	BSE799	Ph. D. Thesis
2584	CE	LABORATORY COURSE IN INFRASTRUCTURE ENGINEERING AND MANAGEMENT	CE642	Geological hazards and environmental impact; Landslides: cause, classification,zonation and protection; Earthquakes: historical seismicity, classification, interplateand intraplate earthquakes, effect on ground motion; Floods: causes, extentof flood hazard, flood hazard zoning, flood control and protection; Landsubsidence; Snow avalanches; Rock bursts; Mapping, monitoring and managementof geological hazards.
2585	BSBE	RESEARCH METHODS IN BIOLOGICAL SCIENCES	BSE671	
2586	CE	PALEOSEISMOLOGY AND TECTONIC GEOMORPHOLOGY	CE647	Plate tectonic and its relation to earthquakes; Historical and modern seismicity;Mapping of active tectonic landforms in different tectonic environments; Fieldtechniques in paleoseismology, identification of paleoseismic features, faults,prediction of future earthquake, identification of paleoliquefaction features;Dating techniques; Correlation of paleoseismic data with existing geodetic andgeophysical data; Delineation of seismogenic zones.
2587	CSE	FUNDAMENTALS OF ERROR CORRECTING CODES	CS698Z	
2588	CSE	HUMAN-AI INTERACTION	CS698Y	
2589	CSE	TOPICS IN PROBABILISTIC MODELING AND INFERENCE	CS698X	
2590	PHY	SUPERCONDUCTIVITY AND APPLICATIONS	PHY690T	
2591	MTH	STATISTICAL & AI TECHNIQUES IN DATA MINING	MTH552	
2592	BSBE	BIOMATERIALS	BSE640	
2593	HSS	AN INTRODUCTION TO THE JAPANESE & KOREAN CINEMATIC ART	ART410	
2594	PHY	ADVANCED GENERAL	PHY690M	

S.No.	Branch	Course Title	Courses ID	Content
		RELATIVITY AND BLACK HOLES		
2595	CSE	PARALLEL EXECUTION OF PROGRAMS	CS629	
2596	CSE	TOPICS IN EMBEDDED SYSTEMS	CS624	Current topics in the design, specifications and analysis of embedded systems.Contemporary topics such as specifications of embedded systems, analysisof embedded systems, interface to the realtime systems like formal verification, cosimulation, etc., estimationof hardware and software costs, partitioning, synthesis (hardware, software,memory, bus), retargetable usage of the software, specification and
2597	CSE	UNDER GRADUATE PROJECT-II	CS497	
2598	EE	PEER TO PEER NETWORKS	EE698C	
2599	CHM	RESEARCH	CHM799	Ph. D. Thesis
2600	HSS	NATURAL LANGUAGE SEMANTICS	ENG443	Philosophical and linguistic approaches to semantics; Semanticstructure and its computational modeling; Lexical semantics; Logical form of naturallanguage; Anaphoric dependencies; Pragmatic structure and its computational modeling; 5Semantic Structure and its Computational Modelling: Semantics 6Pragmatics Distinction, SyntaxSemantics Interface: Syntactic andSemantic Parsing, HPSG, FrameNet SemanticsLexical Semantics: Lexical Semantics, Event Structure and Qualia Structure, WordNet and Generative LexiconsLogical Form of Natural Language, . Propositional Structure, 6Quantification, QuasiquantificationAnaphoric Dependencies and Pragmatic Modelling Pragmatic 6Knowledge, Discourse Representation Theory/Situation SemanticsCognitive Grammar and Cognitive Semantics 5
2601	CSE	DESIGN AND ANALYSIS OF ALGORITHMS	CS602	Basic concepts of operating systems, compilers, and data base managementsystems.
2602	CSE	MATHEMATICS FOR COMPUTER SCIENCE	CS601	Programming utilities, lab exercise for developing large system and applicationprograms.
2603	CSE	COMPUTER NETWORKS	CS425	Introduction: Advantages of computer networks, LAN vs. WAN, ISO/OSI sevenlayerarchitecture, networks topologies, Physical Layer: transmission media,data encoding, Data Link Layer: Framing, Error detection and correction, MAC protocols, Other examples of MAC protocols, Network Layer: Internetworking Tunneling, Encapsulation, Fragmentation. Internet Protocol (IP)Header structure, addresses, options, etc. Routing Algorithms and Routing Protocol header, services, connectionmanagement, congestion control, sliding window, timers. User Datagram Protocol.Domain Name Service, Unix network programming, socket abstraction. clientserver programming. 1. AS Tanenbaum, DJ Wetherall, Computer Networks, 5th Ed., PrenticeHall, 2010.2. LL Peterson, BS Davie, Computer Networks: A Systems Approach, 5th Ed., MorganKauffman, 2011.3. JF Kurose, KW Ross, Computer Networks: A Top-Down Approach, 5th Ed., Addison Wesley, 2010. Security, Principles and Practice, 5th Ed., PrenticeHall, 2010
2604	CSE	COMPUTER ARCHITECTURE	CS422	Introduction: Overview of Computer Architecture, Performance evaluation ofprocessors, Pipelining, Superpipelines, Advanced pipelines, static and dynamicscheduling, Instruction level parallelism, loop unrolling, cache organization, cache coherence, Symmetric multiprocessors (SMP),NUMAMPs, Massively parallel processors, Cache coherence protocols,Interconnection networks, I/O processing, parallel programming, I/O organization. 1. JL Hennessy, DA Patterson, Computer Architecture: A Quantitative Approach, 4th Ed., Morgan Kaufmann/ElsevierIndia, 2006.
2605	EEM	M.TECH THESIS	EEM699	M. Tech. Thesis
2606	CSE	TOPICS IN DATA COMPRESSION	CS698D	
2607	CSE	INTRODUCTION TO INTERNET OF THINGS AND ITS INDUSTRIAL APPLICATIONS	CS698T	
2608	PHY	FUNDAMENTALS OF ASTROPHYSICS	PHY690C	
2609	AE	INTRODUCTION TO FINITE ELEMENT METHODS	AE675	Discussion on mathematical models, reliability of computer aided engineeringanalysis. Model problem of linear elastostatics in onedimension, principle ofminimum potential energy, beam bending problem of twodimensional code. Convergence analysis in two dimensions.Characterizational of solution smoothness, rate of convergence in energy norm,a posteriori error estimation. Direct computation of stress and strain
2610	CHM	FRONTIERS IN INORGANIC CHEMISTRY	CHM691	Developing facets of Inorganic Chemistry (2 Lectures)Oxidative generation of molecular oxygen from water during photosynthesis (8 Lectures)Its importance from the standpoint of nonconventional energy storage and conversion novel organic transformations including methane tomethanol performed by a large number of Fe containing metalloenzymes (8 Lectures)Reductive cleavage of dioxygen bond and novel organic transformations Bioinorganic Chemistry, I. Bertini, H. B. Gray, S. J. Lippard and J. S. Valentine, UniversityScience Books, Mill Valley, 2006.
2611	EE	WIDE-BANDGAP SEMICONDUCTORS: PHYSICS, TECHNOLOGY AND APPLICATIONS	EE698M	
2612	EE	INTRODUCTION TO FLEXIBLE ELECTRONICS	EE698N	
2613	CSE	MODERN CRYPTOLOGY	CS641	Basics of finite fields. Private and Publickey cryptography, existing cryptosystemsand their security. Cryptanalysis of existing systems. Zeroknowledge protocols,Oneway functions. Advanced protocols for secure communication
2614	MSE	SEMINAR PRESENTATION	MSE691	Seminar Participation
2615	MSE	SEMINAR PARTICIPATION	MSE690	Seminar Participation
2616	HSS	TWENTIETH CENTURY CONTINENTAL PHILOSOPHY	PHI784	
2617	ECO	DYNAMIC MACROECONOMICS	ECO409	
2618	BSBE	METHODS IN BIOTECHNOLOGY-I EXPERIMENTAL BIOSEPERATION	BSE492	
2619	CSE	TOPICS IN GAME THEORY	CS698W	

S.No.	Branch	Course Title	Courses ID	Content
		AND COLLECTIVE CHOICE		
2620	CSE	QUANTUM COMPUTING	CS682	
2621	ME	INTRODUCTION TO ROD THEORY	ME698H	
2622	ME	DEEP LEARNING IN MECHANICAL SCIENCES	ME698G	
2623	ME	CALCULUS OF VARIATIONS	ME624	
2624	ME	ACOUSTICS	ME698E	
2625	HSS	PHD THESIS	ENG799	Ph. D. Thesis
2626	BSBE	BIOINFORMATICS & COMPUTATIONAL BIOLOGY	BSE322	Biological Databases; Global and local alignment; pair wise and multiple sequencealignment; Pattern searching in DNA & Protein sequences; Alignment tools, BLAST,FASTA, phylogenetic prediction, evolu predictionfrom sequence features, Comparative genomics.
2627	EE	ECONOMIC OPERATION & CONTROL OF POWER SYSTEMS	EE632A	Economic load dispatch, loss formula, introduction to mathematical programming,hydrothermal scheduling systems, power system security, optimal real andreactive power dispatch, state estimation, load
2628	BSBE	DEVELOPMENTAL BIOLOGY I	BSE630	
2629	BSBE	NEUROBIOLOGY	BSE629	Cell and molecular biology of neurons. Membrane potential, local signaling andgeneration of action potential. Mechanism of synaptic transmission. Sensoryperception : Vision, hearing, taste, touch and smell. Learning and memory system and the Cerebral cortex learning and memory, sleep and dream.
2630	AE	THEORY OF COMBUSTION	AE753	Introduction to combustion : Types of flames, role of chemical kineticsChemical Kinetics : Formulation of chemical kinetics equations, reactionmechanisms, steady state approximation, Arrhenius Law : Factors affecting rate of branching explosions, Chemical equilibrium Conservationequations for reacting flows : Shvab Zeldovich formulation. Laminar premixedcombustion : Flame speed Thermal theory (Mallard and Le Chatellier) Limits limits micro combustion. Detonation andDeflagration : Chapman Hugoniot relations, Chapman Jouguet points Liminarnon premixed conbustion: Burke Schuman Analysis, Phenomenological AnalysisIgnition delay stirred reactor. Turbulent nonpremixedcombustion : Conserved scalar approach, two variable approach,flamelet model, direct closure.
2631	CHM	INTRODUCTION TO BIONANOTECHNOLOGY	CHM676	
2632	CE	MASONARY STRUCTURES	CE625	Properties of constituents: units burnt clay, concrete blocks, mortar, grout,reinforcement; Masonry bonds and properties: patterns, shrinkage, differentialmovement, masonry properties compression strength tensile strength shear strength inplane, outofplane; Behaviour of masonry walls and piers:axial and flexure, axial shear and flexure, Behaviour of Masonry Buildings:unreinforced masonry buildings importance of bands and corner & vertical infills in RC frames: strut action; Structuraldesign of masonry in buildings: methods of design WSD, USD, seismic design seismic loads, code provisions, infills, connectors, ties; Seismic evaluationand strengthening
2633	ME	ELEMENTS OF SHELL THEORY	ME698C	Condensed Syllabus: Flapping foils in nature, airfoil theory, vortex dynamics, governing equations and parameters, kinematics, vortex wakes, drag to thrust transition, lift laws, thrust generation in quiescent ambient, deflected wakes, unsteady mechanisms, flexible flapping foils, fluid-flexible-structure interaction, 3D foils, foils in tandem, biomimetic applications. (6 Lectures) Unsteady Mechanisms: Thrust and Lift Generation by Flapping Foil: Unsteady Mechanisms: Flapping Flexible Foil: Biomimetic Propulsion and Other Topics: References: 1. Theory of wing sections - I. H. Abbott (Dover, 1959) 2. Fluid Mechanics & Aerodynamics - S. P. Kuethe (McGraw-Hill, 1992) 3. Aerodynamic Theory – W. F. Durand (Julius Springer, 1935) 4. Aerodynamic Theory – W. F. Durand (Julius Springer, 1935) 5. An introduction to flapping wing aerodynamics – W. Shyy, H. Aono, C. Kang, H. Liu (Cambridge University Press, 2008)
2634	BSBE	BIOMATERIALS	BSE452	Introduction to Materials Science: Bulk and surface properties of materials;Polymeric materials; synthesis, characterization, and fabrication methodsInert, biodegradable, hydrogels, Natural, Genetically engineered materialsProtein structure, interaction of proteins with synthetic materialsCharacterization of cellmaterial interactions; inflammatory responses; acuteinflammation, chronic inflammation, foreign body response, as well as drug delivery
2635	BSBE	INTRODUCTION TO TISSUE ENGG.	BSE454	What is tissue engineering? Scope and objective of tissue engineering; Principlesof tissue engineering; Essential components of tissue engineering; MaterialsScience/Engineering aspects (degradable materials); Biologicalaspects (choice of cell type, progenitor cells and cell differentiations); Molecularbiology aspects (cell signaling moleculesgrowth factors, cell attachmentintegrins);Drug delivery in tissue engineering
2636	DES	INTEGRATED PRODUCT DEVELOPMENT-I	DES633	
2637	MME	MANUFACTURING PROCESS	MME470	
2638	IME	MANAGEMENT SEMINARS	MBA698.	Management Seminars
2639	EE	EMI/EMC IN POWER ELECTRONICS	EE698A	
2640	CSE	OPERATING SYSTEMS	CS330	Introduction: Role of operating System; System Calls; Processes and ThreadsConcepts of Processes, Threads; Process Control Block. CPU Scheduling: Examplesfrom contemporary Oses (UNIX and NT).Process Synchronization Critical Section Problem; Hardware Mechanism forsynchronization; Semaphores and Mutex objects; Classical Problems (ProducerConsumer, dining philosophers etc.); Monitoring and avoidance mechanisms.Virtual Memory: Address binding process (compilation and linking); DynamicLinking; Segmentation; Paging Protection; Demand Paging; Page Replacementpolicies Thrashing and its prevention. File Systems: File organization in directories; File attributes; Operationson files; Directory attributes and operations on directories; Directory organizations;File and directory protections.File system implementation Concepts of file systems;caching;Examples of files systems from one or more of DOS, BSD, Linux, HPFSand NTFS;Device Drivers:Storage management Disk scheduling; Disk Management; Swapand swap management.Security and Access Control Trojan Horses etc.
2641	EE	ARCHITECTURE OF MICRO-PROCESSORS & MICROCOMPUTERS	EE674	
2642	MSE	PH D THESIS	MSE799	Ph. D. Thesis
2643	MSE	M.TECH THESIS	MSE699	M. Tech. Thesis

S.No.	Branch	Course Title	Courses ID	Content
2644	ME	COMPUTATIONAL TURBULENCE	ME698M	
2645	ME	ATOMISTIC SIMULATIONS IN MECHANICAL ENGINEERING	ME698I	
2646	MTH	FRACTAL INTERPOLATION AND APPLICATIONS	MTH622	
2647	CHE	CHEMICAL PROCESS CALCULATIONS	CHE251	Guidelines for Problem Solving; Review of Basic concepts Process variables& properties, Degree of Freedom, Steady State Material Balances in nonreactingsystems and reacting system, Recycle & purge, Steady State Energy Balances in nonreacting & reacting systems,DeCoupled & coupled mass & energy balances, Calculations for network of unitswith recycle & bypass, Process Flow sheeting with sequence
2648	BSBE	HUMAN MOLECULAR GENETICS	BSE636	Simple Mendelian traits; Lossoffunction mutations; Gainoffunction mutations;Gene interactions; Dynamic mutations; Genetics of neoplasia; Genomic imprintingand human disease; Xinactivation and DNA disorders;Pharmacogenetics and biochemical genetics; Animal models in human genetics:Methods used for diagnosis and detection of gene mutations; Gene Therapy.
2649	BSBE	STRUCTURAL BASIC OF PROTEIN FUNCTION	BSE632	Conformational studies of biomolecules, protein secondary structure, 3Dstructuresof proteins, DNA structures, membrane proteins. Xray and NMR techniques,electron crystallography, other spectroscopic computational studies to understand the structurefunction relationshipsof proteins, molecular mechanics and dynamics, force fields, biomolecularsimulation programs. Structural genomics, highthroughput
2650	CSE	INTRODUCTION TO COMPUTER GRAPHICS	CS360	Introduction to Picture Synthesis and Analysis. Conceptual Framework of anInteractive Graphical Simulation System, Graphics hardware. Basic RasterGraphics Algorithms. Introduction to Simple Raster, Rasterization Techniques, Geometric Modeling in 3D. Viewing in 3D. Concept of Synthetic Camera. Dialogue Design. Graphics User Interfaces. WindowingSystems Graphi cal Modeling of Discrete events. Simulation of Virtual Reality in Graphical Simulation.
2651	CE	GEOENVIRONMENTAL DESIGN ASPECTS OF SOLID WASTE MANAGEMENT	CE733	Identification, characterization and regulatory requirements for disposal ofhazardous, nonhazardous and domestic wastes. Waste ManagementRecycling,composting, incineration and various disposal methods. Design practices of solid wastes. Tailing dams for disposal of flyash,coal, copper, iron and other metal wastes. Single and double lined landfill design,linear material clay, geosynthetics amended soils and performance monitoring. Application of geosynthetics in wastedisposal design.
2652	CSE	INTRODUCTION TO MATHEMATICAL LOGIC	CS202	
2653	IME	SIMULATION AND SCHEDULING OF MANUFACTURING	IME684	
2654	CHE	PROCESS ENGINEERING	CHE654	Introduction to the elements of process design, process development, processevaluation ; flow sheeting; Pilot plant; Optimization and economic considerations,process design engineering, project engineering
2655	CHE	OPTIMIZATION	CHE652	Mathematical formulation of optimization problems; single variable problems:search techniques; Multivariable problems without constraints: direct methods,first and second order methods; Multivariable
2656	CSE	PROGRAMMING TOOLS AND TECHNIQUES	CS355	Software management tools, CVS, Scripting tools, GUI programming tools,Language processing tools, Web programming tools.
2657	BSBE	UG PROJECT (UGP-I)	BSE398A	UG PROJECT (UGPI)
2658	NET	SAFETY ANALYSIS OF NUCLEAR POWER REACTORS	NT661A	
2659	HSS	INTRODUCTION TO LITERATURE	ENG123A	The course will introduce students to literature through exposure to differentliterary genres within the three main categories of prose, poetry, and drama. The focus of thecourse will be on how to read a literary text, to understand the different approaches to literaryanalysis as a way of understanding, interpreting, and evaluating specific literary texts.
2660	CHM	CHEMICAL KINETICS	CHM622	Phenomenological kinetics: Simple and complex systems including opposing, concurrent andconsecutive reactions (8)Rate law and mechanism, relation with thermodynamics (6)Precision in rate measurement andrelaxation techniques (6)Oscillatory reactions (4)Theories of reaction rates: bimolecular reactions, rate coefficient, activation energy, potentialenergy surfaces, trajectory methods and transition state theory (6) heterogeneous catalysisand enzyme kinetics (6) 1. K. J.Laidler, Chemical Kinetics, 3rd Ed. Harper & Row, New York, 19872. R. D. Levine and R. B. Bernstein, Molecular Reaction Dynamics and ChemicalReactivity, Oxford Univ. Press, Oxford, 19873. J. I. Steinfeld, Chemical Kinetics and Dynamics, Wiley, New York, 1989
2661	CE	URBAN TRANSPORTATIONS SYSTEMS	CE684	Dimensions of the widening role of urban transportation system planning; theplanning process; land use and transport system models; comparison andevaluation of various models; transportation impact analysis and plan implementation; Regional Analysisand development concepts; the role of transportation planning in the overallregional system; methodology and models for regional transportation system
2662	CHM	PHYSICAL CHEMISTRY I	CHM421	Atomic Structure, chemical binding and molecular structure. Elements ofmolecular spectroscopy.
2663	CHM	ORGANIC CHEMISTRY LAB.	CHM404	
2664	CHM	INORGANIC CHEMISTRY LAB	CHM343	
2665	CSE	PHD THESIS	CS799	Ph. D. Thesis
2666	CHE	PRINCIPLES OF POLYMER PROCESSING	CHE672	Review of equations of motion, constitutive equations; Calendaring, extrusion;molding; mixing; fibre spinning.
2667	CHE	INTRODUCTION TO POLYMER SCIENCE AND TECHNOLOGY	CHE670	Polymer Fundamentals: Chemistry of Polymer synthesis, Polymer Reaction Kinetics:Stepgrowth polymerization, freeradical chain growth polymerization, EmulsionPolymerization, Ionic and cationic polymerization
2668	CSE	ALGORITHMIC INFORMATION THEORY	CS687	
2669	CSE	BIOMETRIC RECOGNITION	CS686	
2670	MTH	SET THEORIES AND LOGICS	MTH677	
2671	MTH	ECONOMETRICS	MTH676	
2672	CE	ADVANCED MATHEMATICS FOR CIVIL ENGINEERS	CE602	Linear differential equations; Fourier integrals and transforms; Partial differentialequations; Numerical methods in general; Numerical methods for differentialequations; Linear algebra; Numerical methods for ordinary differential equations
2673	AE	UG PROJECT (UGP-I)	AE391A	UG PROJECT (UGPI)

S.No.	Branch	Course Title	Courses ID	Content
2701	PHY	QUANTUM PHYSICS	PSO201A	
2702	MTH	COMPLEX VARIABLES	MSO202A	
2703	ECO	INTRODUCTON TO MACROECONOMIC PROBLEMS AND POLICIES	ECO112A	
2704	CHE	INTRODUCTION TO MOLECULAR SIMULATIONS	CHE622	Theory methods, and application of molecular simulation. Elementary satisticalmechanics. Molecular modeling. Basic Monte Carlo and molecular dynamicstechniques and ensemble averaging. Evaluation of molecular simulation for complex fluids and solids. Commercial simulation software.
2705	CHE	THERMODYNAMICS	CHE621	Postulational Thermodynamics: Postulates; Equilibrium criteria; Gibbs Duhemrelation; Energy minimum principle; Thermodynamic potentials; Stability andphase Transition. Statistical Thermodynamics: Statistical mechanics, equilibriumconstant in reacting systems.
2706	CHE	NEW SEPARATION PROCESSES	CHE618	Separation Factors for Rate Governed Separation Processes, MembraneCharacterization; Reverse Osmosis: Models of Solvent and Solute Transport,Concentration Polarization; Ultrafiltration: Types of Transmembrane Processes; Diafiltration: Process Design; Dialysis: Solute Transport analysisof dialyzer operation, Mode of Dialysis; Electrodialysis: Types of Electro dialysis Ion Transport Fundamentals, Concept of Limiting Current Density.
2707	CHE	TRANSPORT PHENOMENA	CHE611	Kinematics, Transport theorem, constitutive relations, Equations of motion andtheir solutions, Boundary layer theory, Turbulence; Energy equation and its exactsolutions, Continuity equation for multicomponent systems.
2708	CHE	INTRODUCTION TO PROFESSION AND COMMUNICATION SKILLS FOR CHEMICAL ENGINEERS	CHE888	Definition and nature of research, Motivation for research, different types andstyles of research in sciences, role of serendipity, scientific temperament, Isscience necessary? Working of some of the great scientists, Role of science ininnovation, mind mapping; Development of problem solving skills,scaling and orders of magnitude analysis, role of simple models in thinking andin developing an understanding. Scientific and critical reasoning, Scientific assumptions, literature reviews. Professional attitudesand goals, concept of excellence, ethics in science and engineering, some famousfrauds in science. Scientific communication, Organisation of ideas and presentation to semitechnical and or/nontechnical audience/participating in public debates on scientific issues.
2709	CE	ANALYSIS AND DESIGN OF TRANSPORTATION INFRASTRUCTURE	CE682	Introduction to supply and demand sides of transportation engineering, analysisof transportation demand (including topics such as category analysis, gravitymodel, entropy models, choice models, user equilibrium, etc.). Design of transportation infrastructure (includingtopics such as route development, schedule development, pricing strategies,etc.). Concept of structural, functional and drainage design of pavementstructure. Design of flexible and rigid pavement structures. Pavement condition evaluation, maintenance measures, and network level maintenance strategy.
2710	BSBE	EVOLUTION OF BIOLOGICAL MACHINES	BSE441	Living body as an examples of finest designs for diverse activities, functionse.g., flying, swimming, reproduction, sensing, eating, etc. Evolution and naturalselection as the means of optimization of biological systems. Theories of evolution and Darwanian selection. Principles of generatingdiverse body plan and design in nature.
2711	BSBE	BSBE LAB III STRUCTURAL BIOL. & BIOINFORMATICS	BSE392	De novo protein design and artificial proteins: approaches used in designingand constructing novel proteins. Structure based drug design: Virtual screeningtechniques, designing pharmacophore models, Molecular docking, X-ray crystallography;protein purification, current methods in protein crystallization. Evaluating thequality of crystals, crystal freezing at low temperature for data collection. Ademonstration session on Xray diffraction. Protein structure prediction. Exercisesinvolving homology modeling of protein structures, Pairwise and multiplesequence alignments using tools such as BLAST, FASTA, CLUSTAL and from GCGsuite of programs.
2712	BSBE	BSBE LAB-II BIOCHEMISTRY & BIOCHEM. ENGG.	BSE391	Estimation of proteins by UV, Bradford and Lowry methods. SDSPAGE separationof proteins Enzyme Kinetics Salivary amylase different pH buffers andtemperature Chromatography Ion exchange/ HPLC/ GC. Bioreactors andbioprocess engineering use computer simulation to explore mass transferphenomena and cell growth kinetics. Biosensors and their applications. Bioscreeningmethodology to obtain high affinity antibodies.
2713	MTH	ROBUST STATISTICAL METHODS	MTH673	
2714	PHY	GROUP THEORY AND ITS APPLICATIONS TO PARTICLE PHYSICS	PHY690J	
2715	ME	UNDER GRADUATE PROJECT I	ME398A	UG PROJECT (UGPI)
2716	DES	ELECTRONICS FOR DESIGNERS	DES634	Contents upload in PDF file.
2717	CHM	SURFACES INTERFACES THIN FILMS & RELATED ANALYTICAL TECHNIQUES	CHM683	
2718	CHM	ORBITAL INTERACTIONS IN CHEMISTRY	CHM645	
2719	ECO	CONTRACT THEORY	ECO755	
2720	MTH	AN INTRODUCTION TO MATHEMATICAL MODELLING	MTH426	
2721	MTH	DISTRIBUTIONS IN STATISTICS	MTH425	
2722	CSE	SPECIAL TOPICS IN : GAME THEORY AND ITS APPLICATIONS IN COMPUTER SCIENCE	CS698WA	
2723	CSE	SPECIAL TOPICS IN : GAME THEORY AND ITS APPLICATIONS IN COMPUTER SCIENCE	CS698AW	
2724	MTH	NUMERICAL ANALYSIS & SCIENTIFIC COMPUTING	MTH706	
2725	CHM	METHODS OF ELECTRONIC STRUCTURE CALCULATION	CHM627	

S.No.	Branch	Course Title	Courses ID	Content
2726	MTH	WAVELET METHODS FOR ELLIPTIC PARTIAL DEFFERENTIAL EQUATIONS	MTH705	
2727	ME	COMPLIANT MECHANISMS	ME851	Genesis of compliant mechanisms; glimpse of applications of compliant mechanisms; mobilityanalysis; large deformation analysis: continuum and discrete perspectives: finite element analysis,pseudo rigid body models; synthesis of compliant mechanisms: synthesis based on linkage synthesis methods, selectionbased methods, and structural optimizationbased methods; kinetoelastic maps and understanding limits of compliant mechanisms; static balancing of compliant mechanisms
2728	MSE	UNDER GRADUATE PROJECT I	MSE349A	UG PROJECT (UGPI)
2729	BSBE	DEVELOPMENTAL BIOLOGY I	BSE630A	
2730	CSE	COMPUTATIONAL ARITHMETIC-GEOMETRY & APPLICATIONS	CS688	
2731	CSE	INTRODUCTION TO MACHINE LEARNING	CS771A	MACHINE LEARNNING IS THE DISCIPLINE OF DESIGNING ALGORITHMS THAT MACHIENES TO LEARN PATTERNS AND CONEPTS FROM DATA WITHOUT BEING EXPLICITLY PROGRAMMED. THIS COURSE WILL COVER THE BASICS OF MACHINE LEARNING, INCLUDING LINEAR AND NON-LINEAR MODELS, AND FOCUSING ON RECENT ADVANCES,AND EXAMPLE OF REAL WORLD APPLICATION OF MACHINE LEARINING ALGORITHMS.INTRODUCTION TO PRELIMINARIES IN LINEAR ALGEBRA, AND PROBABILITY THEORY (THE TOPICS TO BE COVERED DEPENDING ON INTEREST AND TIME)
2732	CSE	INTRODUCTION TO MACHINE LEARNING	CS771	
2733	CSE	COMPUTATIONAL ARITHMETIC-GEOMETRY & APPLICATIONS	CS683A	
2734	CHM	UNDER GRADUATE PROJECT-I	CHM391A	5TH SEMESTER UNDERGRADUATE PROJECT
2735	BSBE	METHODS IN BIOTECHNOLOGY-I EXPERIMENTAL BIOSEPERATION	BSE492A	
2736	EE	CONTROL SYSTEMS ANALYSIS	EE250A	Negative feedback control systems, Linear timeinvariant dyynamic systems, Mason's gainformula, transfer function, frequency and time domain analysis, performance analysis,Nyquist stability criterion, root locus, circle criterion, antiwindup schemes.Statespace models.
2737	LT	PHD THESIS	LT799	Ph. D. Thesis
2738	PHY	M SC RESEARCH PROJECT-II	PHY697	
2739	PHY	M SC RESEARCH PROJECT -I	PHY697A	
2740	PHY	MATHEMATICAL METHODS II	PHY663	
2741	DES	PH.D THESIS	DES799	Ph. D. Thesis
2742	CHE	UNIT OPERATION LAB II	CHE492	Mass Transfer, Heat transfer, chemical reactors, Computer aided data acquisitionand online analysis, Process dynamics and control.
2743	CSE	COMPUTATIONAL COMPLEXITY	CS640	Complexity Classes. NP and coNP, Results on structure of NPcomplete sets,Sparse NPhard sets, Basic Inclusions and Separations, Nondeterministic SpaceClasses, Logarithmic Space, A PSPACE complete problem, PSPACE vs. NP, PSPACE vs. EXPTIME, ComplexityClasses,SchwartzZippel Lemma and BPP, BPP and its relationship with other ComplexityClasses
2744	CHE	PROCESS DESIGN CAPSTONE PROJECT	CHE453	Introduction to Design Process Development, process alternatives, ProcessFlowsheeting and simulation using ASPEN PLUS, Conceptual Process Synthesis,Conceptual design of reactors, pressure vessel design, process profitabilityanalysis, Heat Exchange Network Analysis, Scaleup & pilot plant studies, HAZOPand Safety in design, Batch process Design for sequential processing usingSUPERPRO, Continuous plant design
2745	BSBE	FUNCTIONAL GENOMICS	BSE634	Introduction; The impact of genomics on biological research; Identification ofa large set of genes involved in a biological process; Highthroughput expressionanalysis; Genomewide search for interacting proteins; Proteomics; Functional genomics in model organisms; Highthroughputphenotypic analysis; Recent developments in genomics.
2746	BSBE	BIOINFORMATICS AND COMPUTATIONAL BIOLOGY	BSE633	Introduction to Molecular Genetics, The Human Genome Project, Biologicaldatabases, Gene prediction, Analysis of genomic sequences, Pair wise andmultiple sequence alignments, Bioinformatics tools, Hidden Markov Models, HMM problem, Homology modeling, Simulation studies of proteins and nucleicacids.
2747	CSE	SPECIAL TOPICS IN COMPUTER SCIENCE	CS397	This course is meant for a 3rd year BTech (CSE) student to study a topicof their interest, somewhat independently. A student may also carry out aproject in this course. In this course, there will be a faculty member who will monitor the progress of the student, examine thestudent, and give a grade at the end of the semester.
2748	BSBE	MODERN INSTRUMENTAL METHODS IN BIOLOGICAL SCIENCES	BSE611	1. Applications of spectroscopic and other techniques to the study of biomolecules: UVVis spectroscopy, Circular dichroism, Fluorescence, NMR, Mass, IR and Raman spectroscopy, XRay diffraction. 2. Analytical techniques (e.g. electrophoretic analysis), detection (staining, blotting and immunedetection) specialized applications (in vitro synthesis of protein, labeling and Mass spectrometry), ultracentrifugation. 3. Techniques with applications in biology: Autoradiography, Biological Uses of Radioisotopes, Tracer Dilution Technique, Radioimmunoassay4. Cellular Imaging Techniques: Microscopy: Phase contrast, Fluorescence, Atomic Force and Confocal Microscopy
2749	BSBE	BSBE LAB IV BIOMECHANICS & BIOMATERIALS	BSE491	Biomechanics: Human body motion analysis. Control of artificial arms/legs usingbiological principles. Rehabilitation applications of biomechanics. Biosensors andMEMS: Development of sensors for medical applications. Nanotechnology: Nanolithography.Testing of synthetic materialsBiomaterials: Biodegradable polymers synthesis, fabrication test theircharacteristic, computational simulation of biomaterial for their mechanicalstrength, tissue engineering and human tissue and tissuebiomaterial interactions.
2750	PHY	QUANTUM DYNAMICS INFORMATION AND COMPUTATION	PHY690K	
2751	PHY	QUANTUM OPTICS II	PHY690L	
2752	PHY	SPECIAL & GENERAL RELATIVITY	PHY659	
2753	MSE	FUNDAMENTALS OF SPRAY TECHNIQUES	MSE634	Introduction Different Spray Techniques and their need Combustion Spraying:0 Flame Spraying0 DGunThermal Spraying Techniques: 0 High Velocity oxyfuelPrinciple and Working Arc and Plasma Spraying0 Laser Spraying0 Plasma Primary/ Secondary/ Carrier gases Power rating Feed rate Standoff distance Substrate preparationPowder Powder size and distribution Powder Injection Reaction of particlesCoating Formation

S.No.	Branch	Course Title	Courses ID	Content
				splat Splay layering and deposition Microstructure and densification of deposited coatingsDiagnostics and Coating Reliability Thermal and Kinetic Profiles Inflight particle sensor Control of Deposition parameters Mandrel choice Mandrel removal Lectures Case Studies: Thermal Barrier/ Ultra high temperatureceramics Reference Books:1. Handbook of thermal spray technology, Joseph R. Davis, ASM International. Thermal Spray SocietyTraining Committee (2004).2. Advanced Structural Ceramics, Bikramjit Basu and Karim
2754	MTH	FREDHOLM THEORY	MTH715	
2755	CSE	RIEMANN HYPOTHESIS AND ITS APPLICATIONS	CS746	
2756	AE	MECHANICS OFHIGHLY DEFORMABLE STRUCTURES	AE698B	
2757	PHY	NONLINEAR SYSTEM & DYNAMICS	PHY695	
2758	MTH	INTRODUCTION TO FOURIER SERIES	MTH205	
2759	MTH	SPATIO-TEMPORAL MODELS IN MATHEMATICAL BIOLOGY	MTH679	
2760	MTH	INTRODUCTION TO FOURIER SERIES	MTH205A	
2761	MTH	FRACTAL INTERPOLATION AND APPLICATIONS	MTH622A	
2762	MTH	SPATIO-TEMPORAL MODELS IN MATHEMATICAL BIOLOGY	MTH679A	
2763	MTH	ALGEBRIC TOPOLOGY II	MTH759A	
2764	EE	ELECTROMAGNETIC INTERFERENCE AND COMPATIBILITY TECHNIQUES	EE644A	
2765	ME	ROBOT MANIPULATORS: DYNAMICS AND CONTROL	ME763A	Review of robot manipulators. Manipulator kinematics, dynamics and control.Singularity and workspace analysis. Introduction to manipulator design.
2766	CSE	MOBILE COMPUTING	CS634	Introduction: Mobile computing a vision for future, ubiquitous computing versusvirtual reality, software models for mobile computing. Data managementIssues. Distributed algorithms and mobility: structure of mobile computing in the air: pull and push based data transfers, data disseminationby broadcast, treating air as cache, energy efficient indexing in air. Handoffmanagement: handoff detection, failures, channel assignments. Location management. Approximatequery processing: concept hierarchy, summary database, updates and viewmaintenance, approximate query processing. Mobile Transaction Models. MobileComputing Models. Mobile transaction models. Mobile transaction models. Mobile voice and data serviceson 3G networks, battery problem and power dissipation, low energy processors.File system support for mobile computing: Coda and Bayou file systems.Adhoc network routing protocols.
2767	CHE	BIOCHEMICAL ENGINEERING	CHE362	Cell Structure and Cell Types, Chemicals of Life (RNA, DNA, enzymes etc.),Kinetics of Enzyme Reactions, Applied Enzyme Catalysis, Metabolic Stoichiometricand Energetics, Molecular Genetics and Control of Gene Expression, Fermentors,Downstream Product Recovery and Purification, Interaction of Mixed MicrobialPopulations, Biological Wastewater Treatment.
2768	CHE	CHEMICAL PROCESS INDUSTRIES	CHE361	Role of Chemical Engineer, Elementary Process Flowsheeting P & I diagrams,Inorganic IndustryFertilizers, Chlor alkalies, Natural Products Pulp & Paper, Oils,Soaps, Herbs, Petroleum & Petrochemicals: Refining, Polymers, Intermediates, dyes & paints, Pharmaceuticals, Environmental Pollution& Waste Treatment.
2769	CHE	SEPARATION PROCESSES	CHE313	Mass transfer equipment: continuous contact and staged contact units forabsorption, extraction, distillation, adsorption, humidification, drying, Phaseequilibria: phase diagrams, VLE, GLE, SLE; estimation of thermodynamic properties: unsteady state: (distillation, drying, adsorption[blow down]), Separations without reflux: isothermal cases: stage contactor:McCabe Thiele method, stage efficiency, matrix method, Thomas method forsolubility, humidification): McCabeThiele method, numerical solutions, Separations with reflux: distillation, extractionand adsorption, McCabeThiele and matrix method; continuous contact, Membraneseparation processes: continuous contact,Estimation of stage efficiencies, Shortcut methods for distillation. Aspen shouldbe used as much as possible
2770	CHE	HEAT AND MASS TRANSFER	CHE312	Heat conduction, Molecular diffusion, Convective heat transfer (laminar &turbulent), convective mass transfer (laminar & Turbulent), simultaneous heatand mass transfer, wetbulb and adiabatic saturation temperature, heat and mass condensation, correlations, Radiation: thermal radiation, radiationproperties, view factors, heat exchange between surfaces, Heat exchangerdesign: Shellandtube, compact exchangers, reboiler, and condenser design, growth, types of equipment, design, Aspen/Matlab should beused wherever possible.
2771	CHE	CHEMICAL ENGINEERING THERMODYNAMICS	CHE221	Review of I and II Laws of Thermodynamics, PVT Relations of Pure FluidsGraphical,Tabular and Mathematical representaion; Generalized compressibilitychart; Generalized EOS; Thermodynamic Potentials; Mixtures, mixtures, Properties of solutions, Phase Equilibrium (VLE, LLE,VLLE), Review of Thermochemistry; Chemical reaction equilibria.
2772	CSE	PRINCIPLES OF DATA BASE SYSTEMS	CS315	Overview of file organisation techniques: sequential, direct, indexed, hashed,inverted, Btrees, Data models: relational, network, hierarchical, Relationalmodel: algebra, calculus, normal forms. Implementation of relational model: introduction to distributed data base systems, case studies. 1. H GarciaMolina, JD Ullman and Widom, Database Systems: The Complete Book,2nd Ed., PrenticeHall, 2008.2. A Silberschatz, H Korth and S Sudarshan, Database System Concepts, 6th Ed., McGrawHill, 2006.3. J Gehrke, Database Management Systems, 3rd Ed., McGrawHill, 2002.
2773	AE	INTRODUCTION TO THE THEORY OF SMART STRUCTURES	AE689	Introduction to smart materials, piezo, pyro and ferro electric effects; hysteresiseffects:electric field in solids: fundamentals of continuum mechanics; basicconservation laws; thermodynamic principles; constitutive equations; smart structures; applications to aerospace vehicles.
2774	CE	TRANSPORTATION ENGINEERING	CE382A	<p>Course Contents: Introduction, transportation engineering elements, geometric design, traffic flow fundamentals, uninterrupted traffic flow, pavement analysis, highway maintenance, traffic engineering, "Transportation Engineering",Prentice Hall of India, 2003</p> <p>*Garder, N.J. and Hoel, L.A., "Traffic & Highway Engineering", 3rd Ed.,Brooks/Cole, Pacific Grove, 2001</p> <p>* McShane W.R, Roess R.P.,
2775	AE	AEROMODEL DESIGN & FABRICATION	AE361A	Design and fabrication of aero models/components; Balsa, Styrofoam, wood,parchment, composites based model making; model upgradation; design,fabrication and testing of components; use of flight simulation software.*Introduction to Flight: J.D. Anderson, Jr., McGrawHill International Editions.*Handouts.*Internet resources.
2776	CSE	ALGORITHMIC INFORMATION THEORY	CS687A	
2777	BSBE	BIOELECTRICITY	BSE615A	

S.No.	Branch	Course Title	Courses ID	Content
		SECURITY		access control: models for access control, some confidentiality,integrity, and hybrid models of access control such as BellLa Padula, Biba,Chinese Wall etc., discretionary v/s mandatory access control.Ca which have security implicationssuch as buffer overflows, race conditions etc. Malicious code: Viruses, worms,Trojan horses; how they work and how to defend against them.Network Security: problems i VPN,SSL, S/MIME etc. Protocol vulnerabilities: examples of protocol vulnerabilitiessuch as in TCP/IP, denial of service attacks etc. Tools for network securitysuch as firewalls and intrusion detection syste
2803	CHE	UNIT OPERATION LABORATORY-I	CHE391	Fluid flow, Fluid particle systems, Thermodynamics, Heat transfer, Mass transfer.
2804	CHE	PROCESS DYNAMICS AND CONTROL	CHE381	Introduction, Process models, Linearization, Laplace transforms. Process dynamics,Time delay, Feedback control, Instrumentation, Stability (Routh array & rootlocus), Frequency response analysis (Bode a adaptive,inferential, ratio, override etc.), MIMO systems, Digital control.
2805	CSE	ADVANCED COMPUTER NETWORKS	CS625	Introduction: Overview of computer networks, sevenlayer architecture, TCP/IP suite of protocols, etc. MAC protocols for highspped LANs, MANs, andwireless LANs. (for example, FDDI, DQDB, HIPPI, Gigab ATM layer. ATM Adaptation Layers. Congestion control.Signalling, Routing, QoS support, Neighbourdisconery, Autoconfiguration. Changesto other protocols. Application Programming Interface for IPv6. M session discovery, etc. TCP extensions forhighspeed networks, transactionoriented applications. Other new optionsin TCP. Network security at various layers. SecureHTTP, SSL, ESP, Authenticationheader,
2806	CSE	ADVANCED COMPUTER ARCHITECTURE	CS622	Singlethreaded execution, traditional microprocessors, DLP, ILP, TLP, memorywall, Parallel programming and performance issues, Shared memorymultiprocessors, Synchronization, smallscale symmetric coherence, Interconnection network, Memory consistencymodels, Software distributed shared memory, multithreading in hardware, Chipmultiprocessing, Current research and future trends.
2807	MTH	UG PROJECT (UGP-III)	MTH393A	UG PROJECT (UGPIII)
2808	EE	ELECTRICAL ENGINEERING COMMUNICATION SKILLS	EE401A	
2809	MTH	INTRODUCTION TO HOMOGENIZATION	MTH719A	
2810	PHY	INTRODUCTION TO SPINTRONICS	PHY693	
2811	CSE	INTRODUCTION TO COGNITIVE SEIENCE	SE367A	
2812	HSS	COMMUNICATION SKILLS: COMPOSITION	COM200A	
2813	ECO	GAME THEORY	ECO754	
2814	CHE	ANALYTICAL & MATERIAL CHARACTERIZATION TECHNIQUES OF ENGINEERS	CHE661	1. Introduction: Various terms in measurement. Accuracy, precision, selectivity, limit of detection, sensitivity, response time, reliability, signal/noise and sources of noise/errors (2 lectures).2. Ch materials. Xray photoelectron spectroscopy (XPS), mass spectrometry (MS) and other spectroscopy techniques at atomic and molecular levels (12lectures).3. Microscopy and diffraction: Optical mic practical aspects of the microscopy and diffraction techniques will be introduced. Imaging using scanning tunneling microscope (STM) will also be considered (12 lectures).4. Thermal analysis: Applica 21-SEP-2014
2815	BSBE	BIOPHAMACEUTICALS	BSE680A	
2816	BSBE	UNDER GRADUATE PROJECT IV	BSE499A	UP PROJECT (UGP II)
2817	EE	UNDER GRADUATE PROJECT II	EE392A	UG PROJECT (UGP II)
2818	CHE	PETROLEUM REFINERY ENGINEERING	CHE662A	Petroleum refining in India; refinery tests and crude oil evaluation ; crudedistillation column design; Delayed Coking; Catalytic cracking; Catalytic reforming;Catalytic isomerization; Alkylation; Polymerization
2819	MTH	UNDER GRADUATE PROJECT II	MTH392A	UG PROJECT (UGPII)
2820	PHY	NONLINEAR SYSTEM & DYNAMICS	PHY695A	
2821	CSE	RIEMANN HYPOTHESIS AND ITS APPLICATIONS	CS746A	
2822	CSE	QUANTUM COMPUTING	CS682A	THIS COURSE WILL FOCUSE ON THE COMPUTER SCIENCE ASPECTS OF QUANTUM COMPUTING.FOUNDATION: HILBERT SPACE(FINIT DIMENSIONAL).AXIOMS OF QUANTAM PROBABILITY. QUANTAM CIRCUITS,UNIVERSALITY. SIMON'S PROBLEM,PHASE FINDING,SHOR'S ALGORITHM,GROVERS ALGORITHM,PROBABIIY AMPLIFICATION.SOME APPLICATIONS.QUANTAM INFORMATION PR INFORMATION THEORY.ADDITIONAL TOPICS: BASEED ON THE INTERESTS OF THE AUDIENCE.
2823	CE	PROJECT MANAGEMENT	CE641A	
2824	CE	FOUNDATION ANALYSIS AND DESIGN	CE632A	Settlement and bearing capacity; shallow spread footings, mats, and deep foundations.Foundation models, Contact pressure distribution for footings, Rafts, Piles, RetainingStructures; Soilstructure interme Bowles, J. E.: Foundation Analysis and Design (5nd Edition),The McGraw Hill Companies Inc. (2001)
2825	CE	UNDER GRADUATE RESEARCH -III	CE492A	UNDER GRADUATE RESEARCH II
2826	BSBE	BIOCHEMICAL ENGINEERING	BSE341	Fundamentals of Biochemical engineering: Stoichiometry and energetics ofmicrobial metabolism, Transport phenomena, Enzyme catalyzed reactions andprocesses, Bioreactor design and applications, St biological materials, pretreatment methods; separation of cell biomass,adsorption, filtration, reverse osmosis, isoelectroc focusing, affinity basedseparation, polishingcrystallization, drying, case studies; monitoring and process economics; Environmental bioprocesses Interaction of mixed microbial population, applications, biological wastewatertreatment, anaerobic, digesters, bioremediation
2827	BSBE	BIOCHEMISTRY	BSE216	Properties of water, Acids, Bases and Buffers. First and Second law ofthermodynamics; Free energy as an indicator of spontaneity; Chemical Equilibria.Amino Acids of Proteins; Optical Activity. Primary St Proteins; Quaternary Structure. Polysaccharides; Glycoproteins. Lipidsand Membranes; Mechanisms of enzyme action; Substrate Specificity; Coenzymes;Regulation of Enzymatic Activity: enzyme kinetic
2828	BSBE	MOLECULAR BIOLOGY	BSE212	Genome as the store house of information, DNA as the carrier of encodedmessages. Genomic diversity across organisms as a source of embeddedintelligence in nature. Intracellular transmission of gene messages regulation of gene expression; logic of building body plansdevelopmental translation of encoded messages. Techniques of DNA manipulationand engineering.
2829	MSE	UNDER GRADUATE PROJECT IV	MSE449A	UG PROJECT (UGPIV)

S.No.	Branch	Course Title	Courses ID	Content
2830	MSE	UNDER GRADUATE PROJECT II	MSE398A	UP PROJECT (UGPII)
2831	SDS	PROBABILITY THEORY	MTH309	
2832	MTH	AN INTRODUCTION TO MATHEMATICAL MODELLING	MTH426A	
2833	CHM	UG PROJECT II	CHM392A	UG PROJECT II
2834	CSE	MICROPROCESSOR DESIGN METHODOLOGY	CS620	
2835	CHM	CHEMISTRY AND OUR ENVIRONMENT	SE344	
2836	CHM	CHEMISTRY AND OUR ENVIRONMENT	CHM482	
2837	HSS	MODERN LOGIC	PHI753	Propositional Logic; Decision Procedures; Quantification Theory; AxiomaticMethod; Philosophical Problems.
2838	CE	LABORATORY COURSE IN TRANSPORTATION ENGINEERING	CE690	Experiments to characterize pavement materials like, viscosity tests, ageing tests, skid tests, etc. Experiments to characterize bituminous mixes, like mixdesign related experiments, moisture sensitivity related experiments, etc. Design of a road studio (students will learn to use geometric design softwareand video data analysis software). Demonstrations of various equipmentsincluding possible visits to advanced labs and road systems.
2839	PHY	M SC REVIEW PROJECT II	PHY501	M.Sc. Review Project II
2840	PHY	SUMMER PROJECT	PHY500	M.Sc. Review Project I
2841	CE	GEOTECHNICAL MEASUREMENTS AND EXPLORATION	CE638	Subsurface exploration planning, drilling and sampling techniques, field andlaboratory tests, instrumentation and monitoring of field data, report preparation.
2842	CE	EXPERIMENTAL METHODS IN STRUCTURL ENGINEERING	CE623	Similitude and Structural Models: Dimensional analysis, Buckingham’s Pi theorem, Scale factors and dynamicsimilitudeUses and Applications of Models: Types of model investigation, Indirect and Direct methods, Error and uncertainty in experiment, Measurement systems, Accuracy in models and reliability of resultsTest Planning, Design and Implementation: Testing sequence and experimental plan, Loading system calibration, various types of sensors for displacement, velocity, acceleration, pressure, loads, strains, etc, fullfield measurements, .Data Acquisition System and Data Processing: Analog systems, digital systems, Exercises: Experiments to illustrate buckling of structural members; loaddeflection behaviour of beams, columns, joints, and frames under various loads; mode shapes, natural frequency, damping factor, etc. (Dally, J. W. and Riley, W. F. (1978). Experimental Stress Analysis, McGraw HillHarris H. G. and Sabnis, G. M. (1999). Structural Modeling and Experimental Techniques, CRC Press 1999Nachtigal, C.L. (1999). Structural Testing, Prentice Hall I Fairmont PressHolman, J. P. (2001). Experimental Methods for Engineers, McGraw Hill
2843	NET	SPECIAL TOPICS	NT698	
2844	PHY	PHYSICS II	PHY103	
2845	CSE	SUBLINEAR ALGORITHMS FOR PROCESSING MASSIVE DATA SETS	CS718	
2846	CSE	SMALL SPACE BOUNDED COMPUTATIONS	CS745	
2847	CSE	MULTIAGENT SYSTEMS : GAMES ALGORITHMS EVOLUTION	CS785	
2848	BSBE	BIOPHARMACEUTICALS	BSE680	
2849	MTH	ALGEBRIC TOPOLOGY II	MTH759	
2850	MTH	NUMERICAL SOLUTION OF INTEGRAL EQUATIONS	MTH720	
2851	ECO	EMPIRICAL METHODS IN APPLIED MICROECONOMICS	ECO444	
2852	NET	SAFETY ANALYSIS OF NUCLEAR POWER REACTORS	NT661	
2853	HSS	INTRODUCTORY SOCIOLOGY	SOC171AA	
2854	HSS	DISASTER MENTAL HEALTH	PSY794	
2855	ECO	HISTORY OF ECONOMIC DEVELOPMENT	ECO540	
2856	MTH	MATHEMATICS I	MTH101A	Real numbers, Sequences; Series; Power series, Limit, Continuity; Differentiability,Mean value theorems and applications; Linear Approximation, Newton and Picard method, Taylor’s theorem (one variable), integral calculus, Improper integrals, Trapezoidal and Simpson’s rule; error bounds, Space coordinates, lines and planes, Polar coordinates, Graphs of polar equations; Cylinders, Quadric surfaces, Volume and Surface Integrals; Functions of two or more variables, partial derivatives Statement only, of Taylor’s theorem and criteria for maxima/Minima/saddle points, Double, triple integrals, Jacobians; Surfaces, integrals, Vector Calculus
2857	CSE	INDEXING AND SEARCHING TECHNIQUES IN DATABASE	CS618	Index structures Rtree, Mtree, Vfile, etc., Spacepartitioning versus datapartitioningmethods; Similarity queries Range search, kNN search, Selfjoin;Retrieval techniques Fagin’s Algorithm, Thresholding, etc.; FastMap, Wavelets, Fourier transform, etc.; Distance measures Lpnorm, Mahalanobis distance, KullbackLeibler divergence measure, Earth Mover’sDistance, etc.; Data compression Wavelets, Fourier transform, etc.
2858	CE	DURABILITY OF CONCRETE	CE628	Concrete and the environment: interaction; Overview of concrete deterioration:alkaliaggregate reaction, corrosion, carbonation; Permeability of concrete andits measurement: penetration of carbon dioxide

S.No.	Branch	Course Title	Courses ID	Content
		STRUCTURES		corrosion cells and currents, role of concrete, prevention ofcorrosion; Corrosion induced longitudinal cracks: nature and properties ofcorrosion products; Alkali aggregate reaction: reactive minerals, mech rehabilitation of structures.
2859	ME	ENGINE DESIGN	ME698F	
2860	CE	AIRPORT SYSTEMS PLANNING AND DESIGN	CE688	Air Transportstructure and organization, the challenges and the issues, Forecastingair travel demandtrend forecasts and analytical methods; Air freight demand,Characteristics of the aircraft as they affect the airport, andsigning; Airport capacity and configuration; Geometric design of runway, taxiwayand aprons; passenger terminal functions, passenger and baggage flow, designconcepts, analysis of flow through terminal buildings; Airport systems;Airport drainage and pavement design; Airport access problem; Environmentalimpact of airports.
2861	CE	ROCK MECHANICS	CE630	Physical properties and classification of intact rock and rock masses, rockexploration, engineering properties of rock, stresses in rock near undergroundopenings; Rock tunneling, rock slope stability, bolted rock masses.
2862	PHY	BACHELOR OF SCIENCE PROJECT -III	PHY557A	
2863	BSBE	BIOMATERIALS	BSE640A	
2864	EE	ARCHITECTURE OF MICRO-PROCESSORS & MICROCOMPUTERS	EE674A	
2865	CE	CONSTITUTIVE MODELING OF FRICTIONAL MATERIALS	CE637	Role of constitutive modeling; Importance of laboratory testing with relationto constitutive modeling; Elasticity: linear, quasilinear, anisotropic; Plasticitybasics: yield criteria, flow rule, plastic potential, hardening; Critical state soil mechanics: critical state concept,camclay models, simulation of single element test using camclay, consolidation,drained and undrained triaxial test; Stressdilatancy theory; Work hardening; hypoeasticityplasticity, disturbed state concept.
2866	CE	ADVANCED DESIGN OF STEEL STRUCTURES	CE627	Properties of steel: mechanical properties, hysteresis, ductility; HotRolledSections: compactness and noncompactness, slenderness, residual stresses;Design of steel structures: inelastic bending curvature, compression flange & web, lateraltorsional buckling, stability ofcolumns slenderness ratio of columns, local buckling of flanges and web,bracing of column about weak axis, method of design allowable stress design, moment magnification factor, effective length,PM interaction, biaxial bending, joint panel zones; Drift criteria: P effect,deformationbased design; Connections: types welded, bolted, location beamcolumn, moment-resisting joints.
2867	BSBE	METHODS IN BIO-TECHNOLOGY-1	BSE462	
2868	BSBE	TRENDS IN BIOLOGY AND BIOENGINEERING RESEARCH	BSE460	
2869	PHY	PH D THESIS	PHY799A	Ph. D. Thesis
2870	CHE	PROJECT-I	CHE495	Student will work on design/research projects to carry out theoretical/experimentalwork on problems in chemical engineering as assigned by the project advisor.
2871	PHY	QUANTUM DYNAMICS: COMPUTATION & INFORMATION	PHY690F	
2872	PHY	THERMAL PHYSICS AT NANO-SCALE	PHY690Y	
2873	PHY	HIGH PERFORMANCE COMPUTING & MACHINE LEARNING	PHY690W	
2874	AE	FUNDAMENTALS OF LIQUID ATOMIZATION	AE751	Introduction to atomization, Physical processes in atomization, Types of atomizers,Classical Theories of atomization, Numerical modeling of atomization process,Theory of multiphase flows, Atomizer design, Applicationsof Atomizers Metal forming Chemical Industry Combustion.
2875	NET	PHD THESIS	NT799	Ph. D. Thesis
2876	EE	MIXED -SIGNAL IC DESIGN	EE698I	
2877	PHY	PHYSICS AT LARGE HADRON COLLIDER	PHY690B	
2878	AE	AIRCRAFT PROPULSION	AE652	Gas turbine engines, performance analysis, subsonic and supersonic diffusers,centrifugal and axial compressors, stage dynamics, compressor stall, axialturbines, compressorturbine matching, gas turbine engine design.
2879	CHM	PROJECT	CHM700	PROJECT
2880	EE	CIRCUIT DESIGN FOR FREQUENCY & PHASE SYNTHESIS	EE698G	
2881	EE	OPERATION AND CONTROL OF THE POWER SYSTEM WITH INVERTER BASED RESOURCES	EE698H	
2882	CSE	TOPICS IN HOMOTOPY TYPE THEORY	CS698H	
2883	CHM	MATHEMATICS FOR CHEMISTRY	CHM600	Error Analysis, Scalars, vectors, curl, divergence and gradient, ordinary3000[4] differential equations, symmetry and group theory, matrices, etc.
2884	CSE	Introduction to Constructive Types Theory and its Applications	CS698J	
2885	AE	THERMAL TURBOMACHINERY	AE653	Axial compressors, stage dynamics, degree of reaction, pressure rise limitations,secondary flows, performance: design and offdesign, starting problems, centrifugalcompressors; inlet flow, slip, sweep, diffuser, compressor design.

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2886	CE	GROUNDWATER HYDROLOGY & POLLUTANT TRANSPORT	CE717	Groundwater as a resource, general problems of chemical contamination ingroundwater; Fluid potential, hetrogenity and anisotropy; Aquifiers, aquitradsand general geology, well hydraulics, parameter estimation and transport processes, numerical modeling and solution, breakthrough curves; Seawater intrusion in coastal aquifiers; Modelling of pollutanttransport in the unsaturated zone; Optimization models for resource management; Conjective management of surface and groundwater;Special topics.
2887	CE	STOCHASTIC HYDROLOGY	CE614	The hydrologic processes: precipitation, evaporation, infiltration, groundwater,and stream flow; Hydrologic measurements and networks; Analysis of discreteand continuous hydrologic data: harmonic analysis, spectral analysis, time series analysis and its applications; System analysis and synthesis; Linearand non linear, lumped and distributed parameter systems; Queing models,simulation analysis; Hydrologic design of water resource projects.
2888	CHM	ADVANCED INORGANIC CHEMISTRY I	CHM641	An advanced course on the physical principles of inorganic chemistry withillustrations from the chemistry of transition and nontransition elements.
2889	DES	METHODS FOR DESIGN RESEARCH	DES635	
2890	CHM	MOLECULAR REACTION DYNAMICS	CHM679	1. Introduction (5)Review of kinetic theory of gases, collisions atomic and molecular.2. Rate theories (6)Transition state theory and RRKM theory, scattering classical and quantum.3. Reactive Collisions (6)Rate constants, cross rates and cross sections.4. Dynamics in gas phase (8)Photodissociation, energy transfer, stereodynamics, chemistry in real time with lasers,control.5. Dynamics in condensed phase (8)Solvation, diffusion and reaction on surfaces, spatiotemporal aspects of pattern formation, ultrafast dynamics. 1. Levine, Molecular Reaction Dynamics, 2005.2. Henriksen & Hansen, Theories of Molecular Reaction Dynamics, 2008.3. Schinke, Photodissociation Dynamics, 1993.4. Manz & Wste, Femtosecond Chemistry, 1994.
2891	CHE	SPECIAL TOPICS	CHE679	Course contents vary from time to time
2892	EE	MIXED -SIGNAL IC DESIGN	EE607	Basics of functional Analysis; Basics of Fourier Analysis; Spectral Theory; TimeFrequency representations; Nonstationary Processes; Continuous WaveletTransforms; Discrete TimeFrequency Transforms
2893	BSBE	BIOPHARMACEUTICALS	BSE616	
2894	IME	SUPPLY CHAIN MANAGEMENT	VLM602.	
2895	IME	PRODUCT DESIGN AND NEW PRODUCT MANAGEMENT	VLM601.	
2896	CHM	MOLECULAR SPECTROSCOPY	CHM637	1. Group theory (6)Review of point groups, permutation inversion and molecular symmetry groups.2. Interaction of radiation and matter (5)Qualitative aspects, Einstein A, B coefficients, absorptionemission coefficients.3. Vibrational Spectroscopy (8)Normal modes, selection rules, Fermi and Coriolis perturbations, polyatomic molecules.5. Electronic Spectroscopy (6)Introduction, symmetry aspects, vibronic effects.6. Advanced Topics (6)WilsonHowardWatson Hamiltonian, timedependent viewpoint. 1. Bernath, Spectra of Atoms and Molecules, 1995.2. Bunker & Jensen, Molecular Symmetry & Spectroscopy, 1998.3. Papousek & Aliev, Molecular VibrationalRotational Spectra, 1982.4. Hollas, Modern Spectroscopy, 1996.
2897	CSE	RELATIONAL STRUCTURES IN GAMES	CS698I	
2898	EE	POWER CONVERTERS FOR EV CHARGING	EE698E	
2899	PHY	QUANTUM MECHANICS II	PHY432	Bound State Perturbation Theory, TimeDependent PerturbationTheory, Semiclassical Treatment of Radiation, Scattering Theory, Relativistic Wave Equations,Foundational Issues in Quantum Mechanics, Quantum Field Theory (1) Powell and Crasemaun(2) Feymnan Lectures vol. III(3) Merzbacher(4) Gasiorowicz(5) Schiff(6) Sakurai(7) LandauLifshitz(8) CohenTaunoudji(9) Griffiths
2900	MTH	PROJECT- II	MTH599	PROJECT II
2901	EE	INTRODUCTION TO QUANTUM OPTICS	EE698Y	
2902	SDS	PROJECT-I	MTH598	PROJECT I
2903	EE	ARCHITECTURE AND HARDWARE FOR QUANTUM COMPUTING	EE698T	
2904	EE	HIGH VOLTAGE POWER TRANSMISSION ENGINEEERING	EE698S	
2905	EE	ADVANCED TOPICS IN MACHINE LEARNING	EE698R	
2906	PHY	ASTROPHYSICS	PHY690R	
2907	PHY	COOPERATIVE PHENOMENA IN CLASSICAL & QUANTUM SYSTEMS	PHY690Q	
2908	EE	GRAPH THEORY AND APPLICATIONS	EE698V	
2909	EE	INFORMATION THEORY: CLASSICAL TO MODERN APPLICATIONS	EE698U	
2910	HSS	ENGLISH LANGUAGE AND COMPOSITION	ENG121B	
2911	DES	FUNDAMENTALS OF GRAPHICS	DES630	
2912	CSE	CONCURRENT DATA STRUCTURES AND ALGORITHMS	CS651	

S.No.	Branch	Course Title	Courses ID	Content
2913	HSS	ENGLISH LANGUAGE AND COMPOSITION	ENG121A	
2914	CSE	SELECTED AREAS OF MECHANISM DESIGN	CS698A	
2915	AE	COMPOSITE MATERIALS	AE681	Introduction, Definition, classification, behaviors of unidirectional composites:prediction of strength, stiffness, factors influencing strength & stiffness, failuremodes, analysis of lamina; constitutive classic
2916	AE	THEORY OF VIBRATIONS	AE678	Introd. Hamiltons principles, Lagranges eqn, Eigenvalue problem (EVP), discrete& continuous system. Boundary value problem formulation. General EVP, positivedefinite system, self adjoint property. Vibra
2917	AE	AEROELASTICITY	AE676	Influence coefficients and function. Formulation of static and dynamic aeroelasticequations. Static aeroelasticity; divergence, aileron reversal & controleffectiveness, solutions by matrix and energy metho
2918	BSBE	BIOMATERIALS	BSE613	Introduction to Materials Science: Bulk and surface properties of materials; Polymeric materials; synthesis, characterization, and fabrication methods Inert, biodegradable, hydrogels, Natural, Genetically e Biomaterials: Protein structure, interaction of proteins with synthetic material; characterization of cellmaterial interactions; inflammatory responses; acute inflammation, chronic inflammation, foreign bod
2919	EE	PROJECT II	EE492	PROJECT II
2920	EE	PROJECT I	EE491	PROJECT I
2921	AE	INTRO TO VIRTUAL INSTRUMENTATION	AE698	Introduction to VI, typical applications, functional systems, graphical programming,data flow techniques, advantages of VI techniques. VI programming techniques;VIs and subVIs, loops and charts, arrays interfacenodes and DLL links. Sensors, transducers and signal conditioning; commontransducers for displacement, temperature, load, pressure, flow etc. Singleended, floating and differential inputs, grou
2922	AE	AEROSPACE STRUC ANALYSIS II	AE687	General loads on aircraft, load factor, Vn diagram, effect of gust loading. Energyprinciples, potential and complementary potential; deflection analysis,indeterminate structures. Analysis of plates, Kirchhof
2923	CHE	PHD THESIS	CHE799	Ph. D. Thesis
2924	EE	ELECTRICAL ENGINEERING LAB II	EE381	Experiments from various areas of Electrical Engineering with emphasis on digitalelectronics, communication, machines, drives and power systems, and electromagnetics.This course has three labs: Elec
2925	MME	INDUSTRIAL TOUR	MME390	Visit to industries in and around Kanpur or elsewhere primarily of interest toMaterials and Metallurgical Engineering.
2926	AE	COMPUTATIONAL FLUID DYNAMICS	AE622	Eqns of fluid dynamics & its classifications. Boundary conditions. Stabilityanalysis & concept of feedback. Various explicit & implicit schemes. Gridgeneration. Solving parabolic, elliptic PDEs by explicit, in
2927	CSE	INTRODUCTION TO PROFESSION	CS100	
2928	AE	ADVANCED COMPUTATIONAL METHODS IN CFD	AE615	Main issues of spacetime resolution: Computing timeaveraged and unsteadyproblems. Discretization with operators. Problems in physical and transformedplane: Jacobians and flux vector splitting. Gene
2929	AE	VISCOUS FLOWS	AE614	Basic concepts of BL theory; similar flows: generalized techniques of solvingBL eqns. for incompressible fluids : thermal BL. BL control. Intro. to turbulentshear flows.
2930	IME	KNOWLEDGE STRATEGIES & MANAGEMENT	MBA644.	
2931	EE	SIMULATION STUDY OF POWER SYSTEM WITH MICROGRID	EE698Z	
2932	AE	ADVANCES IN WIND ENERGY CONVERSION	AE629	Earths atmosphere & rotation, Coriolis force, geostrophic winds. Terrain, mettheories, measuring technique. Wind power site. Atmosph BL. Aerodyn in windpower. Sails, airscrews. Vertical & horizontal axis
2933	AE	CONTINUUM HYPERSONIC AERODYNAMICS	AE628	Continuum hypersonics in entry flight. Gen features, Mach no. Small disturbtheory, similitude. Large diflecn similitude; wedge, cone, wing. Unified similitude.Lighthill & other piston analogies, NewtonBuse
2934	AE	AERODYNAMICS II	AE612	Thin aerofoil theory, finite wing theory, basic thermodynamics, one and twodimensionalflows, isentropic flows, waves (shock, expansion, characteristicsetc.), potential flows, perturbation equation, subson
2935	AE	AERODYNAMICS-I	AE610	Basic fluid mechanics Navier stokes equation, vorticity kinematics, Basicpotential flows viscous flows including boundary layer theory, turbulence(introduction)
2936	COM	COMMUNICATION SKILLS: COMPOSITION	COM200	
2937	BSBE	UNDER GRADUATE PROJECT IV	BSE499	PROJECT II
2938	BSBE	BIOMATERIALS	BSE352	
2939	BSBE	BSBE LAB-I MOLECUTAR BIOLOGY	BSE292	Use of Transgenic organisms in the study of gene expression methodology toobtain high affinity antibodies.
2940	BSBE	INTRODUCTION TO BIOLOGICAL SCIENCES & BIOENGG.	BSE100	Biology in the 21st Century The brave new world in the post genome era. Past,present and future of our society, industry and life style: impact of discoveriesand technological innovations in biology. Chall theintersection of biology, engineering, physics and chemistry. Career opportunitiesin biotechnology, biomedical engineering, pharmaceutical industry, agrobiotechnologyand in the diverse areas of basic s
2941	EE	RFIC DESIGN	EE698L	
2942	BSBE	BIOLOGY LABORATORY	BSE681	
2943	CSE	INTRODUCTION TO COGNITIVE SEIENCE	SE367	

S.No.	Branch	Course Title	Courses ID	Content
2944	BSBE	STRUCTURAL BASIS OF PROTEIN FUNCTION	BSE638	
2945	PHY	INTRODUCTION TO BIOPHOTONICS	PHY610	
2946	CE	EARTHQUAKE ANALYSIS AND DESIGN OF STRUCTURES	CE629	Characteristics of earthquakes; Earthquake response of structures; Concept ofearthquake resistant design; Code provisions of design of buildings; Design ofliquid storage tanks; Liquefaction; Nonenginee
2947	CE	STRUCTURAL DYNAMICS	CE620	Loading: nature of dynamic loading, harmonic, random, types of dynamic loading;Continuous systems: rods (axial vibrations), beams (shear, axial and axialshearflexuralvibrations); Discrete mass systems analysis); Introduction of random vibration: stochastic processes, stochasticanalysis of linear dynamical systems to Gaussian inputs, SDOF, MDOF.
2948	CHM	PRINCIPLES OF ORGANIC CHEMISTRY	CHM609	Stereochemistry, mechanisms of selected reactions, Reactive intermediates,oxidation, Reduction, CC bond formations, synthesis of some useful naturalproducts
2949	CSE	INTRODUCTION TO MATHEMATICAL LOGIC	CS302	
2950	PHY	CONCEPTS OF PLASMA PHYSICS	PHY682	
2951	MME	PHD THESIS	MME799	Ph. D. Thesis
2952	EE	PHD THESIS	EE799	Ph. D. Thesis
2953	HSS	MODERN DRAMA	ENG433	1. Introduction to Drama2. An Overview of the Theatre:3. European Drama & EuropeanSociety4. Ibsen's The Wild Duck5. Strindberg's The Father6. Russian Society in the 191h Century7. Chekov or Godot 11. An Overview of American Drama 12. Miller's Death of a Salesman 13. Williams's A Streetcar Named Desire 1. Weiss. Drama in the Modern World.2. Gassner, J. Treasury of the Theatre.